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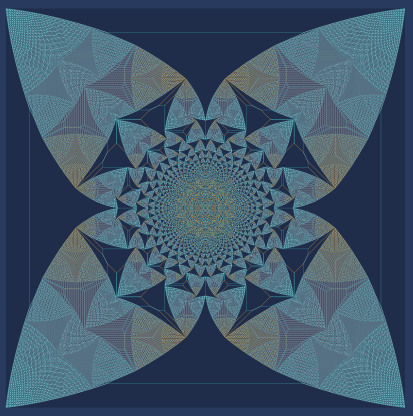
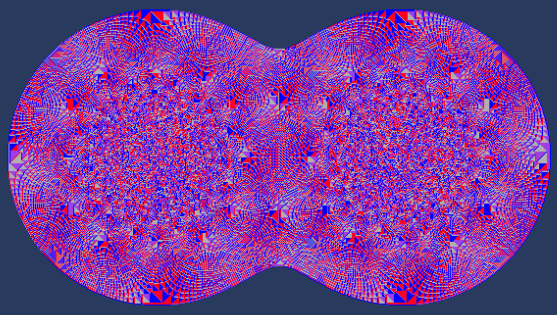
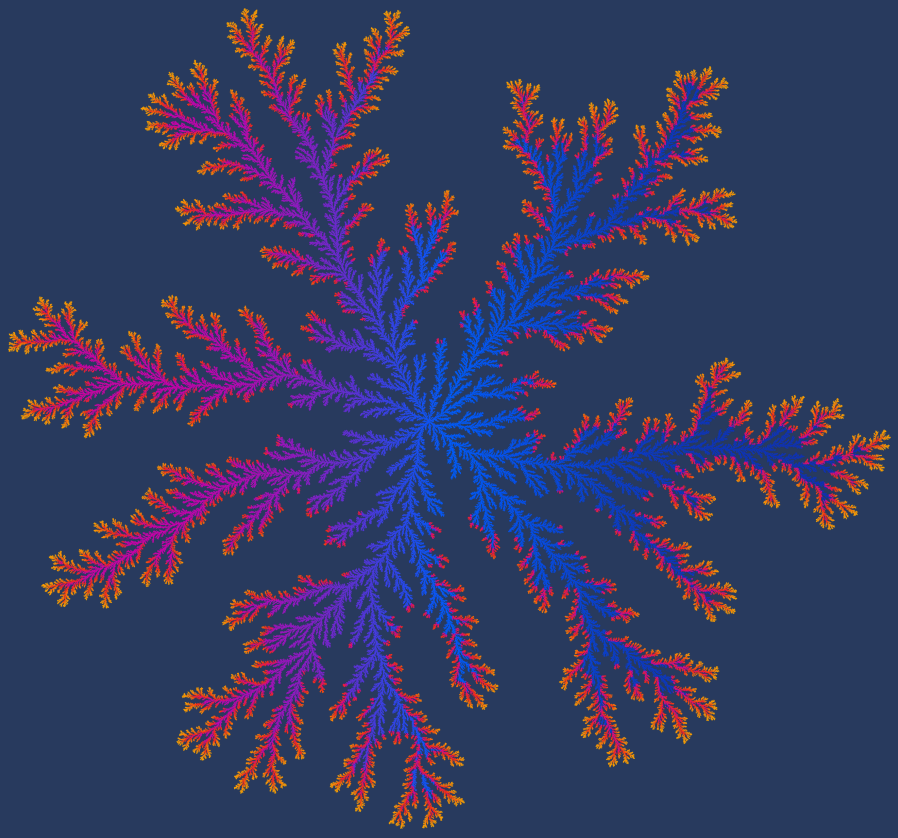
Discrete Laplacian Growth

Sandpiles, Rotors, and DLA







Ahmed Bou-Rabee, Lionel Levine, Yuval Peres

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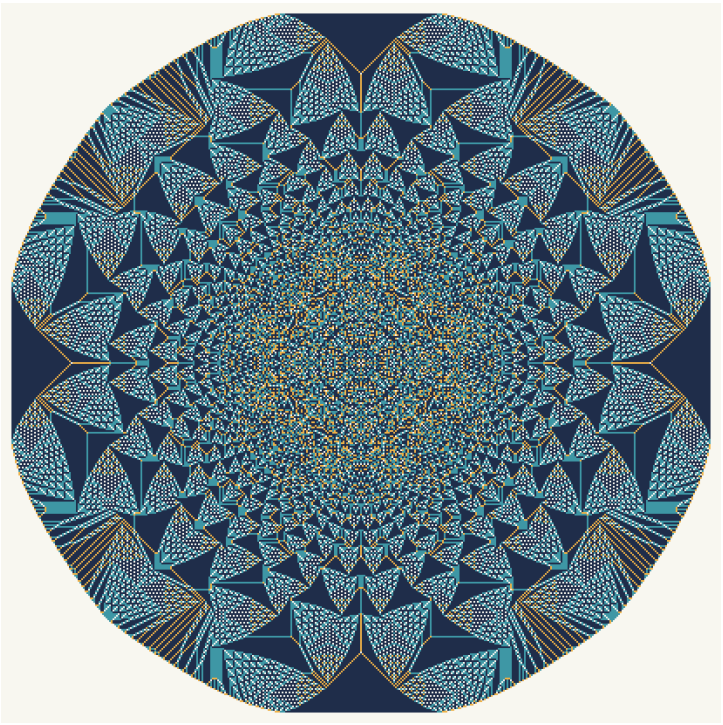

The abelian sandpile formed by stabilizing 3×10^5 grains at the origin of $\mathbb{Z}^2$. Each site is colored by the number of grains it retains: off-white for 0, orange for 1, teal for 2, and navy for 3.

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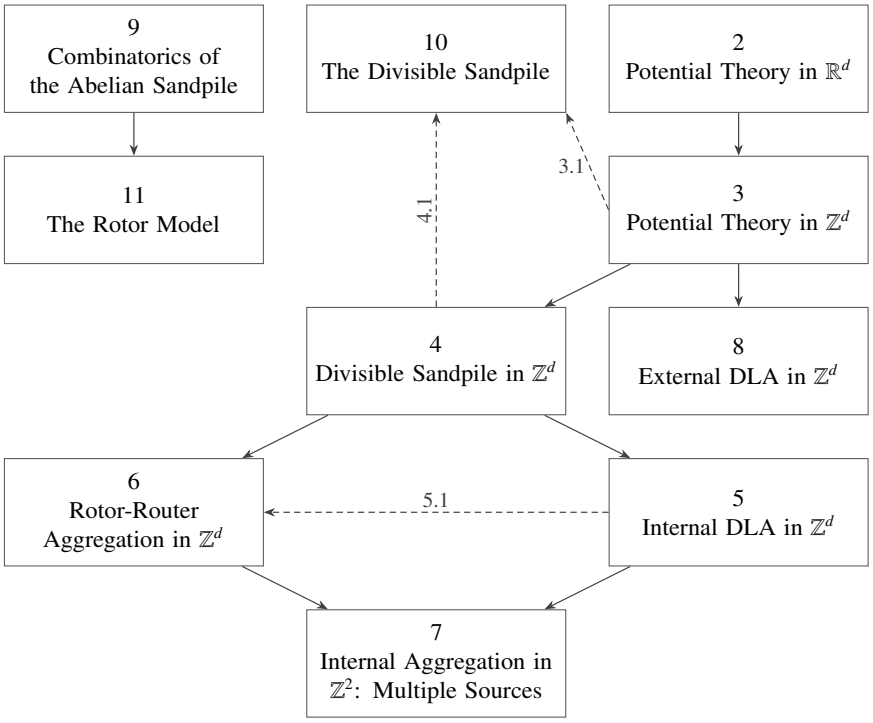

Figure 1: Prerequisites for the main text of Chapters 2–11: an arrow points from a chapter to a later one that uses it, and a prerequisite implied by a chain of solid arrows is not drawn. A dashed arrow lists the sections that suffice. Historical comparisons in the Notes are not prerequisites.

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Preface

Inject a viscous fluid between two closely spaced glass plates, and the region it fills grows at each point of its boundary at a rate given by harmonic measure, the distribution of the point where a Brownian motion started at the injection point first reaches the boundary. Growth of this kind is called Laplacian growth, and the tools used to analyze it come from potential theory, the study of harmonic and superharmonic functions. Here we study lattice models of Laplacian growth, in which particles move on $\mathbb{Z}^d$ by simple local rules, and ask what shapes they form.

In Chapter 1 we describe the models, the abelian and divisible sandpiles, internal DLA, rotor walk and rotor-router aggregation, and external DLA, and state their main properties. The rest of the book has three parts. In the first we develop potential theory from the mean value property, on $\mathbb{R}^d$ in Chapter 2 and on $\mathbb{Z}^d$ in Chapter 3, where the central objects are the Green function of simple random walk and, in the plane, the harmonic measure of a finite set from infinity.

In the second part we prove the shape theorems. In Chapter 4 we define the divisible sandpile, prove its least action principle, show by way of an obstacle problem that a point source fills a ball up to a bounded error in the radius, and deduce that an abelian sandpile from a point source topples a set between two Euclidean balls (Section 4.3). In Chapters 5 and 6 we compare internal DLA and rotor-router aggregation with the divisible sandpile, and in Chapter 7 we start the three models from two equal point sources in the plane and prove that they have the same limit shape, two disks or Neumann's oval, depending on the distance between the sources. In Chapter 8 we prove the bound of Kesten (1987b, p. L29) that a planar external DLA cluster of n particles has radius at most of order $n^{2/3}$, almost surely for all large n, from a discrete Beurling estimate and the harmonic measure from infinity of Chapter 3; a starred section gives the corresponding bounds of order $n^{2/(d+1)}$ in dimensions $d \geq 4$ and $n^{1/2}(\log n)^{1/4}$ in $\mathbb{Z}^3$, where the capacity of the cluster replaces the Beurling estimate.

The third part is on general graphs. In Chapter 9 we develop the combinatorics of the abelian sandpile, including the sandpile group and the least action principle. In Chapter 10 we characterize the final divisible sandpile by quadrature inequalities, relate its odometer to optimal stopping for random walk, and ask when a random initial configuration stabilizes. In Chapter 11

we study rotor walk, whose abelian property leads to a proof of the matrix tree theorem, and ask for which initial rotors it keeps returning to its starting point. A preview of Volume II ends the volume.

We assume familiarity with measure-theoretic probability (conditional expectation, martingales and stopping times), linear algebra and multivariable calculus. No previous knowledge of potential theory or of the models is needed. A few passages use more, and each says what it uses:

- Section 9.4 uses quotients of abelian groups;
- the planar arguments of Subsection 2.2.1 and Section 7.2 use basic complex analysis, and Section 2.6 uses planar Brownian motion together with its strong Markov property and the facts that it almost surely hits every disk and never hits a given point;
- Exercises 2.49–2.53 use Brownian motion;
- the solution of Exercise 2.42 uses the identity theorem for real analytic functions;
- Subsection 8.2.3 uses recurrence and the potential kernel of planar random walks with bounded steps.

A star after the title of a section, or after the number of an exercise, marks material that can be skipped on a first reading. We write $\mathbb{N} := \{0, 1, 2, \ldots\}$.

After Chapter 1, the chapters can be read in order, and the prerequisites for the main text are in Figure 1, before this preface. The reader interested in the shape theorems can read Chapters 2 and 3, omitting the harmonic measure from infinity in Section 3.7, and then Chapters 4 to 7, and may skip the proof in Section 3.6 that the Green function is radial to leading order (Theorem 3.9). The reader interested in external DLA can pass from Chapters 2 and 3, including Section 3.7, directly to Chapter 8. The third part can be read afterward.

Each chapter ends with exercises and Notes. Some exercises supply arguments that the text points to, and others give examples and generalizations. We solve the exercises the text uses, and a separate document solves them all. The Notes give the sources of the results and suggest further reading.

The research of Yuval Peres was supported by National Natural Science Foundation of China grant RFIS-W2531011.

Chapter One

Models of Laplacian Growth

Pile n grains of sand on the origin of the square grid $\mathbb{Z}^2$. A site that holds four grains or more is unstable, and we let it topple by sending one grain to each of its four neighbors (North, East, South and West). A toppling can make its neighbors unstable in turn, so we keep toppling until every site holds at most three grains. The rule is the same in all four directions, yet the pile of 3×10^5 grains in the frontispiece is far from round, and it is covered with self-similar patterns that are still only partly understood.

Change the rule a little and the patterns disappear. If the sand is replaced by a liquid that can be divided as finely as we like, or if each grain in turn performs a random walk from the origin until it finds an empty site, then the occupied region is very nearly a disk (Figure 1.1). Why should a cluster grown on the square lattice be round? The answer, which takes up much of this book, is that harmonic functions describe these processes on large scales, and the harmonic functions of the lattice are nearly as symmetric as those of the plane.

Processes of this kind are lattice models of **Laplacian growth**, in which a region expands at each point of its boundary at a rate given by harmonic measure (the distribution of the point where a Brownian motion started at the source first reaches the boundary). In the continuum, Laplacian growth describes the Hele-Shaw flow of a viscous fluid injected between two plates, the growth of bacterial colonies, and the melting of ice in the quasistatic Stefan problem. The tools used to analyze it come from potential theory, which is the study of harmonic and superharmonic functions, and which grew out of gravitation and electrostatics. Three themes run through this chapter. First, in the abelian and divisible sandpiles the outcome does not depend on the order in which the sites topple, and the number of topplings at each site is the smallest among all ways of reaching a stable state (Sections 1.1 and 1.2); internal DLA has the same freedom in the order of its particles (Section 1.3). Second, internal DLA, rotor-router aggregation and the divisible sandpile, started from n particles at the origin of $\mathbb{Z}^d$ with $d \geq 2$, fill a Euclidean ball in the scaling limit, and from several sources in the plane they still have one common shape, but the abelian sandpile does not (Section 1.6). Third, in external DLA, where the particles come from infinity, the radius of a planar cluster of n particles is almost surely at most of order $n^{2/3}$ for all large n

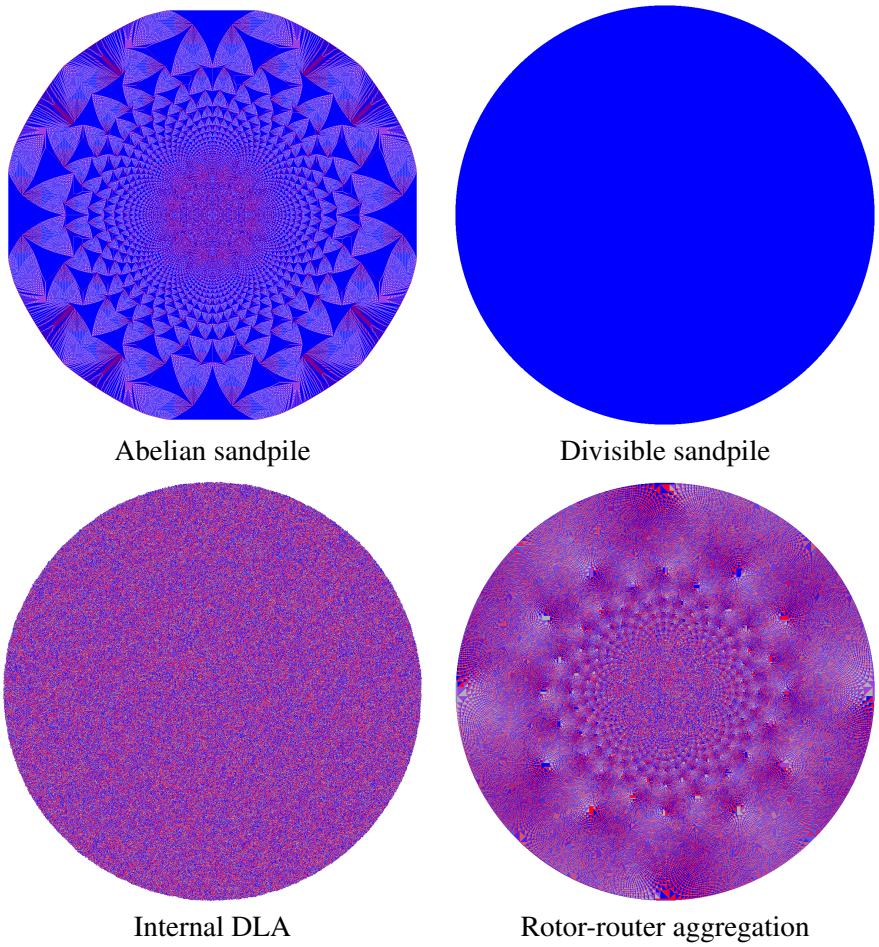


Figure 1.1: Four discrete Laplacian growth models, each started from a point source at the origin of $\mathbb{Z}^2$ and drawn at the same scale: the abelian sandpile of 2.3×10^6 grains, colored by the number of grains at each site, the divisible sandpile of mass 10^6, and internal DLA and rotor-router aggregation (all rotors initially North) of 10^6 particles each, colored by the direction of the last step taken from each site, or as North if there was none. We will prove that three of the four models have a circular scaling limit.

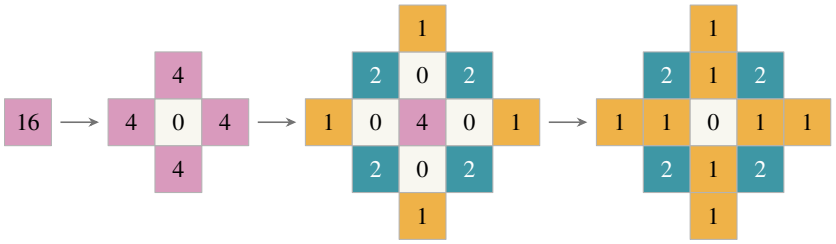


Figure 1.2: Sixteen grains at the origin of $\mathbb{Z}^2$ (left) reach the stable sandpile (right) in three rounds, in each of which every unstable site (pink) topples as many times as its grains allow. The numbers count the grains at each site.

(Section 1.5).

1.1 THE ABELIAN SANDPILE

We view the set $\mathbb{Z}^d$ of points with integer coordinates in $\mathbb{R}^d$ as a graph in which points x and y are neighbors, written $x \sim y$, if $x - y = \pm\mathbf{e}_i$ for some i, where $\mathbf{e}_1, \ldots, \mathbf{e}_d$ are the standard basis vectors. We write $\mathbf{0}$ for the origin and $|x|$ for the Euclidean norm. In the **abelian sandpile** on $\mathbb{Z}^d$, a site with fewer than $2d$ grains is **stable**. Sites with at least $2d$ grains are **unstable**, and can **topple** by sending one grain to each of the $2d$ neighboring sites. Physicists study the model as an example of self-organized criticality, and combinatorialists, who found it independently, call it **chip-firing**. Start with 16 grains at the origin of $\mathbb{Z}^2$ and topple unstable sites until none is left. The origin topples four times, each of its neighbors then topples once and returns a grain to the origin, the origin topples once more, and the grains come to rest on 12 sites (Figure 1.2).

Does it matter which unstable site we topple first? We prove in Chapter 9 that, on every finite connected graph with a sink (defined below), the final sandpile does not depend on the order of legal topplings. In fact, more is true: the **odometer** $u(x)$, the number of times x topples, does not depend on the order either. This **abelian property** explains the name of the model. Let σ be the initial sandpile and σ° the stable sandpile that it reaches. A site that topples $u(x)$ times sends $u(x)$ grains to each neighbor and receives $u(y)$ grains from each neighbor y, so

$$\sigma^\circ = \sigma + \Delta^{::} u\,,$$

where $\Delta^{::}$ is the **discrete Laplacian**

$$\Delta^{::} f(x) := \sum_{y \sim x} (f(y) - f(x))\,.$$

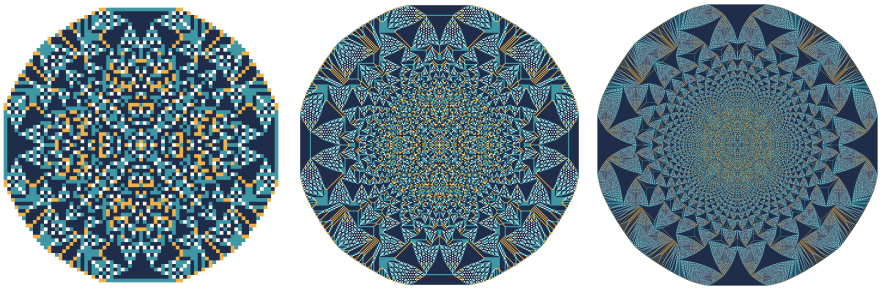
The **least action principle** says that, if we allow illegal topplings as well, of sites holding fewer than $2d$ grains, then every sequence of topplings that ends in a stable sandpile topples each site x at least $u(x)$ times. It is stronger than the abelian property, and we use it again and again, since each such sequence gives an upper bound for the odometer.

On a finite connected graph the sand needs somewhere to go, so one vertex is a **sink**, where arriving grains disappear. Adding grains one at a time and stabilizing after each addition leads from one stable sandpile to another. The sandpiles that can be reached in this way from every sandpile are called **recurrent**, and under addition followed by stabilization they form a finite abelian group, the **sandpile group** (which is the second reason for the name of the model), whose order is the determinant of the reduced Laplacian (the Laplacian matrix of the graph with the row and column of the sink deleted). The burning algorithm of Dhar (1990, p. 1615) recognizes a recurrent sandpile by a single further stabilization. We prove these facts in Chapter 9, the last one on every finite strongly connected directed graph in which each vertex has as many edges entering as leaving, and we also give criteria for two kinds of backgrounds on $\mathbb{Z}^d$: **robust** ones, on which adding finitely many grains makes finitely many sites topple, and **explosive** ones, on which it can make every site topple. The simplest example of an explosive background is three grains at every site of $\mathbb{Z}^2$, to which the addition of a single grain causes every site to topple infinitely many times (Section 9.10). In Section 4.3 we use the divisible sandpile of the next section to show that, for $d \geq 2$, the sites toppled by n grains at the origin of $\mathbb{Z}^d$ lie between two Euclidean balls of volume of order n (see Figure 4.7 for $n = 10^5$).

These bounds do not determine the shape (Figure 1.3). Pegden and Smart (2013, Theorem 1.1 and Remark 4.3) proved that the sandpile of n grains at the origin of $\mathbb{Z}^d$, rescaled by $n^{1/d}$, converges in the sense of local averages as $n \to \infty$, and Volume II proves that the limit is not a ball (see the preview on page 379). What happens if the grains can be divided as finely as we like?

1.2 THE DIVISIBLE SANDPILE

Now replace the grains by a liquid. Pour $n > 1$ grams of maple syrup onto the middle square of a very large waffle, each square of which holds one gram before it overflows and shares the excess equally among its four neighbors. Which squares fill up? On $\mathbb{Z}^d$ this process is the **divisible sandpile**, in which a site with mass $s > 1$ topples by distributing the excess mass $s - 1$ equally among its $2d$ neighbors. A site can receive mass again after it has toppled, so the process usually needs infinitely many topplings (Figure 1.4), but the masses converge, and in the limit every partially filled site has a filled



$n = 10^4$ $n = 10^5$ $n = 10^6$

Figure 1.3: The stable sandpiles reached from n grains at the origin of $\mathbb{Z}^2$, scaled to the same width and colored by the number of grains as in Figure 1.2. The same large patches appear at each size, with finer patterns between them as n grows.

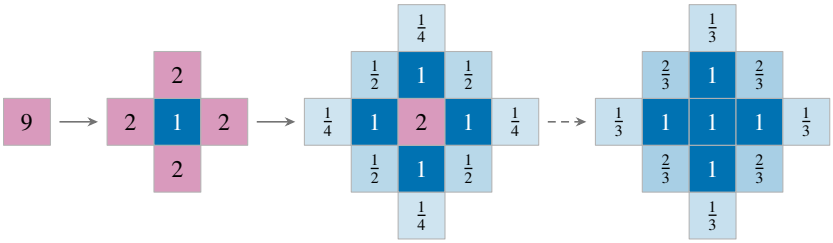


Figure 1.4: Mass 9 at the origin of $\mathbb{Z}^2$ (left), after one and after two rounds in which every site with mass greater than one (pink) topples, and the limit (right), with the full sites dark blue. The second round returns mass to the origin, and the dashed arrow stands for infinitely many further rounds.

neighbor.

The odometer $u(x)$ is now the total mass that x sends to each of its neighbors, and as for the abelian sandpile, the final configuration is $\sigma + \Delta^{::}u$. By the least action principle, the odometer is the smallest nonnegative function f with $\sigma + \Delta^{::}f \le 1$. The odometer of the abelian sandpile solves the same problem among nonnegative integer-valued functions, with the threshold 1 replaced by $2d - 1$ (the largest number of grains that a stable site holds). So the divisible sandpile, after all masses are divided by $2d - 1$, is the abelian sandpile without its integrality constraint, and it is much easier to analyze. How close to a ball are the fully occupied sites? Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r = (n/\omega_d)^{1/d}$, so that the ball of radius r has volume n. For $d \ge 2$, the fully occupied sites contain every lattice point at distance less than $r - C$ from the origin and none at distance more than $r + C$, where C depends only on d (Figure 1.5). So the error in the radius stays bounded however much syrup we pour.

On an infinite graph a new question appears: does the redistribution stop at all? Start from mass one at every vertex of a locally finite connected graph,

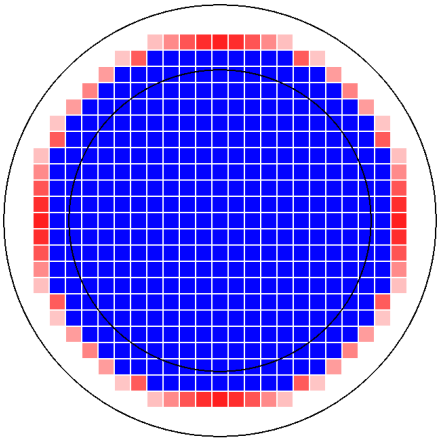


Figure 1.5: The divisible sandpile in $\mathbb{Z}^2$ started from mass 400 at the origin: a square is blue if its site is full, and red if it is partially filled, the deeper the red the larger the mass. Every red square lies between the circles of radii $r-2$ and $r+2$, where $\pi r^2 = 400$.

and add an extra mass at one vertex. This configuration stabilizes if and only if simple random walk on the graph is transient (see Figure 10.2 in Chapter 10 for the extra mass spreading like the walk), so on $\mathbb{Z}^2$ the extra mass cannot be absorbed, while on $\mathbb{Z}^3$ it can. If the masses are independent, identically distributed and bounded, with mean μ, on an infinite connected graph of bounded degree, then they stabilize almost surely if $\mu < 1$, and almost surely do not stabilize if $\mu \geq 1$ and the masses are not almost surely equal to one. We prove the least action principle (Section 4.1) and the shape theorem in Chapter 4, and the results on infinite graphs in Chapter 10. What if the liquid is replaced by particles that move at random?

1.3 INTERNAL DLA

Starting with n particles at the origin of $\mathbb{Z}^d$, where $d \geq 2$, let each particle in turn perform a simple random walk until reaching an unoccupied site, where it stays (Figure 1.6). This procedure is called **internal diffusion-limited aggregation**, or **internal DLA**. The resulting set of n occupied sites is random, but Lawler, Bramson, and Griffeath (1992, Theorem 1) proved that after rescaling by $n^{1/d}$ it converges almost surely to a Euclidean ball. The cluster of 10^6 particles in Figure 1.1 (bottom left) contains every site at distance less than 560.7 from the origin and no site at distance more than 567.5, while the disk of area 10^6 has radius 564.2.

Internal DLA has an abelian property, which holds deterministically in a certain sense. Put an infinite stack of random arrows at each site, each

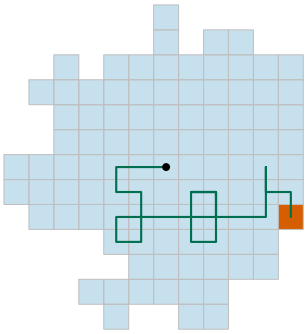


Figure 1.6: Internal DLA in $\mathbb{Z}^2$: the 101st particle walks from the origin (dot) through the cluster of the first 100 particles (light blue) until it reaches an empty site (vermilion), where it stays.

pointing to a neighbor, and let the kth particle to leave a site follow the kth arrow there. From the same stacks, every complete sequence of legal moves (each moving a particle from a site with at least two) produces the same cluster, so we may release the particles in every order, or pause some of them and restart them later, without changing the law of the cluster. With this freedom we show that, with $r = (n/\omega_d)^{1/d}$ as in Section 1.2, the cluster of n particles contains the ball of radius r minus an error of order $r^{1/3}$ and lies in the ball of radius r plus an error of order $r^{1-2/(3d)}$, both up to logarithmic factors, with high probability and also almost surely for all large n. In the plane the fluctuations are in fact at most logarithmic in the radius (Figure 1.9, left), as Jerison, Levine, and Sheffield (2012, Theorem 1) and Asselah and Gaudillière (2013b, Remark 1.2) proved. In Chapter 5 we prove the abelian property, and the bounds with polynomial errors by comparison with the divisible sandpile. On $\mathbb{Z}$ the cluster is an interval, which is off center by order $\sqrt{n}$ sites in root mean square (Exercise 1.3). Can we remove the randomness from the walks altogether?

1.4 ROTOR WALK AND ROTOR-ROUTER AGGREGATION

At each site of the square grid $\mathbb{Z}^2$ is a **rotor**, an arrow pointing North, East, South or West. In **rotor walk**, during each time step the rotor at the walker's location rotates 90 degrees clockwise, and then the walker takes a step in the direction of the newly rotated rotor. Rotor walk has been introduced several times: as a model of self-organized criticality, under the name Eulerian walkers; as a way for robots to explore an unknown network; and as a way to take the randomness out of random walk. If the walker has left a site m times, then it has moved from there to each neighbor either $\lfloor m/4 \rfloor$ or $\lceil m/4 \rceil$ times, as a random walk does on average. On $\mathbb{Z}^d$ each site serves its $2d$ neighbors

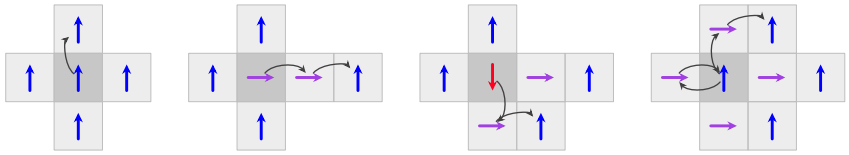


Figure 1.7: Rotor-router aggregation of 5, 6, 7 and 8 particles started at the origin (dark square) of $\mathbb{Z}^2$, with rotors turning clockwise and all initially North. Each arrow is the rotor at its site, blue for North, purple for East, red for South and gray for West, and the curved arrows trace the walk of the last particle.

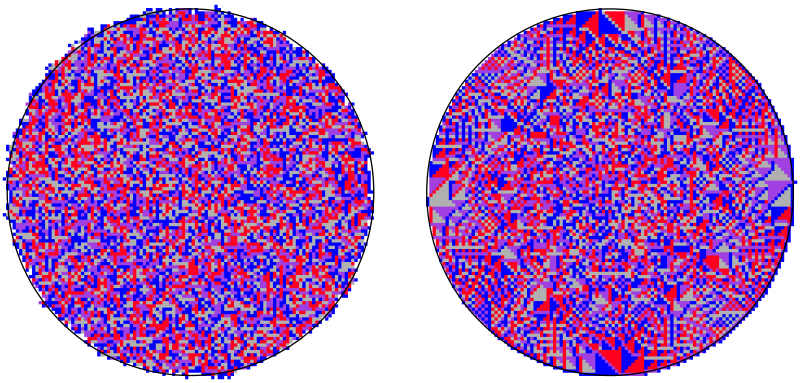


Figure 1.8: Internal DLA (left) and rotor-router aggregation (right) of 10^4 particles, started and colored as in Figure 1.1 (by the last step from each site: North blue, East purple, South red, West gray), with the circle of area 10^4. The internal DLA cluster contains every site within distance 54.5 of the origin and none beyond 58.9; for the rotor-router aggregate these radii are 55.7 and 57.3.

in a prescribed cyclic order, the **rotor mechanism** at that site.

In **rotor-router aggregation**, we start with n particles at the origin, and each in turn performs rotor walk until it reaches a site not occupied by an earlier particle. The rotors are not reset between walks, so the steps of all the particles from each site cycle through its neighbors. For example, if all rotors initially point North, then the first particle stays at the origin, the second turns the rotor at the origin to East and stops at $(1, 0)$, and the third turns it to South and stops at $(0, -1)$. After five particles every rotor points North again, and the sixth particle turns the rotor at the origin to East, moves to $(1, 0)$, turns the rotor there to East as well, and stops at $(2, 0)$ (Figure 1.7).

Simulations show that these aggregates are extraordinarily close to circular (Figure 1.8), and it is natural to ask why. In the simulations of Figure 1.9, with 10^6 particles and all rotors initially North, the distance of the jth particle from the origin minus the radius of the disk of area j ranges from -0.95 to 0.96 for rotor-router aggregation, and from -5.44 to 5.34 for internal DLA. Part of the answer is that the particles leaving each site are split equally among its neighbors, as in the divisible sandpile, up to less than one particle

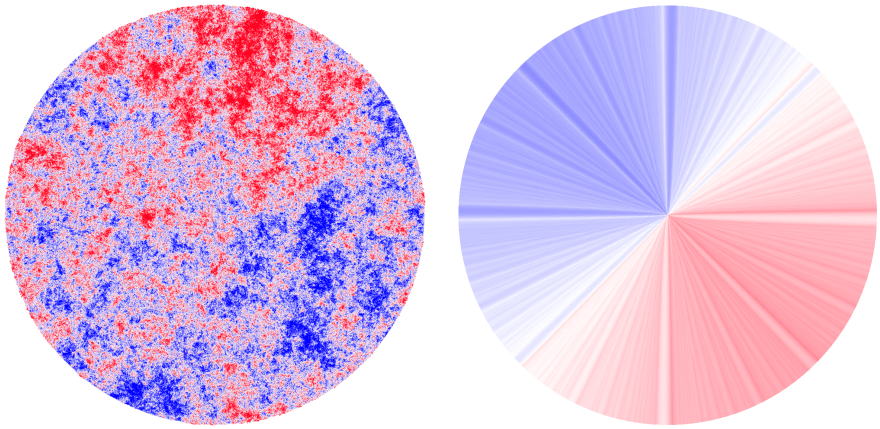


Figure 1.9: The internal DLA cluster (left) and the rotor-router aggregate (right) of 10^6 particles of Figure 1.1, with the site of the jth particle colored by its distance from the origin minus the radius of the disk of area j (white for 0, the deepest blue for -2 or less, the deepest red for 2 or more). The rotor-router panel is blue in its upper left half and red in its lower right half because that aggregate is very nearly a disk centered at $(1/2, -1/2)$ rather than at the origin.

per edge. We prove in Chapter 6 that there is a constant C depending only on d such that, for $d \geq 2$, all initial rotors and all rotor mechanisms on $\mathbb{Z}^d$, the aggregate of n particles contains the ball of radius $r - C \log r$ and lies in the ball of radius $r + C(\log r)^2$, for all sufficiently large $r = (n/\omega_d)^{1/d}$ (with ω_d as in Section 1.2).

What happens if a single rotor walk runs forever? Every time the walker returns to a site, it turns the rotor it left there and follows it, so the initial rotors can have a lasting effect. On $\mathbb{Z}^d$ in every dimension, a suitable choice of initial rotors makes rotor walk return to the origin infinitely often, while it escapes to infinity if the rotors along an infinite path point back toward the start (see Figure 11.6 in Chapter 11 for both on $\mathbb{Z}^2$). For all initial rotors, rotor walk on $\mathbb{Z}^d$ visits at least a constant times $t^{d/(d+1)}$ sites in its first t steps, and the clockwise rotor walk on $\mathbb{Z}^2$ with independent uniform initial rotors is almost surely recurrent (Figure 11.15). Rotor walk also leads to a proof of the matrix tree theorem. On a finite connected graph with a sink, the sandpile group acts freely and transitively on the spanning trees oriented toward the sink, so the number of these trees is the order of the group, the determinant of the reduced Laplacian. We prove these statements in Chapter 11, which studies rotor walk on general graphs. In all of the models so far the particles start inside the cluster; what if they come from outside?

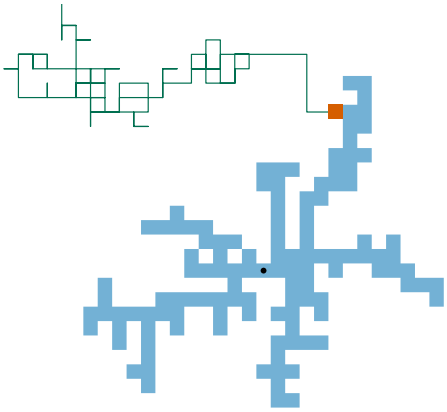


Figure 1.10: External DLA in $\mathbb{Z}^2$: a walk from far away (green, last steps only) first reaches a site next to a cluster of 120 sites (light blue) at the vermilion site, near the end of an arm, which joins the cluster.

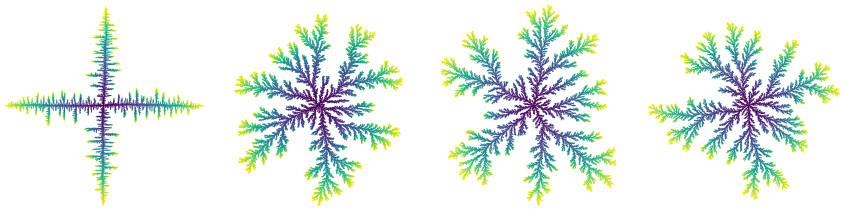


Figure 1.11: External DLA clusters of 10^9 sites on four planar lattices, each site colored from dark to light by the time at which it joined the cluster. On the square lattice the arms follow the two axes, and on the other three lattices they spread in more directions.

1.5 EXTERNAL DLA

Start with a single occupied site at the origin of $\mathbb{Z}^2$, release a particle very far away, and let it perform simple random walk until it first reaches a site next to the cluster, where it stays; then release the next particle (Figure 1.10). This process is called **external diffusion-limited aggregation**, or usually just **DLA**. It is a model of irreversible aggregation driven by diffusion, and its clusters grow long branching arms. The tip of an arm is much easier for an incoming walk to reach than a site deep between two arms, and growth at a tip exposes it further, so the arms keep growing while the sites between them are screened. On $\mathbb{Z}^2$ the arms of very large clusters grow along the two coordinate axes (Figure 1.11; see Figure 8.4 in Chapter 8 for a cluster of 10^{10} sites). Of all the models in this book it is the one we understand least. Off-lattice simulations suggest that a cluster of n particles has radius of order $n^{0.585}$, but it is not even known that the radius grows like a power of n.

The probability that a walk started at infinity first hits a finite set K at a site x is the **harmonic measure from infinity** $\mathcal{H}_K(x)$, and the next site of a DLA cluster is chosen according to the harmonic measure of its outer boundary. To grow quickly, a cluster must put much of its harmonic measure on a few tips, and the **discrete Beurling estimate** limits how much. It states that there is an absolute constant C such that, for every finite connected set $K \subset \mathbb{Z}^2$ of diameter $\mathrm{diam}(K)$ and every $x \in K$,

$$\mathcal{H}_K(x) \le \frac{C}{\sqrt{1 + \mathrm{diam}(K)}} .$$

The exponent $\frac{1}{2}$ cannot be improved, since the endpoint of a straight segment of n sites has harmonic measure of order $n^{-1/2}$ (see Figure 3.16 in Chapter 3 for harmonic measure on a segment of 11 sites). Kesten (1987b) deduced from it that, almost surely, the radius of the DLA cluster of n particles is at most a constant times $n^{2/3}$ for all large n, and the bound that we prove has an exponential tail. This does not identify the exponent, since simulated clusters grow more slowly. In Chapter 8 we prove the Beurling estimate and the bound on the radius. Why do the other models, unlike this one, produce balls?

1.6 UNIVERSALITY

The divisible sandpile, internal DLA and rotor-router aggregation move mass by three different rules, one by exact averaging, one at random and one by rotation, yet from one source each fills a Euclidean ball (Sections 1.2–1.4). The Euclidean ball as a limiting shape is not too hard to guess, although proving it is another matter. But what if the particles start at two different points? Start n particles at each of two sites $(\pm a, 0)$ of $\mathbb{Z}^2$, where a is of order $\sqrt{n}$. Each source alone would fill a disk of area about n. In the scaling limit, if the sources are far apart, then the two disks stay disjoint, and if they are close, then the disks merge and the merged shape is no longer round (Figure 1.12), but the three models still have the same limit shape (Figure 1.13). We prove this in Chapter 7 for two equal sources, where the limit is either two disks or a domain bounded by a quartic curve, Neumann's oval, depending on the distance between the sources (see Figure 7.2 in that chapter for internal DLA from two sources with the quartic boundary of its limit). Levine and Peres (2010, Theorem 1.4) proved that the three models have a common limit for every finite collection of point sources, and for distinct sources its boundary lies on an algebraic curve of degree twice the number of sources (Gustafsson and Shapiro, 2005, Section 4).

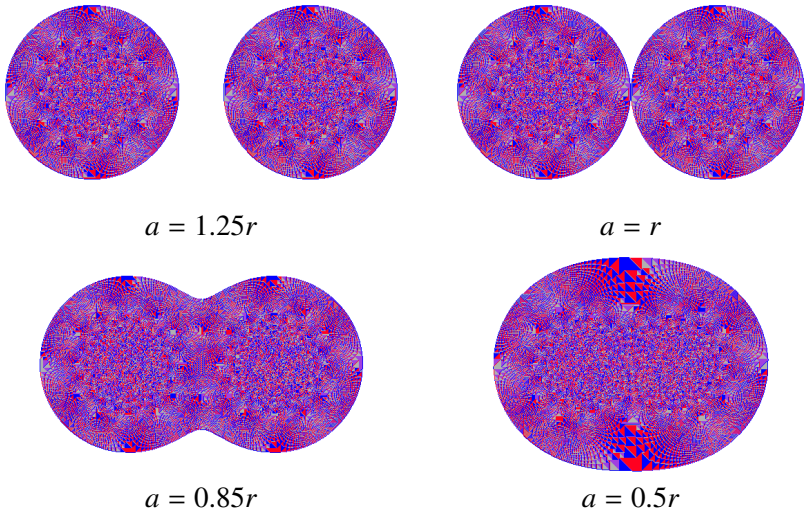


Figure 1.12: Rotor-router aggregation of $n = 5 \times 10^4$ particles from each of the sites $(\pm a, 0)$, with all rotors initially North, at one scale and colored as in Figure 1.1; here $r = \sqrt{n/\pi}$ is the radius of a disk of area n. The disks touch for $a = r$, and for $a < r$ the cluster lies within about one site of Neumann's oval, bounded by the quartic $(x^2 + y^2)^2 - 2r^2(x^2 + y^2) - 2a^2(x^2 - y^2) = 0$ in lattice coordinates.

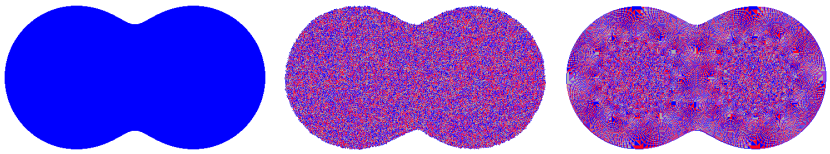


Figure 1.13: The divisible sandpile of mass 5×10^4, and internal DLA and rotor-router aggregation of 5×10^4 particles, at each of the sites $(\pm a, 0)$ with $a = 0.85r$ as in Figure 1.12, drawn at one scale and colored as in Figure 1.1. The three clusters have the same shape, Neumann's oval.

To see why the three models agree and why their clusters are balls, we start with harmonic functions. A function f on $\mathbb{Z}^d$ is **discrete harmonic** at x if $\Delta^{::} f(x) = 0$, that is, if $f(x)$ is the average of f over the $2d$ neighbors of x, and **discrete superharmonic** at x if $\Delta^{::} f(x) \le 0$. If h is discrete superharmonic at x, then a divisible sandpile toppling at x cannot increase the total weight $\sum_y h(y)\sigma(y)$ of a mass σ, and if h is harmonic at x, then the toppling leaves this weight unchanged. Hence, if σ is a finitely supported nonnegative initial mass with final configuration σ_∞, and $h : \mathbb{Z}^d \to \mathbb{R}$ is discrete superharmonic at every site, then

$$\sum_{x \in \mathbb{Z}^d} h(x)\sigma_\infty(x) \le \sum_{x \in \mathbb{Z}^d} h(x)\sigma(x)\,, \tag{1.1}$$

with equality when h is harmonic at every site (Exercise 4.2 asks for the details). The continuous analogue of (1.1) is the mean value property, by which spreading a point mass $\omega_d r^d$, the volume of the ball of radius r, uniformly over this ball leaves its integral against a function harmonic in a neighborhood of the closed ball unchanged, and cannot increase its integral against a function superharmonic there. The divisible sandpile spreads mass in the same way on the lattice, which suggests that the region it fills should be nearly a ball. The random walks of internal DLA spread the particles in the same way on average, and the rotors do so up to the error described in Section 1.4. From two equal sources the limit shape is called a quadrature domain, because the integral over it of every function that is harmonic near its closure equals the sum of the values of the function at the two sources (Chapter 7).

The abelian sandpile is the exception (Figure 1.1). Its odometer is characterized by the same least action principle, but only among integer-valued functions (Section 1.2), and the integrality matters on large scales, since the scaling limit of Section 1.1 is not a ball, as Volume II proves.

To prove that the three models fill balls we need functions with a prescribed Laplacian, such as the odometer. The key tool for this purpose is the **Green function**, whose Laplacian is zero except at the origin. In dimensions $d \geq 3$ it is the expected number of visits to x by a simple random walk started at the origin, divided by $2d$, and it decays like $|x|^{2-d}$. The case $d = 2$ is more delicate because simple random walk is recurrent, and the role of the Green function is played by the potential kernel, which grows like $\frac{1}{2\pi} \log |x|$. These leading terms are spherically symmetric, although the lattice is not, and the agreement is already close at small distances (see Figure 3.5 in Chapter 3 for level sets of the potential kernel and circles). For example, at every site $x \neq \mathbf{0}$ of $\mathbb{Z}^2$ with $|x| \leq 10$, the potential kernel differs from $\frac{1}{2\pi} \log |x| + \kappa$, where $\kappa \approx 0.257$ is the constant term in its asymptotics, by less than 0.008. This near symmetry is why the three models fill a Euclidean ball, and why with several sources they fill the same quadrature domain. We develop potential theory in $\mathbb{R}^d$ in Chapter 2 and in $\mathbb{Z}^d$ in Chapter 3.

EXERCISES

Exercise 1.1. For an abelian sandpile $\sigma : \mathbb{Z}^d \to \mathbb{N}$ with finitely many grains, where $d \geq 1$ (Section 1.1), prove that every legal toppling sequence started from σ topples only finitely many sites.

Exercise 1.2. For the abelian sandpile on $\mathbb{Z}^d$, where $d \geq 1$, let $\sigma : \mathbb{Z}^d \to \mathbb{N}$ be a sandpile that stabilizes, and let σ° be the stable sandpile that it reaches. Prove that if $A \subset \mathbb{Z}^d$ is a finite set of sites all of which topple, then the number

of grains remaining in A in σ° is at least the number of edges of $\mathbb{Z}^d$ with both endpoints in A.

Exercise 1.3. Consider internal DLA as in Section 1.3, but on $\mathbb{Z}$ and started from the origin. After $n \geq 1$ particles have stopped, the cluster is an interval $A_n = \{-a_n, \ldots, b_n\}$, where $a_n + b_n = n - 1$.

(a) Show that, given A_n, the next particle stops at $-a_n - 1$ with probability $(b_n + 1)/(n + 1)$.

(b) Show that $\mathbb{E}[(a_n - b_n)^2] = (n + 1)/3$ for every $n \geq 2$.

Thus the cluster is off center by order $\sqrt{n}$ sites in root mean square, whereas the divisible sandpile on $\mathbb{Z}$ is symmetric about the origin, and rotor-router aggregation with all rotors initially pointing West is symmetric up to one site (Exercise 6.4).

Exercise 1.4. Here is one precise meaning of the convergence of rescaled internal DLA clusters to a ball in Section 1.3. Let $d \geq 1$ be an integer. Given $D_r \subset r^{-1}\mathbb{Z}^d$ for each real $r \geq 1$, and an open set $\Omega \subset \mathbb{R}^d$, we write $D_r \to \Omega$ if for all $\epsilon > 0$ and all sufficiently large r, the set D_r contains every point $x \in r^{-1}\mathbb{Z}^d$ such that $\overline{\mathrm{B}}(x, \epsilon) \subset \Omega$, and every point of D_r lies at distance less than ϵ from Ω. Show that if Ω_1 and Ω_2 are open subsets of $\mathbb{R}^d$ such that $D_r \to \Omega_1$ and $D_r \to \Omega_2$, then $\bar{\Omega}_1 = \bar{\Omega}_2$. The limit need not be unique: if D_r is the set of points of $r^{-1}\mathbb{Z}^2$ in the open unit disk, then D_r converges both to this disk and to the disk with the segment $[0, 1) \times \{0\}$ removed.

NOTES

The two-dimensional sandpile model was introduced by Bak, Tang, and Wiesenfeld (1987, pp. 381–384) as a model of self-organized criticality. Its fundamental properties were developed by Dhar (1990, pp. 1613–1614), who formulated it on arbitrary finite systems, proved that stabilization does not depend on the toppling order, and introduced the burning algorithm on p. 1615. The same toppling rule had appeared earlier in the probabilistic abacus of Engel (1975, 1976), who used it to compute absorption probabilities of finite Markov chains exactly, and combinatorialists later studied it independently as the chip-firing games of Björner, Lovász, and Shor (1991) and Björner and Lovász (1992). Fey, Levine, and Peres (2010, Section 1) defined robust and explosive backgrounds and observed that the background of three grains at every site of $\mathbb{Z}^2$ is explosive; their Lemma 2.3 is the least action principle for sandpiles on $\mathbb{Z}^d$ that stabilize, and their Theorems 4.1 and 4.2 give criteria for explosion and for robustness. Pegden and Smart (2013, Theorem 1.1 and Remark 4.3) proved that the abelian sandpile formed by n grains at the

origin of $\mathbb{Z}^d$, rescaled by $n^{1/d}$, converges in the sense of local averages as $n \to \infty$, and identified the limit as the Laplacian of the solution of an obstacle problem.

Scott Sheffield suggested the divisible sandpile model to Levine and Peres in a conversation around 2006, and Levine and Peres (2009) introduced it. For the waffle metaphor we thank Sara Romik; our waffle follows Section 2 of the survey of Levine and Peres (2017). The stabilization criterion for one extra mass on a graph with mass one at every vertex is Lemma 2.9 of Levine, Murugan, Peres, and Ugurcan (2016a). The phase transition for independent initial masses on graphs of bounded degree is a special case of Theorems 1.2–1.3 of Bou-Rabee, Peres, and Sava-Huss (2026), which assume less: below mean one, a finite pth moment of the positive part for at least one $p > 3$, and at mean one, either a positive finite variance or a law symmetric about one, not a point mass. The shape theorem for the divisible sandpile is Theorem 1.3 of Levine and Peres (2009).

Internal DLA appears on pp. 96–97 of Diaconis and Fulton (1991), and the stacks on pp. 97–98. The shape theorem for internal DLA from a point source is Theorem 1 of Lawler, Bramson, and Griffeath (1992, p. 2119). In the plane, Theorem 1 of Jerison, Levine, and Sheffield (2012) shows that, for every $\gamma > 0$, the probability that the fluctuations of internal DLA about the disk of radius r exceed a constant times $\log r$ is at most $r^{-\gamma}$, where the constant depends on γ; this bound also holds almost surely for all large n. By another method, Asselah and Gaudillière (2013a, Theorem 1.2) bound the inner error by a constant times $\log r$ and the outer error by a constant times $(\log r)^2$ in every dimension $d \geq 2$, and Asselah and Gaudillière (2013b, Remark 1.2) obtain a constant times $\log r$ on both sides in the plane; the Notes to Chapter 5 give the precise statements.

Rotor walk was introduced by Priezzhev, Dhar, Dhar, and Krishnamurthy (1996) under the name Eulerian walkers, in connection with self-organized criticality. With one walker and fixed tie-breaking, the algorithm ANT-WALK-1 of Wagner, Lindenbaum, and Bruckstein (1996, Section 2.1), for a robot exploring an unknown network, is the same walk. In their Introduction, Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) list later rediscoveries, including Propp's derandomization of random walk, and Levine and Peres (2009, p. 2) record that Propp introduced rotor-router aggregation, observed from simulations that its aggregates are extraordinarily close to circular, and asked why. Theorem 1.1 of Levine and Peres (2009) gives the inner bound of the shape theorem for rotor-router aggregation (Theorem 6.6) and an outer error of order $r^{1-1/d} \log r$; the Notes to Chapter 6 describe the later improvements of the outer error. The lower bound $t^{d/(d+1)}$ on the range is the case $\mathbb{Z}^d$ of Theorem 1.1 of Florescu, Levine, and Peres (2016), whose Theorem 1.2 gives the transience from an infinite chain of rotors directed

toward the starting point (theorem numbers refer to arXiv:1408.5533). The recurrence of the clockwise rotor walk on $\mathbb{Z}^2$ with independent uniform initial rotors (Theorem 11.42) is the square-lattice case of Theorem 1.1(i) of Bou-Rabee and Peres (2026). The action of the sandpile group on oriented spanning trees, and the resulting proof of the matrix tree theorem, follow Sections 3–4 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008).

Diffusion-limited aggregation was introduced by Witten and Sander (1981, 1983). The discrete Beurling estimate is equivalent, by translation and comparison of radius with diameter, to the planar part of the theorem on page 167 of Kesten (1987a), and the growth bound for planar DLA is due to Kesten (1987b), who proved that, almost surely, the cluster of n particles lies in the ball of radius $Cn^{2/3}$ about the origin for all large n, for an absolute constant C. The Notes to Chapter 8 discuss the numerical estimates of the fractal dimension.

The multiple-source shape theorem for all three models, in every dimension $d \geq 2$, is Theorem 1.4 of Levine and Peres (2010); the Notes to Chapter 7 discuss the planar case, Neumann's oval, the algebraic boundary and the quadrature domains. The values of the potential kernel behind the bound 0.008 in Section 1.6 are determined by its exact values on the diagonal (Lemma 3.7), its value $\frac{1}{4}$ at the neighbors of the origin, its symmetries and its harmonicity away from the origin.

The simulations in the figures of this chapter and in Section 1.4 follow the rules in the text, with one exception: in the external DLA simulations, as is customary, each walk starts at a uniform point of a circle around the cluster, a walk far from the cluster jumps to a uniform point of a circle around it that does not meet the cluster, and a walk that wanders past a larger circle starts again on the first one. The first two shortcuts are exact for Brownian motion and approximate for random walk, and the third is an approximation for both.

For further reading, Levine and Peres (2017) survey the growth models of this book and their scaling limits, Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) develop the algebra of chip-firing and rotor-routing on directed graphs, and Lawler (1996) and Lawler and Limic (2010) develop the estimates for random walk that Chapters 3 and 8 use.

PART 1

Potential Theory

Chapter Two

Potential Theory in $\mathbb{R}^d$

Potential theory is the study of harmonic and superharmonic functions. Historically it arose out of the theory of gravity, in studying the gravitational field (or potential) of massive bodies; and later, in the theory of electromagnetism, the Coulomb potential of charged bodies. Harmonic functions are also of fundamental importance as the steady-state solutions to the heat equation $\partial u/\partial t = \Delta u$. Discrete potential theory also describes hitting probabilities and the expected occupation times of a random walk.

The connections between harmonic functions, gravitational potential and random walks are hardly apparent from the outset. In this chapter and the next we will illuminate these connections as we develop the potential theory of $\mathbb{R}^d$ and $\mathbb{Z}^d$. In this chapter we prove the minimum principle (Lemma 2.12), which implies that a harmonic function on a bounded domain is determined by its boundary values, the Poisson integral formula that recovers a harmonic function on a ball from its boundary values (Section 2.3), and the Harnack inequality that controls how much a positive harmonic function can vary (Section 2.4). We then construct the potential of a mass distribution, which inverts the negative Laplacian (Section 2.5), and prove the Beurling estimate, which bounds the chance that planar Brownian motion started near the origin reaches a distant circle without meeting a connected set that joins the origin to that circle (Section 2.6).

2.1 HARMONIC FUNCTIONS

The simplest definition of a harmonic function is a function u satisfying

$$\Delta u = 0\,,$$

where $\Delta = \sum_i \partial^2/\partial x_i^2$ is the Laplacian. The drawback of this definition is that it requires taking two derivatives. For this reason we will define harmonic via the **mean value property**: the average value of a harmonic function on a ball equals its value at the center. In Section 2.2 we prove the equivalence of the two definitions.

Definition 2.1 (Averaging Operator) Let u be a locally integrable function

on an open set $\Omega \subset \mathbb{R}^d$. If $\mathrm{B}(x,r) \subset \Omega$ and $u \in L^1(\mathrm{B}(x,r))$, write

$$\mathbf{A}_r u(x) := \fint_{\mathrm{B}(x,r)} u(y)\,dy := \frac{1}{\omega_d r^d} \int_{\mathrm{B}(x,r)} u(y)\,dy\,,$$

where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

Remark. Since u is locally integrable, $\mathbf{A}_r u$ is continuous on Ω_r, the set of points x with $\overline{\mathrm{B}}(x,r) \subset \Omega$. Indeed, if $x_n \to x$ then

$$|\mathbf{A}_r u(x_n) - \mathbf{A}_r u(x)| \le \frac{1}{\omega_d r^d} \int |u(y)| \left|\mathbf{1}_{\mathrm{B}(x_n,r)} - \mathbf{1}_{\mathrm{B}(x,r)}\right| dy\,,$$

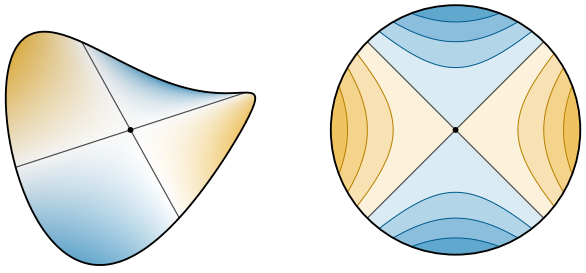
and the right side tends to 0 as $n \to \infty$ by the dominated convergence theorem.

Definition 2.2 (Harmonic Function) Let Ω be an open set in $\mathbb{R}^d$. A function u defined on Ω is **harmonic** at a point $x \in \Omega$ if there exists a neighborhood $x \in U \subset \Omega$ such that $u \in L^1(U)$ and

$$u(y) = \mathbf{A}_r u(y) \qquad \text{for all } \mathrm{B}(y,r) \subset U\,.$$

We say that u is harmonic on Ω if it is harmonic at x for all $x \in \Omega$.

For example, $x^2 - y^2$ is harmonic on $\mathbb{R}^2$ (see Figure 2.1), since its Laplacian vanishes, and in Section 2.2 we show that for twice differentiable functions this is equivalent to the mean value property.



(a) (b)

Figure 2.1: The harmonic function $u(x,y) = x^2 - y^2$ on the unit disk: its graph (a), and the regions where $u > c$ (orange) and $u < -c$ (blue) for $c = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ (b), darker for larger c. The rotation by a right angle about the origin turns u into $-u$, so the average of u over every disk centered at the origin is $0 = u(\mathbf{0})$.

The first thing we will prove about harmonic functions is that they are infinitely differentiable. We use the standard technique of **mollifying**, which starts with the observation that there exist nonnegative, spherically symmetric

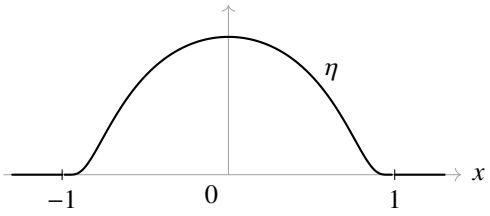


Figure 2.2: The mollifier η in (2.1) when $d = 1$. Observe that $\eta^{(k)}(\pm 1) = 0$ for all $k \geq 0$.

and infinitely differentiable functions supported on the unit ball $\mathrm{B}(\mathbf{0}, 1)$. An example is the function

$$\eta(x) := \begin{cases} C_d e^{1/(|x|^2-1)}, & |x| < 1 \\ 0, & |x| \geq 1\,, \end{cases} \tag{2.1}$$

with the normalizing constant C_d chosen so that $\int_{\mathbb{R}^d} \eta \, \mathrm{d}x = 1$. A visualization is given in Figure 2.2.

The **convolution** of two functions u and v on $\mathbb{R}^d$ is the function

$$(u * v)(x) := \int_{\mathbb{R}^d} u(y) v(x - y) \, \mathrm{d}y\,,$$

defined when the integrand is in $L^1(\mathbb{R}^d)$. The averaging operator (Definition 2.1) can be expressed as a convolution, namely

$$\mathbf{A}_r u = u * \beta_r\,,$$

where $\beta_r := \frac{1}{\omega_d r^d} \mathbf{1}_{\mathrm{B}(\mathbf{0},r)}$. The advantage of considering more general averages defined by convolution is that $u * v$ is at least as smooth as v. Indeed, let u be locally integrable and v be n times continuously differentiable with bounded support. Then for every differential operator $D = \partial^n / \partial x_{i_1} \cdots \partial x_{i_n}$, we can bring the derivative inside the integral (by the dominated convergence theorem, since for x in a compact set the difference quotients are bounded by $C|u(y)|\mathbf{1}_K(y)$ for a compact set K) to obtain

$$D(u * v)(x) = \int_{\mathbb{R}^d} u(y) Dv(x - y) \, \mathrm{d}y = (u * Dv)(x)\,.$$

Now consider a spherically symmetric mollifier $v(y) = V(|y|)$ with integral 1, where V is smooth, weakly decreasing, and vanishes on $[\epsilon, \infty)$. In this case, $u * v$ can be expressed as a mixture of averaging operators applied to u. To see this, note that by the fundamental theorem of calculus

$$v(y) = \int_0^\epsilon (-V'(r)) \mathbf{1}_{\mathrm{B}(\mathbf{0},r)}(y) \, \mathrm{d}r\,,$$

hence by Fubini's theorem, for locally integrable u,

$$(u * v)(x) = \int_0^\epsilon (-V'(r))\omega_d r^d \mathbf{A}_r u(x)\, \mathrm{d}r\,. \tag{2.2}$$

Taking $u \equiv 1$ shows that the nonnegative weights in this integral have total mass 1. Armed with these observations we turn to our main goal.

Theorem 2.3 *Let $\Omega \subset \mathbb{R}^d$ be open. If u is harmonic on Ω, then $u \in C^\infty(\Omega)$.*

Proof. Fix $x_0 \in \Omega$. Since u is harmonic on Ω, by Definition 2.2 there is a ball $U = \mathrm{B}(x_0, 2\epsilon)$ such that $u \in L^1(U)$ and $\mathbf{A}_r u(x) = u(x)$ whenever $\mathrm{B}(x, r) \subset U$. Extend u by zero outside U, and let $v(y) = \epsilon^{-d}\eta(y/\epsilon)$, with η as in (2.1). For $x \in \mathrm{B}(x_0, \epsilon)$ the convolution $(u * v)(x)$ uses only values of u in U, so by (2.2) and the normalization of its weights we have $(u * v)(x) = u(x)$. Since $v \in C^\infty(\mathbb{R}^d)$, the convolution $u * v$ is infinitely differentiable, and hence u is infinitely differentiable at x_0. □

2.2 SUPERHARMONIC FUNCTIONS

Superharmonic functions are defined by replacing the mean value equality $\mathbf{A}_r u(x) = u(x)$ by the inequality $\mathbf{A}_r u(x) \le u(x)$. In the previous section we deduced from the mean value equality that harmonic functions are infinitely differentiable. The situation with superharmonic functions is quite different. The function $u(x) = -|x|$ on $\mathbb{R}$ is superharmonic but not differentiable. Later we will see that a superharmonic function can even be discontinuous (Example 2.41).

Definition 2.4 (Lower Semicontinuity) Let $\Omega \subset \mathbb{R}^d$ be an open set, and u a function on Ω taking values in $\mathbb{R} \cup \{+\infty\}$. We say that u is **lower semicontinuous** at a point $x \in \Omega$ if for all sequences $x_n \to x$, we have $u(x) \le \liminf u(x_n)$. We say that u is lower semicontinuous in Ω if it is lower semicontinuous at every point of Ω.

Exercise 2.1 gives an equivalent definition in the case that u takes values in $\mathbb{R}$.

The inverse image of an open interval under a continuous function is open. The next lemma says that lower semicontinuous functions satisfy "one half" of this condition: the inverse image of an open half-line (a, ∞) is open. The proof is left as an exercise (Exercise 2.2).

Lemma 2.5 *Let $\Omega \subset \mathbb{R}^d$ be open, and let $u : \Omega \to \mathbb{R} \cup \{+\infty\}$. Then u is lower semicontinuous in Ω if and only if the set*

$$L_a := \{x \in \Omega \mid u(x) > a\}$$

is open for all $a \in \mathbb{R}$.

Lower semicontinuity is preserved by taking pointwise suprema (Exercise 2.4). Exercise 2.5 asks you to show that every lower semicontinuous function is a pointwise supremum of continuous functions. (This is how lower semicontinuous functions often arise in practice.)

We say that u is **upper semicontinuous** if $-u$ is lower semicontinuous. Note that if both u and $-u$ are lower semicontinuous, then u is continuous.

Definition 2.6 (Superharmonic Function) Let $\Omega \subset \mathbb{R}^d$ be an open set, and u a function on Ω taking values in $\mathbb{R} \cup \{+\infty\}$. We say that u is **superharmonic** at $x \in \Omega$ if there is a neighborhood W of x such that $u \in L^1(W)$ and

$$u(y) \geq \mathbf{A}_r u(y) \qquad \text{for all } \mathrm{B}(y, r) \subset W\,. \tag{2.3}$$

We say that u is **subharmonic** if $-u$ is superharmonic. Subharmonic functions take values in $\mathbb{R} \cup \{-\infty\}$. Note that if u is both superharmonic and subharmonic, then u takes values in $\mathbb{R}$ and is harmonic (Definition 2.2).

Note that we do not include lower semicontinuity in the definition of superharmonic. For example, $\mathbf{1}_{\{\mathbf{0}\}}$ is superharmonic on $\mathbb{R}^d$, since it is nonnegative and its ball averages vanish, but it is not lower semicontinuous at $\mathbf{0}$; since it attains its minimum without being constant, the strong minimum principle (Lemma 2.8) needs lower semicontinuity as a separate hypothesis.

Example 2.7 The function $|x| = \left(\sum_{i=1}^d x_i^2\right)^{1/2}$ is subharmonic in $\mathbb{R}^d$. Indeed,

$$\mathbf{A}_r|x| = \fint_{\mathrm{B}(x,r)} |y|\,\mathrm{d}y \geq \left| \fint_{\mathrm{B}(x,r)} y\,\mathrm{d}y \right| = |x|\,,$$

where the inequality is the triangle inequality for integrals. Exercise 2.6 asks you to generalize this example, by showing that $f \circ h$ is subharmonic for every convex $f : \mathbb{R} \to \mathbb{R}$ and harmonic h.

Note that (2.3) asserts the mean value inequality only for "small balls," i.e. those in a neighborhood of x. In Section 2.3 we will prove the stronger "all balls" mean value inequality: if u is superharmonic on Ω, then $\mathbf{A}_r u(y) \leq u(y)$ for all $\overline{\mathrm{B}}(y, r) \subset \Omega$ (Lemma 2.23 for continuous u, and Corollary 2.27 in general). Along the way we will prove some fundamental properties of superharmonic functions. The advantage of the "small balls" definition is that it is local: it makes clear that if u is superharmonic on both Ω_1 and Ω_2, then u is superharmonic on $\Omega_1 \cup \Omega_2$.

If u is lower semicontinuous on $\Omega \subset \mathbb{R}^d$, then u is bounded below on every compact subset $K \subset \Omega$ (Exercise 2.3). In this case, for each ball with $\overline{\mathrm{B}}(x, r) \subset \Omega$ the integral $\mathbf{A}_r u$ is always defined in $\mathbb{R} \cup \{+\infty\}$.

The most important property of superharmonic functions is the minimum principle. Next we prove several versions of this principle.

Lemma 2.8 (Strong Minimum Principle) *Let Ω be a connected open set in $\mathbb{R}^d$. If u is superharmonic and lower semicontinuous in Ω, and the infimum $\inf_{x\in\Omega} u(x)$ is attained, then u is constant in Ω.*

Proof. Let $m := \inf_{x\in\Omega} u(x)$, and let

$$\Omega_{\min} := \{x \in \Omega \mid u(x) = m\}.$$

We will argue that $\Omega_{\min}$ is both closed and open (in the induced topology on Ω). Since Ω is connected, if the infimum is attained, so that $\Omega_{\min}$ is nonempty, then $\Omega_{\min} = \Omega$.

By Lemma 2.5, since u is lower semicontinuous in Ω, the set

$$\Omega^c_{\min} = \{x \in \Omega \mid u(x) > m\}$$

is open, so $\Omega_{\min}$ is closed.

To argue that $\Omega_{\min}$ is open, we fix $x \in \Omega_{\min}$ and show that $B(x, \epsilon) \subset \Omega_{\min}$ whenever $B(x, \epsilon)$ lies in a neighborhood of x where the mean value inequality (2.3) holds. Otherwise, some $y \in B(x, \epsilon)$ satisfies $u(y) > m$, so $u(y) > m+\delta$ for some $\delta > 0$. Since u is lower semicontinuous, there is a neighborhood $W \subset B(x, \epsilon)$ of y such that $u > m + \delta$ on W (see Figure 2.3); since $u \geq m$ on Ω, this would imply $\mathbf{A}_\epsilon u(x) > m = u(x)$, a contradiction to superharmonicity of u. □

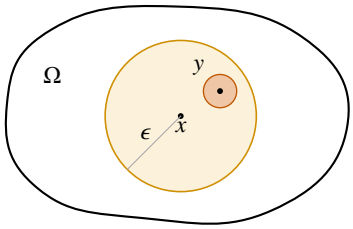


Figure 2.3: In the proof of Lemma 2.8, if $u(x) = m$ and some $y \in B(x, \epsilon)$ had $u(y) > m$, then $u > m + \delta$ on a small ball about y (shaded) and $u \geq m$ on the rest of $B(x, \epsilon)$, so $\mathbf{A}_\epsilon u(x) > m = u(x)$.

A corollary of the strong minimum principle is that a lower semicontinuous superharmonic function has no strict local minima.

The following topological fact is useful when applying Lemma 2.8.

Lemma 2.9 *Let Y be a connected component of an open set $\Omega \subset \mathbb{R}^d$. Then $\partial Y \subset \partial\Omega$.*

Proof. Since $\overline{Y} \cap \Omega = Y$, we have $(\partial Y) \cap \Omega = \emptyset$, hence $\partial Y \subset \overline{\Omega} \setminus \Omega = \partial\Omega$. □

Lemma 2.10 (Unbounded Minimum Principle) *Let $\Omega \subset \mathbb{R}^d$ be open. If u is superharmonic and lower semicontinuous in Ω, then there is a sequence $(x_n)_{n\geq 0}$ in Ω converging either to a point in $\partial\Omega$ or to ∞, such that $u(x_n) \downarrow \inf_\Omega u$.*

Proof. Let $m = \inf_\Omega u$, and let $(y_n)_{n\geq 0}$ be a sequence in Ω such that $u(y_n) \downarrow m$. If y_n has a subsequence converging either to a point in $\partial\Omega$ or to ∞, then we may take x_n to be that subsequence. Otherwise, the sequence y_n has a limit point $y \in \Omega$. In this case $u(y) = m$ by lower semicontinuity. By Lemma 2.8, u is constant on the connected component Y of Ω containing y. By Lemma 2.9, $\partial Y \subset \partial\Omega$, so we may take x_n to be a sequence in Y converging either to a point in ∂Y or to ∞. □

Corollary 2.11 *Let $\Omega \subset \mathbb{R}^d$ be open. If u is superharmonic and lower semicontinuous in Ω and h is subharmonic and upper semicontinuous in Ω, and $\liminf (u(x_n) - h(x_n)) \geq 0$ for all sequences $(x_n)_{n\geq 0}$ in Ω converging either to a point in $\partial\Omega$ or to ∞, then $u \geq h$ in Ω.*

Proof. Apply Lemma 2.10 to the function $u - h$. □

Next we state the most commonly used version of the minimum principle.

Lemma 2.12 (Minimum Principle) *Let Ω be a bounded open set in $\mathbb{R}^d$ and let $u : \overline{\Omega} \to \mathbb{R} \cup \{+\infty\}$ be lower semicontinuous on $\overline{\Omega}$ and superharmonic in Ω. Then the following hold.*

(a) $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$.

(b) If h is harmonic in Ω, continuous in $\overline{\Omega}$, and $u \geq h$ on $\partial\Omega$, then $u \geq h$ in $\overline{\Omega}$.

Proof. Since $\overline{\Omega}$ is compact, the infimum $m := \inf_{x\in\overline{\Omega}} u(x)$ is attained (Exercise 2.3). If this infimum is attained by a point $x \in \partial\Omega$, then the proof is complete. Otherwise, it is attained by a point $x \in \Omega$. By Lemma 2.8, we have $u \equiv m$ on the connected component X of Ω containing x. By lower semicontinuity, $u \equiv m$ on $\overline{X}$. By Lemma 2.9 we have $\partial X \subset \partial\Omega$, which completes the proof of (a). To prove (b), apply (a) to $u - h$. □

A consequence of the minimum principle is that a harmonic function on a bounded domain is determined by its boundary values: if two functions u and h are harmonic on Ω, continuous on $\overline{\Omega}$, and $u = h$ on $\partial\Omega$, then $u = h$ on Ω.

The following gives two canonical examples of harmonic functions.

Lemma 2.13 *Let $b_i, M_{ij} \in \mathbb{R}$ for $1 \le i, j \le d$. Let*

$$L(x) := b^T x = \sum_{i=1}^{d} b_i x_i \qquad \text{and} \qquad Q(x) := x^T M x = \sum_{i=1}^{d}\sum_{j=1}^{d} M_{ij} x_i x_j \,.$$

Then the following hold.

(a) The function L is harmonic on $\mathbb{R}^d$.

(b) The function Q is harmonic on $\mathbb{R}^d$ if and only if $\sum_{i=1}^d M_{ii} = 0$.

(c) The function Q is superharmonic on $\mathbb{R}^d$ if and only if $\sum_{i=1}^d M_{ii} \le 0$.

Proof. Replacing M by $\frac{1}{2}(M + M^T)$ does not change Q, so we may assume that M is symmetric. By a change of variables, and since $\int_{\mathrm{B}(\mathbf{0},r)} x_i \,\mathrm{d}x = \int_{\mathrm{B}(\mathbf{0},r)} x_i x_j \,\mathrm{d}x = 0$ for $i \ne j$, we have

$$\mathbf{A}_r L(y) = L(y) \qquad \text{and} \qquad \mathbf{A}_r Q(y) = Q(y) + \frac{r^2}{d+2}\sum_{i=1}^{d} M_{ii} \,.$$

(a) By the first identity, L has the mean value property on every ball.

(b) By the second identity, Q has the mean value property if and only if $\sum_i M_{ii} = 0$.

(c) Likewise, Q satisfies the mean value inequality if and only if $\sum_i M_{ii} \le 0$. □

We have defined harmonic functions by the mean value property in balls $\mathrm{B}(x, r)$. Next we will show that they also enjoy a mean value property on spheres $\partial\mathrm{B}(x, r)$.

The surface measure $S(A)$ of a Borel set $A \subset \partial\mathrm{B}(\mathbf{0}, r) \subset \mathbb{R}^d$ can be defined using the cone $\mathrm{Cone}(A) := \{Rx : R \ge 0 \text{ and } x \in A\}$. Namely, let $S(A) := f_A'(r)$, where

$$f_A(t) := \mathcal{L}(\mathrm{Cone}(A) \cap \mathrm{B}(\mathbf{0}, t)) \,,$$

and $\mathcal{L}$ denotes Lebesgue measure on $\mathbb{R}^d$. Observe that by scaling, $f_A(t) = t^d f_A(1)$, so $S(A) = dr^{d-1} f_A(1)$.

Definition 2.14 (Spherical Averaging Operator) Let u be a Borel function defined on an open set $\Omega \subset \mathbb{R}^d$. Let

$$\mathbf{a}_r u(x) := \fint_{\partial\mathrm{B}(x,r)} u(y) \,\mathrm{d}S(y) := \frac{1}{d\omega_d r^{d-1}} \int_{\partial\mathrm{B}(x,r)} u(y) \,\mathrm{d}S(y) \,,$$

whenever these integrals are defined.

The ball average $\mathbf{A}_R u$ can be expressed in terms of $\mathbf{a}_r u$ by integrating over $\mathrm{B}(x,R)$ in spherical shells:

$$\begin{aligned}\mathbf{A}_R u(x) &= \frac{1}{\omega_d R^d} \int_0^R \int_{\partial \mathrm{B}(x,r)} u(y)\, \mathrm{d}S(y)\, \mathrm{d}r \\ &= \frac{d}{R^d} \int_0^R r^{d-1} \mathbf{a}_r u(x)\, \mathrm{d}r\,. \end{aligned} \tag{2.4}$$

Now we prove equivalence of Definition 2.6 to the usual definition, $\Delta u \le 0$, in the case that our function u is twice continuously differentiable. Recall that

$$\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}\,.$$

Lemma 2.15 *Let $\Omega \subset \mathbb{R}^d$ be open, and let $u \in C^2(\Omega)$. Then u is superharmonic in Ω if and only if $\Delta u \le 0$ in Ω.*

Proof. For every ball with $\overline{\mathrm{B}}(x,r) \subset \Omega$, we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}r}\mathbf{a}_r u(x) &= \frac{\mathrm{d}}{\mathrm{d}r} \fint_{\partial \mathrm{B}(\mathbf{0},1)} u(x+ry)\, \mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(\mathbf{0},1)} \frac{\mathrm{d}}{\mathrm{d}r}[u(x+ry)]\, \mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(\mathbf{0},1)} (\nabla u)(x+ry)\cdot y\, \mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(x,r)} \frac{\partial u}{\partial \mathbf{n}}(z)\, \mathrm{d}S(z)\,, \end{aligned} \tag{2.5}$$

where $\mathbf{n}$ is the outward unit normal. By the divergence theorem, the right side equals $\frac{1}{d\omega_d r^{d-1}} \int_{\mathrm{B}(x,r)} \Delta u(y)\, \mathrm{d}y$. Moreover, $\mathbf{a}_r u(x) \to u(x)$ as $r \downarrow 0$, since u is continuous.

Suppose first that u is superharmonic in Ω. If $\Delta u(x) > 0$ for some $x \in \Omega$, then since Δu is continuous, there is $s > 0$ such that $\overline{\mathrm{B}}(x,s) \subset \Omega$ and $\Delta u > 0$ on $\overline{\mathrm{B}}(x,s)$. Then $\mathbf{a}_r u(x)$ is strictly increasing for $0 < r \le s$, so $\mathbf{a}_r u(x) > u(x)$, and by (2.4) $\mathbf{A}_r u(x) > u(x)$ for all $0 < r \le s$, contradicting the mean value inequality (2.3) at x. Hence $\Delta u \le 0$ in Ω.

Conversely, suppose that $\Delta u \le 0$ in Ω. Then $\mathbf{a}_r u(x)$ is weakly decreasing in r whenever $\overline{\mathrm{B}}(x,r) \subset \Omega$. In particular, $\mathbf{a}_r u(x) \le u(x)$, which implies $\mathbf{A}_r u(x) \le u(x)$ by (2.4). Thus u is superharmonic in Ω. □

An alternative proof which avoids the use of the divergence theorem is described in Exercise 2.29.

The normal derivative in (2.5) appears precisely because the integral is taken over a sphere. In Exercises 2.16–2.21 we explore what classes of functions result if the ball average $\mathbf{A}_r u$ in the definition of harmonic is replaced by other types of averages.

Recalling that harmonic functions are infinitely differentiable (Theorem 2.3) we see from Lemma 2.15 that u is harmonic on Ω if and only if $\Delta u = 0$ on Ω.

2.2.1 Interlude: Harmonic Functions in Two Dimensions

Identify $\mathbb{R}^2$ with the complex plane $\mathbb{C}$, and let Ω be a bounded connected open set in $\mathbb{C}$. A holomorphic function $\varphi = u + \mathbf{i}v : \Omega \to \mathbb{C}$ satisfies the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}\,.$$

Since holomorphic functions are infinitely differentiable, $u, v \in C^2(\Omega)$, and by the Cauchy–Riemann equations

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} = -\frac{\partial^2 u}{\partial y^2}\,,$$

so $\Delta u = 0$ in Ω. A similar calculation shows that $\Delta v = 0$. Thus, the real and imaginary parts of a holomorphic function are harmonic.

Alternatively, the mean value property for u and v follows from the Cauchy integral formula: if $\varphi : \Omega \to \mathbb{C}$ is holomorphic, then for every ball $\overline{\mathrm{B}}(a, r) \subset \Omega$

$$\varphi(a) = \frac{1}{2\pi\mathbf{i}} \int_{\partial \mathrm{B}(a,r)} \frac{\varphi(w)}{w - a}\, \mathrm{d}w = \frac{1}{2\pi} \int_0^{2\pi} \varphi(a + re^{\mathbf{i}\theta})\, \mathrm{d}\theta = \mathbf{a}_r \varphi(a)\,.$$

Taking real and imaginary parts we see that $\mathbf{a}_r u = u$ and $\mathbf{a}_r v = v$.

Conversely, if Ω is simply connected, then every harmonic function on Ω is the real part of a holomorphic function φ on Ω. Indeed, note that if $\Delta u = 0$, then the function

$$g := \frac{\partial u}{\partial x} - \mathbf{i}\frac{\partial u}{\partial y}$$

satisfies the Cauchy–Riemann equations. Now fix a base point $z_0 \in \Omega$ and define

$$\varphi(z) := u(z_0) + \int_{z_0}^{z} g(w)\, \mathrm{d}w\,,$$

by integrating along a path in Ω from z_0 to z. Since Ω is simply connected and g is holomorphic in Ω, the integral of $g\,dz$ around every closed path is zero, which shows that φ is well-defined.

By using a path that ends with a horizontal or vertical segment, we deduce that $\nabla\varphi = (g, \mathbf{i}g)$. Hence, $\operatorname{Re}\nabla\varphi = (\operatorname{Re} g, -\operatorname{Im} g) = \nabla u$. Since $u - \operatorname{Re}\varphi$ vanishes at z_0 and its gradient is identically zero, we conclude that $u = \operatorname{Re}\varphi$.

Lemma 2.16 *Let $\Omega_1, \Omega_2 \subset \mathbb{C}$ be open. If u is harmonic on Ω_2 and $\psi : \Omega_1 \to \Omega_2$ is holomorphic, then $u \circ \psi$ is harmonic on Ω_1.*

Proof. Fix $z \in \Omega_1$, and choose a ball $B \subset \Omega_2$ about $\psi(z)$ and a neighborhood U of z such that $\psi(U) \subset B$. We have $u = \operatorname{Re}\varphi$ in B for a holomorphic function φ. Since $\varphi \circ \psi$ is holomorphic, $u \circ \psi = \operatorname{Re}(\varphi \circ \psi)$ is harmonic on U. Since z was arbitrary, $u \circ \psi$ is harmonic on Ω_1. □

Exercise 2.25 asks you to extend this lemma to superharmonic u and nonconstant ψ.

2.3 POISSON KERNEL: FROM SMALL BALLS TO ALL BALLS

Given a bounded open set $\Omega \subset \mathbb{R}^d$ and a continuous function F on $\partial\Omega$, the **Dirichlet problem** asks for a continuous function h in $\overline{\Omega}$ such that

$$\begin{aligned} \Delta h &= 0 && \text{in } \Omega \\ h &= F && \text{on } \partial\Omega\,. \end{aligned}$$

Given two solutions h_1 and h_2, the difference $h_1 - h_2$ is identically zero by the minimum principle. So the Dirichlet problem has at most one solution. A solution need not exist for every Ω. For example, on the punctured disk $\mathrm{B}(\mathbf{0}, 1) \setminus \{\mathbf{0}\}$ in $\mathbb{R}^2$, no solution takes the value 0 on the unit circle and 1 at the origin. Indeed, if h were such a solution, then given $\epsilon > 0$ we could choose $0 < \delta < 1$ with $\epsilon \log(1/\delta) \geq \max h$. The harmonic function $\epsilon \log(1/|x|)$ is then at least h on both boundary circles of the annulus $\{\delta < |x| < 1\}$ (see Figure 2.4), so by the minimum principle $h(x) \leq \epsilon \log(1/|x|)$ in the annulus. Letting $\delta \downarrow 0$ and then $\epsilon \downarrow 0$ we obtain $h \leq 0$ off the origin, so $h(\mathbf{0}) \leq 0$ by continuity, a contradiction. In this section we construct a solution when Ω is a ball.

Consider first the case when F is a polynomial, and Ω is the unit ball. Write V_m for the vector space of polynomials in $\mathbb{R}[x_1, \ldots, x_d]$ of total degree at most m, with $V_m = \{0\}$ when $m < 0$. Note that Δ maps V_m to V_{m-2}, and that multiplication by $|x|^2 = \sum x_i^2$ is a linear map $V_{m-2} \to V_m$.

In this case, we will show that the harmonic extension is also a polynomial. For example, when $F(x, y) = 2x^2$ on the unit circle, the harmonic extension of F to the unit disk is $h(x, y) = x^2 - y^2 + 1$.

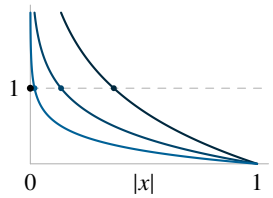


Figure 2.4: The barriers $\epsilon \log(1/|x|)$ as functions of $|x|$, for $\epsilon = 1, \frac{1}{2}, \frac{1}{4}$ (darker for smaller ϵ): each exceeds the level $\max h$, drawn as 1 (dashed), inside a circle of radius δ (dots), and they tend to 0 at every $x \neq \mathbf{0}$ as $\epsilon \downarrow 0$.

Lemma 2.17 *Let $B = \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$, and let $F \in \mathbb{R}[x_1, \ldots, x_d]$ be a polynomial of total degree m. Then there is a polynomial h of total degree at most m such that $\Delta h = 0$ on $\mathbb{R}^d$ and $h|_{\partial B} = F|_{\partial B}$. Moreover*

$$F = (1 - |x|^2)g + h$$

for a polynomial g of total degree at most $m - 2$.

Proof. Note that if $g(x)$ is a polynomial such that $(1 - |x|^2)g$ is harmonic in B, then $(1 - |x|^2)g$ is identically zero in B by the minimum principle, so g is the zero polynomial. Thus $g \mapsto \Delta((1 - |x|^2)g)$ defines an injective linear map from V_{m-2} to itself. Since V_{m-2} is finite-dimensional, this map is also surjective. In particular, there exists $g \in V_{m-2}$ such that $\Delta((1 - |x|^2)g) = \Delta F$. Now set $h = F - (1 - |x|^2)g$. Then h has total degree at most m, we have $\Delta h = 0$ by the choice of g, and $h = F$ on ∂B since $1 - |x|^2$ vanishes there. □

An analogue of the lemma for paraboloids is given in Exercise 2.28.

For $x \in \mathrm{B}(\mathbf{0}, 1)$ and $y \in \partial\mathrm{B}(\mathbf{0}, 1)$, the function

$$P(x, y) := \frac{1 - |x|^2}{|x - y|^d} \tag{2.6}$$

is called the **Poisson kernel** for the unit ball (see Figure 2.5). In general, for a set X and a measure space Y, a "kernel" is a two-variable function $P : X \times Y \to \mathbb{R}$ which defines an operator acting on a function $f : Y \to \mathbb{R}$ to produce a function $X \to \mathbb{R}$ according to the integral formula $\int_Y P(x, y) f(y)\, \mathrm{d}y$.

Theorem 2.18 (Poisson Integral Formula) *Let $d \geq 1$, let $B = \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$, and let $P : B \times \partial B \to \mathbb{R}$ be the Poisson kernel*

$$P(x, y) := \frac{1 - |x|^2}{|x - y|^d} \qquad \text{for } x \in B \text{ and } y \in \partial B\,.$$

Let $F : \partial B \to \mathbb{R}$ be an integrable function with respect to the surface measure $\mathrm{d}S$ on ∂B (counting measure on $\partial B = \{-1, 1\}$ when $d = 1$). The function

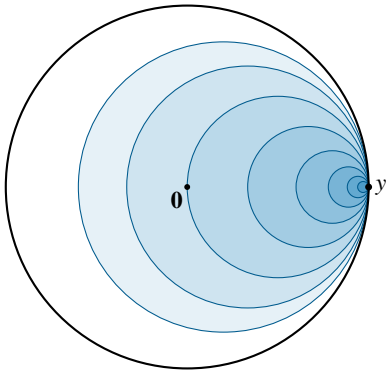


Figure 2.5: Level sets of the Poisson kernel (2.6) of the unit disk for a fixed point y of the unit circle. The shaded disks are the sets where $P(\cdot, y) > c$ for $c = 2^k$ and $-2 \le k \le 5$, darker for larger c; each is the disk of radius $1/(1+c)$ tangent to the unit circle at y. Points approaching y with $1 - |x|$ at most a constant times $|x-y|^2$ stay outside the small disks, so the kernel stays bounded along such paths.

$h : B \to \mathbb{R}$ given by

$$h(x) := \fint_{\partial B} P(x,y) F(y)\, \mathrm{d}S(y)$$

is harmonic in B. Moreover, if F is continuous at $y_0 \in \partial B$, then

$$\lim_{x \to y_0} h(x) = F(y_0)\,.$$

Proof. For fixed $y \in \partial B$, an explicit calculation shows that $P(\cdot, y)$ is harmonic in B. Indeed, since $x \neq y$ for $x \in B$, we have

$$\nabla_x |x-y|^{-d} = -d\,(x-y)\,|x-y|^{-d-2} \qquad \text{and} \qquad \Delta_x |x-y|^{-d} = 2d\,|x-y|^{-d-2}\,,$$

and since $\nabla_x(1-|x|^2) = -2x$ and $\Delta_x(1-|x|^2) = -2d$, the product rule gives

$$\Delta_x P(x,y) = 2d\,|x-y|^{-d-2}(1-|x|^2+2x\cdot(x-y)-|x-y|^2) = 2d\,|x-y|^{-d-2}(1-|y|^2)\,,$$

where the second equality follows by expanding $|x-y|^2$; the right side vanishes since $|y| = 1$. (For $d \ge 2$, see Exercise 2.48 for a proof of this fact that does not require calculating second derivatives.) On each compact subset of B the distance $|x-y|$ is bounded below uniformly in $y \in \partial B$, so the first two x-derivatives of $P(x,y)F(y)$ are bounded by $C|F(y)|$, which is integrable. Differentiating twice under the integral sign (by the dominated convergence theorem), it follows that $h \in C^2(B)$ and $\Delta h = 0$ in B, so h is harmonic in B by Lemma 2.15.

Next note that $h_1(x) := \fint_{\partial B} P(x,y)\, \mathrm{d}S(y)$ is harmonic in B, by the case $F \equiv 1$, and spherically symmetric, since $P(Qx, Qy) = P(x,y)$ for every

orthogonal matrix Q and Q preserves surface measure on ∂B. Hence h_1 is constant on each sphere $\partial\mathrm{B}(\mathbf{0}, r)$ with $0 < r < 1$, so by the minimum principle (Lemma 2.12(a), applied on $\mathrm{B}(\mathbf{0}, r)$ to h_1 and to $-h_1$) it is constant on $\overline{\mathrm{B}}(\mathbf{0}, r)$. Since $P(\mathbf{0}, y) = 1$ for all $y \in \partial B$, we have $h_1(\mathbf{0}) = 1$, and we conclude that $h_1 \equiv 1$.

Now suppose that F is continuous at y_0, and fix $\epsilon > 0$. Choose $\delta > 0$ such that $|F(y) - F(y_0)| < \epsilon$ for all $y \in \partial B$ with $|y - y_0| < \delta$. If $|x - y_0| < \delta/2$, then every $y \in \partial B$ with $|y - y_0| \geq \delta$ satisfies $|x - y| \geq \delta/2$ (see Figure 2.6), so $P(x, y) \leq (1 - |x|^2)(2/\delta)^d$. Since the kernel is positive with mean one, and the points y with $|y - y_0| < \delta$ contribute at most ϵ to the integral below, we obtain

$$\begin{aligned} |h(x) - F(y_0)| &\leq \fint_{\partial B} P(x, y)\, |F(y) - F(y_0)|\, \mathrm{d}S(y) \\ &\leq \epsilon + (1 - |x|^2)\Big(\frac{2}{\delta}\Big)^d \fint_{\partial B} |F(y) - F(y_0)|\, \mathrm{d}S(y) \end{aligned} .$$

The last integral is finite since F is integrable, and $1 - |x|^2 \to 0$ as $x \to y_0$, hence $\limsup_{x \to y_0} |h(x) - F(y_0)| \leq \epsilon$. Since ϵ was arbitrary we conclude that $h(x) \to F(y_0)$. □

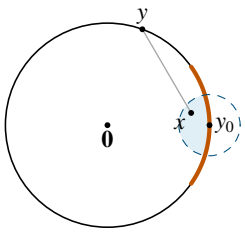


Figure 2.6: In the proof of Theorem 2.18, for x within $\delta/2$ of y_0 (dashed circle), every $y \in \partial B$ outside the cap $|y - y_0| < \delta$ (vermilion) satisfies $|x - y| \geq \delta/2$.

Corollary 2.19 *Let $B = \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$ and $F \in C(\partial B)$. There is a continuous function $h : \overline{B} \to \mathbb{R}$ such that $\Delta h = 0$ on B and $h = F$ on ∂B.*

Proof. Let h be the Poisson integral of F, extended to ∂B by $h = F$. By Theorem 2.18, h is harmonic in B and $h(x) \to F(y_0)$ as $x \to y_0$ for every $y_0 \in \partial B$, so h is continuous on $\overline{B}$. □

In the plane, identify $\mathbb{R}^2$ with $\mathbb{C}$ and parametrize the unit circle by $e^{\mathbf{i}\theta}$. The normalized surface measure becomes $\mathrm{d}\theta/(2\pi)$, and the Poisson integral of F becomes

$$h(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z|^2}{|e^{\mathbf{i}\theta} - z|^2}\, F(e^{\mathbf{i}\theta})\, \mathrm{d}\theta \qquad \text{for } |z| < 1\,. \tag{2.7}$$

Every function u that is harmonic in the unit ball $B = \mathrm{B}(\mathbf{0}, 1)$ and continuous on $\overline{B}$ is the Poisson integral of its boundary values. Indeed, by Theorem 2.18 the Poisson integral h of $u|_{\partial B}$ is harmonic in B and extends continuously to $\overline{B}$ with $h = u$ on ∂B, so by Lemma 2.12(b), applied to $u - h$ and to $h - u$, we have $u = h$.

If u is a continuous function on $\Omega \subset \mathbb{R}^d$ and $W = \mathrm{B}(x_0, r)$ with $\overline{W} \subset \Omega$, we denote by $\mathbf{P}_W u$ the function on W that extends $u|_{\partial W}$ harmonically inside W. Then $\mathbf{P}_W = T^{-1} \circ \mathbf{P}_B \circ T$, where $Tu(x) := u(x_0 + rx)$ and $B := \mathrm{B}(\mathbf{0}, 1)$.

Note that if $u \geq \mathbf{P}_W u$ for all sufficiently small balls $W = \mathrm{B}(y, r)$ near each point of Ω (that is, for each $x \in \Omega$ there is $r_0 > 0$ such that this holds whenever $r < r_0$, $|y - x| < r_0$ and $\overline{\mathrm{B}}(y, r) \subset \Omega$), then $u(y) \geq \mathbf{P}_W u(y) = \mathbf{a}_r u(y)$ for these balls, so u is superharmonic in Ω by (2.4).

Recalling that the definition of superharmonic only requires the mean value inequality to hold for sufficiently small balls, our next goal is to show that in fact a superharmonic function obeys the mean value inequality for all balls (Lemma 2.23 for continuous functions, and Corollary 2.27 in general). We begin with the case of a harmonic function.

Lemma 2.20 (Mean Value Equality for All Spheres and Balls) *If h is harmonic in an open set Ω and $\overline{\mathrm{B}}(x, R) \subset \Omega$, then for every $r \in (0, R]$ we have*

$$h(x) = \mathbf{a}_r h(x) = \mathbf{A}_r h(x)\,.$$

Proof. Since h is smooth and $\Delta h = 0$ (by Theorem 2.3 and Lemma 2.15), the calculation (2.5) and the divergence theorem show that $\mathbf{a}_r h(x)$ has derivative zero in r for $0 < r \leq R$. Since it tends to $h(x)$ as $r \downarrow 0$, we have $\mathbf{a}_r h(x) = h(x)$. The equality for $\mathbf{A}_r h$ now follows from (2.4). □

The next lemma shows that we can choose uniformly a radius δ such that the mean value inequality holds for all balls of radius at most δ. You are asked to prove it in Exercise 2.30.

Lemma 2.21 (Mean Value Inequality for All Small Balls) *If Ω is a bounded domain and u is superharmonic in a neighborhood of $\overline{\Omega}$, then there exists $\delta > 0$ such that $\mathbf{A}_r u \leq u$ on $\overline{\Omega}$ for all $0 < r \leq \delta$.*

Now we show that if u is superharmonic, then $\mathbf{a}_r u$ and $\mathbf{A}_r u$ are themselves superharmonic.

Lemma 2.22 *Let $x \in \mathbb{R}^d$ and $r > 0$, let $\Omega \subset \mathbb{R}^d$ be an open set that contains $\overline{\mathrm{B}}(x, r)$, and let u be superharmonic in Ω. Then the following hold.*

(a) The function $\mathbf{A}_r u$ is superharmonic at x.

(b) If u is continuous, then $\mathbf{a}_r u$ is superharmonic at x.

Proof. By compactness there exists $r_1 > r$ such that $\overline{\mathrm{B}}(x, r_1) \subset \Omega$. By Lemma 2.21 there exists $\delta > 0$ such that $\mathbf{A}_\rho u \le u$ on $\mathrm{B}(x, r_1)$ for all $\rho \le \delta$. Now let U be the open ball of radius $\min(\delta, r_1 - r)$ centered at x.

(a) By Fubini's theorem, for all balls $\mathrm{B}(y, \rho) \subset U$,

$$\mathbf{A}_\rho \mathbf{A}_r u(y) = \mathbf{A}_r \mathbf{A}_\rho u(y) \le \mathbf{A}_r u(y)\,,$$

so $\mathbf{A}_r u$ is superharmonic at x.

(b) If u is continuous, then its spherical averages are defined pointwise and vary continuously with the center. Fubini's theorem and the mean value inequality give

$$\mathbf{A}_\rho \mathbf{a}_r u(y) = \mathbf{a}_r \mathbf{A}_\rho u(y) \le \mathbf{a}_r u(y)\,,$$

so $\mathbf{a}_r u$ is superharmonic at x. □

Our next lemma shows that if u is superharmonic and continuous on Ω, then the mean value inequality holds for all balls with $\overline{\mathrm{B}}(x, r) \subset \Omega$, and moreover, the mean value $\mathbf{A}_r u(x)$ is a weakly decreasing function of the radius r.

Lemma 2.23 *If u is superharmonic and continuous on an open set $\Omega \subset \mathbb{R}^d$, then for all x and $0 < r < R$ with $\overline{\mathrm{B}}(x, R) \subset \Omega$ we have*

$$\mathbf{a}_R u(x) \le \mathbf{a}_r u(x) \le u(x)\,,$$

and

$$\mathbf{A}_R u(x) \le \mathbf{A}_r u(x) \le u(x)\,.$$

Proof. Let $B = \mathrm{B}(x, R)$. By Theorem 2.18 there is a continuous function h on $\overline{B}$ that is harmonic on B and agrees with u on ∂B. By Lemma 2.12(b) we have $h \le u$ on $\overline{B}$. Since h is the Poisson integral of $u|_{\partial B}$ and the Poisson kernel at the center of the ball is identically 1, we have $h(x) = \mathbf{a}_R u(x)$. Hence by the mean value property for spheres, Lemma 2.20, applied in B, we have

$$\mathbf{a}_R u(x) = h(x) = \mathbf{a}_r h(x) \le \mathbf{a}_r u(x)\,,$$

which proves the first inequality. The same argument applied to the harmonic extension of $u|_{\partial \mathrm{B}(x,r)}$ shows that $\mathbf{a}_r u(x) \le u(x)$, which proves the second.

By (2.4) the inequalities for $\mathbf{A}$ follow from the inequalities for $\mathbf{a}$. □

Lemma 2.24 *Let $\Omega_n \uparrow \Omega$ be open sets. If u_n is superharmonic and continuous in Ω_n, and $u_n \to u$ uniformly on compact subsets of Ω, then u is superharmonic and continuous in Ω.*

Proof. Since u is a uniform limit of continuous functions on compact subsets of Ω, u is continuous. To show that u is superharmonic, fix $y \in \Omega$ and $\rho > 0$ such that $\overline{\mathrm{B}}(y,\rho) \subset \Omega$. Then $\overline{\mathrm{B}}(y,\rho) \subset \Omega_n$ for all sufficiently large n. By Lemma 2.23, $\mathbf{A}_\rho u_n(y) \le u_n(y)$ for all sufficiently large n. Since integration can be interchanged with uniform limits, we obtain

$$\mathbf{A}_\rho u(y) = \lim_{n\to\infty} \mathbf{A}_\rho u_n(y) \le \lim_{n\to\infty} u_n(y) = u(y)\,,$$

so u is superharmonic. □

Theorem 2.25 (Harnack Principle) *Let $(u_n)_{n\ge1}$ be uniformly bounded harmonic functions that converge pointwise in an open set $\Omega \subset \mathbb{R}^d$. Then the convergence is uniform on compact subsets of Ω, and the limit is harmonic in Ω.*

Proof. Let $|u_n| \le M$ for all n, and let $u = \lim_{n\to\infty} u_n$, which is Borel with $|u| \le M$. If $\overline{\mathrm{B}}(y,\rho) \subset \Omega$, then $u_n(y) = \mathbf{A}_\rho u_n(y)$ by Lemma 2.20, and $\mathbf{A}_\rho u_n(y) \to \mathbf{A}_\rho u(y)$ by the dominated convergence theorem, so $u(y) = \mathbf{A}_\rho u(y)$. Thus u is harmonic in Ω.

To show that the convergence is uniform on a compact set $K \subset \Omega$, choose $\rho > 0$ such that $K_\rho := \{z : \operatorname{dist}(z,K) \le \rho\}$ is contained in Ω. For $y \in K$ we have $u_n(y) - u(y) = \mathbf{A}_\rho(u_n - u)(y)$, so

$$\sup_{y\in K} |u_n(y) - u(y)| \le \frac{1}{\omega_d \rho^d} \int_{K_\rho} |u_n - u|\,\mathrm{d}z \longrightarrow 0\,,$$

by the dominated convergence theorem, since $|u_n - u| \le 2M$ on the bounded set K_ρ. □

2.3.1 Fine Properties of Superharmonic Functions

Next we show that every superharmonic function u has a lower semicontinuous representative $\tilde{u}$ that agrees with u almost everywhere.

Theorem 2.26 *Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$. Let $R > 0$, and let*

$$\Omega_R := \{x \in \Omega : \overline{\mathrm{B}}(x,R) \subset \Omega\}\,.$$

Then the following hold.

(a) For every $0 < r < R$,

$$\mathbf{A}_R u(x) \le \mathbf{A}_r u(x) \qquad \text{for all } x \in \Omega_R\,.$$

(b) The function

$$\tilde{u}(x) := \lim_{\epsilon \downarrow 0} \mathbf{A}_\epsilon u(x)$$

is an increasing limit by (a), is lower semicontinuous and superharmonic in Ω, and equals u almost everywhere in Ω.

Proof. (a) By the remark after Definition 2.1, $\mathbf{A}_\epsilon u$ is continuous in Ω_ϵ for each $\epsilon > 0$, and it is superharmonic there by Lemma 2.22(a). Thus, given $0 < r < R$, by Lemma 2.23 we have

$$\mathbf{A}_R \mathbf{A}_\epsilon u \le \mathbf{A}_r \mathbf{A}_\epsilon u \qquad \text{on } \Omega_{R+\epsilon}\,. \tag{2.8}$$

By Fubini's theorem, the two sides of (2.8) equal $\mathbf{A}_\epsilon(\mathbf{A}_R u)$ and $\mathbf{A}_\epsilon(\mathbf{A}_r u)$. Since $\mathbf{A}_R u$ and $\mathbf{A}_r u$ are continuous, taking $\epsilon \downarrow 0$ we obtain $\mathbf{A}_R u \le \mathbf{A}_r u$ on Ω_R.

(b) As an increasing limit of continuous functions, $\tilde{u}$ is lower semicontinuous (Exercise 2.4). The mean value inequality (2.3) gives $\tilde{u} \le u$. Every compact subset of Ω has a neighborhood on which, for a suitable $\delta > 0$, $\mathbf{A}_\delta u$ is continuous and $\mathbf{A}_\delta u \le \tilde{u} \le u$; hence $\tilde{u}$ is locally integrable.

Next we show that $\tilde{u} = u$ almost everywhere. Let $\overline{\mathrm{B}}(x, R) \subset \Omega$. Averaging $\mathbf{A}_\epsilon u \le \tilde{u} \le u$ over $\mathrm{B}(x, R)$ and commuting the two averages by Fubini's theorem gives, for all sufficiently small $\epsilon > 0$,

$$\mathbf{A}_\epsilon(\mathbf{A}_R u)(x) = \mathbf{A}_R(\mathbf{A}_\epsilon u)(x) \le \mathbf{A}_R \tilde{u}(x) \le \mathbf{A}_R u(x)\,.$$

Since $\mathbf{A}_R u$ is continuous at x, the left side tends to the right side as $\epsilon \downarrow 0$, so $\mathbf{A}_R \tilde{u}(x) = \mathbf{A}_R u(x)$. Hence the nonnegative difference $u - \tilde{u}$, defined almost everywhere, vanishes almost everywhere on $\mathrm{B}(x, R)$, and countably many such balls cover Ω.

Finally, fix $R > 0$. Since $u = \tilde{u}$ almost everywhere, $\mathbf{A}_R \tilde{u} = \mathbf{A}_R u$, and by (a), $\mathbf{A}_R u \le \mathbf{A}_r u$ for every $0 < r < R$. Taking $r \to 0$, this implies $\mathbf{A}_R \tilde{u} \le \tilde{u}$ on Ω_R, so $\tilde{u}$ is superharmonic. □

In particular, we obtain the mean value inequality on all balls, without assuming continuity.

Corollary 2.27 *Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$, and let $\overline{\mathrm{B}}(x, R) \subset \Omega$. Then $\mathbf{A}_R u(x) \le u(x)$.*

Proof. By Definition 2.6, $\mathbf{A}_r u(x) \le u(x)$ for all sufficiently small $r > 0$. Since $x \in \Omega_R$, by Theorem 2.26(a) we have $\mathbf{A}_R u(x) \le \mathbf{A}_r u(x)$. □

Theorem 2.26 shows the importance of lower semicontinuous superharmonic functions. Next we show that every such function is a pointwise limit of continuous superharmonic functions.

Lemma 2.28 *Let u be a lower semicontinuous superharmonic function on an open set $\Omega \subset \mathbb{R}^d$. Then*

$$u(x) = \lim_{r \downarrow 0} \mathbf{A}_r u(x) \qquad \text{for all } x \in \Omega\,. \tag{2.9}$$

Proof. Since u is superharmonic at $x \in \Omega$, there exists $\delta > 0$ such that $u(x) \geq \mathbf{A}_r u(x)$ for all $r \leq \delta$. Given a real number $a < u(x)$, since u is lower semicontinuous at x, there exists $\delta_1 < \delta$ such that $u > a$ on $\mathrm{B}(x, \delta_1)$ (Lemma 2.5). Thus for all $r \leq \delta_1$ we have

$$a \leq \mathbf{A}_r u(x) \leq u(x)\,.$$

Letting $a \uparrow u(x)$ proves (2.9), also when $u(x) = +\infty$. □

By Theorem 2.26(a), the convergence in (2.9) is monotone. Hence, if $K \subset \Omega$ is compact and $r_n \downarrow 0$ satisfies $\overline{\mathrm{B}}(x, r_n) \subset \Omega$ for all $x \in K$ and all n, then the continuous superharmonic functions $\mathbf{A}_{r_n} u$ increase pointwise to u on K (see Figure 2.7 for the case $u(x) = -|x|$).

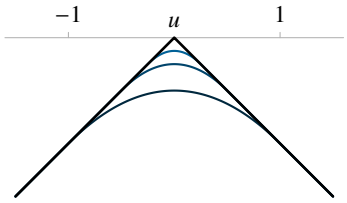


Figure 2.7: The ball averages $\mathbf{A}_\epsilon u$ of the superharmonic function $u(x) = -|x|$ on $\mathbb{R}$, for $\epsilon = 1, \frac{1}{2}, \frac{1}{4}$ (darker for smaller ϵ): they are continuous, lie below u, and increase to u as $\epsilon \downarrow 0$.

We conclude this section by giving one more characterization of superharmonic functions in the continuous case.

Theorem 2.29 *Let u be continuous on an open set $\Omega \subset \mathbb{R}^d$. Then u is superharmonic in Ω if and only if*

$$\int u\, \Delta\psi \, \mathrm{d}x \leq 0\,, \tag{2.10}$$

for all nonnegative C^∞ compactly supported functions ψ on Ω.

Proof. For $\epsilon > 0$ let $u_\epsilon := u * \varphi_\epsilon$, where $\varphi_\epsilon(z) := \epsilon^{-d}\eta(z/\epsilon)$ is a spherically symmetric smooth mollifier, with η as in (2.1); we consider u_ϵ on the set Ω_ϵ of points x with $\overline{\mathrm{B}}(x, \epsilon) \subset \Omega$. As in Section 2.1, u_ϵ is C^∞ on Ω_ϵ and $\Delta u_\epsilon = u * \Delta\varphi_\epsilon$. Since $|u_\epsilon(y) - u(y)| \leq \sup_{|z|<\epsilon} |u(y-z) - u(y)|$ and u is uniformly continuous on compact sets, $u_\epsilon \to u$ uniformly on compact subsets of Ω as $\epsilon \downarrow 0$.

Suppose first that u is superharmonic in Ω. By Lemma 2.23, $\mathbf{A}_r u \le u$ on Ω_r for every $r > 0$, so by Fubini's theorem $\mathbf{A}_r u_\epsilon = (\mathbf{A}_r u) * \varphi_\epsilon \le u_\epsilon$ on $\Omega_{r+\epsilon}$. Hence u_ϵ is superharmonic in Ω_ϵ, and $\Delta u_\epsilon \le 0$ there by Lemma 2.15. Now let ψ be a nonnegative C^∞ function with compact support in Ω. For all sufficiently small ϵ the support of ψ lies in Ω_ϵ, and integrating by parts twice (there are no boundary terms since ψ has compact support) we obtain

$$\int u_\epsilon \, \Delta\psi \, \mathrm{d}x = \int \psi \, \Delta u_\epsilon \, \mathrm{d}x \le 0 \,.$$

Since $u_\epsilon \to u$ uniformly on the support of ψ, taking $\epsilon \downarrow 0$ proves (2.10).

Conversely, suppose that (2.10) holds. For $x \in \Omega_\epsilon$, the function $\psi(y) := \varphi_\epsilon(x - y)$ is nonnegative and C^∞ with support in $\overline{\mathrm{B}}(x, \epsilon) \subset \Omega$, so

$$\Delta u_\epsilon(x) = \int u(y) \, (\Delta\varphi_\epsilon)(x - y) \, \mathrm{d}y = \int u \, \Delta\psi \, \mathrm{d}y \le 0 \,.$$

By Lemma 2.15, u_ϵ is superharmonic and continuous in the open set Ω_ϵ. Since these sets increase to Ω as $\epsilon \downarrow 0$, and $u_\epsilon \to u$ uniformly on compact subsets of Ω (here we use the assumption that u is continuous), we conclude from Lemma 2.24, applied along a sequence $\epsilon_n \downarrow 0$, that u is superharmonic in Ω. □

2.4 HARNACK INEQUALITY

The Harnack inequality says that a nonnegative harmonic function on the unit ball cannot vary much in a smaller ball. We will give two proofs: the first, using the Poisson kernel, gives sharp constants, while the second uses only the mean value property.

Theorem 2.30 *Let $d \ge 1$ and $0 \le r < 1$. Let h be a nonnegative harmonic function on $\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. Then for every $x, y \in \overline{\mathrm{B}}(\mathbf{0}, r)$,*

$$h(x) \le \left(\frac{1 + r}{1 - r}\right)^d h(y) \,.$$

Proof of Theorem 2.30 using the Poisson kernel. Fix $w \in \overline{\mathrm{B}}(\mathbf{0}, r)$, and choose ρ with $|w| < \rho < 1$. Since the function $z \mapsto h(\rho z)$ is harmonic in $\mathrm{B}(\mathbf{0}, 1)$ and continuous on its closure, it is the Poisson integral of its boundary values (see the remarks after Theorem 2.18), so scaling the surface measure we obtain

$$h(w) = \fint_{\partial \mathrm{B}(\mathbf{0}, \rho)} \rho^{d-2} \frac{\rho^2 - |w|^2}{|w - v|^d} h(v) \, \mathrm{d}S(v) \,.$$

By the previous display, the inequalities $\rho - |w| \le |w - v| \le \rho + |w|$ for $|v| = \rho$ (see Figure 2.8), and the fact that the spherical average of h equals $h(\mathbf{0})$, we obtain

$$\rho^{d-2}\frac{\rho^2 - |w|^2}{(\rho + |w|)^d}h(\mathbf{0}) \le h(w) \le \rho^{d-2}\frac{\rho^2 - |w|^2}{(\rho - |w|)^d}h(\mathbf{0})\,.$$

Letting $\rho \uparrow 1$ and taking the worst case over $|w| \le r$ gives

$$\frac{1 - r}{(1 + r)^{d-1}}h(\mathbf{0}) \le h(w) \le \frac{1 + r}{(1 - r)^{d-1}}h(\mathbf{0})\,. \tag{2.11}$$

Applying the upper bound to x and the lower bound to y proves the theorem. □

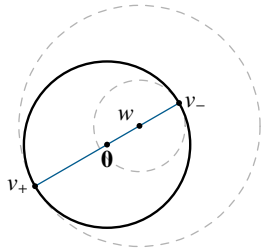


Figure 2.8: For $|w| < \rho$, every point v of the sphere of radius ρ lies between the circles about w of radii $\rho - |w|$ and $\rho + |w|$ (dashed), which touch it at the points v_- and v_+ nearest to and farthest from w. This gives the two bounds on the Poisson kernel in the proof of Theorem 2.30.

2.4.1 Alternative Proof via the Mean Value Property

Using only the mean value property, we now prove that $h(x) \le Ch(y)$ for all $x, y \in \mathrm{B}(\mathbf{0}, r)$, where C depends only on d and r but is not sharp.

Proof of a weaker form of Theorem 2.30. The idea is to join x to y by a chain of equally spaced points, and to compare consecutive values using two nested balls (see Figure 2.9). Let $N := 1 + \lfloor 4r/(1 - r) \rfloor$. If $x = y$ there is nothing to prove, so let $a := |x - y|/N > 0$, and let $z_i := (1 - i/N)x + (i/N)y$ for $0 \le i \le N$. Each point z_i lies in $\mathrm{B}(\mathbf{0}, r)$, consecutive points are at distance a, and $2a < 4r/N < 1 - r$, so $\mathrm{B}(z_{i-1}, a) \subset \mathrm{B}(z_i, 2a)$ and $\overline{\mathrm{B}}(z_i, 2a) \subset \mathrm{B}(\mathbf{0}, 1)$. Hence for $1 \le i \le N$, by Lemma 2.20, since $h \ge 0$ we have

$$h(z_{i-1}) = \frac{1}{\omega_d a^d}\int_{\mathrm{B}(z_{i-1},a)} h(v)\,\mathrm{d}v \le \frac{1}{\omega_d a^d}\int_{\mathrm{B}(z_i,2a)} h(v)\,\mathrm{d}v = 2^d h(z_i)\,.$$

Iterating along the chain, we obtain $h(x) \le 2^{dN}h(y)$, where N depends only on r. □

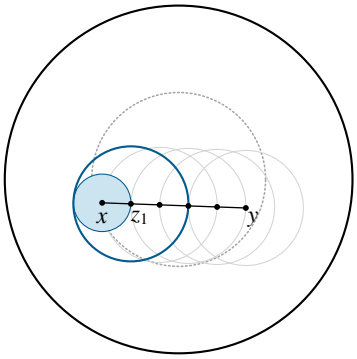


Figure 2.9: The chain $x = z_0, z_1, \dots, z_N = y$ of equally spaced points in $\mathrm{B}(\mathbf{0}, r)$ (dotted circle), drawn for $r = 1/2$, so that $N = 5$. The shaded ball $\mathrm{B}(z_0, a)$ lies in $\mathrm{B}(z_1, 2a)$ (blue), and every ball $\overline{\mathrm{B}}(z_i, 2a)$ lies in the unit ball, so consecutive values of h along the chain differ by at most a factor 2^d.

Exercise 2.35 shows that the bounds (2.11) are optimal, and Exercise 2.36 extends the result to other domains. An important corollary of Theorem 2.30, which Exercise 2.37 asks you to prove, is the following.

Corollary 2.31 *Every nonnegative harmonic function on $\mathbb{R}^d$ is constant.*

The Harnack inequality has another consequence: a nonnegative function on $\mathbb{R}^d$ with bounded Laplacian can grow at most quadratically.

Proposition 2.32 *Let $d \ge 1$, let $R > 0$, let $1/2 \le \beta < 1$, and let $C := (1+\beta)/(1-\beta)^{d-1}$. Let f be a nonnegative continuous function on $\mathrm{B}(\mathbf{0}, R)$, let $\lambda > 0$, and let $f_\pm(x) := f(x) \pm \lambda|x|^2$. If f_+ is subharmonic and f_- is superharmonic in $\mathrm{B}(\mathbf{0}, R)$, then, for every $x \in \mathrm{B}(\mathbf{0}, \beta R)$,*

$$f(x) \le C(f(\mathbf{0}) + 8\lambda|x|^2)\,.$$

Proof. The case $x = \mathbf{0}$ is immediate since $C \ge 1$. Given $x \ne \mathbf{0}$ in $\mathrm{B}(\mathbf{0}, \beta R)$, let $r = |x|/\beta < R$, and let h be the harmonic function on the ball $B = \mathrm{B}(\mathbf{0}, r)$, continuous on $\overline{B}$, which agrees with f on ∂B. Since $f \ge 0$, we have $h \ge 0$ in B by the minimum principle. Since the harmonic functions $h \pm \lambda r^2$ agree with $f_\pm$ on ∂B, by Lemma 2.12(b) applied to the superharmonic functions f_- and $-f_+$ we have

$$|f(z) - h(z)| \le \lambda(r^2 - |z|^2) \qquad \text{for all } z \in B\,. \tag{2.12}$$

Since $|x|/r = \beta$, by the Harnack inequality in the form of the upper bound in (2.11) (after rescaling) we have $h(x) \le Ch(\mathbf{0})$. Hence by (2.12) at x and at $\mathbf{0}$,

$$f(x) \le Ch(\mathbf{0}) + \lambda r^2 \le Cf(\mathbf{0}) + (C+1)\lambda r^2 \le C(f(\mathbf{0}) + 8\lambda|x|^2)\,,$$

where in the last step we have used $r = |x|/\beta \le 2|x|$ and $C \ge 1$. □

We remark that in dimensions $d \ge 2$, the upper and lower bounds on the Laplacian are both necessary: see Exercise 2.39. In dimension 1, just the upper bound suffices; and in all dimensions, an upper bound on the Hessian suffices (Exercise 2.40).

2.5 SUPERHARMONIC POTENTIALS

Throughout this section we assume $d \ge 2$. Since the Laplacian is invariant with respect to rotation, radial harmonic functions play an important role in potential theory. In Exercise 2.41 you will find their form, which we will use.

For a bounded measurable function σ on $\mathbb{R}^d$ with compact support, write

$$\mathbf{G}\sigma(x) := \int_{\mathbb{R}^d} \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\,, \tag{2.13}$$

where

$$\mathcal{G}(x) := \begin{cases} -\frac{1}{2\pi}\log|x|, & d=2, \\ a_d|x|^{2-d}, & d \ge 3\,. \end{cases} \tag{2.14}$$

We use the convention $\mathcal{G}(\mathbf{0}) = +\infty$. Here $a_d := \frac{1}{d(d-2)\omega_d}$, where ω_d is the volume of the unit ball in $\mathbb{R}^d$. Exercise 2.41 shows that $\mathcal{G}$, which we call the **Green function**, is harmonic on $\mathbb{R}^d \setminus \{\mathbf{0}\}$.

The operator $\mathbf{G}$ has both physical and mathematical significance. If $\sigma \ge 0$ and we interpret $\sigma(y)\,\mathrm{d}y$ as the density of mass near $y \in \mathbb{R}^d$, then $\mathbf{G}\sigma$ has the physical interpretation of minus the **gravitational potential** of σ. An observer located at x will feel a gravitational force proportional to $\nabla(\mathbf{G}\sigma)(x)$. We could also interpret $\sigma(y)\,\mathrm{d}y$ as a charge density, in which case the requirement $\sigma \ge 0$ is not needed since charge may be negative; $\mathbf{G}\sigma$ is then called the **Coulomb potential** of σ.

The mathematical significance of $\mathbf{G}$ derives from the identity

$$\mathbf{G} = (-\Delta)^{-1}\,,$$

made precise by Theorem 2.38 and its corollaries below. Given a function σ, formula (2.13) tells us how to construct a function whose Laplacian is $-\sigma$.

Our first result, Newton's shell theorem, has a simple interpretation in terms of gravity. The gravitational potential produced by a uniform spherical shell ∂B (where B is a ball centered at the origin) is constant in the interior of B, and agrees outside $\overline{B}$ with the potential produced by a point mass at the center of B. In other words, an observer constrained to stay in the interior of B feels no net gravitational force from the mass on ∂B, and an observer constrained to stay in the exterior cannot tell the difference (using

gravitational measurements alone) between a sphere of uniform density and a point mass at its center.

Lemma 2.33 (Newton's Shell Theorem) *Let $d \geq 2$, let $r > 0$, let $B = \mathrm{B}(\mathbf{0}, r) \subset \mathbb{R}^d$, and let $\mathbf{e}_1$ be the first standard basis vector. Then*

$$\fint_{\partial B} \mathcal{G}(x-y)\,\mathrm{d}S(y) = \begin{cases} \mathcal{G}(r\mathbf{e}_1), & |x| < r, \\ \mathcal{G}(x), & |x| \geq r\,. \end{cases}$$

Proof. Since $\mathcal{G}(x-y)$ is a function of $|x-y|$ (and surface measure is rotation invariant), the function $u(x) \coloneqq \fint_{\partial B} \mathcal{G}(x-y)\,\mathrm{d}S(y)$ depends only on $|x|$. Moreover, u is harmonic on B, since $\mathcal{G}(\cdot - y)$ is harmonic on B for each $y \in \partial B$ and is bounded on compact subsets of B uniformly in $y \in \partial B$ (so that by Fubini's theorem we may average in x before integrating in y). Hence for $|x| < r$ we have by the mean value property for spheres, Lemma 2.20,

$$u(x) = \mathbf{a}_{|x|}u(\mathbf{0}) = u(\mathbf{0}) = \mathcal{G}(r\mathbf{e}_1)\,.$$

Now fix $x \in \mathbb{R}^d$ with $|x| > r$. Since $y \mapsto \mathcal{G}(x-y)$ is harmonic on $\mathbb{R}^d \setminus \{x\}$, we have $u(x) = \mathbf{a}_r[\mathcal{G}(x-\cdot)](\mathbf{0}) = \mathcal{G}(x)$.

Finally, if $|x| = r$, then $\mathcal{G}(tx-y) \uparrow \mathcal{G}(x-y)$ as $t \downarrow 1$. These functions are nonnegative when $d \geq 3$, and when $d = 2$ they become nonnegative for $1 < t \leq 2$ after adding a constant independent of y, so the monotone convergence theorem concludes the proof. □

Corollary 2.34 *Let $d \geq 2$. For fixed $x \in \mathbb{R}^d$ the function $y \mapsto \mathcal{G}(x-y)$ is superharmonic in $\mathbb{R}^d$.*

Proof. For every ball $\mathrm{B}(z, r) \subset \mathbb{R}^d$, we have by translation invariance

$$\mathbf{a}_r[\mathcal{G}(x-\cdot)](z) = \fint_{\partial \mathrm{B}(\mathbf{0},r)} \mathcal{G}(x-z-y)\,\mathrm{d}S(y)\,.$$

By Lemma 2.33, the right side is at most $\mathcal{G}(x-z)$; here we have used the fact that $\mathcal{G}(w)$ is decreasing in $|w|$. From (2.4) we conclude that $\mathbf{A}_r[\mathcal{G}(x-\cdot)](z) \leq \mathcal{G}(x-z)$. □

In dimensions $d \geq 3$, the Green function $\mathcal{G}(x)$ is positive and superharmonic on $\mathbb{R}^d$. The next lemma shows that the situation is different on $\mathbb{R}^2$ (see Figure 2.10).

Lemma 2.35 *If $u \geq 0$ is superharmonic in $\mathbb{R}^2$, then u is constant almost everywhere.*

Proof. By Theorem 2.26(b), replacing u by its lower semicontinuous representative changes u only on a null set, and the representative is superharmonic and nonnegative (being a limit of averages of u), so we may assume that u is lower semicontinuous. Fix $x \neq y$. If $u(y) = 0$ then trivially $u(x) \geq u(y)$,

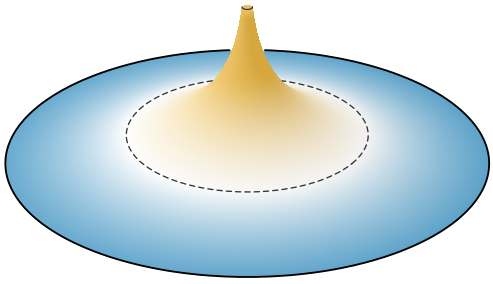


Figure 2.10: The graph of $\min(\mathcal{G}(x), \frac{1}{2})$ over the disk $\mathrm{B}(\mathbf{0}, 2)$ in the plane, where $\mathcal{G}(x) = -\frac{1}{2\pi}\log|x|$ is positive (orange) inside the unit circle (dashed) and negative (blue) outside it. The Green function must take negative values: by Lemma 2.35, a nonnegative superharmonic function on $\mathbb{R}^2$ is constant almost everywhere.

so suppose $u(y) > 0$, and fix a real number M with $0 \le M < u(y)$. By lower semicontinuity, there exists $0 < \delta < |x - y|$ such that $u > M$ on $\overline{\mathrm{B}}(y, \delta)$. For $R > |x - y|$, the function

$$h_R(z) := M\frac{\log(R/|z - y|)}{\log(R/\delta)}$$

is harmonic in the annulus $A_R := \mathrm{B}(y, R) \setminus \overline{\mathrm{B}}(y, \delta)$, which contains x, and satisfies $u \ge h_R$ on ∂A_R (it equals M on the inner circle and 0 on the outer circle). By Lemma 2.12(b) we have $u \ge h_R$ in A_R, so $u(x) \ge \lim_{R\to\infty} h_R(x) = M$ (see Figure 2.11). Hence $u(x) \ge u(y)$, and by symmetry $u(x) = u(y)$. The constant is finite because u is locally integrable. □

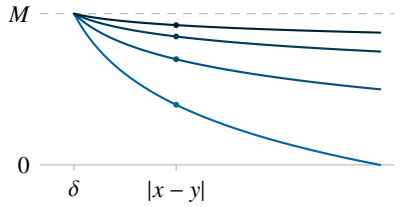


Figure 2.11: The barriers h_R of the proof of Lemma 2.35 as functions of $|z - y|$, for $\delta = M = 1$ and $R = 10, 10^2, 10^4, 10^8$ (lighter for larger R): each equals M on the circle of radius δ about y, and at a fixed point x (dots) they increase to M as $R \to \infty$.

For an alternative proof using Brownian motion, see Exercise 2.49.

Lemma 2.36 *Let $d \geq 2$, and let σ be a bounded measurable function on $\mathbb{R}^d$ with compact support.*

(a) If $\sigma \geq 0$ on an open set $\Omega \subset \mathbb{R}^d$, then $\mathbf{G}\sigma$ is superharmonic in Ω.

(b) The potential $\mathbf{G}\sigma$ is harmonic off the closed support of σ.

Proof. Choose $L > 0$ so that σ vanishes outside $\mathrm{B}(\mathbf{0}, L)$. For every $M > 0$ and $|z| \leq M$, since $\mathcal{G}$ is locally integrable we have

$$\int_{\mathbb{R}^d} |\sigma(y)\mathcal{G}(z-y)|\,\mathrm{d}y \leq \|\sigma\|_\infty \int_{\mathrm{B}(\mathbf{0},L+M)} |\mathcal{G}(w)|\,\mathrm{d}w < \infty\,.$$

Thus $\mathbf{G}\sigma$ is locally bounded, and Fubini's theorem applies to the integrals below.

(a) Given $\overline{\mathrm{B}}(x,r) \subset \Omega$, by Fubini's theorem

$$\mathbf{G}\sigma(x) - \mathbf{A}_r(\mathbf{G}\sigma)(x) = \int_{\mathbb{R}^d} \sigma(y)\left(\mathcal{G}(x-y) - \fint_{\mathrm{B}(x,r)} \mathcal{G}(z-y)\,\mathrm{d}z\right)\mathrm{d}y\,.$$

By Corollary 2.34, the term in parentheses is nonnegative, and by Newton's shell theorem (Lemma 2.33) and (2.4), it vanishes when $|x-y| \geq r$. Since $\sigma \geq 0$ on $\mathrm{B}(x,r)$, the integral is nonnegative. Hence $\mathbf{G}\sigma$ obeys the mean value inequality on Ω. This shows part (a).

(b) Part (b) follows by applying part (a) to σ and to $-\sigma$, both of which vanish off the closed support of σ. □

Next we give an explicit formula for the potential of a ball.

Lemma 2.37 *Let $d \geq 2$, let $r > 0$, and let $B = \mathrm{B}(\mathbf{0}, r)$. In dimensions $d \geq 3$,*

$$\mathbf{G}\mathbf{1}_B(x) = \begin{cases} \frac{r^2}{2(d-2)} - \frac{1}{2d}|x|^2, & |x| < r, \\ \frac{r^2}{d(d-2)}\left(\frac{|x|}{r}\right)^{2-d}, & |x| \geq r\,. \end{cases} \tag{2.15}$$

In dimension $d = 2$,

$$\mathbf{G}\mathbf{1}_B(x) = \begin{cases} \frac{r^2}{4}(1 - 2\log r) - \frac{1}{4}|x|^2, & |x| < r, \\ -\frac{r^2}{2}\log|x|, & |x| \geq r\,. \end{cases} \tag{2.16}$$

Proof. The case $d \geq 3$ is Exercise 2.44, and the case $d = 2$ is Exercise 2.45. □

The potentials of a uniform ball, of its boundary sphere and of a point mass at its center are compared in Figure 2.12.

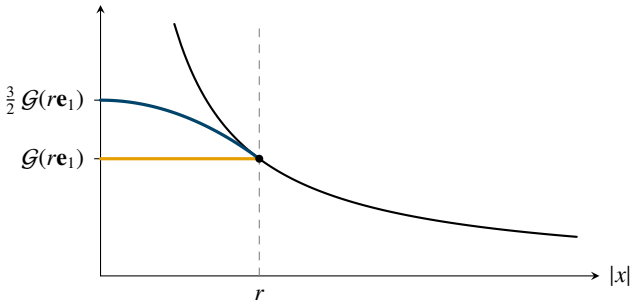


Figure 2.12: The potentials in $d = 3$, as functions of $|x|$, of unit mass at the origin (black), spread uniformly on $\mathrm{B}(\mathbf{0}, r)$ (blue), and spread uniformly on the sphere $\partial\mathrm{B}(\mathbf{0}, r)$ (orange). Outside the ball the three agree, as Newton's shell theorem asserts; inside, the potential of the sphere is constant and that of the ball is a parabola that meets the outer curve with the same slope at $|x| = r$.

Theorem 2.38 (Green Function and Inverse Laplacian) *Let $d \geq 2$, and let σ be a continuous function on $\mathbb{R}^d$ with compact support. If the convolution $\mathbf{G}\sigma$ is C^2, then*

$$\Delta(\mathbf{G}\sigma) = -\sigma\,. \tag{2.17}$$

Proof. Fix $z \in \mathbb{R}^d$. Given $\epsilon > 0$, choose $\delta > 0$ so that $|\sigma - \sigma(z)| \leq \epsilon$ on $\mathrm{B}(z, \delta)$, and let $q := \mathbf{G}\mathbf{1}_{\mathrm{B}(z,\delta)}$. By Lemma 2.37 (translated to z), q is C^2 and $\Delta q = -1$ in $\mathrm{B}(z, \delta)$. The functions

$$\mathbf{G}\sigma - (\sigma(z) - \epsilon)q \qquad \text{and} \qquad (\sigma(z) + \epsilon)q - \mathbf{G}\sigma$$

are potentials of sources that are nonnegative on $\mathrm{B}(z, \delta)$, so they are superharmonic there by Lemma 2.36(a). Since they are also C^2 there, their Laplacians are nonpositive by Lemma 2.15, which gives

$$-\sigma(z) - \epsilon \leq \Delta\mathbf{G}\sigma(z) \leq -\sigma(z) + \epsilon\,.$$

Since ϵ was arbitrary, we conclude that $\Delta\mathbf{G}\sigma(z) = -\sigma(z)$. □

Remark. If σ is C^2 with compact support, then it is immediate that $\mathbf{G}\sigma$ is C^2. Indeed, writing

$$\mathbf{G}\sigma(x) = \int_{\mathbb{R}^d} \mathcal{G}(z)\sigma(x - z)\,\mathrm{d}z\,,$$

the derivatives of σ are bounded and, locally in x, the functions $z \mapsto \sigma(x - z)$ have a common compact support, so since $\mathcal{G}$ is locally integrable we may differentiate twice under the integral sign by the dominated convergence theorem. Also, if σ is bounded and measurable and vanishes outside $\mathrm{B}(\mathbf{0}, R)$, then $\mathbf{G}\sigma$ is locally bounded. For $d \geq 3$ this follows from $\mathcal{G} > 0$, since

$|\mathbf{G}\sigma| \le \|\sigma\|_\infty \mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},R)}$, which is bounded by Lemma 2.37. For $d = 2$, the identity $|\mathcal{G}(w)| = \mathcal{G}(w) + \frac{1}{\pi}\log^+|w|$ gives

$$|\mathbf{G}\sigma(x)| \le \|\sigma\|_\infty \Big(\mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},R)}(x) + R^2 \log^+(|x| + R)\Big).$$

In fact $\mathbf{G}\sigma$ is continuous, since

$$|\mathbf{G}\sigma(x) - \mathbf{G}\sigma(x')| \le \|\sigma\|_\infty \int_{\mathrm{B}(\mathbf{0},R)} |\mathcal{G}(x-y) - \mathcal{G}(x'-y)|\,\mathrm{d}y$$

and, for $|x' - x| < 1$, the right side is at most $\|\sigma\|_\infty$ times the L^1 distance between the integrable function $\mathcal{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},|x|+R+1)}$ and its translate by $x' - x$, which tends to 0 as $x' \to x$ since translation is continuous in L^1.

Corollary 2.39 *Let $d \ge 2$, and let $\psi : \mathbb{R}^d \to \mathbb{R}$ be a C^2 function with compact support. Then $\mathbf{G}(\Delta\psi) = -\psi$.*

Proof. Differentiating twice under the integral sign, as in Remark 2.5, we see that $\mathbf{G}\psi$ is C^2 and $\Delta\mathbf{G}\psi = \mathbf{G}(\Delta\psi)$. By Theorem 2.38, the left side equals $-\psi$. □

If $\mathbf{G}\sigma$ is not C^2, then we cannot interpret (2.17) in the classical sense. This leads to the notion of weak solutions: if f and u are locally integrable functions on $\mathbb{R}^d$, we say that u is a **weak solution** of $\Delta u = f$ if for every compactly supported C^∞ function ψ,

$$\int_{\mathbb{R}^d} u(x)\Delta\psi(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} f(x)\psi(x)\,\mathrm{d}x\,.$$

The next corollary demonstrates that $\mathbf{G}\sigma$ is a weak solution of $\Delta u = -\sigma$ when σ is only bounded, measurable and compactly supported rather than differentiable. This formulation transfers the differentiability requirement from the solution u to the test function ψ through integration by parts.

Corollary 2.40 *Let $d \ge 2$, and let $\sigma : \mathbb{R}^d \to \mathbb{R}$ be bounded, measurable, and compactly supported. Then for every compactly supported C^2 function $\psi : \mathbb{R}^d \to \mathbb{R}$,*

$$\int_{\mathbb{R}^d} (\mathbf{G}\sigma)(x)\Delta\psi(x)\,\mathrm{d}x = -\int_{\mathbb{R}^d} \sigma(x)\psi(x)\,\mathrm{d}x\,.$$

Proof. By Fubini's theorem,

$$\int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\right)\Delta\psi(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \mathbf{G}(\Delta\psi)(y)\sigma(y)\,\mathrm{d}y\,.$$

Now use Corollary 2.39. □

Example 2.41 We construct a superharmonic function that is not continuous. In dimensions $d \ge 2$, let $y_n \in \mathbb{R}^d \setminus \{\mathbf{0}\}$ be a sequence converging to $\mathbf{0}$, and let

m_n be a sequence of positive real numbers satisfying

$$\sum_{n=1}^{\infty} m_n \mathcal{G}(y_n) < 1 \,.$$

(Note that $\sum_n m_n < \infty$, since $\mathcal{G}(y_n) \to \infty$.) Then the function

$$u(x) := \min\left(1, \sum_{n=1}^{\infty} m_n \mathcal{G}(x - y_n)\right)$$

is superharmonic on $\mathbb{R}^d$ (Exercise 2.46) and discontinuous at $\mathbf{0}$, since $u(y_n) = 1$ for all n while $u(\mathbf{0}) = \sum_n m_n \mathcal{G}(y_n) < 1$.

On the other hand, Exercise 2.47 shows that every lower semicontinuous superharmonic function is continuous in dimension $d = 1$.

2.5.1 The Arcsine Law

We end the section with one potential that we can compute exactly. It is the source of the square root in the Beurling estimates of Section 2.6 and Chapter 8, which bound the probability that Brownian motion (or random walk) avoids a long connected set.

For real numbers $\alpha < \beta$, the **arcsine law** on $[\alpha, \beta]$ is the probability measure

$$\text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) := \frac{\mathrm{d}t}{\pi\sqrt{(t-\alpha)(\beta-t)}} \qquad \text{for } \alpha < t < \beta \,.$$

Its density is symmetric about the midpoint of the interval, and of order the inverse square root of the distance to the nearer endpoint. (By Exercise 2.52, the arcsine law is the harmonic measure from infinity of the segment for planar Brownian motion.) Next we compute its logarithmic potential, using the fact that the arcsine law on $[0, 1]$ is the vertical projection of the uniform distribution on a circle (see Figure 2.13).

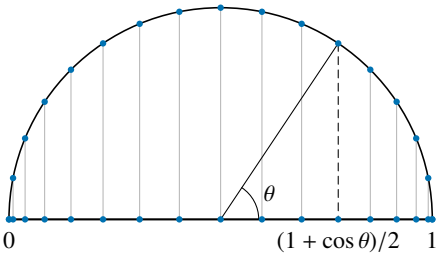


Figure 2.13: Vertical projection maps the uniform distribution on the upper half of the circle with diameter $[0, 1]$ to the arcsine law $\text{Arcsine}_{0,1}$: if θ is uniform on $[0, \pi]$, then $(1 + \cos\theta)/2$ has law $\text{Arcsine}_{0,1}$. The points at angles $\pi k/16$ cut the half-circle into arcs of equal length, so their projections cut $[0, 1]$ into 16 intervals of equal mass, which are shortest near the endpoints.

Lemma 2.42 *Let $\alpha < \beta$ be real numbers, and let $\text{Arcsine}_{\alpha,\beta}$ be the arcsine law on $[\alpha, \beta]$.*

(a) For every real number s,

$$\int \log|s - t| \, \text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) \geq \log\frac{\beta - \alpha}{4},$$

with equality when $\alpha \leq s \leq \beta$.

(b) For every $q \geq 0$,

$$\int \log\Big(1 + \frac{q}{t - \alpha}\Big) \, \text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) \leq 2\sqrt{\frac{q}{\beta - \alpha}}.$$

Since $\log(1 + q/(t - \alpha)) = \log|\alpha - q - t| - \log|\alpha - t|$, the integral in part (b) is the difference between the logarithmic potential at the point $\alpha - q$, at distance q from the interval, and its value on the interval. In other words, the potential rises only like the square root of the distance as the point leaves the interval through an endpoint (see Figure 2.14).

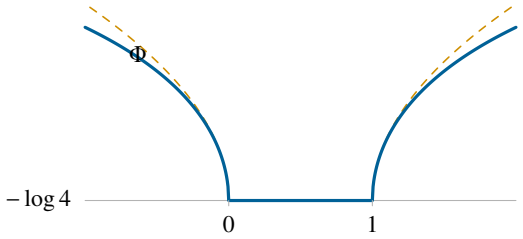


Figure 2.14: The logarithmic potential $\Phi(s) = \int \log|s - t| \, \text{Arcsine}_{0,1}(\,\mathrm{d}t)$ of the arcsine law on $[0, 1]$: it equals $-\log 4$ on the interval and, outside it, stays below $-\log 4 + 2\sqrt{\text{dist}(s, [0, 1])}$ (dashed), which it follows near the endpoints.

Proof of Lemma 2.42. Since the map $t \mapsto \alpha + (\beta - \alpha)t$ sends the arcsine law on $[0, 1]$ to the arcsine law on $[\alpha, \beta]$, it adds $\log(\beta - \alpha)$ to the potential in (a) and replaces q by $q/(\beta - \alpha)$ in (b), so it suffices to consider $\alpha = 0$ and $\beta = 1$.

We first write the potential as an average over a circle. If θ is uniformly distributed on $[0, 2\pi]$, then $(1 + \cos\theta)/2$ has law $\text{Arcsine}_{0,1}$, since each $t \in (0, 1)$ has two preimages and $|\,\mathrm{d}t/\,\mathrm{d}\theta| = |\sin\theta|/2 = \sqrt{t(1-t)}$. Given a real number s, the roots of $w^2 - (4s - 2)w + 1 = 0$ have product 1, so one of them, w, satisfies $|w| \geq 1$ (when $0 \leq s \leq 1$, both roots have modulus 1). Expanding the product, we find that

$$s - \frac{1 + \cos\theta}{2} = \frac{(w - e^{\mathbf{i}\theta})(w - e^{-\mathbf{i}\theta})}{4w}\,.$$

We identify $\mathbb{R}^2$ with $\mathbb{C}$. Since $\mathcal{G}$ is a negative multiple of the logarithm in the plane and $|w| \geq 1$, by Newton's shell theorem (Lemma 2.33) for the unit circle we have $\frac{1}{2\pi}\int_0^{2\pi} \log|w - e^{\pm\mathbf{i}\theta}|\,\mathrm{d}\theta = \log|w|$. Hence

$$\int \log|s - t|\,\text{Arcsine}_{0,1}(\,\mathrm{d}t) = \log|w| - \log 4\,. \tag{2.18}$$

(a) Since $|w| \geq 1$, with equality when $0 \leq s \leq 1$, part (a) follows from (2.18).

(b) Take $s = -q$. Then the root of modulus at least one is $w = -(\sqrt{1+q} + \sqrt{q})^2$, whose reciprocal is $-(\sqrt{1+q} - \sqrt{q})^2$. Since $\log(1 + q/t) = \log|-q - t| - \log|t|$, subtracting (2.18) at $s = 0$ from (2.18) at $s = -q$ gives

$$\begin{aligned}\int \log\Bigl(1 + \frac{q}{t}\Bigr)\,\text{Arcsine}_{0,1}(\,\mathrm{d}t) &= 2\log(\sqrt{1+q} + \sqrt{q}) \\ &= 2\int_0^{\sqrt{q}} \frac{\mathrm{d}b}{\sqrt{1+b^2}} \leq 2\sqrt{q}\,.\end{aligned}$$

The second equality follows by differentiating $\log(b + \sqrt{1+b^2})$. □

In the last section of this chapter we use planar Brownian motion (which the rest of the book does not need) to prove the Beurling estimate. In Chapter 8 we prove the corresponding estimate for random walk on $\mathbb{Z}^2$ from Lemma 2.42, without Brownian motion.

2.6 HARMONIC MEASURE AND THE BEURLING ESTIMATE*

How likely is planar Brownian motion, started near the origin, to reach a distant circle without meeting a set that joins the origin to that circle? If the

set is a segment from the origin and the starting point lies on the opposite ray, the probability is of order the square root of the ratio of the two radii. The Beurling estimate says that, up to a constant factor, no connected set is easier to avoid. The square root comes from the arcsine law on an interval (Subsection 2.5.1), whose logarithmic potential is constant on the interval and whose density blows up like the inverse square root of the distance to an endpoint. The idea of the proof is to consider, for one point of the set at each radius, the logarithm of the distance to that point, to use the triangle inequality, and then to average over the radii with respect to the arcsine law. In Subsection 2.6.1 we prove the stronger Beurling projection theorem, which says that among all compact sets with the same distances from the origin, the easiest to avoid is the set of these distances on the ray opposite the starting point.

Throughout this section $Z = (Z_t)_{t\geq 0}$ is planar Brownian motion, and $\tau_F := \inf\{t \geq 0 : Z_t \in F\}$ is its hitting time of a Borel set $F \subset \mathbb{R}^2$. Write $\mathfrak{C}_r := \{z : |z| = r\}$ for the circle of radius r, and abbreviate $\tau_r := \tau_{\mathfrak{C}_r}$. Since planar Brownian motion is neighborhood-recurrent, it hits every circle almost surely, so the hitting times τ_r are almost surely finite. The **harmonic measure** of a set $A \subseteq \partial D$, seen from a point x in a domain D, is the probability $\mathbb{P}_x\{Z_{\tau_{\partial D}} \in A\}$ that Brownian motion started at x first leaves D through A (defined whenever ∂D is hit almost surely).

The next lemma says that if a bounded harmonic function on a domain tends to given values at the boundary, then its value at a point is the expected boundary value at the place where Brownian motion from that point first leaves the domain. The proof uses a sequence of circles, and only the strong Markov property and rotation invariance of Brownian motion.

Lemma 2.43 *Let $D \subset \mathbb{R}^2$ be open, and suppose that $\mathbb{P}_z\{\tau_{\partial D} < \infty\} = 1$ for every $z \in D$. Let $E \subset \partial D$ be finite, let $f : \partial D \to \mathbb{R}$ be bounded and Borel measurable, and let h be a bounded harmonic function on D such that $h(w) \to f(y)$ as $w \to y$ within D, for every $y \in \partial D \setminus E$. Then, for every $z \in D$,*

$$h(z) = \mathbb{E}_z\big[f(Z_{\tau_{\partial D}})\big]\,.$$

Proof. Fix $z \in D$, and write $\tau := \tau_{\partial D}$. Let $T_0 := 0$, and for $k \geq 0$ let T_{k+1} be the first time after T_k at which Z reaches the circle of radius $\frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$ about Z_{T_k} (see Figure 2.15). These circles bound closed disks contained in D, so $T_k < \tau$ for all k. By the strong Markov property at T_k and rotation invariance, $Z_{T_{k+1}}$ is uniformly distributed on that circle given the path up to time T_k, so by the mean value property of h we have $\mathbb{E}_z h(Z_{T_{k+1}}) = \mathbb{E}_z h(Z_{T_k})$. Hence $\mathbb{E}_z h(Z_{T_k}) = h(z)$ for all k.

The times T_k increase to a limit $T \leq \tau$, and $Z_{T_k} \to Z_T$ by continuity of the path. If Z_T were in D, then the steps $|Z_{T_{k+1}} - Z_{T_k}| = \frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$ would

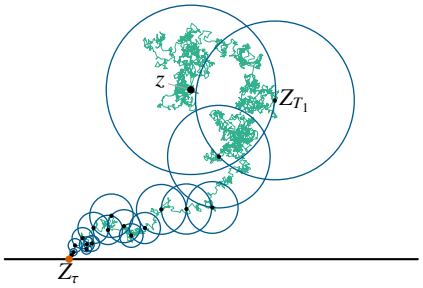


Figure 2.15: The first eighteen circles in the proof of Lemma 2.43, for D the upper half-plane and a simulated Brownian path from z: the circle about Z_{T_k} has radius $\frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$, and the path first reaches it at $Z_{T_{k+1}}$. The points Z_{T_k} converge to the exit point Z_τ.

tend to $\frac{1}{2}\mathrm{dist}(Z_T, \partial D) > 0$, which is impossible for a convergent sequence. So $Z_T \in \partial D$, hence $T = \tau$ and $Z_{T_k} \to Z_\tau$. Since planar Brownian motion almost surely never hits a given point, we have $Z_\tau \notin E$ almost surely, and then $h(Z_{T_k}) \to f(Z_\tau)$. Since h is bounded, by the bounded convergence theorem we conclude that $h(z) = \mathbb{E}_z f(Z_\tau)$. □

For the unit disk, Lemma 2.43 gives the harmonic measure explicitly (we use this in Exercises 2.50 and 2.51). Indeed, let f be continuous on $\mathfrak{C}_1$, and let h be its Poisson integral (2.7). By Theorem 2.18, h is harmonic in the unit disk and $h(z) \to f(y)$ as $z \to y \in \mathfrak{C}_1$. Since Brownian motion leaves the disk almost surely, by Lemma 2.43 we have $\mathbb{E}_z f(Z_{\tau_1}) = h(z)$ for $|z| < 1$. Since two finite measures on $\mathfrak{C}_1$ that give every continuous function the same integral are equal, it follows that for every Borel set $A \subseteq \mathfrak{C}_1$,

$$\mathbb{P}_z\{Z_{\tau_1} \in A\} = \frac{1}{2\pi} \int_0^{2\pi} \mathbf{1}_A(e^{\mathbf{i}\theta})\, P(z, e^{\mathbf{i}\theta})\, \mathrm{d}\theta\,, \tag{2.19}$$

where P is the Poisson kernel (2.6). By scaling, Brownian motion started at z with $|z| < s$ leaves the disk of radius s with density $(s^2 - |z|^2)/|y - z|^2$ at $y \in \mathfrak{C}_s$ with respect to normalized arc length. The law (2.19) concentrates near the point of the circle closest to z. In fact the chords through z from equally spaced points of the circle end at points that cut the circle into arcs of equal harmonic measure (see Figure 2.16). To see this, let a chord through z have ends at distances a and b from z. Then $ab = 1 - |z|^2$, and by symmetry about the perpendicular bisector of the chord, the chord meets the circle at the same angle at both ends, so when the chord turns about z, its end at distance b moves along the circle b/a times as fast as its end at distance a. Since the density of $\mathbb{P}_z\{Z_{\tau_1} \in \cdot\}$ with respect to normalized arc length at the end at distance b is $(1 - |z|^2)/b^2 = a/b$, the end at distance b sweeps out harmonic measure at the same rate as the end at distance a sweeps out normalized arc length.

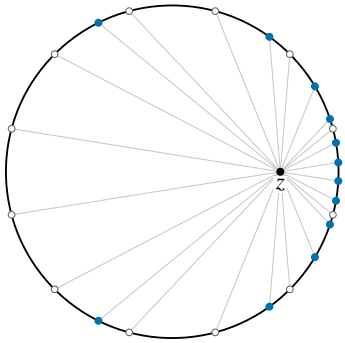


Figure 2.16: The chords through $z = 0.65$ from 12 equally spaced points of the unit circle (open dots) end at points (blue) that cut the circle into 12 arcs of equal harmonic measure seen from z. The arcs are shortest near the point of the circle closest to z, where Brownian motion from z is most likely to leave the disk.

Now we can prove the Beurling estimate, using the logarithmic potential of the arcsine law computed in Lemma 2.42. Recall that τ_F is the hitting time of a set F, and τ_R that of the circle $\mathfrak{C}_R$ of radius R.

Theorem 2.44 (Beurling Estimate) *Let $0 < r < R$, and let $K \subset \mathbb{R}^2$ be a compact set with $[r, R] \subseteq \{|y| : y \in K\}$. Then*

$$\mathbb{P}_z\{\tau_R < \tau_K\} \le 4\sqrt{\frac{r}{R}} \qquad \text{for every } z \text{ with } |z| \le r\,.$$

Note that the set K need not be connected, and it may wind around the origin many times; only its set of distances from the origin enters the hypothesis.

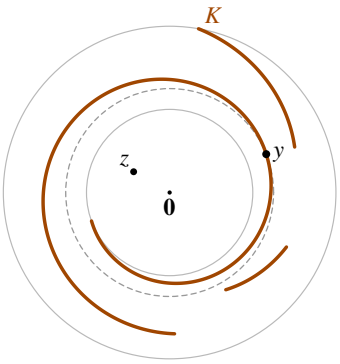


Figure 2.17: The part of a compact set K between the circles of radii r and $R/4$ (gray), for $R = 8r$: three disjoint pieces, one winding around the origin, whose distances from the origin cover $[r, R/4]$. For each t in $[r, R/4]$, Step 1 of the proof of Theorem 2.44 uses only one point y of K on the circle of radius t (dashed); the starting point z has $|z| \le r$.

Proof. If $r > R/8$, then $4\sqrt{r/R} > 1$ and there is nothing to prove, and if $z \in K$, then the probability in the theorem is zero. So we may assume that

$r \le R/8$ and $z \notin K$. The idea is to apply Lemma 2.43 to $\log|w-y|$ for one point $y \in K$ at each distance t from the origin, to eliminate y by the triangle inequality, and only then to average over t with respect to the arcsine law $\mathrm{Arcsine}_{r,R/4}$. Let

$$\Phi(s) := \int \log|s-t| \,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t) \qquad \text{for } s \in \mathbb{R}\,,$$

and let $c := \log\frac{R/4-r}{4}$, so that $\Phi \ge c$ on $\mathbb{R}$ by Lemma 2.42(a). For each integer $n > 1/\mathrm{dist}(z,K)$, let T_n be the exit time of Z from the open set $\{w : |w| < R,\ \mathrm{dist}(w,K) > 1/n\}$, which contains z; note that $T_n \le \tau_R$.

Step 1: We fix one distance $t \in [r, R/4]$ and a point $y \in K$ with $|y| = t$ (see Figure 2.17). Since the closure of the set defining T_n does not meet K, the function $w \mapsto \log|w-y|$ is bounded and harmonic there, so by Lemma 2.43 with $E = \emptyset$ we have $\mathbb{E}_z \log|Z_{T_n} - y| = \log|z-y|$. Since $\big||w| - t\big| \le |w-y|$ for every w and $|z-y| \le r+t$, it follows that

$$\mathbb{E}_z \log\big||Z_{T_n}| - t\big| \le \log(r+t)\,. \tag{2.20}$$

The point y no longer appears.

Step 2: Next we average over the distances. For $0 \le s \le R$ and $r \le t \le R/4$ the function $\log R - \log|s-t|$ is nonnegative, and $|Z_{T_n}| \le R$, so by Tonelli's theorem we may integrate (2.20) against the arcsine law $\mathrm{Arcsine}_{r,R/4}$ and exchange the expectation and the integral, which gives

$$\mathbb{E}_z \Phi(|Z_{T_n}|) \le \int \log(r+t)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t)\,.$$

By Lemma 2.42(a) at $s = r$ we have $c = \int \log(t-r)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t)$, so by Lemma 2.42(b) with $q = 2r$,

$$\mathbb{E}_z[\Phi(|Z_{T_n}|) - c] \le \int \log\Big(1+\frac{2r}{t-r}\Big)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t) \le 2\sqrt{\frac{2r}{R/4-r}} \le 8\sqrt{\frac{r}{R}}\,,$$

where in the last step we have used $R/4 - r \ge R/8$.

Step 3: Finally we let $n \to \infty$. The random variable $\Phi(|Z_{T_n}|) - c$ is nonnegative, and on the event $\{T_n = \tau_R\}$ we have $|Z_{T_n}| = R$ and $R - t \ge 3R/4$ for all $t \le R/4$, so

$$\Phi(|Z_{T_n}|) - c \ge \log\frac{3R}{4} - \log\frac{R}{16} = \log 12\,.$$

Hence $(\log 12)\,\mathbb{P}_z\{T_n = \tau_R\} \le 8\sqrt{r/R}$. On the event $\{\tau_R < \tau_K\}$, the compact path $\{Z_t : 0 \le t \le \tau_R\}$ has positive distance from K, so $T_n = \tau_R$ for all large

n. By Fatou's lemma it follows that

$$(\log 12)\,\mathbb{P}_z\{\tau_R < \tau_K\} \le \liminf_{n\to\infty} (\log 12)\,\mathbb{P}_z\{T_n = \tau_R\} \le 8\sqrt{\frac{r}{R}}\,.$$

Since $\log 12 > 2$, this proves the theorem. □

For a connected set the hypothesis of Theorem 2.44 holds automatically, since the distance from the origin is continuous, and the continuous image of a connected set is connected. This gives the Beurling estimate in its usual form.

Corollary 2.45 *Let $R > 0$, and let $K \subset \mathbb{R}^2$ be a compact connected set that contains the origin and meets the circle $\mathfrak{C}_R$. Then for every $z \in \mathbb{R}^2$,*

$$\mathbb{P}_z\{\tau_R < \tau_K\} \le 4\sqrt{\frac{|z|}{R}}\,.$$

Proof. If $z = \mathbf{0}$, then $z \in K$ and the probability is zero. If $|z| \ge R/16$, then the right side is at least 1. Otherwise $0 < |z| < R$, and the set $\{|y| : y \in K\}$ is a connected subset of the real line that contains 0 and R, so it contains the interval $[|z|, R]$, and the bound follows from Theorem 2.44 with $r = |z|$. □

The exponent $\frac{1}{2}$ cannot be improved. Let $0 < r < R$, let K be the segment $[0, R]$ on the positive real axis, and start the Brownian motion at $-r$. We write the points of $\mathrm{B}(\mathbf{0}, R) \setminus K$ as $w = \rho e^{\mathbf{i}\theta}$ with $0 < \rho < R$ and $0 < \theta < 2\pi$. The function

$$w \mapsto \sqrt{\frac{\rho}{R}}\,\sin\frac{\theta}{2}$$

is the imaginary part of the branch of $\sqrt{w/R}$ with argument $\theta/2$, so it is harmonic on $\mathrm{B}(\mathbf{0}, R) \setminus K$. It extends continuously by zero to K, it takes values between 0 and 1 on $\mathfrak{C}_R$, and it equals $\sqrt{r/R}$ at $-r$ (see Figure 2.18). By Lemma 2.43 it follows that $\mathbb{P}_{-r}\{\tau_R < \tau_K\} \ge \sqrt{r/R}$, so the segment K attains the bound of Theorem 2.44 up to the factor 4.

In Chapter 8 we prove the discrete Beurling estimate for random walk on $\mathbb{Z}^2$ by the same argument, with the potential kernel of the walk in place of the logarithm and the masses that the arcsine law gives to unit intervals in place of its density. There we must also show that this discretization changes the potential by at most the mass at an endpoint.

2.6.1 The Beurling Projection Theorem*

The proof of Theorem 2.44 uses only one point of K at each distance from the origin. The Beurling projection theorem says that, among the sets with the

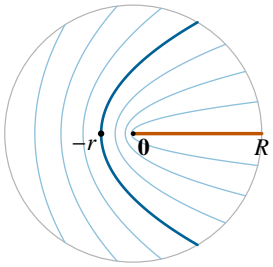


Figure 2.18: Level curves of $\sqrt{\rho/R}\,\sin(\theta/2) = \mathrm{Im}\,\sqrt{w/R}$ on the disk of radius R slit along $K = [0, R]$ (vermilion), at the values $1/8, \ldots, 7/8$: they are parabolas with focus $\mathbf{0}$. The level $\sqrt{r/R}$ (dark) passes through $-r$, drawn for $r = R/4$.

same distances from the origin, the easiest to avoid is the circular projection, which has all of them on a single ray. We prove it for planar Brownian motion started at a point a of the negative real axis, by showing that replacing a compact set by its circular projection on the positive real axis can only increase the probability of reaching a small circle about the origin before the set (see Figure 2.19).

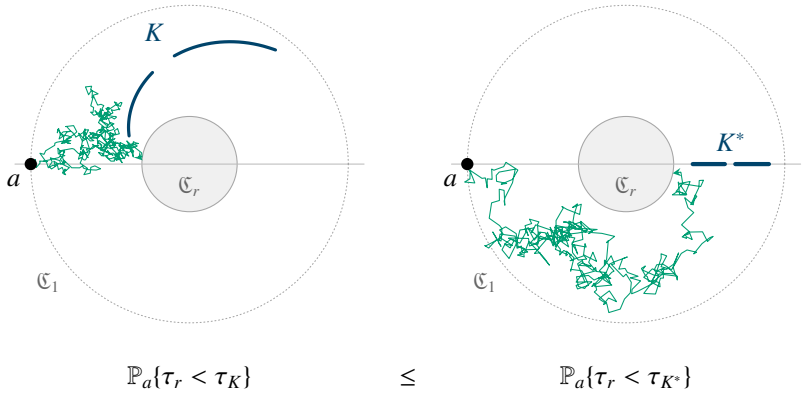


Figure 2.19: A Brownian path from a that reaches the circle $\mathfrak{C}_r$ of radius r, the boundary of the shaded disk, before the set K (left), and the circular projection $K^* = \{|z| : z \in K\}$ on the positive real axis (right). Projecting can only increase the chance of reaching $\mathfrak{C}_r$ first: $\mathbb{P}_a\{\tau_r < \tau_K\} \le \mathbb{P}_a\{\tau_r < \tau_{K^*}\}$, where τ_F denotes the hitting time of a set F and τ_r that of $\mathfrak{C}_r$.

The **circular projection** of a compact set $K \subset \mathbb{R}^2$ is the set $K^* := \{|z| : z \in K\} \subseteq [0, \infty)$ of distances from the origin to points of K. Recall that τ_F is the hitting time of a set F by the planar Brownian motion Z, and τ_r that of the circle $\mathfrak{C}_r$ of radius r.

Theorem 2.46 (Beurling Projection Theorem) *Let $r > 0$ and let $K \subset \{|z| \ge r\}$ be compact with circular projection K^*. Then for every real $a < -r$,*

$$\mathbb{P}_a\{\tau_r < \tau_K\} \le \mathbb{P}_a\{\tau_r < \tau_{K^*}\}\,.$$

Again the set K need not be connected. For a compact connected set that contains the origin and meets the unit circle, the theorem gives the following square-root estimate.

Corollary 2.47 *Let $E \subset \mathbb{R}^2$ be compact and connected, with $\mathbf{0} \in E$ and $E \cap \{|z| \geq 1\} \neq \emptyset$. Then for every $0 < r < 1$ and every starting point a with $|a| \geq 1$,*

$$\mathbb{P}_a\{\tau_r < \tau_E\} \leq \frac{4\sqrt{2}}{\pi}\, r^{1/2}\,.$$

This is the counterpart of Corollary 2.45 for a Brownian motion that starts outside the unit circle and must reach a small circle, now with the explicit constant $4\sqrt{2}/\pi \approx 1.80$, which cannot be improved. To deduce it, we stop the Brownian motion at its first visit to the unit circle, rotate that point to -1, and project the set E. Since the circular projection of E contains the segment joining $\mathfrak{C}_r$ to the unit circle, it suffices to bound the chance of reaching $\mathfrak{C}_r$ before that segment, which we compute exactly with two conformal maps.

We prove Theorem 2.46 by successively folding K across lines through the origin. Let $\mathcal{R}z := \bar{z}$ denote reflection across the real axis, acting on sets by $\mathcal{R}K := \{\mathcal{R}z : z \in K\}$. In the next lemma we reflect the part of K below the real axis into the upper half-plane (see Figure 2.20), and show that this can only increase the chance of reaching the inner circle before the set. Since every circle centered at the origin is symmetric across every line through the origin, the case of a general line follows by a rotation.

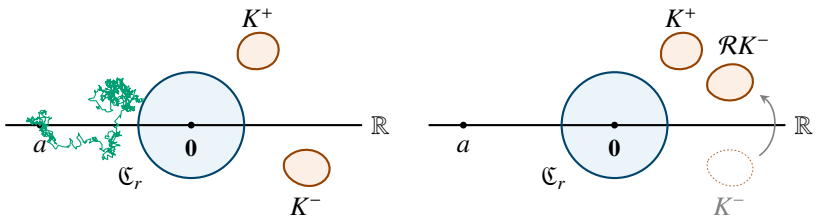


Figure 2.20: (a) The set $K = K^+ \cup K^-$ and a simulated Brownian path from a on the event $\{\tau_r < \tau_K\}$. (b) Reflecting K^-, dotted in its original position, gives $\widetilde{K} = K^+ \cup \mathcal{R}K^-$, which lies in the closed upper half-plane. Here $K^+ = K \cap \{\operatorname{Im} z > 0\}$, $K^- = K \cap \{\operatorname{Im} z \leq 0\}$, and $\mathcal{R}$ is reflection in the real axis. This reflection can only increase the chance of reaching $\mathfrak{C}_r$ before the set.

Lemma 2.48 *Let $r > 0$, let $K \subset \{|z| \geq r\}$ be compact, let $K^+ := K \cap \{\operatorname{Im} z > 0\}$ and $K^- := K \cap \{\operatorname{Im} z \leq 0\}$, and let $\widetilde{K} := K^+ \cup \{\bar{z} : z \in K^-\}$ be the set obtained by reflecting K^- across the real axis. Then for every real a with $|a| > r$,*

$$\mathbb{P}_a\{\tau_r < \tau_K\} \leq \mathbb{P}_a\{\tau_r < \tau_{\widetilde{K}}\}\,.$$

Proof. The idea is to compare the two hitting probabilities first at reflected pairs of points of $K \cup \mathcal{R}K$, and then to use reflection symmetry on the real axis to show that reflecting K^- cannot increase the probability of hitting the set before the circle. For $|z| \geq r$, let

$$h(z) := \mathbb{P}_z\{\tau_K \leq \tau_r\} \qquad \text{and} \qquad \widetilde{h}(z) := \mathbb{P}_z\{\tau_{\widetilde{K}} \leq \tau_r\}\,,$$

so that $h = \mathbf{1}_K$ and $\widetilde{h} = \mathbf{1}_{\widetilde{K}}$ on $\mathfrak{C}_r$. Since τ_r is finite almost surely, we must show that $\widetilde{h}(a) \leq h(a)$. Let $g := \widetilde{h} - h$, let $S := K \cup \mathcal{R}K$, which contains K and $\widetilde{K}$, and let

$$\delta := \max\Big\{0,\ \sup\{g(x) : x \in \mathbb{R},\ |x| \geq r\}\Big\} \leq 1\,.$$

Step 1: We show that $g(z) + g(\bar{z}) \leq \delta$ for every $z \in S$ with $\operatorname{Im} z > 0$ and $|z| > r$. From z or $\bar{z}$ the path stays outside the disk of radius r until time τ_r, so on the event $\{\tau_{\mathbb{R}} \leq \tau_r\}$, where $\tau_{\mathbb{R}}$ is the hitting time of the real axis, $Z_{\tau_{\mathbb{R}}}$ is a real point with $|Z_{\tau_{\mathbb{R}}}| \geq r$. Since reflection across the real axis maps Brownian motion from z to Brownian motion from $\bar{z}$ (see Figure 2.21) and fixes the real axis and $\mathfrak{C}_r$, the law of $Z_{\tau_{\mathbb{R}}}$ on this event is the same from z and from $\bar{z}$. The point z lies in $\widetilde{K}$, since either $z \in K$ or $\bar{z} \in K$, so $\widetilde{h}(z) = 1$. Since $\widetilde{K}$ lies in the closed upper half-plane, a path from $\bar{z}$ can reach it only at or after time $\tau_{\mathbb{R}}$, so by the strong Markov property

$$\widetilde{h}(z) + \widetilde{h}(\bar{z}) = 1 + \mathbb{E}_{\bar{z}}\big[\widetilde{h}(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \leq \tau_r\big]\,.$$

On the other hand, at least one of z and $\bar{z}$ lies in K, and h equals 1 there; from the other point, reaching K after time $\tau_{\mathbb{R}}$ but by time τ_r is one way to hit K by time τ_r, so by the strong Markov property and the symmetry of the law of $Z_{\tau_{\mathbb{R}}}$,

$$h(z) + h(\bar{z}) \geq 1 + \mathbb{E}_{\bar{z}}\big[h(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \leq \tau_r\big]\,.$$

(This inequality also holds when both points lie in K, since $h \leq 1$.) Subtracting the two displays, we obtain $g(z) + g(\bar{z}) \leq \mathbb{E}_{\bar{z}}[g(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \leq \tau_r] \leq \delta$.

Step 2: Next we show that $\delta = 0$. Let a be real with $|a| > r$, and let T be the first time at which the path from a meets $S \cup \mathfrak{C}_r$; then $T \leq \tau_r < \infty$. Since both K and $\widetilde{K}$ lie in S, by the strong Markov property at time T we have $g(a) = \mathbb{E}_a g(Z_T)$. We bound g at the possible values of Z_T. At a real point x of S, both $h(x)$ and $\widetilde{h}(x)$ equal 1, since the real points of S, of K and of $\widetilde{K}$

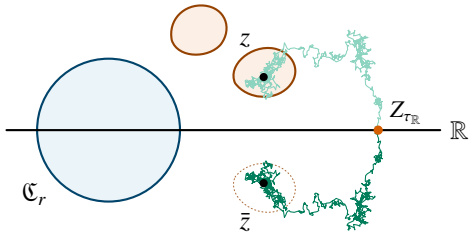


Figure 2.21: Step 1 of the proof of Lemma 2.48, for the set K of Figure 2.20 and a point $z \in \mathcal{R}K^-$, so that $\bar{z} \in K^-$ (dotted). Reflection in the real axis maps the Brownian path from $\bar{z}$ (dark), up to time $\tau_{\mathbb{R}}$, to a path from z (light) with the same endpoint $Z_{\tau_{\mathbb{R}}}$, and the path from $\bar{z}$ cannot meet $\overline{K}$ (shaded) before $\tau_{\mathbb{R}}$.

coincide; at a real point of $\mathfrak{C}_r$, both equal $\mathbf{1}_K(x)$. So $g = 0$ at the real values of Z_T. For a point w of $\mathfrak{C}_r$ with $\operatorname{Im} w > 0$, we have $\mathbf{1}_{\widetilde{K}}(w) = \max\{\mathbf{1}_K(w), \mathbf{1}_K(\bar{w})\}$ and $\mathbf{1}_{\widetilde{K}}(\bar{w}) = 0$, so $g(w) + g(\bar{w}) \le 0$. Every other value of Z_T lies in a pair as in Step 1. Since the law of Z_T is symmetric under reflection across the real axis, it gives probability at most $1/2$ to the upper half-plane, and pairing each point there with its reflection we obtain

$$g(a) = \mathbb{E}_a[g(Z_T) + g(\overline{Z_T});\ \operatorname{Im} Z_T > 0] \le \frac{\delta}{2}\,.$$

Since also $g = 0$ at the two real points of $\mathfrak{C}_r$, taking the supremum over real a with $|a| \ge r$ we obtain $\delta \le \delta/2$, so $\delta = 0$. Hence $\widetilde{h}(a) \le h(a)$ for every real a with $|a| > r$, which proves the lemma. □

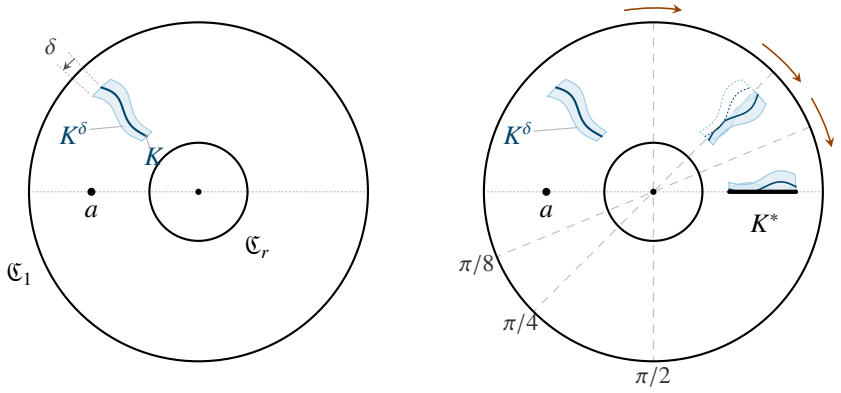
For a line ℓ through the origin other than the real axis, let $\mathbb{H}_\ell$ be the open half-plane bounded by ℓ that contains the negative real axis, and let $\mathcal{R}_\ell$ be the reflection in ℓ. The **fold** across ℓ maps a compact set E to the compact set

$$\mathcal{F}_\ell E := (E \setminus \mathbb{H}_\ell) \cup \mathcal{R}_\ell(E \cap \mathbb{H}_\ell)\,,$$

which lies in the closed half-plane opposite $\mathbb{H}_\ell$ and satisfies

$$\{|z| : z \in \mathcal{F}_\ell E\} = \{|z| : z \in E\}\,.$$

Next we show that finitely many folds map the set K^δ of rotations of points of K by angles at most δ onto a set containing K^* (see Figure 2.22).



(a) (b)

Figure 2.22: (a) A compact set K and the larger set K^δ, which contains the point at angle $3\pi/4$ at each distance in K^*. (b) The folds across the dashed lines at angles $\pi/2$, $\pi/4$ and $\pi/8$ send the ray at angle $3\pi/4$ onto the positive real axis, so the image of K^δ contains K^* (heavy segment); those for the angles $-\pi/4$ and $-\pi/8$ do not move the pictured sets.

Lemma 2.49 *Let K be a compact set in the plane, let $\delta > 0$, and let $K^\delta := \{e^{\mathbf{i}\varphi} z : z \in K,\ |\varphi| \le \delta\}$. Then there exist an integer $N \ge 0$ and lines $\ell_1, \dots, \ell_{2N+1}$ through the origin, none equal to the real axis, with*

$$K^* \subseteq \mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta .$$

Proof. Choose an integer $N \ge 0$ with $\pi/2^{N+1} < \delta$. Let ℓ_1 be the imaginary axis. For $1 \le j \le N$, let ℓ_{2j} and ℓ_{2j+1} be the lines at angles $\pi/2^{j+1}$ and $-\pi/2^{j+1}$, respectively.

We prove by induction on m that $\mathcal{F}_{\ell_{2m+1}} \circ \cdots \circ \mathcal{F}_{\ell_1}$ maps a ray of angle θ to a ray whose angle has absolute value

$$\operatorname{dist}\left(\theta, \frac{\pi}{2^m}\mathbb{Z}\right) .$$

The case $m = 0$ follows from the definition of $\mathcal{F}_{\ell_1}$. For the inductive step, by the induction hypothesis the image of the ray has angle in $[-\pi/2^{m+1}, \pi/2^{m+1}]$. The folds across the lines at angles $\pi/2^{m+2}$ and $-\pi/2^{m+2}$ replace the absolute value s of this angle by $\min\{s, \pi/2^{m+1} - s\}$, which equals the distance from θ to $(\pi/2^{m+1})\mathbb{Z}$. This proves the inductive step. In particular, the image of every ray lies in $|\arg z| \le \pi/2^{m+1}$, and every ray with angle $k\pi/2^m$ is mapped to the positive real axis.

The point $\mathbf{0}$, if it lies in K^*, lies in K^δ and is fixed by every fold. Now let $x \in K^*$ be positive and choose θ with $xe^{\mathbf{i}\theta} \in K$. There is an angle $\psi = k\pi/2^N$ whose circular distance from θ is at most $\pi/2^{N+1} < \delta$. Hence $xe^{\mathbf{i}\psi} \in K^\delta$, and

by the case $m = N$, $\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1}$ maps $xe^{\mathbf{i}\psi}$ to x. Therefore

$$K^* \subseteq (\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1})(K^\delta)\,.$$

□

Now we can prove the Beurling projection theorem.

Proof of Theorem 2.46. For a point x and a compact set $E \subset \{|z| \geq r\}$, let $h_x(E) := \mathbb{P}_x\{\tau_E \leq \tau_r\}$ be the probability that the Brownian motion from x meets E by the time it reaches $\mathfrak{C}_r$. Note that h_x is increasing in E, and since $\mathbb{P}_a\{\tau_r < \tau_E\} = 1 - h_a(E)$ we must show that $h_a(K) \geq h_a(K^*)$. If $a \in K$ then $\tau_K = 0$, so $h_a(K) = 1$ and the inequality is immediate; we therefore assume $a \notin K$.

We first show that, for every line ℓ through the origin other than the real axis,

$$h_a(\mathcal{F}_\ell E) \leq h_a(E)\,.$$

Since a lies on the negative real axis, $a \in \mathbb{H}_\ell$, while $\mathcal{F}_\ell E$ lies in the closed half-plane $\mathbb{C}\setminus\mathbb{H}_\ell$; hence the Brownian motion from a can meet $\mathcal{F}_\ell E$ only at or after time τ_ℓ. Let μ be the law of Z_{τ_ℓ} on the event $\{\tau_\ell \leq \tau_r\}$, a sub-probability measure on $\ell \cap \{|z| \geq r\}$. By the strong Markov property at τ_ℓ, we have

$$h_a(\mathcal{F}_\ell E) = \int_\ell h_y(\mathcal{F}_\ell E)\ \mathrm{d}\mu(y)\,.$$

To bound $h_a(E)$ from below, note that on $\{\tau_\ell \leq \tau_r\}$, by the strong Markov property at τ_ℓ,

$$\mathbb{P}_a(\tau_E \leq \tau_r \,\big|\, (Z_{t\wedge\tau_\ell})_{t\geq 0}) = \mathbf{1}\{\tau_E \leq \tau_\ell\} + \mathbf{1}\{\tau_E > \tau_\ell\}\, h_{Z_{\tau_\ell}}(E) \geq h_{Z_{\tau_\ell}}(E)\,.$$

Hence

$$h_a(E) \geq \int_\ell h_y(E)\ \mathrm{d}\mu(y)\,.$$

For each $y \in \ell$ with $|y| > r$, if we rotate coordinates so that $\ell = \mathbb{R}$ and $\mathbb{H}_\ell$ becomes the lower half-plane, then $\mathcal{F}_\ell E$ is the set $\widetilde{K}$ of Lemma 2.48 for $K = E$, so $h_y(E) \geq h_y(\mathcal{F}_\ell E)$. If $|y| = r$, then $h_y(E) = \mathbf{1}_E(y) = \mathbf{1}_{\mathcal{F}_\ell E}(y) = h_y(\mathcal{F}_\ell E)$, because ℓ meets neither $\mathbb{H}_\ell$ nor its reflection. Integrating against μ proves the inequality.

Now fix $\delta > 0$ small enough that $a \notin K^\delta$. The set K^δ is compact, satisfies $(K^\delta)^* = K^*$, and lies in $\{|z| \geq r\}$, as do its images under compositions of folds, since rotations and folds preserve the modulus. Let N and the lines

$\ell_1, \dots, \ell_{2N+1}$ be as in Lemma 2.49. Applying the inequality $h_a(\mathcal{F}_\ell E) \le h_a(E)$ successively, we obtain

$$h_a(K^\delta) \ge h_a(\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta)\,.$$

By the inclusion of Lemma 2.49 and monotonicity of h_a, we have

$$h_a(\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta) \ge h_a(K^*)\,.$$

It remains to let $\delta \downarrow 0$. The compact sets K^δ decrease, with $\bigcap_{\delta>0} K^\delta = K$. Almost surely the path $\{Z_t : 0 \le t \le \tau_r\}$ is compact. If it meets $K^{1/n}$ at a point z_n for every $n \ge 1$, then a limit point of (z_n) lies on the path and in every K^δ, hence in K. Thus the decreasing events $\{\tau_{K^\delta} \le \tau_r\}$ have intersection $\{\tau_K \le \tau_r\}$, and $h_a(K^\delta) \downarrow h_a(K)$ by continuity from above of $\mathbb{P}_a$. Letting $\delta \downarrow 0$ in $h_a(K^\delta) \ge h_a(K^*)$, we conclude that $h_a(K) \ge h_a(K^*)$. □

Next we compute the probability of reaching $\mathfrak{C}_r$ before the segment $[r, 1]$. A Möbius map sends the outside of $\mathfrak{C}_r$ into the unit disk and the segment onto a radius, and the square root maps the image into a half-disk (see Figure 2.23), where the probability is a sum of two arctangents.

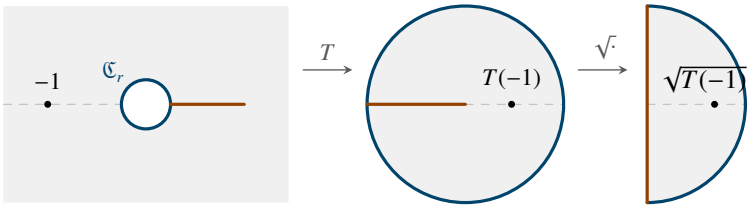


Figure 2.23: The map T in the proof of Lemma 2.50 sends the outside of $\mathfrak{C}_r$ slit along $[r, 1]$ (a) into the unit disk slit along $(-1, 0]$ (b), and the square root maps this onto the right half-disk (c); drawn for $r = 1/4$. The circle (blue) is mapped to the arc, both sides of the slit (vermilion) to the diameter, and the starting point -1 to $\sqrt{T(-1)} = \sqrt{2r/(1+r^2)}$.

Lemma 2.50 *For every* $0 < r < 1$,

$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} = \frac{4}{\pi} \arctan \sqrt{\frac{2r}{1+r^2}}\,.$$

Proof. Let $D := \{|z| > r\} \setminus [r, 1]$. Brownian motion started at -1 hits $\mathfrak{C}_r$ almost surely, so it leaves D almost surely, and the point r is hit with probability zero. We will construct a bounded harmonic function on D with boundary values 1 on $\mathfrak{C}_r \setminus \{r\}$ and 0 on $(r, 1]$.

We first map D into a half-disk. Let

$$T(z) := r\,\frac{z-1}{z-r^2} \qquad \text{for } z \in D\,.$$

The pole r^2 lies outside D, and

$$|z-r^2|^2 - r^2|z-1|^2 = (1-r^2)(|z|^2-r^2) \qquad \text{and} \qquad \operatorname{Im} T(z) = \frac{r(1-r^2)\operatorname{Im} z}{|z-r^2|^2}\,.$$

Hence $|T| < 1$ on D. The value $T(z)$ is real only for real z, and for real z it lies in $(-\infty, 0]$ only when $r^2 < z \le 1$, so T takes no value in $(-\infty, 0]$ on D. On $\mathbb{C} \setminus (-\infty, 0]$, let $\sqrt{w}$ be the square root with positive real part, $\sqrt{\rho e^{\mathbf{i}\theta}} = \sqrt{\rho}\, e^{\mathbf{i}\theta/2}$ for $\rho > 0$ and $-\pi < \theta < \pi$. It is continuous, and it is holomorphic, since $(\sqrt{w+\eta} - \sqrt{w})/\eta = 1/(\sqrt{w+\eta} + \sqrt{w}) \to 1/(2\sqrt{w})$ as $\eta \to 0$. Hence $\sqrt{T}$ is holomorphic on D, with $\operatorname{Re}\sqrt{T} > 0$ and $|\sqrt{T}| < 1$.

Next we define the harmonic function on the half-disk. For $x > 0$ and $x^2 + y^2 < 1$, let

$$v(x + \mathbf{i}y) := \frac{2}{\pi}\Big(\arctan\frac{x}{1-y} + \arctan\frac{x}{1+y}\Big)\,.$$

The function $(s,t) \mapsto \arctan(s/t)$ is harmonic for $t > 0$, since its second derivatives in s and in t are $\mp 2st/(s^2+t^2)^2$, so v is harmonic. The two arctangents are positive, and the product of their tangents is $x^2/(1-y^2) < 1$, so their sum lies in $(0, \pi/2)$ and $0 < v < 1$. By Lemma 2.16, $h := v \circ \sqrt{T}$ is a bounded harmonic function on D (see Figure 2.24 for the level curves of v and of h).

Now we check the boundary values. On $\mathfrak{C}_r$ we have $|T| = 1$, and $T = -1$ only at $z = r$. Hence $\sqrt{T(z)}$ tends to a point of the unit circle with positive real part as z tends to a point of $\mathfrak{C}_r \setminus \{r\}$. There the product of the two tangents equals 1, so the sum of the arctangents is $\pi/2$, and $h(z) \to 1$. On $(r, 1]$ the map T takes values in $(-1, 0]$, so as z tends to a point of $(r, 1]$, from either side of the segment, every limit point of $\sqrt{T(z)}$ lies on the imaginary axis strictly between $-\mathbf{i}$ and $\mathbf{i}$, where v tends to 0, and $h(z) \to 0$. By Lemma 2.43 with $E = \{r\}$ and f the indicator function of $\mathfrak{C}_r$, we conclude that

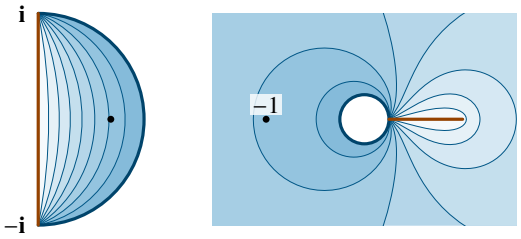
$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} = h(-1) = \frac{4}{\pi}\arctan\sqrt{\frac{2r}{1+r^2}}\,,$$

since $T(-1) = 2r/(1+r^2)$, so $\sqrt{T(-1)} = \sqrt{2r/(1+r^2)}$ is real and the two arctangents in v coincide there. □

Since $\arctan s \le s$ for $s \ge 0$ and $1 + r^2 \ge 1$, by Lemma 2.50 we have

$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} \le \frac{4\sqrt{2}}{\pi}\, r^{1/2} \qquad \text{for every } 0 < r < 1\,, \tag{2.21}$$

and since $\arctan s / s \to 1$ as $s \downarrow 0$, the ratio of the two sides of (2.21) tends to 1 as $r \downarrow 0$. Moreover, for the segment $E = [0, 1]$ and the starting point



(a) (b)

Figure 2.24: Level curves of v on the right half-disk (a) and of $h = v \circ \sqrt{T}$ outside $\mathfrak{C}_r$ slit along $[r, 1]$ (b) at the values $1/8, \ldots, 7/8$, for $r = 1/4$; $h(z)$ is the probability that Brownian motion from z reaches $\mathfrak{C}_r$ (blue) before $[r, 1]$ (vermilion). All level curves of v pass through $\pm\mathbf{i}$, and all those of h pass through r, where the boundary values jump.

$a = -1$, the probability in Corollary 2.47 equals that of Lemma 2.50, so the constant $4\sqrt{2}/\pi$ there cannot be replaced by a smaller number.

Proof of Corollary 2.47. If $a \in E$ the probability is zero, so we may assume that $a \notin E$. Since $|a| \geq 1 > r$, the path must cross $\mathfrak{C}_1$ before reaching $\mathfrak{C}_r$; on $\{\tau_r < \tau_E\}$ it reaches $\mathfrak{C}_1$, at time τ_1, before $\mathfrak{C}_r$ and before E, so by the strong Markov property at τ_1,

$$\mathbb{P}_a\{\tau_r < \tau_E\} \leq \sup_{y \in \mathfrak{C}_1} \mathbb{P}_y\{\tau_r < \tau_E\}\,.$$

For each $y \in \mathfrak{C}_1$, multiplication by $-\bar{y}$ is a rotation that maps y to -1, so by rotational invariance $\mathbb{P}_y\{\tau_r < \tau_E\} = \mathbb{P}_{-1}\{\tau_r < \tau_{-\bar{y}E}\}$, where $-\bar{y}E$ is compact and connected, contains $\mathbf{0}$, and meets $\{|z| \geq 1\}$. It therefore suffices to prove the estimate for $a = -1$. Let $K := E \cap \{|z| \geq r\}$. Since the part of E inside $\{|z| < r\}$ cannot be met before $\mathfrak{C}_r$, we have $\mathbb{P}_{-1}\{\tau_r < \tau_E\} = \mathbb{P}_{-1}\{\tau_r < \tau_K\}$. Since E is connected, contains $\mathbf{0}$, and meets $\{|z| \geq 1\}$, its circular projection contains $[0, 1]$, so K^* contains the segment $[r, 1]$. Hence by Theorem 2.46,

$$\mathbb{P}_{-1}\{\tau_r < \tau_K\} \leq \mathbb{P}_{-1}\{\tau_r < \tau_{K^*}\} \leq \mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\}\,,$$

and by (2.21) the last probability is at most $(4\sqrt{2}/\pi)\, r^{1/2}$. □

EXERCISES

Exercise 2.1. Show that if u is a function on an open set $\Omega \subset \mathbb{R}^d$ taking values in $\mathbb{R}$, then u is lower semicontinuous at $x \in \Omega$ if and only if for every $\epsilon > 0$ there is a neighborhood $U \subset \Omega$ of x such that

$$u(y) \geq u(x) - \epsilon \qquad \text{for every } y \in U\,.$$

Exercise 2.2. Prove Lemma 2.5.

Exercise 2.3. Show that if $u : K \to \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous (in the relative topology) on a nonempty compact set $K \subset \mathbb{R}^d$, then the infimum $\inf_{x\in K} u(x)$ is attained.

Exercise 2.4. Let $\mathcal{F}$ be a nonempty set of functions $\Omega \to \mathbb{R} \cup \{+\infty\}$ defined on an open set $\Omega \subset \mathbb{R}^d$, and fix a point $x_0 \in \Omega$. Show that if each $f \in \mathcal{F}$ is lower semicontinuous at x_0, then the function $s(x) := \sup_{f\in\mathcal{F}} f(x)$ is lower semicontinuous at x_0.

Exercise 2.5. (a) Show that if $u : K \to \mathbb{R}$ is lower semicontinuous (in the relative topology) and $K \subset \mathbb{R}^d$ is compact, then there is a family $\mathcal{F}$ of continuous functions $K \to \mathbb{R}$ such that for all $x \in K$

$$u(x) = \sup_{f\in\mathcal{F}} f(x)\,.$$

(b) Show that part (a) remains true if u takes values in $\mathbb{R} \cup \{+\infty\}$, and that the compactness assumption is not needed.

Exercise 2.6. Prove that if h is a harmonic function on an open set $\Omega \subset \mathbb{R}^d$, then $|h|$ and h^2 are subharmonic on Ω. More generally, prove that $f \circ h$ is subharmonic for every convex function $f : \mathbb{R} \to \mathbb{R}$.

Exercise 2.7. Prove that a superharmonic function on an open set $\Omega \subset \mathbb{R}^d$ (not necessarily lower semicontinuous) is bounded below on every compact subset $K \subset \Omega$. Show by example that it need not be bounded above on K, even if it is real-valued.

Exercise 2.8. Let $d \geq 2$. Show that each of the functions

$$x_1, \quad x_1x_2, \quad x_1^2 - x_2^2$$

is harmonic in $\mathbb{R}^d$. Use this to give a second proof of Lemma 2.13.

Exercise 2.9. Is the function

$$f(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0 \end{cases}$$

superharmonic on $\mathbb{R}$? Why or why not?

Exercise 2.10. Let μ be a Borel probability measure on $\partial\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. Define, for $r > 0$ and every Borel set A on the sphere,

$$\mu_r(A) := \int_A \frac{\mu(\mathrm{B}(x, r))}{S\,(\mathrm{B}(x, r) \cap \partial\mathrm{B}(\mathbf{0}, 1))}\,\mathrm{d}S\,(x)\,.$$

Equivalently,

$$\mu_r(A) = \int_A \frac{1}{S\,(\mathrm{B}(x, r) \cap \partial\mathrm{B}(\mathbf{0}, 1))} \int \mathbf{1}_{\{|x-y|<r\}}\,\mathrm{d}\mu(y)\,\mathrm{d}S\,(x)\,,$$

where S is the surface measure on the sphere (Section 2.2). Show that μ_r converges weakly to μ as $r \to 0$.

Exercise 2.11. Let $d \geq 2$ and let μ be a Borel probability measure on the sphere $\partial \mathrm{B}(\mathbf{0}, 1)$. Show that if μ is rotationally invariant, then μ is the normalized surface measure on the sphere.

Exercise 2.12. Let $\Omega \subset \mathbb{R}^d$ be open, and let $u : \Omega \to \mathbb{R}$ be locally integrable. Prove or disprove the following "local" version of Lemma 2.15: if u is twice differentiable at $x \in \Omega$, then u is superharmonic at x if and only if $\Delta u(x) \leq 0$.

Exercise 2.13. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfy $\Delta u \leq 0$. Show without using Lemmas 2.12 and 2.15 that $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$.

Hint: first reduce to the case where $\Delta u < 0$ by considering $u(x) - \epsilon|x|^2$.

Exercise 2.14. Consider the semi-infinite strip

$$\Omega := \{(x, y) : |x| < 1 \text{ and } y > 0\}.$$

Suppose that u is harmonic on Ω, bounded above, and extends continuously to the open boundary edges, with value 1 on the bottom edge and 0 on the two vertical edges. Assume also that the limit superior of u at each bottom corner is at most 1. Show that $u(x, y) \leq C \exp(-y)$ for some constant $C < \infty$ (use Corollary 2.11). Hint: compare u with a multiple of $\cos(x)e^{-y}$. The bounded-above hypothesis cannot be dropped.

Exercise 2.15. In two dimensions, we have the following somewhat stronger version of the maximum principle (Lemma 2.12 applied to $-h$). Let $\Omega \subset \mathbb{R}^2$ be a domain which is not necessarily bounded but not equal to all of $\mathbb{R}^2$. Suppose h is subharmonic on Ω, upper semicontinuous on $\overline{\Omega}$, and satisfies $h(x) = o(\log|x|)$ as $x \to \infty$. Show that $\sup_\Omega h \leq \sup_{\partial\Omega} h$.

Hint: fix $y \in \partial\Omega$ and compare with $\log(|x - y|/r)$ outside a small circle about y.

A natural question is what happens if we replace the ball average $\mathbf{A}_r u$ in Definition 2.6 by some other average. Let μ be a compactly supported probability measure on $\mathbb{R}^d$. For an open set $\Omega \subset \mathbb{R}^d$, a point $x \in \Omega$, and a radius $r > 0$ satisfying $x + r \operatorname{supp} \mu \subset \Omega$, write

$$\mathbf{A}_r^\mu u(x) := \int_{\mathbb{R}^d} u(x + ry)\, d\mu(y).$$

A lower semicontinuous function $u : \Omega \to \mathbb{R} \cup \{+\infty\}$ is **μ-superharmonic** if

$$\mathbf{A}_r^\mu u(x) \leq u(x) \qquad \text{whenever } x + r \operatorname{supp} \mu \subset \Omega. \tag{2.22}$$

A continuous function is **μ-harmonic** if equality holds for every such x and r. When μ is uniform on $\mathrm{B}(\mathbf{0}, 1)$, the operator $\mathbf{A}_r^\mu$ equals the ball average $\mathbf{A}_r$ of Definition 2.1.

Exercise 2.16. Suppose that $\operatorname{supp}\mu$ contains a neighborhood of $\mathbf{0}$. Let $\Omega \subset \mathbb{R}^d$ be a bounded open set, and let u be μ-superharmonic in Ω. Show that if u is lower semicontinuous on $\overline{\Omega}$, then it obeys the minimum principle $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$ of Lemma 2.12.

Exercise 2.17. Let $d \geq 2$. Show that if μ is symmetric under reflection across the hyperplanes $x_i = 0$ and $x_i = x_j$, then the functions x_i, and $x_i x_j$ and $x_i^2 - x_j^2$ for $i \neq j$, are μ-harmonic (compare Lemma 2.13).

Exercise 2.18. Let $\mathrm{d}\mu = 2^{-2}\mathbf{1}_{[-1,1]^2}\,\mathrm{d}x$. Show that all harmonic polynomials of degree at most 3 are μ-harmonic, and find a μ-harmonic polynomial of degree 4.

Exercise 2.19. Let $\mu * \mu$ be the law of a sum of two independent vectors with law μ. Show that if h is μ-harmonic on $\mathbb{R}^d$, then it is also $\mu * \mu$-harmonic on $\mathbb{R}^d$.

Exercise 2.20. Let $(X_n)_{n\geq 0}$ be a random walk in $\mathbb{R}^d$ with $X_0 = \mathbf{0}$ whose increments are independent with distribution μ. Let T be a bounded integer-valued stopping time, and let μ_T be the distribution of X_T. Show that every μ-harmonic function on $\mathbb{R}^d$ is μ_T-harmonic.

Exercise 2.21*. Suppose that $\int x_i\,\mathrm{d}\mu = 0$ and $\int x_i x_j\,\mathrm{d}\mu = c\delta_{i,j}$ for $1 \leq i, j \leq d$, where $c > 0$. Show that if u is μ-harmonic on an open set Ω, then

$$\mathbf{a}_r u(x) = u(x) \qquad \text{whenever } \overline{\mathrm{B}}(x, r) \subset \Omega\,,$$

where $\mathbf{a}_r$ denotes the ordinary spherical average. Deduce that all μ-harmonic functions are harmonic. Hint: differentiate $\mathbf{A}_r^{\mu} u = u$ twice in r in the distributional sense; use Theorem 2.29.

Exercise 2.22. Prove that 1 and the real and imaginary parts of $(x + \mathbf{i}y)^k$ for $k \geq 1$ form a basis for the space of harmonic polynomials in $\mathbb{R}^2$.

Exercise 2.23. Find a function u which is harmonic in the punctured disk $\Omega = \mathrm{B}(\mathbf{0}, 1) \setminus \{\mathbf{0}\}$ in $\mathbb{R}^2$ but is not the real part of a holomorphic function in Ω.

Exercise 2.24. Let $\Omega_1, \Omega_2 \subset \mathbb{C}$ be open. Show that if $\psi : \Omega_1 \to \Omega_2$ is holomorphic and $u \in C^2(\Omega_2)$, then

$$\Delta(u \circ \psi)(z) = |\psi'(z)|^2 (\Delta u)(\psi(z))\,.$$

Exercise 2.25. Let $\Omega_2 \subset \mathbb{C}$ be open and $\Omega_1 \subset \mathbb{C}$ a domain. Show that if u is superharmonic on Ω_2, and $\psi : \Omega_1 \to \Omega_2$ is holomorphic and nonconstant, then $u \circ \psi$ is superharmonic on Ω_1.

Exercise 2.26. Let $d \geq 2$. Show that if $p \in \mathbb{R}[x_1, \ldots, x_d]$ is a nonzero polynomial, then $p(x)|x|^2$ is not harmonic. Hint: otherwise, $p(x)|x|^2$ would

be a harmonic extension of p from the unit sphere, but we know another harmonic extension (Lemma 2.17).

Exercise 2.27. Let $d \geq 2$ and $m \geq 0$ be integers. Let $\mathcal{P}_m(d)$ denote the vector space of homogeneous polynomials of degree m in $\mathbb{R}^d$ and $\mathrm{Harm}_m(d)$ the subspace of $\mathcal{P}_m(d)$ which consists of harmonic polynomials. Use the convention $\mathcal{P}_m(d) = \{0\}$ when $m < 0$.

(a) Show that $\mathcal{P}_m(d) = \mathrm{Harm}_m(d) \oplus |x|^2 \mathcal{P}_{m-2}(d)$.

(b) Show that $\dim(\mathrm{Harm}_m(d)) = \dim(\mathcal{P}_m(d)) - \dim(\mathcal{P}_{m-2}(d))$.

(c) Show that $\dim(\mathcal{P}_m(d)) = \binom{d+m-1}{d-1}$.

(d) For $m \geq 1$, deduce that $\mathrm{Re}\,((x + \mathbf{i}y)^m)$ and $\mathrm{Im}\,((x + \mathbf{i}y)^m)$ form a basis for $\mathrm{Harm}_m(2)$.

Exercise 2.28. This exercise extends Lemma 2.17 from spheres to paraboloids. Let $d \geq 2$ and let $p(x) = x_d - \sum_{i=1}^{d-1} x_i^2$. Consider the superlevel set $W := \{x \in \mathbb{R}^d : p(x) > 0\}$. Let $F \in \mathbb{R}[x_1, \ldots, x_d]$ be a polynomial of total degree $m \geq 0$. Show that there is a polynomial h of total degree at most m such that $\Delta h = 0$ on $\mathbb{R}^d$ and $h|_{\partial W} = F|_{\partial W}$. Moreover, establish that $F = pg + h$ for a polynomial g of total degree at most $m - 2$, with $g = 0$ when $m < 2$.

(a) For $M > 0$, let $W_M := W \cap \{x_d < M\}$. Suppose that u is harmonic in W_M, continuous on $\overline{W_M}$, nonpositive on $\partial W \cap \{x_d \leq M\}$, and at most 1 on $W \cap \{x_d = M\}$. Show that

$$u(x) \leq \frac{1}{\cos 1} \exp\left(-\frac{M - x_d}{\sqrt{M}}\right) \qquad \text{for } x \in W_M\,,$$

by comparing u with a harmonic function depending only on x_1 and x_d.

(b) Let u be harmonic in W and continuous on $\overline{W}$. Suppose that $u = 0$ on ∂W and $|u(x)| \leq C(1 + |x|)^k$ for constants $C < \infty$ and $k \geq 0$. Use part (a) to prove that $u = 0$ in W.

(c) Conclude by a suitable modification of the proof of Lemma 2.17.

Exercise 2.29. This exercise guides the reader through an alternative proof of the converse implication in Lemma 2.15 which does not use the divergence theorem. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and let $u \in C^2(\Omega)$ satisfy $\Delta u \leq 0$. We will verify the mean value inequality $\mathbf{a}_r u(x) \leq u(x)$ for all balls with $\overline{\mathrm{B}}(x, r) \subset \Omega$. By translation and scaling, it suffices to consider the ball $B := \mathrm{B}(\mathbf{0}, 1)$.

(a) Define $w := \mathbf{P}_B u$ on B (Theorem 2.18) and $w := u$ on ∂B. Show that $w \in C(\overline{B})$ and $\Delta w = 0$ in B.

(b) Apply Exercise 2.13 to $u - w$ to deduce $\mathbf{a}_1 u(\mathbf{0}) = w(\mathbf{0}) \le u(\mathbf{0})$.

Hint for (a): to avoid Lemma 2.15, show that the Poisson kernel has mean one by applying Exercise 2.13 to the Poisson integral h_1 of the constant function 1 from the proof of Theorem 2.18.

Exercise 2.30. Use a compactness argument to prove Lemma 2.21.

Exercise 2.31. (a) Let $d \ge 1$ and $R > 0$. For $t > 0$ when $d \ne 2$, and for $t \in \mathbb{R}$ when $d = 2$, let

$$r(t) = \begin{cases} t, & d = 1, \\ e^t, & d = 2, \\ t^{-1/(d-2)}, & d \ge 3\,. \end{cases}$$

Prove that if u is subharmonic and upper semicontinuous in the ball $\mathrm{B}(\mathbf{0}, R)$, then the spherical average $\mathbf{a}_{r(t)} u(\mathbf{0})$ is a convex function of t on the interval where $0 < r(t) < R$.

(b) Prove the same statement for u subharmonic and upper semicontinuous in an annular region $\mathrm{B}(\mathbf{0}, R_1) \setminus \overline{\mathrm{B}}(\mathbf{0}, R_0)$ with $0 < R_0 < R_1$, on the interval where $R_0 < r(t) < R_1$.

Hint: $r(t)$ solves a differential equation of the form

$$\frac{\mathrm{d}r}{\mathrm{d}t} = c_d r^{d-1}\,,$$

where c_d is a positive constant if $d = 1, 2$ and a negative constant if $d \ge 3$. Assume first that u is smooth, express $\frac{\mathrm{d}}{\mathrm{d}r}\mathbf{a}_r u(\mathbf{0})$ as an integral involving ∇u and apply the divergence theorem.

Exercise 2.32. Let $d \ge 2$, let $\Omega \subset \mathbb{R}^d$ be a bounded open set, and let $F : \partial\Omega \to \mathbb{R}$ be a bounded function. Consider the family of continuous superharmonic functions u on Ω with $\liminf_{x \to z} u(x) \ge F(z)$ for every $z \in \partial\Omega$. If u_n is a sequence of functions belonging to this family, is the pointwise infimum $\inf u_n$ a member of the family? (Compare Lemma 2.24.)

Exercise 2.33. Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$.

(a) Prove or disprove a uniform version of Lemma 2.28: $\mathbf{A}_r u \to u$ uniformly on compact subsets of Ω.

(b) Suppose that u is also lower semicontinuous. Does $\mathbf{A}_r u \to u$ uniformly on compact subsets of Ω? Consider $d = 1$ and $d \ge 2$ separately.

Exercise 2.34. Let $\Omega \subset \mathbb{R}^d$ be open. Show that for every superharmonic function u in Ω there exist C^∞ superharmonic functions u_n in $\Omega_{1/n}$ such that $u_n \le u$ in $\Omega_{1/n}$ and $u_n \to u$ almost everywhere in Ω. Recall that Ω_ϵ consists

of the points of Ω at distance $> \epsilon$ from the boundary of Ω. Moreover, show that if u is continuous then the convergence is uniform on compact subsets of Ω.

Hint: use mollifiers as in Theorem 2.3.

Exercise 2.35. Let $d \geq 1$ and $0 \leq r < 1$. Show that the constants $(1-r)/(1+r)^{d-1}$ and $(1+r)/(1-r)^{d-1}$ in the bounds (2.11) from the first proof of Theorem 2.30 are optimal.

Exercise 2.36. Let $d \geq 2$, and let Ω_1 and Ω_2 be bounded domains in $\mathbb{R}^d$ where $\overline{\Omega}_1 \subset \Omega_2$. Show that there exists C depending only on Ω_1 and Ω_2 such that for every positive harmonic function $h : \Omega_2 \to \mathbb{R}$, we have $\sup_{\Omega_1} h \leq C \inf_{\Omega_1} h$ (compare Theorem 2.30).

Exercise 2.37. Deduce from Theorem 2.30 that every nonnegative harmonic function in $\mathbb{R}^d$ is constant.

Exercise 2.38. Let $d \geq 1$ and $k \geq 0$. Show that if h is harmonic in $\mathbb{R}^d$ and $h(x) = O(|x|^k)$ as $|x| \to \infty$, then h must be a polynomial function of $x_1, \ldots, x_d$, of degree at most $\lfloor k \rfloor$.

Hint: by Lemma 2.20, h equals its convolution with every mollifier of radius R; differentiate the mollifier.

Exercise 2.39. Let $d \geq 2$. True or false: if f is a nonnegative continuous function on $\mathrm{B}(\mathbf{0}, 2) \subset \mathbb{R}^d$ with $f(\mathbf{0}) = 0$, and $f(x) - |x|^2$ is superharmonic in $\mathrm{B}(\mathbf{0}, 2)$, then $f(x) \leq 100|x|^2$ in $\mathrm{B}(\mathbf{0}, 1)$. (Compare Proposition 2.32.)

Exercise 2.40. Let $d \geq 1$, let I be the $d \times d$ identity matrix, and write $D^2 f$ for the Hessian matrix of a function f. True or false: if $f \in C^2(\mathrm{B}(\mathbf{0}, 1))$ is nonnegative with $f(\mathbf{0}) = 0$, and $I - D^2 f$ is nonnegative definite in $\mathrm{B}(\mathbf{0}, 1)$, then $f(x) \leq \frac{1}{2}|x|^2$ for every $x \in \mathrm{B}(\mathbf{0}, 1)$.

Exercise 2.41. Let $d \geq 2$, and suppose $f : (0, \infty) \to \mathbb{R}$ is twice continuously differentiable. Show that $x \mapsto f(|x|^2)$ is harmonic in $\mathbb{R}^d \setminus \{\mathbf{0}\}$ if and only if

$$2sf''(s) + df'(s) = 0 \qquad \text{for all } s > 0\,.$$

Check that the solution to this ODE is $f'(s) = cs^{-d/2}$ and deduce that

$$f(s) = \begin{cases} c_1 \log s + c_2, & d = 2, \\ \frac{c_3}{s^{(d-2)/2}} + c_4, & d \geq 3\,. \end{cases}$$

Exercise 2.42. Prove Earnshaw's theorem: given $n \geq 2$ static point charges $q_1, \ldots, q_n$ at distinct locations $y_1, \ldots, y_n$ in $\mathbb{R}^d$, where $d \geq 2$, no charge can be in stable equilibrium. Here a charge $q_i \neq 0$ at y_i is in stable equilibrium if $q_i \sum_{j \neq i} q_j \mathcal{G}(x - y_j)$ has a local minimum at $x = y_i$, and we assume that $q_j \neq 0$ for at least one $j \neq i$. You may use Lemma 2.8 and the identity theorem: a real analytic function that is constant on a nonempty open subset of a connected open set is constant.

Exercise 2.43. Let n be a positive integer. (a) Prove that for every set of points $z_1, \dots, z_n$ on the unit circle in $\mathbb{R}^2$, there exists a point z on the unit circle such that $\prod_{k=1}^n |z - z_k| \geq 1$.

(b) Show that for every n, the lower bound can be improved to 2.

(c) Show that the lower bound of 2 is sharp.

Hint for (a): by Lemma 2.33 with $d = 2$, the average of $\log|z - z_k|$ over z on the unit circle is 0.

Exercise 2.44. Prove the case $d \geq 3$ of Lemma 2.37.

Exercise 2.45. Prove the case $d = 2$ of Lemma 2.37.

Exercise 2.46. Verify that the discontinuous function $u(x)$ of Example 2.41 is superharmonic in $\mathbb{R}^d$ for $d \geq 2$.

Exercise 2.47. Prove that every lower semicontinuous superharmonic function on $\mathbb{R}$ is continuous, and that the set of points where it is not differentiable is at most countable. Show by example that such a function need not be differentiable.

Exercise 2.48. Let $d \geq 2$ and let $B := \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. For a fixed $y \in \partial\mathrm{B}(\mathbf{0}, 1)$ let $p(x) := P(x, y)$ where P is the Poisson kernel for the unit ball. Show, without computing a second derivative, that $\Delta p = 0$ in $\mathrm{B}(\mathbf{0}, 1)$.

Hint: for $0 < r < 1$, the function $x \mapsto \mathcal{G}(x - ry) - \mathcal{G}(rx - y)$ is harmonic on $B \setminus \{ry\}$, and it vanishes identically when $r = 1$; show that its derivative in r at $r = 1$ is $-p/(d\omega_d)$.

Exercise 2.49. Let $D \subset \mathbb{R}^d$ be a bounded domain, and let $Z = (Z_t)_{t \geq 0}$ be Brownian motion started at a point $x \in D$. Let $\tau_{\partial D}$ be its exit time. Suppose that f is continuous and superharmonic in a neighborhood of $\overline{D}$. Show that $f(Z_{t \wedge \tau_{\partial D}})$ is a supermartingale, and deduce Lemma 2.35 from neighborhood recurrence of planar Brownian motion. Hint: mollify f, and differentiate the Gaussian density to show that $g(Z_t) - \frac{1}{2}\int_0^t \Delta g(Z_s)\,\mathrm{d}s$ is a martingale for every $g \in C_c^2(\mathbb{R}^d)$; stop it at $t \wedge \tau_{\partial D}$.

Exercise 2.50*. Let $0 < x < s$. Show that planar Brownian motion started at x hits the circle of radius s centered at the origin before hitting the imaginary axis with probability

$$\frac{4}{\pi} \arctan \frac{x}{s},$$

and deduce that this probability is at most $4x/(\pi s)$. Hint: reflecting the path across the imaginary axis after its first visit to the imaginary axis shows that this probability equals the difference between the probabilities of exiting $\mathrm{B}(\mathbf{0}, s)$ through the right and the left half-circles; compute both with the harmonic measure (2.19) of the disk.

Exercise 2.51*. Let $Z = (Z_t)_{t\ge 0}$ be planar Brownian motion, with hitting times τ_F as in Section 2.6. Let $K \subset \mathbb{R}^2$ be compact, and suppose that $\mathbb{P}_z\{\tau_K < \infty\} = 1$ for every $z \in \mathbb{R}^2 \setminus K$. Show that, for every continuous function f on K, the following limit exists:

$$\lim_{|z|\to\infty} \mathbb{E}_z[f(Z_{\tau_K})]\,.$$

The limit, as a function of f, is given by integration against a probability measure on K, called **harmonic measure from infinity** on K. Hint: first prove convergence of the harmonic measure of a disk from z, by inversion and (2.19). Then stop Brownian motion on entering a disk that contains K.

Exercise 2.52*. Show that harmonic measure from infinity (Exercise 2.51) on $[-1, 1]$ has density

$$\frac{1}{\pi\sqrt{1-x^2}} \qquad \text{for } -1 < x < 1\,.$$

Hint: use $\varphi(w) = \frac{1}{2}(w + w^{-1})$.

Exercise 2.53*. In the notation of Section 2.6, let $0 < r < R$, and let $K \subset \mathbb{R}^2$ be a compact set with $[r, R] \subseteq \{|y| : y \in K\}$. Show that for every z with $|z| \le r$,

$$\mathbb{P}_z\{\tau_R < \tau_K\} \le \frac{4}{\pi} \arctan \sqrt{\frac{2rR}{r^2+R^2}} \le \frac{4\sqrt{2}}{\pi}\sqrt{\frac{r}{R}}\,,$$

with equality in the first inequality when $K = [r, R]$ and $z = -r$. Deduce that Theorem 2.44 holds with $4\sqrt{2}/\pi$ in place of 4, and that the constant $4\sqrt{2}/\pi$ cannot be replaced by a smaller number. Hint: use the inversion $w \mapsto r/w$ and approximate compact sets from outside by finite unions of disks.

NOTES

Potential theory grew out of Newton's theory of gravitation and Coulomb's electrostatics. Two good expositions on harmonic and superharmonic functions can be found in Axler, Bourdon, and Ramey (2001) and (Lieb and Loss, 2001, §9). The treatment in the first is the more elementary, although only continuous superharmonic functions are considered there. For the origin of the term "harmonic," see (Axler et al., 2001, p. 25). The Poisson kernel of the disk first appears in §4 of Poisson (1820), pp. 422–423: Poisson sums the series $\frac{1}{2} + \sum_{i\ge 1} r^i \cos i\theta$ in closed form, obtaining one half of the kernel (2.6) for $d = 2$, and shows that integrating it against a function f recovers f as $r \uparrow 1$. Harnack (1887) proved the bounds (2.11) for $d = 2$

in §19, pp. 61–62, by the argument of our first proof of Theorem 2.30, and deduced the comparison of Exercise 2.36 for planar domains by a chain of disks. In §20, pp. 67–68, he showed that if a series of harmonic functions of one sign converges at one point of a domain, then it converges at every point of the domain to a harmonic function; this monotone form is usually called Harnack's principle.

The most exhaustive general reference on potential theory is Doob (1984). Other classical texts are Helms (2009) and Wermer (1974).

For more on mollifiers, see (Evans, 1998, Appendix C.4).

Some authors include lower semicontinuity in the definition of superharmonic; we have followed Lieb and Loss (2001) in not doing so. By Theorem 2.26(b) every superharmonic function has a lower semicontinuous representative. Helms (2009) discusses several variations on the definition, among them one that asks for the mean value inequality only for small balls centered at each point, together with local integrability and lower semicontinuity; see (Helms, 2009, §2.3).

The Green function (2.14) is the fundamental solution of the negative Laplacian, normalized so that $-\Delta_x \mathcal{G}(x-y) = \delta_y$.

Exercises 2.16–2.21 explore what happens if the ball average in Definition 2.6 is replaced by a general average $\mathbf{A}_r^\mu$.

For another proof of Theorem 2.26, see (Lieb and Loss, 2001, Theorem 9.3).

The sphere averages $\mathbf{a}_r u$ of a continuous superharmonic function are decreasing in the radius r by Lemma 2.23. For $d \geq 2$ they are also concave after a suitable dimension-dependent change of variables, namely as functions of $\log r$ when $d = 2$ and of r^{2-d} when $d \geq 3$; see (Doob, 1984, §I.II.10).

Exercise 2.28 gives a paraboloid version of Lemma 2.17. Flatto, Newman, and Shapiro (1966) study level curves of harmonic functions in two dimensions.

Exercise 2.51 introduces harmonic measure from infinity as the limiting Brownian hitting distribution seen from points tending to infinity; see also (Mörters and Peres, 2010, Theorem 3.46). Exercise 2.52 computes this distribution on a line segment, giving the arcsine density; this is the interval example in (Mörters and Peres, 2010, Example 7.24), after an affine change of variables.

The Beurling estimate and the Beurling projection theorem, Theorems 2.44 and 2.46, are due to Beurling. Our proof of the projection theorem follows the proofs of Lemma 1 and Theorem 1 of Øksendal (1983). Werner (1996) gave another proof, based on estimates for the times at which one-dimensional Brownian motion hits moving boundaries, for compact sets that are countable unions of closed path-connected sets (Section 4 of his paper). The proof of Theorem 2.44 uses the arcsine law, which is the equilibrium measure of

a segment for the logarithmic kernel: its logarithmic potential is constant on the segment, as in Lemma 2.42, and it is the harmonic measure from infinity of the segment, as in Exercise 2.52. In Chapter 8 we apply the same argument to random walk on $\mathbb{Z}^2$.

The proof of Theorem 2.44 uses neighborhood recurrence (Theorem 3.20 of Mörters and Peres (2010)), the strong Markov property and rotation invariance, which give Lemma 2.43 when $E = \emptyset$.

The argument of Subsection 2.6.1 uses in addition the invariance of planar Brownian motion under reflection across a line, which maps Brownian motion from a point to Brownian motion from the reflected point, and the fact that planar Brownian motion almost surely never hits a given point. In the computation of Lemma 2.50, Lemmas 2.16 and 2.43 take the place of the conformal invariance of Brownian motion and harmonic measure, Theorems 7.20 and 7.23 of Mörters and Peres (2010). The solution of Exercise 2.53, in the separate document of solutions, uses the same two lemmas with an inversion, after approximating the image of the compact set under the inversion from outside by finite unions of disks.

Chapter Three

Potential Theory in $\mathbb{Z}^d$

Having developed potential theory in $\mathbb{R}^d$, we now turn to the lattice $\mathbb{Z}^d$. A function is discrete harmonic if its value at each site is the average of its values at the $2d$ neighbors, and such a function evaluated along simple random walk is a martingale, so identities for random walk take the place of the integral formulas of $\mathbb{R}^d$.

We first prove the discrete minimum principle, Pólya's theorem that simple random walk is recurrent in dimensions one and two and transient in higher dimensions, and the existence of the Green function (the fundamental solution of the negative discrete Laplacian). In the starred Section 3.6 we compare the discrete and continuous Green functions to find the precise asymptotics, which are logarithmic in dimension two and of order $|x|^{2-d}$ in dimensions $d \geq 3$. Together with the theory in $\mathbb{R}^d$, these asymptotics also give the discrete Harnack inequality (Lemma 3.16), which bounds the variation of a nonnegative discrete harmonic function on a lattice ball over the concentric ball of half the radius.

The other main result of the chapter, in Section 3.5, is that every discrete harmonic function of polynomial growth is a polynomial. The proof couples two random walks started at neighboring sites. We also compute the dimension of the space of discrete harmonic polynomials of degree at most k.

We finish in the plane, where simple random walk is recurrent, so a walk started far away still hits every nonempty finite set K. In Section 3.7 we prove that the hitting distribution on K of a walk started at y converges as $|y| \to \infty$ to a probability measure on K, the harmonic measure of K from infinity, and that this is the unique probability measure on K whose potential (formed with the potential kernel) is constant on K. It gives the attachment probabilities of external DLA in Chapter 8. In the next chapter we begin the study of lattice growth models with the divisible sandpile, and use the asymptotics of the Green function to show that it fills a ball.

3.1 DISCRETE HARMONIC FUNCTIONS

Throughout this chapter, $d \geq 1$ is an integer, and two sites $x, y \in \mathbb{Z}^d$ are neighbors, written $x \sim y$, if they differ by one in exactly one coordinate. For

$A \subset \mathbb{Z}^d$, the **outer boundary** ∂A is defined as the set of $x \in \mathbb{Z}^d \setminus A$ such that $y \in A$ for some neighbor $y \sim x$. The **closure** $\bar{A}$ is given by $A \cup \partial A$. For $x \in \mathbb{R}^d$ and real r, the **discrete ball** is $\mathrm{B}^{::}(x, r) := \{y \in \mathbb{Z}^d : |y - x| < r\}$, and we abbreviate $\mathrm{B}^{::}_r := \mathrm{B}^{::}(\mathbf{0}, r)$ (see Figure 3.1).

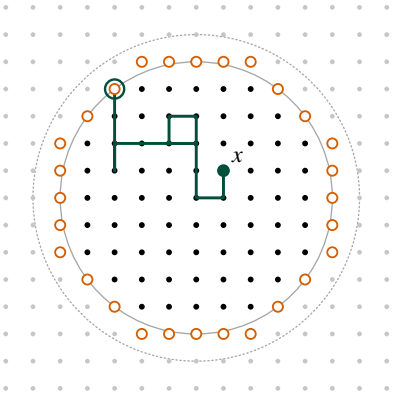


Figure 3.1: The discrete ball $\mathrm{B}^{::}_5$ in $\mathbb{Z}^2$ (black dots) and its outer boundary (open dots), which lies in the annulus $5 \le |x| < 6$ between the two circles. A simple random walk from x is stopped at its first visit to the outer boundary, as in Lemmas 3.4 and 3.5.

The **discrete Laplacian** of a function $f : \mathbb{Z}^d \to \mathbb{R}$ is the function $\Delta^{::} f : \mathbb{Z}^d \to \mathbb{R}$ given by

$$\Delta^{::} f(x) := \sum_{y \sim x} (f(y) - f(x)) . \tag{3.1}$$

If $\Delta^{::} f(x) = 0$, then we say that f is **discrete harmonic** at x. If $\Delta^{::} f(x) \le 0$ (resp. ≥ 0), then we say that f is **discrete superharmonic** (resp. **discrete subharmonic**) at x.

On a locally finite undirected simple graph $G = (V, E)$, we use the same normalization, so the Laplacian of $f : V \to \mathbb{R}$ is

$$\Delta f(v) := \sum_{w \sim v} (f(w) - f(v)) , \tag{3.2}$$

where $w \sim v$ means $\{v, w\} \in E$ (on $\mathbb{Z}^d$ this is $\Delta^{::}$), and f is harmonic (resp. superharmonic, subharmonic) at v if $\Delta f(v) = 0$ (resp. ≤ 0, ≥ 0). We will also need the maximum principle on finite subsets of a graph.

Lemma 3.1 *Let $G = (V, E)$ be a locally finite connected graph, let A be a nonempty finite proper subset of V, and let $h : V \to \mathbb{R}$ satisfy $\Delta h \ge 0$ on A. Then*

$$\max_{x \in A} h(x) \le \max_{\substack{x \notin A \\ x \sim y \text{ for at least one } y \in A}} h(x) .$$

Proof. Suppose that the maximum of h on A is attained at $x \in A$ and is larger than the value of h at every vertex outside A adjacent to A. Since $\Delta h(x) \geq 0$, the maximum is then also attained at every neighbor of x. Applying this along a path from x to $V \setminus A$ gives a contradiction. □

Applying Lemma 3.1 to the negative of a superharmonic function gives the following.

Lemma 3.2 (Discrete Minimum Principle) *Let $A \subset \mathbb{Z}^d$ be a nonempty finite set and f a function defined on $\bar{A}$. If f is discrete superharmonic on A, then*

$$\min_{\bar{A}} f = \min_{\partial A} f .$$

Proof. If $\min_{\bar{A}} f$ is attained at $x_0 \in A$, then it is also attained at every neighbor of x_0 in $\bar{A}$. Applying this along a nearest-neighbor path $x_0, x_1, \ldots, x_m$ from x_0 to the boundary (see Figure 3.2), we find that it is attained at $x_m \in \partial A$. □

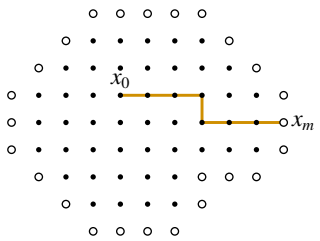


Figure 3.2: Proof of Lemma 3.2. Along a nearest-neighbor path from a minimum point $x_0 \in A$ (dots) to its first vertex x_m in ∂A (open dots), f is equal to $f(x_0)$, since f is superharmonic.

The minimum principle implies that for all functions $F : A \to \mathbb{R}$ and $b : \partial A \to \mathbb{R}$, the problem

$$\begin{aligned} \Delta^{::} u &= F && \text{on } A, \\ u &= b && \text{on } \partial A \end{aligned} \tag{3.3}$$

has a unique solution. (Indeed, by Lemma 3.2 the linear map $u \mapsto \Delta^{::} u|_A$ on functions on $\bar{A}$ vanishing on ∂A is one-to-one, hence onto, since both spaces have dimension $|A|$.) In the next section, we will construct the solution explicitly using random walks when $F = 0$; Exercise 3.20 treats the general case, and Exercise 3.5 shows that the solution with $F = 0$ has the least energy among functions with its boundary values.

3.2 RANDOM WALK AND MARTINGALES

A sequence of random variables $(X_n)_{n \geq 0}$ is called a **random walk** if the **increments** (also known as steps) $X_{n+1} - X_n$ are independent and identically

distributed (**i.i.d.**). A **simple random walk** in $\mathbb{Z}^d$ is the case where each step is uniformly distributed on the set $\{\pm\mathbf{e}_1, \ldots, \pm\mathbf{e}_d\}$ of standard basis vectors and their negatives. For $x \in \mathbb{Z}^d$, we write $\mathbb{P}_x$ for the law of simple random walk started at $X_0 = x$, and $\mathbb{E}_x$ for the corresponding expectation.

Martingales are a generalization of random walks with mean zero steps, dropping the requirement that the steps be independent, but requiring each step to have mean zero when conditioned on the entire past. A **filtration** $(\mathcal{F}_n)_{n\geq 0}$ is an increasing sequence of σ-algebras. A sequence of random variables $(M_n)_{n\geq 0}$ is called an $(\mathcal{F}_n)$-**martingale** if for all $n \geq 0$, M_n is measurable with respect to $\mathcal{F}_n$, $\mathbb{E}|M_n| < \infty$, and $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ almost surely. If we have $\leq$ (resp. $\geq$) in the last condition, then (M_n) is called an $(\mathcal{F}_n)$-**supermartingale** (resp. $(\mathcal{F}_n)$-**submartingale**). Note that $\mathbb{E}M_{n+1} \leq \mathbb{E}M_n$ for a supermartingale, and $\mathbb{E}M_{n+1} = \mathbb{E}M_n$ for a martingale. When no filtration is mentioned we use the **natural filtration** given by $\mathcal{F}_n = \sigma(M_0, \ldots, M_n)$; see Exercise 3.7.

Loosely speaking, a stopping time is a random time T such that knowledge of the past and present suffices to determine whether T has happened already. Formally, a random variable T with values in $\{0, 1, 2, \ldots\} \cup \{\infty\}$ is called an $(\mathcal{F}_n)$-**stopping time** if $\{T \leq n\} \in \mathcal{F}_n$ for all $n \geq 0$. If $(M_n)_{n\geq 0}$ is a supermartingale and T is a stopping time, then $(M_{T\wedge n})_{n\geq 0}$ is also a supermartingale, because $M_{T\wedge(n+1)} - M_{T\wedge n} = (M_{n+1} - M_n)\mathbf{1}\{T > n\}$ and the event $\{T > n\} = \{T \leq n\}^c$ is $\mathcal{F}_n$-measurable. If (M_n) is a martingale, then both (M_n) and $(-M_n)$ are supermartingales, and hence

$$\mathbb{E}[M_{T\wedge n}] = \mathbb{E}[M_0] \qquad \text{for every } n \geq 0\,.$$

An important example of a (super)martingale is $M_n = h(X_n)$, where X_n is simple random walk in $\mathbb{Z}^d$ and $h : \mathbb{Z}^d \to \mathbb{R}$ is a discrete (super)harmonic function. A stopping time that we will use often is the **hitting time** of a set $B \subset \mathbb{Z}^d$ by a random walk $(X_t)_{t\geq 0}$, defined as

$$\tau_B := \inf\{t \geq 0 : X_t \in B\}\,. \tag{3.4}$$

Doob's optional stopping theorem gives sufficient criteria under which we have $\mathbb{E}[M_T] = \mathbb{E}[M_0]$.

Theorem 3.3 (Doob's Optional Stopping Theorem) *Let $(M_n)_{n\geq 0}$ be a martingale with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$, and let T be a stopping time that is finite almost surely. Suppose that at least one of the following conditions holds:*

(a) $|M_{n\wedge T}| \leq Z$ almost surely for every $n \geq 0$, where Z is an integrable random variable;

(b) $\mathbb{E}[T] < \infty$ *and there is* $K > 0$ *such that* $\sup_{n\geq 1} |M_n - M_{n-1}| \leq K$ *almost surely.*

Then M_T *is integrable and*

$$\mathbb{E}[M_T] = \mathbb{E}[M_0]\,.$$

Proof. Recall that $\mathbb{E}[M_{T\wedge n} - M_0] = 0$. If condition (a) holds, then we may take $n \to \infty$, using dominated convergence to interchange limits. In case (b), observe that

$$|M_{T\wedge n} - M_0| \leq \sum_{k=1}^{T\wedge n} |M_k - M_{k-1}| \leq KT\,.$$

By assumption KT is integrable, and dominated convergence can be used again. □

See Exercise 3.8 for a simple application of Theorem 3.3 to random walk on $\mathbb{Z}$.

To see why some integrability condition is needed in Theorem 3.3, let (X_n) be a simple random walk on $\mathbb{Z}$ started at the origin, and recall that $\tau_{\{1\}}$ denotes the hitting time of 1 (which is finite almost surely, by letting $\ell \to \infty$ in Exercise 3.8(a)). Then $\mathbb{E}[X_n] = 0$ for all n, but $\mathbb{E}[X_{\tau_{\{1\}}}] = 1$. See Exercise 3.10 for an example of a stopping time T such that X_T is not integrable.

It is sometimes hard to verify the sufficient conditions of the optional stopping theorem, but the proof method can be used. We illustrate this to compute the expected time for simple random walk to exit a ball.

Lemma 3.4 *Fix* $r > 0$ *and let* $B = \mathrm{B}^{\cdot\cdot}(\mathbf{0}, r) \subset \mathbb{Z}^d$. *Then for* $x \in B$,

$$r^2 - |x|^2 \leq \mathbb{E}_x[\tau_{\partial B}] < (r+1)^2 - |x|^2\,.$$

Proof. Since $|X_t|^2 - t$ is a martingale, optional stopping at the bounded time $\tau_{\partial B} \wedge n$ gives, for every $n \in \mathbb{N}$,

$$|x|^2 = \mathbb{E}_x[|X_{\tau_{\partial B}\wedge n}|^2] - \mathbb{E}_x[\tau_{\partial B} \wedge n]\,.$$

Since $|X_{\tau_{\partial B}\wedge n}| < r + 1$, it follows that

$$\mathbb{E}_x[\tau_{\partial B} \wedge n] \leq (r+1)^2 - |x|^2\,,$$

so by monotone convergence $\mathbb{E}_x[\tau_{\partial B}] < \infty$, and in particular $\tau_{\partial B} < \infty$ almost surely. The proof is completed by using dominated and monotone convergence to take $n \to \infty$, observing that $X_{\tau_{\partial B}} \in \partial B$ and hence $r \leq |X_{\tau_{\partial B}}| < r + 1$. □

We can also use simple random walk to construct harmonic functions with prescribed boundary values.

Lemma 3.5 *Let $A \subset \mathbb{Z}^d$ be finite and let $h_0 : \partial A \to \mathbb{R}$. Then $h : \bar{A} \to \mathbb{R}$ given by $h(x) = \mathbb{E}_x[h_0(X_{\tau_{\partial A}})]$ agrees with h_0 on ∂A and is discrete harmonic in A.*

Proof. Choose $r > 0$ such that $\bar{A} \subset \mathrm{B}^{\ddots}(\mathbf{0}, r)$. A walk started in $\bar{A}$ must hit ∂A before $\partial \mathrm{B}^{\ddots}(\mathbf{0}, r)$, so $\tau_{\partial A} < \infty$ almost surely by Lemma 3.4. Conditioning on the first step, we find that for every $x \in A$,

$$h(x) = \mathbb{E}_x[h_0(X_{\tau_{\partial A}})] = \frac{1}{2d} \sum_{y \sim x} \mathbb{E}_y[h_0(X_{\tau_{\partial A}})] = \frac{1}{2d} \sum_{y \sim x} h(y) \,.$$

Thus h is discrete harmonic in A. For $x \in \partial A$ we have $\tau_{\partial A} = 0$, so $h(x) = h_0(x)$. □

For a finite set $A \subset \mathbb{Z}^d$, $x \in \bar{A}$ and $B \subset \partial A$, the probability $\mathbb{P}_x\{X_{\tau_{\partial A}} \in B\}$ that the walk started at x first leaves A through B is called the **harmonic measure** of B seen from x. In other words, Lemma 3.5 says that the harmonic function with boundary values h_0 is the average of h_0 with respect to harmonic measure.

3.2.1 Conditioned Random Walks

Harmonic measure tells us where a walk leaves a finite set. How does the walk move if we know where it will leave?

Let $A \subset \mathbb{Z}^d$ be finite and connected, and let B be a nonempty subset of ∂A. By Lemma 3.5, the probability $h(x) := \mathbb{P}_x\{X_{\tau_{\partial A}} \in B\}$ of leaving A through B is discrete harmonic in A and equals $\mathbf{1}_B$ on ∂A, and $h > 0$ on A since A is connected. The **Doob h-transform** is the walk on $\bar{A}$, stopped on reaching ∂A, with transition probabilities

$$p^h(x, y) := \frac{h(y)}{2d\, h(x)} \qquad \text{for } x \in A \text{ and } y \sim x \,. \tag{3.5}$$

These sum to one since h is harmonic at x, and they vanish if $y \in \partial A \setminus B$. Along a nearest-neighbor path $x_0, \dots, x_n$ with $x_0, \dots, x_{n-1} \in A$, the product telescopes to give

$$\prod_{j=1}^{n} p^h(x_{j-1}, x_j) = (2d)^{-n} \frac{h(x_n)}{h(x_0)} \,. \tag{3.6}$$

By the Markov property at time n, the right side is also the probability that simple random walk started at x_0 follows this path, conditioned on $X_{\tau_{\partial A}} \in B$. Hence the h-transform is simple random walk conditioned to leave A through

B (see Figure 3.3). In Exercises 3.13 and 3.31 you are asked to condition a walk never to leave a half-space and never to hit the origin.

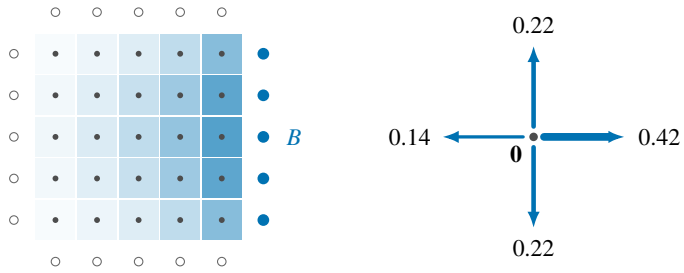


Figure 3.3: The probability h of leaving $A = \{-2, \dots, 2\}^2$ through its right side B (left), and the transition probabilities $p^h(\mathbf{0}, y) = h(y)$ from the center, where $h(\mathbf{0}) = \frac{1}{4}$ by symmetry (right). The step toward B is about three times as likely as the step away.

3.3 GREEN FUNCTION

Given a function $f : \mathbb{Z}^d \to \mathbb{R}$, it will be important for us later to construct a function F such that $\Delta^{::} F = f$. The most straightforward method is to assign arbitrary values for F on a pair of adjacent parallel hyperplanes, from which the relation $\Delta^{::} F = f$ determines the other values of F uniquely (see Exercise 3.14). This method suffers from the drawback that the growth rate of F is hard to control. A better method uses what is called the **Green function** or **fundamental solution** for the discrete Laplacian. This is a function $g^{::} : \mathbb{Z}^d \to \mathbb{R}$ whose discrete Laplacian is zero except at the origin where it is -1,

$$\Delta^{::} g^{::}(x) = -\delta_{\mathbf{0}}(x) = \begin{cases} -1, & x = \mathbf{0}, \\ 0, & x \neq \mathbf{0}. \end{cases} \tag{3.7}$$

If f has finite support, then we can construct F as a convolution $F(x) := -\sum_{y \in \mathbb{Z}^d} f(y) g^{::}(x-y)$, in which only finitely many terms are nonzero. Then for all $x \in \mathbb{Z}^d$, we have $\Delta^{::} F(x) = \sum_{y \in \mathbb{Z}^d} f(y) \delta_{\mathbf{0}}(x-y) = f(x)$ as desired. Moreover, if we can control the growth rate of the Green function, then we can control the growth rate of F. The minus sign in (3.7) is a convention: as we will now see, with this sign convention $g^{::}$ has a natural definition in terms of random walk.

Write $(X_k)_{k \geq 0}$ for simple random walk in $\mathbb{Z}^d$ and consider the function

$$g_n^{::}(x) := \frac{1}{2d} \mathbb{E}_{\mathbf{0}} \sum_{k=0}^{n} \mathbf{1}\{X_k = x\}.$$

By conditioning on the first step, we find that

$$\begin{aligned} 2d g_{n+1}^{::}(x) &= \delta_{\mathbf{0}}(x) + \mathbb{E}_{\mathbf{0}} \sum_{k=1}^{n+1} \mathbf{1}\{X_k = x\} \\ &= \delta_{\mathbf{0}}(x) + \sum_{z \sim \mathbf{0}} g_n^{::}(x - z)\,. \end{aligned}$$

We deduce that for all $x \in \mathbb{Z}^d$,

$$\Delta^{::} g_n^{::}(x) = -\delta_{\mathbf{0}}(x) + \mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\}\,. \tag{3.8}$$

Whether $g_n^{::}(\mathbf{0})$ stays bounded as $n \to \infty$ is the question of recurrence. Let $b_m := 4^{-m}\binom{2m}{m}$ be the probability that one-dimensional simple random walk is back at its starting point at time $2m$. By Stirling's formula,

$$b_m \sim \frac{1}{\sqrt{\pi m}} \qquad \text{as } m \to \infty\,. \tag{3.9}$$

In the plane, rotating by 45° makes the two coordinates independent: the diagonal coordinates $U_k := X_k \cdot (\mathbf{e}_1 + \mathbf{e}_2)$ and $V_k := X_k \cdot (\mathbf{e}_1 - \mathbf{e}_2)$ change by ± 1 at each step, and the four pairs of signs of $(U_{k+1} - U_k, V_{k+1} - V_k)$ are equally likely (see Figure 3.4(a)), so U and V are independent one-dimensional simple random walks started at 0, and

$$\mathbb{P}_{\mathbf{0}}\{X_k = x\} = \mathbb{P}\{U_k = x_1 + x_2\}\,\mathbb{P}\{V_k = x_1 - x_2\} \qquad \text{for } x \in \mathbb{Z}^2\,. \tag{3.10}$$

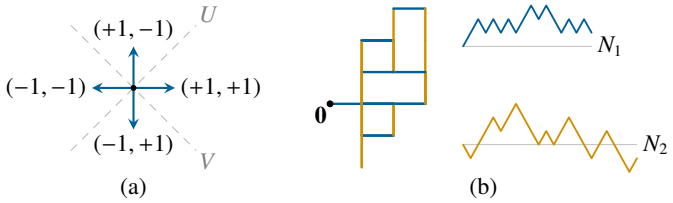


Figure 3.4: (a) The four steps of simple random walk in $\mathbb{Z}^2$, labeled by the increments of $U = x_1 + x_2$ and $V = x_1 - x_2$ (dashed diagonals), as in (3.10). (b) A walk of 40 steps with its horizontal (blue) and vertical (orange) steps, and each coordinate as a one-dimensional walk run for its own number N_i of steps, as in the proof of Lemma 3.6.

Lemma 3.6 *Simple random walk in $\mathbb{Z}^d$ is recurrent in dimensions $d = 1, 2$ and transient in dimensions $d \geq 3$.*

Proof. By the strong Markov property, the probability of another return is the same after each visit to the origin, so the number of visits is geometric and

its expectation $2d \lim_{n\to\infty} g_n^{::}(\mathbf{0})$ is infinite if and only if the return probability equals one.

The return probability at time $2m$ is b_m for $d = 1$ and, by (3.10), b_m^2 for $d = 2$. By (3.9) these are of order $m^{-1/2}$ and m^{-1}, so their sums diverge and the walk is recurrent.

For $d \geq 3$, we bound the probability of each site at time $n \geq 1$. By the unimodality of binomial coefficients (and, for odd k, conditioning on the last step), one-dimensional simple random walk at time k is at each site with probability at most $b_{\lfloor k/2 \rfloor}$, which is at most $C(k+1)^{-1/2}$ by (3.9). Let N_i be the number of the first n steps that change the ith coordinate. Given $N_1, \ldots, N_d$, the coordinates of X_n are independent, and the ith has the law of one-dimensional simple random walk at time N_i (see Figure 3.4(b)). Therefore, if $N_i \geq n/(2d)$ for every i, then the conditional probability that $X_n = x$ is at most $C_d n^{-d/2}$; and it is always at most $C_d n^{-1/2}$, since some N_i is at least n/d. Each N_i is binomial with parameters n and $1/d$, so by Chebyshev's inequality $\mathbb{P}\{N_i < n/(2d)\} \leq 4d/n$. Hence

$$\sup_{x\in\mathbb{Z}^d} \mathbb{P}_{\mathbf{0}}\{X_n = x\} \leq C_d n^{-d/2} + C_d n^{-1/2} \cdot \frac{4d^2}{n} \leq C_d n^{-3/2} . \tag{3.11}$$

Since these bounds are summable, $g_n^{::}(\mathbf{0})$ is bounded, and the walk is transient. □

For $d \geq 3$ simple random walk is transient, and by (3.11) we may define the Green function by

$$g^{::}(x) := \lim_{n\to\infty} g_n^{::}(x) = \frac{1}{2d} \sum_{k\geq 0} \mathbb{P}_{\mathbf{0}}\{X_k = x\} .$$

Since the terms of this series tend to zero, we have $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ by (3.8).

In dimension $d \leq 2$, simple random walk is recurrent, so $g_n^{::}(x)$ diverges. Instead, we normalize the sum and define the **potential kernel** by

$$a(x) := \lim_{n\to\infty} \big(g_n^{::}(\mathbf{0}) - g_n^{::}(x)\big) . \tag{3.12}$$

In dimension one a direct calculation gives $a(x) = |x|/2$, and in dimension two we will see in the next two lemmas that the limit defining $a(x)$ exists. Hence by (3.8) its discrete Laplacian is $\Delta^{::} a = \delta_{\mathbf{0}}$, since $\mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\} \to 0$ (by (3.10) and the one-dimensional bound in the proof of Lemma 3.6). In order to make dimension-independent statements, we will adopt the notation $g^{::} := -a$ for $d = 1, 2$, so that $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ in all dimensions. Since $g^{::}$ is explicit in dimension one, we focus on the case $d \geq 2$.

We first compute the potential kernel along the diagonal, where the walks U and V of (3.10) make the sums explicit. For integers $m \geq 0$ and j, let

$p_m(j) := 4^{-m}\binom{2m}{m+j}$ (with $p_m(j) = 0$ if $|j| > m$) be the probability that one-dimensional simple random walk is at $2j$ at time $2m$, so that $p_m(0) = b_m$. For integers k, l, the site $x = (k+l, k-l)$ has $x_1 + x_2 = 2k$ and $x_1 - x_2 = 2l$. Since the walk visits such a site only at even times, we have by (3.10)

$$g^{::}_{2M}(\mathbf{0}) - g^{::}_{2M}(x) = \frac{1}{4}\sum_{m=0}^{M}[b_m^2 - p_m(k)p_m(l)]\,, \tag{3.13}$$

and the same identity is true with $2M+1$ in place of $2M$. Every summand is nonnegative, because $p_m(j) \le b_m$ by the unimodality of binomial coefficients.

Lemma 3.7 *In dimension two, the limit* (3.12) *exists at the diagonal sites* (n, n)*, and for every integer* $n \ge 0$*,*

$$a(n,n) = \frac{1}{\pi}\sum_{j=1}^{n}\frac{1}{2j-1}\,.$$

Proof. The site (n, n) corresponds to $k = n$ and $l = 0$ in (3.13). For $j, M \ge 0$, let

$$D_M(j) := \frac{1}{4}\sum_{m=0}^{M} b_m[b_m - p_m(j)]\,,$$

so that $D_M(n) = g^{::}_{2M}(\mathbf{0}) - g^{::}_{2M}(n,n)$. We compute the increments $D_M(j) - D_M(j-1)$ by telescoping. For $j \ge 1$, let $t_m := b_m p_m(j-1)$. For $m \ge j-1$, the ratios of binomial coefficients are

$$\frac{p_m(j)}{p_m(j-1)} = \frac{m-j+1}{m+j} \qquad \text{and} \qquad \frac{t_{m+1}}{t_m} = \frac{(2m+1)^2}{4(m+j)(m-j+2)}\,.$$

The first ratio gives $b_m[p_m(j-1) - p_m(j)] = (2j-1)t_m/(m+j)$, and the second, together with the identity $(2m+1)^2 - 4(m-j+1)(m+j) = (2j-1)^2$, gives

$$b_m[p_m(j-1) - p_m(j)] = \frac{4}{2j-1}[(m-j+2)t_{m+1} - (m-j+1)t_m]\,.$$

Both sides vanish for $m < j-1$, and the last term vanishes at $m = j-1$, so summing over $m \le M$ gives

$$D_M(j) - D_M(j-1) = \frac{(M-j+2)\,t_{M+1}}{2j-1} \qquad \text{for } M \ge j-1\,.$$

For fixed j, the ratio $p_m(j-1)/b_m = \prod_{i=1}^{j-1}(m-i+1)/(m+i)$ tends to 1, so $t_m \sim b_m^2 \sim 1/(\pi m)$ by (3.9). Hence $D_M(j) - D_M(j-1) \to 1/(\pi(2j-1))$

as $M \to \infty$. Since $D_M(0) = 0$, summing over $1 \le j \le n$ shows that $D_M(n)$ converges to the stated value. The odd truncations give the same sums, so the limit (3.12) exists at (n, n). □

Note that the sums $D_M(j)$ increase with M, and so do their increments $D_M(j+1) - D_M(j)$, which are sums of nonnegative terms by the proof above. Hence

$$D_M(j) \le a(j,j) \qquad \text{and} \qquad D_M(j+1) - D_M(j) \le \frac{1}{\pi(2j+1)}\,. \tag{3.14}$$

Using the diagonal values, we can now control the potential kernel everywhere.

Lemma 3.8 *In dimension two, the limit (3.12) exists at every $x \in \mathbb{Z}^2$. The potential kernel vanishes at the origin and satisfies $\Delta^{\cdot\cdot} a = \delta_{\mathbf{0}}$ and $a(\mathbf{e}_1) = \frac{1}{4}$. There is a constant C such that*

$$\Big| a(x) - \frac{1}{2\pi} \log(1 + |x|) \Big| \le C \qquad \textit{for every } x \in \mathbb{Z}^2\,.$$

Proof. We first compare the increments in k of the sums (3.13) with those of D_M. For integers k, l and $M \ge 0$, let

$$A_M(k,l) := \frac{1}{4} \sum_{m=0}^{M} [b_m^2 - p_m(k) p_m(l)]\,.$$

Let $k, l \ge 0$ and $r := \max\{k, l\}$. For $m \ge j \ge 0$, the binomial ratios give

$$p_m(j) - p_m(j+1) = p_m(j)\, \frac{2j+1}{m+j+1}\,,$$

and the fraction increases with j. If $r = k$, then we use $p_m(l) \le b_m$; if $r = l$, then we use $p_m(k) \le b_m$ and compare the fractions at k and l. In both cases

$$0 \le p_m(l)[p_m(k) - p_m(k+1)] \le b_m[p_m(r) - p_m(r+1)] \qquad \text{for } m \ge r\,,$$

and for $m < r$ the middle term vanishes. Summing over $m \le M$ and using (3.14) gives

$$0 \le A_M(k+1,l) - A_M(k,l) \le D_M(r+1) - D_M(r) \le \frac{1}{\pi(2r+1)}\,.$$

By symmetry, $A_M(k,l)$ depends only on $|k|$ and $|l|$ and is symmetric in them. Let $r := \max\{|k|, |l|\}$ and $s := \min\{|k|, |l|\}$. Since $A_M(r,0) = D_M(r)$, adding this bound along the s steps from $(r,0)$ to (r,s) and using (3.14) gives

$$D_M(r) \le A_M(k,l) \le a(r,r) + \frac{s}{\pi(2r+1)} \le a(r,r) + \frac{1}{2\pi}\,. \tag{3.15}$$

Next we pass to the limit. At the site $x = (k+l, k-l)$, the sums (3.13) are $A_M(k,l)$, which increase with M and are bounded by (3.15). Hence the limit (3.12) exists at every site x with $x_1 + x_2$ even. Every other site x is not the origin, and its four neighbors are of this kind. At such a site, equation (3.8) reads

$$4(g_n^{::}(\mathbf{0}) - g_n^{::}(x)) = \sum_{y\sim x}(g_n^{::}(\mathbf{0}) - g_n^{::}(y)) + \mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\}\,,$$

and the last term tends to zero, so the limit exists there as well. The identity $\Delta^{::}a = \delta_{\mathbf{0}}$ was shown above, and $a(\mathbf{0}) = 0$ by definition. At the origin, $\Delta^{::}a(\mathbf{0}) = 1$ is the sum of the four values $a(\pm\mathbf{e}_i)$, which are equal by symmetry, so $a(\mathbf{e}_1) = \frac{1}{4}$.

Finally we compare a with the logarithm. Comparing the sum in Lemma 3.7 with an integral gives $|a(r,r) - \frac{1}{2\pi}\log(1+r)| \le C$ for every $r \ge 0$. At the site $x = (k+l, k-l)$ we have $|x|^2 = 2(k^2+l^2)$, so $\sqrt{2}\,r \le |x| \le 2r$. Thus, letting $M \to \infty$ in (3.15) gives the stated bound at every site with $x_1 + x_2$ even. At every other site, $a(x)$ is the average of its four neighbors, whose norms differ from $|x|$ by at most one, which gives the bound with a larger constant. □

Now we turn to the fundamental question of the growth rate of $g^{::}(x)$ as $|x| \to \infty$.

Theorem 3.9 *In dimension $d \ge 3$, we have, as $|x| \to \infty$,*

$$g^{::}(x) = a_d|x|^{2-d} + O(|x|^{-d})\,, \tag{3.16}$$

where $a_d = 1/(d(d-2)\omega_d)$, and ω_d is the volume of the unit ball $\mathrm{B}(\mathbf{0},1) \subset \mathbb{R}^d$.

In dimension $d = 2$, we have, as $|x| \to \infty$,

$$g^{::}(x) = -a(x) = -\frac{1}{2\pi}\log|x| - \kappa + O(|x|^{-2})\,, \tag{3.17}$$

where $\kappa = (2\gamma_E + \log 8)/(4\pi)$, and γ_E is Euler's constant.

Note that the main term in the expansion of $g^{::}$ is spherically symmetric, even though the lattice $\mathbb{Z}^d$ is not (see Figure 3.5). Moreover, the error term $O(|x|^{-d})$ is two orders lower than the main term (than the constant κ when $d = 2$). To emphasize how small this error is, the value of the main term $|x|^{2-d}$ (or $\log|x|$ when $d = 2$) typically changes on the order of $|x|^{1-d}$ when we move x by just one lattice spacing. To produce a change of order $|x|^{-d}$ in the main term requires moving x by a distance of only $1/|x|$, much less than one lattice spacing. Therefore $g^{::}$ is extraordinarily close to being spherically symmetric. Exercise 3.29 asks you to quantify this by bounding the level sets

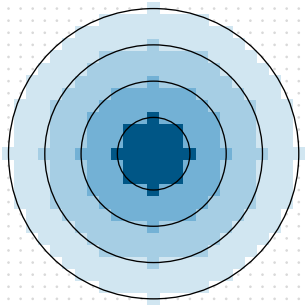


Figure 3.5: The sets $\{x \in \mathbb{Z}^2 : a(x) < \frac{1}{2\pi}\log\rho + \kappa\}$ for $\rho = 3, 6, 9, 12$ (darker for smaller ρ), and the circles of radius ρ. Already at these radii the potential kernel is nearly radial, as predicted by Theorem 3.9.

of $g^{::}$ between two concentric balls. We prove Theorem 3.9 in Section 3.6 by comparing $g^{::}$ with a correction of the continuous Green function.

Let $A \subset \mathbb{Z}^d$ be such that simple random walk killed on exiting A is transient (as it is for every finite A and, when $d \geq 3$, for every A). The **killed Green function** of simple random walk killed on exiting A is defined by

$$g_A^{::}(x,y) := \frac{1}{2d}\sum_{n\geq 0}\mathbb{P}_x\{X_n = y,\ n < \tau_{A^c}\}\,. \tag{3.18}$$

By definition, $g_A^{::}(x,y) = 0$ whenever $x \notin A$ or $y \notin A$. Since for each n, the product $(2d)^n\mathbb{P}_x\{X_n = y,\ n < \tau_{A^c}\}$ is the number of paths of length n in A from x to y, we see (by reversing each path) that $g_A^{::}(x,y) = g_A^{::}(y,x)$. Conditioning on the first step and using this symmetry gives the Poisson equation

$$\Delta_y^{::} g_A^{::}(x,y) = -\delta_x(y) \qquad \text{for } x \in \mathbb{Z}^d \text{ and } y \in A\,. \tag{3.19}$$

When A is finite, Exercise 3.18 asks you to extend this identity to all of $\mathbb{Z}^d$ by including the exit distribution on ∂A.

In the plane, simple random walk killed at the origin is transient, and its killed Green function gives a probabilistic interpretation of the potential kernel.

Lemma 3.10 *For all $x, y \in \mathbb{Z}^2 \setminus \{\mathbf{0}\}$,*

$$g_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}^{::}(x,y) = a(x) + a(y) - a(x-y)\,.$$

Proof. Fix $y \neq \mathbf{0}$ and let $f(x) := a(x) + a(y) - a(x-y)$. By Lemma 3.8, f is bounded, since $\log(1+|x|) - \log(1+|x-y|)$ is bounded for fixed y. The walk is symmetric, so a is even, $f(\mathbf{0}) = 0$, and $\Delta^{::} f = \delta_{\mathbf{0}} - \delta_y$. For a walk started

at $x \neq \mathbf{0}$, since $\Delta^{::} f = -\delta_y$ off the origin, the process

$$f(X_{n\wedge\tau_{\mathbf{0}}}) + \frac{1}{4} \sum_{k<n\wedge\tau_{\mathbf{0}}} \mathbf{1}\{X_k = y\}$$

is a martingale, so

$$f(x) = \mathbb{E}_x f(X_{n\wedge\tau_{\mathbf{0}}}) + \frac{1}{4}\mathbb{E}_x \sum_{k<n\wedge\tau_{\mathbf{0}}} \mathbf{1}\{X_k = y\}\,.$$

Note that the walk hits the origin almost surely, since by Lemma 3.6 it returns to x infinitely often, its excursions from x are independent by the strong Markov property, and each of them visits the origin with the same positive probability. Taking $n \to \infty$, the first term tends to $f(\mathbf{0}) = 0$ by bounded convergence, and by monotone convergence the second tends to $\frac{1}{4}$ times the expected number of visits to y before $\tau_{\mathbf{0}}$ (see Figure 3.6), which is $g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x,y)$ by (3.18). □

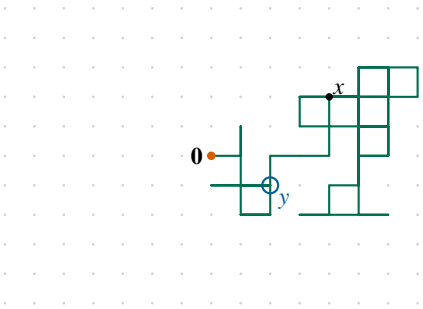


Figure 3.6: A walk from x stopped when it hits the origin (vermilion), with its visits to y (circled). By Lemma 3.10, the expected number of such visits is $4g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x,y) = 4(a(x) + a(y) - a(x-y))$.

We will need the following bounds in the proof of Theorem 3.9 in Section 3.6.

Lemma 3.11 *In dimensions $d \geq 3$, the Green function $g^{::}$ is positive and bounded, and $g^{::}(x) \to 0$ as $|x| \to \infty$. In dimension $d = 2$, there is a constant $C < \infty$ such that*

$$0 \leq a(x) \leq C \log(1 + |x|) \qquad \textit{for every } x \in \mathbb{Z}^2\,.$$

Proof. For $d \geq 3$, if $|x_1| + \cdots + |x_d| > N \geq 0$, then the walk is not at x before time $N+1$, so (3.11) gives $g^{::}(x) \leq C_d \sum_{n>N} n^{-3/2}$, which is finite for $N = 0$ and tends to zero as $N \to \infty$. Since every site is reached with positive probability, $g^{::} > 0$. For $d = 2$, by Lemma 3.10 with $y = x$ and $a(\mathbf{0}) = 0$ we

have $2a(x) = g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x,x) \ge 0$ for $x \ne \mathbf{0}$, and the upper bound follows from Lemma 3.8. □

The next lemma estimates the killed Green function of a discrete ball with pole at its center. In dimension two the first order asymptotic is easy to derive probabilistically; see Exercise 3.24.

Lemma 3.12 *Let $d \ge 2$ and $r \ge 2$, and let $B = \mathrm{B}^{::}(\mathbf{0}, r) \subset \mathbb{Z}^d$.*

(a) For $x \in B \setminus \{\mathbf{0}\}$ we have

$$g^{::}_B(\mathbf{0}, x) = \begin{cases} \dfrac{1}{2\pi} \log \dfrac{r}{|x|} + O(r^{-1} + |x|^{-2}), & d = 2; \\ a_d(|x|^{2-d} - r^{2-d}) + O(r^{1-d} + |x|^{-d}), & d \ge 3\,, \end{cases}$$

where $a_d = 1/(d(d-2)\omega_d)$, and ω_d is the volume of the unit ball $\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$.

(b) At the center,

$$g^{::}_B(\mathbf{0}, \mathbf{0}) = \begin{cases} \dfrac{1}{2\pi} \log r + \kappa + O(r^{-1}), & d = 2; \\ g^{::}(\mathbf{0}) - a_d r^{2-d} + O(r^{1-d}), & d \ge 3\,, \end{cases}$$

where $\kappa = (2\gamma_E + \log 8)/(4\pi)$ and γ_E is Euler's constant.

Proof. Let $v_r := \lceil r \rceil \mathbf{e}_1 \in \partial B$, and consider the function

$$f(x) := g^{::}(x) - g^{::}_B(\mathbf{0}, x) - g^{::}(v_r)\,.$$

Since every vertex of ∂B has norm in $[r, r+1)$, by Theorem 3.9 we have $f = O(r^{1-d})$ on ∂B. Since f is discrete harmonic in B, it follows from the minimum principle that for all $x \in B$,

$$g^{::}_B(\mathbf{0}, x) = g^{::}(x) - g^{::}(v_r) + O(r^{1-d})\,. \tag{3.20}$$

(a) Now the formulas for $x \ne \mathbf{0}$ follow from Theorem 3.9.

(b) Taking $x = \mathbf{0}$ in (3.20), along with the observation that $g^{::}(\mathbf{0}) = 0$ in dimension 2, proves the formulas at the origin. □

3.4 APPROXIMATING DISCRETE HARMONIC FUNCTIONS

In this section we relate discrete harmonic functions in the lattice $\delta\mathbb{Z}^d$ to their continuous counterparts in $\mathbb{R}^d$ as $\delta \downarrow 0$. If A is a domain in $\delta\mathbb{Z}^d$, write

$$A^{\square}_{\delta} := \left(A + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right)^o$$

for the interior of the union of the cubes $x_\delta^\square := x + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d$ for $x \in A$. If A is a domain in $\mathbb{R}^d$, write $A_\delta^{\cdot\cdot\cdot} := A \cap \delta\mathbb{Z}^d$. Given $x \in \mathbb{R}^d$, let $x_\delta^{\cdot\cdot\cdot}$ be the unique point satisfying

$$x_\delta^{\cdot\cdot\cdot} \in \left(x + \left(-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right) \cap \delta\mathbb{Z}^d ,$$

so that $x_\delta^{\cdot\cdot\cdot}$ is the closest lattice point to x, breaking ties to the right. For a function f on $\delta\mathbb{Z}^d$, write

$$f_\delta^\square(x) := f(x_\delta^{\cdot\cdot\cdot})$$

for the corresponding step function on $\mathbb{R}^d$. Likewise, for a function f on $\mathbb{R}^d$, write $f_\delta^{\cdot\cdot\cdot} := f|_{\delta\mathbb{Z}^d}$. These notations are summarized in Table 3.1.

In $\delta\mathbb{Z}^d$, two sites are neighbors if they differ by δ in exactly one coordinate, and outer boundaries and discrete harmonic functions are defined as in Section 3.1. We define the discrete Laplacian of a function f on $\delta\mathbb{Z}^d$ to be

$$\Delta_\delta^{\cdot\cdot\cdot} f(x) := \delta^{-2} \sum_{y \sim x} (f(y) - f(x)) .$$

For $f : \Omega \to \mathbb{R}$, where $\Omega \subset \mathbb{R}^d$ is open, the discrete Laplacian is defined on the set of points of $\Omega_\delta^{\cdot\cdot\cdot}$ whose segments to their $2d$ neighbors lie in Ω,

$$\Omega_\delta^+ := \left\{ x \in \Omega_\delta^{\cdot\cdot\cdot} : x + t\mathbf{e}_i \in \Omega \text{ for every } t \in [-\delta, \delta] \text{ and } 1 \le i \le d \right\} . \quad (3.21)$$

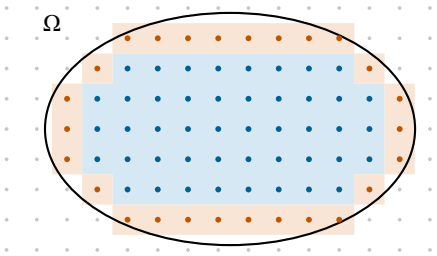


Figure 3.7: An ellipse Ω and the lattice $\delta\mathbb{Z}^2$ for $\delta = 1/6$, with the points of Ω_δ^+ (blue) and of $\Omega_\delta^{\cdot\cdot\cdot} \setminus \Omega_\delta^+$ (vermilion), each with its cube $x_\delta^\square$ shaded (so that the shading is the closure of $(\Omega_\delta^{\cdot\cdot\cdot})_\delta^\square$). The vermilion points are the outer boundary of Ω_δ^+ in $\delta\mathbb{Z}^2$, so $U_\delta = \Omega_\delta^+$ in Theorem 3.14 has $U_\delta \cup \partial U_\delta \subset \Omega$.

By the following lemma, if f is sufficiently smooth, then its discrete Laplacian on $\delta\mathbb{Z}^d$ approximates its Laplacian. Two derivatives suffice for convergence; four derivatives make the error of order δ^2, and more derivatives do not reduce the error further (indeed, part (c) gives an explicit error term of order δ^2).

Lemma 3.13 *Let $\Omega \subset \mathbb{R}^d$ be an open set.*

	$\delta\mathbb{Z}^d$	$\mathbb{R}^d$
points	$x_\delta^{\vdots} \in \left(x + \left(-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right) \cap \delta\mathbb{Z}^d$	$x_\delta^{\square} := x + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d$
sets	$A_\delta^{\vdots} := A \cap \delta\mathbb{Z}^d$	$A_\delta^{\square} := \left(A + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right)^o$
	$\Omega_\delta^+ \subset \Omega_\delta^{\vdots}$, see (3.21)	
functions	$f_\delta^{\vdots} := f\|_{\delta\mathbb{Z}^d}$	$f_\delta^{\square}(x) := f(x_\delta^{\vdots})$

Table 3.1: Notation for passing between $\delta\mathbb{Z}^d$ and $\mathbb{R}^d$.

(a) If $f \in C^2(\Omega)$ has uniformly continuous second derivatives in Ω, then

$$\lim_{\delta\to 0} \sup_{\Omega_\delta^+} \left|\Delta_\delta^{\vdots} f - \Delta f\right| = 0\,.$$

(b) If $f \in C^4(\Omega)$ has uniformly bounded fourth derivatives in Ω, then

$$\sup_{\Omega_\delta^+} \left|\Delta_\delta^{\vdots} f - \Delta f\right| \le \frac{\delta^2}{12} \sum_{i=1}^d \sup_{\Omega} \left|\frac{\partial^4 f}{\partial x_i^4}\right|\,.$$

(c) If $f \in C^6(\Omega)$ has uniformly bounded sixth derivatives in Ω, then

$$\sup_{x\in\Omega_\delta^+} \left|\Delta_\delta^{\vdots} f(x) - \Delta f(x) - \frac{\delta^2}{12} \sum_{i=1}^d \frac{\partial^4 f}{\partial x_i^4}(x)\right| \le \frac{2\delta^4}{6!} \sum_{i=1}^d \sup_{\Omega} \left|\frac{\partial^6 f}{\partial x_i^6}\right|\,.$$

Proof. By uniform continuity of the second derivatives $f_{ii} = \partial^2 f/\partial x_i^2$, the modulus of continuity

$$\omega(\delta) := \max_{1\le i\le d} \sup_{\substack{x,y\in\Omega \\ |x-y|\le\delta}} |f_{ii}(x) - f_{ii}(y)|$$

tends to 0 as $\delta \downarrow 0$.

Applying Taylor's theorem in the directions $\pm\mathbf{e}_i$ gives for each $i = 1, \dots, d$

$$f(x + \delta\mathbf{e}_i) - 2f(x) + f(x - \delta\mathbf{e}_i) = \frac{\delta^2}{2}\left(f_{ii}(x + t_{i,+}\mathbf{e}_i) + f_{ii}(x - t_{i,-}\mathbf{e}_i)\right)\,,$$

for some $t_{i,+}, t_{i,-} \in [0, \delta]$. Summing over i and dividing by δ^2 yields

$$\left|\Delta_\delta^{\vdots} f - \Delta f\right| \le d\,\omega(\delta)\,.$$

The right side tends to 0 as $\delta \downarrow 0$, which proves (a).

Carrying the Taylor expansion to fourth order, the terms of degree three cancel, leaving

$$f(x+\delta\mathbf{e}_i) - 2f(x) + f(x-\delta\mathbf{e}_i) - \delta^2 f_{ii}(x) \\ = \frac{\delta^4}{24}\left(\frac{\partial^4 f}{\partial x_i^4}(x + a_{i,+}\mathbf{e}_i) + \frac{\partial^4 f}{\partial x_i^4}(x - a_{i,-}\mathbf{e}_i)\right), \quad (3.22)$$

for some $a_{i,+}, a_{i,-} \in [0,\delta]$. Summing over i and dividing by δ^2 proves (b).

Carrying the Taylor expansion to sixth order, the terms of odd degree cancel, leaving

$$f(x+\delta\mathbf{e}_i) - 2f(x) + f(x-\delta\mathbf{e}_i) - \delta^2 f_{ii}(x) - \frac{\delta^4}{12}\frac{\partial^4 f}{\partial x_i^4}(x) \\ = \frac{\delta^6}{6!}\left(\frac{\partial^6 f}{\partial x_i^6}(x + \alpha_{i,+}\mathbf{e}_i) + \frac{\partial^6 f}{\partial x_i^6}(x - \alpha_{i,-}\mathbf{e}_i)\right),$$

for some $\alpha_{i,+}, \alpha_{i,-} \in [0,\delta]$. Summing over i and dividing by δ^2 proves (c). □

Next, we find sufficient conditions for solutions of the discrete Dirichlet problem in $\delta\mathbb{Z}^d$ to converge as $\delta \downarrow 0$ to a solution of the continuous Dirichlet problem in $\mathbb{R}^d$. We will use this for the scaling limit of killed Green functions.

Theorem 3.14 *Let $\Omega \subset \mathrm{B}(x_0, R) \subset \mathbb{R}^d$ be a bounded open set and suppose that $h : \Omega \to \mathbb{R}$ is harmonic in Ω. For $\delta > 0$, let $U_\delta \subset \Omega_\delta^+$ and suppose that h_δ is discrete harmonic in U_δ with boundary values $h|_{\partial U_\delta}$.*

(a) If h has uniformly continuous second derivatives in Ω, then

$$\max_{U_\delta} |h_\delta - h| \longrightarrow 0 \qquad \text{as } \delta \downarrow 0\,.$$

(b) If h has uniformly bounded fourth derivatives in Ω, then

$$\max_{U_\delta} |h_\delta - h| \le \frac{R^2\delta^2}{24d} \sum_{i=1}^{d} \sup_{\Omega} \left|\frac{\partial^4 h}{\partial x_i^4}\right|.$$

(c) If h has uniformly bounded sixth derivatives in Ω and there is $\beta > 0$ such that

$$\frac{1}{12}\sum_{i=1}^{d} \frac{\partial^4 h}{\partial x_i^4} \ge 2\beta \qquad \text{in } \Omega\,,$$

then, for all sufficiently small $\delta > 0$,

$$h_\delta(x) - h(x) \ge \frac{\beta}{2d}\delta^2 \mathrm{dist}(x, \partial U_\delta)^2 \qquad \text{for every } x \in U_\delta\,.$$

Proof. We compare $h_\delta - h$ with explicit quadratic functions and apply the discrete minimum principle (Lemma 3.2, after scaling $\delta\mathbb{Z}^d$ to $\mathbb{Z}^d$). By translation we may assume $x_0 = \mathbf{0}$, and we may assume that U_δ is nonempty. Since $U_\delta \subset \Omega_\delta^+$, the finite set $U_\delta \cup \partial U_\delta$ lies in $\Omega \subset \mathrm{B}(\mathbf{0}, R)$ (see Figure 3.7). By definition $h_\delta - h = 0$ on ∂U_δ, and since h_δ is discrete harmonic in U_δ, we have $\Delta_\delta^{::}(h_\delta - h) = -\Delta_\delta^{::} h$ there. Note also that the discrete Laplacian of $|z|^2$ equals $2d$ at every lattice point, since the two neighbors $z \pm \delta \mathbf{e}_i$ change $|z|^2$ by $\pm 2\delta z_i + \delta^2$.

(a) and (b): Consider the function

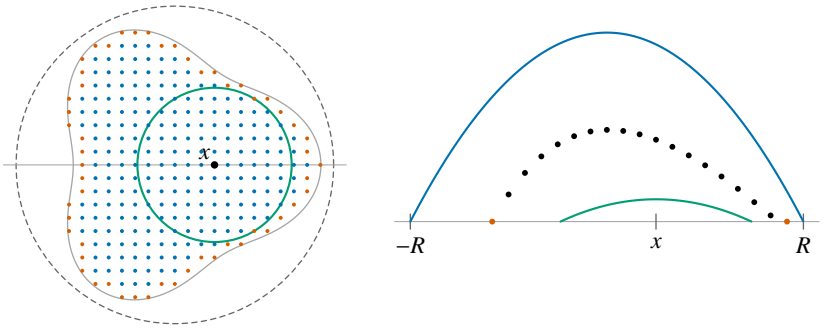
$$b(z) := \frac{R^2 - |z|^2}{2d} \max_{U_\delta} \left|\Delta_\delta^{::} h\right|.$$

Then

$$\Delta_\delta^{::}(b \pm (h_\delta - h)) = -\max_{U_\delta} \left|\Delta_\delta^{::} h\right| \mp \Delta_\delta^{::} h \le 0 \qquad \text{in } U_\delta\,,$$

and both functions $b \pm (h_\delta - h)$ are equal to $b \ge 0$ on ∂U_δ. By the minimum principle $|h_\delta - h| \le b$ in U_δ (see Figure 3.8), and since $b \le R^2 \max_{U_\delta} |\Delta_\delta^{::} h|/(2d)$, we obtain

$$\max_{U_\delta} |h_\delta - h| \le \frac{R^2}{2d} \max_{U_\delta} \left|\Delta_\delta^{::} h\right|. \tag{3.23}$$



(a) (b)

Figure 3.8: Proof of Theorem 3.14 for $h = \mathrm{Re}\,((x_1 + \mathbf{i}x_2)^4)$, $R = 1$ and $\delta = 1/12$. (a) $U_\delta = \Omega_\delta^+$ (blue) and its outer boundary (vermilion) inside Ω (gray) and $\mathrm{B}(\mathbf{0}, R)$ (dashed), and the disk of radius $\mathrm{dist}(x, \partial U_\delta)$ about x (green). Along the horizontal axis (b), the error $h_\delta - h$ (dots) vanishes on ∂U_δ and lies below the barrier b of parts (a) and (b) (blue) and above the barrier b_x of part (c) (green), all three divided by δ^2.

Since h is harmonic, by Lemma 3.13(a) the right side of (3.23) tends to zero as $\delta \downarrow 0$, which proves (a). If h has uniformly bounded fourth derivatives

in Ω, then by Lemma 3.13(b) we have

$$\max_{U_\delta} \left|\Delta_\delta^{\vdots} h\right| \le \frac{\delta^2}{12} \sum_{i=1}^{d} \sup_{\Omega} \left|\frac{\partial^4 h}{\partial x_i^4}\right| ,$$

and (3.23) proves (b).

(c) By Lemma 3.13(c) and the hypothesis on the fourth derivatives, we have

$$\min_{\Omega_\delta^+} \Delta_\delta^{\vdots} h \ge 2\beta\delta^2 - \frac{2\delta^4}{6!} \sum_{i=1}^{d} \sup_{\Omega} \left|\frac{\partial^6 h}{\partial x_i^6}\right| \ge \beta\delta^2$$

for all sufficiently small δ. For $x \in U_\delta$, consider the function

$$b_x(z) := \frac{\beta\delta^2}{2d}(\operatorname{dist}(x, \partial U_\delta)^2 - |z - x|^2) .$$

Then $\Delta_\delta^{\vdots}(h_\delta - h - b_x) = -\Delta_\delta^{\vdots} h + \beta\delta^2 \le 0$ in U_δ, while $h_\delta - h - b_x = -b_x \ge 0$ on ∂U_δ, since $|z - x| \ge \operatorname{dist}(x, \partial U_\delta)$ there. By the minimum principle $h_\delta - h \ge b_x$ in U_δ, and taking $z = x$ gives

$$h_\delta(x) - h(x) \ge \frac{\beta}{2d}\delta^2 \operatorname{dist}(x, \partial U_\delta)^2 ,$$

which proves (c). □

An example of a harmonic function which satisfies the hypotheses of Theorem 3.14(c) is

$$f(x) = \operatorname{Re}\left((x_1 + \mathbf{i}x_2)^4\right) = x_1^4 - 6x_1^2x_2^2 + x_2^4 .$$

Indeed, f is harmonic and $\partial^4 f/\partial x_1^4 = \partial^4 f/\partial x_2^4 = 24$, so

$$\frac{1}{12} \sum_{i=1}^{2} \frac{\partial^4 f}{\partial x_i^4} = 4$$

and the condition is satisfied with $\beta = 2$.

Next we prove a discrete version of the Harnack inequality of Theorem 2.30, by comparing killed Green functions of lattice balls with the Green function $\mathcal{G}$ of $\mathbb{R}^d$. Recall from (2.14) in Chapter 2 that $\mathcal{G}(x) = -\frac{1}{2\pi}\log|x|$ for $d = 2$ and $\mathcal{G}(x) = a_d|x|^{2-d}$ for $d \ge 3$.

We first find the scaling limit of a killed Green function when its two arguments are far from each other and from the boundary, using the asymptotics of Theorem 3.9, the method of images for the ball, and Theorem 3.14(b).

Lemma 3.15 *Let $d \ge 2$ and $\varpi > 0$, and let $B := \mathrm{B}(\mathbf{0}, \varpi)$. For each $y \in B$, let h_y be the harmonic function in B with boundary values $\mathcal{G}(\cdot - y)$, and let*

$$\mathcal{G}_B(x, y) := \mathcal{G}(x - y) - h_y(x) .$$

Let $K \subset B \times B$ be compact and disjoint from the diagonal. In dimension $d = 2$,

$$\lim_{R\to\infty} \sup_{\substack{x,y\in \mathrm{B}^{::}_{\varpi R} \\ (x/R,y/R)\in K}} \left| g^{::}_{\mathrm{B}^{::}_{\varpi R}}(x,y) - \mathcal{G}_B(x/R, y/R) \right| = 0\,,$$

whereas for $d \geq 3$,

$$\lim_{R\to\infty} \sup_{\substack{x,y\in \mathrm{B}^{::}_{\varpi R} \\ (x/R,y/R)\in K}} \left| R^{d-2} g^{::}_{\mathrm{B}^{::}_{\varpi R}}(x,y) - \mathcal{G}_B(x/R, y/R) \right| = 0\,.$$

We interpret a supremum over an empty set of lattice pairs as zero.

Proof. Since the functions $g^{::}(\cdot - y) - g^{::}_{\mathrm{B}^{::}_{\varpi R}}(\cdot, y)$ and $\mathbb{E}_\cdot g^{::}(X_{\tau_{\partial \mathrm{B}^{::}_{\varpi R}}} - y)$ agree on $\partial \mathrm{B}^{::}_{\varpi R}$ and are discrete harmonic in $\mathrm{B}^{::}_{\varpi R}$, by Lemma 3.2 we have

$$g^{::}_{\mathrm{B}^{::}_{\varpi R}}(x,y) = g^{::}(x-y) - \mathbb{E}_x g^{::}(X_{\tau_{\partial \mathrm{B}^{::}_{\varpi R}}} - y)\,. \tag{3.24}$$

By Theorem 3.9 (with the constant κ of (3.17)), we have uniformly for $|u| \geq cR$ with fixed $c > 0$,

$$g^{::}(u) + \frac{1}{2\pi} \log R + \kappa = \mathcal{G}(u/R) + O(R^{-2}) \qquad \text{in } d = 2\,, \tag{3.25}$$

and

$$R^{d-2} g^{::}(u) = \mathcal{G}(u/R) + O(R^{-2}) \qquad \text{for } d \geq 3\,. \tag{3.26}$$

Choose $\epsilon \in (0, \varpi/4)$ with $|\eta| \leq \varpi - 2\epsilon$ and $|\xi - \eta| \geq 2\epsilon$ for all $(\xi, \eta) \in K$. For $\eta \in B \setminus \{\mathbf{0}\}$, let $\eta^* := \varpi^2 \eta/|\eta|^2$ and $h_\eta(z) := \mathcal{G}(|\eta|(z - \eta^*)/\varpi)$, and let $h_{\mathbf{0}} \equiv \mathcal{G}(\varpi \mathbf{e}_1)$. If $|z| = \varpi$, then $|\eta|\,|z - \eta^*|/\varpi = |z - \eta|$ (see Figure 3.9). Since $\mathcal{G}$ is harmonic off the origin (Section 2.5), h_η is harmonic off η^*, so it is the function h_η of the statement. If $|\eta| \leq \varpi - 2\epsilon$, then $|\eta^*| \geq \varpi + 2\epsilon$. Hence h_η is harmonic in $\mathrm{B}(\mathbf{0}, \varpi + \epsilon)$, and there its partial derivatives of orders 1 to 4 are bounded by a constant $C = C(d, \varpi, \epsilon)$, uniformly in η.

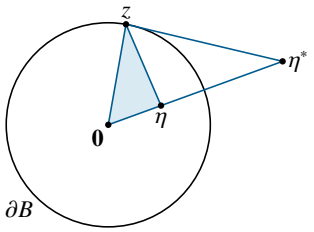


Figure 3.9: The image point $\eta^* = \varpi^2\eta/|\eta|^2$ of $\eta \in B = \mathrm{B}(\mathbf{0}, \varpi)$ in the proof of Lemma 3.15. For $z \in \partial B$ the triangles $\mathbf{0}\eta z$ (shaded) and $\mathbf{0}z\eta^*$ are similar with ratio $|\eta|/\varpi$, so $|z - \eta| = |\eta|\,|z - \eta^*|/\varpi$.

Let $R > 1/\epsilon$, let $x, y \in \mathrm{B}^{::}_{\varpi R}$ with $(\xi, \eta) := (x/R, y/R) \in K$, and let τ be the exit time $\tau_{\partial \mathrm{B}^{::}_{\varpi R}}$. Since each $v \in \partial \mathrm{B}^{::}_{\varpi R}$ satisfies $\varpi R \le |v| < \varpi R + 1$, we have $|v - y| \ge 2\epsilon R$, and also $|x - y| \ge 2\epsilon R$. In dimension $d = 2$, letting $c_R := \frac{1}{2\pi} \log R + \kappa$, we have by (3.24) and (3.25)

$$\begin{aligned} g^{::}_{\mathrm{B}_{\varpi R}}(x, y) &= [g^{::}(x - y) + c_R] - \mathbb{E}_x[g^{::}(X_\tau - y) + c_R] \\ &= \mathcal{G}(\xi - \eta) - \mathbb{E}_x \mathcal{G}(X_\tau/R - \eta) + O(R^{-2}) \,. \end{aligned}$$

For $d \ge 3$, (3.24) and (3.26) give the same expansion of $R^{d-2} g^{::}_{\mathrm{B}_{\varpi R}}(x, y)$.

For $v \in \partial \mathrm{B}^{::}_{\varpi R}$, let $v' := \varpi v/|v| \in \partial B$, so that $|v/R - v'| < 1/R$. The segment $[v', v/R]$ lies in $\{\varpi \le |z| < \varpi + \epsilon\}$, where $|z - \eta| \ge 2\epsilon$. Enlarging C if necessary, both $\mathcal{G}(\cdot - \eta)$ and h_η have gradient at most C there, and since $\mathcal{G}(v' - \eta) = h_\eta(v')$, it follows that $|\mathcal{G}(v/R - \eta) - h_\eta(v/R)| \le 2C/R$. Finally, the function $u(z) := h_\eta(z/R)$ is harmonic in $\Omega := \mathrm{B}(\mathbf{0}, (\varpi + \epsilon)R)$, with fourth derivatives at most CR^{-4}, and $\mathrm{B}^{::}_{\varpi R} \subset \Omega^+_1$ (the set (3.21) with $\delta = 1$), so by Theorem 3.14(b) with $\delta = 1$ we have $|\mathbb{E}_x u(X_\tau) - u(x)| \le C(\varpi + \epsilon)^2 R^{-2}$. Since $u(x) = h_\eta(\xi)$, we conclude that in $d = 2$

$$g^{::}_{\mathrm{B}_{\varpi R}}(x, y) = \mathcal{G}(\xi - \eta) - h_\eta(\xi) + O(R^{-1}) = \mathcal{G}_B(\xi, \eta) + O(R^{-1}) \,,$$

and that the same holds for $R^{d-2} g^{::}_{\mathrm{B}_{\varpi R}}(x, y)$ when $d \ge 3$, uniformly over $(\xi, \eta) \in K$. □

By Lemma 3.15 and the continuous Harnack inequality, killed Green functions from different starting points are comparable. Summing over the last step of the walk, we deduce a comparison of exit distributions, which gives the discrete Harnack inequality.

Lemma 3.16 (Discrete Harnack Inequality) *For each $d \ge 2$ there is $R_0(d) < \infty$ such that the following holds. Let $z \in \mathbb{Z}^d$, let $R \ge R_0(d)$, and let $h : \mathrm{B}^{::}(z, R) \to \mathbb{R}_{\ge 0}$ be discrete harmonic at every lattice point whose nearest neighbors all lie in $\mathrm{B}^{::}(z, R)$. Then*

$$\sup_{\mathrm{B}^{::}(z, R/2)} h \le 4^d \inf_{\mathrm{B}^{::}(z, R/2)} h \,.$$

Proof. By translation we may assume $z = \mathbf{0}$. Let $R \ge 32$ and let

$$A := \mathrm{B}^{::}(\mathbf{0}, 7R/8) \qquad \text{and} \qquad B := \mathrm{B}^{::}(\mathbf{0}, 15R/16) \,.$$

Let $U := \mathrm{B}(\mathbf{0}, 15/16)$, so that $\mathcal{G}_U$ is the kernel of Lemma 3.15 with $\varpi = 15/16$. Then $\bar{A} \subset B$ and $\bar{B} \subset \mathrm{B}^{::}(\mathbf{0}, R)$, so h is discrete harmonic on B (see Figure 3.10).

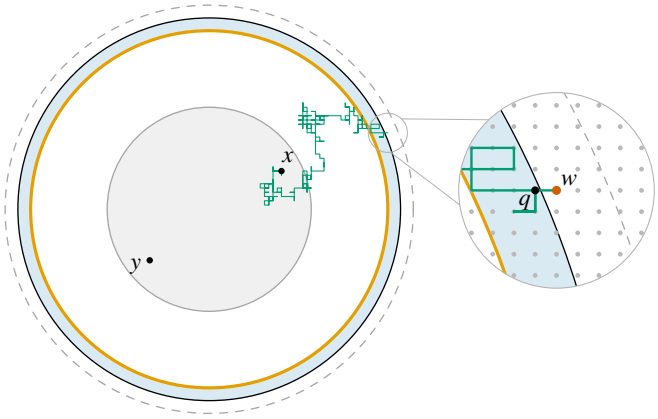


Figure 3.10: The balls in the proof of Lemma 3.16 in $\mathbb{Z}^2$ for $R = 48$. The points x and y lie in $\mathrm{B}^{::}(\mathbf{0}, R/2)$ (gray), the thick circle bounds $A = \mathrm{B}^{::}(\mathbf{0}, 7R/8)$, the thin one $B = \mathrm{B}^{::}(\mathbf{0}, 15R/16)$, and the dashed one $\mathrm{B}^{::}(\mathbf{0}, R)$. A walk from x leaves B at w after a last step from a site q of the shaded annulus $B \setminus A$ (magnified on the right), where the bound $g_B^{::}(x, q) \le 4^d g_B^{::}(y, q)$ extends, by the minimum principle, from the sites of A next to the thick circle.

We first compare the killed Green functions. If $q \in A$ has a neighbor outside A, then

$$\frac{27}{32} \le \frac{7}{8} - \frac{1}{R} \le \frac{|q|}{R} < \frac{7}{8}\,.$$

For $27/32 \le |p| \le 7/8$, the function $\mathcal{G}_U(\cdot, p)$ is positive and harmonic in $\mathrm{B}(\mathbf{0}, 27/32)$. Positivity follows from the minimum and strong minimum principles (Lemmas 2.12 and 2.8), since the kernel vanishes on ∂U and tends to $+\infty$ at its pole. By the continuous Harnack inequality, Theorem 2.30,

$$\mathcal{G}_U(\xi, p) \le \left(\frac{27/32 + 1/2}{27/32 - 1/2}\right)^d \mathcal{G}_U(\eta, p) = \left(\frac{43}{11}\right)^d \mathcal{G}_U(\eta, p)\,,$$

for $|\xi| \le 1/2$ and $|\eta| \le 1/2$. By the formula for h_η in the proof of Lemma 3.15, the kernel is jointly continuous, so it has a positive minimum on the compact set

$$\overline{\mathrm{B}}(\mathbf{0}, 1/2) \times \{v \in \mathbb{R}^d : 27/32 \le |v| \le 7/8\} \subset U \times U\,.$$

This set avoids the diagonal. Since $(43/11)^d < 4^d$, it follows from the uniform convergence in Lemma 3.15 that for all sufficiently large R,

$$g_B^{::}(x, q) \le 4^d g_B^{::}(y, q)\,, \tag{3.27}$$

for every $x, y \in \mathrm{B}^{::}(\mathbf{0}, R/2)$ and every such q.

For such x and y, the function

$$v(q) := 4^d g_B^{::}(y,q) - g_B^{::}(x,q)\,,$$

extended by zero outside B, is discrete harmonic in $B \setminus A$ by (3.19). It is nonnegative on the part of $\partial(B \setminus A)$ in A by (3.27), and zero on the rest, so $v \geq 0$ on $B \setminus A$ by the discrete minimum principle.

It remains to compare exit distributions. For $u \in B$ and $w \in \partial B$, let $\mu_u(w) := \mathbb{P}_u\{X_{\tau_{\partial B}} = w\}$. Summing over the last step before $\tau_{\partial B}$ (and using the normalization in (3.18)), we find that

$$\mu_u(w) = \sum_{\substack{q \in B \\ q \sim w}} g_B^{::}(u,q)\,.$$

Since $\bar{A} \subset B$, every q in this sum lies in $B \setminus A$, so $\mu_x(w) \leq 4^d \mu_y(w)$. By Lemma 3.5 and uniqueness of the solution of the Dirichlet problem,

$$h(x) = \sum_{w \in \partial B} \mu_x(w) h(w) \leq 4^d \sum_{w \in \partial B} \mu_y(w) h(w) = 4^d h(y)\,,$$

since $h \geq 0$ on ∂B. Taking the supremum over x and the infimum over y proves the lemma. □

3.5 DISCRETE HARMONIC POLYNOMIALS

In this section we consider polynomial functions on $\mathbb{Z}^d$, that is, linear combinations of monomials $x_1^{k_1} \cdots x_d^{k_d}$. The degree of this monomial is $k_1 + \cdots + k_d$, and the (total) degree of a polynomial is the maximal degree of its monomials. We will show that in every dimension $d \geq 2$ and for every integer $n \geq 0$, there exist polynomials of degree n on $\mathbb{Z}^d$ that are discrete harmonic. We will first prove that every discrete harmonic function of polynomial growth must actually be a polynomial. The base case is especially simple.

Lemma 3.17 *If $h : \mathbb{Z}^d \to \mathbb{R}$ is discrete harmonic and $h(x) \to 0$ as $|x| \to \infty$, then $h = 0$.*

Proof. Given $x \in \mathbb{Z}^d$ and $\epsilon > 0$, choose R sufficiently large so that $x \in \mathrm{B}^{::}(\mathbf{0}, R)$ and $|h(y)| < \epsilon$ for all $y \in \partial \mathrm{B}^{::}(\mathbf{0}, R)$. By the minimum principle applied to h and $-h$, we obtain $|h(x)| < \epsilon$. Since ϵ was arbitrary, we conclude that $h(x) = 0$. □

The next theorem generalizes the fact that a discrete harmonic function of sublinear growth must be constant. We use the convention that nonzero constants are polynomials of degree 0 and that 0 has degree -1.

Theorem 3.18 *Let $d \geq 2$. If $h : \mathbb{Z}^d \to \mathbb{R}$ is discrete harmonic and there is an integer $n \geq 0$ such that $|h(x)|/|x|^n \to 0$ as $|x| \to \infty$, then h is a polynomial function of the coordinates $x_1, \ldots, x_d$ of degree at most $n-1$.*

The case $n = 1$ of this theorem includes a discrete version of the classical Liouville theorem: bounded discrete harmonic functions on $\mathbb{Z}^d$ (and more generally discrete harmonic functions of strictly sublinear growth) are constant. To prove Theorem 3.18, we characterize polynomials on $\mathbb{Z}^d$ using discrete derivatives, starting with the one-dimensional case.

Lemma 3.19 *Let $n \geq 0$ be an integer. A function $f : \mathbb{Z} \to \mathbb{R}$ is a polynomial of degree at most n if and only if the difference $Df(x) := f(x+1) - f(x)$ is a polynomial of degree at most $n-1$ (where degree at most -1 means the zero polynomial).*

Proof. The case $n = 0$ is immediate, so we assume $n \geq 1$. Let $\mathcal{P}_n$ denote the vector space of real polynomials of degree at most n, and let $\mathcal{P}_n^0 := \{f \in \mathcal{P}_n : f(0) = 0\}$. The binomial theorem shows that $D(\mathcal{P}_n) \subset \mathcal{P}_{n-1}$, which is one direction of the lemma. If $g \in \mathcal{P}_n^0$ has $Dg = 0$, then g vanishes on $\mathbb{Z}$, so $g = 0$. Since $D : \mathcal{P}_n^0 \to \mathcal{P}_{n-1}$ is one-to-one and both vector spaces have dimension n, we infer that $D(\mathcal{P}_n^0) = \mathcal{P}_{n-1}$. Conversely, if $f : \mathbb{Z} \to \mathbb{R}$ satisfies $Df \in \mathcal{P}_{n-1}$, then $Df = Dh$ for some $h \in \mathcal{P}_n^0$, so $f - h$ is constant on $\mathbb{Z}$ and $f \in \mathcal{P}_n$. □

Applying Lemma 3.19 along each coordinate gives the multivariable characterization.

Lemma 3.20 *Let $n \geq 0$ be an integer. A function $f : \mathbb{Z}^d \to \mathbb{R}$ is a polynomial of degree at most n if and only if for every $1 \leq i \leq d$, the difference $D_i f(x) := f(x + \mathbf{e}_i) - f(x)$ is a polynomial of degree at most $n-1$ (where degree at most -1 means the zero polynomial).*

Proof. If f is a polynomial of degree at most n, then by the binomial theorem each $D_i f$ is a polynomial of degree at most $n-1$.

Conversely, suppose that each $D_i f$ is a polynomial of degree at most $n-1$. By the proof of Lemma 3.19, for every integer $k \geq 0$ there is a polynomial g_k of degree at most $k+1$ with $g_k(0) = 0$ and $Dg_k(t) = t^k$ for $t \in \mathbb{Z}$. For $1 \leq i \leq d$, let P_i be the polynomial obtained from $D_i f$ by replacing x_i^k with $g_k(x_i)$ in each monomial. Then P_i has degree at most n, vanishes when $x_i = 0$, and satisfies $D_i P_i = D_i f$, so $f - P_i$ does not depend on x_i. Changing one coordinate at a time along the path from $\mathbf{0}$ to x, we find that

$$f(x) = f(\mathbf{0}) + \sum_{i=1}^{d} P_i(x_1, \ldots, x_i, 0, \ldots, 0)\,,$$

which is a polynomial of degree at most n. (For $n = 0$ all the P_i vanish, and f is constant.) □

3.5.1 Coupling

The previous two lemmas suggest a proof strategy for Theorem 3.18: if h is discrete harmonic and $h(x) = o(|x|^n)$, then its difference $D_i h(x) = h(x + \mathbf{e}_i) - h(x)$ is also discrete harmonic, and if we can show that $D_i h(x) = o(|x|^{n-1})$, then by induction on n and Lemma 3.20, h is a polynomial. One technique for this is coupling: to get an upper bound on the difference $|h(x + \mathbf{e}_i) - h(x)|$, we will show how to couple random walks started at x and $x + \mathbf{e}_i$ so that the walks are likely to meet before traveling very far.

To overcome periodicity, we use the **lazy random walk** $(X_t)_{t\geq 0}$ in $\mathbb{Z}^d$ defined as follows. At each time $t \geq 1$ select a coordinate $j(t)$ uniformly in $\{1, \ldots, d\}$, and let

$$X_t - X_{t-1} = \begin{cases} \mathbf{e}_{j(t)}, & \text{with probability } 1/4; \\ -\mathbf{e}_{j(t)}, & \text{with probability } 1/4; \\ 0, & \text{with probability } 1/2\,. \end{cases} \tag{3.28}$$

The coordinate choices and increments at distinct times are independent.

In general, a **coupling** of two probability measures μ and ν is a probability measure on the product space whose marginals are equal to μ and ν. A coupling of two random variables is a pair of random variables on one probability space whose distribution is a coupling of their distributions.

In our case X is a lazy random walk started at $X_0 = x$, and we define a walk Y started at $Y_0 = x + \mathbf{e}_i$ by specifying its increments for $t \geq 1$ as follows. If $X_{t-1} = Y_{t-1}$, then

$$Y_t - Y_{t-1} = X_t - X_{t-1}\,.$$

Otherwise $Y_{t-1} - X_{t-1} = m\mathbf{e}_i$ for some $m \geq 1$. If $j(t) \neq i$, then again

$$Y_t - Y_{t-1} = X_t - X_{t-1}\,.$$

If $j(t) = i$ and $m > 1$, then the two walks take opposite steps,

$$Y_t - Y_{t-1} = X_{t-1} - X_t\,.$$

Finally, if $j(t) = i$ and $m = 1$, then

$$(X_t - X_{t-1},\, Y_t - Y_{t-1}) = \begin{cases} (\mathbf{e}_i, 0), & \text{with probability } 1/4; \\ (0, -\mathbf{e}_i), & \text{with probability } 1/4; \\ (-\mathbf{e}_i, \mathbf{e}_i), & \text{with probability } 1/4; \\ (0, 0), & \text{with probability } 1/4\,. \end{cases} \tag{3.29}$$

Since $m = 1$ at time $t = 1$, the last rule applies at the first step when $j(1) = i$. In (3.29), the first two outcomes make the walks meet, while the last two change m by 2 or leave it unchanged. Each marginal increment in (3.29) equals $\mathbf{e}_i$ with probability $1/4$, equals $-\mathbf{e}_i$ with probability $1/4$, and equals $\mathbf{0}$ with probability $1/2$, so both X and Y have the law (3.28) of lazy random walk. We call the pair (X, Y) the **reflection coupling**.

Every step that does not make the walks meet changes m by 0 or ± 2 and changes the ith coordinates of X and Y by opposite amounts. Hence, until the walks meet, m is a positive odd integer, and the sum of the ith coordinates stays at its initial value $2x_i + 1$. Equivalently, until the walks meet, Y is the reflection of X in the hyperplane

$$\Pi = \{z \in \mathbb{R}^d : z_i = x_i + \tfrac{1}{2}\}.$$

In particular, the ith coordinate of X never exceeds x_i before the walks meet.

The walks coalesce at the **meeting time**

$$T_{\text{meet}} := \inf\{t \geq 0 : X_t = Y_t\}.$$

See Figure 3.11 for a pair of coupled paths before and after T_{meet}.

We can use this coupling to bound $D_i h$ for a discrete harmonic function h. Fix $R > 1$ and let $h : \mathrm{B}^{::}(x, R) \cup \partial \mathrm{B}^{::}(x, R) \to \mathbb{R}$ be discrete harmonic in $\mathrm{B}^{::}(x, R)$. Let τ_R^X and τ_R^Y be the first exit times of X and Y from $\mathrm{B}^{::}(x, R)$. Since h is also harmonic in $\mathrm{B}^{::}(x, R)$ for lazy random walk, by optional stopping we have

$$h(x) = \mathbb{E}h(X_{\tau_R^X}) \qquad \text{and} \qquad h(x + \mathbf{e}_i) = \mathbb{E}h(Y_{\tau_R^Y}).$$

Let $\tau_R := \tau_R^X \wedge \tau_R^Y$. On the event $\{T_{\text{meet}} \leq \tau_R\}$ the walks have coalesced by time τ_R, whence $\tau_R^X = \tau_R^Y$ and $X_{\tau_R^X} = Y_{\tau_R^Y}$. Therefore,

$$D_i h(x) = \mathbb{E}\left[(h(Y_{\tau_R^Y}) - h(X_{\tau_R^X}))\mathbf{1}\{T_{\text{meet}} > \tau_R\}\right].$$

We conclude that

$$|D_i h(x)| \leq \left(\max_{\partial \mathrm{B}^{::}(x,R)} h - \min_{\partial \mathrm{B}^{::}(x,R)} h\right) \mathbb{P}\{T_{\text{meet}} > \tau_R\}. \tag{3.30}$$

Next we bound the probability in (3.30).

Lemma 3.21 *Let $d \geq 2$, let $x \in \mathbb{Z}^d$, and let $1 \leq i \leq d$. Let (X, Y) be the reflection coupling of lazy random walks started at x and $x + \mathbf{e}_i$. Let T_{meet} be their meeting time. Let $R > 0$, and let τ_R be the first time either walk exits $\mathrm{B}^{::}(x, R)$. Then*

$$\mathbb{P}\{\tau_R < T_{\text{meet}}\} \leq \frac{4d}{R}.$$

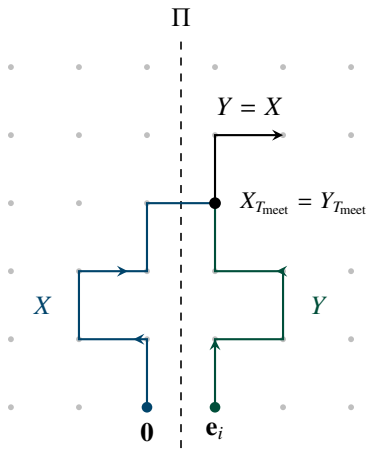


Figure 3.11: The reflection coupling started at $X_0 = \mathbf{0}$ and $Y_0 = \mathbf{e}_i$. Until T_{meet}, the paths are reflections of each other in the hyperplane $\Pi = \{z_i = 1/2\}$; after T_{meet}, the two walks coincide.

The proof uses a lemma about the probability that random walk in $\mathbb{Z}^d$ exits a ball before leaving a half-space. A **half-space** in $\mathbb{Z}^d$ is a set of the form

$$\mathbb{H} = \{x \in \mathbb{Z}^d : b \cdot x \le r\},$$

where $b \in \mathbb{R}^d$ and $r \in \mathbb{R}$ are fixed, and by normalizing we can always take b to be a unit vector. For the present purposes, we will only need half-spaces parallel to the coordinate axes (that is, $b = \mathbf{e}_i$ for some i), but we will use the following lemma again in Chapter 6, where it will be useful to have the extra generality.

Lemma 3.22 *Let $R > 0$ and $r \ge 0$, and let $b \in \mathbb{R}^d$ be a unit vector. Let T_R and $T_{\mathbb{H}}$ be the first exit times of simple random walk from the ball $\mathrm{B}^{\cdot\cdot}(\mathbf{0}, R)$, and from the half-space $\mathbb{H} = \{x \in \mathbb{Z}^d : b \cdot x \le r\}$, respectively. Then*

$$\mathbb{P}_{\mathbf{0}}\{T_R \le T_{\mathbb{H}}\} \le \frac{d(r+1)(R+1)}{R^2}.$$

Proof. Writing $(X_t)_{t \ge 0}$ for simple random walk in $\mathbb{Z}^d$ started at $\mathbf{0}$, consider the quadratic function of X_t

$$M_t = (r + 1 - b \cdot X_t)(R + 1 + b \cdot X_t) + \frac{1}{d}|X_t|^2.$$

Since b is a unit vector, M is a martingale. By optional stopping at time $T = T_{\mathbb{H}} \wedge T_R$ (which applies since $M_{t \wedge T}$ is bounded), we have

$$\begin{aligned}(r+1)(R+1) = M_0 &= \mathbb{E}M_T \\ &= \mathbb{E}\left[M_{T_{\mathbb{H}}}\mathbf{1}\{T_{\mathbb{H}} < T_R\} + M_{T_R}\mathbf{1}\{T_R \le T_{\mathbb{H}}\}\right].\end{aligned}$$

To estimate the right side, note that on $\{T_{\mathbb{H}} < T_R\}$ we have $r < b \cdot X_{T_{\mathbb{H}}} \le r+1$, so $M_{T_{\mathbb{H}}} > 0$. Moreover, on $\{T_R \le T_{\mathbb{H}}\}$ we have $-R-1 < b \cdot X_{T_R} \le r+1$ and $|X_{T_R}| \ge R$ (see Figure 3.12), so the product in M_{T_R} is nonnegative and $M_{T_R} \ge R^2/d$. We conclude that

$$(r+1)(R+1) \ge \mathbb{P}_{\mathbf{0}}\{T_R \le T_{\mathbb{H}}\}\frac{R^2}{d}\,.$$

□

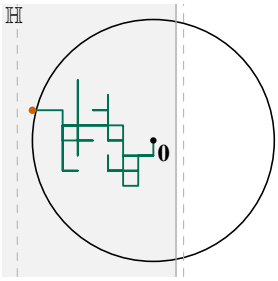


Figure 3.12: A walk from $\mathbf{0}$ that leaves the disk of radius $R = 8$ before the half-plane $\mathbb{H} = \{x_1 \le r\}$ with $r = 1$ (shaded), the event bounded in Lemma 3.22 for $b = \mathbf{e}_1$. The product $(r + 1 - x_1)(R + 1 + x_1)$ in the martingale vanishes on the dashed lines and is nonnegative between them.

We now prove Lemma 3.21.

Proof of Lemma 3.21. Since $d \ge 2$, the bound is trivial if $R \le 8$, so we may assume that $R > 8$. By translation we may assume $X_0 = \mathbf{0}$ and $Y_0 = \mathbf{e}_i$, so that $\Pi = \{z_i = 1/2\}$. Before the walks meet, the ith coordinate of X is at most 0, so X stays in the half-space $\mathbb{H} := \{z \in \mathbb{Z}^d : z_i \le 0\}$, and $|Y_t| \le |X_t| + 1$ (since the walks agree in every coordinate except the ith, and the ith coordinate of Y_t exceeds that of X_t by one in absolute value).

If $\tau_R < T_{\text{meet}}$, then whichever of the two walks exits $\mathrm{B}^{\cdot\cdot}(\mathbf{0}, R)$ at time τ_R, we have $|X_{\tau_R}| \ge R - 1$, so X exits $\mathrm{B}^{\cdot\cdot}(\mathbf{0}, R-1)$ before leaving $\mathbb{H}$. Since the probability of this event is the same for lazy and simple random walk, applying Lemma 3.22 with $b = \mathbf{e}_i$, $r = 0$ and $R - 1$ in place of R, we infer that

$$\mathbb{P}\{\tau_R < T_{\text{meet}}\} \le \frac{dR}{(R-1)^2} \le \frac{4d}{R}\,,$$

since $R \ge 2$. □

Combining Lemma 3.21 with (3.30), we can now show that the difference $D_i h$ grows one order more slowly than h.

Lemma 3.23 *Let $d \geq 2$ and $n \geq 1$, and let h be a discrete harmonic function on $\mathbb{Z}^d$ satisfying $|h(x)|/|x|^n \to 0$ as $|x| \to \infty$. Then for $1 \leq i \leq d$, the difference $D_i h(x) = h(x + \mathbf{e}_i) - h(x)$ satisfies $|D_i h(x)|/|x|^{n-1} \to 0$ as $|x| \to \infty$.*

Proof. Fix $x \in \mathbb{Z}^d$ with $|x| \geq 2$ and let $R = 2|x|$. Since every $y \in \partial \mathrm{B}^{::}(x, R)$ satisfies $|x| \leq |y| < 3|x| + 1$, the growth hypothesis gives

$$\max_{y \in \partial \mathrm{B}^{::}(x,R)} |h(y)| = o(|x|^n) .$$

By Lemma 3.21 and (3.30), it follows that

$$|D_i h(x)| \leq \frac{8d}{R} \max_{y \in \partial \mathrm{B}^{::}(x,R)} |h(y)| = o(|x|^{n-1}) .$$

□

By the same bound, locally bounded discrete harmonic functions on finer and finer lattices have harmonic subsequential limits (see Exercise 3.36). We can now prove Theorem 3.18.

Proof of Theorem 3.18. We proceed by induction on n, with the base case $n = 0$ given by Lemma 3.17. Let h be a discrete harmonic function on $\mathbb{Z}^d$ satisfying $|h(x)|/|x|^n \to 0$ as $|x| \to \infty$ with $n \geq 1$. Then $D_i h$ is also discrete harmonic on $\mathbb{Z}^d$. By Lemma 3.23, we have $|D_i h(x)|/|x|^{n-1} \to 0$ as $|x| \to \infty$. We obtain by the induction hypothesis that $D_i h$ is a polynomial of degree at most $n - 2$ for all $i = 1, \ldots, d$, so h must be a polynomial of degree at most $n - 1$ by Lemma 3.20. □

Let $f(x_1, \ldots, x_d)$ be a polynomial that is harmonic on $\mathbb{R}^d$. Next we give a recipe for constructing a polynomial that closely approximates f and is discrete harmonic on $\mathbb{Z}^d$ (see Table 3.2 for the first examples in $\mathbb{Z}^2$).

For a polynomial g and $1 \leq i \leq d$, let

$$\mathcal{D}_i g(x) := g\left(x + \frac{1}{2}\mathbf{e}_i\right) - g\left(x - \frac{1}{2}\mathbf{e}_i\right) .$$

Then the symmetric second difference of g in direction $\mathbf{e}_i$ is

$$\mathcal{D}_i^2 g(x) = g(x + \mathbf{e}_i) - 2g(x) + g(x - \mathbf{e}_i) .$$

For every integer $k \geq 0$, define the monic polynomial of degree k

$$P_k(x) := \prod_{j=0}^{k-1} \left(x + j - \frac{k-1}{2}\right) .$$

degree k	p_k	q_k
0	1	–
1	x	y
2	$x^2 - y^2$	$2xy$
3	$x^3 - 3xy^2 - \frac{1}{4}x$	$3x^2y - y^3 + \frac{1}{4}y$
4	$x^4 - 6x^2y^2 + y^4 - x^2 - y^2 + \frac{3}{4}$	$4x^3y - 4xy^3$
	...	...

Table 3.2: A basis for the discrete harmonic polynomials in $\mathbb{Z}^2$. There are two basis elements p_k and q_k in each degree $k \geq 1$. The polynomials p_k and q_k equal the real and imaginary parts of $(x + \mathbf{i}y)^k$ plus lower order terms of degree at most $k - 2$.

The empty product gives $P_0 = 1$, and for $k \geq 1$ we observe that

$$P_k(x) = \begin{cases} \displaystyle\prod_{j=1}^{k/2} \left(x^2 - \frac{(2j-1)^2}{4} \right), & k \text{ even;} \\ \displaystyle x \prod_{j=1}^{(k-1)/2} (x^2 - j^2), & k \text{ odd}. \end{cases}$$

Since the products for $P_k(x + 1/2)$ and $P_k(x - 1/2)$ share $k - 1$ factors, an easy calculation shows that

$$\mathcal{D}P_k = kP_{k-1} \tag{3.31}$$

(we omit the subscript from $\mathcal{D}$ here since P_k is a polynomial in only one variable). The similarity to the formula $\frac{d}{dx}x^k = kx^{k-1}$ will be useful.

Consider the linear map $\Xi : \mathbb{R}[x_1, \ldots, x_d] \to \mathbb{R}[x_1, \ldots, x_d]$ defined on monomials by

$$\Xi\left(\prod_{j=1}^{d} x_j^{k_j} \right) = \prod_{j=1}^{d} P_{k_j}(x_j)\,.$$

Since every P_k is monic of degree k, the map Ξ is triangular with ones on its diagonal in the monomial basis, so it is invertible. By (3.31), the map Ξ converts continuous harmonic polynomials into discrete harmonic polynomials.

Lemma 3.24 *Let $f \in \mathbb{R}[x_1, \ldots, x_d]$ be a polynomial of total degree $k \geq 0$ that is harmonic on $\mathbb{R}^d$.*

(a) Then $\Xi(f)$ is a discrete harmonic polynomial on $\mathbb{Z}^d$.

(b) If $k = 0$, then $f - \Xi(f) = 0$; if $k \geq 1$, then $f - \Xi(f)$ has total degree at most $k - 2$.

Proof. (a) By (3.31),

$$\begin{aligned}\mathcal{D}_i \Xi\left(\prod_{j=1}^{d} x_j^{k_j}\right) &= \mathcal{D}_i P_{k_i}(x_i) \prod_{\substack{1\leq j\leq d\\ j\neq i}} P_{k_j}(x_j)\\ &= k_i P_{k_i-1}(x_i) \prod_{\substack{1\leq j\leq d\\ j\neq i}} P_{k_j}(x_j)\\ &= \Xi\left(k_i x_i^{k_i-1} \prod_{\substack{1\leq j\leq d\\ j\neq i}} x_j^{k_j}\right).\end{aligned}$$

It follows that for all polynomials $f \in \mathbb{R}[x_1, \ldots, x_d]$, we have

$$\mathcal{D}_i \Xi(f) = \Xi\left(\frac{\partial f}{\partial x_i}\right).$$

Applying $\mathcal{D}_i$ to both sides gives

$$\mathcal{D}_i^2 \Xi(f) = \Xi\left(\frac{\partial^2 f}{\partial x_i^2}\right). \tag{3.32}$$

Summing (3.32) over $1 \leq i \leq d$ gives

$$\Delta^{::} \Xi(f) = \Xi(\Delta f).$$

Since Ξ is an invertible linear map, it follows that $\Xi(f)$ is discrete harmonic if and only if f is harmonic.

(b) Note that $P_k(x)$ is an even function for k even and an odd function for k odd. In particular, since P_k is monic, $P_k(x) - x^k$ vanishes when $k = 0$ and has degree at most $k - 2$ when $k \geq 1$. Applying this to each factor of every monomial proves the degree statement for $f - \Xi(f)$. □

In dimension $d = 2$, for each $k \geq 1$ the real and imaginary parts of $(x + \mathbf{i}y)^k$ are harmonic functions on $\mathbb{R}^2$. Hence their images under Ξ are discrete harmonic polynomials on $\mathbb{Z}^2$ (Table 3.2). In Exercise 3.37 you are asked to show that these, together with the constant function 1, form a basis for the space of all discrete harmonic polynomials in $\mathbb{Z}^2$.

The next lemma shows that there are lots of discrete harmonic polynomials in dimensions $d \geq 3$.

Lemma 3.25 *Let $k \geq 0$ be an integer. The vector space of discrete harmonic polynomials of degree at most k in $x_1, \dots, x_d$ has dimension*

$$\binom{k+d}{d} - \binom{k+d-2}{d},$$

where the second binomial coefficient is understood to be zero when $k < 2$.

Proof. For every integer j, let $\mathcal{P}_j^d$ be the real vector space of polynomials of degree at most j in $x_1, \dots, x_d$, with $\mathcal{P}_j^d = \{0\}$ when $j < 0$. Let $\mathrm{Harm}_{\leq k}$ and $\mathrm{Harm}_{\leq k}^{\dddot{}}$ be the spaces of harmonic and of discrete harmonic polynomials in $\mathcal{P}_k^d$. By Lemma 2.17, every $F \in \mathcal{P}_k^d$ can be written as $F = (1 - |x|^2)g + h$ with $g \in \mathcal{P}_{k-2}^d$ and $h \in \mathrm{Harm}_{\leq k}$. This decomposition is unique, since a harmonic polynomial $(1 - |x|^2)g$ vanishes on $\partial \mathrm{B}(\mathbf{0}, 1)$, hence on $\mathrm{B}(\mathbf{0}, 1)$ by Lemma 2.12 applied to it and to its negative, so $g = 0$. Thus $\dim \mathrm{Harm}_{\leq k} = \binom{k+d}{d} - \binom{k+d-2}{d}$, because $\dim \mathcal{P}_j^d = \binom{j+d}{d}$ for $j \geq 0$ (the number of ways of allocating j indistinguishable balls to $d+1$ bins). The map Ξ is a bijection from $\mathcal{P}_j^d$ to itself for every $j \geq 0$, and by the proof of Lemma 3.24 we have $\Delta^{\dddot{}}\Xi = \Xi\Delta$. Hence $\Xi(\mathrm{Harm}_{\leq k}) = \mathrm{Harm}_{\leq k}^{\dddot{}}$, which proves the dimension formula. □

By Lemma 3.25 and rank–nullity, for every integer $k \geq 0$ the discrete Laplacian maps the polynomials of degree at most k onto those of degree at most $k - 2$.

The reader willing to take Theorem 3.9 on faith may skip its proof in the next section.

3.6 ASYMPTOTICS OF THE GREEN FUNCTION*

In this section we prove Theorem 3.9. The idea of the proof is to add to the continuous Green function a correction bounded by $C|x|^{-d}$, chosen so that the negative discrete Laplacian of the sum is an even function $q : \mathbb{Z}^d \to \mathbb{R}$ of total mass one with $|q(y)| \leq C(1 + |y|)^{-d-4}$. For $d \geq 3$, convolving the discrete Green function with q gives back the corrected function, and in the plane the same holds up to an additive constant. Up to this constant, the error is therefore a weighted sum of the differences $g^{\dddot{}}(x) - g^{\dddot{}}(x - y)$, and since q is even, we can pair these differences into second differences and apply the bound (3.30) of Section 3.5. The constant κ then comes from the diagonal values of Lemma 3.7.

Throughout this section, $\mathcal{G}$ is the Green function of (2.14) in Chapter 2 (see Section 3.4), and $\partial_i := \partial/\partial x_i$.

We first correct $\mathcal{G}$ to cancel the fourth order term in the Taylor expansion of $\Delta^{\dddot{}}\mathcal{G}$ (without the correction, $-\Delta^{\dddot{}}\mathcal{G}(y)$ is of order $|y|^{-d-2}$, too large for its

absolute second moment to be finite).

Lemma 3.26 *Let $d \geq 2$, and let $K : \mathbb{Z}^d \to \mathbb{R}$ be given by $K(\mathbf{0}) := 0$ and*

$$K(x) := \mathcal{G}(x) - \frac{1}{24} \sum_{i=1}^{d} x_i \, \partial_i^3 \mathcal{G}(x) \qquad \text{for } x \neq \mathbf{0} \,.$$

Then the function $q := -\Delta^{::} K$ is even, $\sum_{y \in \mathbb{Z}^d} q(y) = 1$, and there is a constant C such that

$$|q(y)| \leq C(1 + |y|)^{-d-4} \qquad \text{and} \qquad |K(x) - \mathcal{G}(x)| \leq C|x|^{-d}$$

for every $y \in \mathbb{Z}^d$ and every $x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$.

Proof. For $x \in \mathbb{R}^d \setminus \{\mathbf{0}\}$, let

$$\psi(x) := -\frac{1}{24} \sum_{i=1}^{d} x_i \, \partial_i^3 \mathcal{G}(x) \,,$$

so that $\psi = K - \mathcal{G}$ at every nonzero lattice point. Since every derivative of $\mathcal{G}$ of order $j \geq 1$ is homogeneous of degree $2 - d - j$ (also when $d = 2$), the function ψ is homogeneous of degree $-d$ and its derivatives of order j are $O(|x|^{-d-j})$. In particular $|K - \mathcal{G}| \leq C|x|^{-d}$. Since $\mathcal{G}$ is even and its third derivatives are odd, K is even, and hence so is q.

Next we bound q. Since $\mathcal{G}$ is harmonic off the origin, so is $\partial_i^3 \mathcal{G}$, and by the product rule $\Delta(x_i \, \partial_i^3 \mathcal{G}) = 2\partial_i^4 \mathcal{G}$. Hence

$$\Delta\psi = -\frac{1}{12} \sum_{i=1}^{d} \partial_i^4 \mathcal{G} \,.$$

Let $|x| \geq 2$. Applying Lemma 3.13 with $\delta = 1$ on the ball $\mathrm{B}(x, 3/2)$ (on which every point has norm at least $|x|/4$), part (c) to $\mathcal{G}$, whose sixth derivatives are $O(|x|^{-d-4})$ there, and part (b) to ψ, whose fourth derivatives are $O(|x|^{-d-4})$ there, we obtain

$$\Delta^{::} \mathcal{G}(x) = \frac{1}{12} \sum_{i=1}^{d} \partial_i^4 \mathcal{G}(x) + O(|x|^{-d-4}) \,,$$

$$\text{and} \qquad \Delta^{::} \psi(x) = \Delta\psi(x) + O(|x|^{-d-4}) \,.$$

Adding these, the fourth derivatives cancel, so $|q(x)| \leq C|x|^{-d-4}$ for $|x| \geq 2$, and enlarging C to cover the finitely many remaining sites gives the stated bound.

Finally we compute the total mass of q. For an integer $N \geq 2$, consider the sum of $\Delta^{::} K$ over the lattice cube $[-N, N]^d \cap \mathbb{Z}^d$. Each edge with both

endpoints in the cube contributes two terms that cancel, so only the edges leaving the cube remain (see Figure 3.13), and

$$\sum_{x\in[-N,N]^d\cap\mathbb{Z}^d} \Delta^{::} K(x) = \sum_{j=1}^{d} \sum_{\pm} \sum_{\substack{x\in[-N,N]^d\cap\mathbb{Z}^d \\ x_j=\pm N}} \left(K(x\pm\mathbf{e}_j) - K(x)\right).$$

Since there are $O(N^{d-1})$ edges leaving the cube and the first derivatives of ψ are $O(N^{-d-1})$ near the faces, the differences of ψ contribute $O(N^{-2})$ in total. The gradient of $\mathcal{G}$ is homogeneous of degree $1-d$ (also when $d=2$), and the second derivatives of $\mathcal{G}$ are $O(N^{-d})$ within distance one of the faces, so by Taylor's theorem

$$\mathcal{G}(x+\mathbf{e}_j) - \mathcal{G}(x) = N^{1-d}\,\partial_j\mathcal{G}(x/N) + O(N^{-d})$$

uniformly for x in the cube with $x_j = N$, and the same expansion holds with $-\partial_j\mathcal{G}$ on the face $x_j = -N$. The errors total $O(N^{-1})$, and the main terms are Riemann sums (with mesh $1/N$) for the flux of $\nabla\mathcal{G}$ out of the cube $[-1,1]^d$. Since $\Delta^{::}K = -q$ is absolutely summable, letting $N\to\infty$ gives

$$-\sum_{y\in\mathbb{Z}^d} q(y) = \int_{\partial[-1,1]^d} \nabla\mathcal{G}\cdot n\,\mathrm{d}S\,,$$

where n is the outward unit normal. For $0<s<1$, the function $\mathcal{G}$ is harmonic between the sphere $\partial\mathrm{B}(\mathbf{0},s)$ and the boundary of the cube, so by the divergence theorem the flux of $\nabla\mathcal{G}$ out of the cube equals its flux out of the sphere. By (2.14), the outward normal derivative of $\mathcal{G}$ on the sphere is $-1/(d\omega_d s^{d-1})$, and the sphere has area $d\omega_d s^{d-1}$, so the flux is -1 and $\sum_y q(y) = 1$. □

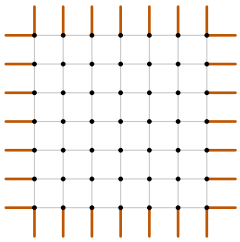


Figure 3.13: Summing $\Delta^{::}K$ over the cube $[-N,N]^2\cap\mathbb{Z}^2$ (here $N=3$) in the proof of Lemma 3.26. The two terms of each edge inside the cube (gray) cancel, and only the edges leaving it (vermilion) remain.

Next we bound second differences of a discrete harmonic function by its oscillation, just as we bounded first differences using the coupling. For $v\in\{\pm\mathbf{e}_1,\dots,\pm\mathbf{e}_d\}$ and a function f on $\mathbb{Z}^d$, let $D_v f(w) := f(w+v) - f(w)$.

Lemma 3.27 *Let $d \geq 2$. There is a constant C, depending only on d, with the following property. Let $z \in \mathbb{Z}^d$, let $R \geq 2$ and $M \geq 0$, and let h be a real-valued function on the closure of $\mathrm{B}^{::}(z, 4R)$, discrete harmonic in $\mathrm{B}^{::}(z, 4R)$. Suppose that $|h(w) - h(w')| \leq M$ for all $w, w' \in \mathrm{B}^{::}(z, 4R) \cup \partial\mathrm{B}^{::}(z, 4R)$. Then*

$$\left|h(z+y) + h(z-y) - 2h(z)\right| \leq \frac{CM|y|^2}{R^2} \qquad \text{for every } y \in \mathbb{Z}^d \text{ with } |y| \leq R\,.$$

Proof. The idea is to apply (3.30) first to h and then to its first differences, and to telescope along a lattice path. By (3.30) and Lemma 3.21, if a function is discrete harmonic in a ball $\mathrm{B}^{::}(w, \rho)$ with $\rho > 1$, then its first differences at w are bounded in absolute value by $4d/\rho$ times its oscillation on $\partial\mathrm{B}^{::}(w, \rho)$. For radius one the same estimate follows directly from the minimum principle (the ball consists of its center, whose value lies between the minimum and maximum of the boundary values). For $w \in \mathrm{B}^{::}(z, 3R)$, the ball $\mathrm{B}^{::}(w, R)$ lies in $\mathrm{B}^{::}(z, 4R)$ (see Figure 3.14) and its boundary lies in $\mathrm{B}^{::}(z, 4R) \cup \partial\mathrm{B}^{::}(z, 4R)$, so $|D_{\mathbf{e}_i} h(w)| \leq 4dM/R$. Since differences commute with $\Delta^{::}$, each first difference $D_{\mathbf{e}_i} h$ is discrete harmonic in $\mathrm{B}^{::}(z, 3R)$. For $w \in \mathrm{B}^{::}(z, 2R)$, the ball $\mathrm{B}^{::}(w, R/2)$ and its boundary lie in $\mathrm{B}^{::}(z, 3R)$, where the oscillation of $D_{\mathbf{e}_i} h$ is at most $8dM/R$, so

$$|D_{\mathbf{e}_j} D_{\mathbf{e}_i} h(w)| \leq \frac{64d^2 M}{R^2} \qquad \text{for } w \in \mathrm{B}^{::}(z, 2R)\,.$$

For $v, v' \in \{\pm\mathbf{e}_1, \dots, \pm\mathbf{e}_d\}$, the second difference $D_v D_{v'} h(w)$ involves the four sites w, $w + v$ and their translates by v', and it equals, up to sign, the second difference $D_{\mathbf{e}_j} D_{\mathbf{e}_i} h(w')$ at the point w' among them with the smallest coordinates.

Now let $|y| \leq R$, and let $p_0 = \mathbf{0}, p_1, \dots, p_L = y$ be a path of length $L = |y_1| + \cdots + |y_d|$ in which every coordinate moves monotonically, with steps $v_k := p_k - p_{k-1}$. Telescoping h along the path and its translate by $-y$, we find that

$$\begin{aligned} &h(z+y) + h(z-y) - 2h(z) \\ &\qquad = \sum_{k=1}^{L} \left[D_{v_k} h(z + p_{k-1}) - D_{v_k} h(z - y + p_{k-1}) \right]. \end{aligned}$$

Telescoping each bracket along the same path translated to $z - y + p_{k-1}$, and using that differences commute, we obtain

$$h(z+y) + h(z-y) - 2h(z) = \sum_{k=1}^{L} \sum_{l=1}^{L} D_{v_k} D_{v_l} h(z - y + p_{k-1} + p_{l-1})\,.$$

The four points involved in each term lie in the coordinate box between $z - y$ and $z + y$, so the corresponding point w' lies within distance $|y| \le R$ of z. Since $L^2 \le d|y|^2$, the sum is at most $64d^3M|y|^2/R^2$ in absolute value. □

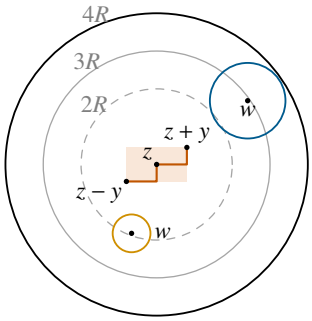


Figure 3.14: The balls in the proof of Lemma 3.27. For $w \in \mathrm{B}^{::}(z, 3R)$ the ball $\mathrm{B}^{::}(w, R)$ (blue) lies in $\mathrm{B}^{::}(z, 4R)$, and for $w \in \mathrm{B}^{::}(z, 2R)$ the ball $\mathrm{B}^{::}(w, R/2)$ (orange) lies in $\mathrm{B}^{::}(z, 3R)$. The second difference at z is telescoped along a monotone path from $z - y$ through z to $z + y$, inside the box between them (shaded).

We will also need a bound on the increments of the potential kernel. By the nonnegativity of the killed Green function in Lemma 3.10, applied with the pair $(u, -v)$, and since a is even, we have

$$a(u + v) \le a(u) + a(v) \qquad \text{for all } u, v \in \mathbb{Z}^2 ,$$

where the cases $u = \mathbf{0}$ and $v = \mathbf{0}$ are immediate. Since $a(\pm\mathbf{e}_i) = \frac{1}{4}$, it follows that $a(y) \le \frac{1}{4}(|y_1| + |y_2|)$, and

$$|a(x) - a(x - y)| \le a(y) \le \frac{1}{4}(|y_1| + |y_2|) \qquad \text{for all } x, y \in \mathbb{Z}^2 . \tag{3.33}$$

Proof of Theorem 3.9. Let K and q be as in Lemma 3.26. By the bound on q in that lemma (counting lattice points in shells), we have

$$\sum_{y \in \mathbb{Z}^d} |q(y)|(1 + |y|^2) < \infty \qquad \text{and} \qquad \sum_{|y| > R} |q(y)|(1 + |y|) \le CR^{-3} \quad \text{for } R \ge 1 .$$

By Lemma 3.11 and (3.33), every convolution below converges absolutely. Let $x \in \mathbb{Z}^d$ with $r := |x|$ large, and let $R := r/12$. Every point of $\mathrm{B}^{::}(x, 4R) \cup \partial\mathrm{B}^{::}(x, 4R)$ has norm at least $r/2$, so the ball does not contain the origin (see Figure 3.15).

Step 1: We prove the transient case $d \ge 3$. By Lemma 3.11 and dominated convergence (with majorant $C|q(y)|$), we have $(g^{::} * q)(w) \to 0$ as $|w| \to \infty$, and also $K(w) \to 0$ by Lemma 3.26. Since $\Delta^{::}(K - g^{::} * q) = 0$, it follows from Lemma 3.17 that $K = g^{::} * q$, and since $\sum_y q(y) = 1$, we obtain

$$g^{::}(x) - K(x) = \sum_{y \in \mathbb{Z}^d} q(y)[g^{::}(x) - g^{::}(x - y)] .$$

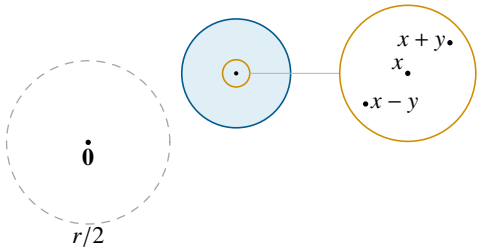


Figure 3.15: Proof of Theorem 3.9 for $|x| = r$ and $R = r/12$. The ball $\mathrm{B}^{::}(x, 4R)$ (shaded) misses the disk of radius $r/2$ (dashed), and the ball $\mathrm{B}^{::}(x, R)$ (orange, enlarged on the right) contains the pairs $x \pm y$ whose second differences are bounded by Lemma 3.27.

Suppose that $g^{::}(v) \le C(1+|v|)^{-\beta}$ for every v, where $0 \le \beta \le d-2$ (by Lemma 3.11 this holds for $\beta = 0$). We split the sum at $|y| = R$. On $\mathrm{B}^{::}(x, 4R)$ and its boundary, the function $g^{::}$ is discrete harmonic, with oscillation at most $Cr^{-\beta}$. Since q is even, we may pair y with $-y$, and by Lemma 3.27 the part with $|y| \le R$ is at most

$$\frac{1}{2}\sum_{|y|\le R} |q(y)| \left|2g^{::}(x) - g^{::}(x-y) - g^{::}(x+y)\right|$$
$$\le \frac{Cr^{-\beta}}{R^2} \sum_{y\in\mathbb{Z}^d} |q(y)||y|^2 \le Cr^{-\beta-2}\,.$$

In the part with $|y| > R$, the terms containing $g^{::}(x)$, or $g^{::}(x-y)$ with $|x-y| > r/2$, total at most $Cr^{-\beta}R^{-3}$, and the others have $|y| \ge r/2$ and total at most $Cr^{-d-4}\sum_{|y|\le r/2} g^{::}(v) \le Cr^{-\beta-4}$. Hence $g^{::}(x) - K(x) = O(r^{-\beta-2})$, and since $K(x) = O(r^{2-d})$, the supposition holds with $\min\{\beta+2, d-2\}$ in place of β. Repeating, we reach $\beta = d-2$, where $g^{::}(x) - K(x) = O(r^{-d})$, and by Lemma 3.26 it follows that $g^{::}(x) = a_d|x|^{2-d} + O(|x|^{-d})$.

Step 2: We prove the planar case. The function $h := K + a * q$ has discrete Laplacian $-q + q = 0$. Since $\sum_y q(y) = 1$, by (3.33) we have

$$|(a*q)(w) - a(w)| \le \sum_{y\in\mathbb{Z}^2} |q(y)|a(y) \le \frac{1}{4}\sum_{y\in\mathbb{Z}^2} |q(y)|(|y_1| + |y_2|) < \infty\,,$$

uniformly in w. By the logarithmic estimate in Lemma 3.8 and by Lemma 3.26, $K + a$ is bounded. Hence h is bounded, so it is equal to a constant κ_0 by Theorem 3.18 with $n = 1$. Since $\sum_y q(y) = 1$,

$$a(x) + K(x) - \kappa_0 = \sum_{y\in\mathbb{Z}^2} q(y)[a(x) - a(x-y)]\,.$$

On $\mathrm{B}^{::}(x, 4R)$ and its boundary, the function a is discrete harmonic, and by Lemma 3.8 its oscillation is bounded, since $\log(1+|w|)$ varies by a

bounded amount there. As in Step 1, the part with $|y| \le R$ is at most CR^{-2} by Lemma 3.27, and by (3.33) the part with $|y| > R$ is at most $\frac{1}{4}\sum_{|y|>R} |q(y)|(|y_1| + |y_2|) \le CR^{-3}$. Since $-K(x) = \frac{1}{2\pi}\log|x| + O(|x|^{-2})$ by Lemma 3.26, we conclude that

$$a(x) = \frac{1}{2\pi}\log|x| + \kappa_0 + O(|x|^{-2})\,.$$

Step 3: We identify κ_0. By Lemma 3.7 and Exercise 3.16,

$$a(n,n) = \frac{1}{2\pi}\log n + \frac{\log 2}{\pi} + \frac{\gamma_E}{2\pi} + O(n^{-2})\,.$$

Since $|(n,n)| = \sqrt{2}\,n$, comparison with Step 2 gives

$$\kappa_0 = \frac{2\gamma_E + 3\log 2}{4\pi} = \frac{2\gamma_E + \log 8}{4\pi}\,,$$

which is the constant κ in (3.17). Negating the asymptotic for $a(x)$ completes the proof. □

Since $g^{::} > 0$, and $a(x) > 0$ for $x \neq \mathbf{0}$ by Lemma 3.10 with $y = x$, it follows from Theorem 3.9 that $c(1+|x|)^{2-d} \le g^{::}(x) \le C(1+|x|)^{2-d}$ for $d \ge 3$, and that $a(x) \ge c\log(1+|x|)$ for $d = 2$ (matching the upper bound of Lemma 3.11). In particular, by Lemma 3.10 with $y = x$, the expected number of visits to x by a walk started at x before hitting the origin is of order $\log(1+|x|)$.

In Chapter 4 we use these asymptotics to show that the divisible sandpile is within a bounded distance of a Euclidean ball, and in Chapter 5 we use them again for internal DLA. In the next section we apply them to random walk in the plane started far from a nonempty finite set.

3.7 HARMONIC MEASURE FROM INFINITY IN THE PLANE

How does a random walk started far from a nonempty finite set $K \subset \mathbb{Z}^2$ first hit K? Simple random walk in the plane is recurrent (Lemma 3.6), so the walk hits K almost surely, and we expect the hitting distribution to depend less and less on the starting point as it moves away. We prove that it converges, and we describe the limit in terms of the potential kernel. This limit gives meaning to releasing a particle from infinity in external DLA (the model of Chapter 8, in which the cluster grows by adding the first site of its outer boundary visited by such a walk).

Throughout this section, $X = (X_n)_{n\ge 0}$ is simple random walk on $\mathbb{Z}^2$, and τ_E is the hitting time (3.4) of $E \subset \mathbb{Z}^2$.

Definition 3.28 Let $K \subset \mathbb{Z}^2$ be finite and nonempty. The **harmonic measure of K from infinity** is

$$\mathcal{H}_K(x) := \lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_K} = x\} \qquad \text{for every } x \in K\,.$$

For a finite nonempty set $A \subset \mathbb{Z}^2$, we write

$$H_A(z) := \mathcal{H}_{\partial A}(z) \qquad \text{for every } z \in \partial A\,,$$

for the harmonic measure of its outer boundary.

We will see that the limit in the definition exists for every nonempty finite set (see Figure 3.16 for two small sets).

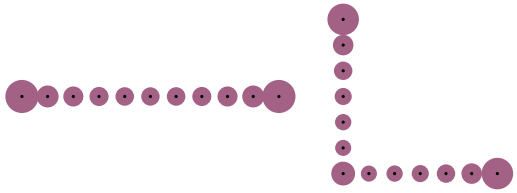


Figure 3.16: Harmonic measure from infinity $\mathcal{H}_K$ (the areas of the dots are proportional to mass) on a segment of 11 sites and on an L-shaped set of 13 sites in $\mathbb{Z}^2$, computed from the potential kernel by Lemma 3.29(a). It is largest at the ends and at the outer corner, like the arcsine law of Figure 2.13.

Recall that the potential kernel a of (3.12) satisfies $\Delta^{\cdot\cdot} a = \delta_{\mathbf{0}}$ and $a(\mathbf{0}) = 0$, that a is even (by the symmetry of the walk), and that by (3.17),

$$a(x) = \frac{1}{2\pi} \log |x| + \kappa + O(|x|^{-2}) \qquad \text{as } |x| \to \infty\,, \tag{3.34}$$

where κ is a constant. In particular, $a(x-v) - a(x) \to 0$ as $|x| \to \infty$, for every $v \in \mathbb{Z}^2$.

We will also use the following consequence of recurrence. Let $A \subset \mathbb{Z}^2$ be finite and nonempty, and let $g : \mathbb{Z}^2 \to \mathbb{R}$ be bounded and harmonic on $\mathbb{Z}^2 \setminus A$. For $y \in \mathbb{Z}^2$, the process $g(X_{k\wedge\tau_A})$ is a bounded martingale under $\mathbb{P}_y$, and $\tau_A < \infty$ almost surely, since by Lemma 3.6 the walk returns to y infinitely often, its excursions from y are independent by the strong Markov property, and each of them visits a fixed point of A with the same positive probability. The optional stopping theorem and the bounded convergence theorem therefore give

$$g(y) = \mathbb{E}_y[g(X_{\tau_A})] \qquad \text{for every } y \in \mathbb{Z}^2\,. \tag{3.35}$$

Next we prove that the limit exists, and we describe harmonic measure in terms of a potential that vanishes on the set. In Chapter 8 we use this potential to prove the discrete Beurling estimate (Section 8.2), which bounds the harmonic measure of a single site of a connected set.

Lemma 3.29 *Let $A \subset \mathbb{Z}^2$ be finite and nonempty.*

(a) For every $z \in A$, the limit $\mathcal{H}_A(z) = \lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_A} = z\}$ exists, and $\mathcal{H}_A$ is the unique probability measure ν on A whose potential $x \mapsto \sum_{v\in A} \nu(v)\, a(x-v)$ is constant on A.

(b) Let U_A be the potential of $\mathcal{H}_A$ minus its value on A. Then U_A is nonnegative, vanishes on A, is harmonic on $\mathbb{Z}^2 \setminus A$, satisfies $\Delta^{::} U_A(v) = \mathcal{H}_A(v)$ for every $v \in A$, and satisfies $U_A(x) = a(x) + O(1)$ as $|x| \to \infty$.

(c) Let $F : \mathbb{Z}^2 \to \mathbb{R}$ be harmonic on $\mathbb{Z}^2 \setminus A$ and such that $F - a$ is bounded. Then $F(x) - U_A(x) = \mathbb{E}_x[F(X_{\tau_A})]$ for every $x \in \mathbb{Z}^2$.

Proof. We use (3.35) in all three parts.

(a) Fix $v, w \in A$. By (3.34), the function $x \mapsto a(x-v) - a(x-w)$ tends to zero as $|x| \to \infty$, so it is bounded, and it is harmonic on $\mathbb{Z}^2 \setminus \{v, w\}$. By (3.35), for every $y \in \mathbb{Z}^2$,

$$a(y-v) - a(y-w) = \sum_{z\in A} \mathbb{P}_y\{X_{\tau_A} = z\}[a(z-v) - a(z-w)]\,. \tag{3.36}$$

Since $\tau_A < \infty$ almost surely, each $\mathbb{P}_y\{X_{\tau_A} = \cdot\}$ is a probability measure on the finite set A, so every sequence y_k with $|y_k| \to \infty$ has a subsequence along which these measures converge to a probability measure ν on A. Letting y run along this subsequence in (3.36), we obtain $\sum_{z\in A} \nu(z)[a(z-v) - a(z-w)] = 0$, and since a is even, the potential of ν takes the same value at v and at w. Hence every subsequential limit is a probability measure on A with constant potential on A.

It remains to show that there is only one such measure. Let ν_1 and ν_2 be two of them, and let $P(x) := \sum_{v\in A}(\nu_1 - \nu_2)(v)\, a(x-v)$. Since $\nu_1 - \nu_2$ has total mass zero, $P(x) = \sum_{v\in A}(\nu_1 - \nu_2)(v)[a(x-v) - a(x)]$ tends to zero as $|x| \to \infty$, so P is bounded. It is harmonic off A and constant on A, so by (3.35) it is constant on $\mathbb{Z}^2$. Hence $\nu_1 - \nu_2 = \Delta^{::} P = 0$, so all subsequential limits coincide, which proves (a).

(b) Since $\Delta^{::} a = \delta_{\mathbf{0}}$, we have $\Delta^{::} U_A = \sum_{v\in A} \mathcal{H}_A(v)\delta_v$, which gives the harmonicity off A and the Laplacian on A. Since $\mathcal{H}_A$ is a probability measure and $a(x-v) - a(x) \to 0$ as $|x| \to \infty$, the function $U_A - a$ is bounded at infinity. Since $a(x) \to \infty$, it follows that $U_A(x) \to \infty$ as $|x| \to \infty$. For $x \notin A$, choose $R > |x|$ so large that $U_A \geq 0$ outside $\mathrm{B}^{::}(\mathbf{0}, R)$. By the minimum principle (Lemma 3.2) applied on the finite set $\mathrm{B}^{::}(\mathbf{0}, R) \setminus A$, whose boundary values are those on A and on $\partial\mathrm{B}^{::}(\mathbf{0}, R)$, we have $U_A(x) \geq 0$.

(c) The function $F - U_A$ is bounded, because $F - a$ and $U_A - a$ are, and it is harmonic off A and equal to F on A. Applying (3.35) to $F - U_A$ proves (c). □

If F is as in part (c), then since $X_{\tau_A} \in A$, we have

$$\min_A F \le F - U_A \le \max_A F \qquad \text{on } \mathbb{Z}^2 . \tag{3.37}$$

We use both inequalities in the proof of the Beurling estimate in Section 8.2. Taking $A = \{\mathbf{0}\}$ in part (b), we have $U_{\{\mathbf{0}\}} = a$, so $a \ge 0$.

Exercise 3.43 asks you to prove that on every recurrent graph, the Laplacian of a bounded function that is harmonic off a finite set sums to zero over that set. In dimensions $d \ge 3$ the hitting probabilities tend to zero, but conditioning the walk to hit the set gives a probability measure (see Exercise 3.46), which we use for external DLA in $\mathbb{Z}^d$ in Section 8.4 of Chapter 8.

In Chapter 4 we turn from potential theory to lattice growth, beginning with the divisible sandpile, whose odometer we estimate with the Green function of this chapter.

EXERCISES

Exercise 3.1. In the setting of Section 3.1, let $A \subset \mathbb{Z}^d$ be finite, and let $h, g : \bar{A} \to \mathbb{R}$ be functions such that h is discrete harmonic on A and g is discrete superharmonic on A. Prove that if $g \ge h$ on ∂A, then $g \ge h$ on A.

Exercise 3.2. In the setting of Section 3.1, let $g : \mathbb{Z}^d \to \mathbb{R}$ be discrete superharmonic, and let $A \subset \mathbb{Z}^d$ be finite. Let $h : \mathbb{Z}^d \to \mathbb{R}$ be the unique function that equals g outside A and is discrete harmonic on A. Prove that h is discrete superharmonic on $\mathbb{Z}^d$.

Exercise 3.3. In the setting of Section 3.1, let $f, g : \mathbb{Z}^d \to \mathbb{R}$ be discrete superharmonic. Determine whether $\min(f, g)$ and $\max(f, g)$ must be discrete superharmonic.

Exercise 3.4. Let $L^2(\mathbb{Z}^d)$ be the space of square-summable functions $\mathbb{Z}^d \to \mathbb{R}$, with the inner product $(f, g) := \sum_{x \in \mathbb{Z}^d} f(x)g(x)$. Prove that the discrete Laplacian (3.1) is self-adjoint on $L^2(\mathbb{Z}^d)$, that is, $(f, \Delta^{::} g) = (\Delta^{::} f, g)$ for all $f, g \in L^2(\mathbb{Z}^d)$.

Exercise 3.5. Let $A \subset \mathbb{Z}^d$ be finite and nonempty. The energy of $u : \bar{A} \to \mathbb{R}$ is the sum of $(u(x) - u(y))^2$ over the edges $\{x, y\}$ of $\mathbb{Z}^d$ that meet A. Show that if h is discrete harmonic in A and v vanishes on ∂A, then the energy of $h + v$ is that of h plus that of v. Deduce that h is the unique function of least energy with its values on ∂A.

Exercise 3.6. Let $G = (V, E)$ be a finite connected simple graph, and let Δ be the Laplacian (3.2) of G.

(a) Show that the image of Δ is the set of functions on V that sum to zero.

(b) Given $v, w \in V$, describe a function $f : V \to \mathbb{R}$ with $\Delta f = \delta_v - \delta_w$.

Exercise 3.7. Show that if $(X_n)_{n\geq 0}$ is a supermartingale with respect to some filtration (in the sense of Section 3.2), then it is also a supermartingale with respect to its natural filtration.

Exercise 3.8. For positive integers ℓ and r, consider random walk $(X_t)_{t\geq 0}$ on $\mathbb{Z}$ started at zero, and let τ be its hitting time of $\{-\ell, r\}$.

(a) Compute $\mathbb{P}_0\{X_\tau = r\}$ when $(X_t)_{t\geq 0}$ is a simple random walk.

(b) Compute $\mathbb{P}_0\{X_\tau = r\}$ when each increment $X_{t+1} - X_t$ equals 1 with probability $p \in (1/2, 1)$ and -1 with probability $1 - p$, independently.

Hint for (b): Check that $((1-p)/p)^{X_{t\wedge\tau}}$ is a martingale.

Exercise 3.9. Let $(U_k)_{k\geq 0}$ be simple random walk from zero on $\mathbb{Z}$, and let τ_b be its hitting time (3.4) of an integer $b \geq 1$. Prove that, for $0 < s < 1$,

$$\mathbb{E}\big[s^{\tau_b}\big] = \left(\frac{1 - \sqrt{1 - s^2}}{s}\right)^b .$$

Hint: The function $u(j) := \mathbb{E}_j[s^{\tau_b}]$ satisfies $u(j) = s(u(j-1) + u(j+1))/2$ for $j < b$, equals 1 at b, and is bounded as $j \to -\infty$.

Exercise 3.10. Let $(X_n)_{n\geq 0}$ be simple random walk on $\mathbb{Z}$ started at zero. Find a stopping time T with respect to the natural filtration so that X_T is not integrable (compare Theorem 3.3).

Hint: Define $T = 2\min\{n \geq 1 : X_n = 0\}$.

Exercise 3.11. Let $d \geq 1$ and $R \geq 1$. Strengthen Lemma 3.4 by proving that, for every $r > 0$,

$$\mathbb{P}_{\mathbf{0}}\{\tau_{\partial \mathrm{B}_R^{::}} \leq rR^2\} \leq 2de^{-1/(2dr)} \qquad \text{and} \qquad \sup_{x\in \mathrm{B}_R^{::}} \mathbb{P}_x\{\tau_{\partial \mathrm{B}_R^{::}} > rR^2\} \leq 2^{1-r/9} .$$

Deduce that $\mathbb{E}_x e^{\theta\tau_{\partial \mathrm{B}_R^{::}}/R^2}$ is bounded uniformly in $R \geq 1$ and $x \in \mathrm{B}_R^{::}$ if $\theta < (\log 2)/9$.

Exercise 3.12. Show that Lemma 3.5 can fail if A is infinite, even if $\mathbb{P}_x\{\tau_{\partial A} < \infty\} = 1$ for all $x \in A$. Can it fail if $\mathbb{E}_x\tau_{\partial A} < \infty$ for all $x \in A$?

Exercise 3.13. Let $d \geq 2$ and $h(x) := x_d$. Show that (3.5) defines transition probabilities from every x with $x_d \geq 1$: to $x \pm \mathbf{e}_j$ with probability $1/(2d)$ for $j < d$, and to $x \pm \mathbf{e}_d$ with probability $(x_d \pm 1)/(2dx_d)$. Show that, on every finite path in $\{x_d \geq 1\}$, simple random walk conditioned to reach height R before height 0 agrees with this h-transform for all large integers R.

Exercise 3.14. Let $f : \mathbb{Z}^d \to \mathbb{R}$ be a function. Show that for every real-valued function F defined on $\{x \in \mathbb{Z}^d : x_1 \in \{0, 1\}\}$, there is a unique extension of F to all of $\mathbb{Z}^d$ satisfying $\Delta^{::} F = f$.

Exercise 3.15. Let $0 \le k < d$. Use Lemma 3.6 to show that simple random walk in $\mathbb{Z}^d$ visits $\{x : x_{k+1} = \cdots = x_d = 0\}$ infinitely often almost surely if $d - k \le 2$, and finitely often almost surely if $d - k \ge 3$.

Exercise 3.16. Prove that the sum in Lemma 3.7 satisfies, as $n \to \infty$,

$$\sum_{j=1}^{n} \frac{1}{2j-1} = \frac{1}{2}\log n + \log 2 + \frac{\gamma_E}{2} + O(n^{-2}),$$

with Euler's constant γ_E.

Hint: Subtract half the nth harmonic number from the $2n$th.

Exercise 3.17. Give a second proof of Lemma 3.7 by reflection. Let $n \ge 1$ be an integer, let U and V be the independent one-dimensional walks of (3.10), let $b := 2n - 1$, and let τ_b be the first time U reaches b.

(a) Show that $\mathbb{P}\{U_{2m} = b - 1\} - \mathbb{P}\{U_{2m} = b + 1\} = 2\mathbb{P}\{\tau_b = 2m + 1\}$ for every $m \ge 0$.

(b) Show that $\sum_{m \ge 0} \mathbb{P}\{V_{2m} = 0\}\mathbb{P}\{\tau_b = 2m + 1\} = 2/(\pi b)$, and deduce that $a(n, n) - a(n - 1, n - 1) = 1/(\pi b)$.

Hint for (b): Use the identity $\mathbb{P}\{V_{2m} = 0\} = \frac{2}{\pi}\int_0^{\pi/2} \cos^{2m} t \, \mathrm{d}t$.

Exercise 3.18. Let $A \subset \mathbb{Z}^d$ be finite and let $x \in A$, and consider the harmonic measure seen from x, extended by zero off ∂A,

$$\mu_x(z) := \begin{cases} \mathbb{P}_x\{X_{\tau_{\partial A}} = z\}, & z \in \partial A, \\ 0, & z \notin \partial A\,. \end{cases}$$

Show that the killed Green function (3.18) satisfies

$$\Delta^{::} g_A^{::}(x, \cdot) = \mu_x - \delta_x\,.$$

Exercise 3.19. Let $A \subset \mathbb{Z}^d$ be finite, and let $x \in A$. Show that the number of visits to x before $\tau_{\partial A}$ by the walk from x is geometric on $\{1, 2, \ldots\}$ with mean $2d\, g_A^{::}(x, x)$, and independent of $X_{\tau_{\partial A}}$.

Exercise 3.20. Let $d \ge 2$, let $A \subset \mathbb{Z}^d$ be finite, and let $F : A \to \mathbb{R}$ and $b : \partial A \to \mathbb{R}$. Let $u : A \cup \partial A \to \mathbb{R}$ be the unique solution of the problem (3.3) from Section 3.1, so that $\Delta^{::} u = F$ in A and $u = b$ on ∂A.

(a) Show that u can be expressed in the form

$$u(x) = g(x) - \sum_{z \in A} g^{::}(x - z) F(z) ,$$

where g is discrete harmonic in A, and $g^{::}$ is the Green function (3.7).

(b) Show that u can be expressed in terms of the killed Green function $g^{::}_A$ of (3.18),

$$u(x) = h(x) - \sum_{z \in A} g^{::}_A(x, z) F(z) ,$$

where h solves the discrete Dirichlet problem in A with boundary values b.

(c) Give a probabilistic interpretation for the solution $u(x)$.

Exercise 3.21. For $d \geq 2$ and $z \in \mathbb{Z}^d$, write $h_z(x) := g^{::}(x - z)$, so that $\Delta^{::} h_z = -\delta_z$ by (3.7). For finitely supported $f : \mathbb{Z}^d \to \mathbb{R}$, write

$$(f, h_z) := \sum_{x \in \mathbb{Z}^d} f(x) h_z(x) .$$

Show that the numbers (f, h_z) for $z \in \mathbb{Z}^d$ determine f uniquely.

Exercise 3.22. For $d \geq 2$ and $z \in \mathbb{Z}^d$, write $h_z(x) := g^{::}(x - z)$ with $g^{::}$ from (3.7). For finitely supported $f : \mathbb{Z}^d \to \mathbb{R}$, write

$$(f, h_z) := \sum_{x \in \mathbb{Z}^d} f(x) h_z(x) .$$

Prove that the following are equivalent:

(a) There is a finitely supported function F such that $f = \Delta^{::} F$.

(b) The number (f, h_z) vanishes for all but finitely many $z \in \mathbb{Z}^d$.

Exercise 3.23. Let $G = (V, E)$ be a finite connected simple graph, and let Δ be the Laplacian (3.2) of G. Let $g : V \times V \to \mathbb{R}$ be the symmetric function with zero row sums and

$$\Delta_y \, g(x, y) = |V|^{-1} - \mathbf{1}\{x = y\} .$$

For distinct vertices $u, v \in V$, let $R(u, v) := g(u, u) + g(v, v) - 2g(u, v)$. Let $(X_t)_{t \geq 0}$ be simple random walk on G. For $w \in V$, write

$$\tau_w := \inf\{t \geq 0 : X_t = w\} \qquad \text{and} \qquad \tau_w^+ := \inf\{t \geq 1 : X_t = w\} .$$

Prove that

$$R(u,v) = \frac{1}{\deg(u)\,\mathbb{P}_u\{\tau_v < \tau_u^+\}}\,,$$

and deduce that

$$\deg(u)\mathbb{P}_u\{\tau_v < \tau_u^+\} = \deg(v)\mathbb{P}_v\{\tau_u < \tau_v^+\}\,.$$

Exercise 3.24. This exercise outlines a probabilistic proof for the first order term of the killed Green function of a disk in $\mathbb{Z}^2$ (Lemma 3.12(b)).

(a) For simple random walk $(X_n)_{n\geq 0}$ in $\mathbb{Z}^2$ started at $\mathbf{0}$, prove using Stirling's formula (or Wallis' integral $\binom{2k}{k}2^{-2k} = \frac{2}{\pi}\int_0^{\pi/2}\cos^{2k}t\,\mathrm{d}t$) that for $k \geq 1$,

$$\mathbb{P}_{\mathbf{0}}\{X_{2k} = \mathbf{0}\} = \frac{1}{\pi k} + O(k^{-2})\,.$$

Hint: Rotating by 45° makes the horizontal and vertical projections independent.

(b) Deduce that

$$\sum_{k=0}^{2n}\mathbb{P}_{\mathbf{0}}\{X_k = \mathbf{0}\} = \frac{1}{\pi}\log n + O(1)\,. \tag{3.38}$$

(c) Let

$$\tau_r := \inf\{n \geq 0 : |X_n| \geq r\}$$

be the time for simple random walk in $\mathbb{Z}^2$ to reach distance r. Show that $\log\tau_r$ is concentrated near $2\log r$, that is, for every $\epsilon > 0$,

$$\mathbb{P}_{\mathbf{0}}\{r^{2-\epsilon} \leq \tau_r \leq r^{2+\epsilon}\} \longrightarrow 1 \qquad \text{as } r \to \infty\,.$$

Conclude from (3.38) that the expected number of visits to $\mathbf{0}$ before time τ_r is $(2/\pi + o(1))\log r$. Since $g^{::}_{\mathrm{B}(\mathbf{0},r)}(\mathbf{0},\mathbf{0})$ is one fourth of this expectation by (3.18), we obtain

$$g^{::}_{\mathrm{B}(\mathbf{0},r)}(\mathbf{0},\mathbf{0}) = \Big(\frac{1}{2\pi} + o(1)\Big)\log r\,.$$

Compare the result with Lemma 3.12(b). Hint: reversing a path of length $2n$ from $\mathbf{0}$ to $\mathbf{0}$ turns an exit from the disk after time n into an exit before time n.

Exercise 3.25. For simple random walk in $\mathbb{Z}^2$, denote by $\tau_{\mathbf{0}}^+$ the first positive hitting time of zero. For $r \geq 2$, use Theorem 3.9 to prove that, as $r \to \infty$,

$$\mathbb{P}_{\mathbf{0}}\{\tau_{\partial \mathrm{B}^{::}(\mathbf{0},r)} < \tau_{\mathbf{0}}^+\} = \frac{\pi/2}{\gamma_E + \frac{3}{2}\log 2 + \log r} + O\left(\frac{1}{r \log^2 r}\right),$$

where γ_E is Euler's constant.

Exercise 3.26. For $d \geq 2$ and real $r \geq 2$, use Lemma 3.12 with $B := \mathrm{B}^{::}(\mathbf{0}, r)$ to prove that, for $x \in B \setminus \mathrm{B}^{::}(\mathbf{0}, r/2)$,

$$g_B^{::}(\mathbf{0}, x) \asymp \frac{r + 1 - |x|}{r^{d-1}},$$

where $\asymp$ means the ratio of the two sides is bounded above and below by positive constants (here the constants may depend on d).

Exercise 3.27. Let $d \geq 2$, let $R \geq 2$, and let $B := \mathrm{B}^{::}(\mathbf{0}, R)$. For $x \in B \setminus \{\mathbf{0}\}$, prove that

$$\mathbb{P}_x\{\tau_{\mathbf{0}} < \tau_{\partial B}\} = \frac{g_B^{::}(\mathbf{0}, x)}{g_B^{::}(\mathbf{0}, \mathbf{0})}.$$

Use Lemma 3.12 to estimate this probability explicitly in dimensions $d = 2$ and $d \geq 3$.

Exercise 3.28. Let $A \subset \mathbb{Z}^d$ be finite, let $z \in A$, and let $x, y \in A \setminus \{z\}$. Prove that

$$g_{A\setminus\{z\}}^{::}(x, y) = g_A^{::}(x, y) - \frac{g_A^{::}(x, z)\, g_A^{::}(z, y)}{g_A^{::}(z, z)},$$

and interpret the last term. Prove that, for $b \in \partial A$, the walk from x leaves $A \setminus \{z\}$ at b with probability $\mathbb{P}_x\{X_{\tau_{\partial A}} = b\} - \mathbb{P}_z\{X_{\tau_{\partial A}} = b\} g_A^{::}(x, z)/g_A^{::}(z, z)$.

Exercise 3.29. For $d \geq 3$, use Theorem 3.9 to prove that for every sufficiently small $\ell > 0$ there is a radius $r > 0$ such that the level set of the Green function

$$L_\ell := \{x \in \mathbb{Z}^d : g^{::}(x) \geq \ell\}$$

satisfies

$$\mathrm{B}^{::}(\mathbf{0}, r - c/r) \subset L_\ell \subset \overline{\mathrm{B}}(\mathbf{0}, r + c/r),$$

where c is a constant depending only on d.

Exercise 3.30. For simple random walk $(X_n)_{n\geq 0}$ in $\mathbb{Z}^2$, let $z \in \mathbb{Z}^2$ satisfy $z \cdot \mathbf{e}_2 > 0$, and let

$$\tau := \inf\{n \geq 0 : X_n \cdot \mathbf{e}_2 = 0\},$$

and write $p(z) := \mathbb{P}_z\{X_\tau \cdot \mathbf{e}_1 = 0\}$.

(a) Find a formula for $p(z)$ in terms of the potential kernel (3.12).

(b) What is $p(0,1)$?

(c) Let $K \subset \mathbb{R} \times (0,\infty)$ be compact. Show that, uniformly for $(x,y) \in K \cap n^{-1}\mathbb{Z}^2$,

$$\mathbb{P}_{(0,ny)}\{X_\tau = (nx,0)\} = \frac{y}{\pi n(x^2+y^2)} + O_K(n^{-2})\,.$$

(d) Let w lie in the upper half-plane, let $w_n \in n^{-1}\mathbb{Z}^2$ converge to w, and let $[t_1,t_2]$ be an interval on the real axis. Show that the probability that simple random walk on $n^{-1}\mathbb{Z}^2$, started at w_n and stopped on first hitting the real axis, lands in $[t_1,t_2]$ converges to $\angle t_1 w t_2/\pi$.

Hint for (a): Consider the function given by p on the upper half-plane, zero on the horizontal axis, and $-p$ on the lower half-plane. What is its discrete Laplacian?

Exercise 3.31*. Show that (3.5), with h the potential kernel a, defines a walk on $\mathbb{Z}^2 \setminus \{\mathbf{0}\}$, that its expected number of visits to its starting point x is $8a(x)$, and that it is transient. Show that, on every finite path avoiding $\mathbf{0}$, it is the limit as $R \to \infty$ of simple random walk conditioned to leave $\mathrm{B}_R^{::}$ before hitting $\mathbf{0}$.

Exercise 3.32. Let $d \geq 3$ and $\alpha > 0$. Use Theorem 3.9 to show that

$$\sum_{x \in \mathbb{Z}^d} g^{::}(x)^\alpha = \infty \quad \text{if and only if} \quad \alpha \leq \frac{d}{d-2}\,.$$

Conclude that $\sum_x g^{::}(x)^\alpha = \infty$ on $\mathbb{Z}^5$ when $\alpha \leq 5/3$ and on $\mathbb{Z}^6$ when $\alpha \leq 3/2$.

Exercise 3.33*. Let $d \geq 2$. Extend Exercise 3.11 by showing that there are $C, c > 0$, depending only on d, such that every finite nonempty $A \subset \mathbb{Z}^d$ satisfies

$$\mathbb{E}_x \tau_{\partial A} \leq C|A|^{2/d} \qquad \text{and} \qquad \mathbb{P}_x\{\tau_{\partial A} > t\} \leq 2e^{-ct/|A|^{2/d}}$$

for $x \in A$ and $t \geq 0$, and that balls attain this scale.

Hint: For $d \geq 3$, use Theorem 3.9; for $d = 2$, use (3.10).

Exercise 3.34. Let $f \in \mathbb{R}[x_1,\dots,x_d]$ be a linear combination of monomials of the form $x_1^{n_1}\cdots x_d^{n_d}$ with $0 \leq n_i \leq 3$ for all i. Prove that $\Delta^{::} f = \Delta f$ (compare Lemma 3.13(b)).

Exercise 3.35. Let $d \geq 2$ and $\epsilon > 0$. Show that there is $R_0 = R_0(d,\epsilon) < \infty$ such that the following holds. Let $z \in \mathbb{Z}^d$, let $R \geq R_0$, and let

$h : \mathrm{B}^{::}(z,R) \to \mathbb{R}_{\geq 0}$ be discrete harmonic at every lattice point whose nearest neighbors all lie in $\mathrm{B}^{::}(z,R)$. Then

$$\sup_{\mathrm{B}^{::}(z,R/2)} h \leq (3+\epsilon)^d \inf_{\mathrm{B}^{::}(z,R/2)} h\,.$$

Hint: In the proof of Lemma 3.16, replace $27/32$, $7R/8$ and $15R/16$ by t, sR and ρR, respectively, where $1/2 < t < s < \rho < 1$ and

$$\frac{t+1/2}{t-1/2} < 3+\epsilon\,.$$

Exercise 3.36*. Let $d \geq 2$, let $\Omega \subset \mathbb{R}^d$ be open, and let $\delta_n \downarrow 0$. Let $h_n : \Omega \cap \delta_n \mathbb{Z}^d \to \mathbb{R}$ have vanishing discrete Laplacian $\Delta^{::}_{\delta_n} h_n$ on the set $\Omega^+_{\delta_n}$ of (3.21), and be bounded on each compact set uniformly in n. Show that a subsequence of their multilinear interpolations converges uniformly on compact subsets of Ω to a harmonic function.

Hint: Use (3.30), the Arzelà–Ascoli theorem and Theorem 2.29.

Exercise 3.37. Let Ξ be the map of Lemma 3.24. Using that lemma and Lemma 3.25, prove that 1 and the polynomials

$$\Xi(\mathrm{Re}\,(x+\mathbf{i}y)^k) \qquad \text{and} \qquad \Xi(\mathrm{Im}\,(x+\mathbf{i}y)^k) \qquad \text{for } k \geq 1\,,$$

form a basis for the discrete harmonic polynomials on $\mathbb{Z}^2$.

Exercise 3.38. Prove that if $f : \mathbb{Z}^d \to \mathbb{R}$ is periodic (that is, there is a positive integer m such that $f(x+my) = f(x)$ for all $x, y \in \mathbb{Z}^d$) with average value zero, that is,

$$\sum_{x \in \{1,\dots,m\}^d} f(x) = 0\,,$$

then there is a periodic function F such that $f = \Delta^{::} F$.

Exercise 3.39. Prove that if $f : \mathbb{Z}^d \to \mathbb{R}$ is periodic (in the sense of Exercise 3.38), then there are a constant $c \in \mathbb{R}$ and a periodic function F such that

$$f = \Delta^{::}(F + c|x|^2)\,.$$

Exercise 3.40. Let $d \geq 2$. Prove that if $u : \mathbb{Z}^d \to \mathbb{R}$ has polynomial growth and $\Delta^{::} u$ is periodic (in the sense of Exercise 3.38), then there are a periodic function F and a polynomial P such that $u = F + P$. Moreover, P can be expressed in the form

$$P(x) = c|x|^2 + h(x)\,,$$

where h is a discrete harmonic polynomial and $c \in \mathbb{R}$.

Exercise 3.41. Let $d \geq 2$. Using Lemma 3.25, prove that if $f : \mathbb{Z}^d \to \mathbb{R}$ has polynomial growth and $\Delta^{::} f$ is a polynomial, then f is a polynomial.

Exercise 3.42. Let $d \geq 2$.

(a) Use Theorem 3.18 to show that $g^{::}$ is the unique function $u : \mathbb{Z}^d \to \mathbb{R}$ of sublinear growth (that is, $u(x)/|x| \to 0$ as $|x| \to \infty$) satisfying $\Delta^{::} u = -\delta_{\mathbf{0}}$, together with the normalization $u(x) \to 0$ as $|x| \to \infty$ when $d \geq 3$ and $u(\mathbf{0}) = 0$ when $d = 2$.

(b) Let $K \subset \mathbb{Z}^d$ be finite and nonempty, and let $u : \mathbb{Z}^d \to \mathbb{R}$ satisfy $u(x)/|x| \to 0$ as $|x| \to \infty$ and $\Delta^{::} u = 0$ off K. Show that $u(x) = b - \sum_{y \in K} g^{::}(x - y) \Delta^{::} u(y)$ for a constant b, so that $u(x) \to b$ if $d \geq 3$, and $u(x) - sa(x) \to b$ if $d = 2$, where $s := \sum_{y \in K} \Delta^{::} u(y)$.

(c) In part (b), show that $u(x) = \mathbb{E}_x[u(X_{\tau_K}) \mathbf{1}\{\tau_K < \infty\}] + b \mathbb{P}_x\{\tau_K = \infty\}$ if $d \geq 3$, and $u(x) = s U_K(x) + \mathbb{E}_x[u(X_{\tau_K})]$ if $d = 2$, with the function U_K of Lemma 3.29(b), which vanishes on K and differs from a by a bounded function.

Exercise 3.43. Let $G = (V, E)$ be a connected recurrent locally finite graph with the Laplacian Δ of (3.2), and let $(X_n)_{n \geq 0}$ denote the simple random walk on G.

(a) Suppose $v \in V$ and $h : V \to \mathbb{R}$ is bounded and harmonic on $V \setminus \{v\}$. Show that h is constant.

(b) Suppose $K \subset V$ is finite and nonempty, and $h : V \to \mathbb{R}$ is bounded and harmonic on $V \setminus K$. Show that

$$\sum_{w \in K} \Delta h(w) = 0 \,.$$

Hint: Let $\tau_K^+ := \inf\{n \geq 1 : X_n \in K\}$ and show

$$\Delta h(w) = \deg(w) \big(\mathbb{E}_w[h(X_{\tau_K^+})] - h(w) \big) \,.$$

Then show

$$\deg(w) \, \mathbb{P}_w\{X_{\tau_K^+} = x\} = \deg(x) \, \mathbb{P}_x\{X_{\tau_K^+} = w\} \qquad \text{for all } w, x \in K \,.$$

Exercise 3.44. This exercise computes hitting probabilities of Section 3.7 for a set of two neighboring sites. Let $K = \{\mathbf{0}, \mathbf{e}_1\} \subset \mathbb{Z}^2$, and for $y \in \mathbb{Z}^2$ let

$$h(y) := \mathbb{P}_y\{X_{\tau_K} = \mathbf{0}\} \,.$$

Show that every neighbor of $2\mathbf{e}_1$ outside K has h-value strictly larger than $h(2\mathbf{e}_1)$:

$$h(2\mathbf{e}_1) < h(3\mathbf{e}_1) \qquad \text{and} \qquad h(2\mathbf{e}_1) < h(2\mathbf{e}_1 + \mathbf{e}_2) = h(2\mathbf{e}_1 - \mathbf{e}_2)\,.$$

The only neighbor of $2\mathbf{e}_1$ with smaller h-value is $\mathbf{e}_1 \in K$, so the restriction of h to $\mathbb{Z}^2 \setminus K$ has a strict local minimum at $2\mathbf{e}_1$.

Hint: The potential kernel satisfies $a(\mathbf{e}_1) = \frac{1}{4}$. Show that the function

$$g(y) := \frac{1}{2} + 2[a(y - \mathbf{e}_1) - a(y)]$$

is bounded, harmonic off K, and equal to h on K, and deduce $h = g$ from (3.35). Then use the harmonicity of a at $\mathbf{e}_1 + \mathbf{e}_2$ and at $2\mathbf{e}_1$, the value $a(\mathbf{e}_1 + \mathbf{e}_2) = \frac{1}{\pi}$ from Lemma 3.7, the value $a(2\mathbf{e}_1) = 1 - \frac{2}{\pi}$ that follows from it by the harmonicity of a at $\mathbf{e}_1$, and the bounds $3 < \pi < 16/5$.

Exercise 3.45. Let $A \subset B \subset \mathbb{Z}^2$ be finite and nonempty, and let U_A be as in Lemma 3.29(b). Prove that the value of the potential of $\mathcal{H}_B$ on B exceeds that of $\mathcal{H}_A$ on A by $\sum_{y \in B} \mathcal{H}_B(y) U_A(y) \geq 0$, which is $\mathcal{H}_B(z) U_A(z)$ if $B = A \cup \{z\}$.

Exercise 3.46*. This exercise identifies the analogue in dimensions $d \geq 3$ of the harmonic measure from infinity of Definition 3.28. Proposition 8.9 in Chapter 8 gives another proof of part (b), through the last-exit decomposition.

Let $d \geq 3$, let X be simple random walk on $\mathbb{Z}^d$, with $\mathbb{P}_y$ and the hitting times τ_E defined as in Section 3.7, let $K \subset \mathbb{Z}^d$ be finite and nonempty, and let

$$\tau_K^+ := \inf\{n \geq 1 : X_n \in K\}$$

denote the first return time to K.

(a) Show that $\lim_{|y| \to \infty} \mathbb{P}_y\{X_{\tau_K} = x\} = 0$ for every $x \in K$.

(b) Show that, for every $x \in K$,

$$\lim_{|y| \to \infty} \mathbb{P}_y\{X_{\tau_K} = x \mid \tau_K < \infty\} = \frac{\mathrm{Es}_K(x)}{\mathrm{cap}(K)}\,,$$

where $\mathrm{Es}_K(x) := \mathbb{P}_x\{\tau_K^+ = \infty\}$ is the escape probability from x and $\mathrm{cap}(K) := \sum_{x \in K} \mathrm{Es}_K(x)$ is the capacity of K.

Hint for (b): Reverse the paths from y that first hit K at x.

NOTES

For an introduction to martingales in discrete time, see (Williams, 1991, Chapter 10, pp. 93–101).

Lemma 3.6, first proved by Pólya (see (Spitzer, 1976, p. 52, Problems 4–5)), spawned the vast field of random walks. For a deeper look at simple random walk in $\mathbb{Z}^d$ (including the Green function, the potential kernel, harmonic measure and capacity), see Spitzer (1976), Chapters 1 and 2 of Lawler (1996) and Chapters 4 and 6 of Lawler and Limic (2010). For symmetric random walks more generally (finite-range, symmetric and irreducible), see (Lawler and Limic, 2010, § 1.1). The proof given here needs no Fourier analysis.

The planar case of Theorem 3.9, with the constant κ and the error $O(|x|^{-2})$, is due to Stöhr in 1950, as Kozma and Schreiber (2004) report on p. 2 and Fukai and Uchiyama (1996) in their Remark 2. For $d \geq 3$, Uchiyama obtained the asymptotic expansion of the Green function in 1998; see p. 2 of Kozma and Schreiber (2004). Their Theorem 1 expands, to arbitrary order, the function $x \mapsto \sum_{n\geq 0}(\mathbb{P}_{\mathbf{0}}\{X_n = \mathbf{0}\} - \mathbb{P}_{\mathbf{0}}\{X_n = x\})$ of every balanced bounded random walk on a lattice; for simple random walk this function is $2d(g^{::}(\mathbf{0}) - g^{::}(x))$ when $d \geq 3$ and $4a(x)$ when $d = 2$. Their Theorem 5 shows that for a reversible walk, apart from the additive constant, only inverse powers whose order has the parity of the dimension appear. For simple random walk this gives the errors $O(|x|^{-d})$ and $O(|x|^{-2})$ of Theorem 3.9. The potential kernel in $\mathbb{Z}^2$ has an asymptotic expansion which begins

$$a(z) = \frac{1}{2\pi}\log|z| + \kappa - \frac{\operatorname{Re}(z^4)}{24\pi|z|^6} + O(|z|^{-4}),$$

where we identify $(x, y) \in \mathbb{Z}^2$ with the Gaussian integer $z = x + \mathbf{i}y$, and κ is as in Theorem 3.9. For a detailed proof, see Theorem 2 and Remark 2 on p. 1980 of Fukai and Uchiyama (1996) (proved in their Section 4); for more terms in the expansion, see Theorem 3 of Kozma and Schreiber (2004). In Spitzer's unnormalized convention, the potential kernel is $4a = -4g^{::}$, and each of its values is an integer plus a rational multiple of $1/\pi$; for example, $-4g^{::}(1) = 1$ and $-4g^{::}(1 + \mathbf{i}) = 4/\pi$. McCrea and Whipple (1940) gave a procedure for computing these values (Sections 7–8); see also (Spitzer, 1976, §15).

Claim 2.1 of Nachmias and Peres (2026) identifies $g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(\cdot, y)$ as the unique nonnegative function u with $\Delta^{::}u = \delta_{\mathbf{0}} - \delta_y$ on $\mathbb{Z}^2$ that vanishes at the origin. Their Remark 3.3 expresses the potential kernel through the diagonal values $g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x, x)$. Lemma 3.10 is the corresponding off-diagonal identity in our normalization.

On a finite connected graph, the kernel of Δ is the space of constants and its range is the space of mean-zero functions, so $\Delta u = -\delta_x$ has no solution u. Projecting δ_x onto the mean-zero functions and inverting $-\Delta$ there gives the x-column of the **Moore–Penrose pseudoinverse**, which

annihilates constants. If $g(x, y)$ are the entries of the pseudoinverse, then the effective resistance between x and y is $g(x, x) + g(y, y) - 2g(x, y)$, a formula proved in Theorem A and Corollary A on pp. 87–88 of Klein and Randić (1993). Exercise 3.23 expresses the resistance through random-walk escape probabilities. The potential kernel in $\mathbb{Z}^2$ plays the analogous role, replacing the Green function, which is infinite by recurrence, and it still satisfies $\Delta^{::} a = \delta_{\mathbf{0}}$. Among the solutions u of $\Delta^{::} u = \delta_{\mathbf{0}}$ with $u(x)/|x| \to 0$ as $|x| \to \infty$, it is unique up to an additive constant, which the normalization $a(\mathbf{0}) = 0$ fixes; Exercise 3.42 asks for the proof, in the equivalent form for $g^{::} = -a$.

For more on coupling, see (Levin and Peres, 2017, Chapter 5) and the lectures of (Lindvall, 1992, Chapter II, Sections 12–18).

Theorem 3.18 is due to Heilbronn. Theorem 6 on p. 199 of Heilbronn (1949), proved on p. 206, shows that a discrete harmonic function of growth $O(|x|^n)$ is a polynomial of degree at most n. The little-o form in Theorem 3.18 follows because a polynomial of degree n cannot be $o(|x|^n)$. Theorem 3 on p. 196 of Heilbronn (1949) computes the dimension of the space of discrete harmonic polynomials.

Discrete harmonic polynomials on $\mathbb{Z}^d$ have been studied extensively, primarily in two variables: see for example Sections 3 and 5 of Duffin (1956), who gives a construction based on discrete contour integration. Another construction is due to Zeilberger (1977) (pp. 97–99). Sections 4.1 and 4.3 of the survey of Lovász (2004) describe discrete analytic polynomials whose real and imaginary parts are discrete harmonic on the even and odd diagonal sublattices. The map Ξ of Lemma 3.24 is defined in Section 2.2 of Jerison, Levine, and Sheffield (2014), whose modification of Lovász's construction makes $f - \Xi(f)$ have degree at most $k - 2$, rather than $k - 1$, when f has degree k. In their Section 2.3, the map Ξ was used in the proof that internal DLA fluctuations, tested against discrete harmonic polynomials, converge to the augmented Gaussian free field. Sections 2–5 of Smirnov (2010) survey discrete harmonic and discrete holomorphic functions as applied to recent research in probability.

Our proof of Theorem 3.9 corrects $\mathcal{G}$ by the term $-\frac{1}{24}\sum_i x_i \partial_i^3 \mathcal{G}$; Section 1 of Kozma and Schreiber (2004) sketches a related planar approach that corrects the logarithm on the lattice by repeated convolution. The discrete Harnack inequality (Lemma 3.16) with an unspecified constant is Theorem 1.7.2 of Lawler (1996); for another proof, see Theorem 6.3.9 of Lawler and Limic (2010).

Theorem 3.9 can also be proved by Fourier analysis. Let $\varphi(\theta)$ be the average of $\cos\theta_1, \ldots, \cos\theta_d$. Extracting Fourier coefficients from the generating

function of the transition probabilities gives, for $d \geq 3$,

$$g^{::}(x) = \frac{1}{2d}\left(\frac{1}{2\pi}\right)^d \int_{(-\pi,\pi)^d} \frac{e^{-\mathbf{i}x\cdot\theta}}{1-\varphi(\theta)}\,\mathrm{d}\theta\,,$$

and, for $d = 2$, the same formula for $-a(x)$ with $e^{-\mathbf{i}x\cdot\theta} - 1$ in place of $e^{-\mathbf{i}x\cdot\theta}$ and $\frac{1}{4}$ in place of $\frac{1}{2d}$. The numerator $e^{-\mathbf{i}x\cdot\theta}$ is a rapidly oscillating function of θ when $|x|$ is large, so the main contribution is near the singularity at $\theta = \mathbf{0}$, and the main term can be found by approximating the denominator $1 - \varphi(\theta)$ by $|\theta|^2/(2d)$. In the plane, P12.3 of Spitzer (1976), on pp. 124–127, concerns random walks whose steps generate $\mathbb{Z}^2$ and that have mean zero, a covariance matrix proportional to the identity and a finite moment of order $2+\delta$ for some $\delta > 0$, such as simple random walk. For them P12.3 shows that the potential kernel minus a multiple of $\log|x|$ converges as $|x| \to \infty$, by replacing the reciprocal of $1 - \varphi$, with φ now the characteristic function of a step, by the reciprocal of its quadratic part.

Spitzer proved the existence of the harmonic measure from infinity of a finite set in the plane, the first assertion of part (a) of Lemma 3.29: Theorem (B′) on p. 595 of Spitzer (1962) applies to every recurrent random walk on $\mathbb{Z}^2$ whose steps generate $\mathbb{Z}^2$, and on p. 603 the existence is derived from the convergence of $a(x - v) - a(x)$ to zero. Theorem 14.1 on pp. 141–142 of Spitzer (1976) expresses the limit through the inverse of the matrix of values $a(x - v)$ with x and v in the set, so that its potential is constant on the set, as in part (a). Other proofs are Theorem 2.1.3 of Lawler (1996), which adds an error estimate, and Proposition 6.6.1 with equation (6.47) of Lawler and Limic (2010).

The h-transform of Subsection 3.2.1 is the h-process of Section 11.1 of Lawler and Limic (2010), pp. 245–248. Their Proposition 11.1.1 on p. 246 sums (3.6) over paths, and on pp. 247–248 they consider the case $d = 2$ of Exercise 3.13 and describe the walk of Exercise 3.31 as a limit of conditioned walks; its transience is part (i) of their Exercise 11.2 on p. 258. Exercise 3.5 is their Exercise 6.8 on p. 163. Exercise 3.11 is their Proposition 2.4.5 on p. 49 for simple random walk, with explicit constants; their proof of the second bound uses the central limit theorem instead of Lemma 3.4. Exercise 3.42(c) is their Proposition 6.4.6 on p. 134 when $d \geq 3$, and extends their Proposition 6.4.8 on p. 135, which assumes $u = 0$ on K, when $d = 2$; they estimate exit distributions instead of using Theorem 3.18. Exercise 3.45 is their Lemma 6.6.6 on p. 149: four times the value of the potential of $\mathcal{H}_A$ on A is their capacity of A in two dimensions, defined on p. 146. Exercise 3.36 extends their Proposition 8.3.2 on p. 193, for functions bounded by one on a ball. Their Exercise 6.9 on p. 163 and Proposition 8.1.1 on pp. 181–182 give the formula of Exercise 3.30(a) by the method of images.

PART 2

Lattice Growth

Chapter Four

Divisible Sandpile in $\mathbb{Z}^d$: Single Source

In the divisible sandpile, each vertex x of a graph starts with a mass $\sigma(x) \in \mathbb{R}$. A positive value of $\sigma(x)$ indicates the amount of mass present at x, and a negative value indicates a "hole" that can be filled by mass. Each vertex x can topple if and only if $\sigma(x) > 1$, and toppling x means splitting the excess mass $\sigma(x) - 1$ equally among the neighbors of x. (This is the maple syrup on the waffle of Section 1.2.) We first develop the model on a general graph and then specialize to a point source on $\mathbb{Z}^d$ for $d \geq 2$.

In Section 4.1 we prove a least action principle (Theorem 4.3) characterizing the outcome without reference to the toppling sequence, which shows that when stabilization occurs, the choice of toppling sequence does not affect the outcome, provided every vertex is chosen infinitely often; each vertex emits the least amount of mass required in order to reach a globally stable configuration. For a point source on $\mathbb{Z}^d$, the odometer is also the gap between an obstacle and its least superharmonic majorant. This connects the model to the obstacle problem, the simplest free boundary problem in partial differential equations, in which the unknown is both a function and the region where it touches a prescribed obstacle.

In Section 4.2 we start with mass m at the origin of $\mathbb{Z}^d$ and show that the set of fully occupied sites is close to a Euclidean ball, with only an additive constant error in the radius, and that the odometer is determined up to a bounded additive error (Theorem 4.7). The idea is to compare the odometer with a quadratic function plus the discrete Green function, using the maximum principle and the asymptotics of the Green function, and to bound the outer radius by following a path along which the odometer increases. The estimates from the proof will prove useful in comparing other aggregation models, internal DLA in Chapter 5 and rotor-router aggregation in Chapter 6, to the divisible sandpile. The same methods give constraints on the shape of the abelian sandpile, and in Section 4.3 we find Euclidean balls inside and outside the set of sites that topple.

4.1 THE DIVISIBLE SANDPILE AND LEAST ACTION

Throughout this section, $G = (V, E)$ is undirected, locally finite, connected and simple, and has at least two vertices (so that every vertex has a neighbor).

In the **divisible sandpile**, each vertex v starts with a mass $\sigma(v) \in \mathbb{R}$. A positive value of $\sigma(v)$ indicates the amount of mass present at v, and a negative value indicates a "hole" that can be filled by mass. A vertex with mass exceeding 1 topples by splitting its excess mass equally among its neighbors. Our first task is to define precisely what it means to stabilize a configuration $\sigma : V \to \mathbb{R}$ by divisible sandpile topplings. We then prove a least action principle (Theorem 4.3) characterizing the outcome without reference to the toppling sequence, which shows that the choice of toppling sequence does not affect the outcome, provided every vertex topples infinitely often. We then show that a finitely supported configuration on an infinite graph stabilizes, with only finitely many vertices ever emitting mass, and relate the odometer to an obstacle problem.

Let $a_+ := \max\{a, 0\}$ denote the positive part of a real number a. The act of **toppling** a vertex v means removing mass $(\sigma(v) - 1)_+$ from v and dividing it equally among its neighbors. Recalling the graph Laplacian (3.2), toppling v changes σ to

$$\sigma + \frac{(\sigma(v) - 1)_+}{\deg(v)} \Delta\delta_v \,.$$

A vertex $v \in V$ is called **unstable** for σ if $\sigma(v) > 1$; otherwise it is **stable**. Note that toppling v leaves σ unchanged unless v is unstable for σ.

We do not require $\sigma \geq 0$.

We would like to perform topplings until all vertices are stable. Note, however, that if neighboring vertices $v \sim w$ satisfy $\sigma(v) > 1$ and $\sigma(w) \geq 1$, then toppling v makes w unstable, so a vertex may need to topple more than once. Let $v_1, v_2, \ldots$ be a sequence of vertices to be toppled, which we call a **toppling sequence**, set $\sigma_0 := \sigma$, and for $k \geq 1$ let σ_k be the configuration obtained from σ_{k-1} by toppling v_k, that is,

$$\sigma_k := \sigma_{k-1} + \frac{(\sigma_{k-1}(v_k) - 1)_+}{\deg(v_k)} \Delta\delta_{v_k} \,. \tag{4.1}$$

Since $(\sigma_{j-1}(v_j) - 1)_+$ is the amount of mass emitted at step j, let

$$u_k := \sum_{j=1}^{k} \frac{(\sigma_{j-1}(v_j) - 1)_+}{\deg(v_j)} \delta_{v_j} \,.$$

Then $u_k(v)$ is the total mass emitted from v to each of its neighbors during the first k topplings, and we call u_k the **odometer** after k topplings. For each

neighbor w of v, the total mass sent from w to v during the first k topplings is $u_k(w)$, so the net change in mass at v is

$$\sigma_k(v) - \sigma_0(v) = \Delta u_k(v) = \sum_{w \sim v} \big(u_k(w) - u_k(v)\big)\,.$$

Note that the increase of the odometer at each toppling depends only on the current odometer. Let

$$\zeta(x) := \frac{\sigma(x) - 1}{\deg(x)} \qquad \text{for } x \in V\,.$$

If $u : V \to [0, \infty)$ is the odometer so far (so that the current configuration is $\sigma + \Delta u$), then toppling a vertex x raises the odometer at x from $u(x)$ to

$$\max\Big\{u(x),\ \frac{1}{\deg(x)} \sum_{y \sim x} u(y) + \zeta(x)\Big\} \tag{4.2}$$

and leaves it unchanged elsewhere, since $((\sigma + \Delta u)(x) - 1)/\deg(x)$ is the second entry of the maximum minus $u(x)$, and $a + (b-a)_+ = \max\{a, b\}$ for real a and b. We will use two properties of (4.2), namely that it is nondecreasing in each value of u, and that it equals $u(x)$ if and only if $(\sigma + \Delta u)(x) \le 1$.

Since u_k is nondecreasing, $u_k \uparrow u_\infty$ for some function $u_\infty : V \to [0, \infty]$, which is called the odometer of σ for the toppling sequence $(v_k)_{k \ge 1}$. The simplest example is a point source $7\delta_{\mathbf{0}}$ on $\mathbb{Z}$, for which the final mass is 1 on each of the sites $-3, \dots, 3$, and the odometer takes the values 6, 3, 1 and 0 at distances 0, 1, 2 and 3 from the source.

The odometer is not always finite. A toppling sequence $(v_k)_{k \ge 1}$ is called **thorough** if every vertex occurs in it infinitely often.

Lemma 4.1 *Let $\sigma : V \to \mathbb{R}$ be a configuration, and let u_∞ be the odometer of σ for a thorough toppling sequence. Then either $u_\infty(v) < \infty$ for all $v \in V$, or $u_\infty \equiv \infty$ on V.*

Proof. If $u_\infty(v) = \infty$ for some v, then v emits an infinite amount of mass to each of its neighbors. Let w be a neighbor of v, and suppose for contradiction that $u_\infty(w) < \infty$. Then

$$\Delta u_k(w) = \sum_{y \sim w} \big(u_k(y) - u_k(w)\big) \xrightarrow[k \to \infty]{} \infty\,,$$

since the contribution from v tends to $+\infty$, while every other term is bounded below by $-u_\infty(w)$. Hence $\sigma_k(w) = \sigma(w) + \Delta u_k(w) \to \infty$. But w topples infinitely often, and its mass is at most 1 after each such toppling, a contradiction. Hence $u_\infty(w) = \infty$, and since G is connected, $u_\infty \equiv \infty$. □

Definition 4.2 We say that a configuration σ on V **stabilizes** (or is **stabilizable**) if there exists a thorough toppling sequence such that $u_\infty(v) < \infty$ for all $v \in V$; otherwise we say that σ **explodes**.

If σ stabilizes, then for a thorough toppling sequence with finite odometer, the final configuration after toppling is (by local finiteness)

$$\sigma_\infty := \lim_{k\to\infty} \sigma_k = \lim_{k\to\infty} (\sigma + \Delta u_k) = \sigma + \Delta u_\infty \,.$$

Since the toppling sequence is thorough, $\sigma_k(v) \le 1$ for infinitely many k, so $\sigma_\infty \le 1$. Stabilizing therefore means that starting from σ it is possible to reach a configuration that is everywhere at most 1 by toppling a finite amount of mass from each vertex. If σ does not stabilize, then by Lemma 4.1 each vertex emits an infinite amount of mass in every thorough toppling sequence, and in this case the limit defining σ_∞ may not exist.

Do the odometer u_∞ and the final configuration σ_∞ depend on the choice of thorough toppling sequence?

Theorem 4.3 (Least Action Principle)
Let $\sigma : V \to \mathbb{R}$ be a configuration, and consider the set of functions

$$\mathcal{F}_\sigma := \{f : V \to \mathbb{R} \mid f \ge 0 \text{ and } \sigma + \Delta f \le 1\}\,.$$

(i) If $f \in \mathcal{F}_\sigma$, then for every toppling sequence, the odometer u_∞ of σ satisfies

$$u_\infty(v) \le f(v) \qquad \text{for every } v \in V\,.$$

(ii) If σ stabilizes, then for every thorough toppling sequence, the odometer u_∞ of σ satisfies $u_\infty \in \mathcal{F}_\sigma$ and hence

$$u_\infty(v) = \min_{f\in\mathcal{F}_\sigma} f(v) \qquad \text{for every } v \in V\,.$$

(iii) The configuration σ stabilizes if and only if $\mathcal{F}_\sigma$ is nonempty.

Proof. (i) Given $f \in \mathcal{F}_\sigma$ and a toppling sequence $v_1, v_2, \ldots$, we will show by induction on k that $u_k \le f$ for all $k \ge 0$. The case $k = 0$ is immediate since $u_0 = 0$. For the inductive step, suppose $u_{k-1} \le f$. By (4.2), its monotonicity and the induction hypothesis,

$$u_k(v_k) \le \max\Big\{f(v_k),\ \frac{1}{\deg(v_k)} \sum_{y\sim v_k} f(y) + \zeta(v_k)\Big\} = f(v_k)\,,$$

where the equality follows from $(\sigma + \Delta f)(v_k) \le 1$. Since u_k and u_{k-1} differ only at v_k, it follows that $u_k \le f$. Since $u_k \uparrow u_\infty$ pointwise, we conclude that $u_\infty \le f$ (see Figure 4.1(a)).

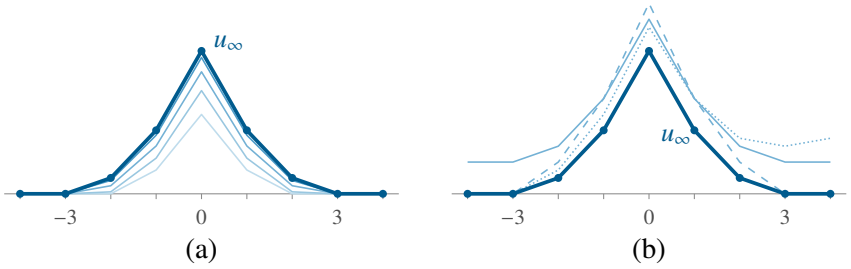


Figure 4.1: The least action principle on $\mathbb{Z}$ for $\sigma = 6\delta_{\mathbf{0}}$, whose odometer u_∞ (thick) takes the values 4.5, 2, 0.5 and 0 at distances 0, 1, 2 and 3 from the source. (a) The odometers after 1, 2, 4 and 8 sweeps of the toppling sequence $0, 1, -1, 2, -2, 3, -3$ (light to dark) increase to u_∞ from below; (b) three other members of $\mathcal{F}_\sigma$ lie above it, namely $u_\infty + 1$ (solid), the odometer of $7\delta_{\mathbf{0}}$ (dashed) and $u_\infty(x) + \frac{1}{4}(x+3)_+$ (dotted).

(ii) If the toppling sequence is thorough and its odometer is finite, then as shown after Definition 4.2, the final configuration $\sigma + \Delta u_\infty$ is at most 1, so $u_\infty \in \mathcal{F}_\sigma$.

Now consider an arbitrary thorough toppling sequence with odometer u_∞. Since σ stabilizes, there is a thorough toppling sequence whose odometer g is finite, and $g \in \mathcal{F}_\sigma$ as we have just seen. By part (i) with $f = g$, we have $u_\infty \le g$, so u_∞ is finite and hence belongs to $\mathcal{F}_\sigma$. By part (i) again, $u_\infty(v) \le f(v)$ for every $f \in \mathcal{F}_\sigma$, which proves the minimum formula.

(iii) If σ stabilizes, then by part (ii) the set $\mathcal{F}_\sigma$ is nonempty. Conversely, if $f \in \mathcal{F}_\sigma$, then by part (i) the odometer of every thorough toppling sequence is at most f, and a thorough toppling sequence exists since V is countable (being connected and locally finite), so σ stabilizes. □

An immediate consequence of part (ii) of the least action principle is the abelian property of the divisible sandpile: if σ stabilizes, then all thorough toppling sequences have the same odometer u_∞ and hence the same final configuration σ_∞.

The name "least action principle" comes from the following interpretation. Starting from σ, and sending mass $f(v) \ge 0$ from each vertex v to each of its neighbors (a total emission of $\deg(v)f(v)$) without requiring that each vertex emit only its current excess, the condition $\sigma + \Delta f \le 1$ says that the resulting configuration is stable. The odometer u_∞ of σ for a thorough toppling sequence is one such stabilizing function, and the least action principle says that it is (pointwise) minimal among all such functions (see Figure 4.1(b)). Thus, each vertex in the divisible sandpile emits the least amount of mass required in order to reach a globally stable configuration.

For finitely supported σ, we will need to know that only finitely many vertices emit mass, and that every vertex that emits mass ends with mass one. The second fact holds because a vertex's mass can never fall below 1 after it

emits mass, and counting such vertices gives the first.

Lemma 4.4 *Let $\sigma : V \to \mathbb{R}$ be a configuration, and assume that V is infinite. Then the following statements hold.*

(i) If σ is stabilizable with odometer u_∞, then the final configuration $\sigma_\infty = \sigma + \Delta u_\infty$ satisfies $\{u_\infty > 0\} \subseteq \{\sigma_\infty = 1\}$.

(ii) If σ has finite support, then σ is stabilizable and its odometer has finite support.

Proof. Fix a thorough toppling sequence $v_1, v_2, \ldots$, and let $\sigma_k = \sigma + \Delta u_k$ be its configurations. Once a vertex emits positive mass, its mass stays at least 1, since it has mass exactly 1 after that toppling and after every later toppling of it, and between these topplings it only receives mass (as at the origin in Figure 1.4). Thus

$$\sigma_j(v) \geq 1 \qquad \text{whenever } u_k(v) > 0 \text{ and } j \geq k\,. \tag{4.3}$$

(i) Suppose σ is stabilizable. By Theorem 4.3(ii), the odometer of the fixed thorough toppling sequence is finite, so $u_k \uparrow u_\infty < \infty$ and $\sigma_k \to \sigma_\infty$. If $u_\infty(v) > 0$, then there is k with $u_k(v) > 0$, so by (4.3) we have $\sigma_j(v) \geq 1$ for every $j \geq k$. Letting $j \to \infty$, we obtain $\sigma_\infty(v) \geq 1$, and since $\sigma_\infty \leq 1$, it follows that $\sigma_\infty(v) = 1$.

(ii) Now suppose σ has finite support, and let $M := \sum_{v\in V} \sigma(v)_+$, which is finite. We divide the proof into three steps: the total positive mass never exceeds M, only finitely many vertices ever emit mass, and the odometer is finite.

Step 1. We start by showing that toppling never increases the total positive mass. Indeed, when a vertex v emits mass $a > 0$, its positive part decreases by a, while each neighbor receives $a/\deg(v)$; since $(b + c)_+ \leq b_+ + c$ for $c \geq 0$, the positive parts at the neighbors increase by at most a in total. Hence $\sum_{v\in V} \sigma_k(v)_+ \leq M$ for every k.

Step 2. Next we show that only finitely many vertices ever emit mass. Let $A_k := \{u_k > 0\}$ be the set of vertices that have emitted positive mass by time k. By (4.3), every vertex of A_k contributes at least 1 to the total positive mass, so

$$|A_k| \leq \sum_{v\in V} \sigma_k(v)_+ \leq M \qquad \text{for every } k\,.$$

Since the sets A_k increase with k and have uniformly bounded cardinality, their union $A := \bigcup_{k\geq 0} A_k$ is finite.

Step 3. It remains to show that u_∞ is finite. If u_∞ were infinite at one vertex, then $u_\infty \equiv \infty$ by Lemma 4.1, so $A = V$, contradicting the assumption that V

is infinite. Therefore u_∞ is finite on V, so σ is stabilizable, and $\operatorname{supp}(u_\infty) = A$ is finite. □

If h is harmonic at v, then toppling v leaves the total "weight" $\sum_w \sigma(w)h(w)$ unchanged, and by summation by parts the same holds for stabilization.

Lemma 4.5 *Let $\sigma : V \to \mathbb{R}$ be a finitely supported stabilizable configuration with odometer u_∞ and final configuration σ_∞, and assume that u_∞ has finite support. Let $h : V \to \mathbb{R}$ be harmonic on $\{u_\infty > 0\}$. Then*

$$\sum_{v \in V} (\sigma_\infty(v) - \sigma(v))h(v) = 0\,.$$

Proof. Since the support of u_∞ is finite, so is that of Δu_∞ (by local finiteness), and summation by parts gives

$$\sum_{v \in V} (\Delta u_\infty)(v)h(v) = \sum_{v \in V} u_\infty(v)\Delta h(v) = 0\,.$$

The last equality follows because $u_\infty = 0$ outside $\{u_\infty > 0\}$ and $\Delta h = 0$ on $\{u_\infty > 0\}$. Since $\sigma_\infty - \sigma = \Delta u_\infty$, this proves the identity. □

In particular, in the setting of the lemma, if $\sigma = m\delta_o$ for a vertex $o \in V$ and a real number $m > 0$, then $\sum_{z \in V} \sigma(z)h(z) = mh(o)$, so the lemma gives

$$\frac{1}{m} \sum_{z \in V} \sigma_\infty(z)h(z) = h(o)\,. \tag{4.4}$$

For finite graphs the support hypothesis on the odometer is automatic; for infinite graphs it follows from Lemma 4.4(ii).

Next we translate the least action principle into an obstacle problem. Recall that by Theorem 4.3(ii), the odometer of a stabilizable configuration σ is the pointwise minimum of the family $\mathcal{F}_\sigma$. Given a function $\gamma : V \to \mathbb{R}$, which we will call the **obstacle**, consider the family

$$\mathcal{S}_\gamma := \{\varphi : V \to \mathbb{R} \mid \varphi \geq \gamma \text{ and } \Delta\varphi \leq 0\}\,.$$

The **least superharmonic majorant** of γ is

$$s(v) := \inf_{\varphi \in \mathcal{S}_\gamma} \varphi(v)\,. \tag{4.5}$$

We adopt the convention that $s \equiv \infty$ if $\mathcal{S}_\gamma$ is empty. As the next lemma shows, subtracting γ maps $\mathcal{S}_\gamma$ onto $\mathcal{F}_\sigma$.

Lemma 4.6 *Let $\sigma, \gamma : V \to \mathbb{R}$ satisfy $\Delta\gamma = \sigma - 1$, and let*

$$\mathcal{F}_\sigma := \{f : V \to \mathbb{R} \mid f \geq 0 \text{ and } \sigma + \Delta f \leq 1\}\,,$$
$$\mathcal{S}_\gamma := \{\varphi : V \to \mathbb{R} \mid \varphi \geq \gamma \text{ and } \Delta\varphi \leq 0\}\,.$$

Then

$$\mathcal{F}_\sigma = \mathcal{S}_\gamma - \gamma\,.$$

Proof. We simply observe that $\varphi \in \mathcal{S}_\gamma$ if and only if $f := \varphi - \gamma \in \mathcal{F}_\sigma$. Indeed, $\varphi \geq \gamma$ if and only if $f \geq 0$, and $\Delta\varphi \leq 0$ if and only if $\Delta f \leq 1 - \sigma$. □

In particular, if $\Delta\gamma = \sigma - 1$ and σ stabilizes, then since the odometer u_∞ of σ is the pointwise minimum of $\mathcal{F}_\sigma$ (Theorem 4.3(ii)), it follows from Lemma 4.6 that

$$u_\infty = s - \gamma\,,$$

where s is the least superharmonic majorant (4.5) of γ. In other words, the odometer is the difference between γ and its least superharmonic majorant (see Figure 4.2 for an example on $\mathbb{Z}$), so finding it is an obstacle problem. In the next section we study this problem for a point source $\sigma = m\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$ with $d \geq 2$, taking for γ the sum of a negative quadratic function and a multiple of the Green function of Chapter 3, and show that the sites that end with mass one fill a Euclidean ball up to an error of at most a constant in the radius.

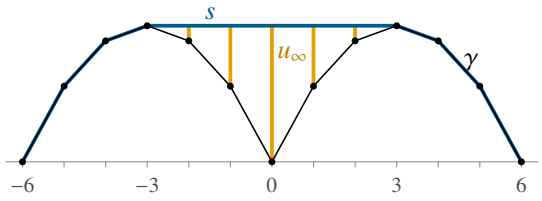


Figure 4.2: The obstacle $\gamma(x) = 3|x| - x^2/2$ on $\mathbb{Z}$ (black), which satisfies $\Delta\gamma = \sigma - 1$ for $\sigma = 6\delta_{\mathbf{0}}$, and its least superharmonic majorant s (blue), which on $\mathbb{Z}$ is its least concave majorant. The gaps $s - \gamma$ (orange) are the odometer u_∞ of Figure 4.1, with values 4.5, 2, 0.5 and 0 at distances 0, 1, 2 and 3 from the source.

4.2 SPHERICAL ASYMPTOTICS FOR A POINT SOURCE

The results of the previous section applied to arbitrary initial configurations. We now turn to a very specific initial condition, the point source $m\delta_{\mathbf{0}}$ of mass $m > 0$ at the origin of $\mathbb{Z}^d$. Divisible sandpile topplings will spread this mass over at least m different sites of the lattice, so where does the mass end up? See Figure 4.3 for several values of m in $\mathbb{Z}^2$, and Figure 4.4 for one in $\mathbb{Z}^3$.

Since $m\delta_{\mathbf{0}}$ has finite support, the divisible sandpile started from it stabilizes, by Lemma 4.4(ii). The next theorem shows that the set of fully occupied sites is close to a Euclidean ball, with only an additive constant error in the radius. Recall the abbreviation $\mathrm{B}^{::}_r = \mathrm{B}^{::}(\mathbf{0}, r)$ from Section 3.1 (it is empty when $r \leq 0$), the discrete Laplacian $\Delta^{::}$ of $\mathbb{Z}^d$ from (3.1), and the Green function

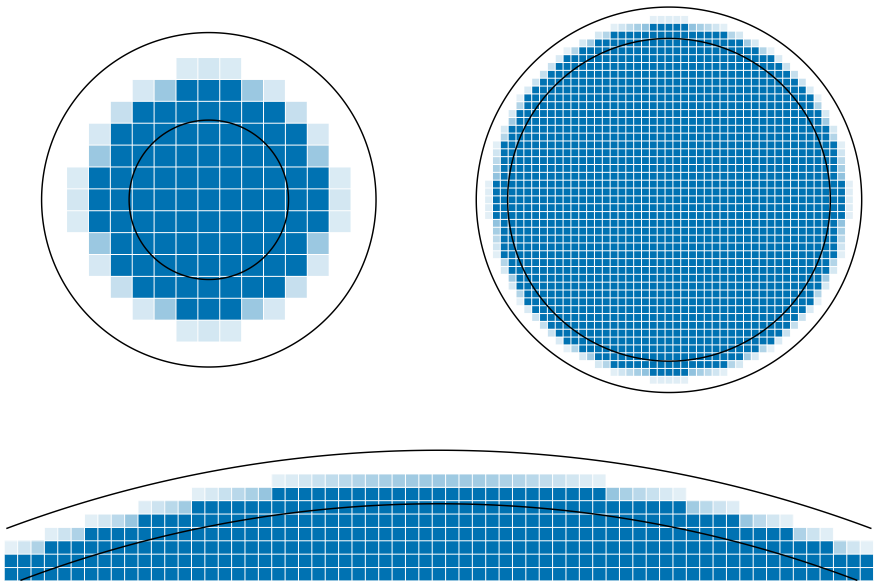


Figure 4.3: Top: Point source divisible sandpiles in $\mathbb{Z}^2$ of mass $m = 100$ and $m = 1600$. Bottom: Part of the boundary of the divisible sandpile of mass $m = 25{,}600$. Dark squares contain mass 1 and light squares contain mass strictly between 0 and 1, darker for larger mass. In each case the black circles are centered at the origin with radii $r \pm 2$, where $r = \sqrt{m/\pi}$.

$\mathcal{G}$ of $\mathbb{R}^d$ from (2.14). As usual, we write $g^{::}$ for the Green function (3.16) of $\mathbb{Z}^d$ when $d \geq 3$, or minus the potential kernel (3.17) when $d = 2$, so that $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ in all dimensions $d \geq 2$. By (3.16) and (3.17),

$$g^{::}(x) = \begin{cases} \mathcal{G}(x) - \kappa + O(|x|^{-2}), & d = 2, \\ \mathcal{G}(x) + O(|x|^{-d}), & d \geq 3, \end{cases} \qquad \text{for } x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}, \tag{4.6}$$

where κ is the constant in the potential kernel asymptotics of Theorem 3.9.

Theorem 4.7 *Let $d \geq 2$ and $m > 0$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Let $\sigma_\infty : \mathbb{Z}^d \to [0, 1]$ and $u_\infty : \mathbb{Z}^d \to [0, \infty)$ be the final configuration and the odometer of the divisible sandpile in $\mathbb{Z}^d$ with starting configuration $m\delta_{\mathbf{0}}$, and let $D_m := \{x \in \mathbb{Z}^d : \sigma_\infty(x) = 1\}$ be the set of fully occupied sites. Then there is a constant $c > 0$ depending only on d such that the following hold.*

(i) The odometer and the set D_m satisfy

$$\mathrm{B}^{::}_{r-c} \subset \{u_\infty > 0\} \subset D_m \subset \mathrm{B}^{::}_{r+c}\,.$$

Figure 4.4: The divisible sandpile in $\mathbb{Z}^3$ started from mass $m = 10^6$ at the origin, with the octant $\{x_1, x_2, x_3 > 0\}$ removed. Full sites are blue, and the partially filled sites, which form a layer one site thick, are red. On the cut faces, alternating shades mark eight bands of values of the odometer, which are nearly spherical shells.

(ii) For every $x \in \mathrm{B}^{::}_{r+c}$, the odometer satisfies

$$u_\infty(x) = mg^{::}(x) - m\mathcal{G}(r\mathbf{e}_1) + \frac{|x|^2 - r^2}{2d} + \begin{cases} m\kappa, & d = 2, \\ 0, & d \geq 3, \end{cases} + O(1)\,, \quad (4.7)$$

where the constant in the $O(1)$ depends only on d.

If $x \neq \mathbf{0}$ receives mass, then some neighbor of x must have emitted mass: $\sigma_\infty(x) > 0$ implies $u_\infty(y) > 0$ for some $y \sim x$, which in turn implies $\sigma_\infty(y) = 1$ by Lemma 4.4(i). So all mass in the final configuration σ_∞ is contained in the set $\{\mathbf{0}\} \cup D_m \cup \partial D_m$, where ∂D_m is the outer boundary of D_m.

The idea of the proof is to compare $u_\infty + \gamma$ with a constant, for an obstacle γ with discrete Laplacian $\Delta^{::}\gamma = m\delta_{\mathbf{0}} - 1$. By mass balance, $u_\infty + \gamma$ is then superharmonic everywhere and harmonic wherever the odometer is positive. Recalling that $|x|^2$ and the discrete Green function $g^{::}$ have discrete Laplacians $2d$ and $-\delta_{\mathbf{0}}$, respectively, a natural choice of obstacle is

$$\gamma(x) := -\frac{|x|^2}{2d} - mg^{::}(x) \qquad \text{for } x \in \mathbb{Z}^d\,. \quad (4.8)$$

We will use two properties of γ: away from the origin it is nearly spherically symmetric (because $g^{::}$ is), and it is maximized near $|x| = r$. For the first

property, by (4.6), replacing $g^{::}$ by $\mathcal{G}$ when $d \geq 3$, or by $\mathcal{G} - \kappa$ when $d = 2$, changes $\gamma(x)$ by $O(m|x|^{-d})$ for $x \neq \mathbf{0}$, which is $O(1)$ when $|x| \geq r/4$, since $m = \omega_d r^d$. The function obtained by this replacement is radial, and its value on the sphere of radius r is

$$\gamma_r := \begin{cases} -r^2/4 - m(\mathcal{G}(r\mathbf{e}_1) - \kappa), & d = 2, \\ -r^2/(2d) - m\mathcal{G}(r\mathbf{e}_1), & d \geq 3\,. \end{cases} \tag{4.9}$$

The claim of (4.7) is that $u_\infty(x)$ is within an additive constant of $\gamma_r - \gamma(x)$, and we will prove this by bounding $u_\infty + \gamma$ above and below by constants close to γ_r. By Lemma 4.6 and Theorem 4.3, $u_\infty = s - \gamma$, where s is the least superharmonic majorant (4.5) of γ (see Figure 4.5 for the continuum version), so we are showing that s is nearly the constant γ_r on $\mathrm{B}_r^{::}$. The estimates below do not use this identity; Exercise 4.5 returns to it on a general graph.

To estimate $\gamma_r - \gamma$ in all dimensions $d \geq 2$, let

$$\psi(t) := \frac{t^2}{2d} + m\,\mathcal{G}(t\mathbf{e}_1) \qquad \text{for } t > 0\,.$$

Although $\mathcal{G}$ is logarithmic when $d = 2$ and a power when $d \geq 3$, differentiating $\mathcal{G}(t\mathbf{e}_1)$ in t gives the single formula $-1/(d\omega_d t^{d-1})$ in both cases. Since $m = \omega_d r^d$, it follows that

$$\psi'(t) = \frac{t}{d}\left[1 - \left(\frac{r}{t}\right)^d\right] \qquad \text{and} \qquad \psi''(t) = \frac{1}{d}\left[1 + (d-1)\left(\frac{r}{t}\right)^d\right].$$

Thus ψ is convex and has its minimum at r.

Comparing (4.8) with (4.9) and using (4.6) to replace $g^{::}$ by $\mathcal{G}$, we obtain

$$\gamma_r - \gamma(x) = \psi(|x|) - \psi(r) + O(m|x|^{-d}) \qquad \text{for } x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}\,.$$

(In dimension two the constant κm occurs in both γ_r and $\gamma(x)$ and cancels.) For $|x| > r/4$ the error term is at most a constant depending only on d, because $m = \omega_d r^d$. Since $\psi'(r) = 0$, the Taylor expansion of ψ about r gives, for some ξ between $|x|$ and r,

$$\gamma_r - \gamma(x) = \frac{1}{2d}\left[1 + (d-1)\left(\frac{r}{\xi}\right)^d\right](r - |x|)^2 + O(1) \qquad \text{for } |x| > r/4\,. \tag{4.10}$$

When $|x| = r$ we take $\xi = r$. We will need the following bounds on the obstacle.

Lemma 4.8 *Let $d \geq 2$ and $m > 0$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Let $\gamma : \mathbb{Z}^d \to \mathbb{R}$ be given by*

$$\gamma(x) := -\frac{|x|^2}{2d} - m g^{::}(x) \qquad \textit{for } x \in \mathbb{Z}^d\,.$$

Let

$$\gamma_r := \begin{cases} -r^2/4 - m(\mathcal{G}(r\mathbf{e}_1) - \kappa), & d = 2, \\ -r^2/(2d) - m\mathcal{G}(r\mathbf{e}_1), & d \geq 3\,. \end{cases}$$

Then there is a constant $C = C(d) < \infty$ with the following properties.

(i) For every $x \in \mathbb{Z}^d$ with $|x| > r/4$,

$$\gamma_r - \gamma(x) \geq \frac{(r - |x|)^2}{2d} - C\,.$$

(ii) For every $b > 0$, the absolute value of $\gamma_r - \gamma(x)$ is bounded by a constant depending only on b and d when $x \in \mathrm{B}^{::}_{r+b} \setminus \mathrm{B}^{::}_{r-b}$.

(iii) For every $x \in \mathbb{Z}^d$,

$$\gamma_r - \gamma(x) \geq \frac{(r - |x|)^2}{8d} - C\,.$$

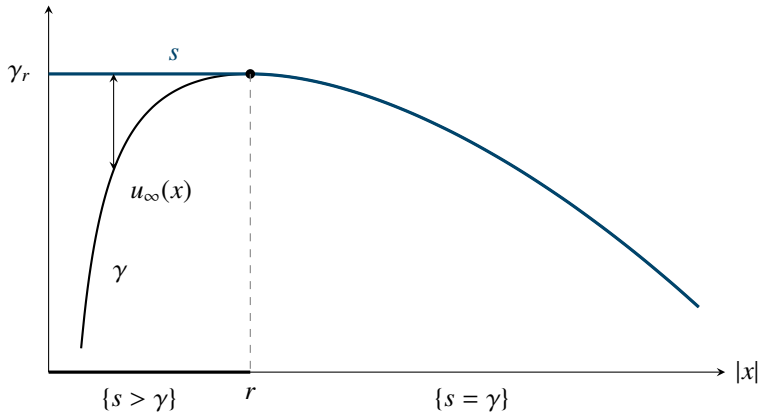


Figure 4.5: Cross-sections in $d = 3$ of the obstacle (4.8) with $g^{::}$ replaced by $\mathcal{G}$ (black) and of its least superharmonic majorant (blue), both shifted by the same constant. The majorant equals γ_r for $|x| \leq r$ and the obstacle for $|x| \geq r$, and the vertical difference of the two curves is the continuum analogue of $u_\infty = s - \gamma$.

Proof. (i) This follows directly from (4.10), after increasing C.

(ii) If $r \geq 2b$, then every point of $\mathrm{B}^{::}_{r+b} \setminus \mathrm{B}^{::}_{r-b}$ has norm at least $r - b \geq r/2$, so part (ii) follows from (4.10). If $r < 2b$, then $m < \omega_d(2b)^d$, the annulus lies in $\mathrm{B}^{::}_{3b}$, and for $x \in \mathrm{B}^{::}_{3b}$ we have

$$\gamma_r - \gamma(x) = \frac{|x|^2 - r^2}{2d} + mg^{::}(x) - m\mathcal{G}(r\mathbf{e}_1) + \begin{cases} m\kappa, & d = 2, \\ 0, & d \geq 3\,. \end{cases}$$

Each term is bounded by a constant depending only on b and d, since $|x|^2$ and r^2 are less than $9b^2$, $g^{::}$ is bounded on the finite set $\mathrm{B}^{::}_{3b}$, and $m\mathcal{G}(r\mathbf{e}_1)$ equals $r^2/(d(d-2))$ for $d \geq 3$ and $-\frac{1}{2}r^2 \log r$ for $d = 2$, which lies between $-2b^2 \max\{\log(2b), 0\}$ and $1/(4e)$. This proves part (ii).

(iii) For $|x| \geq r/3$, part (iii) follows from part (i), since $r/3 > r/4$. To prove (iii) in $\mathrm{B}^{::}_{r/3}$, note that $\gamma(x) + |x|^2/(2d) = -mg^{::}(x)$ is subharmonic (its discrete Laplacian is $m\delta_{\mathbf{0}}$), so by the minimum principle (Lemma 3.2) applied to its negative, it attains its maximum over $\mathrm{B}^{::}_{r/3} \cup \partial\mathrm{B}^{::}_{r/3}$ at a point $z \in \partial\mathrm{B}^{::}_{r/3}$. Thus for every $x \in \mathrm{B}^{::}_{r/3}$ we have $\gamma(x) + |x|^2/(2d) \leq \gamma(z) + |z|^2/(2d)$. Dropping the term $|x|^2/(2d) \geq 0$ and applying part (i) at the point z, where $r/3 \leq |z| < r/3 + 1$, we obtain

$$\gamma_r - \gamma(x) \geq \gamma_r - \gamma(z) - \frac{|z|^2}{2d} \geq \frac{r^2 - 2r|z|}{2d} - C \geq \frac{r^2}{6d} - \frac{r}{d} - C\,.$$

Since $r^2/24 - r = (r-12)^2/24 - 6$ and $(r-|x|)^2 \leq r^2$, the right side is at least $(r-|x|)^2/(8d) - C - 6/d$. Replacing C by $C + 6/d$, which preserves parts (i) and (ii), completes the proof. □

Proof of Theorem 4.7. By Lemma 4.4, the set $A := \{u_\infty > 0\}$ is finite and

$$\Delta^{::}(u_\infty + \gamma) = \sigma_\infty - 1 \leq 0\,,$$

with equality on A.

Step 1: We bound $u_\infty + \gamma$ and prove the inner inclusion in part (i). By Lemma 4.8(iii) we have $\gamma \leq \gamma_r + C$ on $\mathbb{Z}^d$, so $u_\infty + \gamma \leq \gamma_r + C$ off A, where the odometer vanishes. If A is nonempty, then by the maximum principle (Lemma 3.2 applied to $-u_\infty - \gamma$) this bound extends to A, and hence

$$u_\infty(x) \leq \gamma_r + C - \gamma(x) \qquad \text{for } x \in \mathbb{Z}^d\,. \tag{4.11}$$

Every $x \in \partial\mathrm{B}^{::}_r$ satisfies $r \leq |x| < r+1$, so by Lemma 4.8(ii) and $u_\infty \geq 0$, after increasing C, we have $u_\infty + \gamma \geq \gamma_r - C$ on $\partial\mathrm{B}^{::}_r$. Since $u_\infty + \gamma$ is superharmonic, the minimum principle gives

$$u_\infty(x) \geq \gamma_r - C - \gamma(x) \qquad \text{for } x \in \mathrm{B}^{::}_r\,.$$

Choose $c = c(d) > 0$ with $c^2 > 16dC$. By Lemma 4.8(iii), for all $x \in \mathrm{B}^{::}_{r-c}$ we then have

$$u_\infty(x) \geq \gamma_r - C - \gamma(x) \geq \frac{(r-|x|)^2}{8d} - 2C > 0\,.$$

This proves the inner bound $\mathrm{B}^{::}_{r-c} \subset \{u_\infty > 0\}$ (which is vacuous when $r \leq c$), and $\{u_\infty > 0\} \subset D_m$ by Lemma 4.4(i).

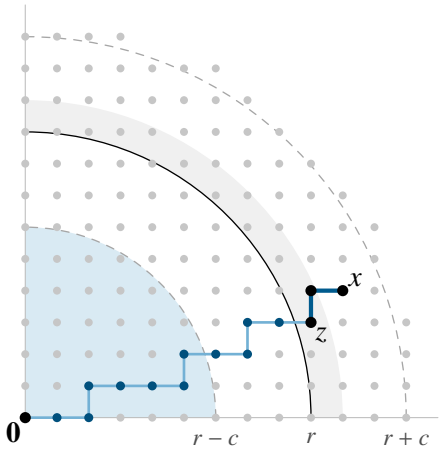


Figure 4.6: Step 2 of the proof of Theorem 4.7, drawn in the quarter plane of $\mathbb{Z}^2$: the odometer is positive on $\mathrm{B}^{::}_{r-c}$ (blue) by Step 1 and at most C_1 on the band $\mathrm{B}^{::}_{r+1} \setminus \mathrm{B}^{::}_r$ (gray). Along the path from a site $x \in D_m \setminus \mathrm{B}^{::}_r$ to $\mathbf{0}$ it increases by at least $1/(2d)$ at each step, so at most $2dC_1$ steps (thick) lead from x to the vertex z before the first entrance into $\mathrm{B}^{::}_r$, and $|x| < r + c$.

Step 2: We prove the outer inclusion in part (i) (see Figure 4.6). By (4.11) and Lemma 4.8(ii), there is a constant $C_1 = C_1(d)$ such that $u_\infty \le C_1$ on $\mathrm{B}^{::}_{r+1} \setminus \mathrm{B}^{::}_r$. If $z \in D_m \setminus \{\mathbf{0}\}$, then

$$\sum_{y \sim z} (u_\infty(y) - u_\infty(z)) = \sigma_\infty(z) = 1\,,$$

so a neighbor y of z maximizing $u_\infty(y)$ satisfies $u_\infty(y) \ge u_\infty(z) + 1/(2d) > 0$, and hence $y \in D_m$. Iterating, we obtain a path in D_m along which the odometer increases by at least $1/(2d)$ at each step, and this path must reach $\mathbf{0}$, since otherwise the odometer would increase without bound along it, contrary to finite support. Now given a point $x \in D_m \setminus \mathrm{B}^{::}_r$, the path from x must enter $\mathrm{B}^{::}_r$. Let z be the vertex before its first entrance, and let k be the number of steps from x to z. Then $r \le |z| < r + 1$ and

$$\frac{k}{2d} \le u_\infty(z) - u_\infty(x) \le C_1\,,$$

so $|x| \le |z| + k < r + 1 + 2dC_1$. Increasing c to at least $1 + 2dC_1$ proves the outer inclusion and preserves the inner inclusion.

Step 3: To prove the odometer estimate (4.7), it suffices, by the definitions of γ and γ_r, to show that

$$u_\infty(x) = \gamma_r - \gamma(x) + O(1) \qquad \text{for } x \in \mathrm{B}^{::}_{r+c}\,. \tag{4.12}$$

On $\mathrm{B}^{::}_r$ this follows from the two bounds of Step 1. On $\mathrm{B}^{::}_{r+c} \setminus \mathrm{B}^{::}_r$ we have

$\gamma_r - \gamma(x) = O(1)$ by Lemma 4.8(ii), so (4.12) follows from $u_\infty \geq 0$ and (4.11). □

How fast does the odometer grow as we move inward from the boundary of the nearly ball-shaped D_m? The answer is quadratically.

Proposition 4.9 *Let $d \geq 2$ and $m > 0$. Let $u_\infty : \mathbb{Z}^d \to \mathbb{R}_{\geq 0}$ be the divisible sandpile odometer for the initial configuration $m\delta_{\mathbf{0}}$ in $\mathbb{Z}^d$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Then there is a constant $C = C(d) < \infty$ such that*

$$u_\infty(x) \geq \frac{(r - |x|)^2}{2} - C \qquad \text{for } x \in \mathrm{B}_r^{::} \setminus \mathrm{B}_{r/3}^{::}\,.$$

Proof. By Theorem 4.7(ii), in the form (4.12), we have $u_\infty(x) \geq \gamma_r - \gamma(x) - C$ for $x \in \mathrm{B}_r^{::}$. For $x \in \mathrm{B}_r^{::} \setminus \mathrm{B}_{r/3}^{::}$, the number ξ in (4.10) satisfies $\xi \leq r$, so $(r/\xi)^d \geq 1$ and the coefficient of $(r - |x|)^2$ there is at least $1/2$ (compare Figure 4.5). Hence $u_\infty(x) \geq (r - |x|)^2/2 - C$, after increasing C. □

A quadratic upper bound on $\mathrm{B}_{r+c}^{::} \setminus \mathrm{B}_{r/3}^{::}$, where $\xi \geq r/3$, also follows from (4.12) along with the Taylor expansion (4.10). It is the discrete analogue of Proposition 2.32, by which a nonnegative function on $\mathbb{R}^d$ with bounded Laplacian grows at most quadratically away from a point where it vanishes. A sharper estimate near the boundary is

$$u_\infty(x) = \frac{(r - |x|)^2}{2} + O\left(1 + \frac{\bigl|r - |x|\bigr|^3}{r}\right) \qquad \text{for } x \in \mathrm{B}_{r+c}^{::} \setminus \mathrm{B}_{r/3}^{::}\,,$$

where the implied constant depends only on d, and Exercise 4.11 asks you to verify this.

4.3 ABELIAN SANDPILE FROM A POINT SOURCE

In the abelian sandpile, each site of $\mathbb{Z}^d$ has an integer number of grains of sand, and if a site has at least $2d$ grains, it topples, sending one grain to each neighbor. Start with n grains at the origin and topple until every site is stable (see Figure 1.3 for 10^4, 10^5 and 10^6 grains in $\mathbb{Z}^2$). The methods used to prove the near-perfect circularity of the divisible sandpile in Theorem 4.7 also give constraints on the shape of the abelian sandpile. We show that the sites that topple contain a Euclidean ball of volume about $n/(2d-1)$ and lie in a Euclidean ball whose volume is as close to n/d as we like, up to additive constants in the radii (Figure 4.7).

Figure 4.7: The stable sandpile for $n = 10^5$ grains started at the origin of $\mathbb{Z}^2$, colored by the number of grains as in Figure 1.3, and the circles of radii $(2d-1)^{-1/d}r$ (white) and $d^{-1/d}r$ (black) for $d = 2$, where $\pi r^2 = n$. These are the radii in Proposition 4.11 without the constants c and C, the second in the limit $a \uparrow d$, and the toppled set S_n lies between them.

Recall from Section 1.1 that the odometer $u(x)$ is the number of times that x topples, and that the stable sandpile σ° obtained from the initial sandpile $n\delta_{\mathbf{0}}$ satisfies

$$\sigma^\circ = n\delta_{\mathbf{0}} + \Delta^{::}u\,, \tag{4.13}$$

where $\Delta^{::}$ is the discrete Laplacian (3.1). Stability of σ° means $\sigma^\circ \le 2d-1$. We call a toppling legal when it takes place at a site that holds at least $2d$ grains. (Chapter 9 develops the abelian sandpile on general graphs; here we prove only what this section needs.) We need three facts about stabilization on the infinite lattice: a set of sites that all topple ends with at least as many grains as it has internal edges, every sequence of legal topplings is finite, and the order of the topplings does not matter.

Lemma 4.10 *Let $d \ge 1$ and $n \ge 1$ be integers. Start the abelian sandpile on $\mathbb{Z}^d$ with n grains at $\mathbf{0}$ and no grains elsewhere. Then the following statements hold.*

(i) Let A be a finite set of sites, and let a finite sequence of legal topplings topple every site of A at least once. Then at the end of the sequence the sites of A hold in total at least as many grains as there are edges of $\mathbb{Z}^d$ with both endpoints in A.

(ii) Every sequence of legal topplings is finite.

(iii) Every two finite sequences of legal topplings that end in stable sandpiles topple each site the same number of times.

Proof. We prove (i) by noting, for each edge, which endpoint topples last; we prove (ii) by combining (i) with the second moment of the sandpile, which grows by $2d$ at every toppling; and we prove (iii) by comparing two sequences one toppling at a time.

(i) Heights are never negative, since the initial heights are nonnegative, a legal toppling leaves at least 0 grains at its site, and it only adds grains elsewhere. Let x and y be neighbors in A such that the last toppling of x comes before the last toppling of y. (Two topplings never occur at the same time, so the last topplings of x and y are distinct.) When y topples for the last time, it sends x a grain which never moves again, since x has already toppled for the last time. Hence at the end of the sequence each site $x \in A$ holds at least as many grains as it has neighbors in A whose last toppling is later than its own. Each edge with both endpoints in A is counted at exactly one of its endpoints, the one whose last toppling comes first (Figure 4.8(a)), and summing over $x \in A$ proves (i).

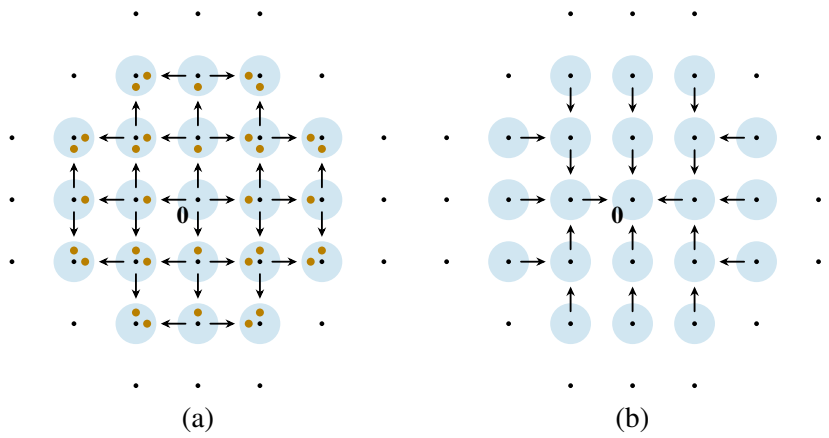


Figure 4.8: Lemma 4.10 for $n = 64$ grains in $\mathbb{Z}^2$, toppled one at a time in first-in, first-out order, with the 21 sites that topple shaded. (a) Each of the 32 edges between them points to the endpoint whose last toppling comes first, and that endpoint holds a grain sent along the edge (orange); (b) each of them other than $\mathbf{0}$ points to the neighbor that sent it its first grain, which toppled before it, so the arrows lead to $\mathbf{0}$.

(ii) Consider a finite sequence of legal topplings, and let W be the set of sites that topple. If W is empty, then no grain moves. Otherwise $\mathbf{0} \in W$, because $\mathbf{0}$ is the only site that holds grains at the start. Every other site of W holds no grains at the start and at least $2d$ grains at its first toppling, so it has received a grain from a neighbor in W that toppled earlier. The first topplings of the sites along a chain of such neighbors occur at strictly earlier times, so the chain ends at $\mathbf{0}$ (Figure 4.8(b)). Thus every site of W is joined to $\mathbf{0}$ by a path in W, and in particular at least $|W| - 1$ edges have both endpoints in W. By (i) the sites of W hold at least $|W| - 1$ grains at the end, and there are only

n grains, so $|W| \le n+1$. A shortest path in W from a site of W to $\mathbf{0}$ has at most $|W|-1 \le n$ steps, and therefore $|x| \le n$ for every $x \in W$. A grain that moves goes from a site of W to a neighbor, and a grain that does not move stays at $\mathbf{0}$, so every grain ends within distance $n+1$ of $\mathbf{0}$.

Applying this to the first t topplings of a legal sequence, we see that the sandpile σ_t after these topplings satisfies

$$\sum_{x\in\mathbb{Z}^d} |x|^2 \sigma_t(x) \le n(n+1)^2 .$$

On the other hand, a toppling at x increases this sum by $2d$, since $|x+\mathbf{e}_i|^2 + |x-\mathbf{e}_i|^2 - 2|x|^2 = 2$ for $1 \le i \le d$, and the sum is 0 when $t=0$, so it equals $2dt$. Hence $t \le n(n+1)^2/(2d)$, which proves (ii).

(iii) Let f be the odometer of a finite sequence of legal topplings that ends in the stable sandpile $\sigma^\circ = n\delta_{\mathbf{0}} + \Delta^{::} f$. Consider a second sequence of legal topplings, and let g_t be its odometer after t topplings. We prove by induction on t that $g_t \le f$. The case $t=0$ is clear. For the inductive step, suppose that $g_{t-1} \le f$, and let x be the site of the tth toppling. If $g_t(x) > f(x)$, then $g_{t-1}(x) = f(x)$. Just before the tth toppling the number of grains at x is

$$n\delta_{\mathbf{0}}(x) + \sum_{y\sim x} g_{t-1}(y) - 2d\, g_{t-1}(x) \le \sigma^\circ(x) \le 2d-1 ,$$

where the first inequality holds because $g_{t-1} \le f$ by the inductive hypothesis, $g_{t-1}(x) = f(x)$, and $\sigma^\circ(x) = n\delta_{\mathbf{0}}(x) + \sum_{y\sim x} f(y) - 2d\, f(x)$. This contradicts the legality of the tth toppling. Hence $g_t \le f$. If the second sequence is finite and ends in a stable sandpile, with final odometer g, then $g \le f$, and exchanging the roles of the two sequences gives $f \le g$. Thus $f = g$. □

We now compare the odometer of the abelian sandpile with the odometers of two divisible sandpiles; just one additional averaging trick is needed. For the inner ball, the number of topplings at each site, divided by $2d-1$, stabilizes the divisible sandpile with mass $n/(2d-1)$ at the origin, so by the least action principle it is at least the odometer of that sandpile. For the outer ball, Lemma 4.10(i) shows that a cube of toppled sites holds nearly d grains per site, and we compare the number of topplings, averaged over such cubes, with the odometer of the second sandpile. By Lemma 4.10, the set of sites that topple is finite and does not depend on the order of the topplings, so the next statement is well posed.

Proposition 4.11 *Let $d \ge 2$ and $n \ge 1$ be integers, and let $0 < a < d$. Let S_n be the set of sites that topple when the abelian sandpile on $\mathbb{Z}^d$ that starts with n grains at $\mathbf{0}$ and no grains elsewhere is stabilized by legal topplings. Let ω_d*

be the volume of the unit ball in $\mathbb{R}^d$*, and let* $r := (n/\omega_d)^{1/d}$*. Then there are constants* $c = c(d) > 0$ *and* $C = C(d, a) > 0$*, independent of* n*, such that*

$$\mathrm{B}^{::}_{(2d-1)^{-1/d}r-c} \subseteq S_n \subseteq \mathrm{B}^{::}_{a^{-1/d}r+C}\,.$$

Proof. We divide the proof into three steps: the inner inclusion, a comparison of averages over cubes, and the outer inclusion.

Let u be the odometer of a stabilization by legal topplings, and let σ° be the stable sandpile that it reaches. By Lemma 4.10, the function u does not depend on the order of the topplings, it has finite support, and $S_n = \{u > 0\}$. By (4.13) and stability,

$$0 \le n\delta_{\mathbf{0}} + \Delta^{::}u = \sigma^\circ \le 2d - 1\,.$$

Step 1: We prove the inner inclusion. The function $u/(2d-1)$ is nonnegative, and

$$\frac{n}{2d-1}\delta_{\mathbf{0}} + \Delta^{::}\Big(\frac{u}{2d-1}\Big) = \frac{\sigma^\circ}{2d-1} \le 1\,.$$

By the least action principle, Theorem 4.3(i), the odometer of the divisible sandpile started from $\frac{n}{2d-1}\delta_{\mathbf{0}}$ is at most $u/(2d-1)$. For this mass, the radius in Theorem 4.7 is $(n/((2d-1)\omega_d))^{1/d} = (2d-1)^{-1/d}r$. Let $c = c(d)$ be the constant of that theorem. By part (i) of the theorem, the odometer of the divisible sandpile is positive on $\mathrm{B}^{::}_{(2d-1)^{-1/d}r-c}$. Hence u is positive there, which proves the inner inclusion.

Step 2: We improve the lower bound on σ° by averaging over a small cube. Fix an integer $k \ge 1$ with $2dk/(2k+1) \ge a$, which is possible because $2dk/(2k+1)$ tends to $d > a$ as $k \to \infty$. For $x \in \mathbb{Z}^d$ let $Q^{::}(x,k) := \{y \in \mathbb{Z}^d : |y - x|_\infty \le k\}$ be the cube of side length $2k+1$ centered at x, and for $f : \mathbb{Z}^d \to \mathbb{R}$ let

$$f^{(k)}(x) := \frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} f(y)$$

be its average over this cube. Let

$$A := \{x \in \mathbb{Z}^d : Q^{::}(x,k) \subseteq S_n\}$$

be the set of centers of cubes all of whose sites topple (Figure 4.9). Then $A \subseteq S_n$ is finite. Since the cube $Q^{::}(x,k)$ has $2dk(2k+1)^{d-1}$ internal edges ($2k$ on each of its $(2k+1)^{d-1}$ lines in each of the d coordinate directions), Lemma 4.10(i) applied at the end of the stabilization gives, for every $x \in A$,

$$\frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} \sigma^\circ(y) \ge \frac{2dk}{2k+1} \ge a\,. \tag{4.14}$$

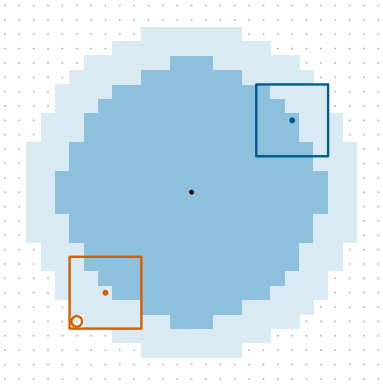


Figure 4.9: Step 2 of the proof of Proposition 4.11 for $n = 1000$ grains in $\mathbb{Z}^2$ and $k = 2$: the toppled set S_n (light blue) and the set A of centers of cubes $Q^{::}(x, 2) \subseteq S_n$ (dark blue). A cube centered in A (blue) holds at least $2dk/(2k+1) = 8/5$ grains per site on average, and a cube centered outside A (vermilion) contains a site where u vanishes (circled), so $u \le C_0$ on it.

Now let $v \ge 0$ be a times the odometer of the divisible sandpile started from $\frac{n}{a}\delta_{\mathbf{0}}$. The final mass of this sandpile is at most 1 at every site, so $\Delta^{::}v \le a - n\delta_{\mathbf{0}}$. Since averaging over cubes preserves inequalities, maps constants to themselves and commutes with $\Delta^{::}$, averaging (4.13) and the inequality for v and subtracting (the averaged sources cancel), we obtain from (4.14)

$$\Delta^{::}(u^{(k)} - v^{(k)})(x) \ge \frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} \sigma^{\circ}(y) - a \ge 0 \qquad \text{for } x \in A\,.$$

Thus $u^{(k)} - v^{(k)}$ is discrete subharmonic on A.

Next we bound $u^{(k)} - v^{(k)}$ outside A. Let x and y be neighbors. Since $u \ge 0$, the identity (4.13) and stability give

$$u(y) \le \sum_{z \sim x} u(z) = 2d\,u(x) + \sigma^{\circ}(x) - n\delta_{\mathbf{0}}(x) \le 2d\,u(x) + 2d - 1\,,$$

so $u(y) + 1 \le 2d(u(x) + 1)$. If $x \notin A$, then $Q^{::}(x, k)$ contains a site where u vanishes, and every site of the cube is joined to it by a path of at most $2dk$ steps. Iterating the last inequality along this path gives $u \le C_0$ on $Q^{::}(x, k)$, where $C_0 := (2d)^{2dk} - 1$. Hence $u^{(k)}(x) \le C_0$, and since $v \ge 0$,

$$u^{(k)} - v^{(k)} \le C_0 \qquad \text{outside } A\,.$$

If A is empty, then this bound holds on all of $\mathbb{Z}^d$. Otherwise $v^{(k)} - u^{(k)}$ is discrete superharmonic on the finite set A and is at least $-C_0$ on the outer boundary ∂A, which lies outside A, so by the minimum principle (Lemma 3.2)

we have $v^{(k)} - u^{(k)} \geq -C_0$ on A as well. In both cases

$$u^{(k)} \leq v^{(k)} + C_0 \qquad \text{on } \mathbb{Z}^d \,.$$

Step 3: We prove the outer inclusion. The divisible sandpile in the definition of v has mass n/a at the origin, so the radius in Theorem 4.7 is $a^{-1/d}r$. By part (i) of that theorem, with the same constant c as in Step 1, the odometer of this divisible sandpile vanishes outside $\mathrm{B}^{::}_{a^{-1/d}r+c}$. Let $R := a^{-1/d}r + c + \sqrt{d}k$. Every site of $Q^{::}(x,k)$ lies within distance $\sqrt{d}k$ of x, so $v^{(k)}(x) = 0$ for $|x| \geq R$. Since $u \geq 0$, we have $u(x) \leq (2k+1)^d u^{(k)}(x)$, so by Step 2, with $C_1 := (2k+1)^d C_0$,

$$u(x) \leq C_1 \qquad \text{for } |x| \geq R \,. \tag{4.15}$$

If S_n is empty, then there is nothing to prove. Otherwise let $x \in S_n$ have maximal Euclidean norm; we show that $|x| < R + C_1 + 1$. If $x = \mathbf{0}$ or $|x| < R$, there is nothing to show, so suppose that $x \neq \mathbf{0}$ and $|x| \geq R$. Changing a nonzero coordinate of x by one unit away from zero gives a neighbor of larger norm, which lies outside S_n by the maximality of $|x|$, and $u = 0$ there. By (4.13), $\Delta^{::}u(z) = \sigma^{\circ}(z) \geq 0$ at every site $z \neq \mathbf{0}$. Hence if u is smaller at a neighbor of z than at z, then u is larger at another neighbor, and so at least $u(z) + 1$ there, because u is integer-valued. Start at $x_0 := x$, where u is larger than at a neighbor. As long as $x_j \neq \mathbf{0}$, the value of u at x_j exceeds its value at a neighbor, namely the neighbor outside S_n when $j = 0$ and x_{j-1} when $j \geq 1$, so we can choose a neighbor x_{j+1} with $u(x_{j+1}) \geq u(x_j) + 1$. Since u is bounded, this path ends, necessarily at $\mathbf{0}$, so there is an index m with $x_m = \mathbf{0}$, as for the divisible sandpile in Figure 4.6.

We now count the steps of this path outside $\mathrm{B}^{::}_R$. Let j be the largest index with $|x_j| \geq R$. Then $j < m$ and $|x_{j+1}| < R$, so $|x_j| < R + 1$. By (4.15),

$$j \leq u(x_j) - u(x_0) \leq C_1 \,,$$

and therefore $|x| \leq |x_j| + j < R + C_1 + 1$. By the maximality of $|x|$, every site of S_n has norm less than $R + C_1 + 1 = a^{-1/d}r + C$, where $C := c + \sqrt{d}k + C_1 + 1$. This proves the outer inclusion, and with Step 1 the proposition. □

The volume n/a of the outer ball can be made as close to n/d as we like, at the cost of a larger constant C, because the cube side $2k+1$ must grow to bring $2dk/(2k+1)$ close to d. The two radii differ because the proof uses only that a stable site holds at most $2d-1$ grains and that cubes of toppled sites hold nearly d grains per site on average; it does not determine how the grains are distributed among the toppled sites.

When every occupied site holds one particle, the inner and outer radii agree to leading order. In Chapter 5 we study internal DLA, a random growth

model with this property, and we show by comparison with the divisible sandpile that its cluster fills a Euclidean ball up to an error of lower order. In Chapter 9 we return to the abelian sandpile and bound its toppled set by combinatorial arguments alone, from inside by a diamond and from outside by a cube, on more general backgrounds.

EXERCISES

Exercise 4.1. Let $n \geq 1$ be an integer. Find the odometer of the divisible sandpile on $\mathbb{Z}$ with initial mass $\sigma = 2\mathbf{1}_{[-n,n]}$. As in Section 4.1, the odometer is the mass sent from each site to each of its neighbors.

Exercise 4.2. In the setting of Section 4.1, let σ be a finitely supported nonnegative divisible sandpile on $\mathbb{Z}^d$ with final configuration σ_∞, and let $h : \mathbb{Z}^d \to \mathbb{R}$ be discrete superharmonic. When we assign weight $h(x)$ to each unit of mass at x, the total weight of a configuration μ is $\sum_x h(x)\mu(x)$. Show that a single toppling cannot increase the total weight. Then prove the quadrature inequality (1.1) of Chapter 1: the total weight of σ_∞ is at most that of σ. Deduce that the two total weights are equal when h is discrete harmonic.

Exercise 4.3. Let $\sigma : \mathbb{Z}^d \to \mathbb{R}$ be a finitely supported nonnegative function. By Lemma 4.4(ii), its divisible sandpile odometer has finite support. Use the discrete minimum principle, Lemma 3.2, to prove that this odometer is the only finitely supported function $u : \mathbb{Z}^d \to \mathbb{R}$ that satisfies the three conditions

$$\begin{aligned} u &\geq 0\,, \\ \sigma + \Delta^{::} u &\leq 1\,, \\ \sigma(x) + \Delta^{::} u(x) &= 1 \qquad \text{if } u(x) > 0\,. \end{aligned} \tag{4.16}$$

Exercise 4.4. In the setting of Section 4.1, let $\sigma \geq 0$ be a finitely supported initial mass on $\mathbb{Z}^d$. By Exercise 4.3, the odometer is the only finitely supported function that satisfies the three conditions in (4.16), and the third condition cannot be dropped: when $\sigma = 0$ the odometer vanishes, while $\mathbf{1}_{\{\mathbf{0}\}}$ satisfies the first two conditions but not the third (its discrete Laplacian at the origin is $-2d$). This exercise gives further examples on $\mathbb{Z}$, each obtained by sampling a continuous function, which satisfy the first two conditions and fail the third at only two sites. For $a \geq 0$, define $\nu_a : \mathbb{R} \to \mathbb{R}$ by

$$\nu_a(x) := \begin{cases} 1 + a - \frac{1}{2}x^2, & |x| \leq 1, \\ \frac{1}{2}(2 - |x|)^2 + a(2 - |x|), & 1 < |x| \leq 2, \\ 0, & |x| > 2\,. \end{cases} \tag{4.17}$$

Each ν_a has second derivative -1 on $(-1,1)$ and 1 on $(-2,-1)\cup(1,2)$, as a continuum odometer should when the initial mass density of 2 on $[-1,1]$ is spread out to mass density 1 on the larger interval $[-2,2]$. Only ν_0 has a continuous derivative. For an integer $n \geq 1$, take

$$\sigma = 2\mathbf{1}_{[-n+1,n-1]} + \mathbf{1}_{\{-n,n\}} \qquad \text{and} \qquad \nu_{n,a}(x) := n^2 \nu_a(x/n) \quad \text{for } x \in \mathbb{Z}\,.$$

(a) Show that $\nu_{n,0}$ is the divisible sandpile odometer for the initial mass σ.

(b) Let $0 < a \leq 1/(2n)$. Show that $\nu_{n,a} \geq 0$ and $\sigma + \Delta^{::}\nu_{n,a} \leq 1$, but that $\nu_{n,a}(n) > 0$ and

$$\sigma(n) + \Delta^{::}\nu_{n,a}(n) = 1 - an < 1\,.$$

Conclude that $\nu_{n,a}$ satisfies the first two conditions in (4.16) but is not the odometer.

Exercise 4.5*. Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $\gamma : V \to \mathbb{R}$ satisfy $\Delta\gamma = \sigma - 1$ and have a superharmonic majorant. Let s be the least superharmonic majorant of γ, let $u_\infty := s - \gamma$, and let $\sigma_\infty := \sigma + \Delta u_\infty$.

(a) Show directly from the definition (4.5) of s that s is superharmonic and that $\Delta s(x) = 0$ whenever $s(x) > \gamma(x)$.

(b) Deduce that $\sigma_\infty \leq 1$ on V and that $\sigma_\infty = 1$ on $\{u_\infty > 0\}$.

By Lemma 4.6 and Theorem 4.3, σ stabilizes and u_∞ is its odometer, so part (b) recovers Lemma 4.4(i) without the toppling dynamics.

Exercise 4.6. Let $d \geq 2$ and $m > 0$, and let σ_∞ and u_∞ be the final configuration and the odometer of the divisible sandpile started from a point source of mass m at the origin in $\mathbb{Z}^d$, as in Lemma 4.5. Prove that, after adjoining independent randomness, there is a stopping time τ for simple random walk $(X_k)_{k\geq 0}$ started at $\mathbf{0}$, such that $\mathbb{E}\tau < \infty$ and

$$\mathbb{P}\{X_\tau = z\} = \frac{\sigma_\infty(z)}{m} \qquad \text{for } z \in \mathbb{Z}^d\,,$$

and that τ can be chosen so that $X_n \in \operatorname{supp}(\sigma_\infty)$ for $n \leq \tau$ and $X_n \in \{u_\infty > 0\}$ for $n < \tau$. Thus simple random walk can be stopped "nearly uniformly" on a discrete ball.

Exercise 4.7. Show that the inclusions of Theorem 4.7(i) hold, after increasing $c = c(d)$, if D_m is replaced by $\{\sigma_\infty > 0\}$. That is, for $\sigma = m\delta_{\mathbf{0}}$ in $\mathbb{Z}^d$,

$$\mathrm{B}^{::}_{r-c} \subset \{u_\infty > 0\} \subset \{\sigma_\infty > 0\} \subset \mathrm{B}^{::}_{r+c}\,.$$

Exercise 4.8. In the setting of Exercise 4.7, let $D_m := \{\sigma_\infty = 1\}$. Show that for every point $x \in D_m \setminus \{\mathbf{0}\}$ there is a simple path $x = x_0, \dots, x_k = \mathbf{0}$ in D_m such that

$$u_\infty(x_{i+1}) \geq u_\infty(x_i) + \frac{1}{2d} \qquad \text{for } 0 \leq i < k\,.$$

Conclude that D_m is empty or nearest-neighbor connected.

Exercise 4.9. Let $d \geq 2$, and let $c = c(d)$ be large enough that the conclusion of Exercise 4.7 holds. Prove the following discrete Harnack inequality: if $R > 0$ and $0 \leq r < R - 3c$, and if $h : \mathbb{Z}^d \to \mathbb{R}_{\geq 0}$ is discrete harmonic on $\mathrm{B}^{::}_R$, then

$$h(x) \leq \left(\frac{R-c}{R-r-3c}\right)^d h(\mathbf{0}) \qquad \text{for } x \in \mathrm{B}^{::}_r\,.$$

This proof is independent of Lemma 3.16 and gives a constant close to the continuum one. (Hint: adapt the proof of the continuous Harnack inequality, replacing balls by point source divisible sandpiles.)

Exercise 4.10. (Exact Mean Value on the Lattice) Let $d \geq 2$. Deduce from Exercise 4.7 and the mean value property (4.4) that there is a constant $c = c(d)$ such that for every $r > 0$ there is a function $\sigma : \mathbb{Z}^d \to [0,1]$ satisfying

(i) $\sigma(x) = 1$ for every $x \in \mathrm{B}^{::}_{r-c}$;

(ii) $\sigma(x) = 0$ for every $x \notin \mathrm{B}^{::}_r$;

(iii) for every $h : \mathbb{Z}^d \to \mathbb{R}$ that is discrete harmonic on $\mathrm{B}^{::}_r$,

$$\sum_{x \in \mathbb{Z}^d} \sigma(x)(h(x) - h(\mathbf{0})) = 0\,.$$

Exercise 4.11. Use the Taylor expansion (4.10), together with (4.12), to verify that the divisible sandpile odometer u_∞ for initial mass $m\delta_{\mathbf{0}}$ in $\mathbb{Z}^d$, where $d \geq 2$ and $m > 0$, satisfies

$$u_\infty(x) = \frac{(r - |x|)^2}{2} + O\left(1 + \frac{\left||x| - r\right|^3}{r}\right) \qquad \text{for } x \in \mathrm{B}^{::}_{r+c} \setminus \mathrm{B}^{::}_{r/3}\,,$$

where ω_d is the volume of the unit ball in $\mathbb{R}^d$, $r = (m/\omega_d)^{1/d}$, $c = c(d)$ is the constant in Theorem 4.7, and the implied constant depends only on d.

The final three exercises relate inner and outer neighborhoods of bounded open sets to Hausdorff distance, in which we state the convergence of rescaled

internal DLA clusters and rotor-router aggregates to the closed unit ball in Chapters 5 and 6.

Exercise 4.12. For $r > 0$, the **inner r-neighborhood** of a set $A \subset \mathbb{R}^d$ (extending Ω_r of Section 2.1) and the **outer r-neighborhood** of A are the sets (Figure 4.10)

$$A_r := \{x \in A : \overline{\mathrm{B}}(x, r) \subset A\} \qquad \text{and} \qquad A^r := \{x \in \mathbb{R}^d : \mathrm{B}(x, r) \cap A \neq \emptyset\}.$$

Show that if A is open, then A_r and A^r are also open.

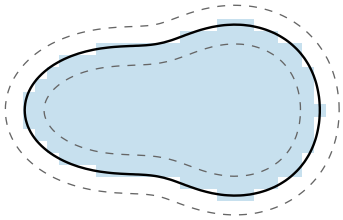


Figure 4.10: A bounded open set $A \subset \mathbb{R}^2$ (black), its inner and outer r-neighborhoods A_r and A^r for $r = 0.16$ (dashed), and the union B of the squares of side $\delta = 0.1$ centered at the points of $\delta\mathbb{Z}^2$ in A (shaded). Since $\delta/\sqrt{2} < r$, we have $A_r \subset B \subset A^r$.

Exercise 4.13. The **Hausdorff distance** between nonempty compact sets K and L in $\mathbb{R}^d$ is defined by

$$d_H(K, L) := \inf\{r > 0 : K \subset L^r \text{ and } L \subset K^r\}.$$

The **inner-outer Hausdorff metric** on nonempty bounded open sets A and B in $\mathbb{R}^d$ is

$$d_{\mathrm{io}}(A, B) := \inf\{r > 0 : A_r \subset B \subset A^r \text{ and } B_r \subset A \subset B^r\},$$

where A_r and A^r are as in Exercise 4.12. Show that d_H is a metric and that, for every compact set F containing $A \cup B$,

$$d_{\mathrm{io}}(A, B) = \max\{d_H(\overline{A}, \overline{B}), d_H(F \setminus A, F \setminus B)\}.$$

Conclude that d_{io} is a metric.

Exercise 4.14. In the notation of Exercise 4.13, find nonempty bounded open sets $A, B \subset \mathbb{R}^d$ such that $d_H(\overline{A}, \overline{B}) = 0$ but $d_{\mathrm{io}}(A, B) = 1$.

NOTES

The divisible sandpile of Section 4.1 was introduced by Levine and Peres (2009). The identity $u_\infty = s - \gamma$ after Lemma 4.6 is (Levine and Peres, 2009, Lemma 3.2), in other sign and normalization conventions, for nonnegative finitely supported configurations on $\mathbb{Z}^d$, and Levine and Peres (2010,

Lemma 3.2) reformulated it as the least action principle. Our odometer is the mass that a vertex sends to each neighbor, as in Levine, Murugan, Peres, and Ugurcan (2016a); in Levine and Peres (2009) and Levine and Peres (2017) it is the total mass sent out, which is $2d$ times larger on $\mathbb{Z}^d$. Theorem 4.3, for real configurations on locally finite graphs, follows Levine, Murugan, Peres, and Ugurcan (2016a, Proposition 2.5), which allows general toppling procedures. Our "thorough" is stronger than the condition in Levine and Peres (2017, Section 2) that a site that attains mass above 1 is eventually toppled.

The inclusions in Theorem 4.7(i) are Theorem 1.3 of Levine and Peres (2009), whose Lemma 3.2 gives the identity $u_\infty = s - \gamma$ (Exercise 4.5) and whose Section 2 and the proof of Theorem 3.3 give the bounds on the odometer. Proposition 4.9 is the point-source analogue of the non-degeneracy estimate in Lemma 5 of Caffarelli (1998), page 388.

Proposition 4.11 is Theorem 1.4 of Levine and Peres (2009), with their $d - \epsilon$ in place of our a. They state it for the set of sites visited by the grains, which lies in the set of toppled sites together with its outer boundary and the origin, and their Section 4 bounds the set of toppled sites from inside and outside by the same radii. They compare these bounds in dimension two with the earlier diamond and square of Le Borgne and Rossin (2002, Theorem 5), and they credit to Le Borgne and Rossin (2002) the count in Lemma 4.10(i) of the grains held by toppled sites, which Le Borgne and Rossin (2002, Section 3.2) attribute to a thesis of Rossin. The proof given here differs from theirs. Levine and Peres (2009) obtain the inner bound by an explicit comparison function, a quadratic function plus the Green function, and the outer bound by averaging a second such function over cubes. Here the inner bound follows from the least action principle for a divisible sandpile, and for the outer bound we compare the odometer averaged over cubes with the odometer of a divisible sandpile, so we need no Green function estimates beyond Theorem 4.7. Both proofs finish the outer bound by following a path along which the odometer increases (Lemma 4.2 of Levine and Peres (2009)). Their Theorem 4.1 also covers a constant background of h grains at every site, with $2d - 1$ and $d - \epsilon$ replaced by $2d - 1 - h$ and $d - \epsilon - h$, for $h \le 2d - 2$ in the inner bound and $h \le d - 1$ and $0 < \epsilon < d - h$ in the outer bound, and it allows holes, that is, negative h. In Section 9.9 we use combinatorial arguments to bound the toppled set on general backgrounds by a diamond and a cube.

On lattices whose spacing tends to zero, Theorem 3.3 of Levine and Peres (2010) proves uniform convergence of the rescaled odometers under their conditions (27)–(31): the initial densities take values between 0 and a common constant and are supported in a common ball, and they converge at every continuity point of a limiting density whose discontinuities form a

null set. Their Theorem 3.9 proves convergence of the fully occupied sets under four further conditions, (48)–(51): the limiting density takes no values strictly between some $\lambda < 1$ and 1; the set where it is at least 1 is the closure of its interior; and, for every $\epsilon > 0$ and fine enough lattices, the densities are at least 1 on the inner ϵ-neighborhood of that set and at most λ outside its outer ϵ-neighborhood. Their Lemma 3.10, used in the proof of Theorem 3.9, is the discrete quadratic-growth estimate, and their Section 6 introduces the inner-outer Hausdorff metric of Exercise 4.13.

Aleksanyan and Shahgholian (2019) introduce a perturbation of the divisible sandpile that depends on a threshold: a site topples its mass in excess of the threshold, and topples all of its mass once its odometer exceeds a level inversely proportional to the threshold. The final mass therefore lies on a sub-level set of the odometer, which links sandpile dynamics to quadrature surfaces. For a point source and a fixed threshold, their Theorem 4.1 and Remark 4.2 give the scaling limits of the odometer and of the set of visited sites. Their Theorem 4.3 shows that, for some threshold tending to infinity slowly enough as the initial mass grows, the rescaled odometers converge to a spherically symmetric function, and the discussion following Remark 4.2 and Appendix A show that in the limit the final mass is uniformly distributed on the boundary of a ball. These limits connect the discrete quadrature inequality (1.1) to free-boundary problems of Bernoulli type.

On $\mathbb{Z}^2$ and for $1 < \alpha < 2$, Frómeta and Jara (2018) study a divisible sandpile in which mass is redistributed according to the transition probabilities of their equation (1.1), those of a truncated α-stable random walk with a nearest-neighbor correction. Their Theorems 1.2 and 1.3 give the scaling limit of this sandpile started from a compactly supported initial density of class C^2 and identify the limiting odometer through an obstacle problem for a truncated fractional Laplacian. Their Theorem 1.4 expands the rescaled Green functions of this walk asymptotically.

Chapter Five

Internal DLA in $\mathbb{Z}^d$: Single Source

Starting with n particles at the origin in $\mathbb{Z}^d$, let each in turn perform simple random walk until reaching an unoccupied site. The resulting random set A_n of n occupied sites is known as an **internal diffusion-limited aggregation (IDLA)** cluster. The name comes from thinking of how the sequence $A_1 \subset A_2 \subset \cdots$ grows. Each cluster A_n is the union of A_{n-1} with a single point, the endpoint of a random walk started at $\mathbf{0}$ and stopped on exiting A_{n-1} (where $A_0 = \emptyset$), so the way in which the new point "aggregates" to the cluster is "limited" by a "diffusion" which is "internal" to the cluster.

Throughout the chapter, $d \geq 2$ is fixed, ω_d is the volume of the unit ball in $\mathbb{R}^d$, and $\mathrm{B}_r^{::} = \mathrm{B}^{::}(\mathbf{0}, r)$ is the discrete ball of Section 3.1. We write c and C for positive constants that depend only on d, unless we say otherwise, and whose values may change from one occurrence to the next.

We are going to prove that the asymptotic shape of A_n is a Euclidean ball, in the sense that for $r := (n/\omega_d)^{1/d}$ the rescaled cluster A_n/r converges almost surely in Hausdorff distance to the closed unit ball. There are two ways to sharpen such a statement: we can seek to reduce the error in the radius to a slower-growing function of r, and we can seek to quantify the probability in terms of r. Theorem 5.9 does both. Up to logarithmic factors, the cluster contains $\mathrm{B}_{r-\varpi}^{::}$ for ϖ of order $r^{1/3}$ and is contained in $\mathrm{B}_{r+m}^{::}$ with m^d of order ϖr^{d-1}, with explicit bounds on the probability that either inclusion fails, and by the Borel–Cantelli lemma both inclusions hold almost surely for all large n (Corollary 5.10). (In Figure 5.1, both inclusions hold with $\varpi = m = 4$.)

In Section 5.1 we construct IDLA from stacks of instructions. For the inner bound, we compare IDLA with the divisible sandpile of Chapter 4, in which continuous mass takes the place of the particles and the fully occupied sites fill a ball up to a bounded error in the radius (Theorem 4.7), using a product over the occupied sites of a positive function harmonic at those sites. For the outer bound, we show that among the particles that reach a cube, the number that reach a smaller concentric cube is smaller by a fixed factor, except with probability exponentially small in the first number.

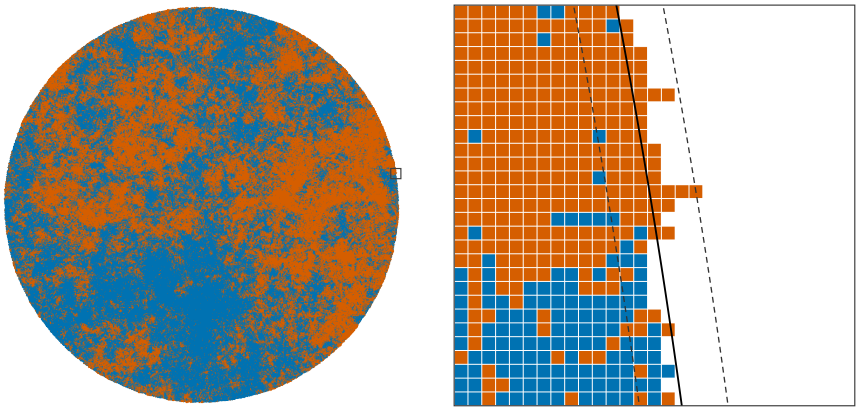


Figure 5.1: An internal DLA cluster of $n = 10^6$ particles in $\mathbb{Z}^2$, with each site x orange if $\pi|x|^2$ exceeds the number of the particle that settled there (it joined early) and blue otherwise (late). Right: a close-up of the small square, with the circle of radius $r = (n/\pi)^{1/2}$ (solid) and the circles through the nearest empty site and the farthest occupied site (dashed).

5.1 STACKS OF INSTRUCTIONS AND MASS BALANCE

An equivalent and very useful way to view random walk is the stack picture, in which the instructions for the walk are indexed by location instead of time. At each $x \in \mathbb{Z}^d$, let $\rho_1(x), \rho_2(x), \ldots$ be a sequence of **arrows**, each a directed edge from x to one of its neighbors. We call this sequence the **stack of instructions** at x. A **legal move** consists of choosing a site x with at least two particles and moving one particle from x along the first unused arrow of the stack at x, which is then discarded. Thus a legal move never takes the last particle from a site. A finite sequence of legal moves is **complete** if it leaves at most one particle at each site (Figure 5.2). By the abelian property, Corollary 5.2 below, we may release, pause and restart the particles in a convenient order without changing the cluster, as we do in Section 5.2.

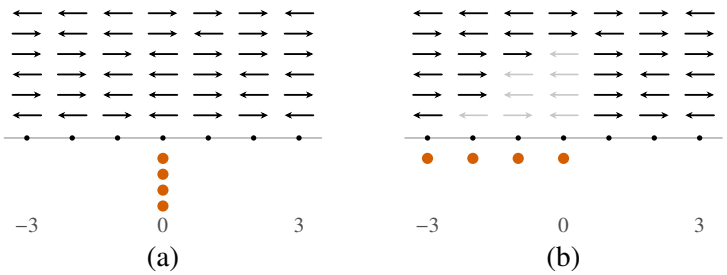


Figure 5.2: (a) Four particles at 0 in $\mathbb{Z}$, and the stack of instructions at each site, read from the bottom up. (b) Every complete sequence of legal moves uses the same arrows (gray), four at 0, three at -1 and one at -2, and leaves one particle at each of the sites $-3, -2, -1, 0$.

The stack picture has several advantages, one of which is generality. If the arrows are independent and uniform, then releasing the particles one at a time and letting each one walk until it reaches an unoccupied site uses every arrow at most once, so the particles perform independent simple random walks and we obtain IDLA. By replacing the i.i.d. random stacks by periodic stacks, which cycle once through the $2d$ neighbors, we obtain rotor-router aggregation, studied in Chapter 6. We first show that a complete sequence makes at least as many moves at each site as every finite sequence of legal moves.

Lemma 5.1 *Let $d \geq 1$ be an integer. At each site of $\mathbb{Z}^d$, fix an infinite stack of arrows, each a directed edge from that site to one of its neighbors, and fix an initial configuration of finitely many particles. Then every finite sequence of legal moves makes at most as many moves at each site as every complete sequence of legal moves.*

Proof. Fix a complete sequence and a finite sequence of legal moves. We show that, throughout the finite sequence, the number of moves it has made at each site is at most the number of moves the complete sequence makes there. Suppose, for contradiction, that this fails, and let x be the site of the first move of the finite sequence that violates it. Just before that move, the finite sequence has made exactly as many moves at x as the complete sequence, and at most as many at every other site. Both sequences use the arrows of each stack in order, so the arrows along which particles have moved into x so far are among the arrows used by the complete sequence. Thus the finite sequence has moved as many particles out of x as the complete sequence, and at most as many into x. It follows that x contains at most as many particles as at the end of the complete sequence, namely at most one, so the move at x is not legal. This contradiction proves the lemma. □

Applying the lemma twice to two complete sequences gives the abelian property.

Corollary 5.2 (Abelian Property) *Let $d \geq 1$ be an integer. At each site of $\mathbb{Z}^d$, fix an infinite stack of arrows, each a directed edge from that site to one of its neighbors, and fix an initial configuration of finitely many particles. Then every two complete sequences of legal moves make the same number of moves at each site and produce the same final configuration.*

Proof. Applying Lemma 5.1 with each of the two sequences in the role of the complete one, we see that they make the same number of moves at each site. These numbers determine the final configuration, since the final number of particles at a site x is the initial number, minus the number of moves at x, plus

the number of arrows pointing to x among those used at the neighbors of x. Hence the two complete sequences produce the same final configuration. □

For independent uniform arrows, almost surely each walk leaves the finite occupied set, so releasing the particles one at a time gives a complete sequence of legal moves, and by Corollary 5.2 every other complete sequence produces the same cluster from the same stacks. Note that the proof of Lemma 5.1 uses only that the moves of the complete sequence use the arrows of each stack in order and leave at most one particle at each site, not that they are legal, so the lemma also holds when the moves of the complete sequence are not required to be legal.

Given stacks ρ, an edge (x, y) of $\mathbb{Z}^d$ and an integer $m \geq 0$, we define the **discrepancy**

$$\mathcal{D}_\rho(x, y, m) := \#\{1 \leq j \leq m : \rho_j(x) = (x, y)\} - \frac{m}{2d}\,. \tag{5.1}$$

It is the number of the first m particles leaving x that move to y, minus $m/(2d)$, so it measures how far the arrows of the stack at x are from being divided equally among the neighbors of x.

Let $\sigma : \mathbb{Z}^d \to \mathbb{N}$ be a finitely supported initial configuration of particles for which there is a finite complete sequence of legal moves. Let A be the set of sites that such a sequence leaves occupied, and let $v(x)$ be the number of moves it makes at x. By the abelian property, Corollary 5.2, neither depends on the choice of complete sequence, and v is the odometer (it counts all moves at a site, so $v/(2d)$ corresponds to the odometer of Chapter 4). The effect of the $v(x)$ moves at x is to send $v(x)/(2d) + \mathcal{D}_\rho(x, y, v(x))$ particles to each neighbor y of x. Counting the particles received at x minus those sent from x, we therefore obtain

$$\mathbf{1}_A(x) - \sigma(x) = \frac{1}{2d}\Delta^{::}v(x) + \sum_{y\sim x} \mathcal{D}_\rho(y, x, v(y))\,.$$

Thus the main term is $\frac{1}{2d}\Delta^{::}v$, as for the divisible sandpile, and the discrepancies measure the error from moving whole particles. In Chapter 6 we sum this identity against a test function (Proposition 6.1) for sequences of moves that need not be complete.

5.2 INNER BOUND

Our starting point is the classical argument that A_n contains a large ball with high probability. For this we need a quadratic lower bound on the solution of a Poisson problem in a lattice ball slightly smaller than the Euclidean ball of volume n, which we prove by comparing this solution with the obstacle (4.8) of Chapter 4. (We use it again for rotor-router aggregation in Chapter 6.)

Lemma 5.3 *Let $d \geq 2$ be an integer, let $\eta > 0$, and let ω_d be the volume of the unit ball in $\mathbb{R}^d$. There is a constant $C = C(\eta, d) < \infty$ with the following property. Let $n \geq 1$ be a real number, let $r := (n/\omega_d)^{1/d} > \eta$, and let $B := \mathrm{B}^{::}_{r-\eta}$. If $f : B \cup \partial B \to \mathbb{R}$ satisfies*

$$\Delta^{::} f = 1 - n\delta_{\mathbf{0}} \quad \text{in } B \qquad \text{and} \qquad f = 0 \quad \text{on } \partial B\,,$$

then

$$f(z) \geq \frac{(r - |z|)^2}{8d} - C \qquad \text{for } z \in B\,.$$

Proof. Recall the obstacle $\gamma(x) = -|x|^2/(2d) - ng^{::}(x)$ of (4.8), with mass n, and its value γ_r at radius r of (4.9). Note that $F := \gamma_r - \gamma$ solves the same equation $\Delta^{::} F = 1 - n\delta_{\mathbf{0}}$ on all of $\mathbb{Z}^d$, since $\Delta^{::}|x|^2 = 2d$ and $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$. Every $x \in \partial B$ satisfies $r - \eta \leq |x| < r - \eta + 1$, so by Lemma 4.8(ii) with $b = \eta + 1$ we have $|F| \leq C$ on ∂B. Hence the difference $f - F$ is discrete harmonic in B and at least $-C$ on ∂B, where $f = 0$. By the minimum principle (Lemma 3.2), $f \geq F - C$ in B, and now the proof is completed by Lemma 4.8(iii), which gives $F(z) \geq (r - |z|)^2/(8d) - C$ for every $z \in \mathbb{Z}^d$. □

We prove the main step of the inner bound for arbitrary starting sites, since we use it again in Chapter 7 for internal DLA from two sources. Recall from (3.18) the killed Green function $g^{::}_B$ of simple random walk killed on exiting B. While a site z stays unoccupied, the function $1 - \lambda g^{::}_B(\cdot, z)$ is harmonic at every occupied site.

Lemma 5.4 *Let $d \geq 2$ and $N \geq 1$ be integers, let $B \subset \mathbb{Z}^d$ be finite, and let $z, s_1, \ldots, s_N \in B$. Release N particles in order, the ith starting at s_i and performing simple random walk, independently of the others. A particle settles at the first unoccupied site of B it reaches, and is paused if it leaves B first. Let I_N be the set of sites where particles settle. Let $\varphi := g^{::}_B(\cdot, z)$, let $L \geq \varphi(z)$ be a positive real number, and let $0 < \lambda \leq 1/(2L)$. Then*

$$\mathbb{P}\{z \notin I_N\} \leq \exp\Big\{-\lambda\Big(\sum_{i=1}^{N} \varphi(s_i) - \sum_{y \in B} \varphi(y)\Big) + \lambda^2 L \sum_{y \in B} \varphi(y)\Big\}\,.$$

Proof. **Step 1:** We bound the expected product of the values of $h := 1 - \lambda\varphi$ over the occupied sites. The function φ is nonnegative, vanishes off B, and satisfies $\Delta^{::}\varphi = -\delta_z$ on B by symmetry and (3.19), so by the minimum principle (Lemma 3.2) applied to $\varphi(z) - \varphi$ on $B \setminus \{z\}$ we have $0 \leq \varphi \leq \varphi(z) \leq L$ (trivially if $B = \{z\}$). Hence $1/2 \leq h \leq 1$, $h = 1$ off B, and h is harmonic on $B \setminus \{z\}$. For $0 \leq i \leq N$, let I_i be the set of sites where the first i particles settle, let $\mathcal{F}_i$ be generated by their walks, and let $W_i := \mathbf{1}\{z \notin I_i\} \prod_{y \in I_i} h(y)$. We claim that $\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] \leq h(s_i) W_{i-1}$ for $1 \leq i \leq N$. If $z \in I_{i-1}$, then both

sides vanish. Otherwise $I_{i-1} \subset B \setminus \{z\}$, so h is harmonic on I_{i-1} (Figure 5.3). Let Y_i be the first site outside I_{i-1} visited by particle i, where it settles or is paused. Then $\mathbb{E}[h(Y_i) \mid \mathcal{F}_{i-1}] = h(s_i)$, which is clear if $s_i \notin I_{i-1}$; otherwise $x \mapsto \mathbb{E}_x h(X_{\tau_{\partial I_{i-1}}})$ is harmonic on I_{i-1} and equals h on ∂I_{i-1} by Lemma 3.5, hence equals h on I_{i-1} by the minimum principle applied to the difference and its negative. If particle i settles at a site other than z, then $W_i = h(Y_i)W_{i-1}$; if it is paused, then $h(Y_i) = 1$ and $W_i = W_{i-1}$; and if it settles at z, then $W_i = 0$. Thus $W_i \le h(Y_i)W_{i-1}$, which proves the claim. Taking expectations and inducting on i from $W_0 = 1$, we obtain $\mathbb{E}W_N \le \prod_{i=1}^{N} h(s_i)$.

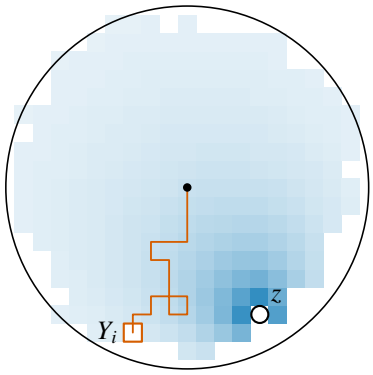


Figure 5.3: Step 1 of the proof of Lemma 5.4 in $B = \mathrm{B}^{\ddot{}}_{10}$: each occupied site y is shaded according to $g^{\ddot{}}_B(y, z)$, where z (circled) is unoccupied, so $h = 1 - \lambda g^{\ddot{}}_B(\cdot, z)$ is harmonic at every occupied site. The next walk from $\mathbf{0}$ stops at Y_i, where it first leaves the occupied sites.

Step 2: We bound $\mathbb{P}\{z \notin I_N\}$. Since $I_N \subset B$ and $0 < h \le 1$, we have $W_N \ge \mathbf{1}\{z \notin I_N\} \prod_{y \in B} h(y)$, so Step 1 gives

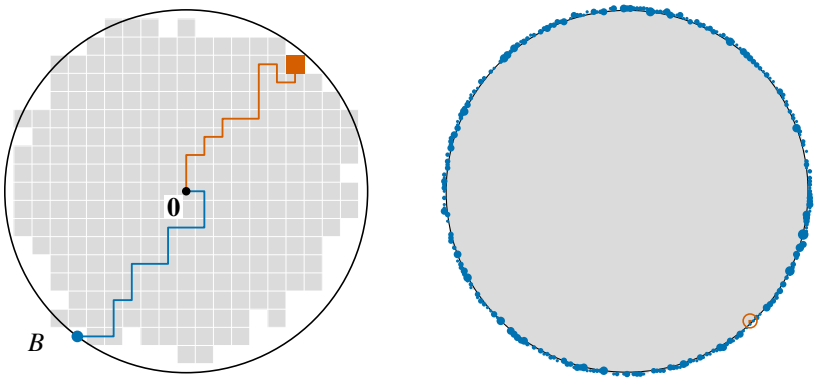
$$\mathbb{P}\{z \notin I_N\} \le \frac{\prod_{i=1}^{N}(1 - \lambda\varphi(s_i))}{\prod_{y \in B}(1 - \lambda\varphi(y))}.$$

Since $\log(1 - t) \le -t$ and $-\log(1 - t) \le t + t^2$ for $0 \le t \le 1/2$, we can take $t = \lambda\varphi \le 1/2$ and use $\varphi^2 \le L\varphi$ to bound the logarithm of the numerator by $-\lambda \sum_{i=1}^{N} \varphi(s_i)$ and minus the logarithm of the denominator by $\lambda \sum_{y \in B} \varphi(y) + \lambda^2 L \sum_{y \in B} \varphi(y)$, which proves the lemma. □

By Lemma 3.12(b), the Green function of a ball of radius $r \ge 2$ at its center is at most a constant times

$$\ell(r) := \begin{cases} \log(e + r), & d = 2, \\ 1, & d \ge 3, \end{cases} \tag{5.2}$$

and we pause each particle that leaves $\mathrm{B}^{\ddot{}}_{r-2}$ before settling, and apply Lemma 5.4 in this ball to bound the probability that a site of $\mathrm{B}^{\ddot{}}_{r-\varpi}$ remains unoccupied.



(a) (b)

Figure 5.4: (a) The construction of I_n in the proof of Theorem 5.5 for $r = 12$, with $B = \mathrm{B}^{::}_{r-2}$ inside the circle: the next particle walks through the occupied sites (gray) until it settles at an unoccupied site of B (orange) or is paused where it first leaves B (blue). (b) The end of the construction for $n = 10^4$: one site of B is still empty (ringed), and the 696 paused particles remain where they first left B (dots of area proportional to their number).

Theorem 5.5 (IDLA Inner Bound) *Let $d \geq 2$ be an integer. There are positive constants c and C depending only on d with the following property. Let n be a positive integer, let $r := (n/\omega_d)^{1/d} \geq 4$, and let $B := \mathrm{B}^{::}_{r-2}$. Release n particles from $\mathbf{0}$ in order, letting them perform independent simple random walks. A particle settles when it first reaches an unoccupied site of B; if it leaves B first, then it is paused at the exit site. Let I_n be the set occupied by the particles that settle in B. Then, for every integer ϖ satisfying $2 \leq \varpi \leq r/2$,*

$$\mathbb{P}\{\mathrm{B}^{::}_{r-\varpi} \not\subset I_n\} \leq Cr^d \exp\left(-c\frac{\varpi^3}{r\ell(r)}\right).$$

Proof of Theorem 5.5. The idea is to apply Lemma 5.4 with every starting site at $\mathbf{0}$ and the Green function of B with pole at a fixed site $z \in B$, and to bound its exponent above using Lemma 5.3.

Step 1: We couple I_n with A_n (Figure 5.4(a)). Informally, I_n is the modified IDLA process in which all particles exiting B are paused. If we restart the paused particles in order, each continuing its own walk until it first reaches an unoccupied site, then the set of occupied sites has the law of A_n. Indeed, every move uses fresh randomness, so we may construct these walks from stacks of instructions as in Section 5.1, and by the abelian property, Corollary 5.2, the outcome is the same as when the particles are released one at a time. We use this coupling to identify the resulting set with A_n, so $I_n \subset A_n$.

Step 2: We bound the probability that a fixed site stays unoccupied. Fix $z \in B$, and let $s := r - |z|$, so that $2 < s \leq r$. We need three facts

about $g_B^{::}$. First, there is a constant $C \geq 1$ such that $L := C\ell(r)$ satisfies $g_B^{::}(z,z) \leq L$. Indeed, B lies in $\mathrm{B}^{::}(z, 2r)$, and the killed Green function (3.18) increases with the domain, so translation invariance and Lemma 3.12(b) give $g_B^{::}(z,z) \leq g_{\mathrm{B}_{2r}}^{::}(\mathbf{0},\mathbf{0}) \leq C\ell(r)$.

Second, let

$$f(x) := n g_B^{::}(\mathbf{0}, x) - \sum_{y \in B} g_B^{::}(y, x) \qquad \text{for } x \in B \cup \partial B\,.$$

Then $\Delta^{::} f = 1 - n\delta_{\mathbf{0}}$ in B and $f = 0$ on ∂B, so Lemma 5.3 with $\eta = 2$ gives $f(z) \geq s^2/(8d) - C$. Third, by symmetry of $g_B^{::}$, the normalization (3.18) and the exit-time estimate (Lemma 3.4),

$$\sum_{y \in B} g_B^{::}(y, z) = \frac{1}{2d} \mathbb{E}_z \tau_{\partial B} \leq \frac{(r-1)^2 - |z|^2}{2d} \leq \frac{rs}{d}\,.$$

Now let $\lambda := s/(16rL)$, which is at most $1/(2L)$ since $s \leq r$. We apply Lemma 5.4 with $N = n$, every $s_i = \mathbf{0}$, and $\varphi = g_B^{::}(\cdot, z)$, which is allowed since $\varphi(z) \leq L$ by the first fact. By the definition of f we have $\sum_{i=1}^n \varphi(s_i) - \sum_{y \in B} \varphi(y) = f(z)$, and $\lambda^2 L r s/d = \lambda s^2/(16d)$, so by the second and third facts

$$\mathbb{P}\{z \notin I_n\} \leq \exp\Big\{-\lambda\Big(\frac{s^2}{8d} - C\Big) + \frac{\lambda s^2}{16d}\Big\}\,.$$

The exponent equals $\lambda C - s^3/(256 d\, rL)$. Since $\lambda \leq 1/16$ and L is a constant times $\ell(r)$, we conclude that

$$\mathbb{P}\{z \notin I_n\} \leq \exp\left(C - c\frac{s^3}{r\ell(r)}\right)\,.$$

Step 3: We take a union bound over $\mathrm{B}_{r-\varpi}^{::}$. Since $\varpi \geq 2$, every $z \in \mathrm{B}_{r-\varpi}^{::}$ lies in B and satisfies $r - |z| > \varpi$. The ball $\mathrm{B}_{r-\varpi}^{::}$ has at most $(2r+1)^d \leq (3r)^d$ sites, so Step 2 gives

$$\mathbb{P}\{\mathrm{B}_{r-\varpi}^{::} \not\subset I_n\} \leq (3r)^d e^C \exp\left(-c\frac{\varpi^3}{r\ell(r)}\right)\,,$$

which proves the theorem. □

In particular, letting $\varpi := \lfloor \epsilon r \rfloor$ for fixed $\epsilon \in (0, 1/2]$ (so that $\varpi \geq \epsilon r/2$ for large r), we obtain from Theorem 5.5 and $I_n \subset A_n$, after absorbing the factor Cr^d into the exponent by decreasing c,

$$\mathbb{P}\left\{\mathrm{B}_{(1-\epsilon)r}^{::} \not\subset A_n\right\} \leq \exp\left(-c\frac{\epsilon^3 r^2}{\ell(r)}\right) \tag{5.3}$$

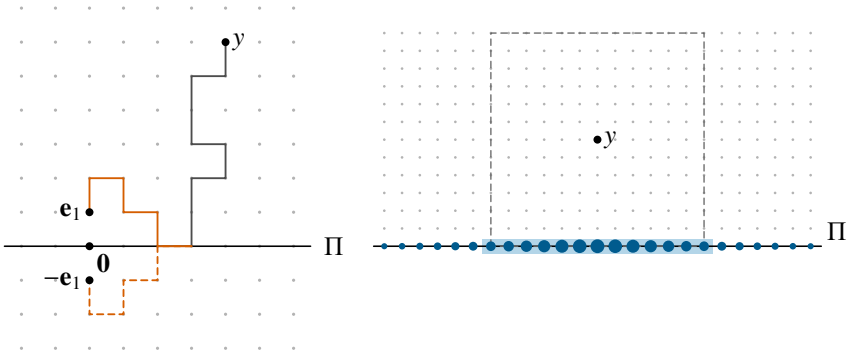


Figure 5.5: (a) Step 1 of the proof of Lemma 5.6: reflecting across Π the part after the first visit to Π (dashed) turns a path from y to $\mathbf{e}_1$ that visits Π into a path from y to $-\mathbf{e}_1$. (b) For $h = 6$ in $\mathbb{Z}^2$, the probability that the walk from y first hits Π at each of its sites (areas of the dots), computed exactly from the identity in Step 1; on the face of $Q^{\cdot\cdot}(y, h)$ (dashed) in Π (shaded), every value is at least $0.15/h$.

for all sufficiently large r. In dimensions $d \geq 3$, Exercise 5.3 asks you to prove a lower bound showing that the r^2 in the exponent is best possible.

5.3 OUTER BOUND

The particles paused in the proof of the inner bound start near the boundary of the occupied region. We show that if at most $b\varpi^d$ particles, where $b > 0$ depends only on d, start outside a cube of side about 6ϖ, then the probability that one of them reaches the concentric cube of side about 2ϖ decays exponentially in ϖ.

Recall the hitting time (3.4) and, from the proof of Proposition 4.11, the cube $Q^{\cdot\cdot}(x, h) = \{y \in \mathbb{Z}^d : |y - x|_\infty \leq h\}$ of side length $2h + 1$ centered at $x \in \mathbb{Z}^d$. According to the next lemma, a simple random walk X started at distance h from a hyperplane is fairly "spread out" by the time it hits the hyperplane, in the sense that its chance of first hitting a point z of the $(d-1)$-dimensional cube of side length $2h + 1$ in the hyperplane, centered at the point nearest to the starting site, is at least of order h^{1-d}.

Lemma 5.6 *Let $d \geq 2$ be an integer. There is a constant $c = c(d) > 0$ with the following property. Let h be a positive integer, let $y \in \mathbb{Z}^d$ satisfy $y_1 = h$, and let $\Pi := \{x \in \mathbb{Z}^d : x_1 = 0\}$. Then*

$$\mathbb{P}_y\{X_{\tau_\Pi} = z\} \geq ch^{1-d} \qquad \text{for every } z \in \Pi \cap Q^{\cdot\cdot}(y, h)\,.$$

Proof. The idea is to write the hitting probability as a difference of two Green function values by reflecting paths across Π (Figure 5.5), and to estimate it

with Theorem 3.9. Fix $z \in \Pi \cap Q^{::}(y,h)$. Translation by $-z$ maps Π to itself, so we may assume that $z = \mathbf{0}$, and then $y_1 = h$ and $|y_i| \le h$ for $2 \le i \le d$.

Step 1: We prove the identity

$$\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} = g^{::}(y - \mathbf{e}_1) - g^{::}(y + \mathbf{e}_1)\,.$$

Let $k \ge 0$. Reflecting across Π the part of a path after its first visit to Π is a bijection from the paths of length k from y to $\mathbf{e}_1$ that visit Π onto the paths of length k from y to $-\mathbf{e}_1$, all of which visit Π (Figure 5.5(a)). The remaining paths from y to $\mathbf{e}_1$, followed by the step to $\mathbf{0}$, are exactly the paths that first hit Π at $\mathbf{0}$ at time $k+1$. Each path of length k has probability $(2d)^{-k}$, so

$$\mathbb{P}_y\{\tau_\Pi = k+1,\ X_{\tau_\Pi} = \mathbf{0}\} = \frac{1}{2d}(\mathbb{P}_y\{X_k = \mathbf{e}_1\} - \mathbb{P}_y\{X_k = -\mathbf{e}_1\})\,.$$

By the symmetry $x \mapsto -x$ of the walk, $\mathbb{P}_y\{X_k = \pm\mathbf{e}_1\} = \mathbb{P}_{\mathbf{0}}\{X_k = y \mp \mathbf{e}_1\}$. Summing over $0 \le k \le t$ and recalling the truncated Green function $g_t^{::}$ of Section 3.3, we obtain

$$\mathbb{P}_y\{\tau_\Pi \le t+1,\ X_{\tau_\Pi} = \mathbf{0}\} = g_t^{::}(y - \mathbf{e}_1) - g_t^{::}(y + \mathbf{e}_1)\,.$$

As $t \to \infty$, the left side increases to $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\}$. If $d \ge 3$, then $g_t^{::} \to g^{::}$. If $d = 2$, then the right side equals $(g_t^{::}(\mathbf{0}) - g_t^{::}(y + \mathbf{e}_1)) - (g_t^{::}(\mathbf{0}) - g_t^{::}(y - \mathbf{e}_1))$, which tends to $a(y + \mathbf{e}_1) - a(y - \mathbf{e}_1)$ by the definition (3.12) of the potential kernel a, and $a = -g^{::}$ in dimension two. This proves the identity (see Figure 5.5(b) for its values when $h = 6$).

Step 2: We bound the right side of this identity from below for large h. Recall the constants a_d and κ of Theorem 3.9, by which $g^{::}(x) = \varphi(|x|^2) + O(|x|^{-d})$ as $|x| \to \infty$, where $\varphi(u) := a_d u^{(2-d)/2}$ when $d \ge 3$ and $\varphi(u) := -\frac{1}{4\pi}\log u - \kappa$ when $d = 2$. In both cases $\varphi'(u) = -u^{-d/2}/(2d\omega_d)$, since $\omega_2 = \pi$. For $h \ge 2$, the squared norms of $y \pm \mathbf{e}_1$ lie between $(h-1)^2 \ge h^2/4$ and $(h+1)^2 + (d-1)h^2 \le (d+2)h^2$, and they differ by $4h$. Since $|\varphi'|$ is decreasing, the mean value theorem gives

$$\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} = \varphi(|y - \mathbf{e}_1|^2) - \varphi(|y + \mathbf{e}_1|^2) + O(h^{-d}) \ge \frac{2h^{1-d}}{d\omega_d(d+2)^{d/2}} - Ch^{-d}\,.$$

Hence there is $h_0 = h_0(d) \ge 2$ such that $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} \ge (d+2)^{-d/2}h^{1-d}/(d\omega_d)$ for $h \ge h_0$.

Step 3: It remains to consider $h < h_0$. In this case a path of at most $(d-1)h$ steps parallel to Π leads from y to $h\mathbf{e}_1$, and h further steps toward Π first hit Π at $\mathbf{0}$. Hence $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} \ge (2d)^{-dh_0} \ge (2d)^{-dh_0}h^{1-d}$, and $c := \min\{(d+2)^{-d/2}/(d\omega_d), (2d)^{-dh_0}\}$ proves the lemma. □

Think of the occupied sites as "safe" and the unoccupied sites as "traps", where a particle settles. Using Lemma 5.6, we show next that if there are

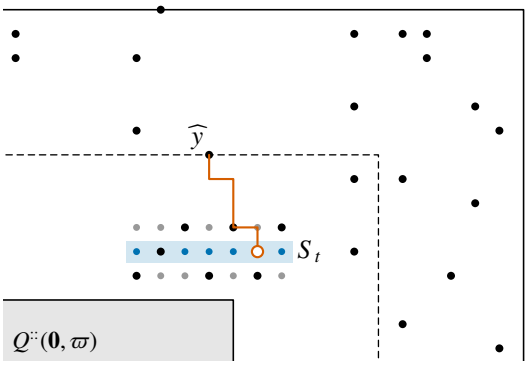


Figure 5.6: The proof of Lemma 5.7 near the site $\widehat{y}$ of first entrance, with the sites of O in black. The walk crosses the hyperplanes of all the sets S_t before it reaches $Q^{::}(\mathbf{0}, \varpi)$, and with probability bounded below it first hits the hyperplane of the set S_t with fewest sites of O (blue) outside O.

few safe sites between two concentric cubes, then a walk that enters the larger cube has a substantial chance of falling into a trap before reaching the smaller cube. For the rest of this section, each released particle follows an independent infinite simple random walk and settles at the first unoccupied site it reaches.

Lemma 5.7 *Let $d \geq 2$ be an integer. There is a constant $p = p(d) \in (0,1)$ with the following property. Let h and ϖ be positive integers with $h \geq 16$, and let $O \subset \mathbb{Z}^d$ satisfy $\#(O \cap Q^{::}(\mathbf{0}, \varpi + h)) \leq 2^{-2d-4}h^d$. Then, for every $x \in \mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, \varpi + h)$,*

$$\mathbb{P}_x\{\tau_{Q^{::}(\mathbf{0},\varpi)} < \infty \text{ and } \tau_{Q^{::}(\mathbf{0},\varpi)} \leq \tau_{\mathbb{Z}^d \setminus O}\} \leq p\,.$$

Proof. The idea is to find, between the two cubes, a $(d-1)$-dimensional cube in a coordinate hyperplane with few sites of O (Figure 5.6).

Let $w := \lfloor h/4 \rfloor$, so that $h/8 \leq w \leq h/4$. If the walk never enters $Q^{::}(\mathbf{0}, \varpi + 2w)$, or reaches $\mathbb{Z}^d \setminus O$ before or at its first entrance, then it does not visit $Q^{::}(\mathbf{0}, \varpi)$ by time $\tau_{\mathbb{Z}^d \setminus O}$. Otherwise let $\widehat{y}$ be the site where it first enters. The permutations of the coordinates and the reflections in the coordinate hyperplanes preserve the cubes and the hypothesis on O, so we may assume that $\widehat{y}_1 = \varpi + 2w$. For $1 \leq t \leq w$, consider the set

$$S_t := \{x \in \mathbb{Z}^d : x_1 = \varpi + t \text{ and } |x_i - \widehat{y}_i| \leq w \text{ for } 2 \leq i \leq d\}\,.$$

These sets lie in $Q^{::}(\mathbf{0}, \varpi + 3w) \subset Q^{::}(\mathbf{0}, \varpi + h)$, are pairwise disjoint, and each contains $(2w+1)^{d-1} \geq (h/4)^{d-1}$ sites. Since O has at most $2^{-2d-4}h^d$

sites in $Q^{::}(\mathbf{0}, \varpi + h)$ and $w \ge h/8$, there is a t with

$$\#(S_t \cap O) \le \frac{2^{-2d-4}h^d}{w} \le \frac{1}{2}\left(\frac{h}{4}\right)^{d-1},$$

so at least half of its sites lie outside O.

The hyperplane containing S_t lies at distance $2w - t$ from $\widehat{y}$, which is between w and h, so by Lemma 5.6, after a translation and a reflection, the walk from $\widehat{y}$ first hits this hyperplane at each site of S_t with probability at least ch^{1-d}. On its way to $Q^{::}(\mathbf{0}, \varpi)$ the walk must pass through this hyperplane, and it can visit $Q^{::}(\mathbf{0}, \varpi)$ by time $\tau_{\mathbb{Z}^d \setminus O}$ only if it first hits the hyperplane at a site of O. The strong Markov property at the first entrance time therefore proves the lemma with $p := 1 - c/(2 \cdot 4^{d-1})$. □

Applying Lemma 5.7 to many nested cubes, we show that a particle reaches the innermost cube only with exponentially small probability.

Lemma 5.8 *Let $d \ge 2$ be an integer. There are positive constants b, c_0, c_1 depending only on d with the following property. Let ϖ and k be positive integers satisfying $k \le b\varpi^d$. Let $O \subset \mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, 3\varpi)$, and place one settled particle at each site of O. Release k additional particles one at a time from specified sites in $\mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, 3\varpi)$, letting them perform independent simple random walks until they first reach unoccupied sites. A particle that never reaches an unoccupied site adds no occupied site. Then the probability that at least one additional particle visits $Q^{::}(\mathbf{0}, \varpi)$ is at most $c_0 e^{-c_1 \varpi}$.*

Proof. The idea is to apply Lemma 5.7 to the cubes of radius 2ϖ plus a constant times $j^{1/d}$, for integers $j \ge 0$, and to take a union bound over j.

Let $p \in (0, 1)$ be the constant in Lemma 5.7, and let $q := (1 + p)/2$ and $b := ((1 - q^{1/d})/18)^d$. For every integer $0 \le j \le k$, let $R_j := 2\varpi + \lfloor (j/b)^{1/d} \rfloor$, and let V_j be the number of additional particles that visit $Q^{::}(\mathbf{0}, R_j)$. Since $k \le b\varpi^d$, we have $2\varpi \le R_j \le 3\varpi$. The numbers V_j are nondecreasing in j, and $V_k \le k$.

Step 1: Let $c_1 := (1 - q)(1 - \log 2) > 0$. We show that, for all integers j and i with $1 \le j \le k$ and $0 \le i \le qj$,

$$\mathbb{P}\{V_j \le j \text{ and } V_i > qj\} \le e^{-c_1 j}. \tag{5.4}$$

Since $\lfloor t \rfloor > t - 1$ for real t and $i \le qj$,

$$R_j - R_i - 1 \ge b^{-1/d}(1 - q^{1/d})j^{1/d} - 2 = 18j^{1/d} - 2 \ge 16j^{1/d},$$

so Lemma 5.7 applies with R_i in place of ϖ and $h := R_j - R_i - 1$, because $h \ge 16$ and $2^{-2d-4}h^d \ge 2^{2d-4}j \ge j$.

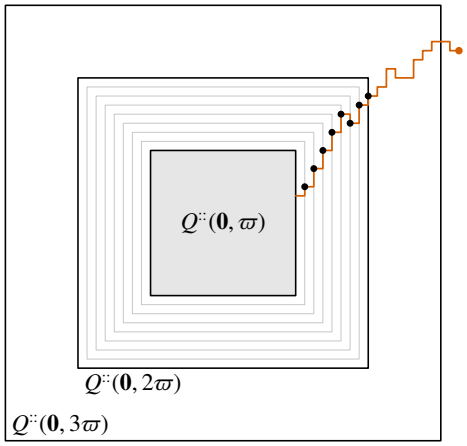


Figure 5.7: Step 2 of the proof of Lemma 5.8. A particle that reaches $Q^{::}(\mathbf{0}, \varpi)$ from outside $Q^{::}(\mathbf{0}, 3\varpi)$ passes over an occupied site x for each value of $|x|_\infty$ between 2ϖ and ϖ (dots), where an earlier particle that visited $Q^{::}(\mathbf{0}, 2\varpi)$ has settled.

Expose the particles in order, and consider the first j of them that enter $Q^{::}(\mathbf{0}, R_j)$. This cube is initially empty, so only earlier particles among these j can have settled in it. Hence fewer than j sites of the cube are occupied when one of these particles enters, and the site where it enters lies outside $Q^{::}(\mathbf{0}, R_j - 1) = Q^{::}(\mathbf{0}, R_i + h)$. By the strong Markov property at the entrance time and Lemma 5.7, the conditional probability that the particle visits $Q^{::}(\mathbf{0}, R_i)$ by its settling time (infinite if it never settles), given everything that happened before its entrance, is at most p.

For $0 \le t \le j$, let M_t be the number of the first t of these particles that visit $Q^{::}(\mathbf{0}, R_i)$, where a particle that does not exist counts as not visiting. Given everything before the entrance of the tth of these particles, M_{t-1} is determined and $M_t - M_{t-1}$ equals 1 with probability at most p, so $\mathbb{E}\, 2^{M_t} \le (1+p)\mathbb{E}\, 2^{M_{t-1}}$, and hence $\mathbb{E}\, 2^{M_j} \le (1+p)^j$. On $\{V_j \le j\}$, every particle that visits $Q^{::}(\mathbf{0}, R_i)$ first enters $Q^{::}(\mathbf{0}, R_j)$ and is among these j, so $M_j = V_i$. Hence $2^{M_j} > 2^{qj}$ on $\{V_j \le j \text{ and } V_i > qj\}$, and Markov's inequality applied to 2^{M_j} gives

$$\mathbb{P}\{V_j \le j \text{ and } V_i > qj\} \le 2^{-qj}(1+p)^j = (q2^{1-q})^j \le e^{-c_1 j},$$

where we used $1 + p = 2q$ and $\log q \le q - 1$.

Step 2: We show that $V_0 > \varpi$ if an additional particle visits $Q^{::}(\mathbf{0}, \varpi)$ (Figure 5.7). Such a particle starts outside $Q^{::}(\mathbf{0}, 3\varpi)$, and each step changes $|x|_\infty$ by at most one, so before its first visit to $Q^{::}(\mathbf{0}, \varpi)$ it visits a site x for each of the values $\varpi + 1, \dots, 2\varpi$ of $|x|_\infty$. These ϖ sites lie in $Q^{::}(\mathbf{0}, 2\varpi)$, so they are initially unoccupied, and they are occupied when visited, since the particle has not settled yet. Hence ϖ distinct earlier particles settled at them. These particles and the particle itself visit $Q^{::}(\mathbf{0}, 2\varpi) = Q^{::}(\mathbf{0}, R_0)$, so

$V_0 \geq \varpi + 1$.

Step 3: We conclude with a union bound. Since $V_k \leq k$, there is a least integer j with $0 \leq j \leq k$ and $V_j \leq j$. If an additional particle visits $Q^{::}(\mathbf{0}, \varpi)$, then Step 2 and the monotonicity of V_j give $j \geq V_j \geq V_0 > \varpi$, so $\lfloor qj \rfloor < j$, and minimality gives $V_{\lfloor qj \rfloor} > \lfloor qj \rfloor$, that is, $V_{\lfloor qj \rfloor} > qj$. Thus this event is contained in the union over $\varpi < j \leq k$ of the events in (5.4) with $i = \lfloor qj \rfloor$, and its probability is at most

$$\sum_{j > \varpi} e^{-c_1 j} = \frac{e^{-c_1 \varpi}}{e^{c_1} - 1} .$$

This proves the lemma with $c_0 := 1/(e^{c_1} - 1)$. □

We apply Lemma 5.8 to the particles paused when they first leave the inner ball. Recall the factor $\ell(r)$ of (5.2).

Theorem 5.9 *Let $d \geq 2$ be an integer. There are positive constants c and C depending only on d with the following property. Let n be a positive integer, let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$, and let $r := (n/\omega_d)^{1/d}$. Let ϖ and m be positive integers satisfying*

$$\varpi^3 \geq C r \ell(r) \log(e + r) , \qquad \varpi \leq r/2 \qquad \text{and} \qquad m^d \geq C \varpi r^{d-1} .$$

Then

$$\mathbb{P}\{(\mathrm{B}^{::}_{r-\varpi} \subset A_n \subset \mathrm{B}^{::}_{r+m})^c\} \leq \exp\left(-c \frac{\varpi^3}{r\ell(r)}\right) + C(r+m)^{d-1} e^{-cm} .$$

Proof. The idea is to bound the probability that $\mathrm{B}^{::}_{r-\varpi} \not\subset I_n$ by Theorem 5.5, and otherwise to apply Lemma 5.8 to the paused particles around each site of $\partial \mathrm{B}^{::}_{r+m}$ (Figure 5.8).

Step 1: We bound the probability that the inner inclusion fails. Let $B := \mathrm{B}^{::}_{r-2}$, and let I_n be the set defined in Theorem 5.5. We take the constant C in the hypotheses large enough that they force $\varpi \geq 2$, hence $r \geq 2\varpi \geq 4$, and $m \geq 12\sqrt{d}$. By Theorem 5.5, the probability that $\mathrm{B}^{::}_{r-\varpi} \not\subset I_n$ is at most a constant times $r^d \exp(-c\varpi^3/(r\ell(r)))$, where c is the constant of that theorem. If the constant in the hypothesis $\varpi^3 \geq C r \ell(r) \log(e + r)$ is large, then the factor in front of the exponential is at most $\exp(c\varpi^3/(2r\ell(r)))$, so this probability is at most $\exp(-c\varpi^3/(2r\ell(r)))$.

Step 2: We prove the outer inclusion when $\mathrm{B}^{::}_{r-\varpi} \subset I_n$. In the coupling of Step 1 of the proof of Theorem 5.5, restarting the paused particles produces A_n. When $\mathrm{B}^{::}_{r-\varpi} \subset I_n$, the number of paused particles (Figure 5.4(b)) satisfies

$$n - \#I_n \leq n - \#\mathrm{B}^{::}_{r-\varpi} \leq C\varpi r^{d-1} ,$$

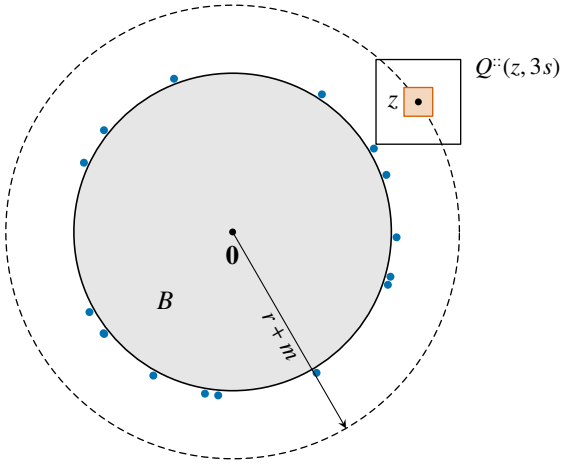


Figure 5.8: Step 2 of the proof of Theorem 5.9. The paused particles (dots) start on the boundary of $B = \mathrm{B}^{::}_{r-2}$, outside the cube $Q^{::}(z,3s)$ around a site z on the outer boundary of $\mathrm{B}^{::}_{r+m}$, and Lemma 5.8 shows that they rarely reach $Q^{::}(z,s)$.

as we see by comparing the unit cubes centered at lattice points with the Euclidean shell between radii $r - \varpi - \sqrt{d}$ and r.

Fix $z \in \partial \mathrm{B}^{::}_{r+m}$, and let

$$s := \left\lfloor \frac{m-1}{3\sqrt{d}} \right\rfloor .$$

We have $s \geq m/(6\sqrt{d})$ and $Q^{::}(z,3s) \subset \mathrm{B}^{::}(z,m)$. The cube $Q^{::}(z,3s)$ is disjoint from B and contains no paused particle, since every paused particle is at a site of norm less than $r-1$, while $|z| \geq r+m$ and every point of $Q^{::}(z,3s)$ lies within distance $m-1$ of z. By the strong Markov property, conditional on I_n and the positions of the paused particles, the restarted particles perform independent walks from these positions. Since $s^d \geq m^d/(6\sqrt{d})^d$, after increasing the constant in the hypothesis on m we have $n - \#I_n \leq bs^d$, where b is the constant of Lemma 5.8. By that lemma, translated to the center z and applied with the initially occupied set I_n, the conditional probability that $z \in A_n$ is at most $c_0 e^{-c_1 s} \leq c_0 e^{-c_1 m/(6\sqrt{d})}$, where c_0 and c_1 are its constants.

The cluster A_n is connected and contains the origin. Hence, if $A_n \not\subset \mathrm{B}^{::}_{r+m}$, then it contains a site of $\partial \mathrm{B}^{::}_{r+m}$, a set of at most $C(r+m)^{d-1}$ sites. A union bound, followed by averaging over I_n and the paused particles, gives

$$\mathbb{P}\{A_n \not\subset \mathrm{B}^{::}_{r+m} \text{ and } \mathrm{B}^{::}_{r-\varpi} \subset I_n\} \leq C c_0 (r+m)^{d-1} e^{-c_1 m/(6\sqrt{d})} .$$

Since $I_n \subset A_n$, combining this with Step 1 proves the theorem, with c the smaller of $c_1/(6\sqrt{d})$ and half the constant c of Theorem 5.5. □

By Theorem 5.9 and the Borel–Cantelli lemma, we obtain almost sure inclusions with explicit widths.

Corollary 5.10 *Let $d \geq 2$ be an integer. For every positive integer n, let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$, and let $r_n := (n/\omega_d)^{1/d}$. There is a constant $C = C(d) < \infty$ such that, almost surely, for all sufficiently large n,*

$$\min_{x \notin A_n} |x| \geq r_n - C(r_n \ell(r_n) \log(e + r_n))^{1/3}$$

and

$$\max_{x \in A_n} |x| < r_n + C r_n^{1-2/(3d)} (\log(e + r_n))^{2/(3d)} .$$

Proof. Let $K \geq 1$ be a constant, depending only on d, to be chosen large, and apply Theorem 5.9 with

$$\varpi = \lceil K(r_n \ell(r_n) \log(e + r_n))^{1/3} \rceil \qquad \text{and} \qquad m = \lceil (K \varpi r_n^{d-1})^{1/d} \rceil .$$

If K is at least the constant C of the theorem, then for large n these choices satisfy its hypotheses. Since $\varpi^3/(r_n \ell(r_n)) \geq K^3 \log(e + r_n)$, the first term in the theorem is at most $(e + r_n)^{-cK^3}$, where c is the constant of the theorem, and this is at most n^{-2} once K is large. Since $\varpi \geq 1$, we have $m \geq r_n^{(d-1)/d}$, so the second term decays faster than every power of n. The Borel–Cantelli lemma therefore shows that, almost surely, $\mathrm{B}^{::}_{r_n - \varpi} \subset A_n \subset \mathrm{B}^{::}_{r_n + m}$ for all large n, that is, $\min_{x \notin A_n} |x| \geq r_n - \varpi$ and $\max_{x \in A_n} |x| < r_n + m$.

It remains to compare ϖ and m with the widths in the statement. We have $\varpi \leq (K + 1)(r_n \ell(r_n) \log(e + r_n))^{1/3}$, and since $\ell(r_n) \leq \log(e + r_n)$, it follows that

$$m \leq ((K(K + 1))^{1/d} + 1) r_n^{1-2/(3d)} (\log(e + r_n))^{2/(3d)} .$$

Taking $C := (K(K + 1))^{1/d} + K + 1$ proves the inclusions. □

Both widths in Corollary 5.10 are small compared with r_n, so A_n/r_n converges almost surely to the closed unit ball in Hausdorff distance. The fluctuations are in fact much smaller. In Figure 5.1 the cluster lies within a few lattice spacings of the circle of radius r, and so do the clusters $A_{kn/8}$ with $k < 8$ (see Figure 5.9). In general the fluctuations of A_n from a ball are at most logarithmic in r (see the Notes).

EXERCISES

Exercise 5.1. Let $d \geq 2$ be an integer, and suppose the stack elements $\rho_j(x)$ for $j \geq 1$ and $x \in \mathbb{Z}^d$ are independent and uniform on the $2d$ directed edges

Figure 5.9: The cluster of Figure 5.1, with the sites of particles $(k-1)n/8+1$ to $kn/8$ shaded alike, alternately dark and light, for $k = 1, \dots, 8$. The outer boundary of the kth band is that of $A_{kn/8}$, which lies between the circles of radii $r_k - 5$ and $r_k + 5$, where $\pi r_k^2 = kn/8$.

leaving x, as for IDLA.

(a) Show that for all real numbers $a, b > 0$ there is a constant $C = C(a, b, d)$ such that, for every real number $R \geq 2$, the discrepancy (5.1) satisfies

$$\mathbb{P}\left(\bigcup_{x \in \mathrm{B}_R^{::}} \bigcup_{y \sim x} \bigcup_{m=1}^{\lfloor R^a \rfloor} \left\{ |\mathcal{D}_\rho(x, y, m)| \geq C \sqrt{m \log R} \right\} \right) \leq R^{-b} .$$

(b) Deduce from part (a) that there is a constant $C = C(d)$ such that, almost surely, for all sufficiently large integers m, every $x \in \mathrm{B}_m^{::}$, every neighbor $y \sim x$ and every integer $1 \leq j \leq m$ satisfy $|\mathcal{D}_\rho(x, y, j)| \leq C \sqrt{m \log(e + m)}$.

Exercise 5.2. Let $d \geq 1$ be an integer, let $B \subset \mathbb{Z}^d$ be finite, and let $g_B^{::}$ be the Green function used in Section 5.2. For $y, z \in B$, prove that

$$\mathbb{P}_y\{\tau_z < \tau_{\partial B}\} = \frac{g_B^{::}(y, z)}{g_B^{::}(z, z)} .$$

Exercise 3.27 is the case $z = \mathbf{0}$ for a ball.

Exercise 5.3. Let $d \geq 3$ be an integer, let $r \geq 1$ be a real number, and let $n := \lfloor \omega_d r^d \rfloor$. Let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$. Show that there is a constant $c = c(d) > 0$ such that, for all sufficiently large r and every $z \in \mathbb{Z}^d$ satisfying $|z - (r/2)\mathbf{e}_1| \leq 1$,

$$\mathbb{P}\{z \notin A_n\} \geq \exp(-cr^2) .$$

Taking r to be an even integer and $z = (r/2 - 1)\mathbf{e}_1$, conclude that the power r^2 in the exponent on the right side of (5.3) cannot be increased.

Exercise 5.4. Let $d \geq 2$ be an integer, let h be a positive integer, let $y \in \mathbb{Z}^d$ satisfy $y_1 = h$, and let $\Pi := \{x \in \mathbb{Z}^d : x_1 = 0\}$. Prove the upper bound

$$\mathbb{P}_y\{X_{\tau_\Pi} = z\} \leq Ch^{1-d} \qquad \text{for } z \in \Pi \cap Q^{::}(y, h)\,,$$

where $C = C(d)$, which matches the lower bound of Lemma 5.6.

Exercise 5.5. This exercise gives a coupling version of the successive conditioning in Step 1 of the proof of Lemma 5.8. Let n be a positive integer, and let $E_1, \dots, E_n$ be events adapted to a filtration $(\mathcal{F}_j)_{j=0}^n$. Let $p \in [0, 1]$, and suppose $\mathbb{P}(E_j \mid \mathcal{F}_{j-1}) \leq p$ for $1 \leq j \leq n$. Show that, on a possibly enlarged probability space, we can couple the indicators $\mathbf{1}\{E_j\}$ with independent Bernoulli(p) random variables $Y_j \geq \mathbf{1}\{E_j\}$ almost surely. Conclude that $\sum_{j=1}^n \mathbf{1}\{E_j\}$ is stochastically dominated by a Binom(n, p) random variable.

Exercise 5.6. Verify the geometric estimates used in the proof of Theorem 5.9. Let $d \geq 2$ be an integer, let n be a positive integer, and let $r := (n/\omega_d)^{1/d}$. Let $\eta \geq 2$ be a real number, and let ϖ be a positive integer satisfying $\eta \leq \varpi \leq r/2$. Suppose $I \subset \mathrm{B}^{::}_{r-\eta}$ contains $\mathrm{B}^{::}_{r-\varpi}$. Prove that $n - \#I \leq C\varpi r^{d-1}$ for a constant $C = C(d) < \infty$.

Let m be a positive integer satisfying $m \geq 12\sqrt{d}$. Let $z \in \partial \mathrm{B}^{::}_{r+m}$, and let $s := \lfloor (m-1)/(3\sqrt{d}) \rfloor$. Verify that $s \geq m/(6\sqrt{d})$ and that $Q^{::}(z, 3s)$ is disjoint from $\mathrm{B}^{::}_{r-\eta} \cup \partial \mathrm{B}^{::}_{r-\eta}$. For every fixed $b > 0$, show that there is a constant $C = C(b, d)$ such that $n - \#I \leq bs^d$ whenever $m^d \geq C\varpi r^{d-1}$.

NOTES

The internal DLA growth model appears on pp. 96–97 of Diaconis and Fulton (1991), and the game with decks of cards appears on pp. 97–98. Their Proposition 4.1 on p. 109 proves termination under a condition on the decks, and that the terminal configuration and the number of moves at each site do not depend on the choice of legal moves; Corollary 5.2 gives the second statement in the form used in this book. The model of Meakin and Deutch (1986, p. 2320) is a related diffusion-limited erosion process with different dynamics. The almost-sure convergence of A_n/r to $\overline{\mathrm{B}}(\mathbf{0}, 1)$ is Theorem 1 of Lawler, Bramson, and Griffeath (1992, p. 2119), where $r = (n/\omega_d)^{1/d}$. The coloring of Figure 5.1 by early and late sites follows Figure 4 of Levine and Peres (2017).

Theorem 1 of Lawler (1995, p. 72) shows that in every dimension $d \geq 2$, almost surely, for all large integers r the cluster A_n with $n = \lfloor \omega_d r^d \rfloor$ contains $\mathrm{B}^{::}(\mathbf{0}, r - r^{1/3}(\log r)^2)$ and is contained in $\mathrm{B}^{::}(\mathbf{0}, r + r^{1/3}(\log r)^4)$. Its outer

bound is much sharper than that of Theorem 5.9. The proof of Theorem 5.9 combines Theorem 5.5 with Lemmas 5.10–5.12 of Levine and Peres (2010), which correspond to Lemmas 5.6–5.8.

Lemma 2 of Jerison, Levine, and Sheffield (2012) rules out thin tentacles in $\mathbb{Z}^2$. It is a variant of Lemma 5.12 of Levine and Peres (2010), on which Lemma 5.8 is based. For every real $m > 0$ and every site z with $|z| \geq m$, the probability that z belongs to A_n while at most a small absolute constant times m^2 sites at distance less than m from z do is exponentially small in m, uniformly in n. Lemma A in their Appendix A.1 improves the exponent to order $m^2/\log m$ in $\mathbb{Z}^2$. It also proves the analogous bound in $\mathbb{Z}^d$ for $d \geq 3$, with m^d in place of m^2 in the count and an exponent of order m^2. Almost surely, for all sufficiently large n, the inner and outer errors of A_n relative to $\mathrm{B}_r^{\ddagger}$ are at most $C \log r$ when $d = 2$, by Theorem 1 of Jerison, Levine, and Sheffield (2012) and the Borel–Cantelli lemma, and at most $C\sqrt{\log r}$ when $d \geq 3$, by Theorem 1 and Corollary 2 of Jerison, Levine, and Sheffield (2013), where C depends only on d. Asselah and Gaudillière (2013a, Theorem 1.2) obtained the inner bound $C \log r$ and the outer bound $C(\log r)^2$ in every dimension $d \geq 2$, and Proposition 1.1 and Remark 1.2 of Asselah and Gaudillière (2013b) give, by a different method, the bound $C \log r$ on both errors when $d = 2$ and the bound $C\sqrt{\log r}$ on both errors when $d \geq 3$. Theorem 1.4 of Asselah and Gaudillière (2014) shows that when $d \geq 3$ the inner error exceeds a constant times $\sqrt{\log r}$ infinitely often, almost surely, so this order is sharp.

Equations (2)–(4) of Friedrich and Levine (2013) give the identity for $\mathbf{1}_A - \sigma$ in Section 5.1. The identity used in the proof of Proposition 13 of Holroyd and Propp (2010) gives the corresponding formula for one rotor walk. The obstacle problem for aggregation with general sources is developed in Theorems 3.3 and 3.9 of Levine and Peres (2010).

Chapter Six

Rotor-Router Aggregation in $\mathbb{Z}^d$: Single Source

Rotor-router aggregation is the deterministic counterpart of internal DLA. In a **rotor walk** on $\mathbb{Z}^d$, each site serves its $2d$ neighbors in a prescribed cyclic order, called the **rotor mechanism** at that site. To visualize a rotor walk, label each site by an arrow ("rotor") pointing toward one of its neighbors, which gives a **rotor configuration**. At each time step, the walker first advances the rotor at its current location to point to the next neighbor in the cyclic order, and then the walker moves to that neighbor. In rotor-router aggregation, we start with n particles at the origin, and each in turn performs rotor walk until it reaches a site not occupied by any other particle (see Figure 1.7 for the first few particles). We do not reset the rotors between walks. Let A_n^{R} denote the resulting set of n occupied sites. Each particle stops after finitely many steps. Indeed, if the walk of a particle stayed forever in the finite set of sites occupied when it was released, then a site would be visited infinitely often, the walker would move from it to every neighbor infinitely often, and since $\mathbb{Z}^d$ is connected every site would be visited infinitely often, contradicting the finiteness of the occupied set.

Simulations in two dimensions show that the aggregates are extraordinarily close to circular (Figure 6.1 shows one with a million particles). The main result of this chapter, Theorem 6.6 in Section 6.3, is that the aggregate is close to a ball in every dimension $d \geq 2$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$. For large $r = (n/\omega_d)^{1/d}$, the aggregate A_n^{R} contains the ball of radius $r - C \log r$ and is contained in the ball of radius $r + C(\log r)^2$, for all initial rotors and rotor mechanisms, so the rescaled aggregates converge to the unit ball. But the near perfect circularity in Figure 6.1, within one lattice spacing of the circle, remains a mystery. Internal DLA, the random model of Chapter 5, has the same limit shape (Figure 1.9 shows both), and Theorem 5.9 gives bounds of the same kind for it that hold with high probability.

The idea of the proof is to compare rotor walk with random walk. For the inner bound, we test a summation by parts for stack moves (Proposition 6.1) against the Green function of a ball. This bounds the number of moves at each site from below by a function that grows quadratically with the distance to the sphere of radius r (the same function appears in the inner bound for internal DLA in Chapter 5), up to an error bounded by the discrepancies of the stacks. The error grows only linearly, up to a logarithm, so every site at distance more than $C \log r$ inside the sphere is occupied. This inner bound,

Theorem 6.5 in Section 6.2, holds for all stacks with bounded discrepancy. For the outer bound, we freeze the particles on a sphere and enlarge the sphere one unit at a time. By testing the same identity against the Green function of the occupied set, with its pole at a site of frozen particles, we show that at most $C \log r$ particles are frozen at each site, so the particles that leave the ball of radius r cannot travel far in a thin tentacle.

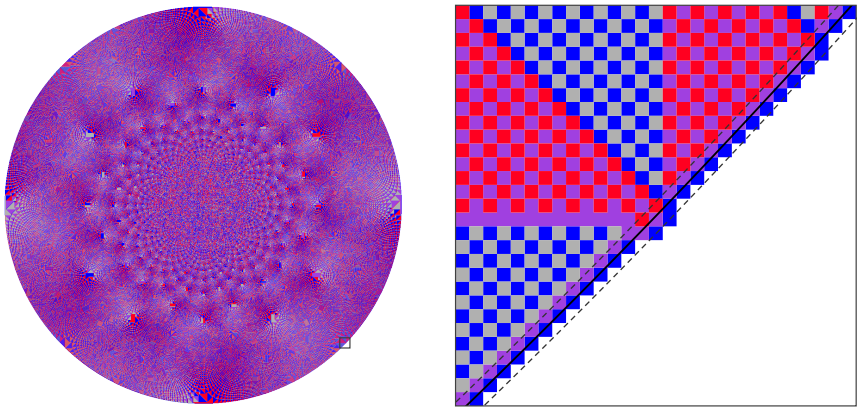


Figure 6.1: The rotor-router aggregate of one million particles started at the origin in $\mathbb{Z}^2$, with the clockwise mechanism and all rotors initially North (left), and the sites in the marked square (right). Each site is colored according to the final direction of its rotor, as in Figure 1.1 (North blue, East purple, South red, West gray). Every site within distance 563.5 of the origin is occupied and none lies beyond 565.1 (dashed circles), while $r = (10^6/\pi)^{1/2} \approx 564.2$ (solid).

6.1 BOUNDED DISCREPANCY

In both halves of the proof we compare rotor walk with random walk, and the comparison is simplest when the arrows at every site are written down in advance. Recall from Section 5.1 that each site $x \in \mathbb{Z}^d$ has an infinite stack of instructions $\rho_1(x), \rho_2(x), \ldots$, each a directed edge from x to one of its neighbors (left half of Figure 6.2), and the discrepancy

$$\mathcal{D}_\rho(x, y, m) = \#\{1 \le j \le m : \rho_j(x) = (x, y)\} - \frac{m}{2d}$$

is the number of the first m arrows at x that point to y, minus the expected number for a random walk. In a stack move at x, one particle follows the next arrow of the stack, the top one in Figure 6.2, so the jth particle to leave x follows $\rho_j(x)$. For aggregation, a move is legal when its site contains at least two particles, and a finite legal sequence is complete when every site contains at most one particle. The odometer is the number of moves at each site.

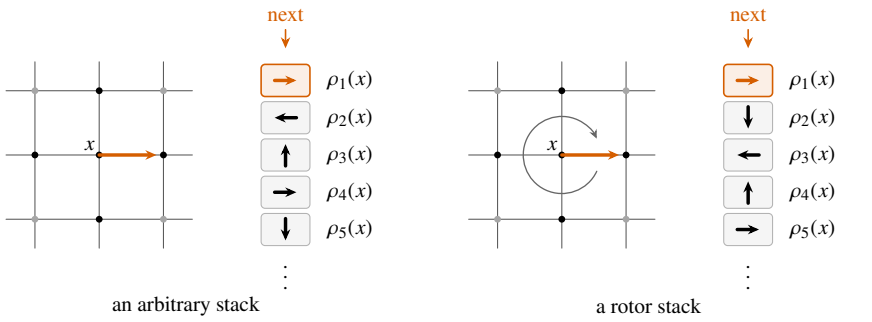


Figure 6.2: Left: the stack at a site x of $\mathbb{Z}^2$, whose top arrow (highlighted) is the one that the next particle to leave x follows. Right: a rotor stack, whose arrows run clockwise through the four neighbors, so that each block of four consecutive arrows uses every neighbor once.

Rotor-router aggregation is the special case in which each stack is periodic and cycles once through the $2d$ neighbors, as in the right half of Figure 6.2. For a rotor stack at x and an integer $m \geq 0$, write $m = 2dq + k$ with integers $q \geq 0$ and $0 \leq k < 2d$. Then k of the neighbors of x have received $q + 1$ of the first m particles to leave x and the others q. Hence $\mathcal{D}_\rho(x, y, m)$ equals $1 - \frac{k}{2d}$ for those k neighbors and $-\frac{k}{2d}$ for the others, and since $0 \leq k \leq 2d - 1$, every rotor stack satisfies

$$|\mathcal{D}_\rho(x, y, m)| \leq 1 - \frac{1}{2d} \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0\,. \tag{6.1}$$

Exercise 6.1 asks you to show that no stack has a smaller bound, and that for a stack of independent uniform arrows the discrepancy has standard deviation of order $\sqrt{m}$.

By Lemma 5.1, every finite legal sequence makes at most as many moves at each site as a complete one, and by Corollary 5.2, all complete legal sequences have the same odometer and the same final configuration.

The next proposition is a summation by parts. Notice that its right side depends on the stacks only via the odometer and the discrepancy. This is particularly powerful when the discrepancy is bounded.

Proposition 6.1 *Let $d \geq 1$ be an integer, and let $\sigma_0, \xi : \mathbb{Z}^d \to \mathbb{N}$ have finite support. Suppose that a finite sequence of stack moves with fixed stacks of instructions ρ converts σ_0 into ξ with odometer v. Then every function*

$f : \mathbb{Z}^d \to \mathbb{R}$ *satisfies*

$$\sum_x f(x)(\xi(x) - \sigma_0(x)) = \frac{1}{2d} \sum_{x:v(x)>0} v(x)\Delta^{::} f(x) + \sum_{x:v(x)>0} \sum_{y\sim x} \mathcal{D}_\rho(x, y, v(x))(f(y) - f(x)) .$$

Proof. Assign weight $f(z)$ to a particle at z, so that a particle sent from x to y changes the total weight by $f(y) - f(x)$. The $v(x)$ moves at x use the first $v(x)$ arrows at x, so by the definition of the discrepancy, their effect is to send $v(x)/(2d) + \mathcal{D}_\rho(x, y, v(x))$ particles to each neighbor y of x, which changes the total weight by

$$\sum_{y\sim x} \left(\frac{v(x)}{2d} + \mathcal{D}_\rho(x, y, v(x)) \right) (f(y) - f(x)) .$$

Summing over the sites x with $v(x) > 0$ and using $\sum_{y\sim x}(f(y)-f(x)) = \Delta^{::} f(x)$ proves the proposition. □

The first term on the right side of Proposition 6.1 is the change in weight for the divisible sandpile with odometer $v/(2d)$. When the discrepancies are small and f varies slowly, the second term can be seen as an error term. In the next section we take f to be a killed Green function to bound the odometer from below, and Exercise 6.3 asks you to compare the odometer with the expected local time of random walk.

6.2 INNER BOUND

In this section we prove that the aggregate contains the ball of radius $r - Cb_\rho \log n$ whenever every stack has discrepancy at most b_ρ (Theorem 6.5). We test Proposition 6.1 against the killed Green function of the ball $\mathrm{B}^{::}(\mathbf{0}, r - 2)$. The main term grows quadratically by Lemma 5.3 of Chapter 5, and the error term is bounded by the gradient of the killed Green function.

Let $d \geq 2$ and $n \geq 2$ be integers, and let ρ be a family of stacks of instructions on $\mathbb{Z}^d$. Let A_n^{R} and v_n be the aggregate and odometer obtained by stack aggregation from $n\delta_{\mathbf{0}}$ using ρ, and as usual let $r := (n/\omega_d)^{1/d}$. Assume now that the stacks have bounded discrepancy, that is, there is a real number $b_\rho \geq 1$ such that

$$|\mathcal{D}_\rho(x, y, m)| \leq b_\rho \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0 .$$

This covers not only rotor stacks, but also certain random stacks: for instance, the stacks could come in blocks of length $2d$, each block consisting of a

random permutation of the $2d$ neighbors. Bounded discrepancy ensures that a particle that follows the successive arrows of the stacks leaves every finite set, since otherwise its infinitely visited sites would form a nonempty finite set closed under taking neighbors, and no such set exists in $\mathbb{Z}^d$ (if it contains x, then it contains $x + k\mathbf{e}_1$ for every $k \geq 1$).

When we test Proposition 6.1 against a killed Green function, the error term is at most b_ρ times the sum of the absolute differences of the Green function along edges. The next two lemmas take care of bounding this sum. Throughout the rest of the chapter, $(X_t)_{t\geq 0}$ denotes simple random walk on $\mathbb{Z}^d$, and for a set $A \subset \mathbb{Z}^d$ we write $\tau_{\partial A}$ for the first time the walk hits the outer boundary ∂A. Recall the Green function $g_A^{::}$ of a walk killed on leaving a finite set A, defined in (3.18). We show first that the sum of its absolute differences along edges starting in a ball is at most a constant times the radius.

Lemma 6.2 *For each integer $d \geq 2$, there is a constant $C = C(d) < \infty$ with the following property. Let $A \subset \mathbb{Z}^d$ be finite, let $x \in A$, let $p \in \mathbb{Z}^d$, and let $s \geq 1$ be real. Then*

$$\sum_{y \in \mathrm{B}^{::}(p,s)} \sum_{z \sim y} |g_A^{::}(x,y) - g_A^{::}(x,z)| \leq Cs\,. \tag{6.2}$$

Proof. The idea is to reflect the Green function in a plane perpendicular to a coordinate axis, halfway between two adjacent sites. By summing the Laplacian of the difference against its sign on one side, we bound the sum of the differences across the plane by the sum of the absolute values of the Laplacian of the Green function, which is two, and a ball of radius s meets at most $2s + 1$ parallel planes.

Step 1: We compute the sum of the absolute values of the Laplacian. Let $f(v) := g_A^{::}(x, v)$ for $v \in \mathbb{Z}^d$, and let $\nu := -\Delta^{::} f$ on $\mathbb{Z}^d$. By (3.19), $\nu = \delta_x$ on A. Outside A we have $f = 0$, and $f \geq 0$ everywhere, so $\nu(v) = -\sum_{w\sim v} f(w) \leq 0$ for $v \notin A$. Since f has finite support, each edge contributes two terms of opposite signs to $\sum_v \Delta^{::} f(v)$, so $\sum_v \nu(v) = 0$. Hence

$$\sum_{v \in \mathbb{Z}^d} |\nu(v)| = 2\,.$$

Step 2: Fix $1 \leq i \leq d$ and an integer k. We show that

$$\sum_{v \in \mathbb{Z}^d:\, v_i = k} |f(v) - f(v + \mathbf{e}_i)| \leq 1\,. \tag{6.3}$$

Let $\mathcal{R}v := v + (2k + 1 - 2v_i)\mathbf{e}_i$ be the reflection of v in the plane $\{v_i = k + \frac{1}{2}\}$ (Figure 6.3), let $Y := \{v \in \mathbb{Z}^d : v_i \leq k\}$, and let $\varphi(v) := f(v) - f(\mathcal{R}v)$.

Reflection commutes with $\Delta^{::}$, so $-\Delta^{::}\varphi = \nu - \nu \circ \mathcal{R}$, and $\varphi \circ \mathcal{R} = -\varphi$. We multiply $-\Delta^{::}\varphi(v)$ by $\operatorname{sgn}\varphi(v)$, with $\operatorname{sgn} 0 = 0$, and sum over $v \in Y$ (all these sums have finitely many nonzero terms). Each edge $\{v, w\}$ with both endpoints in Y contributes

$$(\varphi(v) - \varphi(w))(\operatorname{sgn}\varphi(v) - \operatorname{sgn}\varphi(w)) \geq 0\,.$$

Every other edge at a site $v \in Y$ joins v, with $v_i = k$, to $v + \mathbf{e}_i = \mathcal{R}v$, and contributes $\operatorname{sgn}\varphi(v)(\varphi(v) - \varphi(\mathcal{R}v)) = 2|\varphi(v)|$. Therefore

$$2\sum_{v_i=k}|\varphi(v)| \leq \sum_{v\in Y}\operatorname{sgn}\varphi(v)(\nu(v) - \nu(\mathcal{R}v)) \leq \sum_{v\in Y}(|\nu(v)| + |\nu(\mathcal{R}v)|) = 2\,,$$

where the equality uses that Y and its reflection partition $\mathbb{Z}^d$. Since $\varphi(v) = f(v) - f(v + \mathbf{e}_i)$ when $v_i = k$, this proves (6.3).

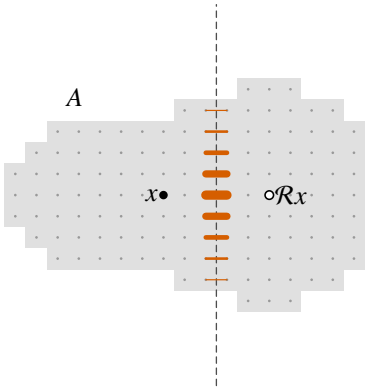


Figure 6.3: Lemma 6.2 in $\mathbb{Z}^2$: for $f = g^{::}_A(x, \cdot)$ on the set A (shaded), the differences $|f(v) - f(v + \mathbf{e}_1)|$ across the line $\{v_1 = k + \frac{1}{2}\}$ (dashed) are drawn as segments of width proportional to them, and the reflection $\mathcal{R}$ in this line maps x to $\mathcal{R}x$ (hollow dot). By (6.3) the differences sum to at most 1; here they sum to 0.22.

Step 3: The ball $\mathrm{B}^{::}(p, s)$ meets at most $2s + 1$ of the planes $\{v_i = k\}$, so (6.3) gives

$$\sum_{v\in \mathrm{B}^{::}(p,s)}|f(v) - f(v + \mathbf{e}_i)| \leq 2s + 1\,.$$

Replacing v by $v - \mathbf{e}_i$ and p by $p - \mathbf{e}_i$ gives the same bound for the differences $f(v) - f(v - \mathbf{e}_i)$. Summing over the $2d$ neighbors of v shows that the left side of (6.2) is at most $2d(2s + 1) \leq 6ds$, since $s \geq 1$. This proves the lemma with $C = 6d$. □

We next bound the sum over all edges when A lies in a ball. The ball lies in the half-space bounded by the tangent plane at the point of the sphere nearest

x, and simple random walk started at x is likely to leave this half-space (and therefore A) before traveling too far from x.

Lemma 6.3 *For each integer $d \geq 2$, there is a constant $C = C(d) < \infty$ with the following property. Let $R > 0$ be real, let $A \subset \mathrm{B}^{\cdot\cdot}(\mathbf{0}, R)$, let $x \in A$, and let $s := 1 + R - |x|$. Then*

$$\sum_{y \in A} \sum_{z \sim y} |g_A^{\cdot\cdot}(x, y) - g_A^{\cdot\cdot}(x, z)| \leq Cs \left(1 + \log \frac{R+1}{s}\right).$$

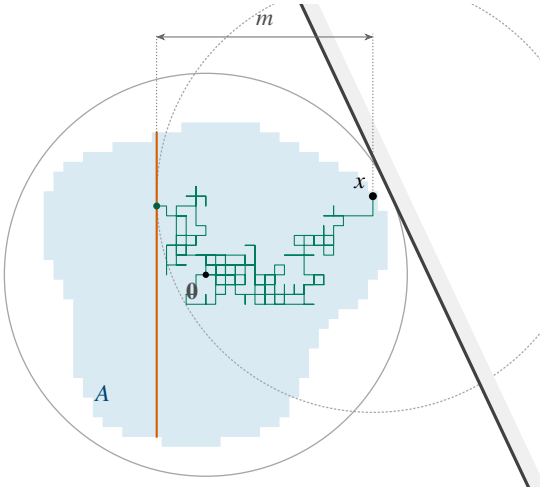


Figure 6.4: The proof of Lemma 6.3 in $\mathbb{Z}^2$: the set A (shaded) lies in the ball of radius R, hence in the half-plane bounded by the tangent line (thick) at the point of the circle nearest x, at distance $s - 1$ from x. A walk from x that reaches the line at distance m from x (vermilion) before leaving A has left $\mathrm{B}^{\cdot\cdot}(x, m)$ (dotted) by the time it leaves this half-plane, which by Lemma 3.22 has probability at most $2ds/m$.

Proof. We bound the differences across each plane perpendicular to a coordinate axis by the probability that the walk from x reaches the plane before leaving A, and then sum over planes.

Fix $e \in \{\pm \mathbf{e}_1, \ldots, \pm \mathbf{e}_d\}$ and an integer $m \geq 0$. Let τ_L be the hitting time of the plane $L := \{v : e \cdot (v - x) = m\}$. Since the coordinate $e \cdot (X_j - x)$ starts at 0 and changes by at most one at each step, all visits to a site v of L and to $v + e$ occur at or after time τ_L. By the strong Markov property at time τ_L, for $w \in \{v, v + e\}$ the value $g_A^{\cdot\cdot}(x, w)$ is the expectation of $g_A^{\cdot\cdot}(X_{\tau_L}, w)$ restricted to the event $\{\tau_L < \tau_{\partial A}\}$, on which $X_{\tau_L} \in A$. The edges $\{v, v + e\}$ with $v \in L$ join L to the parallel plane $L + e$, so by (6.3) the sum of $|g_A^{\cdot\cdot}(y, v) - g_A^{\cdot\cdot}(y, v + e)|$ over $v \in L$ is at most 1 for every $y \in A$. Subtracting the identities for v and

$v + e$ and using this bound with $y = X_{\tau_L}$ gives

$$\sum_{v \in L} |g_A^{::}(x, v) - g_A^{::}(x, v + e)| \le \mathbb{P}_x\{\tau_L < \tau_{\partial A}\}\,.$$

Let $m \ge 1$. Choose a unit vector b with $b \cdot x = |x|$ (take $b = \mathbf{e}_1$ if $x = \mathbf{0}$). Every $y \in A$ satisfies $b \cdot (y - x) \le |y| - |x| < s - 1$, so A lies in the half-space $\{y : b \cdot (y - x) \le s - 1\}$. The plane L is at distance at least m from x, so on $\{\tau_L < \tau_{\partial A}\}$ the walk leaves $\mathrm{B}^{::}(x, m)$ by the time it leaves this half-space (see Figure 6.4). By Lemma 3.22, applied with x translated to the origin, ball radius m and half-space $\{y : b \cdot y \le s - 1\}$, this has probability at most $ds(m + 1)/m^2 \le 2ds/m$.

The two orientations $\{v, v + e\}$ of an edge have integer values of $e \cdot (v - x)$ with sum -1, so exactly one is nonnegative, and every edge occurs in exactly one of the sums above. Both endpoints of an edge in the sum for m are at distance at least m from x, and $|y - x| < 2R$ for $y \in A$, so the terms with $m \ge 2R$ vanish. Each edge occurs at most twice in the sum in the lemma, and there are $2d$ directions e. Since $1 < s \le R + 1$, bounding the terms with $m \le \lceil s \rceil$ by 1 and the others by $2ds \int_{m-1}^{m} \mathrm{d}t/t$ shows that the sum in the lemma is at most

$$4d\Big(1 + \lceil s \rceil + 2ds \log \frac{2(R + 1)}{s}\Big) \le Cs\Big(1 + \log \frac{R + 1}{s}\Big)\,.$$

□

We now test Proposition 6.1 against the killed Green function of the ball $\mathrm{B}^{::}(\mathbf{0}, r - 2)$ to bound the odometer from below. The bound holds for every finite sequence of stack moves that leaves at most one particle at each site of this ball, whether or not the moves are legal, so we state it in that generality.

Lemma 6.4 *For each integer $d \ge 2$, there is a constant $K = K(d) < \infty$ with the following property. Let $n \ge 2$ be an integer, let ρ be a family of stacks of instructions on $\mathbb{Z}^d$, and let $b_\rho \ge 1$ be a real number satisfying*

$$|\mathcal{D}_\rho(x, y, m)| \le b_\rho \qquad \text{for every edge } (x, y) \text{ and every integer } m \ge 0\,.$$

Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, let $r := (n/\omega_d)^{1/d}$, and suppose that $r > 2$. Suppose that finitely many stack moves using ρ convert $n\delta_{\mathbf{0}}$ into a configuration with at most one particle at each site of $\mathrm{B}^{::}(\mathbf{0}, r - 2)$, and let v be their odometer. Then

$$v(z) \ge \frac{(r - |z|)^2}{4} - Kb_\rho(r - |z|) \log n \qquad \text{for every } z \in \mathrm{B}^{::}(\mathbf{0}, r - 2)\,.$$

Proof. Let $B := \mathrm{B}^{::}(\mathbf{0}, r - 2)$, let ξ be the final configuration of the moves, and fix $z \in B$. We apply Proposition 6.1 with $\sigma_0 = n\delta_{\mathbf{0}}$, with ξ, and with

$f(x) := g_B^{::}(x, z)$, where $g_B^{::}$ is the killed Green function (3.18) of B. Note that f is nonnegative, vanishes off B, and satisfies $\Delta^{::} f = -\delta_z$ on B. In the proof of Proposition 6.1 we computed the change in weight caused by the moves at each site x. If $x \notin B$, then $f(x) = 0$, so these moves can only add weight. Let

$$E_z := \sum_{x \in B} \sum_{y \sim x} |f(y) - f(x)| .$$

The moves at sites $x \in B$ contribute $-v(z)/(2d)$ in total through the first term, and at most $b_\rho E_z$ in absolute value through the second. Since $f \geq 0$ vanishes off B and $\xi \leq 1$ on B, it follows that

$$-\frac{v(z)}{2d} - b_\rho E_z \leq \sum_x f(x)(\xi(x) - n\delta_{\mathbf{0}}(x)) \leq \sum_{x \in B} f(x) - n f(\mathbf{0}) .$$

Writing $F(z) := n g_B^{::}(\mathbf{0}, z) - \sum_{x \in B} g_B^{::}(x, z)$, we conclude that

$$v(z) \geq 2d\, F(z) - 2d\, b_\rho E_z . \tag{6.4}$$

We bound the two terms in turn. By (3.19), the function F satisfies $\Delta^{::} F = 1 - n\delta_{\mathbf{0}}$ in B and $F = 0$ on ∂B, so by Lemma 5.3 of Chapter 5, applied to the ball $B = \mathrm{B}^{::}(\mathbf{0}, r - \eta)$ with $\eta = 2$, there is a constant $C_1 = C_1(d)$ such that

$$F(z) \geq \frac{(r - |z|)^2}{8d} - C_1 .$$

Let $\ell := r - |z|$, so that $\ell > 2$. By symmetry of $g_B^{::}$, we may apply Lemma 6.3 with $A = B$, with $r - 2$ in place of R, and with z in place of x, hence with $s = \ell - 1$, which gives

$$E_z \leq C_2(\ell - 1)\Big(1 + \log \frac{r - 1}{\ell - 1}\Big) \leq C_2\, \ell\, (1 + \log r) \leq C_3\, \ell \log n ,$$

because $\ell - 1 > 1$ and $n \geq 2$, where the last step uses $r = (n/\omega_d)^{1/d}$. Substituting both bounds into (6.4) gives $v(z) \geq \ell^2/4 - 2dC_1 - 2dC_3 b_\rho \ell \log n$. Since $b_\rho \geq 1$ and $\ell > 2$, and since $\log n \geq \log 2$, the constant $2dC_1$ is at most a constant times $b_\rho \ell \log n$, and this proves the lemma. □

How far must z be from the sphere of radius r in order to ensure that the quadratic main term in Lemma 6.4 is larger than the error term? Let $K = K(d)$ be the constant of the lemma, let $C := \max\{4K, 2/\log 2\}$, and consider finitely many stack moves using ρ that convert $n\delta_{\mathbf{0}}$ into a configuration with at most one particle at each site of $\mathrm{B}^{::}(\mathbf{0}, r - 2)$, with odometer v. If $z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n)$, then $\ell := r - |z| > Cb_\rho \log n \geq 2$, so $r > 2$ and $z \in \mathrm{B}^{::}(\mathbf{0}, r - 2)$, and the lemma gives $v(z) \geq \ell(\ell/4 - Kb_\rho \log n) > 0$. Hence

$$v(z) > 0 \qquad \text{for every } z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) . \tag{6.5}$$

Now we finish the proof of the inner bound.

THEOREM 6.5 *Let $d \geq 2$ and $n \geq 2$ be integers. Let ρ be a family of stacks of instructions on $\mathbb{Z}^d$, and let $b_\rho \geq 1$ be a real number satisfying*

$$|\mathcal{D}_\rho(x, y, m)| \leq b_\rho \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0\,.$$

Let A_n^{R} be the aggregate obtained from $n\delta_{\mathbf{0}}$ using ρ. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (n/\omega_d)^{1/d}$. Then there is a constant $C = C(d) < \infty$ such that

$$\mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) \subset A_n^{\mathrm{R}}\,.$$

Proof. A complete legal sequence of stack moves using ρ is finite and leaves at most one particle at each site, so (6.5) gives $v_n(z) > 0$ for every $z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n)$. Hence a move is made at each such site, and a site at which a legal move is made is occupied in the final aggregate, so $\mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) \subset A_n^{\mathrm{R}}$. □

It remains to bound the aggregate from outside.

6.3 OUTER BOUND

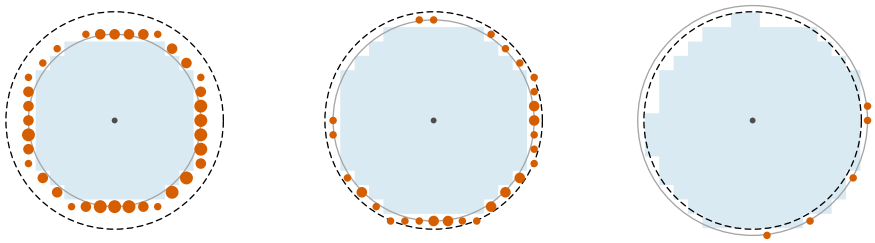
For rotor stacks, which have discrepancy less than one, the aggregate contains the ball of radius $r - C \log r$ by Theorem 6.5, so at most $Cr^{d-1} \log r$ of the n particles stop outside $\mathrm{B}^{::}(\mathbf{0}, r)$. We rule out "thin tentacles" formed by these particles by freezing the particles on a sphere and enlarging the sphere one unit at a time. We test Proposition 6.1 against the Green function of the occupied set with its pole at a site of frozen particles, so there is no odometer term, and we compare with the divisible sandpile of the same mass, whose odometer vanishes near the sphere by Theorem 4.7. It follows that at most $C \log r$ particles are frozen at each site. Each time the radius increases by one, at least one particle of each pile stays inside, so the number of frozen particles is at most $1 - 1/(C \log r)$ times the preceding number, and after $C(\log r)^2$ steps none is left.

THEOREM 6.6 *For each integer $d \geq 2$, there are constants $C = C(d) < \infty$ and $r_0 = r_0(d) < \infty$ with the following property. Let $n \geq 1$ be an integer, let ω_d be the volume of the unit ball in $\mathbb{R}^d$, let $r := (n/\omega_d)^{1/d}$, and suppose that $r \geq r_0$. Then for every initial rotor configuration and every rotor mechanism on $\mathbb{Z}^d$, the rotor-router aggregate A_n^{R} formed from n particles started at $\mathbf{0}$ satisfies*

$$\mathrm{B}^{::}(\mathbf{0}, r - C \log r) \subset A_n^{\mathrm{R}} \subset \mathrm{B}^{::}(\mathbf{0}, r + C(\log r)^2)\,. \tag{6.6}$$

Proof. The inner inclusion follows from Theorem 6.5, because each rotor stack has discrepancy less than one and $\log n = d \log r + O(1)$. We prove the outer inclusion in three steps. Throughout, C denotes a constant that depends only on d and may change from line to line, r is large, and $B_s := \mathrm{B}^{::}(\mathbf{0}, s)$.

Step 1: We freeze the particles on spheres of increasing radius. Let $c = c(d)$ be the constant in Theorem 4.7, and let $t_0 := \lceil r + c \rceil$. Starting from $n\delta_{\mathbf{0}}$, for each integer s with $t_0 \le s \le 2r$ in turn, make legal moves at sites of B_s until every site of B_s has at most one particle. This is possible because we may move the particles one at a time, and a particle that follows the successive arrows of the stacks leaves every finite set (Section 6.2). Let η_s be the resulting configuration, and let N_s be the number of particles outside B_s, all of which lie on ∂B_s, because every move so far is made at a site of B_s. A legal move leaves at least one particle at its site, so every site at which a move has been made is occupied in η_s. (In Figure 6.5 the freezing starts inside the sphere of radius r, so that the piles are visible.)



(a) (b) (c)

Figure 6.5: The configurations η_s in the proof of Theorem 6.6 for $s = 6, 7, 8$, for $n = 180$ particles in $\mathbb{Z}^2$ with the clockwise mechanism and all rotors initially North, and $r \approx 7.57$ (dashed circle): the occupied sites of B_s are shaded, and the frozen particles sit in piles on ∂B_s, just outside the circle of radius s, drawn as dots of area proportional to their size. When s increases by one, at least one particle of each pile stays in B_{s+1}, as in Step 3, so the 71 frozen particles become 35 and then 5.

Step 2: We show that $\eta_s(y) \le C \log r$ for every $y \in \partial B_s$. Let u and σ be the odometer and the stable configuration of the divisible sandpile started from $n\delta_{\mathbf{0}}$. By Theorem 4.7(i) and Lemma 4.4(i), the odometer u vanishes outside B_{r+c}, and $0 \le \sigma \le 1$.

Fix $y \in \partial B_s$ with $\eta_s(y) \ge 1$, let U be the set of sites x with $\eta_s(x) \ge 1$, and let $G := g_U^{::}(y, \cdot)$, which is nonnegative and vanishes outside U (Figure 6.6). By (3.19), $\Delta^{::} G = -\delta_y$ in U. Every site at which a move has been made lies in U and differs from y, because $y \notin B_s$, so $\Delta^{::} G$ vanishes at each such site. Since $U \subset B_{s+1}$ and $s \le |y| < s + 1$, by Lemma 6.3 with $A = U$, $R = s + 1$ and y in place of x we have

$$\sum_{x \in U} \sum_{z \sim x} |G(x) - G(z)| \le C(1 + \log(s + 2)) \le C \log r \,.$$

Since $\Delta^{::}G$ vanishes where moves were made, Proposition 6.1, applied to $n\delta_{\mathbf{0}}$, η_s and G, has no odometer term, and since the rotor stacks have discrepancy less than one, we obtain

$$\sum_x G(x)\eta_s(x) \le nG(\mathbf{0}) + C\log r\,.$$

On the other hand, $\sigma = n\delta_{\mathbf{0}} + \Delta^{::}u$ and u has finite support, so summation by parts gives

$$\begin{aligned}\sum_x G(x)\sigma(x) &= nG(\mathbf{0}) + \sum_x u(x)\Delta^{::}G(x)\\ &= nG(\mathbf{0}) - u(y) + \sum_{x\notin U} u(x)\Delta^{::}G(x) \ge nG(\mathbf{0})\,,\end{aligned}$$

because $u(y) = 0$ by $|y| \ge s \ge r + c$, and because $u \ge 0$ and $\Delta^{::}G(x) \ge 0$ for $x \notin U$, where $G(x) = 0$. Subtracting, we obtain

$$\sum_x G(x)(\eta_s(x) - \sigma(x)) \le C\log r\,.$$

Every term of this sum is nonnegative, because G vanishes outside U and $\eta_s \ge 1 \ge \sigma$ on U. In particular, $(\eta_s(y) - 1)G(y) \le C\log r$. Counting only the visit at time zero in (3.18), we have $G(y) \ge 1/(2d)$. Hence $\eta_s(y) \le C_1 \log r$ for a constant $C_1 = C_1(d)$.

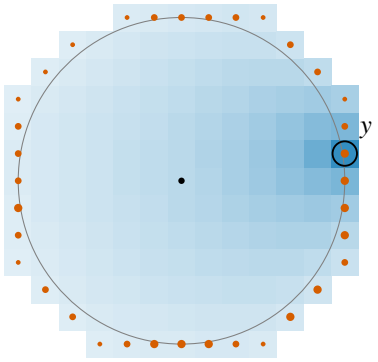


Figure 6.6: Step 2 of the proof of Theorem 6.6 for the configuration η_6 of Figure 6.5: the sites of $U = \{\eta_6 \ge 1\}$ are shaded by $G = g^{::}_U(y,\cdot)$ for a pile at a site y (ringed) of ∂B_6. Since G is harmonic at every site where a move was made, Proposition 6.1 tested against G has no odometer term, and this bounds the number of particles at y.

Step 3: We show that the number of frozen particles decreases geometrically. Let P_s be the number of sites of ∂B_s at which η_s is positive, so that $N_s \le C_1 P_s \log r$ by Step 2. Let $s+1 \le 2r$. The configuration η_{s+1} is obtained from η_s by legal moves at sites of B_{s+1}, and $\partial B_s \subset B_{s+1}$. A legal move leaves

at least one particle at its site, and $\eta_s \le 1$ on B_s, so η_{s+1} is positive at $n - N_s$ sites of B_s and at P_s sites of ∂B_s. Since $\eta_{s+1} \le 1$ on B_{s+1}, at least $n - N_s + P_s$ particles lie in B_{s+1}, and therefore

$$N_{s+1} \le N_s - P_s \le \Big(1 - \frac{1}{C_1 \log r}\Big) N_s \,.$$

Let $k := \lceil C_1 \log r \log n \rceil + 1$ and $t_1 := t_0 + k$, so that $t_1 \le 2r$ for large r. Iterating the last inequality gives

$$N_{t_1} \le n\Big(1 - \frac{1}{C_1 \log r}\Big)^k \le n e^{-k/(C_1 \log r)} < 1 \,,$$

so $N_{t_1} = 0$. Then every site has at most one particle, so the moves made form a complete legal sequence, whose final configuration is $\mathbf{1}_{A_n^{\mathrm{R}}}$ by Corollary 5.2. Hence $A_n^{\mathrm{R}} \subset B_{t_1}$. Since $t_1 \le r + c + 3 + C_1 \log r \log n \le r + C(\log r)^2$, this proves the outer inclusion. □

In particular, by (6.6), the rescaled aggregates $r^{-1} A_n^{\mathrm{R}}$ converge to the closed unit ball in Hausdorff distance, with relative error $O((\log r)^2/r)$ (see Figure 6.7 for an aggregate in $\mathbb{Z}^3$). In Chapter 7 we start particles at two sources in $\mathbb{Z}^2$, where the limit shape need not be a disk.

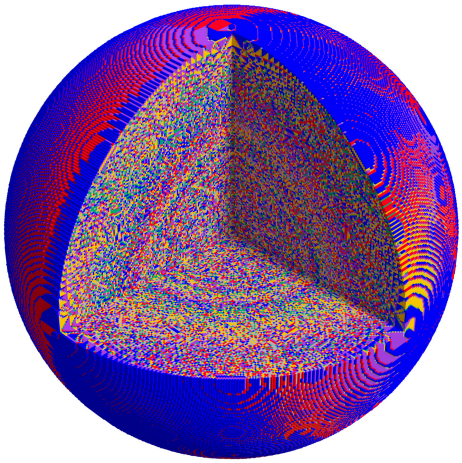


Figure 6.7: Rotor-router aggregation of $n = 10^7$ particles started at the origin of $\mathbb{Z}^3$, with the octant $\{x_1, x_2, x_3 > 0\}$ removed and each site colored by the final direction of its rotor (North, East, South, West, Up, Down: blue, purple, red, gray, gold, green); the rotors turn in the order North, South, East, West, Up, Down, and all start North. The aggregate lies between the spheres of radii 132.6 and 134.7 about the origin, close to the sphere of radius $r = (n/\omega_3)^{1/3} \approx 133.7$.

EXERCISES

Exercise 6.1. Let $d \geq 1$ be an integer, let $x \in \mathbb{Z}^d$, let $\rho_1(x), \rho_2(x), \ldots$ be a stack at x, each a directed edge from x to one of its neighbors, and let $\mathcal{D}_\rho(x, y, m)$ be its discrepancy, defined in (5.1).

(a) Show that $\max_{y\sim x} \mathcal{D}_\rho(x, y, 1) = 1 - 1/(2d)$ for every stack.

(b) Show that a stack that runs through the $2d$ neighbors of x in a fixed cyclic order minimizes $\sup_{m\geq 1} \max_{y\sim x} |\mathcal{D}_\rho(x, y, m)|$ among all stacks.

(c) For a stack of independent arrows, each pointing to a uniformly random neighbor, and for every neighbor y of x, show that $\mathbb{E}\,\mathcal{D}_\rho(x, y, m) = 0$ and

$$\mathbb{E}[\mathcal{D}_\rho(x, y, m)^2] = \frac{m(2d-1)}{(2d)^2}$$

for every integer $m \geq 1$, and that $\sup_{m\geq 1} |\mathcal{D}_\rho(x, y, m)| = \infty$ almost surely.

Exercise 6.2. This exercise is the abelian property, Corollary 5.2, for rotor stacks. Let $d, n \geq 1$ be integers. Fix rotor stacks on $\mathbb{Z}^d$, and start n particles at $\mathbf{0}$. Use Lemma 5.1 to show that the aggregate and the final rotor configuration do not depend on the order in which the particles are released or on which active particle is advanced at each step.

Exercise 6.3. Let $d \geq 1$ be an integer, let $A \subset \mathbb{Z}^d$ be finite, and let $\sigma_0 : \mathbb{Z}^d \to \mathbb{N}$ be supported in A. Fix stacks of instructions ρ on A. Denote by $v_A^\rho(z)$ the number of visits to $z \in A$ if $\sigma_0(x)$ particles start at each site x and follow the successive arrows of the stacks until exiting A, and suppose that this ends after finitely many moves. Denote by $v_A(z)$ the expected number of visits to z if they instead perform independent simple random walks stopped on exiting A. Use Proposition 6.1 to show that, for every $z \in A$,

$$v_A^\rho(z) - v_A(z) = 2d \sum_{x\in A} \sum_{y\sim x} \mathcal{D}_\rho(x, y, v_A^\rho(x))(g_A^{\cdot\cdot}(y, z) - g_A^{\cdot\cdot}(x, z)) .$$

Taking absolute values, in the case of bounded discrepancy we obtain a bound on $|v_A^\rho(z) - v_A(z)|$ that does not depend on the number of particles.

Exercise 6.4. Theorem 6.6 assumes $d \geq 2$; in one dimension the aggregate can be found exactly. Consider rotor-router aggregation on $\mathbb{Z}$ with rotor mechanism (East, West) and with all rotors initially pointing West. For every integer $n \geq 1$, show that A_n^{R} is an interval containing the origin, and determine A_n^{R} and the odometer v_n. How do the formulas change if all rotors initially point East?

NOTES

Rotor walk burst onto the scene in 1996 from two different directions: the rule ANT-WALK-1 of Wagner, Lindenbaum, and Bruckstein (1996, Section 2.1), in which a robot leaves each vertex along its least recently used edge, and the "Eulerian walkers" of Priezzhev, Dhar, Dhar, and Krishnamurthy (1996). The model was rediscovered several times: by Rabani, Sinclair, and Wanka (1998), whose balancers in load balancing circuits send successive tokens alternately along two wires, as a rotor with two outgoing edges does, and by Dumitriu, Tetali, and Winkler (2003), whose round-robin "whirling tours" on finite trees have length equal to an expected hitting time of random walk (their Theorem 9.2).

Jim Propp proposed rotor-router aggregation in $\mathbb{Z}^2$ around 2001 as a way of derandomizing internal DLA, as recounted by Levine and Peres (2008), who define it in $\mathbb{Z}^d$ by cycling each rotor through the $2d$ directions; the date is from Section 3 of Jerison, Levine, and Sheffield (2012). Levine and Peres (2009) recount that Propp observed from simulations that the two-dimensional aggregates are extremely close to circular, and asked why; Kleber (2005) shows such simulations. The first result on circularity is Theorem 1.1 of Levine and Peres (2008), which proved it in a weak sense: the rescaled unions of the closed unit cubes centered at the sites of A_n^{R} converge to the ball in the sense that the volume of the symmetric difference tends to zero. The proof used an isoperimetric inequality for the expected exit time of random walk. Fey and Redig (2008) also show that the aggregate contains a diamond (their Corollary 4.9, specialized to the rotor-router aggregate studied here). These results did not rule out the possibility of holes far from the boundary or of long tendrils extending far beyond the ball, provided the volume of these features is negligible compared to n.

Theorem 1.1 of Levine and Peres (2009) does rule out such holes and tendrils. It gives the inner inclusion of Theorem 6.6 and an outer error of order $r^{1-1/d} \log r$. They remark, after their Theorem 1.1 and after their Lemma 5.1, that the theorem also holds for stacks of bounded discrepancy; Theorem 6.5 makes the dependence of the inner bound on the discrepancy explicit. Levine and Peres (2009) credit Matt Cook with the idea of studying the shape through the odometer. Theorem 4.1 of Levine and Peres (2017) improves the outer error to order $\log r$. In its proof they first show that for every occupied site z and every real t with $0 < t < |z|$, at least ct^d sites within distance t of z are occupied (their Proposition 4.2). They then test the aggregate against a discrete harmonic function that is nonnegative near the ball and decays rapidly away from a given site just outside it, a variant of the functions of Lemmas 7–8 of Jerison, Levine, and Sheffield (2012) and Lemmas 5–7 of Jerison, Levine, and Sheffield (2013), and combine the inner

bound, the mean value property of that function, and the Holroyd–Propp bound (their Lemma 4.11): for a function h harmonic on A_n^{R}, the sum of h over A_n^{R} differs from $nh(\mathbf{0})$ by at most the sum of $|h(x) - h(y)|$ over $x \in A_n^{\mathrm{R}}$ and $y \sim x$.

Proposition 6.1 is a many-particle form, for simple random walk on $\mathbb{Z}^d$, of the identity for a single stack walk in Proposition 21 of Holroyd and Propp (2010).

Levine and Peres (2009) prove the outer bound in their Section 5.2 by showing that the number of particles that reach successive spheres halves at most $C \log r$ times before it drops below a constant. By the inner bound, at most $Cr^{d-1} \log r$ sites outside the sphere of radius r are occupied, and they bound the total number of shells between successive halvings by Jensen's inequality, which gives their outer error $r^{1-1/d} \log r$. In our proof of the outer bound in Theorem 6.6 we instead bound each pile of frozen particles by $C \log r$ (Step 2), so the number of frozen particles decreases geometrically and reaches zero after $C(\log r)^2$ steps.

Lemmas 6.2 and 6.3 extend Lemmas 5.3 and 5.6 of Levine and Peres (2009), which concern the Green function of a ball. The randomized stacks of Friedrich and Levine (2013, Section 6.4) on $\mathbb{Z}^2$ use independent uniform permutations of the four neighbors in successive blocks of length four, and therefore have bounded discrepancy.

Friedrich and Levine (2013) asked whether the difference between the largest distance from the origin of an occupied site and the smallest distance of an unoccupied site stays bounded as n grows. In their simulations in two dimensions, with the counterclockwise mechanism, it was between 1.3 and 1.7 for $n = 2^{10}, 2^{12}, \ldots, 2^{36}$ (their Section 6.2 and Table 1). Monte Carlo simulations of a single Eulerian walker on $\mathbb{Z}^2$, started from random initial rotors, appear in Kapri and Dhar (2009).

Chapter Seven

Internal Aggregation in $\mathbb{Z}^2$: Two Sources

What if the particles start at two different points? Start n particles at the origin and another n at a lattice site at distance of order $\sqrt{n}$ from it. The particles from either source alone would fill a disk of area about n. If the sources are far apart, then these disks are disjoint, and the particles from one source hardly interact with those from the other. On the other hand, if the sources are very close together relative to $\sqrt{n}$, then the cluster looks like a single disk of area $2n$. In between these extreme cases the two disks overlap, and the particles must spread beyond their union. The first main result of this chapter, in Section 7.1, is that the divisible sandpile, rotor-router aggregation and internal DLA all have the same limit shape after scaling by $\sqrt{n}$: a disk of area two if the sources coincide, two disks of area one if they are far apart, and in between an oval bounded by a quartic curve. This extends the single-source shape theorems of the preceding three chapters.

The second main result, in Section 7.2, is that the logarithmic potential of the two sources minus that of the uniform measure on the shape is positive inside the shape and vanishes together with its gradient outside. (In the language of complex analysis, the limit shape is a quadrature domain: the integral over it of every function harmonic near its closure is the sum of the values of the function at the two sources.) We compute this potential from the complex moments of the shape, compare it with the odometer of the divisible sandpile (Section 7.3), and then turn the odometer bounds into bounds for rotor-router aggregation (Section 7.4) by a deterministic argument and for internal DLA (Section 7.5) by a probabilistic one.

7.1 THE SHAPE THEOREM

We identify $\mathbb{R}^2$ with $\mathbb{C}$ and write $|E|$ for the area of a set E and $\#A$ for the number of lattice sites in a set A. Let $r_0 := \pi^{-1/2}$ be the radius of a disk of area one. For two sources at $\pm a$ on the real axis (Figure 7.1), with $a \geq 0$, define

$$O_a := \begin{cases} \{re^{\mathbf{i}\theta} : \theta \in \mathbb{R} \text{ and } 0 \leq r < \sqrt{2r_0^2 + 2a^2 \cos 2\theta}\}, & 0 \leq a < r_0, \\ \mathrm{B}(-a, r_0) \cup \mathrm{B}(a, r_0), & a \geq r_0 \end{cases}.$$

For sources at $\mathbf{0}$ and $q \in \mathbb{C} \setminus \{\mathbf{0}\}$, we rotate and translate this shape, letting $\Omega(q) := q/2 + (q/|q|)O_{|q|/2}$, and let $\Omega(\mathbf{0}) := \mathrm{B}(\mathbf{0}, \sqrt{2}r_0)$. These are open sets, so the point of contact of tangent disks is excluded. For $\epsilon > 0$, the inner and outer ϵ-neighborhoods of an open set Ω are

$$\Omega_\epsilon := \{z \in \Omega : \mathrm{dist}(z, \mathbb{R}^2 \setminus \Omega) > \epsilon\} \quad \text{and} \quad \Omega^\epsilon := \{z : \mathrm{dist}(z, \Omega) < \epsilon\}.$$

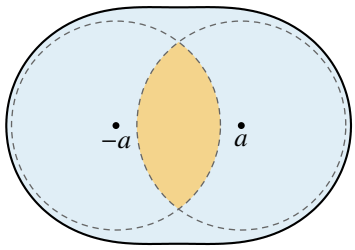


Figure 7.1: The limit shape O_a for $a = 0.6r_0$ (black) and the disks of radius r_0 about the sources $\pm a$ (dashed), which it contains by Lemma 7.4(ii). It has area two, so it exceeds their union by the area of their overlap (orange).

For each positive integer n, we start all three models from $\sigma_n := n\delta_{\mathbf{0}} + n\delta_{b_n}$, where $b_n \in \mathbb{Z}^2$. Let u_n be the odometer of the divisible sandpile (the mass sent to each neighbor), let $\xi_n := \sigma_n + \Delta^{\cdot\cdot} u_n$ be its final mass, and let $S_n := \{u_n > 0\}$ be its toppled set. By Lemma 4.4, S_n is finite and $\xi_n = 1$ on S_n, and $0 \le \xi_n \le 1$ since the final configuration is stable and $\sigma_n \ge 0$. Let A_n^{R} be the rotor-router aggregate and A_n the internal DLA cluster, both of $2n$ sites. Every rotor stack is periodic and cycles once through the four neighbors, as in Section 6.1, in an order and from a first arrow that may vary from site to site, and by the abelian property, Corollary 5.2, the final aggregate does not depend on the order of legal moves.

The following theorem identifies the common limit.

Theorem 7.1 *Let $p \in \mathbb{R}^2$, and let $b_n \in \mathbb{Z}^2$ satisfy $b_n/\sqrt{n} \to p$ as $n \to \infty$. Then the following hold.*

(i) For every $\epsilon > 0$ and all sufficiently large n,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset \{\xi_n = 1\} \subset \{\xi_n > 0\} \subset \sqrt{n}\Omega(p)^\epsilon.$$

(ii) For every $\epsilon > 0$ and all sufficiently large n, uniformly over the rotor stacks,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset A_n^{\mathrm{R}} \subset \sqrt{n}\Omega(p)^\epsilon.$$

(iii) Under every coupling of the clusters $(A_n)_{n\ge1}$, with probability one, for every $\epsilon > 0$ and all sufficiently large n,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset A_n \subset \sqrt{n}\Omega(p)^\epsilon.$$

If $p = \mathbf{0}$, the limit is the disk of area two, and if $|p| \geq 2r_0$, it consists of two disks of area one. For $0 < |p| < 2r_0$ its boundary is the quartic curve (7.1) of Section 7.2 with $a = |p|/2$, in coordinates centered at $p/2$ with the real axis through both sources (see Figure 7.2 for internal DLA in this regime, Figure 1.13 for all three models, and Figure 7.3 for their boundaries near that of the limit shape).

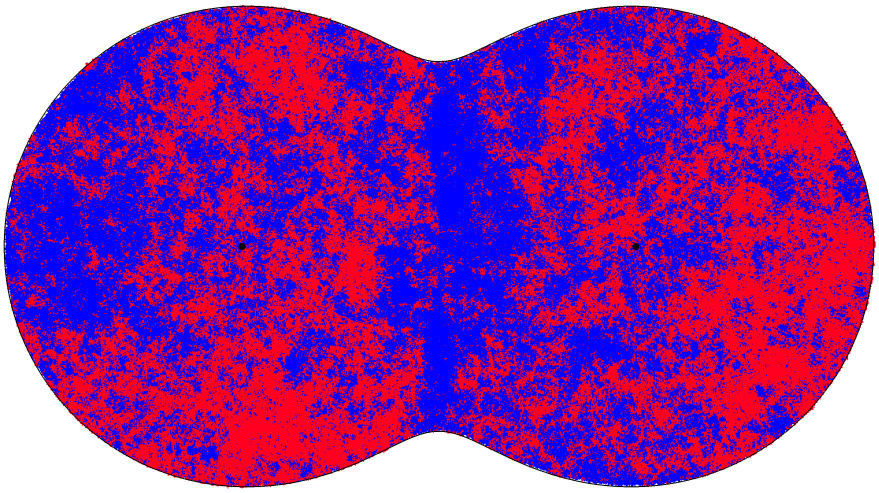


Figure 7.2: Internal DLA from $n = 282{,}743$ particles at each of the sites $(\pm 250, 0)$ (black dots), with the site of the jth particle red if it lies outside $\sqrt{t}\,O_{250/\sqrt{t}}$, the limit shape of $t = j/2$ particles at each source (Theorem 7.2), and blue otherwise. The black curve is the boundary of the limit shape for $t = n$, the quartic $(x^2 + y^2)^2 - \frac{2n}{\pi}(x^2 + y^2) - 2 \cdot 250^2(x^2 - y^2) = 0$ in lattice coordinates, which is (7.1) with $a = 250/\sqrt{n}$ scaled by $\sqrt{n}$.

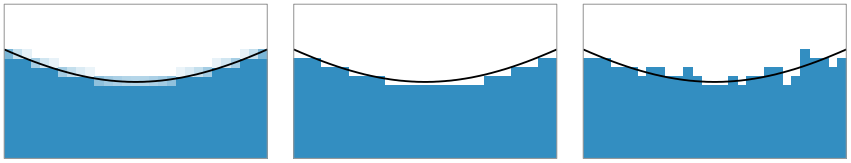


Figure 7.3: The clusters from $n = 70{,}002$ particles at each of $(\pm 127, 0)$ near the top of their limit shape on the y-axis: (a) the divisible sandpile (light squares hold mass strictly between 0 and 1), (b) rotor-router aggregation and (c) internal DLA, with the boundary of the limit shape (black). In each whole cluster, every site farther than 0.6, 0.8 and 3, respectively, inside this boundary holds a particle or mass, and every site farther than that outside it is empty.

The proof uses the logarithmic potential of the limit shape, computed in the next section; its positivity inside the shape leads to the inner bounds, and its vanishing outside to the outer bounds.

7.2 THE POTENTIAL OF THE LIMIT SHAPE

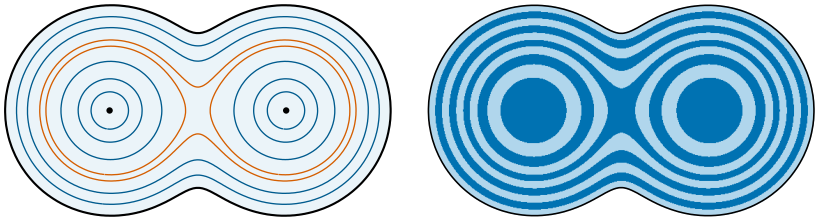
Recall the Green function $\mathcal{G}(z) = -(2\pi)^{-1}\log|z|$ from (2.14), with $\mathcal{G}(\mathbf{0}) = +\infty$, and the potential $\mathbf{G}\sigma(z) = \int \mathcal{G}(z-w)\sigma(w)\,\mathrm{d}w$ from (2.13). For the shapes $\mathcal{O}_a$ of Section 7.1, write

$$w_a(z) := \mathcal{G}(z-a) + \mathcal{G}(z+a) - \mathbf{G}\mathbf{1}_{\mathcal{O}_a}(z)\,.$$

The polar formula for the oval comes from its moments. A boundary $r^2 = A + B\cos 2\theta$ encloses area πA and second complex moment $\pi AB/2$, and equating these with the mass 2 and the second moment $2a^2$ of the two sources gives $A = 2r_0^2$ and $B = 2a^2$. In the next theorem we check that all the moments agree and that the potential has the sign we need.

Theorem 7.2 *Let $a \geq 0$. Then $\mathcal{O}_a$ has area two, and w_a is nonnegative, positive exactly on $\mathcal{O}_a$, and continuously differentiable away from the two sources. Moreover w_a and ∇w_a vanish on $\mathbb{C}\setminus\mathcal{O}_a$. If $a < r_0$, then the boundary of $\mathcal{O}_a$ consists of the points $x + \mathbf{i}y \neq \mathbf{0}$ with $x, y \in \mathbb{R}$ satisfying*

$$(x^2+y^2)^2 - 2r_0^2(x^2+y^2) - 2a^2(x^2-y^2) = 0\,. \tag{7.1}$$



(a) (b)

Figure 7.4: (a) Level curves of the potential w_a for $a = 0.85r_0$: two loops about the sources join through the saddle at the origin (the vermilion curves lie just below and just above its value there), and w_a vanishes with its gradient on and outside $\partial\mathcal{O}_a$ (black). (b) The odometer of the divisible sandpile started from $n = 70{,}002$ at each of the two sources $(\pm 127, 0)$, in ten bands of values with equal numbers of sites, alternately light and dark, with the boundary of $\sqrt{n}\,\mathcal{O}_{127/\sqrt{n}}$ (black), where $127/\sqrt{n} \approx 0.85r_0$.

The level curves of w_a for $a = 0.85r_0$ are in Figure 7.4(a). For $a \geq r_0$ the theorem follows from the potential of a disk, Lemma 2.37, applied to each disk. The differentiability assertion needs only the local integrability of the gradient of the logarithm, by the next lemma.

Lemma 7.3 *Let σ be a bounded measurable function on $\mathbb{R}^2$ with bounded support. Then $\mathbf{G}\sigma$ is continuously differentiable, and*

$$\nabla\mathbf{G}\sigma(x) = \int \nabla\mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\,.$$

Proof. The kernel satisfies $|\nabla \mathcal{G}(z)| = (2\pi|z|)^{-1}$, which is locally integrable. Let $F(x) := \int \nabla \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y$. For continuity of F, we split this integral into a small disk and its complement, as in Remark 2.5; the disk contributes at most $C\|\sigma\|_\infty$ times its radius, and the kernel is uniformly continuous on the bounded remainder. Integrating along a segment and using Fubini's theorem (which applies because the integral of $|\nabla\mathcal{G}|$ over a bounded set is bounded uniformly over translates in a bounded set), we obtain

$$\mathbf{G}\sigma(x+h) - \mathbf{G}\sigma(x) = \int_0^1 h \cdot F(x+th)\,\mathrm{d}t = h \cdot F(x) + o(|h|)\,,$$

where the last equality uses the continuity of F. Thus F is the continuous gradient of $\mathbf{G}\sigma$. □

Proof of Theorem 7.2. We compute the potential for disks, then prove vanishing and positivity for the oval.

Step 1: Suppose first that $a \geq r_0$, so that the two disks are disjoint and each has area one. Write $W := \mathcal{G} - \mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},r_0)}$. By Lemma 2.37, W vanishes for $|z| \geq r_0$ and satisfies

$$W(z) = \frac{|z|^2 - r_0^2}{4} - \frac{r_0^2}{2}\log\frac{|z|}{r_0} \qquad (0 < |z| < r_0)\,.$$

The radial derivative of W is $(|z|^2 - r_0^2)/(2|z|) < 0$, so $W > 0$ inside. Hence $w_a(z) = W(z-a) + W(z+a)$ has the required sign and zero boundary gradient, including when the disks are tangent.

Step 2: Now let $a < r_0$. We show that w_a and ∇w_a vanish outside O_a. The oval contains both sources, since $2r_0^2 + 2a^2 > a^2$. For integers $j \geq 0$, polar coordinates give, with area measure $\mathrm{d}A$,

$$\int_{O_a} z^{2j}\,\mathrm{d}A(z) = \frac{1}{2j+2}\int_0^{2\pi} e^{2j\mathbf{i}\theta}(2r_0^2 + a^2e^{2\mathbf{i}\theta} + a^2e^{-2\mathbf{i}\theta})^{j+1}\,\mathrm{d}\theta\,.$$

Only the terms of the expanded power that contain $e^{-2j\mathbf{i}\theta}$ contribute. If such a term has n_- factors $a^2e^{-2\mathbf{i}\theta}$, n_+ factors $a^2e^{2\mathbf{i}\theta}$ and n_0 factors $2r_0^2$, then $n_- - n_+ = j$ and $n_- + n_+ + n_0 = j+1$. Thus $2n_+ + n_0 = 1$, so the coefficient is $(j+1)2r_0^2a^{2j}$. Odd moments vanish by symmetry. Consequently

$$\int_{O_a} z^\ell\,\mathrm{d}A(z) = a^\ell + (-a)^\ell \qquad (\ell \geq 0)\,, \tag{7.2}$$

where the zeroth moment is 2, also when $a = 0$. For $|z| > \sup_{w\in O_a}|w|$, the uniformly convergent expansion

$$\log|z-w| = \log|z| - \mathrm{Re}\sum_{\ell\geq 1}\frac{w^\ell}{\ell z^\ell}\,,$$

and (7.2) give $w_a(z) = 0$. The exterior of $\overline{O_a}$ is connected, since O_a is star-shaped about the origin. There w_a is harmonic, and $\partial_x w_a - \mathbf{i}\partial_y w_a$ is holomorphic by Subsection 2.2.1. This function vanishes near infinity, so by the identity theorem it vanishes on the entire exterior, and w_a is constant there, hence zero. By Lemma 7.3, w_a and its gradient are continuous, so they also vanish on the boundary.

Step 3: Still with $a < r_0$, we prove positivity by showing that $\partial_y w_a$ is negative in the upper half of the oval and integrating down from the boundary, where w_a vanishes. By Corollaries 2.39 and 2.40, $\Delta w_a = 1$ inside the oval away from the sources. More precisely, $w_a - |z|^2/4$ is harmonic there by Theorem 2.29, and hence smooth by Theorem 2.3. Let $V := \partial_y w_a$ and $O^+ := O_a \cap \{y > 0\}$. Then V is harmonic on the connected set O^+, and $V = 0$ on the boundary of O^+ except at the sources (on the curved boundary by Step 2, and on the real axis by reflection symmetry). For a source $s \in \{-a, a\}$, let m_s be its mass, which is one, or two if $a = 0$. The function $w_a - m_s \mathcal{G}(\cdot - s) - |z - s|^2/4$ is harmonic and even in y on a small disk about s, so

$$V(x, y) = -\frac{m_s y}{2\pi((x - s)^2 + y^2)} + O(y)\,.$$

Therefore $V \le 0$ on sufficiently small upper half-circles about the sources. Applying the minimum principle, Lemma 2.12(a), to $-V$ with these half-disks removed and letting their radii decrease to zero gives $V \le 0$ on O^+. The strict inequality near each source and the strong minimum principle, Lemma 2.8, give $V < 0$ throughout O^+. From a point $(x, y) \in O^+$, follow the upward vertical segment to its first boundary point (x, t_1) (Figure 7.5). Since $w_a(x, t_1) = 0$,

$$w_a(x, y) = -\int_y^{t_1} V(x, t)\,\mathrm{d}t > 0\,.$$

Integrating in the same way shows that $w_a > 0$ on the real axis away from the sources; at the sources w_a is infinite. By reflection, $w_a > 0$ in the lower half-plane. Finally, multiplying the polar boundary equation by r^2 gives (7.1); the only added point is the origin, which lies inside the oval. □

As a increases to r_0, the polar radii of O_a converge uniformly to $2r_0|\cos\theta|$, the polar equation of the two tangent circles, so the oval turns into two disks (Figure 7.6).

For sources at $\mathbf{0}$ and $q \in \mathbb{C}$, write

$$U_q(z) := \mathcal{G}(z) + \mathcal{G}(z - q) - \mathbf{G}\mathbf{1}_{\Omega(q)}(z)\,.$$

By Theorem 7.2, after rotation and translation, U_q is nonnegative, and U_q

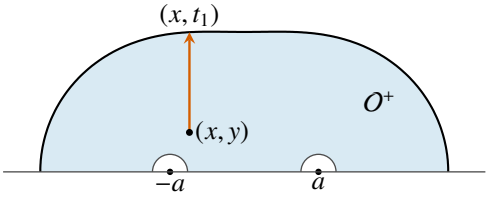


Figure 7.5: Step 3 of the proof of Theorem 7.2: $V = \partial_y w_a$ is harmonic on O^+ with the half-disks about $\pm a$ removed and is negative there, so integrating V along the vertical segment from (x, y) to the boundary point (x, t_1), where $w_a = 0$, gives $w_a(x, y) > 0$.

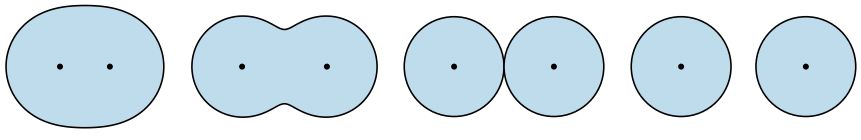


Figure 7.6: The limit shape O_a of two sources of mass one at $\pm a$, for $a/r_0 = 0.5$, 0.85, 1 and 1.25, drawn to the same scale. Every shape has area two, and the oval pinches into two tangent disks of area one when $a = r_0 = \pi^{-1/2}$.

and ∇U_q vanish outside $\Omega(q)$. Before passing to the lattice we also need continuity in q and a quadratic bound near the boundary.

Lemma 7.4 *Let $q \in \mathbb{C}$.*

(i) If $q_n \to q$ in $\mathbb{C}$, then for every $\eta > 0$ and all sufficiently large n we have $\Omega(q)_\eta \subset \Omega(q_n) \subset \Omega(q)^\eta$, and $U_{q_n} \to U_q$ uniformly on compact sets disjoint from $\{\mathbf{0}, q\}$.

(ii) The set $\Omega(q)$ contains the disks of radius r_0 about $\mathbf{0}$ and q.

(iii) For every $w \in \partial\Omega(q)$ and every z with $|z - w| < r_0/4$,

$$0 \le U_q(z) \le 6|z - w|^2 . \tag{7.3}$$

Proof. (i) For ovals, the inclusions follow from uniform convergence of the polar radii, also at tangency, since compact subsets of the tangent disks avoid their point of contact. For two disks they follow from convergence of the centers. Translations and rotations preserve these conclusions. At $q = \mathbf{0}$, the radii tend uniformly to $\sqrt{2}r_0$, for every rotation. Polar integration also gives

$$|O_a \triangle O_b| = \frac{1}{2}\int_0^{2\pi} 2|a^2 - b^2||\cos 2\theta|\,\mathrm{d}\theta = 4|a^2 - b^2|\,,$$

for $0 \le a, b \le r_0$, which tends to zero as $a - b \to 0$; at r_0 the contact point does not affect area. For disks the same conclusion follows by enclosing the

symmetric difference in annuli of vanishing width. Continuity of translations and rotations in L^1 now gives $|\Omega(q_n) \triangle \Omega(q)| \to 0$. To compare potentials, we split their integrals at $|z-y| = t$ with $0 < t < 1$. The near part is at most $Ct^2(1 + |\log t|)$; on a fixed bounded remainder the kernel is bounded, so the far part tends uniformly to zero. The source terms converge away from $\mathbf{0}$ and q.

(ii) In the oval case (Figure 7.1), put $z = x + \mathbf{i}y$ and $r = |z| > 0$. If $|z-a| < r_0$, then the left side of (7.1) is strictly less than $-(r^2 - 2ax)^2$, hence negative. The origin also belongs to O_a. By reflection, the same holds for the disk about $-a$; for $a \geq r_0$ the inclusion is part of the definition. Thus every boundary point is at distance at least r_0 from the sources.

(iii) Fix $w \in \partial\Omega(q)$. By part (ii), no source lies in the disk of radius $r_0/2$ centered at w, so $\Delta U_q = \mathbf{1}_{\Omega(q)}$ there distributionally. Hence $U_q + |z-w|^2/4$ is subharmonic and $U_q - |z-w|^2/4$ is superharmonic, by Theorem 2.29. Now (7.3) follows from Proposition 2.32, centered at the zero w of U_q, with $\lambda = 1/4$ and $\beta = 1/2$. □

We next discretize this potential and compare it with the divisible sandpile odometer.

7.3 THE DIVISIBLE SANDPILE

Throughout the rest of the chapter, we work in the setting of Theorem 7.1 and write $p_n := b_n/\sqrt{n}$, so that $p_n \to p$ and the sets $\Omega(p_n)$ lie in a fixed disk. Recall that $\Delta^{::} f(x) = \sum_{y\sim x}(f(y) - f(x))$, and that $g^{::}$ (minus the potential kernel of the plane) satisfies $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ and $g^{::}(w) = \mathcal{G}(w) - \kappa + O(|w|^{-2})$ for $w \neq \mathbf{0}$ by (4.6), where κ is the constant of Theorem 3.9.

We give each site the area of its cell inside $\sqrt{n}\Omega(p_n)$ (Figure 7.7) by writing

$$Q_y := y + [-\tfrac{1}{2}, \tfrac{1}{2})^2 \quad \text{and} \quad \mu_n(y) := |Q_y \cap \sqrt{n}\Omega(p_n)|\,.$$

Then $0 \leq \mu_n \leq 1$ and $\sum_y \mu_n(y) = 2n$. Define the comparison function

$$F_n(x) := ng^{::}(x) + ng^{::}(x - b_n) - \sum_y \mu_n(y) g^{::}(x-y)\,.$$

Then $\Delta^{::} F_n = \mu_n - \sigma_n$. The next lemma compares F_n with the potential U_{p_n} of Section 7.2, also near the sources. (As Figure 7.4(b) suggests, the odometer u_n itself is close to $nU_{p_n}(x/\sqrt{n})$.)

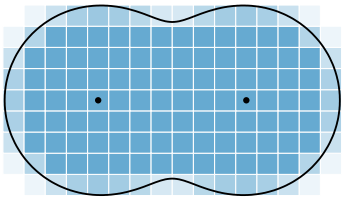


Figure 7.7: The cells Q_y shaded by $\mu_n(y)$, the area of Q_y inside $\sqrt{n}\,\Omega(p_n)$ (black), for $n = 60$ and $b_n = (7, 0)$: the cells with $\mu_n = 1$ (dark) form the set E_n of Step 2 of the proof of Theorem 7.1(i), and the values of μ_n sum to $2n$.

Lemma 7.5 *There is a constant C, independent of n, with the following properties.*

(i) For every fixed $\eta > 0$, uniformly over sites x at distance at least $\eta\sqrt{n}$ from both sources,

$$F_n(x) = nU_{p_n}(x/\sqrt{n}) + O_\eta(\sqrt{n})\,. \tag{7.4}$$

(ii) Let x be a site at distance $r \le \sqrt{n}$ from the nearer source. Then

$$F_n(x)/n \ge \frac{1}{2\pi}\log\frac{\sqrt{n}}{1+r} - C\,. \tag{7.5}$$

(iii) The function F_n satisfies $F_n \ge -C\sqrt{n}$ everywhere.

Proof. (i) We compare cell integrals first and then handle the source terms. If $|x - y| \ge 2$, then the asymptotics (4.6) and $|\nabla\mathcal{G}(z)| = (2\pi|z|)^{-1}$ give

$$\left|\mu_n(y)(g^{::}(x-y)+\kappa) - \int_{Q_y\cap\sqrt{n}\Omega(p_n)} \mathcal{G}(x-w)\,\mathrm{d}w\right| \le \frac{C}{1+|x-y|}\,.$$

After increasing C, the same bound holds for $|x - y| < 2$, by integrability of the logarithm. Summing over cells in a disk of radius $C\sqrt{n}$ gives error at most $C\sqrt{n}$, uniformly in x. For x in the disk of twice that radius, sum over shells of integer radii j about x, each containing $O(j + 1)$ sites. For more distant x, all denominators are at least a constant times $|x|$, giving a bound $Cn/|x| \le C\sqrt{n}$. The errors in the source terms are $O_\eta(1)$ by the assumption on x. The constants κ cancel because the masses agree. After the change of variables $w = \sqrt{n}z$, the terms involving $\log n$ cancel for the same reason. This proves (7.4).

(ii) The asymptotics (4.6) also give, for every lattice point w,

$$g^{::}(w) + \kappa + \frac{\log n}{4\pi} \ge \frac{1}{2\pi}\log\frac{\sqrt{n}}{1+|w|} - C\,.$$

Since $r \le \sqrt{n}$, the point $x/\sqrt{n}$ stays bounded. After adding $\kappa + (\log n)/(4\pi)$ to every kernel in F_n/n, the term of the other source is bounded below, and the sum over y, divided by n, is bounded above by the estimate of part (i) and local integrability of the logarithm, which proves (7.5).

(iii) Choose $\eta > 0$ small enough that the lower bound (7.5) is positive whenever $r \le \eta\sqrt{n}$, for large n. Elsewhere (7.4) and $U_{p_n} \ge 0$ give $F_n \ge -C\sqrt{n}$. Enlarging C covers the finitely many remaining n, since each F_n tends to zero at infinity by cancellation of its total mass. □

For the outer bound we also need the fact that a positive odometer far from the sources is large at a site nearby. We allow an error term in the Laplacian, for use in the next section.

Lemma 7.6 *There is an absolute constant $C_0 \ge 1$ with the following property. Let $z \in \mathbb{Z}^2$, let $r \ge 1$ and $\lambda \ge 0$, and let e be a function on ordered pairs of neighboring sites with $|e| \le \lambda$ and $\sum_{w\sim v} e(v,w) = 0$ for every site v. Let $f : \mathbb{Z}^2 \to \mathbb{R}$ satisfy $f(z) > C_0\lambda r$ and*

$$\Delta^{::} f(v) = 1 - \sum_{w\sim v} e(w,v) \qquad \text{for every } v \in \mathrm{B}^{::}(z,r) \text{ with } f(v) > 0\,.$$

Then there is a site $y \in \partial \mathrm{B}^{::}(z,r)$ with $f(y) > r^2/4$.

Proof. We compare f with $|v-z|^2/4$, whose Laplacian is one, by testing against a Green function. Let S be the set of sites $v \in \mathrm{B}^{::}(z,r)$ with $f(v) > 0$, and let $\varphi := g^{::}_S(z,\cdot)$. Then $\Delta^{::}\varphi = -\delta_z$ on S, and on ∂S the values $\Delta^{::}\varphi(v) = \sum_{w\sim v}\varphi(w)$ are nonnegative with sum one, since $\sum_v \Delta^{::}\varphi(v) = 0$. The function $h(v) := f(v) - |v-z|^2/4$ has $\Delta^{::}h(v) = -\sum_{w\sim v} e(w,v)$ on S, so summation by parts and $\sum_{w\sim v} e(v,w) = 0$ give

$$\sum_{v\in\partial S} h(v)\Delta^{::}\varphi(v) - h(z) = \sum_v \varphi(v)\Delta^{::}h(v) = \sum_v \sum_{w\sim v} e(v,w)(\varphi(v) - \varphi(w))\,.$$

Every edge with an endpoint in S has $v \in \mathrm{B}^{::}(z, r+1)$, so by the gradient bound of Lemma 6.2 the right side is at most $C\lambda(r+1) \le C_0\lambda r$ in absolute value, with $C_0 := \max\{1, 2C\}$. Hence some $y \in \partial S$ has $h(y) \ge f(z) - C_0\lambda r > 0$. As $f(y) > 0$ and $y \notin S$, it lies outside $\mathrm{B}^{::}(z,r)$ and next to it, so $f(y) > |y-z|^2/4 \ge r^2/4$. □

We bound u_n above by the least action principle, which gives the outer inclusion of part (i), and below by the minimum principle on the set where $\mu_n = 1$, which gives the inner inclusion and a margin of order n for the particle models.

Proof of Theorem 7.1(i). **Step 1:** We bound the odometer from above. The function $F_n + C\sqrt{n}$ is nonnegative by Lemma 7.5(iii), and its Laplacian

plus σ_n equals $\mu_n \le 1$, so $u_n \le F_n + C\sqrt{n}$ by the least action principle, Theorem 4.3. Outside $\sqrt{n}\Omega(p_n)$, both sources are at distance at least $r_0\sqrt{n}$ by Lemma 7.4(ii), and $U_{p_n} = 0$ there by Theorem 7.2, so (7.4) gives $u_n \le C\sqrt{n}$. If $x \in S_n$ is at distance $r > 4$ from $\sqrt{n}\Omega(p_n)$, then we apply Lemma 7.6 with $f = u_n$ and $e = \lambda = 0$ on the disk $\mathrm{B}^{::}(x, r/2)$, which contains no source; $\Delta^{::}u_n = 1$ on $\{u_n > 0\}$ by Lemma 4.4. The boundary of this disk lies outside $\sqrt{n}\Omega(p_n)$, so $r^2/16 < C\sqrt{n}$. Consequently

$$S_n \subset \{x : \operatorname{dist}(x, \sqrt{n}\Omega(p_n)) \le Cn^{1/4}\} \subset \mathrm{B}^{::}(\mathbf{0}, C\sqrt{n})\,. \tag{7.6}$$

Step 2: We bound the odometer from below. Let $E_n := \{x : \mu_n(x) = 1\}$. On this finite set, $\Delta^{::}(u_n - F_n) = \xi_n - 1 \le 0$, since $\mu_n = 1$ there. A site on ∂E_n lies within distance 2 of $\partial(\sqrt{n}\Omega(p_n))$, since its cell has positive area outside and a neighboring cell has full area inside, so it is at distance at least $r_0\sqrt{n} - 2$ from both sources. By the quadratic bound (7.3) and (7.4), $|F_n| \le C\sqrt{n}$ there. Since $u_n \ge 0$, the minimum principle for the superharmonic function $u_n - F_n$ gives $u_n \ge F_n - C\sqrt{n}$ on E_n.

Let $K \subset \Omega(p)$ be compact. By the inclusions of Lemma 7.4(i), applied to a small closed neighborhood of K, the set $\sqrt{n}K \cap \mathbb{Z}^2$ lies in E_n for large n. Away from small fixed neighborhoods of $\mathbf{0}$ and p, U_{p_n} converges uniformly to the strictly positive function U_p. Near the sources choose those neighborhoods small enough that (7.5) is at least one, for large n. We conclude that there is $c_K > 0$ such that

$$u_n(x) \ge c_K n \qquad (x \in \sqrt{n}K \cap \mathbb{Z}^2, n \text{ sufficiently large})\,. \tag{7.7}$$

Step 3: We pass to the fixed domain $\Omega(p)$. By (7.7) with $K = \overline{\Omega(p)_\epsilon}$, every site of $\sqrt{n}K$ topples and hence has final mass one. For $n > 1$ the sources topple, so every site with positive final mass belongs to S_n or its outer boundary. By (7.6), and since $\Omega(p_n) \subset \Omega(p)^{\epsilon/2}$ for large n, this set lies in $\sqrt{n}\Omega(p)^\epsilon$. □

For the particle models we use only the bounds (7.7) and (7.6), starting with rotor-router aggregation.

7.4 ROTOR-ROUTER AGGREGATION

Let ρ be the rotor stacks and $\mathcal{D}_\rho$ their discrepancy (5.1). For a finite lattice set S, recall that the killed Green function $g^{::}_S$ is symmetric, vanishes when either argument is outside S, and satisfies $\Delta^{::}_y g^{::}_S(x, y) = -\delta_x(y)$ on S, by (3.19). For $n > 1$ both sources belong to S_n, so the sandpile odometer has

the representation

$$u_n(z) = n g^{::}_{S_n}(\mathbf{0}, z) + n g^{::}_{S_n}(b_n, z) - \sum_{y \in S_n} g^{::}_{S_n}(y, z)\,. \tag{7.8}$$

Indeed, both sides vanish off S_n and have Laplacian $1 - \sigma_n$ on S_n, so they are equal by the minimum principle. We use this identity to prove the inner bound.

Lemma 7.7 *Let K be a compact subset of $\Omega(p)$. Then $\sqrt{n}K \cap \mathbb{Z}^2 \subset A_n^{\mathrm{R}}$ for all sufficiently large n, uniformly over the rotor stacks.*

Proof. Let m be the odometer of rotor-router aggregation from σ_n. We compare m with u_n by testing Proposition 6.1 against a killed Green function of S_n.

Step 1: We show that $m(z)/4 \geq u_n(z) - C\sqrt{n}$ for every $z \in S_n$. Let $\varphi := g^{::}_{S_n}(\cdot, z)$. Proposition 6.1 for the aggregation gives

$$\sum_x \varphi(x)(\mathbf{1}_{A_n^{\mathrm{R}}}(x) - \sigma_n(x)) = -\frac{m(z)}{4} + \sum_{x \in S_n} \sum_{y \sim x} \mathcal{D}_\rho(x, y, m(x))(\varphi(y) - \varphi(x)) + \sum_{x \notin S_n} \sum_{y \sim x} \Big(\frac{m(x)}{4} + \mathcal{D}_\rho(x, y, m(x))\Big)\varphi(y)\,.$$

The last sum is nonnegative, since each bracket counts the particles sent from x to y. The middle sum has absolute value at most $C\sqrt{n}$, by (6.1), symmetry of $g^{::}_{S_n}$ and the gradient bound of Lemma 6.2, applied in the disk $\mathrm{B}^{::}(\mathbf{0}, C\sqrt{n}) \supset S_n$ of (7.6). So the left side is at least $-m(z)/4 - C\sqrt{n}$. Since $\mathbf{1}_{A_n^{\mathrm{R}}} \leq 1$ and $\varphi \geq 0$, it is also at most $\sum_{x \in S_n} \varphi(x) - n\varphi(\mathbf{0}) - n\varphi(b_n) = -u_n(z)$ by (7.8).

Step 2: We deduce the inclusion. For $z \in \sqrt{n}K \cap \mathbb{Z}^2$, the bound (7.7) gives $u_n(z) \geq c_K n > 0$ for large n, so $z \in S_n$, and Step 1 gives $m(z)/4 \geq c_K n - C\sqrt{n} > 0$ uniformly over the rotor stacks. A site at which a legal move is made stays occupied, so $z \in A_n^{\mathrm{R}}$. □

To prove the outer inclusion we must rule out "thin tentacles" of occupied sites. Namely, according to the following lemma, if no source lies within distance $2s$ of an occupied site, then a constant fraction of the sites within distance s of it are occupied.

Lemma 7.8 (No Thin Tentacles) *There is an absolute constant $c > 0$ with the following property. Let $\sigma : \mathbb{Z}^2 \to \mathbb{N}$ have finite support, let A^{R} be the rotor-router aggregate of σ for fixed rotor stacks, and let $x \in A^{\mathrm{R}}$ and $s \geq 1$ satisfy $\sigma = 0$ on $\mathrm{B}^{::}(x, 2s)$. Then*

$$\#(A^{\mathrm{R}} \cap \mathrm{B}^{::}(x, s)) \geq c s^2\,.$$

Proof. We find a site near x with a large odometer and show that a disk about it is occupied (Figure 7.8). We may take s larger than a fixed absolute constant, since x is occupied, so decreasing c handles bounded s. Let m be the odometer of rotor-router aggregation from σ, and $f := m/4$. For neighbors v, w, let $e(v,w) := \mathcal{D}_\rho(v,w,m(v))$, so that $|e| \le 3/4$ by (6.1), and $\sum_{w\sim v} e(v,w) = 0$ since $m(v)$ particles leave v in all. Since $m(w)/4 + e(w,v)$ particles move from w to v, counting the particles that arrive at and leave v gives

$$\Delta^{::} f(v) = \mathbf{1}_{A^R}(v) - \sigma(v) - \sum_{w\sim v} e(w,v)\,.$$

Hence $|\Delta^{::} f| \le 4$ where $\sigma = 0$, and $f = 0$ off A^R, since a site at which a legal move is made stays occupied. So Lemma 7.6, with $\lambda = 3/4$, applies to f on every disk without sources; let C_0 be its constant.

Step 1: We find a site y near x with $f(y) > s^2/64$. Let T be the maximum of m on $\mathrm{B}^{::}(x, s/4)$, and for integers $0 \le j < s/4$ let a_j count the occupied sites at distance at most j from x. Let $1 \le j < s/4$. Since $\sigma = 0$ on $\mathrm{B}^{::}(x, 2s)$, every particle that ends within distance $j-1$ of x entered that ball by a move from an occupied site at distance in $(j-1, j]$, and at most T moves are made at each such site. Hence $a_{j-1} \le T(a_j - a_{j-1})$. Since $a_0 = 1$, this gives $T \ge 1$, and multiplying the inequalities $a_j/a_{j-1} \ge 1 + 1/T$ gives $(1+1/T)^j \le a_j \le (2j+1)^2$. Taking $j = \lceil s/4 \rceil - 1 \ge s/8$ and using $\log(1+1/T) \ge 1/(2T)$, we get $T \ge s/(32\log(2+s))$. Let z_1 attain T, and let $r_1 := s/(512 C_0 \log(2+s))$. Then $f(z_1) \ge 4C_0 r_1$, and for large s we have $1 \le r_1 \le s/4 - 2$ and $r_1^2 > C_0 s$. By Lemma 7.6 with center z_1 and radius r_1, there is a site y_1 at distance less than $r_1 + 1$ from z_1 with $f(y_1) > r_1^2/4 > C_0 s/4$. By the lemma with center y_1 and radius $s/4$, there is a site y with

$$|y - x| < s/2 + r_1 + 2 \le 3s/4 \qquad \text{and} \qquad f(y) > s^2/64\,. \tag{7.9}$$

Both disks used here lie in $\mathrm{B}^{::}(x, 2s)$, where $\sigma = 0$.

Step 2: We show that $\mathrm{B}^{::}(y, \theta s) \subset A^R$ for $\theta := 1/(8\sqrt{153})$. Suppose that a vacant site w satisfies $|w - y| < \theta s$, and let $B := \mathrm{B}^{::}(w, 2\theta s)$. Since $\theta < 1/8$, the bound (7.9) gives $B \subset \mathrm{B}^{::}(x, 2s)$, so $|\Delta^{::} f| \le 4$ on B. The function $g := f + \sum_{v\in B} g_B^{::}(\cdot, v)\Delta^{::} f(v)$ equals f off B and, by symmetry and (3.19), is harmonic in B, so $g \ge 0$ by the minimum principle, Lemma 3.2. By the normalization (3.18) and the exit-time bound, Lemma 3.4, for $z \in B$

$$|g(z) - f(z)| \le 4 \sum_{v\in B} g_B^{::}(z,v) = \mathbb{E}_z[\tau_{\partial B}] \le (2\theta s + 1)^2 \le 9\theta^2 s^2\,.$$

As $f(w) = 0$, we get $g(w) \le 9\theta^2 s^2$. For large s, the discrete Harnack inequality, Lemma 3.16, gives $g \le 16 g(w) \le 144\theta^2 s^2$ on $\mathrm{B}^{::}(w, \theta s) \ni y$.

Hence $f(y) \le 153\theta^2 s^2 = s^2/64$, contradicting (7.9). So $\mathrm{B}^{::}(y, \theta s)$ is occupied. It lies in $\mathrm{B}^{::}(x, s)$ and contains the lattice points v with $|v - y|_\infty \le \theta s/2$, at least $(\theta s)^2/4$ of them once $\theta s \ge 2$. □

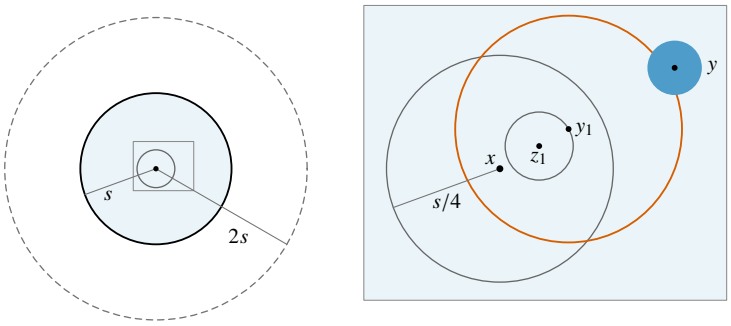


Figure 7.8: The disks in the proof of Lemma 7.8, with the small ones magnified on the right and r_1 and θs enlarged: $\mathrm{B}^{::}(x, 2s)$ (dashed) contains no source; in Step 1, the odometer is largest on $\mathrm{B}^{::}(x, s/4)$ at z_1, the site y_1 lies just outside $\mathrm{B}^{::}(z_1, r_1)$ and y just outside $\mathrm{B}^{::}(y_1, s/4)$ (vermilion). By Step 2, $\mathrm{B}^{::}(y, \theta s)$ (blue) is occupied, so $\mathrm{B}^{::}(x, s)$ (shaded) contains at least cs^2 occupied sites.

We now prove part (ii), using that each aggregate has $2n$ sites.

Proof of Theorem 7.1(ii). The inner inclusion follows from Lemma 7.7 with $K = \overline{\Omega(p)}_\epsilon$. For the outer inclusion, let c be the constant in Lemma 7.8. Choose $\delta > 0$ so that $|\Omega(p)_{2\delta}| > 2 - c\epsilon^2/128$. This is possible by continuity of area from below, since as $\delta \downarrow 0$ the sets $\Omega(p)_{2\delta}$ increase to $\Omega(p)$, which has area two by Theorem 7.2. For large n, each point of $\sqrt{n}\overline{\Omega(p)}_{2\delta}$ lies in a cell centered in $\sqrt{n}\Omega(p)_\delta$, so by Lemma 7.7 with $K = \overline{\Omega(p)}_\delta$, at least $n|\Omega(p)_{2\delta}|$ sites of $\sqrt{n}\Omega(p)_\delta$ are occupied. If an occupied site x has $\mathrm{dist}(x, \sqrt{n}\Omega(p)) \ge \epsilon\sqrt{n}$, then $\mathrm{B}^{::}(x, \epsilon\sqrt{n}/4)$ contains no source for large n, since the rescaled sources tend to points in $\Omega(p)$. By Lemma 7.8 with $s = \epsilon\sqrt{n}/8$, at least $c\epsilon^2 n/64$ sites of $\mathrm{B}^{::}(x, \epsilon\sqrt{n}/8)$, which lies outside $\sqrt{n}\Omega(p)$, are occupied, giving more than $2n$ sites. This contradiction proves the outer inclusion, with constants independent of the rotor stacks. □

The boundary of the rotor-router aggregate from two sources follows the quartic (7.1) (Figure 7.9). In the next section we turn to internal DLA, where bounds on probabilities replace the bound on the discrepancy.

7.5 INTERNAL DLA

We again use the toppled set S_n and the representation (7.8) of u_n. By the abelian property for stacks of instructions (Section 5.1), we may pause particles when they leave S_n and restart them later. (It also permits the

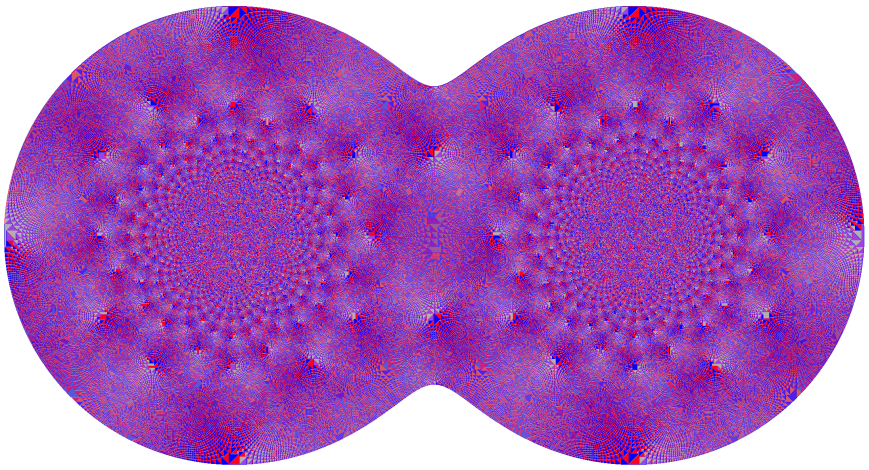


Figure 7.9: Rotor-router aggregation of $n = 10^6$ particles from each of the sites $(\pm 500, 0)$, colored by the final direction of each rotor as in Figure 1.1. Its boundary lies within half a site of the quartic $(x^2+y^2)^2 - \frac{2n}{\pi}(x^2+y^2) - 2\cdot 500^2(x^2-y^2) = 0$ in lattice coordinates, which is (7.1) with $a = 500/\sqrt{n}$ scaled by $\sqrt{n}$.

two-stage construction of Figure 7.10.)

Proof of Theorem 7.1(iii). We first bound the probability that a site of S_n is vacant, then restart the few particles that leave S_n to obtain the outer bound.

Step 1: List the $2n$ starting sites, with multiplicity, as $s_1, \dots, s_{2n}$. Release particles in this order, settling and pausing them as in Lemma 5.4 with $B = S_n$, and let I_{2n} be the set of sites where particles settle. Restarting the paused walks and completing aggregation gives A_n, so $I_{2n} \subset A_n$ in this coupling.

Fix $z \in S_n$, and let $\varphi(y) := g^{\cdots}_{S_n}(y,z)$ and $L := \max_{y\in S_n} g^{\cdots}_{S_n}(y,y)$. By the containment in (7.6), domain monotonicity of the killed Green function, and Lemma 3.12(b) (the value of the killed Green function of a disk at its center), applied in a disk of radius $2C\sqrt{n}$ centered at each $y \in S_n$, we have $L \le C\log n$. By symmetry and the normalization (3.18),

$$\sum_{y\in S_n} \varphi(y) = \tfrac{1}{4}\mathbb{E}_z[\tau_{\partial S_n}] \le Cn\,,$$

where the last bound follows from Lemma 3.4 for a disk containing S_n.

Step 2: We bound the probability that z is vacant. For $n > 1$ both sources lie in S_n, so Lemma 5.4 applies with $d = 2$, $B = S_n$, $N = 2n$, the starting sites of Step 1, and $0 < \lambda \le 1/(2L)$. By (7.8), $\sum_{i=1}^{2n}\varphi(s_i) - \sum_{y\in S_n}\varphi(y) = u_n(z)$, so the bounds of Step 1 give

$$\log \mathbb{P}\{z \notin I_{2n}\} \le -\lambda u_n(z) + C\lambda^2 n\log n\,.$$

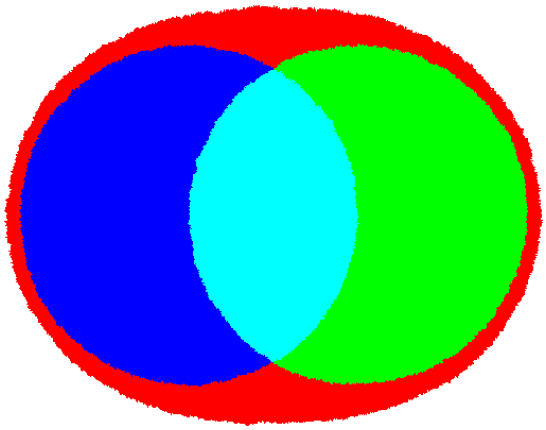


Figure 7.10: Internal DLA from 8×10^4 particles at each of the sites $(\pm 80, 0)$, built in two stages: the particles of each source perform internal DLA alone (blue and green, overlapping in cyan), and then one particle from each cyan site walks until it reaches an unoccupied site (red). The result has the law of internal DLA from both sources.

For a compact set $K \subset \Omega(p)$, we may choose $\lambda = c_K/(2C \log n)$, after increasing C, and then (7.7) gives

$$\mathbb{P}\left\{ \sqrt{n}K \cap \mathbb{Z}^2 \not\subset I_{2n} \right\} \le Cn \exp(-c_K^2 n/(4C \log n)) . \tag{7.10}$$

Here Cn bounds the number of sites of $\sqrt{n}K$. The bound is summable, and it gives the inner inclusion.

Step 3: We restart the paused particles to prove the outer inclusion. Fix $\epsilon > 0$, and let b, c_0, c_1 be the constants of Lemma 5.8 for $d = 2$. Choose $\delta > 0$ with $|\Omega(p)_{2\delta}| > 2 - b\epsilon^2/400$. On the event $G_n := \{ \sqrt{n}\Omega(p)_\delta \cap \mathbb{Z}^2 \subset I_{2n}\}$, counting as in the proof of part (ii) shows that at most $b\epsilon^2 n/400$ particles are paused. The complementary probabilities are summable by (7.10) with $K = \overline{\Omega(p)_\delta}$.

Let $\varpi := \lfloor \epsilon \sqrt{n}/10 \rfloor$. For a site x at distance at least $\epsilon \sqrt{n}$ from $\sqrt{n}\Omega(p)$, the cube $Q^{::}(x, 3\varpi)$ of sites y with $|y - x|_\infty \le 3\varpi$ lies within distance $\epsilon \sqrt{n}/2$ of x. By (7.6) and the convergence of $\Omega(p_n)$ to $\Omega(p)$ from Lemma 7.4(i), this cube contains neither settled nor paused particles for large n. On G_n there are at most $b\varpi^2$ paused particles. Conditionally on the paused walks, their continuations are independent simple random walks. By Lemma 5.8, translated to x, the probability that a restarted particle reaches x is therefore at most $c_0 e^{-c_1 \varpi}$. If there are no paused particles, the probability is zero. Every occupied site is joined to a source by a path of at most $2n$ occupied sites, since a newly occupied site is a source or adjacent to an earlier occupied site. Hence $A_n \subset \mathrm{B}^{::}(\mathbf{0}, Cn)$, and a union bound over the at most Cn^2 sites of

$\mathrm{B}^{::}(\mathbf{0}, Cn)$ gives

$$\mathbb{P}\left\{A_n \not\subset \sqrt{n}\Omega(p)^{\epsilon}\right\} \le \mathbb{P}(G_n^c) + Cn^2 c_0 e^{-c_1 \varpi} .$$

This bound is summable as well, so the Borel–Cantelli lemma gives both inclusions eventually almost surely for each fixed ϵ, without using independence between clusters with different n. Intersecting these events of probability one for $\epsilon = 1/j$ with $j \ge 1$ and using monotonicity of the neighborhoods gives every $\epsilon > 0$ simultaneously. □

We end with two consequences of the explicit form of the common limit.

7.6 TANGENT DISKS AND QUADRATURE DOMAINS

The width of the oval O_a along the imaginary axis is $2\sqrt{2r_0^2 - 2a^2}$. It shrinks to zero at $a = r_0$, when the oval becomes two disks of area one, tangent at the origin (see Figure 1.12). This case corresponds to two clusters that just barely interact, whose interaction is small enough that we do not see it in the limit. Thus if $b_n/\sqrt{n} \to p$ with $|p| = 2r_0$, then Theorem 7.1 gives the same pair of tangent disks whether $|b_n|/\sqrt{n}$ stays below $2r_0$ or above it, with no condition on the rate of approach. Likewise, if $|b_n| = o(\sqrt{n})$, then the limit is the disk of area two even though the two sources $\mathbf{0}$ and b_n may remain distinct.

Theorem 7.2 also gives an integration formula, and Exercise 7.2 asks you to prove that the integral over $\Omega(q)$ of every function h harmonic near the closure of $\Omega(q)$ is $h(\mathbf{0}) + h(q)$. A domain on which the integral of every integrable harmonic function is a combination of its values at finitely many points is called a **quadrature domain**, and such domains link these aggregation models to complex analysis. In Chapter 8, particles start far from the cluster instead of at sources inside it, and the clusters have long branches.

EXERCISES

Exercise 7.1. Let $q \in \mathbb{C}$, and recall the potential U_q of Section 7.2. Use its positivity on $\Omega(q)$ and its vanishing on the boundary of $\Omega(q)$, from Theorem 7.2, rather than the explicit boundary, to prove that every connected component of $\Omega(q)$ contains $\mathbf{0}$ or q. Conclude that $\Omega(q)$ has at most two components.

Exercise 7.2. Let $q \in \mathbb{C}$, and let h be harmonic near $\overline{\Omega(q)}$. Prove from Theorem 7.2 that $\int_{\Omega(q)} h(z)\, \mathrm{d}A(z) = h(\mathbf{0}) + h(q)$. Hint: test the distributional Laplacian of U_q against h multiplied by a smooth cutoff equal to one near $\overline{\Omega(q)}$.

NOTES

Theorem 7.1 is the planar case, with two equal sources, of Levine and Peres (2010, Theorem 1.4). Their theorem allows finitely many sources with positive masses in every dimension $d \geq 2$ and places the sources at the lattice points nearest to fixed locations, whereas Theorem 7.1 allows an arbitrary second source b_n with $b_n/\sqrt{n} \to p$. Their almost sure statement for internal DLA assumes that the nth lattice spacing is at most $1/n$; here the spacing is $n^{-1/2}$, and the summable bounds of Section 7.5 give convergence along the whole sequence under every coupling. Lemma 5.8 is based on Lemma 5.12 of Levine and Peres (2010). Lemma 7.8 takes its name from Proposition 4.2 of Levine and Peres (2017): in $\mathbb{Z}^d$, for n particles started at the origin, every initial rotor configuration and every simple rotor mechanism, at least a constant times s^d of the sites within distance s of each occupied site x with $|x| > s$ are occupied; the counting of occupied sites in its proof follows Lemmas 4.9 and 4.10 of Levine and Peres (2010). Our lemma is the planar version for initial configurations vanishing on $\mathrm{B}^{\cdot\cdot}(x, 2s)$. Its internal DLA analogue is Lemma 2 of Jerison, Levine, and Sheffield (2012), described in the Notes to Chapter 5.

Theorem 7.2 gives the logarithmic potential of the oval O_a; the survey Gustafsson and Shapiro (2005) traces the theory of quadrature domains back to papers of Neumann on this potential from the beginning of the twentieth century. Equation (11) of Levine and Peres (2010) gives the quartic (7.1) for the smash sum of two equal disks and credits it to Shapiro, and their Figure 3 is reproduced in Figure 7.9. The two-stage construction of Figure 7.10 is the smash sum introduced by Diaconis and Fulton (1991), also described in Section 1 of Levine and Peres (2010). The quadrature identity of Exercise 7.2 is the harmonic case of Proposition 6.6 of Levine and Peres (2010), a quadrature inequality for superharmonic functions on smash sums of balls.

For finitely many planar point sources with distinct locations and positive masses, the boundary of the common limit lies, up to finitely many points, on an algebraic curve of degree $2k$ for k sources. This follows from the quadrature identity and the work of Aharonov and Shapiro (1976) and of Gustafsson on quadrature domains, discussed in Sections 3 and 4 of Gustafsson and Shapiro (2005) and applied to aggregation in Section 1 of Levine and Peres (2010).

Chapter Eight

External DLA in $\mathbb{Z}^d$

Starting with a single particle at the origin in $\mathbb{Z}^2$, let particles arrive one at a time, each performing simple random walk from far away until it first visits a site adjacent to the cluster, where it sticks (in the limit as its starting point tends to infinity). The resulting random set $\mathcal{X}_n$ is known as an external diffusion-limited aggregation (DLA) cluster. Its long branching arms, separated by deep gaps, resemble the patterns of electrodeposition, viscous fingering and dielectric breakdown (see Figure 8.1).

How fast does the radius grow? Off-lattice simulations suggest that a cluster of radius r has about r^{d_f} particles, where $d_f \approx 1.71$ is its fractal dimension, so a cluster of n particles has radius about n^{1/d_f}, and $1/d_f \approx 0.585$. On the square lattice, large clusters align their arms with the axes (Figures 8.2, 8.3 and 8.4), and this anisotropy may change the growth rate. No one has proved that the radius grows like a power of n.

The main result of this chapter is that, with high probability, the planar cluster $\mathcal{X}_n$ lies in a ball of radius of order $n^{2/3}$. This exponent is larger than the off-lattice value 0.585, but even this upper bound needs a sharp estimate for planar random walk.

We first show that if each boundary site of a cluster of radius r is added next with conditional probability at most $Cr^{-\beta}$, then with high probability a cluster of n particles has radius at most a constant times $n^{1/(1+\beta)}$, and the discrete Beurling estimate gives $\beta = \frac{1}{2}$. These probabilities come from the harmonic measure from infinity. (Recall from Section 3.7 that the harmonic measure $\mathcal{H}_K$ of a finite set $K \subset \mathbb{Z}^2$ from infinity is the limiting hitting distribution on K of a walk started far away, and that it is the unique probability measure on K whose potential, formed with the potential kernel, is constant on K.) We bound the radius for general β in Section 8.1, prove the discrete Beurling estimate in Section 8.2, where we also show that $\beta = \frac{1}{2}$ is sharp on a horizontal segment, and combine the two in Section 8.3 (Theorem 8.8). In dimensions $d \geq 3$ a walk from far away may miss the cluster, so in the starred Section 8.4 we condition it to hit, and the capacity of the cluster takes the place of the Beurling estimate (Theorems 8.14 and 8.16).

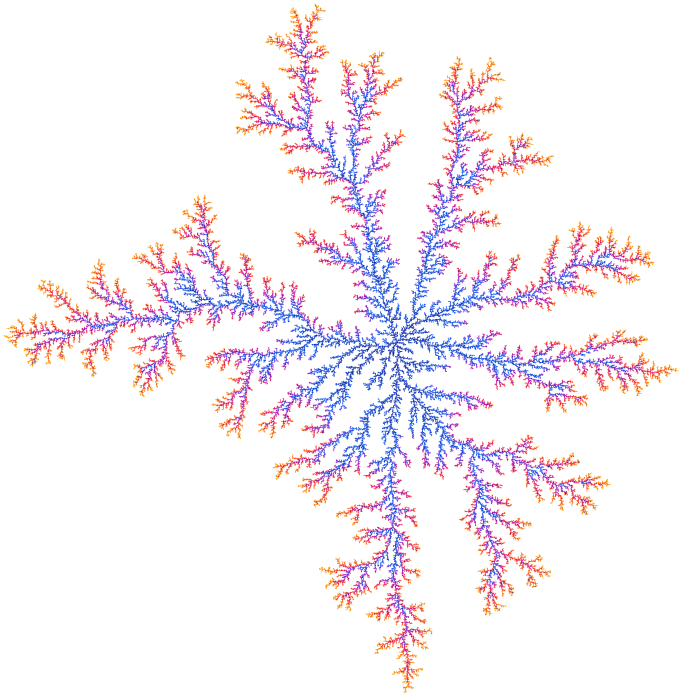


Figure 8.1: An external DLA cluster of 150,000 sites on $\mathbb{Z}^2$, each site colored by the rank of its harmonic measure from infinity, the probability that a walk from far away first hits the cluster there, from navy (smallest, including zero) through blue, magenta and red to orange (largest). Harmonic measure is largest at the tips of the arms and small deep in the gaps between them.

Internal DLA, by contrast, has a spherical limit shape, and in Chapter 5 we prove inner and outer bounds on its fluctuations (which in the plane are only logarithmic; see the Notes). Those proofs use martingales, which exist because every particle starts at the origin, and the nearly spherical shape of the cluster. External DLA has neither, since its particles come from infinity and its clusters have no such shape.

8.1 RADIUS FROM ATTACHMENT PROBABILITIES

Recall from Definition 3.28 that H_A is the harmonic measure from infinity of the outer boundary ∂A of a finite set $A \subset \mathbb{Z}^2$. Until Section 8.4, X is simple random walk on $\mathbb{Z}^2$, $\mathbb{P}_y$ and $\mathbb{E}_y$ denote its law and expectation started at y, and τ_E is its hitting time of $E \subset \mathbb{Z}^2$, as in Section 3.7.

The next site added to $\mathcal{X}_n$ is the first site of $\partial \mathcal{X}_n$ visited by a walk released from infinity (Figure 8.5), so its law is $H_{\mathcal{X}_n}$.

The radius of a finite nonempty set $A \subset \mathbb{R}^d$ is $\mathrm{rad}(A) := \max\{|u| : u \in A\}$.

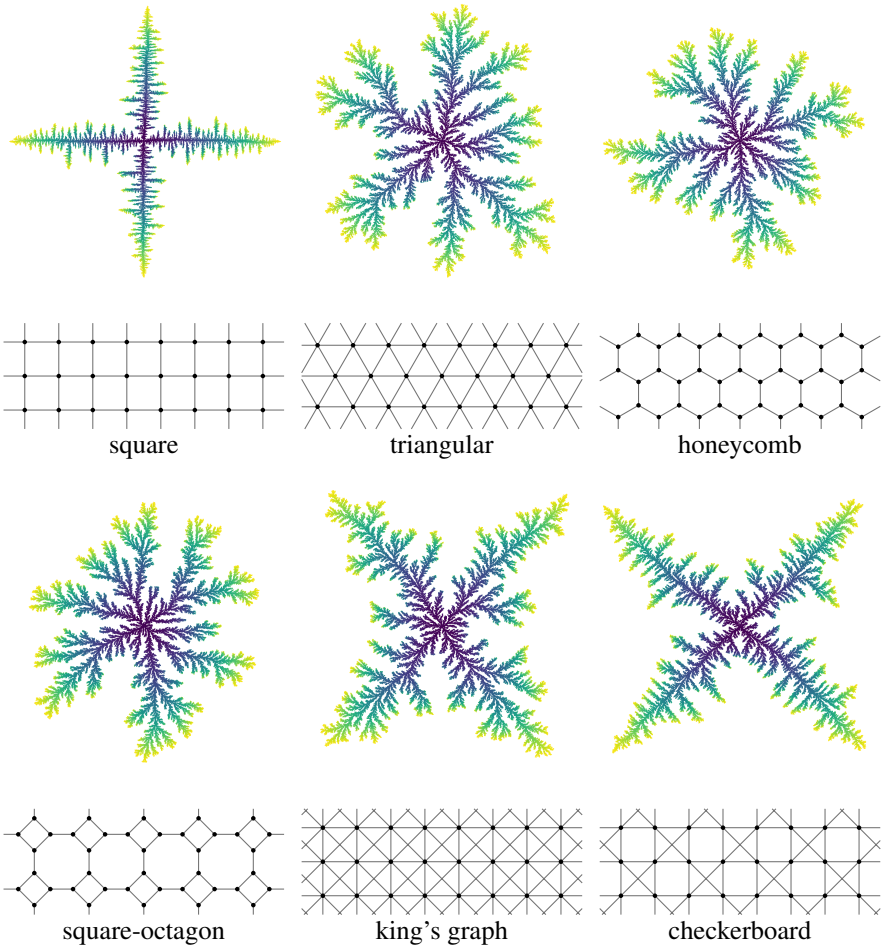


Figure 8.2: External DLA clusters above a few cells of their lattices, colored from dark (early) to light (late) by time of addition: 10^9 sites on the four planar lattices and 10^8 on the two lattices whose edges cross. The arms grow along the axes on the square lattice and along the diagonals on the king's graph and the checkerboard lattice.

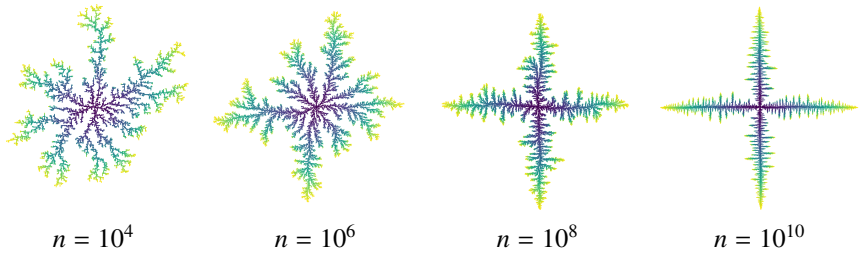


Figure 8.3: External DLA clusters on the square lattice with 10^4, 10^6, 10^8 and 10^{10} sites, each scaled to the same width; the first two are one cluster at two times. The small clusters branch in every direction, but the larger ones form a cross of four arms along the axes, with shorter side branches.

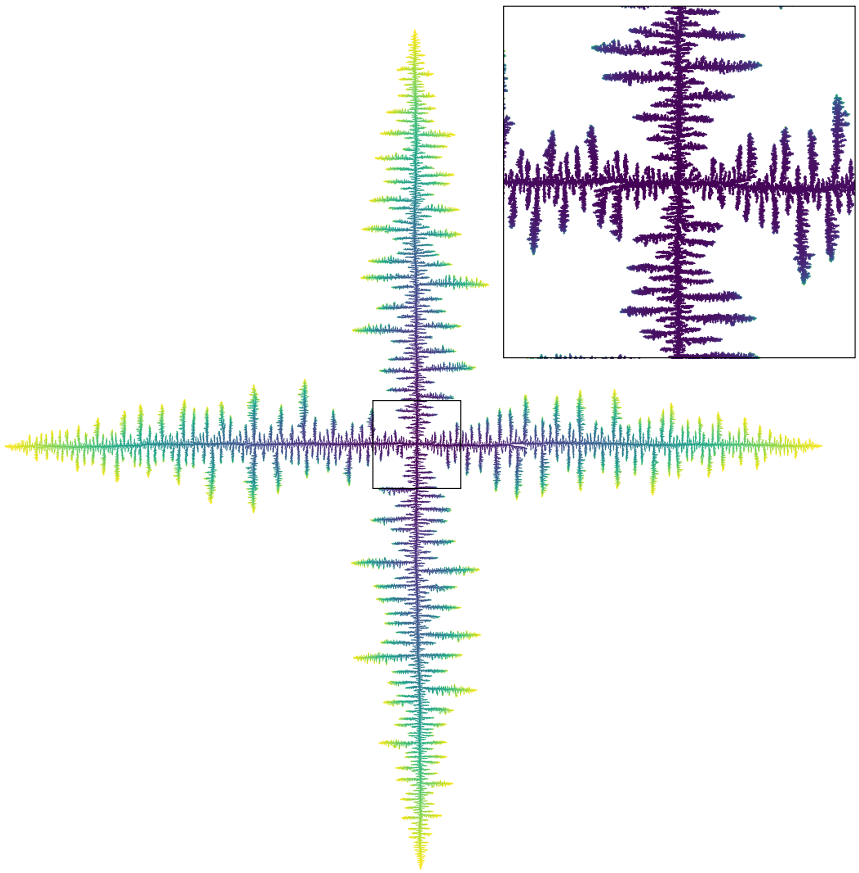


Figure 8.4: The external DLA cluster of 10^{10} sites of Figure 8.3, drawn in blocks of 512×512 sites, where each block that meets the cluster is colored from dark to light by the time at which the first of its sites joined. Its arms reach about 1.07×10^6 sites from the origin, and the inset magnifies the small central square fourfold.

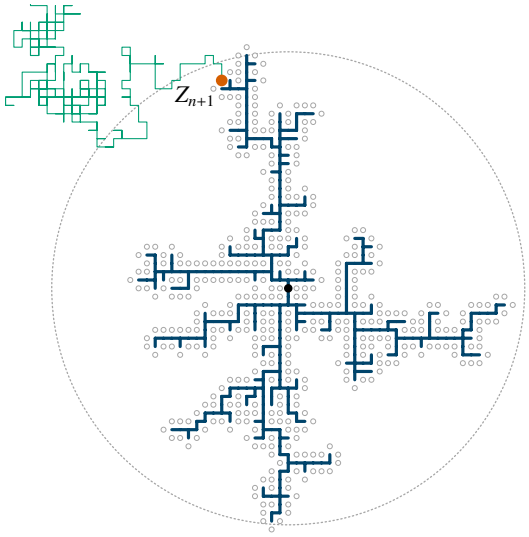


Figure 8.5: One step of external DLA: a walk started far away (green, last steps only) first visits the outer boundary (open dots) of a cluster $\mathcal{X}_n$ of 320 sites, drawn as a tree, at the site Z_{n+1} that joins the cluster. Before then it avoids the connected set $\mathcal{X}_n \cup \{Z_{n+1}\}$, which reaches the dotted circle of radius $\mathrm{rad}(\mathcal{X}_n)$, so by the Beurling estimate each boundary site is added with probability at most $C(1 + \mathrm{rad}(\mathcal{X}_n))^{-1/2}$, as in (8.4).

Definition 8.1 (External DLA) Let $\mathcal{X}_0 = \{\mathbf{0}\}$, and let $\mathcal{F}_n$ be the σ-algebra generated by $\mathcal{X}_0, \dots, \mathcal{X}_n$. **External DLA** on $\mathbb{Z}^2$ is the process $(\mathcal{X}_n)_{n\ge 0}$ defined by

$$\mathcal{X}_{n+1} := \mathcal{X}_n \cup \{Z_{n+1}\},$$

where $Z_{n+1} \in \partial\mathcal{X}_n$ has conditional law

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} = H_{\mathcal{X}_n}(z) \qquad \text{for every } z \in \partial\mathcal{X}_n .$$

To bound the radius we need only an upper bound on the probability of each attachment, so we state the bound for a general nearest-neighbor growth process in $\mathbb{Z}^d$.

Proposition 8.2 *Let $d \geq 1$ be an integer, let $\beta > 0$, and let $0 < J < \infty$. Let $(\mathcal{X}_n)_{n\geq 0}$ be a process of finite subsets of $\mathbb{Z}^d$ with $\mathcal{X}_0 = \{\mathbf{0}\}$, and let $(Z_n)_{n\geq 1}$ be random sites such that, for every $n \geq 0$,*

$$Z_{n+1} \in \partial \mathcal{X}_n \qquad \text{and} \qquad \mathcal{X}_{n+1} = \mathcal{X}_n \cup \{Z_{n+1}\}\,.$$

Let $\mathcal{F}_n$ be the σ-algebra generated by $\mathcal{X}_0, \dots, \mathcal{X}_n$. Suppose that, for every $n \geq 0$ and every $z \in \partial \mathcal{X}_n$,

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} \leq J(1 + \mathrm{rad}(\mathcal{X}_n))^{-\beta}\,.$$

Then there exist $C_0, C, c > 0$, depending only on d, β, J, such that, for every $n \geq 1$ and every $R \geq C_0 n^{1/(1+\beta)}$,

$$\mathbb{P}\left\{\mathcal{X}_n \not\subset \mathrm{B}^{\ddots}(\mathbf{0}, R)\right\} \leq C \exp(-cR)\,.$$

We bound the expected number of sites beyond each integer radius by induction on time, using Pascal's identity, and then apply Markov's inequality.

Proof of Proposition 8.2. For integers $t \geq 0$ and $k \geq 1$, let $N_k(t)$ be the number of sites $x \in \mathcal{X}_t$ with $|x| \geq k$, and let $C_1 := \max\{1, 2dJ\}$. We first show that

$$\mathbb{E}N_k(t) \leq \frac{C_1^k}{(k!)^\beta}\binom{t}{k}, \tag{8.1}$$

where $\binom{t}{k} = 0$ when $k > t$.

To bound the increment in one step, let $k \geq 2$, and note that each boundary site z with $|z| \geq k$ has a neighbor in $\mathcal{X}_t$ of norm at least $k-1$ (see Figure 8.6), so there are at most $2dN_{k-1}(t)$ such sites. Each of them satisfies $1 + \mathrm{rad}(\mathcal{X}_t) \geq |z| \geq k$, so it is added with conditional probability at most $Jk^{-\beta}$. Since $N_k(t+1) - N_k(t) = \mathbf{1}\{|Z_{t+1}| \geq k\}$, summing these probabilities gives

$$\mathbb{E}[N_k(t+1) - N_k(t) \mid \mathcal{F}_t] \leq \frac{C_1}{k^\beta} N_{k-1}(t)\,.$$

We now prove (8.1) by induction on t. For $t = 0$ both sides vanish, since $\mathcal{X}_0 = \{\mathbf{0}\}$, and for $k = 1$ the bound holds since $N_1(t) \leq t \leq C_1 t$. Suppose the bound holds at time t for every $k \geq 1$, and let $k \geq 2$. Taking expectations in the previous display and using the induction hypothesis, we find

$$\mathbb{E}N_k(t+1) \leq \frac{C_1^k}{(k!)^\beta}\left[\binom{t}{k} + \binom{t}{k-1}\right] = \frac{C_1^k}{(k!)^\beta}\binom{t+1}{k},$$

where the equality is Pascal's identity. This completes the induction.

It remains to bound the tail. Let $n \geq 1$ be an integer and $R \geq 2$ real, and let $m := \lfloor R \rfloor$, so that $R/2 \leq m \leq R$. If $\mathcal{X}_n \not\subset \mathrm{B}^{\ddots}(\mathbf{0}, R)$, then $N_m(n) \geq 1$, so by

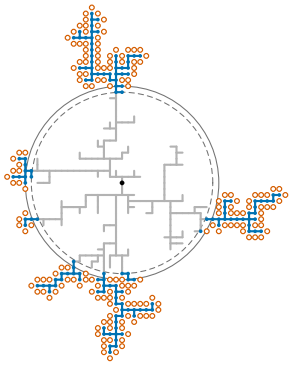


Figure 8.6: The cluster $\mathcal{X}_{320}$ of Figure 8.5 and the circles of radii $k-1$ (dashed) and k, for $k = 16$. Each site $z \in \partial\mathcal{X}_{320}$ with $|z| \geq k$ (vermilion) has a neighbor in the cluster of norm at least $k-1$ (blue), so there are at most $4N_{k-1}(320)$ of them, and each is added next with conditional probability at most $Jk^{-\beta}$.

Markov's inequality, (8.1), $\binom{n}{m} \leq n^m/m!$ and $m! \geq (m/e)^m$ (which follows from $e^m \geq m^m/m!$), we have

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{\cdot\cdot}(\mathbf{0}, R)\} \leq \mathbb{E}N_m(n) \leq \frac{C_1^m n^m}{(m!)^{1+\beta}} \leq \left(\frac{(2e)^{1+\beta} C_1 n}{R^{1+\beta}}\right)^m , \tag{8.2}$$

where in the last step we used $m \geq R/2$.

Finally, choose $C_0 \geq 2$ so that $C_0^{1+\beta} \geq e(2e)^{1+\beta}C_1$. If $R \geq C_0 n^{1/(1+\beta)}$, then the base in (8.2) is at most e^{-1}, so the probability is at most $e^{-m} \leq e^{-R/2}$. This proves the proposition with $C = 1$ and $c = 1/2$. □

For R much larger than $n^{1/(1+\beta)}$, Exercise 8.1 asks you to deduce the stronger bound $C\exp[-cR(1 + \log(R/n^{1/(1+\beta)}))]$ from (8.2).

For planar external DLA, the hypothesis of Proposition 8.2 holds with $\beta = \frac{1}{2}$ by a sharp bound on the harmonic measure of a connected set, the discrete Beurling estimate, and this gives radius of order $n^{2/3}$. In the next section we prove the Beurling estimate using the potential kernel, and show that the exponent $\frac{1}{2}$ is sharp at the endpoint of a horizontal segment.

8.2 THE DISCRETE BEURLING ESTIMATE

How likely is a random walk started far from a finite connected set K to hit K first at a given site x? Let $\mathrm{diam}(K) := \max\{|y - z| : y, z \in K\}$ be the diameter of K. The set K contains a nearest-neighbor path from x to a point at distance at least $\mathrm{diam}(K)/2$ from x, and a walk that hits K first at x must avoid every other site of this path. The discrete Beurling estimate says that this is unlikely when the path is long, in the sense that the harmonic measure

from infinity satisfies

$$\mathcal{H}_K(x) \le C(1 + \operatorname{diam}(K))^{-1/2} .$$

The exponent $\frac{1}{2}$ is the best possible, since it is attained at the endpoint of a straight segment (Proposition 8.7).

8.2.1 The Upper Bound

Our proof is the lattice version of the proof of the Beurling estimate for Brownian motion (Theorem 2.44 in the starred Section 2.6), with the potential kernel a in place of the logarithm, but it does not depend on that section. We use the potential U_A of Lemma 3.29 and the asymptotic expansion $a(x) = \frac{1}{2\pi}\log|x| + \kappa + O(|x|^{-2})$ of (3.34), where κ is a constant. Recall from part (b) of the lemma that the harmonic measure of a finite set A is the Laplacian of U_A, and that U_A vanishes on A, is harmonic off A and grows like a at infinity. If F is harmonic off A and $F - a$ is bounded, then by (3.37) we have

$$U_A \le F - \min_A F \qquad \text{and} \qquad \mathcal{H}_A(x) \le \sum_{u\sim x}(F(u) - \min_A F) \quad \text{for } x \in A .$$

We will build such an F from the arcsine law $\text{Arcsine}_{0,n}$ on $[0, n]$, whose logarithmic potential is constant on this interval by Lemma 2.42(a). For a set with a point at each distance between $j-1$ and j from x, for $1 \le j \le n$, we give that point the mass that $\text{Arcsine}_{0,n}$ gives to the interval $[j-1, j]$. Replacing $a(z)$ by $\frac{1}{2\pi}\log r + \kappa$, for r within two of $|z|$, changes it by at most a constant times $\log(1 + 1/r)$, and the average of this error under the arcsine law is of order $n^{-1/2}$. We bound this average first, using only Lemma 2.42(b).

Lemma 8.3 *Let $n > 0$ be real, and let $\text{Arcsine}_{0,n}$ be the arcsine law on $[0, n]$. Then, for every real number s,*

$$\int \log\Bigl(1 + \frac{1}{|s-t|}\Bigr)\,\text{Arcsine}_{0,n}(\,\mathrm{d}t) \le \frac{4}{\sqrt{n}} .$$

Proof. We parametrize the arcsine law by an angle, which turns $|s-t|$ into the geometric mean of two variables with the arcsine law. Since moving s to the nearest point of $[0, n]$ can only increase the integrand, we may assume that $s = n\sin^2\varphi$ with $0 \le \varphi \le \pi/2$. Let Θ be uniform on $[0, \pi]$. By the substitution $t = n\sin^2\theta$, the variable $T := n\sin^2\Theta$ has law $\text{Arcsine}_{0,n}$ (the projection of the uniform distribution on a circle; see Figure 2.13), and since $\sin^2$ has period π, so do $T_\pm := n\sin^2(\Theta \pm \varphi)$. By the identity $\sin^2\theta - \sin^2\varphi = \sin(\theta+\varphi)\sin(\theta-\varphi)$ we have $|T - s| = \sqrt{T_+T_-}$, and expanding shows that $(1 + 1/\sqrt{T_+T_-})^2 \le (1 + 1/T_+)(1 + 1/T_-)$ almost surely

(the difference is $(T_+^{-1/2} - T_-^{-1/2})^2$). Taking logarithms and expectations, we bound the integral in the lemma by $\int \log(1 + 1/t)\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t)$, which is at most $2/\sqrt{n}$ by Lemma 2.42(b) with $q = 1$. □

The next lemma compares the potential kernel with the logarithm of an approximate distance.

Lemma 8.4 *There is a constant C_1 such that the following holds. Let $r > 0$ be real, and let $z \in \mathbb{Z}^2$. If $r \le |z| + 2$, then*

$$a(z) \ge \frac{1}{2\pi} \log r + \kappa - C_1 \log\Bigl(1 + \frac{1}{r}\Bigr).$$

If $\bigl||z| - r\bigr| \le 2$, then

$$\Bigl|a(z) - \frac{1}{2\pi} \log r - \kappa\Bigr| \le C_1 \log\Bigl(1 + \frac{1}{r}\Bigr).$$

Proof. Since $\log(2 + |z|) - \log|z| \le 2/|z|$ for $z \ne \mathbf{0}$, the expansion (3.34) gives a constant C_0 with $\bigl|a(z) - \frac{1}{2\pi}\log(2 + |z|) - \kappa\bigr| \le C_0/(2 + |z|)$ for every $z \in \mathbb{Z}^2$ (the finitely many sites near the origin only enlarge C_0). Both assertions follow with $C_1 = 2C_0 + 2/\pi$, since if $r \le |z| + 2$, then $\log r \le \log(2 + |z|)$ and $1/(2 + |z|) \le 2/(1 + r) \le 2\log(1 + 1/r)$, and if moreover $|z| \le r + 2$, then $\log(2 + |z|) \le \log(r + 4) \le \log r + 4\log(1 + 1/r)$. □

We can now bound the harmonic measure at a point x of a finite set that has a point at every distance from x up to $n - 1$, within one unit. Only these distances matter (the set need not be connected).

Theorem 8.5 (Discrete Beurling Estimate) *There is an absolute constant C such that the following holds. Let $n \ge 1$ be an integer, let $K \subset \mathbb{Z}^2$ be finite, and let $x \in K$. Suppose that, for every integer $1 \le j \le n$, the set K contains a point whose distance from x is at least $j - 1$ and less than j. Then*

$$\mathcal{H}_K(x) \le \frac{C}{\sqrt{n}}.$$

Proof. The idea is to give n points of K, one at each distance, the masses of the arcsine law, and to compare the potential of these masses with the potential U_A of the set A of these points (Lemma 3.29). By translation we may take $x = \mathbf{0}$. For $1 \le j \le n$ choose $y_j \in K$ with $j - 1 \le |y_j| < j$; these points are distinct, and $y_1 = \mathbf{0}$. Let $A := \{y_1, \dots, y_n\}$, let $\mathrm{Arcsine}_{0,n}$ be the arcsine law on $[0, n]$, and let

$$F(w) := \sum_{j=1}^{n} \mathrm{Arcsine}_{0,n}([j-1, j])\, a(w - y_j) \qquad \text{for } w \in \mathbb{Z}^2.$$

Since the masses add up to one, F is harmonic off A and $F - a$ is bounded, so (3.37) applies to F. For $j - 1 < t < j$ let $y(t) := y_j$. Since $\text{Arcsine}_{0,n}$ has no atoms, we have $F(w) = \int a(w - y(t)) \, \text{Arcsine}_{0,n}(\, \mathrm{d}t)$, and $\big||y(t)| - t\big| < 1$ for $\text{Arcsine}_{0,n}$-almost every t.

We first bound F from below on A. Let $v \in A$. For $\text{Arcsine}_{0,n}$-almost every t with $0 < t < n$, we have $t \neq |v|$ and $\big||v| - t\big| \le |v - y(t)| + 1$, so we may bound $a(v - y(t))$ from below by the first assertion of Lemma 8.4 with $r = \big||v| - t\big|$. Since $0 \le |v| < n$, integrating this bound against $\text{Arcsine}_{0,n}$ and using Lemma 2.42(a) and Lemma 8.3, we get

$$F(v) \ge \frac{1}{2\pi} \log \frac{n}{4} + \kappa - \frac{4C_1}{\sqrt{n}} \, .$$

Next we bound F from above at the neighbors of the origin. Let $u \sim \mathbf{0}$. For $\text{Arcsine}_{0,n}$-almost every t with $0 < t < n$, we have $\big||u - y(t)| - t\big| < 2$, because $|u| = 1$. Hence, by the second assertion of Lemma 8.4 with $r = t$, Lemma 2.42(a) at $s = 0$, and Lemma 8.3,

$$F(u) \le \frac{1}{2\pi} \log \frac{n}{4} + \kappa + \frac{4C_1}{\sqrt{n}} \, .$$

By (3.37), it follows that $U_A(u) \le F(u) - \min_A F \le 8C_1/\sqrt{n}$. By part (b) of Lemma 3.29, $\mathcal{H}_A(\mathbf{0})$ is the sum of U_A over the four neighbors of the origin, so $\mathcal{H}_A(\mathbf{0}) \le 32C_1/\sqrt{n}$.

Finally, since $\mathbf{0} \in A \subset K$, a walk that first hits K at $\mathbf{0}$ also first hits A at $\mathbf{0}$. Letting its starting point tend to infinity, we get $\mathcal{H}_K(\mathbf{0}) \le \mathcal{H}_A(\mathbf{0}) \le 32C_1/\sqrt{n}$. □

For a connected set, we may take the points required by Theorem 8.5 on a path to a farthest point (see Figure 8.7), since one step of the path changes the distance from x by at most one. This gives the discrete Beurling estimate in the form stated at the start of the section.

Corollary 8.6 *There is an absolute constant C such that the following holds. For all finite connected sets $K \subset \mathbb{Z}^2$ and all $x \in K$, the harmonic measure from infinity satisfies*

$$\mathcal{H}_K(x) \le C(1 + \text{diam}(K))^{-1/2} \, .$$

Proof. Choose $w \in K$ with $|w - x| \ge \text{diam}(K)/2$; it exists because two points of K at distance $\text{diam}(K)$ cannot both lie within distance less than $\text{diam}(K)/2$ of x. Let $n := \lfloor |w - x| \rfloor + 1$. Since K is connected, it contains a nearest-neighbor path from x to w. For $1 \le j \le n$, the path has a first point at distance at least $j - 1$ from x, because $|w - x| \ge n - 1$, and this point lies at distance less than j from x, because one step changes the distance from x

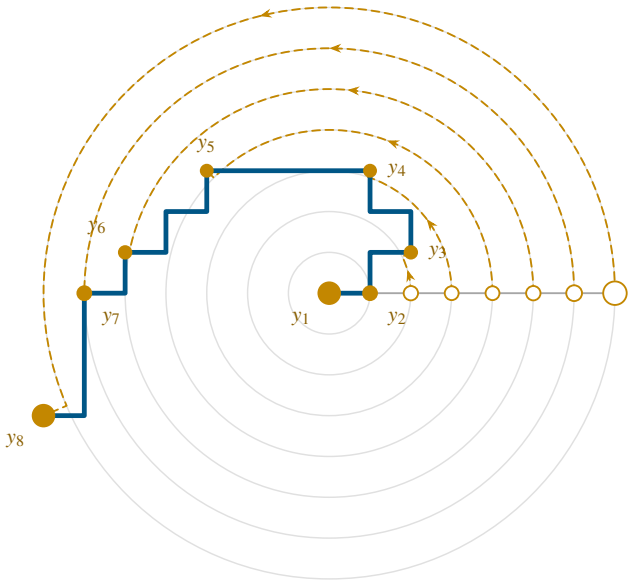


Figure 8.7: In the proof of Theorem 8.5, with x at the origin, each point $y_j \in K$ is given the mass that the arcsine law on $[0, n]$ gives to $[j-1, j]$. Here y_j is the first point of a path in K at distance at least $j-1$ from the origin, so its distance is less than j. Each dot has area proportional to its mass, and the circles have radii $1, 2, \ldots, 7$.

by at most one. By Theorem 8.5, it follows that $\mathcal{H}_K(x) \le Cn^{-1/2}$, and since $n \ge \max\{1, \operatorname{diam}(K)/2\} \ge \frac{1}{4}(1 + \operatorname{diam}(K))$, this proves the corollary after increasing C. □

8.2.2 Sharpness on a Horizontal Segment

The harmonic measure at the endpoint of a straight segment shows that the exponent $\frac{1}{2}$ in Theorem 8.5 cannot be improved (see Figure 3.16 for this harmonic measure on a segment of 11 sites). We put the masses of the arcsine law on the segment, and this time we bound the potential of the segment from below.

Proposition 8.7 *There are absolute constants $c, C > 0$ such that the following holds. For an integer $n \ge 1$, let $\mathcal{J}_n := \{\mathbf{0}, \mathbf{e}_1, \ldots, (n-1)\mathbf{e}_1\}$. Then*

$$\frac{c}{\sqrt{n}} \le \mathcal{H}_{\mathcal{J}_n}(\mathbf{0}) = \mathcal{H}_{\mathcal{J}_n}((n-1)\mathbf{e}_1) \le \frac{C}{\sqrt{n}}\,.$$

Proof. We use the potential of the proof of Theorem 8.5, now on the segment

itself. Let $\mathrm{Arcsine}_{0,n}$ be the arcsine law on $[0,n]$, and let

$$F(w) := \sum_{j=1}^{n} \mathrm{Arcsine}_{0,n}([j-1,j])\, a(w-(j-1)\mathbf{e}_1) \qquad \text{for } w \in \mathbb{Z}^2 .$$

As there, F is harmonic off $\mathcal{J}_n$ and $F-a$ is bounded. Let k be an integer. For $j-1 < t < j$ we have $\big||k-(j-1)| - |k-t|\big| < 1$, so integrating the second assertion of Lemma 8.4 with $r = |k-t|$ against $\mathrm{Arcsine}_{0,n}$ and using Lemma 8.3, we obtain

$$\Big| F(k\mathbf{e}_1) - \frac{1}{2\pi} \int \log|k-t|\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t) - \kappa \Big| \le \frac{4C_1}{\sqrt{n}} . \tag{8.3}$$

For $0 \le k \le n-1$, the integral equals $\log(n/4) = \int \log t\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t)$, by Lemma 2.42(a).

Next we bound $U_{\mathcal{J}_n}$ from below at the point $-M\mathbf{e}_1$, where $M \ge 2$ is an integer to be chosen. By (3.37), and by (8.3) at $k = -M$ and on $\mathcal{J}_n$,

$$\begin{aligned} U_{\mathcal{J}_n}(-M\mathbf{e}_1) &\ge F(-M\mathbf{e}_1) - \max_{\mathcal{J}_n} F \\ &\ge \frac{1}{2\pi} \int \log\Big(1+\frac{M}{t}\Big)\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t) - \frac{8C_1}{\sqrt{n}} \\ &\ge \frac{\log(1+M)/\pi^2 - 8C_1}{\sqrt{n}} . \end{aligned}$$

In the last step we kept only $0 < t < 1$, on which the density $1/(\pi\sqrt{t(n-t)})$ of $\mathrm{Arcsine}_{0,n}$ is at least $1/(\pi\sqrt{nt})$ and $\log(1+M/t) \ge \log(1+M)$, and used $\int_0^1 t^{-1/2}\,\mathrm{d}t = 2$. Now choose M so that the numerator is at least one.

Since $U_{\mathcal{J}_n}$ is nonnegative and harmonic off $\mathcal{J}_n$, its value at $-k\mathbf{e}_1$ is at least a quarter of its value at $-(k+1)\mathbf{e}_1$, for every $k \ge 1$. By part (b) of Lemma 3.29, $\mathcal{H}_{\mathcal{J}_n}(\mathbf{0})$ is the sum of $U_{\mathcal{J}_n}$ over the four neighbors of the origin, so iterating for $1 \le k < M$ gives

$$\mathcal{H}_{\mathcal{J}_n}(\mathbf{0}) \ge U_{\mathcal{J}_n}(-\mathbf{e}_1) \ge 4^{1-M} U_{\mathcal{J}_n}(-M\mathbf{e}_1) \ge \frac{4^{1-M}}{\sqrt{n}} .$$

The reflection through the midpoint of the segment exchanges its endpoints, so their harmonic measures are equal. The upper bound follows from Theorem 8.5, since $\mathcal{J}_n$ has a point at distance $j-1$ from the origin for every $1 \le j \le n$. □

8.2.3 Other Lattices*

The proofs of Theorem 8.5 and Corollary 8.6 use only a few properties of the square lattice, namely recurrence, the identity $\Delta^{\cdot\cdot} a = \delta_{\mathbf{0}}$, the symmetry

$a(-x) = a(x)$, the expansion (3.34) with an error of order $|x|^{-1}$, and the bounded length of a step. Hence the discrete Beurling estimate extends to every irreducible symmetric random walk with bounded steps on a two-dimensional lattice (such as simple random walk on the triangular lattice), with $|x|$ replaced in the proof by the norm in which the covariance matrix of a step of the walk is a multiple of the identity. If L bounds the step lengths in this norm, then a path of allowed steps cannot skip an annulus of width L before reaching its farthest point, so we may select points in these annuli and replace the errors of one and two units in the proof by L and $2L$ (the constants may then depend on the walk). We prove recurrence and these properties of the potential kernel only for simple random walk on $\mathbb{Z}^2$; for the other walks, see the Notes.

The discrete Beurling estimate also applies to the hexagonal lattice, with a different constant. Let K be a finite connected set of vertices of the hexagonal lattice, let $x \in K$, and let K_0 be the set of points of K in the same class of the bipartition as x. This class is a triangular lattice, in which two points are neighbors when they have a common neighbor (see Figure 8.8), and a walk started in this class and observed at even times is the random walk on this triangular lattice that stays put with probability $\frac{1}{3}$ and moves to each of the six neighbors with probability $\frac{1}{9}$. A walk that first hits K at x also first hits K_0 at x. The points of a path in K that lie in the class of x form a path in this triangular lattice, so K_0 is connected there and has a point at distance at least $\operatorname{diam}(K)/2 - 1$ from x. Since a walk started in the other class enters this class after one step, the estimate on the triangular lattice applies to the harmonic measure of K from infinity at x.

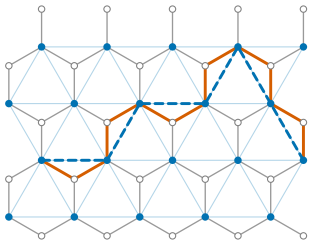


Figure 8.8: The vertices of one class of the hexagonal lattice (blue) form a triangular lattice (light blue), in which two vertices are neighbors when they have a common neighbor. The vertices of a path (vermilion) that lie in this class form a path of this triangular lattice (dashed).

8.3 THE RADIUS OF THE CLUSTER

We now return to external DLA. Let $A \subset \mathbb{Z}^2$ be finite and connected with $\mathbf{0} \in A$, and let $z \in \partial A$. A walk from far away must visit ∂A before it enters A, so a walk whose first visit to ∂A is at z first hits the connected set $A \cup \{z\}$ at z (like the walk in Figure 8.5). Since $\operatorname{diam}(A \cup \{z\}) \geq \operatorname{rad}(A)$, it follows from Corollary 8.6 that

$$H_A(z) \leq \mathcal{H}_{A\cup\{z\}}(z) \leq C(1 + \operatorname{rad}(A))^{-1/2} . \tag{8.4}$$

Combining this bound with Proposition 8.2, we obtain the main result of the chapter.

Theorem 8.8 *Let $(\mathcal{X}_n)_{n\geq 0}$ be external DLA on $\mathbb{Z}^2$. There are constants $C_0, C, c > 0$ such that for all integers $n \geq 1$ and all real $R \geq C_0 n^{2/3}$ we have*

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{\ddots}(\mathbf{0}, R)\} \leq C \exp(-cR) .$$

Proof. The external DLA cluster $\mathcal{X}_n$ is connected and contains $\mathbf{0}$ (each new site is adjacent to an earlier one), so by Definition 8.1 and (8.4),

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} = H_{\mathcal{X}_n}(z) \leq C(1 + \operatorname{rad}(\mathcal{X}_n))^{-1/2} \qquad \text{for every } z \in \partial\mathcal{X}_n .$$

This is the hypothesis of Proposition 8.2 with $d = 2$ and $\beta = \frac{1}{2}$, and its conclusion is the theorem, since $1/(1+\beta) = 2/3$. □

With $R = C_0 n^{2/3}$, the probabilities $C\exp(-cC_0 n^{2/3})$ are summable in n, so by the Borel–Cantelli lemma, almost surely $\mathcal{X}_n \subset \mathrm{B}^{\ddots}(\mathbf{0}, C_0 n^{2/3})$ for all sufficiently large n. In other words, the radius of the cluster grows at most like $n^{2/3}$, while the off-lattice simulations suggest the smaller exponent $1/d_f \approx 0.585$. In the next section we turn to dimensions $d \geq 3$, where a walk from far away may miss the cluster altogether.

8.4 EXTERNAL DLA IN HIGHER DIMENSIONS*

Throughout this section, X is simple random walk on $\mathbb{Z}^d$ with $d \geq 3$, and $\mathbb{P}_y$ and $\mathbb{E}_y$ denote its law and expectation started at y. For $E \subset \mathbb{Z}^d$, the hitting time τ_E is as in (3.4), and $\tau_E^+ := \inf\{n \geq 1 : X_n \in E\}$ is the return time. The constants c and C depend only on d, and their values may change from line to line.

Since simple random walk in dimensions $d \geq 3$ is transient (Lemma 3.6), a walk released far from a finite set will probably never hit it, and the limit in Definition 3.28 is zero. We therefore condition the walk to hit the set

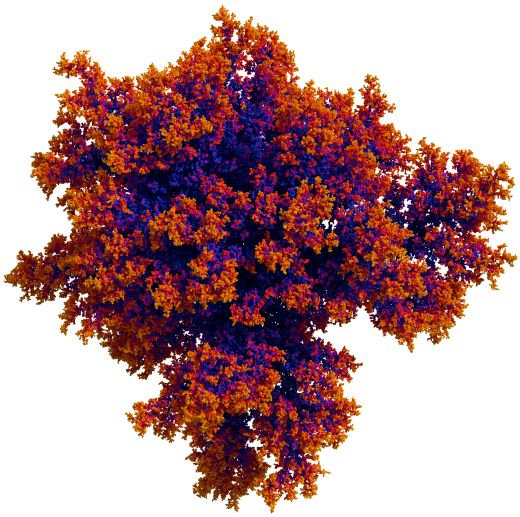


Figure 8.9: An external DLA cluster of 10^6 sites in $\mathbb{Z}^3$, each site colored by the rank of its harmonic measure from infinity, estimated from 4×10^7 walks, from navy (smallest, including zero) through blue, magenta and red to orange (largest). As in the plane, harmonic measure is largest at the tips of the arms.

before letting its starting point tend to infinity, and the resulting harmonic measure of a site is its escape probability divided by the capacity of the set (see Proposition 8.9 and Figure 8.9). We prove that, almost surely, the radius of $\mathcal{X}_n$ is at most $Cn^{2/(d+1)}$ when $d \geq 4$ (Theorem 8.14) and at most $Cn^{1/2}(\log n)^{1/4}$ when $d = 3$ (Theorem 8.16), for all large n. In both proofs we bound the probability that the sites of a path are added in order, instead of bounding each attachment separately.

8.4.1 Harmonic Measure and Capacity

Recall from Section 3.3 that $2d\, g^{::}(y - x)$ is the expected number of visits to y of a walk from x, so $g^{::} \leq g^{::}(\mathbf{0})$ by the strong Markov property at the first visit. Since $g^{::}$ is even and positive, and has the expansion $g^{::}(x) = a_d|x|^{2-d} + O(|x|^{-d})$ with the constant $a_d > 0$ of (3.16), we have, for all $x, v \in \mathbb{Z}^d$,

$$c(1+|x|)^{2-d} \leq g^{::}(x) \leq C(1+|x|)^{2-d} \qquad \text{and} \qquad \lim_{|y|\to\infty} \frac{g^{::}(y-v)}{g^{::}(y)} = 1\,. \tag{8.5}$$

As in Exercise 3.46 of Chapter 3, the **escape probability** of a site x of a finite nonempty set $K \subset \mathbb{Z}^d$ is $\mathrm{Es}_K(x) := \mathbb{P}_x\{\tau_K^+ = \infty\}$, and the **capacity** of K is $\mathrm{cap}(K) := \sum_{x\in K} \mathrm{Es}_K(x)$. We set $\mathrm{Es}_K(x) := 0$ for $x \notin K$.

In the next proposition, the main tool of this section, we express hitting

probabilities through escape probabilities (part (a), the last-exit decomposition) and find the limit of the law of X_{τ_K} given $\tau_K < \infty$.

Proposition 8.9 *Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty.*

(a) For every $y \in \mathbb{Z}^d$,

$$\mathbb{P}_y\{\tau_K < \infty\} = 2d \sum_{x \in K} g^{::}(x - y)\ \mathrm{Es}_K(x)\,. \tag{8.6}$$

In particular, $\mathrm{cap}(K) \geq 1/(2d\, g^{::}(\mathbf{0}))$.

(b) For every $x \in K$,

$$\lim_{|y| \to \infty} \mathbb{P}_y\{X_{\tau_K} = x \mid \tau_K < \infty\} = \frac{\mathrm{Es}_K(x)}{\mathrm{cap}(K)}\,.$$

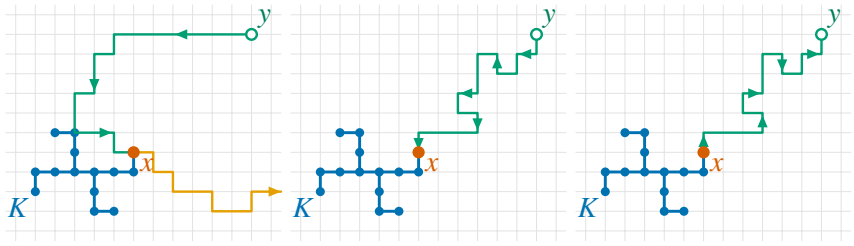
Proof. (a) The expected number of visits to K of a walk from y is finite, so on the event $\{\tau_K < \infty\}$ the walk makes a last visit to K (Figure 8.10(a)). By the Markov property, the walk is at $x \in K$ at time j and never returns to K afterwards with probability $\mathbb{P}_y\{X_j = x\}\ \mathrm{Es}_K(x)$. Summing over j and x gives (8.6). For $y \in K$ the left side equals 1, and since $g^{::} \leq g^{::}(\mathbf{0})$, we get the bound on the capacity.

(b) Fix $x \in K$ and $y \notin K$. Reversing a path from y that first enters K at x, at time j, gives a path from x that is at y at time j and avoids K at the times $1, \dots, j$, and both paths have probability $(2d)^{-j}$ (see Figure 8.10(b) and (c)). Summing over these paths, and then splitting the visits to y at the return time τ_K^+ by the strong Markov property, we obtain

$$\begin{aligned}\mathbb{P}_y\{X_{\tau_K} = x,\ \tau_K < \infty\} &= \mathbb{E}_x \sum_{0 \leq j < \tau_K^+} \mathbf{1}\{X_j = y\} \\ &= 2d\, g^{::}(y - x) - 2d\, \mathbb{E}_x[g^{::}(y - X_{\tau_K^+});\ \tau_K^+ < \infty]\,.\end{aligned}$$

Now divide by $2d\, g^{::}(y)$ and let $|y| \to \infty$. Since the last expectation is a finite sum over the sites of K, by the limit in (8.5) the right side tends to $1 - \mathbb{P}_x\{\tau_K^+ < \infty\} = \mathrm{Es}_K(x)$. By (8.6) and (8.5), $\mathbb{P}_y\{\tau_K < \infty\}/(2d\, g^{::}(y))$ tends to $\mathrm{cap}(K)$, which is positive by (a). Dividing the two limits proves (b). □

We call the limit $\mathcal{H}_K(x)$ in part (b) the harmonic measure of K from infinity. In the plane the walk hits K almost surely, so this agrees with Definition 3.28. As there, $H_A(z) := \mathcal{H}_{\partial A}(z)$ for $z \in \partial A$ is the harmonic measure of the outer boundary of a finite nonempty set $A \subset \mathbb{Z}^d$, and external DLA on $\mathbb{Z}^d$, with Z_{n+1} and $\mathcal{F}_n$ as there, is the process of Definition 8.1 with $\mathbb{Z}^d$ in place of $\mathbb{Z}^2$. A walk started outside the closure $\bar{A} = A \cup \partial A$ of Section 3.1 can enter $\bar{A}$



(a) (b) (c)

Figure 8.10: (a) A path from y that visits K for the last time at x, after which (orange) it never returns. (b) A path from y that first enters K at x, and (c) its reversal, a path from x to y that does not return to K, which has the same probability.

only through ∂A, so it hits $\bar{A}$ and ∂A at the same time and at the same site. Hence, by Proposition 8.9(b),

$$H_A(z) = \mathcal{H}_{\bar{A}}(z) = \frac{\mathrm{Es}_{\bar{A}}(z)}{\mathrm{cap}(\bar{A})} \qquad \text{for every } z \in \partial A\,. \tag{8.7}$$

Every neighbor of a site of A lies in $\bar{A}$, so $\mathrm{Es}_{\bar{A}}$ vanishes on A.

The next estimate, on the potential of sites with first coordinates $0, 1, \dots, r$, is different for $d = 3$ and $d \geq 4$. Let $r \geq 1$ be an integer, and let $x_0, \dots, x_r \in \mathbb{Z}^d$ have first coordinates $(x_i)_1 = i$. For every $z \in \mathbb{Z}^d$ we have $|x_i - z| \geq |i - z_1|$, so by (8.5),

$$2d \sum_{i=0}^{r} g^{::}(x_i - z) \leq C \sum_{i=0}^{r} (1 + |i - z_1|)^{2-d} \leq \begin{cases} C \log(r+2), & d = 3, \\ C, & d \geq 4. \end{cases} \tag{8.8}$$

Summing (8.6) over such sites, we bound the capacity of a connected set from below.

Proposition 8.10 *Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be a finite connected set with at least two sites. Then*

$$\mathrm{cap}(K) \geq \begin{cases} c\ \mathrm{diam}(K)/\log(\mathrm{diam}(K)+2), & d = 3, \\ c\ \mathrm{diam}(K), & d \geq 4. \end{cases}$$

Proof. Choose a coordinate in which the sites of K take the largest range of values, and let r be the length of this range. Then $r \geq \mathrm{diam}(K)/\sqrt{d}$, and $r \geq 1$. After a symmetry of the lattice, the first coordinates of the sites of K range over $\{0, 1, \dots, r\}$. Since the first coordinate changes by at most one along a path in K, the connected set K contains sites $x_0, \dots, x_r$ with $(x_i)_1 = i$ (which need not be neighbors; see Figure 8.11). Summing (8.6)

over $y = x_0, \dots, x_r$, where its left side equals one, we get

$$r + 1 = 2d \sum_{z \in K} \mathrm{Es}_K(z) \sum_{i=0}^{r} g^{::}(x_i - z) \le \mathrm{cap}(K) \max_{z \in \mathbb{Z}^d} 2d \sum_{i=0}^{r} g^{::}(x_i - z)\,.$$

By (8.8), $\mathrm{cap}(K) \ge cr/\log(r+2)$ when $d = 3$ and $\mathrm{cap}(K) \ge cr$ when $d \ge 4$. Since $t \mapsto t/\log(t+2)$ increases on $[0, \infty)$ and $r \ge \mathrm{diam}(K)/\sqrt{d}$, the proposition follows. □

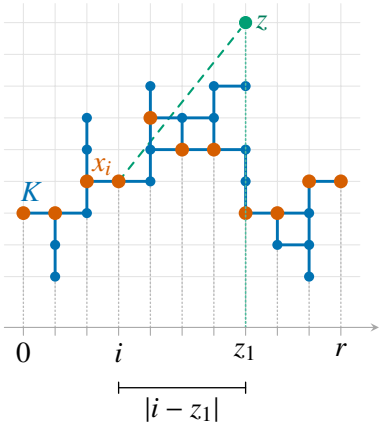


Figure 8.11: A connected set K contains a site x_i with first coordinate i for each $0 \le i \le r$, and every site z satisfies $|x_i - z| \ge |i - z_1|$.

Since $\mathrm{Es}_K \le 1$, by Proposition 8.9(b) we have $\mathcal{H}_K \le 1/\mathrm{cap}(K)$. The connected set $\overline{\mathcal{X}_n}$ contains the neighbors of $\mathbf{0}$ and a site at distance $\mathrm{rad}(\mathcal{X}_n)$ from $\mathbf{0}$, so its diameter is at least $(1 + \mathrm{rad}(\mathcal{X}_n))/2$. Hence for $d \ge 4$, by (8.7) and Proposition 8.10, the hypothesis of Proposition 8.2 holds with $\beta = 1$, which gives radius at most of order $n^{1/2}$. (Bounding the attachment probabilities by the number of sites instead gives order $n^{2/d}$; see Exercise 8.9.) We do better in the next two subsections.

8.4.2 The Radius in Dimensions Four and Higher

We measure the cluster in the maximum norm $|x|_\infty := \max_i |x_i|$, and for an integer $r \ge 1$ we let T_r be the first time n at which $\mathcal{X}_n$ contains a site of maximum norm r. Each new site is a neighbor of the cluster, so $T_1 < T_2 < \cdots$, and no site of $\mathcal{X}_{T_r}$ has maximum norm larger than r. The T_r sites of $\mathcal{X}_{T_r - 1}$ lie in a cube of $(2r-1)^d$ sites, so T_r is a bounded stopping time. We say that a site is added at time t if it equals Z_t. Each site of $\mathcal{X}_n$ other than $\mathbf{0}$ was added next to an earlier site (the neighbor to which it attached when it joined the cluster), so moving from $w \in \mathcal{X}_n$ to an earlier neighbor, and repeating,

gives a nearest-neighbor path in $\mathcal{X}_n$ from $\mathbf{0}$ to w whose sites were added in increasing order of time.

Next we find r sites added in order after time T_r on a path from maximum norm r to $2r$, and count such paths.

Lemma 8.11 *Let $(\mathcal{X}_n)_{n\ge 0}$ be external DLA on $\mathbb{Z}^d$ with $d \ge 3$, and let $r \ge 1$ be an integer. Then there is a nearest-neighbor path $(v_0, v_1, \dots, v_r)$ with $|v_0|_\infty = r$ and $|v_i|_\infty > r$ for $1 \le i \le r$, whose sites $v_1, \dots, v_r$ are added at times $t_1 < t_2 < \cdots < t_r$ with $T_r < t_1$ and $t_r \le T_{2r}$. At most $Cr^{d-1}(2d)^r$ nearest-neighbor paths $(v_0, \dots, v_r)$ have $|v_0|_\infty = r$.*

Proof. At time T_{2r} the cluster contains a site w with $|w|_\infty = 2r$. Take a path from $\mathbf{0}$ to w whose sites were added in order, and let v_0 be its last site of maximum norm at most r (see Figure 8.12). Since each step changes the maximum norm by at most one, $|v_0|_\infty = r$, and at least r sites follow v_0 on the path; let $v_1, \dots, v_r$ be the first r of them. Their maximum norms exceed r, so they are not in $\mathcal{X}_{T_r}$, and they were added after time T_r, in order, by time T_{2r}. For the count, v_0 has at most $(2r+1)^d - (2r-1)^d \le 2d(2r+1)^{d-1}$ choices, and each later site at most $2d$. □

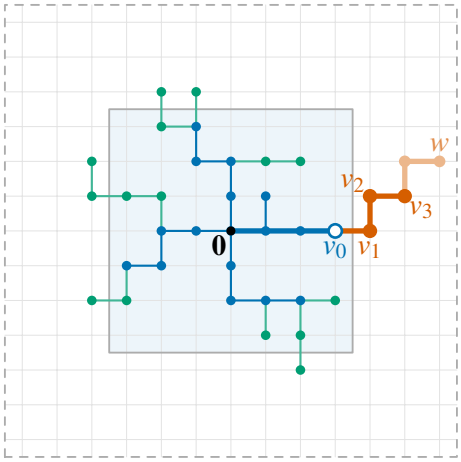


Figure 8.12: The proof of Lemma 8.11 in the plane, with $r = 3$: a path from $\mathbf{0}$ to w whose sites were added in order leaves the shaded square $\{|x|_\infty \le r\}$, which contains $\mathcal{X}_{T_r}$ (blue), for the last time at v_0. The next r sites were added after time T_r.

At time T_r the connected set $\mathcal{X}_{T_r}$ contains $\mathbf{0}$ and a site of maximum norm r. As in the proof of Proposition 8.10, it therefore contains sites $x_0, \dots, x_r$ with $(x_i)_k = \pm i$ for one coordinate k and one sign; we choose them by a fixed rule, so that they are determined by $\mathcal{X}_{T_r}$, and to simplify notation we take

$k = 1$ and the sign positive. For $m \geq T_r$, let

$$\Psi_m := 2d \sum_{i=0}^{r} \sum_{y \in \mathcal{X}_m \setminus \mathcal{X}_{T_r}} g^{::}(x_i - y)$$

be the potential at $x_0, \dots, x_r$ of the sites added after time T_r. We will use two consequences of (8.7), namely that for every $n \geq 0$, every site v and every $x \in \mathcal{X}_n$,

$$\mathbb{P}\{Z_{n+1} = v \mid \mathcal{F}_n\} \leq \frac{1}{\operatorname{cap}(\overline{\mathcal{X}}_n)} = 2d\, \mathbb{E}[g^{::}(x - Z_{n+1}) \,\big|\, \mathcal{F}_n]\,. \tag{8.9}$$

The inequality follows from $\mathrm{Es}_{\overline{\mathcal{X}}_n} \leq 1$, and the equality is (8.6) at the site $y = x$ of $\overline{\mathcal{X}}_n$, where its left side equals one. Thus no site is more likely to be added next than the conditional mean of the potential of the new site at a site of the cluster, so we expect potential of order r at each x_i once the r sites of Lemma 8.11 are added.

Proposition 8.12 *Let $(\mathcal{X}_n)_{n\geq 0}$ be external DLA on $\mathbb{Z}^d$ with $d \geq 4$. Then there are constants $c_0 > 0$ and C such that, for every integer $r \geq 1$,*

$$\mathbb{P}\{\Psi_{T_{2r}} < c_0 r(r+1) \,\big|\, \mathcal{F}_{T_r}\} \leq C r^{d-1} e^{-r}\,.$$

Consequently, almost surely $\Psi_{T_{2r}} \geq c_0 r(r+1)$ for all sufficiently large r.

We combine the path count of Lemma 8.11 with an exponential supermartingale built from the potential and the number of sites of a path added in order.

Proof. The increments $\xi_m := \Psi_{m+1} - \Psi_m$, which equal $2d \sum_i g^{::}(x_i - Z_{m+1})$, satisfy, by (8.9) and (8.8),

$$0 \leq \xi_m \leq C_1 \qquad \text{and} \qquad \mathbb{E}[\xi_m \mid \mathcal{F}_m] = \frac{r+1}{\operatorname{cap}(\overline{\mathcal{X}}_m)} \qquad \text{for } m \geq T_r\,.$$

Let $\lambda := \log(2d) + 2$ and $\theta := (e^\lambda - 1)/((1 - e^{-1})(r+1))$. We assume that r is so large that $\theta C_1 \leq 1$; the finitely many other values of r only enlarge C.

Given a nearest-neighbor path $(v_0, \dots, v_r)$ with $|v_0|_\infty = r$, let $N_{T_r} := 0$, and for $m \geq T_r$ let $N_{m+1} := N_m + 1$ if $N_m < r$ and $Z_{m+1} = v_{N_m+1}$, and $N_{m+1} := N_m$ otherwise. Thus N_m counts how many of the sites $v_1, v_2, \dots$ have been added, in this order, by time m. By (8.9),

$$\mathbb{E}[e^{\lambda(N_{m+1} - N_m)} \,\big|\, \mathcal{F}_m] \leq 1 + \frac{e^\lambda - 1}{\operatorname{cap}(\overline{\mathcal{X}}_m)}\,.$$

Since $e^{-t} \le 1-(1-e^{-1})t$ for $0 \le t \le 1$, and $0 \le \theta\xi_m \le 1$,

$$\mathbb{E}[e^{-\theta\xi_m} \,\big|\, \mathcal{F}_m] \le 1-(1-e^{-1})\frac{\theta(r+1)}{\operatorname{cap}(\overline{X}_m)} = 1-\frac{e^{\lambda}-1}{\operatorname{cap}(\overline{X}_m)}\,.$$

If $u \ge 1$ and $0 < v \le 1$, then $uv \le u+v-1$, since $(u-1)(v-1) \le 0$. Applying this with $u = e^{\lambda(N_{m+1}-N_m)}$ and $v = e^{-\theta\xi_m}$ and adding the last two displays, we find (setting $N_m := \Psi_m := 0$ for $m < T_r$) that $Y_m := \exp(\lambda N_m - \theta\Psi_m)$ is a supermartingale with $Y_m = 1$ for $m \le T_r$. Since $Y_m \le e^{\lambda r}$ and T_{2r} is bounded, we may sum the increments $Y_{m+1} - Y_m$ over $T_r \le m < T_{2r}$, each of which has conditional mean at most zero given $\mathcal{F}_m$, to get $\mathbb{E}[Y_{T_{2r}} \mid \mathcal{F}_{T_r}] \le 1$.

Let $c_0 := (1-e^{-1})/(e^{\lambda}-1)$, so that $\theta c_0(r+1) = 1$. On the event that $N_{T_{2r}} = r$ and $\Psi_{T_{2r}} < c_0 r(r+1)$,

$$Y_{T_{2r}} > e^{\lambda r - \theta c_0 r(r+1)} = e^{(\lambda-1)r} = (2d)^r e^r\,,$$

so by Markov's inequality the conditional probability of this event is at most $(2d)^{-r}e^{-r}$. By Lemma 8.11, $N_{T_{2r}} = r$ for at least one of at most $Cr^{d-1}(2d)^r$ paths. Summing over these paths, we obtain the estimate, and since $\sum_r r^{d-1}e^{-r} < \infty$, the final assertion follows from the Borel–Cantelli lemma. □

The remaining step is deterministic. Pairs (x_i, y) at distance less than a length L of order $\sqrt{r}$ contribute at most half of the potential, while each site y contributes little to the rest.

Lemma 8.13 *Let $d \ge 4$ and $c_0 > 0$. There are constants $c > 0$ and r_0, depending only on d and c_0, such that the following holds. Let $r \ge r_0$ be an integer, let $x_0, \dots, x_r \in \mathbb{Z}^d$ have first coordinates $(x_i)_1 = i$, and let $F \subset \mathbb{Z}^d$ be a finite set with*

$$2d \sum_{i=0}^{r} \sum_{y \in F} g^{::}(x_i - y) \ge c_0 r(r+1)\,.$$

Then $|F| \ge cr^{(d+1)/2}$.

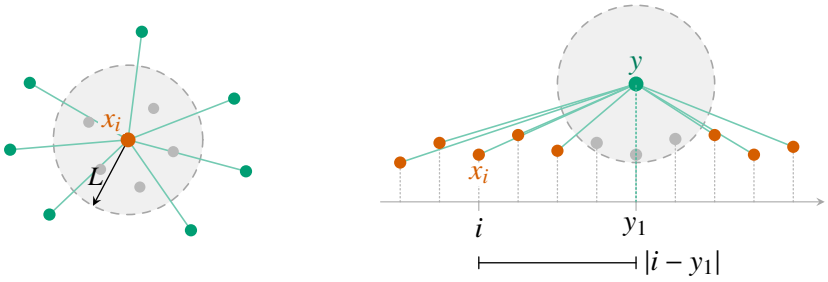
Proof. Summing (8.5) over $|z| < L$, we get $2d\sum_{|z|<L} g^{::}(z) \le C_2 L^2$ for $L \ge 1$. Let $L := (c_0 r/(2C_2))^{1/2}$, and let r_0 be so large that $L \ge 1$. Even if F contains every site within distance L of x_i, these sites contribute at most $c_0 r/2$ to the potential at x_i (see Figure 8.13(a)). Therefore

$$2d \sum_{y \in F} \sum_{i\colon |x_i - y| \ge L} g^{::}(x_i - y) \ge c_0 r(r+1) - (r+1)\frac{c_0 r}{2} = \frac{c_0}{2}\, r(r+1)\,. \quad (8.10)$$

Now fix $y \in F$. If $|x_i - y| \geq L$, then $|x_i - y| \geq \max\{L, |i - y_1|\}$, and at most $2L + 1$ indices i have $|i - y_1| < L$. Hence, by (8.5),

$$2d \sum_{i:\,|x_i - y| \geq L} g^{::}(x_i - y) \leq C(2L+1)L^{2-d} + 2C \sum_{k \geq L} k^{2-d} \leq C_3 L^{3-d},$$

where the last sum is at most a constant times L^{3-d} because $d \geq 4$ (see Figure 8.13(b)). Summing over $y \in F$ and comparing with (8.10), we find $c_0 r(r+1)/2 \leq C_3 |F| L^{3-d}$, so $|F| \geq c r^2 L^{d-3} \geq c r^{(d+1)/2}$. □



(a) (b)

Figure 8.13: (a) The sites within distance L of x_i (gray) contribute at most CL^2 there; the green pairs give the rest. (b) Since $|x_i - y| \geq \max\{L, |i - y_1|\}$ for each x_i joined to y, the site y contributes at most CL^{3-d} to these pairs.

By Proposition 8.12 and Lemma 8.13, doubling the maximum norm of the cluster from r to $2r$ takes at least of order $r^{(d+1)/2}$ particles, and this gives the following bound on the radius.

Theorem 8.14 *Let $(\mathcal{X}_n)_{n \geq 0}$ be external DLA on $\mathbb{Z}^d$ with $d \geq 4$. Then there is a constant C such that, almost surely,* $\mathrm{rad}(\mathcal{X}_n) \leq C n^{2/(d+1)}$ *for all sufficiently large n.*

Proof. The set $F := \mathcal{X}_{T_{2r}} \setminus \mathcal{X}_{T_r}$ has $T_{2r} - T_r$ sites, and its potential at $x_0, \ldots, x_r$ is $\Psi_{T_{2r}}$. By Proposition 8.12 and Lemma 8.13, almost surely $T_{2r} \geq |F| \geq c r^{(d+1)/2}$ for all sufficiently large r. Let $n \geq T_2$, and let r be the integer with $T_{2r} \leq n < T_{2r+2}$; it tends to infinity with n. Every site of $\mathcal{X}_n$ has maximum norm at most $2r + 1 \leq 3r$, and $n \geq T_{2r} \geq c r^{(d+1)/2}$ once n is large, so $\mathrm{rad}(\mathcal{X}_n) \leq 3\sqrt{d}\, r \leq C n^{2/(d+1)}$. □

8.4.3 The Radius in Dimension Three

In $\mathbb{Z}^3$, by (8.8), the increments $\xi_m = \Psi_{m+1} - \Psi_m$ are bounded only by $C \log(r+2)$, so the argument of Section 8.4.2 loses a logarithm. Instead, we bound the product of the attachment probabilities along a path. By the next

lemma, adding a site of harmonic measure p lowers the reciprocal of the capacity of the cluster by at least p^2, so by Proposition 8.10 the squares of the attachment probabilities after the cluster reaches diameter r add up to at most $C \log(r+2)/r$.

Lemma 8.15 *Let $d \geq 3$, and let $A \subset E$ be finite nonempty subsets of $\mathbb{Z}^d$. Then*

$$\operatorname{cap}(E) - \operatorname{cap}(A) = \sum_{z \in E \setminus A} \operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A = \infty\} \geq 0\,.$$

If, moreover, $E = A \cup \{z\}$ for a site $z \notin A$, then

$$\frac{1}{\operatorname{cap}(A)} - \frac{1}{\operatorname{cap}(E)} \geq \mathcal{H}_E(z)^2\,.$$

Proof. Each $w \in A$ lies in E, so the left side of (8.6) for E at $y = w$ equals one. Applying (8.6) to E at $y = w$, exchanging the sums, and applying (8.6) to A at $y = z$, we obtain, since $g^{::}$ is even,

$$\operatorname{cap}(A) = 2d \sum_{w \in A} \operatorname{Es}_A(w) \sum_{z \in E} g^{::}(z-w)\, \operatorname{Es}_E(z) = \sum_{z \in E} \operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A < \infty\}\,.$$

Subtracting this from $\operatorname{cap}(E) = \sum_{z \in E} \operatorname{Es}_E(z)$, and noting that $\mathbb{P}_z\{\tau_A < \infty\} = 1$ for $z \in A$, we obtain the identity. If $E = A \cup \{z\}$, then a walk from z that never returns to E never hits A, so $\mathbb{P}_z\{\tau_A = \infty\} \geq \operatorname{Es}_E(z)$. Using this together with $\operatorname{cap}(A) \leq \operatorname{cap}(E)$ and $\mathcal{H}_E = \operatorname{Es}_E / \operatorname{cap}(E)$ (Proposition 8.9(b)), we deduce from the identity that

$$\frac{1}{\operatorname{cap}(A)} - \frac{1}{\operatorname{cap}(E)} = \frac{\operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A = \infty\}}{\operatorname{cap}(A) \operatorname{cap}(E)} \geq \frac{\operatorname{Es}_E(z)^2}{\operatorname{cap}(E)^2} = \mathcal{H}_E(z)^2\,.$$

□

For $j \geq 1$, let $p_j := H_{\mathcal{X}_{j-1}}(Z_j)$. Since $\mathcal{X}_j \subset \overline{\mathcal{X}}_{j-1}$, by Lemma 8.15 we have $\operatorname{cap}(\mathcal{X}_j) \leq \operatorname{cap}(\overline{\mathcal{X}}_{j-1})$, and $\operatorname{Es}_{\overline{\mathcal{X}}_{j-1}}(Z_j) \leq \operatorname{Es}_{\mathcal{X}_j}(Z_j)$, so by (8.7), $p_j \leq \mathcal{H}_{\mathcal{X}_j}(Z_j)$. Applying Lemma 8.15 with $A = \mathcal{X}_{j-1}$ and $E = \mathcal{X}_j$ and summing over $j > k$, we obtain

$$\sum_{j>k} p_j^2 \leq \frac{1}{\operatorname{cap}(\mathcal{X}_k)} \qquad \text{for every } k \geq 0\,. \tag{8.11}$$

To bound the radius, we sum over the paths of Lemma 8.11 and the times when their sites are added, using (8.11) to bound the product of the attachment probabilities along each path.

Theorem 8.16 *Let $(\mathcal{X}_n)_{n \geq 0}$ be external DLA on $\mathbb{Z}^3$. Then there is a constant C such that, almost surely, $\operatorname{rad}(\mathcal{X}_n) \leq C n^{1/2} (\log n)^{1/4}$ for all sufficiently large n.*

Proof. We start with a deterministic bound. Let $r \geq 1$, and recall from Section 8.4.2 that T_r is the first time at which $\mathcal{X}_n$ contains a site of maximum norm r. The connected set $\mathcal{X}_{T_r}$ contains $\mathbf{0}$ and a site of maximum norm r, so its diameter is at least r, and by Proposition 8.10 and (8.11), $\sum_{j>T_r} p_j^2 \leq C \log(r+2)/r$. By the inequality between the arithmetic and geometric means, all times $T_r < t_1 < \cdots < t_r$ satisfy

$$\prod_{i=1}^{r} p_{t_i} \leq \Big(\frac{1}{r}\sum_{i=1}^{r} p_{t_i}^2\Big)^{r/2} \leq \Big(\frac{C_4\sqrt{\log(r+2)}}{r}\Big)^r . \tag{8.12}$$

Next, let $m \geq 1$ be an integer. If $T_{2r} \leq T_r + m$, then by Lemma 8.11 there is a path $(v_0, \dots, v_r)$ with $|v_0|_\infty = r$ whose sites $v_1, \dots, v_r$ are added at times $T_r + j_1 < \cdots < T_r + j_r$, where $1 \leq j_1 < \cdots < j_r \leq m$. Fix such a path and such $j_1, \dots, j_r$, and for $1 \leq i \leq r$ let E_i be the event that $Z_{T_r+j_i} = v_i$, and let $q_i := \mathbb{P}(E_i \mid \mathcal{F}_{T_r+j_i-1})$. Since $\mathbb{E}[\mathbf{1}_{E_i}/q_i \mid \mathcal{F}_{T_r+j_i-1}] \leq 1$, with $0/0 := 0$, by conditioning successively on $\mathcal{F}_{T_r+j_r-1}, \dots, \mathcal{F}_{T_r+j_1-1}$ we see that $W := \prod_i \mathbf{1}_{E_i}/q_i$ satisfies $\mathbb{E}[W \mid \mathcal{F}_{T_r}] \leq 1$. On the intersection E of the events E_i we have $q_i = p_{T_r+j_i}$, so by (8.12), $\mathbf{1}_E \leq W(C_4\sqrt{\log(r+2)}/r)^r$, and therefore $\mathbb{P}(E \mid \mathcal{F}_{T_r}) \leq (C_4\sqrt{\log(r+2)}/r)^r$.

Finally we sum over paths and times. There are at most $Cr^2\,6^r$ paths and $\binom{m}{r} \leq (em/r)^r$ choices of $j_1 < \cdots < j_r$, so

$$\mathbb{P}\{T_{2r} - T_r \leq m \mid \mathcal{F}_{T_r}\} \leq Cr^2\Big(\frac{C_5 m\sqrt{\log(r+2)}}{r^2}\Big)^r .$$

For $m := \lfloor r^2/(2C_5\sqrt{\log(r+2)})\rfloor$ the right side is at most $Cr^2\,2^{-r}$. By the Borel–Cantelli lemma, almost surely $T_{2r} > m \geq cr^2/\sqrt{\log(r+2)}$ for all sufficiently large r. As in the proof of Theorem 8.14, let r be the integer with $T_{2r} \leq n < T_{2r+2}$. Then $\mathrm{rad}(\mathcal{X}_n) \leq 3\sqrt{3}\,r$, and since $r \leq T_{2r} \leq n$, we have $n \geq cr^2/\sqrt{\log(n+2)}$ once n is large. Hence $\mathrm{rad}(\mathcal{X}_n)^2 \leq Cn\sqrt{\log(n+2)}$ for all sufficiently large n. □

The simulations described in the Notes suggest that neither exponent is sharp.

EXERCISES

Exercise 8.1. This exercise sharpens the tail bound of Proposition 8.2. In the setting of that proposition, show that there are constants $C_0, C, c > 0$, depending only on d, β and J, such that, for every integer $n \geq 1$ and every

real number $R \geq C_0 n^{1/(1+\beta)}$,

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{::}(\mathbf{0}, R)\} \leq C \exp\left[-cR\left(1 + \log \frac{R}{n^{1/(1+\beta)}}\right)\right].$$

Hint: Use (8.2).

Exercise 8.2. This exercise computes the harmonic measure of a diagonal segment explicitly, using the values of the potential kernel on the diagonal in Lemma 3.7. For $k \geq 0$, let $b_k := 4^{-k}\binom{2k}{k}$. Let $n \geq 1$ be an integer, and for $1 \leq j \leq n$ let $z_j := (j-1)(\mathbf{e}_1 + \mathbf{e}_2)$.

(a) Show that the rational function

$$P(s) := \frac{\prod_{r=0}^{n-2}(s - r - \frac{1}{2})}{\prod_{k=0}^{n-1}(s-k)}$$

satisfies

$$P(s) = \sum_{k=0}^{n-1} \frac{b_k b_{n-1-k}}{s-k},$$

and deduce that $\sum_{j=1}^{n} b_{j-1} b_{n-j} = 1$.

(b) Show that the function $x \mapsto \sum_{j=1}^{n} b_{j-1} b_{n-j}\, a(x - z_j)$ takes the same value at each of the points $z_1, \ldots, z_n$.

(c) Deduce that the harmonic measure of $\{z_1, \ldots, z_n\}$ from infinity gives mass $b_{j-1} b_{n-j}$ to z_j, and that its mass at the endpoint z_1 is between two constant multiples of $n^{-1/2}$.

Hint for (b): By Lemma 3.7 and the evenness of a,

$$a((r+1)(\mathbf{e}_1 + \mathbf{e}_2)) - a(r(\mathbf{e}_1 + \mathbf{e}_2)) = \frac{1}{\pi(2r+1)} \qquad \text{for every integer } r\,.$$

Evaluate P at $s = i - \frac{1}{2}$ for $1 \leq i \leq n-1$.

Exercise 8.3. This exercise gives the probabilistic meaning of the weights of Exercise 8.2, a discrete counterpart of the arcsine law of Lemma 2.42. For $k \geq 0$, let $b_k := 4^{-k}\binom{2k}{k}$, and let $n \geq 1$ be an integer. Let $(S_m)_{m \geq 0}$ be simple random walk on $\mathbb{Z}$ started at 0, and let T be the largest time $m \leq 2n-2$ with $S_m = 0$.

(a) Show that the probability that S does not return to 0 during its first 2ℓ steps equals b_ℓ, for every integer $\ell \geq 0$. Hint: Use the reflection principle to count the paths that stay positive after time 0.

(b) Deduce that $\mathbb{P}\{T = 2j-2\} = b_{j-1}b_{n-j}$ for $1 \le j \le n$, and hence that $\sum_{j=1}^{n} b_{j-1}b_{n-j} = 1$.

Exercise 8.4. This exercise shows that Corollary 8.6 fails without connectedness, and that the hypothesis of Theorem 8.5 cannot be weakened to the existence of a single point of K at distance at least $n-1$ from x. For an integer $L \ge 2$, let $K_L := \{\mathbf{0}, L\mathbf{e}_1\} \subset \mathbb{Z}^2$.

(a) Show that $\mathcal{H}_{K_L}(\mathbf{0}) = \frac{1}{2}$.

(b) Conclude that there are no constants $C < \infty$ and $\beta > 0$ such that $\mathcal{H}_K(x) \le C(1 + \operatorname{diam}(K))^{-\beta}$ for every finite set $K \subset \mathbb{Z}^2$ and every $x \in K$.

Hint for (a): Two-dimensional simple random walk is recurrent, and K_L is invariant under reflection across the perpendicular bisector of the segment from $\mathbf{0}$ to $L\mathbf{e}_1$.

Exercise 8.5. This exercise proves the independence used in an alternative proof of Theorem 8.5, described in the Notes. Let $(X_n)_{n\ge 0}$ be simple random walk in $\mathbb{Z}^2$ started at the origin, let $0 < \lambda < 1$, and let T have a geometric distribution independent of the walk,

$$\mathbb{P}\{T = k\} = \lambda(1-\lambda)^k \qquad \text{for } k = 0, 1, 2, \dots .$$

Let V_+ and V_- be the events that $X_1, \dots, X_T$ avoid the nonnegative axis Λ_+ and the negative axis Λ_-, respectively, where

$$\Lambda_+ := \{k\mathbf{e}_1 : k \ge 0\} \qquad \text{and} \qquad \Lambda_- := \{k\mathbf{e}_1 : k < 0\}.$$

Prove that V_+ and V_- are independent events. Hint: Decompose the complement of V_- according to the last visit to the rightmost visited site on the negative axis, and reverse the path up to that visit.

Exercise 8.6. This exercise applies the proof of Proposition 8.2 on the triangular lattice. Let $\mathbb{T} \subset \mathbb{R}^2$ be the lattice with six nearest neighbors per site, generated by unit vectors at angles 0 and $\pi/3$. Define external DLA on $\mathbb{T}$ as in Definition 8.1, using simple random walk on $\mathbb{T}$ to define harmonic measure, and assume that the limits defining harmonic measure from infinity exist on $\mathbb{T}$ for every finite nonempty set. Assume also that there is a constant $0 < J < \infty$ such that every finite connected set $A \subset \mathbb{T}$ containing $\mathbf{0}$ satisfies

$$H_A(z) \le J(1 + \operatorname{rad}(A))^{-1/2} \qquad \text{for every } z \in \partial A .$$

By Section 8.2.3, the discrete Beurling estimate holds on $\mathbb{T}$, and the argument before (8.4) then gives this bound. Show that there are constants $C_0, C, c > 0$ such that, for every $n \ge 1$ and every $R \ge C_0 n^{2/3}$,

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathbb{T} \cap \mathrm{B}(\mathbf{0}, R)\} \le C\exp(-cR).$$

Hint: Repeat the proof of the proposition with six neighbors per site.

Exercise 8.7. This exercise expresses the capacity of Section 8.4.1 as a maximum. Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty. Show that $\mathrm{cap}(K)$ is the maximum of $\sum_{x \in K} \mu(x)$ over $\mu : K \to [0, \infty)$ with $2d \sum_{x \in K} g^{::}(y - x)\mu(x) \leq 1$ for $y \in K$, and that Es_K attains it.

Hint: Multiply the constraint by $\mathrm{Es}_K(y)$ and sum over y.

Exercise 8.8*. In the setting of Exercise 8.7, let

$$\mathcal{I}(\mu) := 2d \sum_{x,y \in \mathbb{Z}^d} \mu(x) g^{::}(x - y) \mu(y)$$

for finitely supported $\mu : \mathbb{Z}^d \to \mathbb{R}$. Show that $\mathcal{I}(\mu) \geq 0$, and that $\mathrm{Es}_K / \mathrm{cap}(K)$ minimizes $\mathcal{I}$ over the probability measures on K, with minimum $1/\mathrm{cap}(K)$.

Hint: Write $2d\, g^{::}$ as a sum over times $n \geq 0$, and group the times $2j$ and $2j + 1$.

Exercise 8.9. This exercise bounds harmonic measure by the number of sites, and applies the bound to the external DLA of Section 8.4. Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty with $|K| = n$.

(a) Show that $2d \sum_{x \in K} g^{::}(x - y) \leq C n^{2/d}$ for every $y \in \mathbb{Z}^d$, and deduce that $\mathcal{H}_K(x) \leq 1/\mathrm{cap}(K) \leq C n^{-(d-2)/d}$ for every $x \in K$.

(b) Show that $\mathbb{P}\{Z_j = z \mid \mathcal{F}_{j-1}\} \leq C j^{-(d-2)/d}$ for every $j \geq 1$ and every site z, and that $\mathbb{P}\{\mathrm{rad}(\mathcal{X}_n) \geq \ell\} \leq (C n^{2/d}/\ell)^\ell$ for all integers $n, \ell \geq 1$. Conclude that, almost surely, $\mathrm{rad}(\mathcal{X}_n) \leq C n^{2/d}$ for all sufficiently large n.

Hint: For (a), split the sum according to whether $|x - y| < n^{1/d}$. For (b), the sum of $\prod_i j_i^{-(d-2)/d}$ over $1 \leq j_1 < \cdots < j_\ell \leq n$ is at most $(\sum_{j \leq n} j^{-(d-2)/d})^\ell / \ell!$.

Exercise 8.10. This exercise shows that, when $d \geq 4$, the exponent $(d - 2)/d$ in Exercise 8.9(a) cannot be improved, even for connected sets. For integers $s, L \geq 2$, let $K := \mathrm{B}_s^{::} \cup \{j\mathbf{e}_1 : 0 \leq j \leq Ls\}$ and $x := Ls\,\mathbf{e}_1$.

(a) Show that $\mathrm{cap}(\mathrm{B}_s^{::}) \leq C s^{d-2}$, and that $\mathrm{cap}(E \cup F) \leq \mathrm{cap}(E) + |F \setminus E|$ for all finite nonempty sets $E, F \subset \mathbb{Z}^d$.

(b) Show that there is L, depending only on d, such that $\mathrm{Es}_K(x) \geq c$ for every s. Conclude that $\mathcal{H}_K(x) \geq c|K|^{-(d-2)/d}$.

Hint for (b): The last $d - 1$ coordinates of the walk perform a lazy random walk on $\mathbb{Z}^{d-1}$, which is transient because $d - 1 \geq 3$.

NOTES

Diffusion-limited aggregation was introduced by Witten and Sander (1981) as a model for irreversible aggregation under diffusion-limited transport, closely related to dendritic growth, and studied further by Witten and Sander (1983). Off-lattice simulations consistently estimate the fractal dimension d_f of the planar cluster near 1.71; Tolman and Meakin (1989) obtained 1.715 ± 0.004 from off-lattice clusters of up to 10^6 particles, refining earlier estimates of Meakin (1983). The rigorous results for planar DLA concern mainly the growth of the cluster radius, the subject of this chapter. On the square lattice the measurement of d_f is further complicated by anisotropy, since a small cluster grows nearly isotropically whereas a large cluster aligns its arms with the lattice axes, and the crossover between the two growth regimes shifts numerical estimates of d_f (for this reason, these estimates usually use off-lattice clusters). Halsey (2000), in a survey of DLA, notes that the fractal dimension is sensitive to the lattice. Grebenkov and Beliaev (2017) simulated square-lattice clusters of up to 1.45×10^8 particles and measured the growth of the radius of gyration (the root-mean-square distance of the particles from the origin). For one hundred clusters of 4.1×10^7 to 5.2×10^7 particles, the exponent of the radius of gyration had mean 0.596 and standard deviation 0.004. Restricted to the particles in narrow sectors around the lattice axes, the radius of gyration grew with exponent about 0.612, and around the diagonals with exponent about 0.535. On the basis of these simulations they argued, without proof, that a square-lattice cluster grows fastest along the axes, so that, rescaled by its diameter, it approaches a deterministic cross-like shape.

The discrete Beurling estimate, Corollary 8.6, is equivalent, by translation and comparison of radius with diameter, to the planar part of the theorem on page 167 of Kesten (1987a). Lawler (1996) proves Theorem 2.5.2 of his book by selecting points of the connected set at regularly spaced radii, as in the proof of Corollary 8.6, and comparing the Green function killed on leaving a large ball, between these points, with the same Green function between the points of a horizontal segment, both weighted by the escape probabilities of the segment, which he estimates in Section 2.4. We also select points, but we work in infinite volume with the potential kernel and compare with the arcsine law, the equilibrium measure of a segment for the logarithmic kernel, whose density has a square-root singularity at the endpoints of the segment. Our proof is the lattice version of the proof of the Beurling estimate for Brownian motion in Section 2.6. Since it uses only the potential theory of the walk (Section 8.2.3), it applies to every irreducible symmetric random walk with bounded steps on a two-dimensional lattice; the recurrence of these walks and the properties of their potential kernels used here are proved in Chapter 4 of Lawler and Limic (2010), and the

expansion in its Theorem 4.4.4. For the diagonal segment, the exact values of Lemma 3.7 and a partial-fraction identity give the harmonic measure explicitly (Exercise 8.2). These weights are also the law of the time of the last visit to zero of a one-dimensional simple random walk of length $2n-2$ (Exercise 8.3).

In another proof of the bound at the endpoint of a horizontal segment, given by Kesten (1987a) on pages 172–175, the walk is observed only at its successive visits to the horizontal axis, where it performs a one-dimensional random walk. The Wiener–Hopf identities and the Green function of the half-line for this walk are developed in Sections 17–19 of Spitzer (1976), and Kesten bounds the harmonic measure of the segment with Propositions 19.3 and 19.5 there. By Proposition 5.2.1 of Lawler and Limic (2010), for a one-dimensional walk with a symmetric step distribution that is killed with a fixed probability at each step, the events that the walk stays nonnegative and that it stays negative, at all positive times before it is killed, are independent. In Section 5.3 they apply this to the walk on the horizontal axis, killed at a geometric time, to bound the probability of avoiding a half-line up to a fixed time (Proposition 5.3.1), and killed when the planar walk leaves a horizontal strip, to bound it up to the exit from a large square (Proposition 5.3.2). Lawler (1996) proves the same independence for the planar walk in Proposition 2.4.6, by the path reversal of Exercise 8.5, and in Section 2.4 converts the resulting bound for the walk killed at a geometric time into a bound for the walk stopped on leaving a ball. Lawler and Limic (2010) then prove the Beurling estimate in Section 6.8 as a bound on the probability that a walk started at the origin reaches distance n before returning to an infinite connected set that contains the origin (Theorem 6.8.1), also by selecting one point of the set in each annulus of unit width (Theorem 6.8.2).

We prove the matching lower bound on a horizontal segment, Proposition 8.7, by placing the arcsine law on the segment and bounding its potential from below. For a classical proof, see Section 2.4 of Lawler (1996), which gives upper and lower bounds of order $n^{-1/2}$ for the harmonic measure $\mathcal{H}_{\mathcal{J}_n}(\mathbf{0})$ at the endpoint of the segment.

The bound of Theorem 8.8 is due to Kesten (1987b), whose theorem on page L29 states that, almost surely, $\mathrm{rad}(\mathcal{X}_n) \le Cn^{2/3}$ for all sufficiently large n on $\mathbb{Z}^2$. He deduced it from the planar Beurling estimate, whose proof appeared in Kesten (1987a), and the bound (8.4), by counting paths. He bounded the probability that the sites of a fixed path are added in order during a time interval $[\ell+1, 2^{k+1}]$ with $2^k \le \ell < 2^{k+1}$ by an exponential martingale inequality, obtained with the Borel–Cantelli lemma a bound on the growth of the radius over each such interval, and deduced the bound $n^{2/3}$ by a deterministic recursion, which is written out in Section 4 of Kesten (1987a) (pages 181–182). We prove Proposition 8.2 instead by solving a recursion

for the expected numbers of sites beyond integer radii with Pascal's identity, and then apply Markov's inequality at each fixed n. The Beurling estimate is used only through (8.4), and in every dimension, by Proposition 8.2, a bound $J(1 + \mathrm{rad}(\mathcal{X}_n))^{-\beta}$ on the conditional probability of each attachment gives an exponential tail for the radius above $n^{1/(1+\beta)}$.

Jerison, Levine, and Sheffield (2012, Theorem 1) proved that the fluctuations of internal DLA in $\mathbb{Z}^2$ are logarithmic, as described in the Notes to Chapter 5. In Section 3 of the same paper, they observe that the martingales behind their logarithmic fluctuation bound remain martingales for a version of external DLA grown from the origin. In this version the initial cluster is the set of sites outside a large ball, and each particle, released at the origin, attaches to the cluster from inside. These martingales are not available for the model of this chapter, whose particles are released at infinity. Jerison, Levine, and Sheffield (2012) add that Kesten (1990) used a closely related martingale to bound the growth rate of external DLA (see below), and that for external DLA the difficulty is to control the quadratic variation of these martingales, since the cluster has no deterministic limit shape.

Theorem 1 of Kesten (1990) concerns external DLA on $\mathbb{Z}^d$, where for $d \geq 3$ each new site has the law (8.7). It shows that, eventually almost surely, the radius of the cluster of n sites is $O(n^{2/(d+1)})$ for $d \geq 2$ with $d \neq 3$, and $O((n \log n)^{1/2})$ for $d = 3$; Theorem 8.14 is the case $d \geq 4$. The earlier bound $O(n^{2/d})$ of Exercise 8.9 is the theorem on p. L29 of Kesten (1987b), proved on pp. 181 and 183 of Kesten (1987a). For $d \geq 4$, Kesten combines an exponential bound for a martingale formed from Green potentials and the reciprocals of the capacities of the clusters (Lemma 1 there), an adaptation of the path count of Kesten (1987b) showing that these reciprocals add up to at least a constant times r while the radius grows from r to $2r$ (Lemma 2 there), and a deterministic selection of shells, between balls of radii ρ and 2ρ, around a path from $\mathbf{0}$ to distance r. In our proof we replace both lemmas by the supermartingale of Proposition 8.12 at sites with first coordinates $0, \ldots, r$, and the selection of shells by Lemma 8.13. Whether the exponent $2/(d+1)$ is sharp is not known, and the simulations of Meakin (1983) in dimensions two through six suggest a smaller exponent. Theorem 8.16 is Theorem 2.6.1 of Lawler (1996), whose proof uses Theorem 2.5.2 there, which says that a site of a connected set of radius $n \geq 2$ in $\mathbb{Z}^3$ that contains $\mathbf{0}$ has harmonic measure at most a constant times $(\log n)^{1/2}/n$. In Lemma 2.5.4 there, the analogue of Proposition 8.10, a factor $(\log n)^{1/2}$ is lost when the escape probabilities are bounded by one. In Section 8.4.3 we avoid this loss by following the proof of Theorem 4 of Losev and Smirnov (2025), described at the end of these Notes; Lemma 8.15 is, with constant one, the form of the lower bound in their Lemma 10 that the proof needs. The last-exit decomposition (8.6) is Proposition 4.6.4 of Lawler and Limic (2010), and Exercise 8.7 is their

Proposition 6.5.5, which imposes the constraint on all of $\mathbb{Z}^d$.

The dielectric breakdown model, introduced by Niemeyer, Pietronero, and Wiesmann (1984), is a one-parameter family of lattice growth models. In the form of Definition 1 of Losev and Smirnov (2025), a finite cluster $A \subset \mathbb{Z}^2$ grows by one site $z \in \partial A$, chosen with probability proportional to $H_A(z)^\eta$ for a parameter $\eta \geq 0$, with the convention $0^0 = 1$. The case $\eta = 1$ is external DLA. The case $\eta = 0$ is the Eden model, in which every site of ∂A is equally likely, including a site enclosed by the cluster, where H_A vanishes. Niemeyer, Pietronero, and Wiesmann (1984) put these probabilities on the bonds that join the cluster to its outer boundary, in proportion to the η-th power of the local electric field, so at $\eta = 0$ their model favors a boundary site with several neighbors in the cluster.

Hastings and Levitov (1998) introduced a family HL(α) of off-lattice aggregation models built by composing conformal maps. At each step a small particle is attached to the unit circle at a uniformly random angle and carried to the cluster by the current aggregate map, so its position on the cluster is distributed according to harmonic measure from infinity. The particle on the unit circle has linear size of order the square root of a parameter, which Hastings and Levitov take to be a fixed constant times the power $-\alpha$ of the modulus of the derivative of the aggregate map at the attachment angle. Near that angle the aggregate map stretches lengths by this modulus, so to leading order the new particle has linear size proportional to the modulus raised to the power $1-\alpha/2$. For $\alpha = 2$ the particles therefore have approximately constant size, and HL(2) is the proposed off-lattice version of DLA. Carleson and Makarov (2001) defined a deterministic analogue of planar DLA through the Loewner equation and proved, in their Theorem A, a deterministic counterpart of the growth bound of Theorem 8.8; their proof also uses a form of the Beurling estimate.

Rohde and Zinsmeister (2005) proved that normalized HL(0) clusters with particles of fixed size have a scaling limit whose boundary has Hausdorff dimension 1. Norris and Turner (2012) proved that small-particle HL(0) clusters converge to a disk, and Silvestri (2017) obtained Gaussian fluctuations around the disk limit. Sola, Turner, and Viklund (2019) introduced the aggregate Loewner evolution ALE(α, η). Its particle sizes are chosen as in HL(α), but the attachment angle is no longer uniform, and its density is proportional to the power $-\eta$ of the modulus of the derivative of the aggregate map, evaluated at a small distance $\varrho > 0$ outside the unit circle. On the boundary of the cluster, harmonic measure from infinity has density with respect to length proportional to the reciprocal of this modulus. Formally, with $\varrho = 0$, ALE(α, η) therefore chooses its attachment point by the rule of the dielectric breakdown model with parameter $\eta + 1$, and ALE($2, \eta$), whose particles have approximately constant size, is its formal off-lattice analogue.

For HL(α), which is ALE($\alpha, 0$), Hastings and Levitov (1998) instead match the expected local rate of area growth with that of the lattice model, which gives the parameter $\alpha - 1$, and they expect that the local area growth law determines the large-scale behavior.

The rigorous results on these models concern the small-particle limit, and they depend on how fast ϱ tends to zero compared with the particle size, which is of order $\lambda^{1/2}$ when the conformal map that attaches one particle to the unit disk has capacity e^{λ}. For $\eta > 1$, slit-shaped particles, and ϱ below a power of λ whose exponent exceeds $5/4$ and depends on η, Sola, Turner, and Viklund (2019) proved that, for each fixed $T > 0$, with probability tending to one each of the first $\lfloor T/\lambda \rfloor$ particles attaches to the previous one, so that the ALE($0, \eta$) cluster after these particles converges to the unit disk with a single radial slit of logarithmic capacity T attached; for $\eta < 1$ and every ϱ, the probability that each of the first $N \geq 2$ particles attaches to its predecessor tends to zero. For ϱ at least $\lambda^{1/2-\epsilon}$, Norris, Silvestri, and Turner (2023) proved that ALE($0, \eta$) clusters converge to disks for every $\eta < 1$, and for ϱ at least $\lambda^{1/4-\epsilon}$ that their fluctuations are Gaussian; at $\eta = 1$ they need ϱ at least $\lambda^{1/5-\epsilon}$ and $\lambda^{1/6-\epsilon}$. Norris, Silvestri, and Turner (2024) extended these limits to ALE(α, η) with $\alpha + \eta \leq 1$, which includes regularized HL(α) for $\alpha \leq 1$. The model HL(2) proposed for off-lattice DLA lies outside the range of all these theorems, and none of them concerns the fractal dimension $d_f \approx 1.71$.

Losev and Smirnov (2025) proved growth bounds of Kesten's type for the dielectric breakdown model with parameter η. On $\mathbb{Z}^2$ with $0 \leq \eta < 2$, their Theorem 3 shows that, eventually almost surely, the radius of the cluster of n sites is at most $Cn^{2/(4-\eta)}$ times a power of $\log n$ that is absent at $\eta = 1$. On $\mathbb{Z}^3$, with H_A given by (8.7), their Theorem 4 gives $Cn^{\eta/(1+\eta)}(\log n)^{\eta/(2(1+\eta))}$ for $\eta \geq 1$ and $Cn^{1/2}(\log n)^{1/4}$ for $\eta < 1$. At $\eta = 1$ these recover the bounds of Theorems 8.8 and 8.16, by a proof that does not use the Beurling estimate. They bound the geometric mean of the harmonic measures of the sites added through the growth of the capacity of the cluster, and then sum over the paths from radius R to radius $2R$ and over the times when their sites are added.

PART 3

Sandpiles and Rotor Walks

Chapter Nine

Combinatorics of the Abelian Sandpile

The abelian sandpile models how a pile of grains relaxes on a graph. Place a nonnegative integer number of grains at each vertex. Whenever a vertex holds at least as many grains as its out-degree, the vertex topples and sends one grain along each outgoing edge. A toppling can make neighboring vertices unstable, so adding one grain may cause an avalanche before the sandpile becomes stable (see Figure 9.7 in Section 9.5). The model was introduced as a prototype of **self-organized criticality**, in which the dynamics drive the system toward a critical state without tuning an external parameter, and its avalanche distributions were predicted to follow power laws (see the Notes). Defined by this one local rule, the model nevertheless produces intricate self-similar patterns that still call for an explanation. Combinatorialists discovered the same dynamics independently and call it **chip-firing**. The graph Laplacian describes the relaxation and connects the model to random walk, to the arithmetic of the graph, and, through the matrix tree theorem of Chapter 11, to spanning trees.

We first consider general graphs and then specialize to the cubic lattice $\mathbb{Z}^d$. By the abelian property, the stabilization and the number of topplings do not depend on the toppling order (Section 9.1), and on a finite digraph with a global sink the recurrent sandpiles form the sandpile group, with Dhar's burning algorithm as a test for recurrence on Eulerian digraphs (Sections 9.4–9.6). The least action principle characterizes the odometer (Section 9.7). On $\mathbb{Z}^d$, we use monotonicity and the least action principle to bound how far topplings spread, and we distinguish robust backgrounds, which stabilize after every addition of finitely many grains, from explosive ones, on which some such addition makes every site topple (Sections 9.9–9.10). The least action principle has counterparts for the divisible sandpile (Theorem 4.3) and for the stacks of instructions of internal DLA and rotor-router aggregation (Lemma 5.1), which we used in the shape theorems of Chapters 4 to 6.

9.1 TOPPLING AND THE ABELIAN PROPERTY

Let $G = (V, E)$ be a finite or infinite directed graph (a **digraph**). A directed edge $e \in E$ points from the vertex tail(e) to the vertex head(e). Self-loops,

where head(e) = tail(e), as well as multiple edges (distinct edges with the same head and the same tail) are permitted. The **out-degree** deg(v) of a vertex v is the number of edges e with tail(e) = v, and the **in-degree** indeg(v) of v is the number of edges e with head(e) = v. We assume that deg(v) is finite for all $v \in V$. A vertex is a **sink** if its out-degree is zero. A **global sink** is a sink s such that from every other vertex there is a directed path leading to s. Note that a global sink is necessarily unique. The **adjacency matrix** $A = (a_{vw})_{v,w \in V}$ of G is the $V \times V$ square matrix with a_{vw} indicating the number of edges from vertex v to w, and the **graph Laplacian** of G is the $V \times V$ matrix $\mathsf{L} = \mathrm{diag}(\deg(v)) - A$. That is,

$$\mathsf{L}_{vw} = \begin{cases} -a_{vw}, & v \neq w, \\ \deg(v) - a_{vv}, & v = w\,. \end{cases}$$

Note that the entries in each row of L sum to zero. If the vertex v is a sink, then the vth row of L is zero.

A **sandpile** σ on G is a function from the set of non-sink vertices of G to the integers, where $\sigma(v)$ represents the number of grains, or **height**, at vertex v (usually $\sigma(v) \geq 0$). A sandpile σ is **stable** if $\sigma(v) < \deg(v)$ for every non-sink vertex v. We call a vertex v **unstable** in σ if v is not a sink and $\sigma(v) \geq \deg(v)$. An unstable vertex v can **topple**, resulting in a new sandpile $\widetilde{\sigma}$ obtained by moving one grain along each of the deg(v) edges emanating from v; that is, for non-sink vertices w,

$$\widetilde{\sigma}(w) = \begin{cases} \sigma(w) + a_{vw}, & w \neq v, \\ \sigma(v) - \deg(v) + a_{vv}, & w = v\,. \end{cases}$$

Grains sent to a sink disappear. We call the sandpile $\widetilde{\sigma}$ a **successor** of σ. The same formula defines the toppling of a sandpile at a non-sink vertex v that need not be unstable; we say that this toppling is **legal** if v is unstable, and is otherwise **illegal**.

If a sandpile σ yields a stable sandpile after a finite sequence of legal topplings, we say that σ **stabilizes**. This may not always happen, as Figures 9.1 and 9.2 and the following example show. A legal toppling sequence is **maximal** if it is not the initial segment of a longer legal toppling sequence, so a finite maximal sequence ends at a stable sandpile.

Example 9.1 Consider the complete directed graph on three vertices (without self-loops). Then placing three grains at a vertex gives a sandpile that stabilizes in one move, while placing four grains at a vertex gives a sandpile that never stabilizes, since topplings conserve the four grains and a stable sandpile on this graph has at most one grain at each vertex.

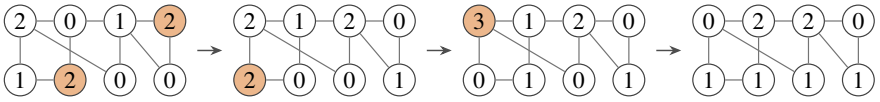


Figure 9.1: Stabilizing a sandpile on an undirected graph without sink by parallel toppling. The number within each vertex indicates the number of grains. Unstable vertices are shaded, and in each step all unstable vertices are toppled; the sandpile stabilizes after three steps.

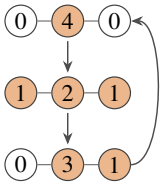


Figure 9.2: A sandpile on the path with three vertices which never stabilizes: a stable sandpile there holds at most one grain, and toppling conserves the four grains. In each step one unstable vertex (shaded) is toppled, and after three steps the sandpile returns to its initial state.

If a sandpile stabilizes, then the order in which we topple unstable vertices does not affect the final stable sandpile (see Figure 9.3 for an example on $\mathbb{Z}$). We derive this from a useful extension, the least action principle, in which we allow illegal topplings. The **odometer** $u : V \to \mathbb{N}$ of a finite sequence of topplings at non-sink vertices $x_1, \ldots, x_n$ counts the number of times a vertex appears in the sequence,

$$u(v) := \#\{1 \le i \le n : x_i = v\}\,,$$

so u vanishes at every sink. Since a toppling at v subtracts from the sandpile the row of L indexed by v, the sequence takes the initial sandpile σ to $\sigma - u\mathsf{L}$.

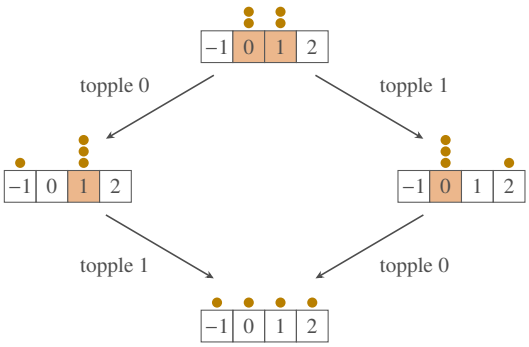


Figure 9.3: The abelian sandpile on $\mathbb{Z}$ started from two grains at each of the sites 0 and 1, with the unstable sites shaded. Both orders of toppling end in the same stable sandpile.

Lemma 9.2 (Least Action Principle) *Let σ be a sandpile on a digraph G. Consider two sequences of topplings at non-sink vertices starting from σ: $x_1, \ldots, x_n$, which ends at a stable sandpile but may include illegal topplings, and $y_1, \ldots, y_m$, which consists of legal topplings. Then $y_1, \ldots, y_m$ is a permutation of a subsequence of $x_1, \ldots, x_n$.*

Proof. Let u be the odometer of $x_1, \ldots, x_n$, and for $0 \le j \le m$ let w_j be the odometer of $y_1, \ldots, y_j$. We must show that $w_m \le u$. Suppose not, let j be the first index with $w_j \not\le u$, and let $v := y_j$. Then $w_{j-1} \le u$ and $w_{j-1}(v) = u(v)$. Since $a_{xv} \ge 0$ for $x \ne v$, the height of v just before the jth toppling of the second sequence is

$$\begin{aligned} &\sigma(v) - (\deg(v) - a_{vv})\, w_{j-1}(v) + \sum_{x \ne v} a_{xv}\, w_{j-1}(x) \\ &\qquad \le \sigma(v) - (\deg(v) - a_{vv})\, u(v) + \sum_{x \ne v} a_{xv}\, u(x) = (\sigma - u\mathsf{L})(v)\,. \end{aligned}$$

The sandpile $\sigma - u\mathsf{L}$ is the stable sandpile reached by the first sequence, and v is not a sink, so this height is less than $\deg(v)$. Thus the jth toppling of the second sequence is not legal, a contradiction. Hence $w_m \le u$. □

The abelian property follows by applying the least action principle to two legal sequences.

Lemma 9.3 *Let G be a digraph, let $\sigma_0, \sigma_1, \ldots, \sigma_n$ be a sequence of sandpiles on G, each of which is a successor of the one before, and let $\eta_0, \eta_1, \ldots, \eta_m$ be another such sequence with $\eta_0 = \sigma_0$.*

(a) If σ_n is stable, then for each vertex, the number of its topplings in $\eta_0, \ldots, \eta_m$ is at most the corresponding number in $\sigma_0, \ldots, \sigma_n$.

(b) If σ_n and η_m are both stable, then each vertex topples the same number of times in both sequences.

Proof. (a) The topplings of the first sequence take σ_0 to the stable sandpile σ_n, and the topplings of the second sequence are legal, so this follows from Lemma 9.2.

(b) Clearly (a), applied in both directions, implies (b). □

Summing over the vertices, part (a) gives $m \le n$, and under the hypotheses of part (b) the two sequences have the same odometer u, so $m = n$ and $\sigma_n = \sigma_0 - u\mathsf{L} = \eta_m$.

Definition 9.4 If a sandpile σ stabilizes, then we denote the resulting stable sandpile by σ° and call it the **stabilization** of σ.

9.2 STABILIZATION TIME WITH SINK

In this section we study when sandpiles stabilize, and how long it takes them to, in the case when the digraph has a global sink. On a finite digraph, stabilization also follows from the hitting-time bound of Proposition 9.7 below, since the expected hitting time of the sink is then finite from every vertex; the proof of the next lemma also applies to infinite digraphs.

Lemma 9.5 *If a digraph G has a global sink, then every nonnegative sandpile with a finite number of grains on G stabilizes.*

Proof. Let s be the global sink of G. For an integer h, let $h^+ := \max\{h, 0\}$. We prove more generally that a sandpile σ, possibly with negative entries, stabilizes whenever its positive mass $N := \sum_{v \neq s} \sigma(v)^+$ is finite. The idea is to sort the distances from the sink of the N grains counted in the positive mass. Every legal toppling moves grains from one vertex and sends at least one of them closer to the sink, so this sorted list decreases.

For each vertex w, let $\ell(w)$ be the length of a shortest directed path from w to the sink s, so that $\ell(s) = 0$; it is finite because s is a global sink, and every vertex $v \neq s$ has an out-neighbor w with $\ell(w) = \ell(v) - 1$. A legal toppling at v lowers the positive part of its height by $\deg(v)$, because the height is at least $\deg(v)$, and each grain it sends raises the positive mass by at most one, so the positive mass never increases. For a sandpile η reached from σ by legal topplings, let $\vec{\ell}(\eta) \in \mathbb{Z}^N_{\geq 0}$ be the list obtained by taking $\eta(x)^+$ copies of $\ell(x)$ for each vertex $x \neq s$, adding zeros to give N entries, and sorting in nondecreasing order.

We claim that each legal toppling decreases $\vec{\ell}$ in the lexicographic order. Indeed, toppling v removes $\deg(v)$ entries equal to $\ell(v)$ from the list, and for each edge out of v it inserts one entry, namely $\ell(w)$ if the grain arrives at a vertex $w \neq s$ and raises its positive part, and 0 otherwise. For the edge to an out-neighbor w with $\ell(w) = \ell(v) - 1$, the inserted entry is less than $\ell(v)$, while no entry less than $\ell(v)$ is removed, so the first coordinate in which the old and the new sorted lists differ has decreased.

Finally, no infinite sequence of N-tuples of nonnegative integers decreases strictly in the lexicographic order. Indeed, the first coordinates form a nonincreasing sequence of nonnegative integers, which is eventually constant; from then on the second coordinates are eventually constant, and so on through all N coordinates, so the tuples are eventually constant. Therefore every legal toppling sequence from σ is finite, and toppling unstable vertices for as long as possible stabilizes σ. □

On a finite graph, sandpiles with negative entries also stabilize; we will use this when studying recurrent sandpiles.

Corollary 9.6 *Let G be a finite digraph with a global sink. Then every sandpile on G stabilizes.*

Proof. On a finite graph every sandpile has finite positive mass, and the proof of Lemma 9.5 shows that every sandpile with finite positive mass stabilizes. □

In the next proposition we present a bound which is useful for finite graphs. Note that toppling a vertex v consists of $\deg(v)$ grain moves.

Proposition 9.7 *Let G be a digraph with a global sink s. Let $(X_k)_{k\geq 0}$ be the random walk that chooses an outgoing edge uniformly at each step, and let $\tau_s := \inf\{k \geq 0 : X_k = s\}$. Let $\sigma \geq 0$ be a sandpile on G satisfying*

$$\sum_{x\in\mathrm{supp}(\sigma)} \sigma(x)\mathbb{E}_x\tau_s < \infty\,.$$

Then σ stabilizes, and the total number of grain moves needed to stabilize σ is at most

$$\sum_{x\in\mathrm{supp}(\sigma)} \sigma(x)\mathbb{E}_x\tau_s\,.$$

Proof. Let $\sigma = \sigma_0, \sigma_1, \sigma_2, \ldots$ be a sequence of sandpiles, finite or infinite, with σ_{i+1} obtained from σ_i by toppling a single unstable vertex x_i. Define the weight of x to be $\mathrm{w}(x) := \mathbb{E}_x\tau_s$, the expected time for the walk started at x to hit the sink (so $\mathrm{w}(s) = 0$), and define the weight of σ_i to be $\mathrm{w}(\sigma_i) := \sum_{x\in\mathrm{supp}(\sigma_i)} \sigma_i(x)\,\mathrm{w}(x)$. By conditioning on the first step of the walk started at $x \neq s$, we compute

$$\mathrm{w}(x) = 1 + \frac{1}{\deg(x)} \sum_{y\in V} a_{xy}\,\mathrm{w}(y)\,.$$

Hence every vertex reached from $\mathrm{supp}(\sigma)$ has finite weight, so all the sums below are finite, and toppling the vertex x_i decreases the total weight by $\deg(x_i)$. Thus

$$\mathrm{w}(\sigma_i) - \mathrm{w}(\sigma_{i+1}) = \deg(x_i)\,. \tag{9.1}$$

Since the weight is nonnegative and every toppling decreases it by at least 1, the toppling sequence has at most $\mathrm{w}(\sigma) < \infty$ terms. In particular, every maximal sequence is finite and ends at a stable sandpile, so σ stabilizes. Summing (9.1) over i, we obtain that the total number of grain moves needed to stabilize σ is

$$\sum_i \deg(x_i) = \mathrm{w}(\sigma) - \mathrm{w}(\sigma^\circ) \leq \sum_{x\in\mathrm{supp}(\sigma)} \sigma(x)\mathbb{E}_x\tau_s\,.$$

□

On a digraph with a global sink, define the **grain addition operator** E_v as the map on nonnegative sandpiles with finitely many grains that adds a single grain at vertex v and then lets the system stabilize. In symbols,

$$E_v(\sigma) := (\sigma + \delta_v)^\circ .$$

Lemma 9.8 *On a digraph with a global sink, the grain addition operators commute, that is, $E_v \circ E_w = E_w \circ E_v$ for all non-sink vertices v and w.*

Proof. Given a sandpile σ and two vertices v and w, vertices that are unstable in $\sigma+\delta_v$ are also unstable in the sandpile $\sigma+\delta_v+\delta_w$. Applying to $\sigma+\delta_v+\delta_w$ a sequence of topplings that stabilizes $\sigma+\delta_v$, we obtain the sandpile $E_v(\sigma)+\delta_w$. Stabilizing this latter sandpile yields $(E_w \circ E_v)(\sigma)$, so $(E_w \circ E_v)(\sigma)$ is a stabilization of $\sigma + \delta_v + \delta_w$. Interchanging the roles of v and w, the sandpile $(E_v \circ E_w)(\sigma)$ is also a stabilization of $\sigma + \delta_v + \delta_w$. From Lemma 9.3(b) we conclude that $(E_w \circ E_v)(\sigma) = (E_v \circ E_w)(\sigma)$. □

Lemmas 9.3 and 9.8 are called the **abelian property**, which explains the word abelian in the model's name. From the above proof we also deduce the following.

Corollary 9.9 *Let G be a digraph with a global sink, let $k \geq 1$ be an integer, and let $v_1, \ldots, v_k$ be non-sink vertices. Let σ be a nonnegative sandpile on G with finitely many grains. Then*

$$(E_{v_k} \circ \cdots \circ E_{v_1})(\sigma) = \left(\sigma + \sum_{i=1}^{k} \delta_{v_i}\right)^\circ .$$

Proof. If we move every grain addition to the beginning of the successive stabilizations, then the topplings remain legal and their final sandpile is stable. Hence $(E_{v_k} \circ \cdots \circ E_{v_1})(\sigma)$ is a stabilization of $\sigma + \sum_{i=1}^{k} \delta_{v_i}$, and the formula follows from Lemma 9.3(b). □

9.3 STABILIZATION TIME WITHOUT SINK

In this section we consider stabilization questions on connected simple graphs without sink. We will need the following dichotomy.

Lemma 9.10 *Let G be a finite connected simple graph with at least two vertices, and let σ_0 be a nonnegative sandpile on G.*

(a) If σ_0 stabilizes, then there are two adjacent vertices of G that do not topple during the stabilization of σ_0.

(b) If σ_0 does not stabilize, then every vertex topples infinitely often in every maximal toppling sequence.

Proof. (a) Fix a toppling sequence which stabilizes σ_0. If every vertex topples, consider the first vertex w to stop toppling. Every neighbor of w must topple later, so w receives at least $\deg(w)$ grains after its last toppling and is unstable at the end. Thus there exists a vertex x which never topples, and one of its neighbors must never topple as well, since otherwise x would end with at least $\deg(x)$ grains (recall that $\sigma_0 \geq 0$).

(b) A maximal toppling sequence cannot end at an unstable sandpile, so it is infinite when σ_0 does not stabilize, and since G is finite, some vertex topples infinitely often. If v topples infinitely often, then every neighbor w of v receives infinitely many grains, and since the number of grains is conserved, w topples infinitely often. Therefore, since G is connected, every vertex topples infinitely often. □

The next lemma bounds the number of edges at the vertices that topple by the number of grains; we will use it on finite and infinite graphs, with and without sinks. Let G be an undirected multigraph without self-loops in which every vertex has finite degree. We regard G as a digraph by replacing each edge by two directed edges, one in each direction. Given a set S of vertices, we may also delete the directed edges leaving S, so that the vertices of S become sinks: they never topple, and grains sent to them disappear.

Lemma 9.11 *Let G be an undirected multigraph without self-loops, finite or infinite, in which every vertex has finite degree, and let S be a set of vertices of G that are made sinks. Let $\sigma_0 \geq 0$ be a sandpile on G with N grains. Fix a legal toppling sequence from σ_0 and a time $t \geq 0$, and let W be the set of vertices that topple at least once by time t. Then at most N edges of G have an endpoint in W.*

Proof. For a vertex v, let t_v be the time of its last toppling by time t, with $t_v = -\infty$ if v does not topple by time t. Let e be an edge with endpoints v and w, at least one of which lies in W. Distinct topplings occur at distinct times, so $t_v \neq t_w$; say $t_v < t_w$. At time t_w the vertex w sent a grain along e to v, and v has not toppled since. If $v \in S$, this grain has disappeared into v. Otherwise, the height of v is nonnegative just after its last toppling by time t, or at time 0 if v does not topple by time t, and from then until time t it can only increase.

Hence at time t every vertex $v \notin S$ holds at least as many grains as there are edges at v whose other endpoint toppled more recently than v, and every $v \in S$ has absorbed at least that many grains. Each edge with an endpoint in W is counted at exactly one of its endpoints, the one whose last toppling is earlier (see Figure 9.4). The heights at time t and the absorbed grains together total N, so at most N edges have an endpoint in W. □

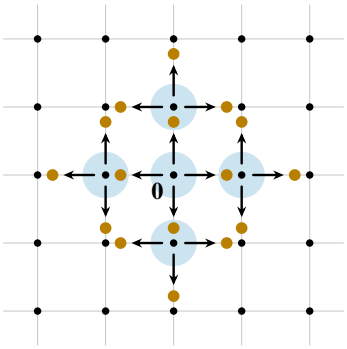


Figure 9.4: The sandpile $16\delta_{\mathbf{0}}$ on $\mathbb{Z}^2$ stabilizes after the origin topples four times, each of its neighbors once, and the origin once more, so five sites topple (shaded). Each of the 16 edges with an endpoint in this set points to the endpoint that last toppled earlier, or never toppled, and the grain sent along the edge (orange) stays there, so Lemma 9.11 holds here with equality.

If the number of grains in a sandpile on $G = (V, E)$ is strictly greater than $\sum_{v \in V}(\deg(v) - 1) = 2|E| - |V|$, then the sandpile never stabilizes. The next lemma gives a converse.

Lemma 9.12 *Let $G = (V, E)$ be a finite connected simple graph with at least two vertices.*

(a) If σ_0 is a nonnegative sandpile on G with $N < |E|$ grains, then σ_0 stabilizes.

(b) There exists a sandpile on G with $|E|$ grains that does not stabilize.

Proof. (a) We prove the contrapositive. Suppose σ_0 does not stabilize, and fix a maximal toppling sequence, so by Lemma 9.10 every vertex topples. Consider the first time that each vertex has toppled at least once. Every edge has an endpoint that has toppled by then, so by Lemma 9.11 with $S = \emptyset$, we have $|E| \leq N$.

(b) Fix an ordering of the vertices, $v_1, \ldots, v_n$. For each vertex v_i, define

$$\sigma_0(v_i) := \left|\{j > i : \{v_j, v_i\} \in E\}\right|.$$

Observe that every vertex $v_1, \ldots, v_n$ may be toppled in sequence, since when v_i topples, the grains received from earlier vertices raise its height to $\deg(v_i)$. Each vertex v_i then has received one grain from each neighbor and lost $\deg(v_i)$ grains, so the resulting sandpile is σ_0. These topplings can be repeated forever, so by Lemma 9.3(a), σ_0 does not stabilize. □

We next bound the number of topplings required to stabilize.

Lemma 9.13 *Let $G = (V, E)$ be a finite connected simple graph with at least two vertices. If a nonnegative sandpile σ on G with N grains stabilizes, then the number of topplings is at most $N \operatorname{diam}(G)|V|$.*

Proof. For each vertex v, let $u(v)$ be the number of times v topples when stabilizing σ (the odometer). It suffices to show

$$|u(v) - u(w)| \le N \qquad \text{for all edges } \{v, w\} \in E\,. \tag{9.2}$$

Indeed, Lemma 9.10 ensures that there is some vertex z where $u(z) = 0$. Once we prove (9.2), it follows that $u(x) \le \operatorname{dist}(x, z)N \le N \operatorname{diam}(G)$ for all $x \in V$, and this bound may be summed over $x \in V$.

To prove (9.2), fix an edge $\{v, w\}$ and suppose that $u(v) > u(w)$. Consider the set $U := \{y \in V : u(y) \le u(w)\}$. Since $\sigma^\circ - \sigma = -u\mathsf{L}$, summing over U and canceling the contributions of edges internal to U shows that the net influx of grains into U from its complement is

$$\sum_{\substack{y \in U,\ x \notin U \\ x \sim y}} (u(x) - u(y)) = \sum_{y \in U} (\sigma^\circ - \sigma)(y)\,. \tag{9.3}$$

Every term on the left of (9.3) is nonnegative, and the edge $\{v, w\}$ contributes $u(v) - u(w)$, so the net influx is at least $u(v) - u(w)$. Since the net influx of grains is at most the total number of grains, this shows that $u(v) - u(w) \le N$, completing the proof. □

Since stabilization preserves the number of grains and a stable sandpile has at most $2|E| - |V|$ grains, a sandpile with N grains that stabilizes has $N < 2|E|$, so the number of topplings in Lemma 9.13 is less than $2 \operatorname{diam}(G)|V| \cdot |E|$.

The next bound is useful for finitely supported sandpiles on graphs of large (or infinite) diameter and does not require the existence of a global sink.

Lemma 9.14 *Let G be a connected locally finite simple graph. Let σ_0 be a nonnegative sandpile with N grains on G, and let B be the set of vertices at graph distance at most N from* $\operatorname{supp}(\sigma_0)$*. If B is not the whole vertex set of G, then no grain ever leaves B, the sandpile σ_0 stabilizes, and the total number of grain moves needed to stabilize σ_0 is at most*

$$\sum_{x \in B} \sigma_0(x) \mathbb{E}_x \tau_{B^c}\,,$$

where $\mathbb{E}_x \tau_{B^c}$ is the expected time for a simple random walk started at x to exit B.

Proof. Since G is locally finite, B is finite, and $G \setminus B$ is nonempty by hypothesis. Form a finite multigraph on $B \cup \{s\}$ by keeping the edges with both endpoints in B, replacing each edge from B to $G \setminus B$ by an edge to s, and making s a sink. Since G is connected, s is a global sink.

Stabilize σ_0 on this finite graph, using Lemma 9.5. We claim that no grain reaches s. Suppose otherwise, let T be the time of the first toppling that

sends a grain to s, and let x be the vertex that topples at time T. A vertex outside $\mathrm{supp}(\sigma_0)$ first topples only after receiving a grain from a neighbor that toppled earlier. If we follow such neighbors backward from x, the first toppling times strictly decrease, so we obtain a path $x_0, x_1, \dots, x_k = x$ of distinct vertices that topple by time T, with $x_0 \in \mathrm{supp}(\sigma_0)$. Its k edges and the edge along which x sent a grain to s are $k+1$ distinct edges with an endpoint that topples by time T. In G they form a path from x_0 to a vertex outside B, so $k+1 \geq N+1$. This contradicts Lemma 9.11 with $S = \{s\}$.

Thus no grain leaves B, and the same topplings stabilize σ_0 on G. We may then conclude by Proposition 9.7 applied to the finite graph, since the hitting time of s in the finite graph is τ_{B^c}. □

The hypothesis on B holds whenever G is infinite, since B is finite.

9.4 SANDPILE GROUP

Let G be a digraph on n vertices with global sink s, and recall the graph Laplacian L from Section 9.1. The **reduced Laplacian** L' of G is obtained by deleting from the Laplacian matrix L the row and column corresponding to the sink. Note that toppling a non-sink vertex v transforms a sandpile σ into the sandpile $\sigma - \mathsf{L}'_v$, where L'_v is the row of the reduced Laplacian corresponding to v. Since we want to view the sandpiles before and after toppling as equivalent, we are led to consider the group quotient $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'$, where $\mathbb{Z}^{n-1}\mathsf{L}'$ is the integer row-span of L'.

Definition 9.15 Let G be a digraph on n vertices with global sink s. The **sandpile group** of G is the group quotient

$$\mathcal{S}(G) := \mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)\,.$$

We first observe that the lattice $\mathbb{Z}^{n-1}\mathsf{L}'$ is $(n-1)$-dimensional, that is, $\mathsf{L}'(G)$ is nonsingular.

Lemma 9.16 *Let G be a finite digraph with a global sink s. Then the reduced Laplacian $\mathsf{L}'(G)$ is nonsingular.*

Proof. Suppose a real column vector u, indexed by the non-sink vertices, satisfies $\mathsf{L}'u = 0$, and extend it by $u(s) = 0$. Suppose that the maximum M of u on V is positive. Of those vertices which achieve the maximum, let v be the vertex with the shortest directed path to the sink. Since $\mathsf{L}'u = 0$, we have

$$\deg(v)\, u(v) = \sum_{w \in V} a_{vw} u(w)\,,$$

so every out-neighbor of v has the value M, which is impossible for the next vertex on a shortest path from v to s. Thus $u \le 0$. Applying the same argument to $-u$ shows $u \equiv 0$, which completes the proof. □

A function $f : V(G) \to [0, 1)$ is **harmonic modulo 1** if for all vertices v,

$$\deg(v)\, f(v) = \sum_w a_{vw} f(w) \pmod 1 .$$

In Exercise 9.3 below, you will show that the sandpile group is isomorphic to the additive group of harmonic functions modulo 1 on G that vanish at the sink. For example, in the graph in Figure 9.5, with non-sink vertices v_1 and v_2, the three harmonic modulo 1 functions that vanish at the sink have $(f(v_1), f(v_2))$ equal to $(0, 0)$, $(1/3, 2/3)$ and $(2/3, 1/3)$.

Figure 9.5: Two stable sandpiles in the equivalence class of the identity, on the complete digraph on three vertices with one vertex made into the sink s (square) by deleting its outgoing edges. They differ by the sum of the two rows of the reduced Laplacian; only the one on the right is recurrent.

The connection between the sandpile group and the dynamics of sandpiles on G is made explicit in Corollary 9.24, where we show that the recurrent sandpiles, under addition followed by stabilization, form a group isomorphic to $\mathcal{S}(G)$. For the graph in Figure 9.5, the sandpile group is the cyclic group of order 3. The group structure of $\mathcal{S}(G)$ when G is a tree, with its leaves collapsed to a single sink, has a recursive description; see the Notes.

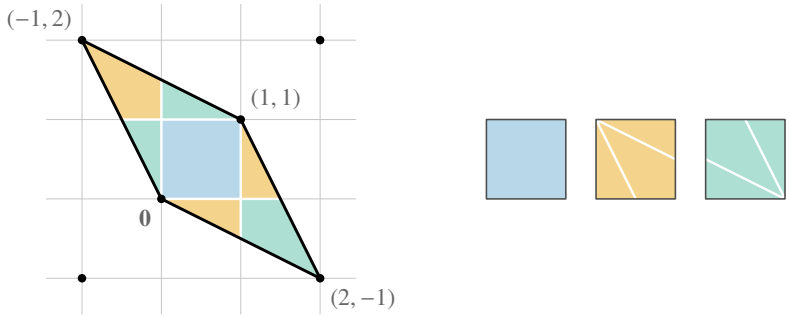
Lemma 9.17 *Let G be a finite digraph with a global sink. Then the order of $\mathcal{S}(G)$ is the absolute value of the determinant of the reduced Laplacian $\mathsf{L}'(G)$ of G.*

The proof is illustrated in Figure 9.6 for the graph of Figure 9.5.

Proof. Let $m := n - 1$. If $m = 0$, then both sides equal 1, so let $m \ge 1$. By Lemma 9.16, the map $x \mapsto x\mathsf{L}'$ of row vectors is a linear bijection of $\mathbb{R}^m$. Let P be the image of $[0, 1)^m$, which has volume $|\det \mathsf{L}'|$. The map sends the translates $[0, 1)^m + k$ with $k \in \mathbb{Z}^m$, which partition $\mathbb{R}^m$, onto the translates $P + k\mathsf{L}'$, so the translates of P by the vectors of $\mathbb{Z}^m\mathsf{L}'$ partition $\mathbb{R}^m$.

We cut P along the unit cubes $[0, 1)^m + z$ with $z \in \mathbb{Z}^m$ and translate the piece in $[0, 1)^m + z$ by $-z$ into $[0, 1)^m$. Fix a coset of $\mathbb{Z}^m\mathsf{L}'$ in $\mathbb{Z}^m$ and a representative z_0. The sets $P - z$ with z in this coset are translates of $P - z_0$ by the vectors of $\mathbb{Z}^m\mathsf{L}'$, so they partition $\mathbb{R}^m$. Thus their intersections with

$[0,1)^m$ partition the cube and have total volume 1. Summing over the cosets, the volume $|\det \mathsf{L}'|$ of P equals the number of cosets, which is the order of $\mathcal{S}(G)$. □



(a) (b)

Figure 9.6: (a) For the graph of Figure 9.5 the reduced Laplacian L' has rows $(2,-1)$ and $(-1,2)$, and the unit squares cut the image P of $[0,1)^2$, of area $\det \mathsf{L}' = 3$, into pieces colored by the coset of $\mathbb{Z}^2\mathsf{L}'$ (black dots) that contains the corner z of their square $[0,1)^2 + z$. (b) Translated by $-z$ into $[0,1)^2$, the pieces of each coset tile it exactly once, so the area of P equals the number of cosets.

It turns out that the determinant of the reduced Laplacian $\mathsf{L}'(G)$ is positive, so the absolute value is not needed in Lemma 9.17. To see this, we remove the edges between non-sink vertices continuously. Let $A_0 := (a_{vw})_{v,w\neq s}$, and let $L_t := \operatorname{diag}(\deg(v)) - tA_0$ for $0 \le t \le 1$. Every matrix L_t is nonsingular: for $t = 1$ this is Lemma 9.16, and if $t < 1$ and $L_t u = 0$ for a nonzero vector u, then at a coordinate v where $|u(v)|$ is maximal we have

$$\deg(v)\,|u(v)| = t\left|\sum_{w\neq s} a_{vw}u(w)\right| \le t\sum_{w\neq s} a_{vw}|u(w)| \le t\deg(v)\,|u(v)|\,,$$

which is impossible. Since $\det L_t$ is continuous and never vanishes, its sign is the sign of

$$\det L_0 = \prod_{v\neq s} \deg(v) > 0\,.$$

Lemma 9.18 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at least one nonnegative stable sandpile.*

Proof. Let $\boldsymbol{\delta}$ be the sandpile given by $\boldsymbol{\delta}(v) = \deg(v)$ for all non-sink vertices v, and let $\boldsymbol{\delta}^\circ$ be its stabilization (see Lemma 9.5). Then $\boldsymbol{\delta}^\circ(v) < \deg(v)$ for all $v \neq s$, so $\boldsymbol{\delta} - \boldsymbol{\delta}^\circ$ is a positive vector equivalent to the zero sandpile. Given

$\alpha \in \mathbb{Z}^{n-1}$, let m denote the minimum of all the coordinates of α together with 0 (so that $m \leq 0$). Then the vector

$$\beta := \alpha + (-m)(\boldsymbol{\delta} - \boldsymbol{\delta}^\circ)$$

is nonnegative and equivalent to α. Hence β° is a stable sandpile in the same equivalence class as α, and it is nonnegative since legal topplings preserve nonnegativity. □

Example 9.19 An equivalence class may contain more than one stable sandpile. For example, consider the complete directed graph on three vertices, with one of the vertices made into a sink by deletion of its two outgoing edges (see Figure 9.5). The reduced Laplacian of this graph is given by

$$\mathsf{L}' = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

The stable sandpile on the right of Figure 9.5 is obtained from the one on the left by adding the two rows of L', so both sandpiles are in the equivalence class of the identity. It might seem natural that, if either of these two stable sandpiles is to be preferred as a representative of the identity element in the sandpile group, it should be the one on the left. However, this instinct is misleading, as we explain below.

9.5 RECURRENT SANDPILES

Throughout this section, G is a finite digraph with a global sink, with reduced Laplacian $\mathsf{L}'(G)$ and sandpile group $\mathcal{S}(G)$ from Section 9.4.

Definition 9.20 A sandpile σ on G is **accessible** if from every sandpile on G it is possible to obtain σ by a combination of adding grains and selectively toppling unstable vertices. A sandpile that is both stable and accessible is called **recurrent**.

Adding grains and stabilizing preserves recurrence.

Lemma 9.21 *Let G be a finite digraph with a global sink, and let σ be a recurrent sandpile on G. Then $(\sigma + \xi)^\circ$ is recurrent for every sandpile $\xi \geq 0$ on G.*

Proof. Since σ is recurrent it is accessible, and adding ξ and then stabilizing leads from σ to $(\sigma + \xi)^\circ$, so $(\sigma + \xi)^\circ$ is also accessible. Since $(\sigma + \xi)^\circ$ is stable, this completes the proof. □

See Figure 9.7 for the topplings caused by adding one grain to a large recurrent sandpile.

We will see shortly (in Lemmas 9.22 and 9.23) that each equivalence class in $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'$ contains a unique recurrent sandpile. It is customary to represent each element of the sandpile group by its unique recurrent element. In Example 9.19 the all-1 sandpile is accessible (every sandpile stabilizes by Corollary 9.6, after which we may add grains to reach the all-1 sandpile), but by uniqueness the all-0 sandpile is not. Therefore the all-1 sandpile is taken as the canonical representative of the identity.

Lemma 9.22 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at least one recurrent sandpile of G.*

Proof. Write $\deg_{\max}$ for the maximum out-degree of a vertex in G, and recall from the proof of Lemma 9.18 that the sandpile $\boldsymbol{\delta} - \boldsymbol{\delta}^\circ$, where $\boldsymbol{\delta}(v) = \deg(v)$, is equivalent to zero and has all entries positive. Given $\alpha \in \mathbb{Z}^{n-1}$, choose an integer $k \geq 0$ so large that the sandpile

$$\beta := \alpha + k(\boldsymbol{\delta} - \boldsymbol{\delta}^\circ)$$

satisfies $\beta \geq \deg_{\max}$; it is equivalent to α. (Inequalities between two vectors or between a vector and a scalar are interpreted componentwise.) In particular, β is accessible, since every sandpile can first be stabilized (by Corollary 9.6), so that each vertex has fewer than $\deg_{\max}$ grains, and then supplemented with extra grains to obtain β. Therefore every sandpile obtained from β by toppling is also accessible. In particular the stabilization β° is recurrent and equivalent to α. □

Next we show, using the least action principle, that every equivalence class modulo L' contains at most one recurrent sandpile.

Lemma 9.23 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at most one recurrent sandpile of G.*

Proof. Let σ and ξ be recurrent and equivalent modulo $\mathbb{Z}^{n-1}\mathsf{L}'(G)$, so that $\sigma - \xi = z\mathsf{L}'$ for an integer row vector z indexed by the non-sink vertices, and let $b(v) := |z(v)| \deg(v)$. Since σ is accessible, it can be obtained from b by adding grains and legally toppling unstable vertices, and making all the additions at the start keeps every toppling legal, because the grains added early only increase the heights. Thus there exist a sandpile $\alpha \geq b$ and a legal toppling sequence that takes α to σ; let u be its odometer, so that $\sigma = \alpha - u\mathsf{L}'$.

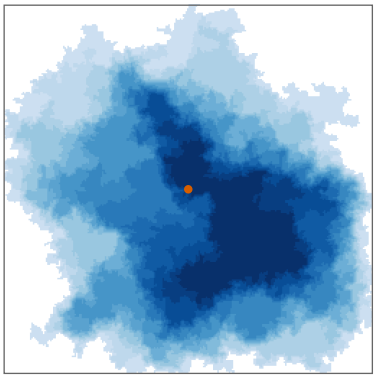


Figure 9.7: An avalanche: the sites that topple when one grain is added at the center (vermilion) of a recurrent sandpile on the 256×256 square grid with wired boundary conditions, shaded from light (once) to dark (15 times). The sandpile was reached from the all-3 sandpile by adding 2×10^6 grains at random sites, and of 2000 grains then added one at a time at the center, this one caused the most topplings.

A toppling at v removes at most $\deg(v)$ grains from v, and topplings at other vertices only add grains there. Since σ is stable,

$$\deg(v) > \sigma(v) \geq \alpha(v) - u(v)\ \deg(v) \geq (|z(v)| - u(v))\deg(v)\,,$$

so $u(v) > |z(v)| - 1$, and since both are integers, $u \geq |z|$. Hence $u + z \geq 0$, and

$$\alpha - (u+z)\mathsf{L}' = \sigma - z\mathsf{L}' = \xi\,.$$

Toppling each non-sink vertex v exactly $u(v) + z(v)$ times, legally or not, therefore takes α to the stable sandpile ξ, so by Lemma 9.2 we have $u \leq u + z$, that is, $z \geq 0$. Exchanging the roles of σ and ξ gives $-z \geq 0$. Hence $z = 0$ and $\sigma = \xi$. □

Corollary 9.24 *Let G be a digraph on n vertices with a global sink. The set of all recurrent sandpiles on G with the operation $(\sigma, \xi) \mapsto (\sigma + \xi)^\circ$ is an abelian group, and it is isomorphic to the sandpile group $\mathcal{S}(G)$ via the map that sends each recurrent sandpile to its class.*

Proof. Every recurrent sandpile is nonnegative because it is accessible from the zero sandpile, and grain additions and legal topplings preserve nonnegativity, so by Lemma 9.21 the set is closed under the operation. By Lemmas 9.22 and 9.23, each class contains exactly one recurrent sandpile, so the map is a bijection, and it preserves addition. □

In view of this isomorphism, we will henceforth use the term "sandpile group" to refer to the group of recurrent sandpiles (see Figure 9.8 for a cycle with five vertices).

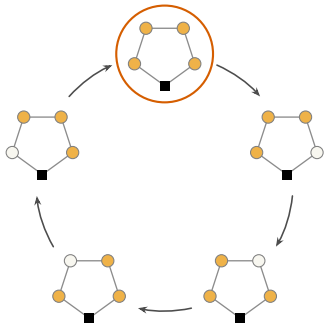


Figure 9.8: The five recurrent sandpiles on the cycle with five vertices, one of them the sink (black square), with zero and one grains in off-white and orange: adding a grain at the vertex just clockwise of the sink and stabilizing moves each to the next (arrows), so the sandpile group is cyclic of order five. Its identity element (circled) has one grain at every vertex.

It is of interest to consider the **identity element** I of the sandpile group, that is, the unique recurrent sandpile equivalent to the all-0 sandpile. Here is one method to compute I. Let σ be the sandpile given by $\sigma(v) = 2\deg(v) - 2$. Since $\sigma^\circ \le \boldsymbol{\delta} - 1$, where $\boldsymbol{\delta}(v) = \deg(v)$ as in the proof of Lemma 9.18, we have $\sigma - \sigma^\circ \ge \boldsymbol{\delta} - 1$, so $\sigma - \sigma^\circ$ is accessible. Since $\sigma - \sigma^\circ$ is equivalent to 0, the identity element is given by $I = (\sigma - \sigma^\circ)^\circ$.

Figure 9.9 shows identity elements for the $L \times L$ square grid with "wired boundary," for several values of L. (To be more precise, the graph G is obtained by replacing each edge of the undirected square grid with a pair of directed edges, and adjoining a sink vertex s along with two edges from each of the four corner vertices to s and one edge from each of the other boundary vertices to s.)

Given two sandpiles σ and ξ, we say that σ is **reachable** from ξ (via excitation-relaxation operations) if there exists a nonnegative sandpile β such that $\sigma = (\xi + \beta)^\circ$. Note that this implies that σ is stable.

A sandpile is recurrent if and only if it is reachable from every sandpile. Indeed, a sandpile reachable from every sandpile is stable, and it is accessible because stabilization topples only unstable vertices. Conversely, let σ be recurrent and let ξ be a sandpile. Since σ is accessible, a sequence of grain additions and legal topplings takes ξ to σ, and since adding grains preserves legality, we may make all grain additions first. This gives a nonnegative sandpile β and a legal toppling sequence from $\xi + \beta$ to σ, and since σ is stable, it follows from Lemma 9.3(b) that $\sigma = (\xi + \beta)^\circ$.

A digraph is **strongly connected** if for every pair of distinct vertices v and w there are directed paths from v to w and from w to v. We write $G \setminus s$ for the graph obtained from G by deleting the vertex s along with all edges incident to s.

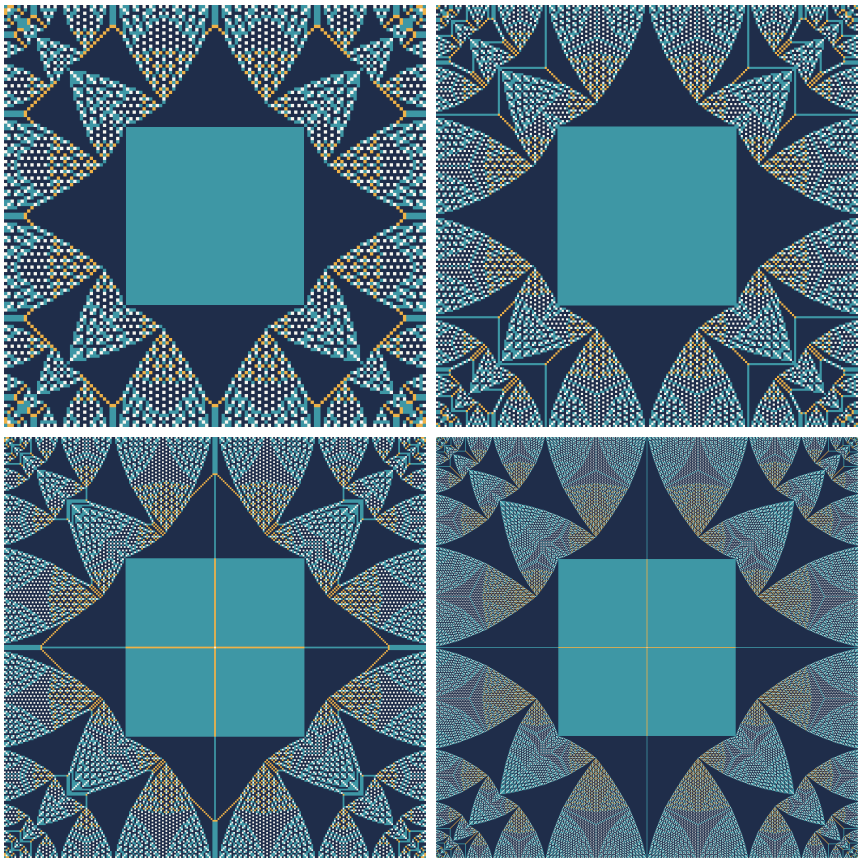


Figure 9.9: The identity element of the sandpile group of the $L \times L$ square grid with wired boundary conditions for $L = 128$ (upper left), 198 (upper right), 243 (lower left) and 521 (lower right), each grid drawn at the same size. Sites with zero, one, two and three grains are off-white, orange, teal and navy, as in the frontispiece.

We next show that σ is recurrent if $(\sigma+\alpha)^\circ = \sigma$ for a sandpile $\alpha \geq 0$ from whose support every vertex can be reached.

Lemma 9.25 *Let G be a finite digraph with a global sink, and let $\sigma \geq 0$ and $\alpha \geq 0$ be sandpiles on G with $(\sigma+\alpha)^\circ = \sigma$. Suppose that, for every non-sink vertex w, there is a directed path from* $\mathrm{supp}(\alpha)$ *to w. Then σ is recurrent.*

Proof. We first find a sandpile β with at least one grain at every non-sink vertex and with $(\sigma+\beta)^\circ = \sigma$. Among all sandpiles $\beta \geq \alpha$ with $(\sigma+\beta)^\circ = \sigma$, choose one whose support has the largest number of vertices; such sandpiles exist, since α is one of them, and the graph is finite. Suppose for sake of contradiction that $\beta(w_0) = 0$ for a non-sink vertex w_0. A directed path from $\mathrm{supp}(\alpha) \subset \mathrm{supp}(\beta)$ to w_0 contains an edge $v \to w$ with $v \in \mathrm{supp}(\beta)$ and w a non-sink vertex outside $\mathrm{supp}(\beta)$. Iterating $(\sigma+\beta)^\circ = \sigma$ gives $(\sigma + (\deg(v)+1)\beta)^\circ = \sigma$. Since $\beta(v) \geq 1$, the sandpile $\sigma + (\deg(v)+1)\beta$ has at least $\deg(v)$ grains at v, so toppling v once is legal. This toppling produces a sandpile of the form $\sigma + \beta'$ with $\beta' \geq \beta$ and $\beta'(w) \geq 1$, since it removes $\deg(v)$ grains from v, where $(\deg(v)+1)\beta$ exceeds β by $\deg(v)\,\beta(v) \geq \deg(v)$ grains, every other vertex only gains grains, and w gains a grain along the edge $v \to w$. By Lemma 9.3(b) we have $(\sigma+\beta')^\circ = \sigma$. Since $\beta' \geq \alpha$ and the support of β' strictly contains that of β, this contradicts the choice of β. Hence $\beta(w) \geq 1$ at every non-sink vertex w.

The sandpile σ equals $(\sigma+\beta)^\circ$, so it is stable. Now let ξ be a sandpile. Since β has at least one grain at each non-sink vertex, we have $\xi \leq \sigma + \ell\beta$ for some positive integer ℓ. Thus we may add grains to ξ and then stabilize to obtain the sandpile $(\sigma+\ell\beta)^\circ$, which equals σ by iterating $(\sigma+\beta)^\circ = \sigma$. Hence σ is accessible, and therefore recurrent. □

We next characterize recurrence in terms of reachability and of the grain addition operators $E_v(\sigma) = (\sigma + \delta_v)^\circ$ of Lemma 9.8.

Lemma 9.26 *Let G be a finite digraph with a global sink s, and let σ be a nonnegative sandpile on G. The following are equivalent.*

(1) The sandpile σ is recurrent.

(2) If ξ is a sandpile reachable from σ, then σ is reachable from ξ.

(3) The sandpile σ is reachable from every sandpile of the form $E_v(\sigma)$, where v is a non-sink vertex of G.

(4) Each strongly connected component of $G \setminus s$ contains a vertex v such that σ is reachable from $E_v(\sigma)$.

Proof. Since a recurrent sandpile is reachable from every sandpile, trivially (1) implies (2), (2) implies (3), and (3) implies (4), so it suffices to show that (4) implies (1).

If (4) holds, then in each strongly connected component we may choose a vertex v and a nonnegative sandpile ξ_v such that $(E_v(\sigma) + \xi_v)^\circ = \sigma$, and since $E_v(\sigma) = (\sigma + \delta_v)^\circ$, by the abelian property we have $(\sigma + \delta_v + \xi_v)^\circ = \sigma$. Let α be the sum of $\delta_v + \xi_v$ over the chosen vertices v. Then repeated stabilization gives $(\sigma + \alpha)^\circ = \sigma$, and α is nonzero on at least one vertex of each strongly connected component of $G \setminus s$, so every non-sink vertex can be reached by a directed path from $\mathrm{supp}(\alpha)$. By Lemma 9.25, σ is recurrent. □

9.6 EULERIAN GRAPHS AND THE BURNING ALGORITHM

A locally finite digraph $G = (V, E)$ is **Eulerian** if it is strongly connected, and for each vertex $v \in V$ the in-degree and the out-degree of v are equal. We call G an **Eulerian digraph with sink** if it is obtained from a finite Eulerian digraph by deleting all the outgoing edges from one vertex; then G has a global sink, and every other vertex has out-degree that is at least as large as its in-degree. An **Eulerian tour** of a finite digraph is a closed directed walk that uses each edge exactly once. If the digraph has no isolated vertices, then such a tour exists if and only if it is Eulerian (see Exercise 9.7). Note that for every connected undirected graph, the corresponding bidirected graph is Eulerian.

Let G be a digraph with global sink s, and let σ be a sandpile on G. A **forbidden subconfiguration** of σ is a nonempty induced subgraph Γ of $G \setminus s$ in which every vertex v has fewer grains than the in-degree of v in Γ.

For an Eulerian digraph with sink, we now give a test for recurrence in which each vertex topples at most once.

Lemma 9.27 (Dhar's Burning Algorithm) *Let G be a finite Eulerian digraph with sink s, and let β be the sandpile on G given by*

$$\beta(v) := \deg(v) - \operatorname{indeg}(v) \qquad \text{for } v \neq s\,.$$

(a) A sandpile σ on G is recurrent if and only if $(\sigma+\beta)^\circ = \sigma$.

(b) If a sandpile σ on G is recurrent, then each non-sink vertex topples exactly once during the stabilization of $\sigma+\beta$.

Proof. We first show that a recurrent sandpile σ has no forbidden subconfiguration; this step uses only that s is a global sink. If σ is recurrent, then there is a nonnegative sandpile α such that $(\boldsymbol{\delta}+\alpha)^\circ = \sigma$, where $\boldsymbol{\delta}(v) = \deg(v)$. Each non-sink vertex has at least $\deg(v)$ grains in $\boldsymbol{\delta}+\alpha$, so it topples at least once in the process of stabilizing $\boldsymbol{\delta}+\alpha$. Fix a toppling sequence that stabilizes $\boldsymbol{\delta}+\alpha$. Given a nonempty induced subgraph Γ of $G \setminus s$, let v be the vertex of Γ which first finishes toppling. During its last toppling, v receives one grain along each self-loop, and afterward it receives at least one grain along every other incoming edge in Γ, so $\sigma(v)$ is at least the in-degree of v in Γ. Thus Γ is not a forbidden subconfiguration of σ.

Now let σ be stable, and stabilize $\sigma+\beta$. For $v \neq s$, $\beta(v)$ counts the deleted edges $s \to v$, so $\beta \geq 0$. Consider a moment at which no vertex has toppled twice, and let U be the set of non-sink vertices that have not yet toppled. For a non-sink vertex v, write $\operatorname{indeg}_U(v)$ for the number of edges from vertices of U to v. Each non-sink vertex outside U has toppled once and has sent one grain along each of its outgoing edges, so the height at v is

$$\begin{aligned}\sigma(v) + \beta(v) - \deg(v)\,\mathbf{1}_{V\setminus U}(v) + \operatorname{indeg}(v) - \operatorname{indeg}_U(v)\\ = \sigma(v) + \deg(v)\,\mathbf{1}_U(v) - \operatorname{indeg}_U(v)\,.\end{aligned} \tag{9.4}$$

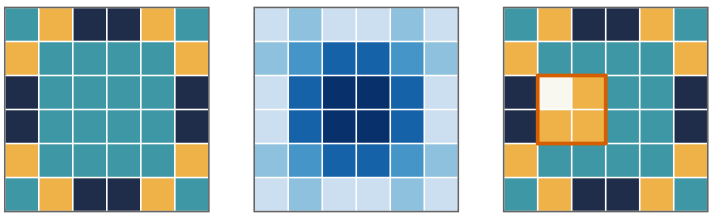
For $v \notin U$, this height is at most $\sigma(v) < \deg(v)$, so no vertex can topple a second time. Hence each non-sink vertex topples at most once during the stabilization of $\sigma+\beta$. From now on, let U be the set of non-sink vertices that never topple. By (9.4), stability of $(\sigma+\beta)^\circ$ gives $\sigma(v) < \operatorname{indeg}_U(v)$ for every $v \in U$, so if U is nonempty, then it induces a forbidden subconfiguration of σ. Moreover, $(\sigma+\beta)^\circ = \sigma$ only if U is empty: otherwise (9.4) would give $\operatorname{indeg}_U(v) = \deg(v)$ for every $v \in U$, so no edge would leave U, which is impossible because s is a global sink. Conversely, if U is empty, then (9.4) gives $(\sigma+\beta)^\circ = \sigma$.

(a) If σ is recurrent, then it is stable and has no forbidden subconfiguration, so U is empty and $(\sigma+\beta)^\circ = \sigma$.

Conversely, suppose that $(\sigma+\beta)^\circ = \sigma$. Then σ is stable, so U is empty and every non-sink vertex topples. Immediately after a non-sink vertex topples, its height is nonnegative, and later topplings at other vertices can only add grains there, so $\sigma \geq 0$. The support of β consists of the non-sink heads of the outgoing edges deleted from the sink. For each non-sink vertex v, choose a directed path from the sink to v in the original Eulerian digraph and keep the part after its last visit to the sink. The first vertex of this part lies in $\mathrm{supp}(\beta)$, and the rest of it stays in $G \setminus s$. Thus every non-sink vertex can be reached by a directed path from $\mathrm{supp}(\beta)$. Since $\beta \geq 0$, by Lemma 9.25 with $\alpha = \beta$, σ is recurrent.

(b) If σ is recurrent, then by (a) the set U is empty, so every non-sink vertex topples exactly once. □

Informally, the burning algorithm can be described as follows: to determine whether a stable sandpile σ is recurrent, first "topple the sink" to obtain the sandpile $\sigma + \beta$. Then σ is recurrent if and only if every non-sink vertex topples in the stabilization of $\sigma + \beta$, and since no vertex topples twice (by the proof of Lemma 9.27), it is enough to record which vertices topple (see Figure 9.10).



(a) (b) (c)

Figure 9.10: (a) The identity element of the sandpile group of the 6×6 square grid with wired boundary conditions, with zero, one, two and three grains in off-white, orange, teal and navy, and (b) the round, light to dark, in which each of its sites topples when the burning algorithm of Lemma 9.27 stabilizes $\sigma+\beta$ by toppling every unstable site in each round. (c) With five grains removed, some sites never topple, and they (outlined in vermilion) form a forbidden subconfiguration, so this stable sandpile is not recurrent.

The burning algorithm has the following equivalent formulation.

Lemma 9.28 *Let G be a finite Eulerian digraph with sink s. A stable sandpile σ on G is recurrent if and only if σ has no forbidden subconfiguration.*

Proof. The first step of the proof of Lemma 9.27 shows that a recurrent sandpile has no forbidden subconfiguration. Conversely, suppose that σ has no forbidden subconfiguration, and let β be the sandpile defined in Lemma 9.27. The proof of that lemma shows that the non-sink vertices that never topple during the stabilization of $\sigma + \beta$ induce a forbidden subconfiguration whenever such vertices exist. Hence every non-sink vertex topples, and

the same proof gives $(\sigma + \beta)^\circ = \sigma$, which implies that σ is recurrent by Lemma 9.27(a). □

The proof of the forward implication uses no property of Eulerian digraphs, but the converse can fail for other digraphs with a global sink. For example, take non-sink vertices v and w and a sink s, with one edge from v to w, two edges from w to v, and one edge from w to s. Then the reduced Laplacian has $\det \mathsf{L}' = 1$, and the only recurrent sandpile has no grain at v and two grains at w, while the stable sandpile with no grain at v and one grain at w has no forbidden subconfiguration.

The sandpile group of a finite Eulerian digraph does not depend on which vertex is chosen as the sink.

Proposition 9.29 *Let G be a finite Eulerian digraph on n vertices, and for a vertex v let G_v be the Eulerian digraph with sink obtained by deleting the outgoing edges from v. Then for all vertices v and w, the sandpile groups $\mathcal{S}(G_v)$ and $\mathcal{S}(G_w)$ are isomorphic.*

Proof. Recall that the sandpile group $\mathcal{S}(G_v)$ is the group $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G_v)$. We argue that it is also isomorphic to $\mathbb{Z}^n_0/\mathbb{Z}^n\mathsf{L}(G)$, where $\mathbb{Z}^n_0$ is the set of vectors in $\mathbb{Z}^n$ whose coordinates sum to 0. Order the coordinates so that v is last. The projection map from $\mathbb{Z}^n_0$ to the first $n-1$ coordinates is an isomorphism to $\mathbb{Z}^{n-1}$. This projection maps $\mathbb{Z}^n\mathsf{L}(G)$ onto $\mathbb{Z}^{n-1}\mathsf{L}'(G_v)$, because the rows other than the vth project to the rows of $\mathsf{L}'(G_v)$, and the vth row is redundant since the sum of the rows of $\mathsf{L}(G)$ is zero in an Eulerian digraph. Since $\mathbb{Z}^n_0/\mathbb{Z}^n\mathsf{L}(G)$ does not depend on the choice of v, this completes the proof. □

9.7 LEAST ACTION PRINCIPLE

Recall the least action principle, Lemma 9.2: no vertex topples more often in a legal toppling sequence than in a sequence of topplings, legal or illegal, that ends at a stable sandpile (see Figure 9.11 for a stabilized sandpile and its odometer). Since a finite sequence with odometer u takes σ to $\sigma - u\mathsf{L}$, this can be used to rephrase the least action principle in terms of the odometer function.

Corollary 9.30 *Let σ be a sandpile on a digraph G. If there exists a nonnegative, finitely supported $g : V \to \mathbb{Z}$, zero at every sink, such that $\sigma - g\mathsf{L}$ is stable, then every legal toppling sequence from σ is finite and has odometer $u \le g$.*

Proof. Topple each non-sink vertex v exactly $g(v)$ times. This finite sequence, which need not be legal, ends at the stable sandpile $\sigma - g\mathsf{L}$, so by Lemma 9.2

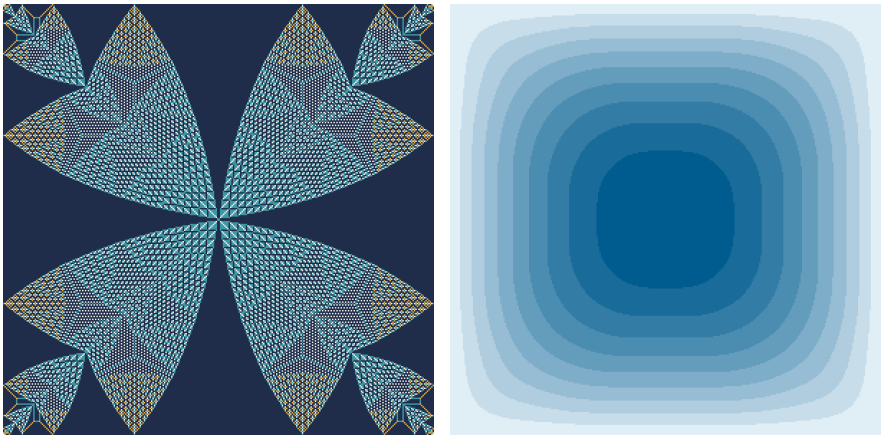


Figure 9.11: The stabilization of four grains at every site of the 300×300 square whose outside is the sink (left), with vertices colored by the number of grains left behind, zero, one, two and three in off-white, orange, teal and navy, and its odometer (right), shaded in ten bands of equal height from light to dark up to its maximum of 11325 topplings at the center. The final sandpile minus 4 is the discrete Laplacian of the odometer.

every legal sequence contains at most $\sum_v g(v)$ topplings and has odometer at most g. □

We call a function g with the properties in Corollary 9.30 a **stabilizing toppling function** for σ. A sandpile with a stabilizing toppling function stabilizes, since by the corollary every maximal legal toppling sequence from it is finite and therefore ends at a stable sandpile.

We say that a sandpile σ on a digraph G is **robust** if $\sigma + \xi$ stabilizes for every finitely supported sandpile $\xi \geq 0$. We will study this notion in detail in Section 9.10. The following family of one dimensional sandpiles is robust. Note that these sandpiles have an infinite number of grains, so Proposition 9.7 does not directly apply.

Example 9.31 Let σ be a periodic sandpile on $\mathbb{Z}$ that takes values 0 or 1 and is not identically 1. Then σ is robust.

Proof. We prove more generally that a sandpile σ on $\mathbb{Z}$ with values 0 and 1 is robust when its zeros are unbounded in both directions; a periodic sandpile that is not identically 1 has this property. We first stabilize σ plus one grain by an explicit sequence of legal topplings.

Add one grain at a site v. If $\sigma(v) = 0$, then the result is stable. Otherwise let $x < v < y$ be the nearest zeros of σ on the two sides of v, so that $\sigma = 1$ strictly between them. Topple the sites $v, v+1, \ldots, y-1$ in this order. Each toppling is legal, since v has two grains, and each later site has two grains once it has received one from its left neighbor. Each toppled site other than

$y-1$ gets one grain back from its right neighbor, the site $y-1$ ends empty, and y receives a grain. The sandpile is now $\sigma+\delta_{v-1}+\delta_y-\delta_{y-1}$, so the extra grain has moved to $v-1$, and the nearest zero on its right has moved from y to $y-1$. While the extra grain is to the right of x, repeat with both endpoints decreased by one, toppling $v-1,\dots,y-2$ next. Each repetition moves both the extra grain and the zero on its right one step to the left (see Figure 9.12). After $v-x$ repetitions the extra grain reaches the zero at x, and the sandpile is

$$\sigma+\delta_x+\delta_y-\delta_{x+y-v}\,.$$

This sandpile takes only the values 0 and 1, so it is stable, and it differs from σ only on $[x,y]$. In particular, its zeros are still unbounded in both directions.

Now let $\xi\geq 0$ be finitely supported. Adding the grains of ξ one at a time, and stabilizing after each addition as above, gives a finite sequence of legal topplings. Making all the additions at the start keeps every toppling legal, because the grains added early only increase the heights, and it leads to the same final sandpile. Hence $\sigma+\xi$ stabilizes. □

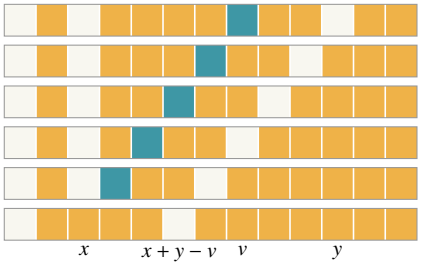


Figure 9.12: Example 9.31 with zeros at x and y and an extra grain at v, with zero, one and two grains in off-white, orange and teal: from one row to the next, topplings from left to right move the extra grain and the nearest zero on its right one step to the left. After $v-x$ rows the grain reaches x, and the only zero left in $[x,y]$ is at $x+y-v$.

In Exercise 9.11 you will find all robust sandpiles on $\mathbb{Z}$ that take only the values zero and one.

9.8 MONOTONICITY OF THE ODOMETER ON THE CUBIC LATTICE

In the rest of this chapter we focus on the sandpile on the integer lattice $\mathbb{Z}^d$. In this case the graph Laplacian is symmetric, and we find it convenient to work with the discrete Laplacian $\Delta^{::}$ of Chapter 3, which satisfies $-u\mathsf{L}=\Delta^{::}u$, and think of it as an operator on functions $f:\mathbb{Z}^d\to\mathbb{R}$,

$$\Delta^{::}f(x)=\sum_{y\sim x}\bigl(f(y)-f(x)\bigr)\,.$$

We will use the pointwise inequality

$$\Delta^{::} \min\{f, g\} \le \max\{\Delta^{::} f, \Delta^{::} g\}\,. \tag{9.5}$$

Indeed, if $f(x) \le g(x)$, then

$$\Delta^{::} \min\{f, g\}(x) = \sum_{y \sim x} (\min\{f(y), g(y)\} - f(x)) \le \Delta^{::} f(x)\,,$$

and symmetrically $\Delta^{::} \min\{f, g\}(x) \le \Delta^{::} g(x)$ when $g(x) \le f(x)$.

Next we observe that the odometer function inherits some of the discrete symmetries of the initial sandpile. One way to see this is with the **parallel toppling procedure**, in which every unstable vertex topples once in each round. In this procedure the evolution of the sandpile and odometer is described by the following recursions,

$$\begin{aligned} \sigma_{t+1} &= \sigma_t + \Delta^{::} \mathbf{1}\{\sigma_t \ge 2d\}, \\ u_{t+1} &= u_t + \mathbf{1}\{\sigma_t \ge 2d\}\,, \end{aligned} \tag{9.6}$$

where $u_0 \equiv 0$ is the initial odometer and $\sigma_0 : \mathbb{Z}^d \to \mathbb{Z}$ is the initial sandpile. By induction on $t \ge 0$, we have that

$$\sigma_t = \sigma_0 + \Delta^{::} u_t\,.$$

This allows us to express the odometer function at time $t+1$ in terms of the odometer function at time t and the initial sandpile,

$$\begin{aligned} u_{t+1}(x) &= u_t(x) + \mathbf{1}\{\sigma_t(x) \ge 2d\} \\ &= \max\left\{u_t(x), \min\left\{\left\lfloor \frac{\sigma_0(x) + \sum_{y \sim x} u_t(y)}{2d} \right\rfloor, u_t(x) + 1\right\}\right\}, \end{aligned} \tag{9.7}$$

where the second equality holds because $\sigma_t = \sigma_0 + \Delta^{::} u_t$, so $\sigma_t(x) \ge 2d$ if and only if the floor is at least $u_t(x) + 1$. We first show that the parallel toppling odometer eventually dominates the odometer of every finite legal toppling sequence; we will also use this to prove explosion in Section 9.10.

Lemma 9.32 *Let $d \ge 1$, let σ be a sandpile on $\mathbb{Z}^d$, and let $(u_t)_{t \ge 0}$ be the parallel toppling odometers started from σ.*

(a) For every finite legal toppling sequence from σ, there is an integer $t \ge 0$ such that u_t is at least the odometer of the finite sequence.

(b) If σ stabilizes with final odometer u, then $u_t = u$ for some integer $t \ge 0$.

Proof. (a) We argue by induction on the length of the legal sequence. The empty sequence is dominated by u_0. For the inductive step, suppose u_t dominates the odometer r of the topplings already performed, and let x be the

next vertex. If $u_t(x) > r(x)$, then already $u_t \geq r + \delta_x$. Otherwise $u_t(x) = r(x)$, while $u_t(y) \geq r(y)$ for every neighbor y of x. Therefore

$$\sigma(x) + \Delta^{::} u_t(x) \geq \sigma(x) + \Delta^{::} r(x) \geq 2d\,.$$

Hence x topples in round $t+1$, and $u_{t+1} \geq r + \delta_x$.

(b) Suppose σ stabilizes with odometer u. We prove by induction on t that $u_t \leq u$. The case $t = 0$ is immediate. For the inductive step, the first t rounds form a finite legal sequence, since $u_t \leq u$. Appending one toppling at a currently unstable vertex is legal, so by Lemma 9.2 we have $u_t + \delta_x \leq u$ for every such vertex x. Hence only finitely many vertices are unstable, and toppling them one at a time is legal, so again $u_{t+1} \leq u$. Applying part (a) to a stabilizing sequence gives an integer t with $u_t \geq u$, so $u_t = u$. □

Under the hypothesis of part (b), parallel toppling reaches σ°, since if $u_t = u$, then after t rounds the sandpile is $\sigma + \Delta^{::} u_t = \sigma + \Delta^{::} u = \sigma^\circ$.

We first observe that because the discrete Laplacian commutes with graph automorphisms, so does parallel toppling.

Lemma 9.33 *Let $d \geq 1$, let σ_0 be a sandpile on $\mathbb{Z}^d$, and let $(\sigma_t, u_t)_{t\geq 0}$ be its parallel toppling process. Let φ be a graph automorphism of $\mathbb{Z}^d$. If $\sigma_0 \circ \varphi = \sigma_0$, then $\sigma_t \circ \varphi = \sigma_t$ and $u_t \circ \varphi = u_t$ for every integer $t \geq 0$.*

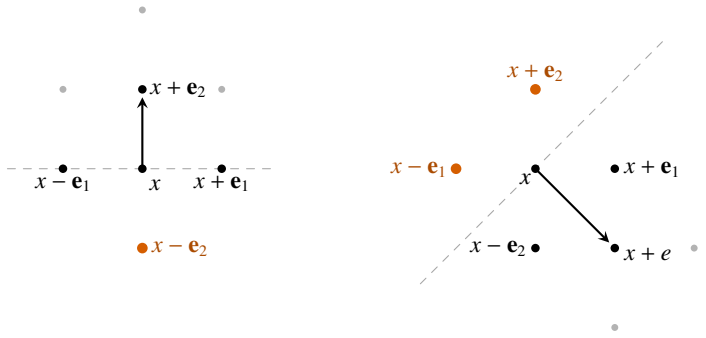
Proof. We argue by induction on $t \geq 0$. The base case $t = 0$ is by assumption, since $u_0 \equiv 0$. For the inductive step, suppose $\sigma_t \circ \varphi = \sigma_t$ and $u_t \circ \varphi = u_t$. Then $\mathbf{1}\{\sigma_t \geq 2d\} \circ \varphi = \mathbf{1}\{\sigma_t \geq 2d\}$, and since the discrete Laplacian commutes with φ, composing both lines of (9.6) with φ gives $\sigma_{t+1} \circ \varphi = \sigma_{t+1}$ and $u_{t+1} \circ \varphi = u_{t+1}$. □

We next show that the parallel toppling process preserves a certain monotonicity of the initial sandpile. A function $f : \mathbb{Z}^d \to \mathbb{R}$ is **decreasing** on a set A in direction $e \in \mathbb{Z}^d$ if

$$f(x) \geq f(x+e) \qquad \text{whenever } x \in A \text{ and } x + e \in A\,.$$

For $e = \mathbf{e}_i$ with $1 \leq i \leq d$ or $e = \mathbf{e}_i - \mathbf{e}_j$ with distinct $1 \leq i, j \leq d$, write φ_e for the reflection across $\{x \in \mathbb{Z}^d : x \cdot e = 0\}$. In particular, when the initial sandpile is invariant with respect to reflection in a hyperplane and is decreasing in the direction orthogonal to the hyperplane, then so is the parallel toppling odometer.

Lemma 9.34 *Let $d \geq 1$, and let $e = \mathbf{e}_i$ for some $1 \leq i \leq d$ or $e = \mathbf{e}_i - \mathbf{e}_j$ for distinct $1 \leq i, j \leq d$. Let σ_0 be a sandpile on $\mathbb{Z}^d$, and let $(\sigma_t, u_t)_{t\geq 0}$ be its parallel toppling process. Suppose $\sigma_0 \circ \varphi_e = \sigma_0$ and σ_0 is decreasing in direction e on the half-space $\{x \in \mathbb{Z}^d : x \cdot e \geq 0\}$. Then for every integer $t \geq 0$,*



(a) $e = \mathbf{e}_2$ (b) $e = \mathbf{e}_1 - \mathbf{e}_2$

Figure 9.13: The two cases $x \cdot e = 0$ in the proof of Lemma 9.34: (a) $e = \mathbf{e}_2$ and (b) $e = \mathbf{e}_1 - \mathbf{e}_2$. The dashed line is the hyperplane $\{y : y \cdot e = 0\}$, the arrow points from x to $x + e$, the neighbors of x outside the half-space $\{y : y \cdot e \geq 0\}$ are orange, and the gray dots are the other neighbors of $x + e$.

the odometer u_t is invariant under φ_e and is decreasing in direction e on this half-space.

Note that if $e = \mathbf{e}_i - \mathbf{e}_j$, then φ_e swaps x_i and x_j.

Proof of Lemma 9.34. By Lemma 9.33, $u_t \circ \varphi_e = u_t$. We prove the directional monotonicity by induction on $t \geq 0$. The base case $t = 0$ is immediate as $u_0 \equiv 0$.

For the inductive step, consider $x \in \mathbb{Z}^d$ with $x \cdot e \geq 0$. If $u_t(x) > u_t(x + e)$, then since u_t is integer-valued and each round increases it by at most one,

$$u_{t+1}(x) \geq u_t(x) \geq u_t(x + e) + 1 \geq u_{t+1}(x + e)\,.$$

We may therefore assume $u_t(x) = u_t(x + e)$.

If $x \cdot e > 0$, then all neighbors $y \sim x$ satisfy $y \cdot e \geq 0$, so by the induction hypothesis

$$\sum_{y \sim x} u_t(y) \geq \sum_{y \sim x} u_t(y + e)\,.$$

Moreover $\sigma_0(x) \geq \sigma_0(x+e)$, and the right-hand side of (9.7) is nondecreasing separately in the three quantities $u_t(x)$, the initial height $\sigma_0(x)$, and the sum $\sum_{y \sim x} u_t(y)$. Hence $u_{t+1}(x) \geq u_{t+1}(x + e)$.

If $x \cdot e = 0$, then we consider two further cases (see Figure 9.13). If $e = \mathbf{e}_i$, then all neighbors $y \sim x$ but one, $x - \mathbf{e}_i$, satisfy $y \cdot e \geq 0$, so by the induction hypothesis $u_t(y) \geq u_t(y + \mathbf{e}_i)$ for all such y. On the other hand, by symmetry,

$$u_t(x - \mathbf{e}_i) = u_t(x + \mathbf{e}_i) = u_t(x)\,,$$

so the sum of u_t over the neighbors of x is at least the corresponding sum at $x + \mathbf{e}_i$.

If $e = \mathbf{e}_i - \mathbf{e}_j$, then the induction hypothesis gives $u_t(y) \geq u_t(y+e)$ for all neighbors $y \sim x$ but two, $x - \mathbf{e}_i$ and $x + \mathbf{e}_j$, and for these two, by symmetry we have

$$u_t(x - \mathbf{e}_i) = u_t(x - \mathbf{e}_j),$$
$$u_t(x + \mathbf{e}_j) = u_t(x + \mathbf{e}_i),$$

so the sum of u_t over the neighbors of x is again at least the corresponding sum at $x + e$. In both cases, $u_t(x) = u_t(x+e)$ and $\sigma_0(x) \geq \sigma_0(x+e)$, so by (9.7) we have $u_{t+1}(x) \geq u_{t+1}(x+e)$. □

9.9 GROWTH RATES

We show that monotonicity of the sandpile on $\mathbb{Z}^d$ gives bounds on the growth of the single source sandpile $\eta + n\delta_{\mathbf{0}}$, where $\eta : \mathbb{Z}^d \to \mathbb{Z}$ is the initial **background** and n is the number of grains at the origin. The idea is to use monotonicity in the case where the background η is zero, and then use a one-dimensional sandpile to extend to larger background heights. For $x \in \mathbb{Z}^d$ we write $|x|_1 := |x_1| + \cdots + |x_d|$ and $|x|_\infty := \max_i |x_i|$ for its ℓ^1-norm and ℓ^∞-norm. The toppled set has diameter at most of order $n^{1/d}$ when $\eta \leq 2d - 2$, and at least of order $n^{1/d}$ when in addition $\eta \geq 0$ (see Figures 9.14 and 9.15 for the toppled sets of 10^6 grains in $\mathbb{Z}^2$ and 10^7 grains in $\mathbb{Z}^3$ when $\eta \equiv 0$). The pattern in $\mathbb{Z}^2$ is shown at three sizes in Figure 1.3 in Chapter 1, and between two balls in Figure 4.7 in Chapter 4.

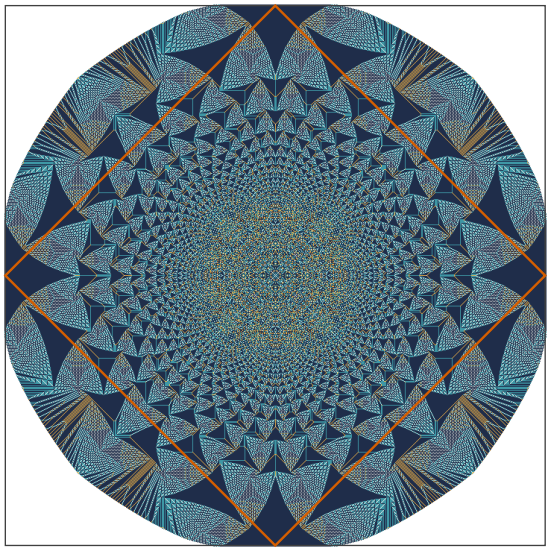


Figure 9.14: The stabilization of $10^6\delta_{\mathbf{0}}$ in $\mathbb{Z}^2$, with toppled sites colored by their final number of grains $0, 1, 2, 3$ in off-white, orange, teal and navy. Here $K = 363$ is the largest ℓ^∞-norm of a toppled site. The toppled set contains the diamond $\{|x|_1 \le K\}$ (vermilion) and lies in the square $\{|x|_\infty \le K\}$ (black), which are the inclusions (9.8) in the proof of Proposition 9.35.

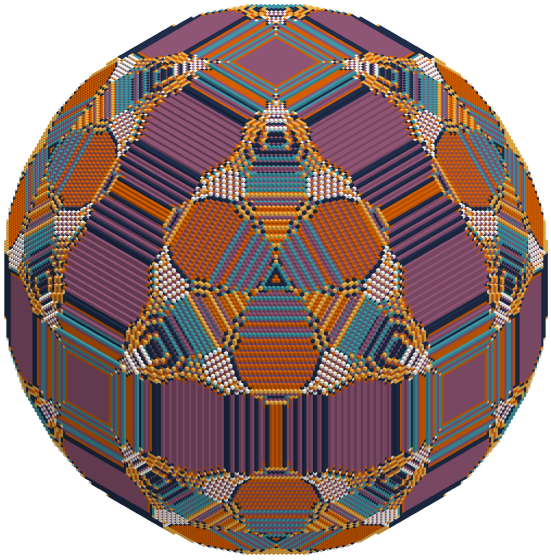


Figure 9.15: The sites of $\mathbb{Z}^3$ that topple when 10^7 grains at the origin are stabilized, each drawn as a unit cube and seen along the direction $(1, 1, 1)$. Each cube is colored by the final number of grains at its site: off-white 0, orange 1, teal 2, navy 3, vermilion 4 and purple 5.

Proposition 9.35 *For every $d \geq 1$ and every integer b with $0 \leq b \leq 2d-2$, there are constants $c = c(d) > 0$ and $C = C(d,b) < \infty$ with the following property. Let $\eta : \mathbb{Z}^d \to \mathbb{Z}$ satisfy $\eta \leq b$.*

(a) For every integer $n \geq 0$, the sandpile $\eta + n\delta_{\mathbf{0}}$ stabilizes.

(b) For every sufficiently large integer n, the final odometer $u_\infty^{(n)}$ of $\eta + n\delta_{\mathbf{0}}$ satisfies

$$\{u_\infty^{(n)} > 0\} \subseteq \{x \in \mathbb{Z}^d : |x|_\infty \leq Cn^{1/d}\}\,.$$

(c) If $\eta \geq 0$, then for every sufficiently large integer n, the final odometer $u_\infty^{(n)}$ of $\eta + n\delta_{\mathbf{0}}$ satisfies

$$\{x \in \mathbb{Z}^d : |x|_1 < cn^{1/d}\} \subseteq \{u_\infty^{(n)} > 0\}\,.$$

Proof. The proof has three steps. We first prove both bounds for $\eta \equiv 0$. We next prove parts (a) and (b) by adding to the odometer for $\eta \equiv 0$ a function built from a one-dimensional sandpile. Finally, we compare with $\eta \equiv 0$ to obtain part (c).

For $\eta \equiv 0$, the sandpile $n\delta_{\mathbf{0}}$ stabilizes by Lemma 9.14. Let u be its final odometer, which equals the parallel toppling odometer u_t for all large t by Lemma 9.32(b). Let $U := \{u > 0\}$, and let K be the largest ℓ^∞-norm of a point in U, with $K := 0$ if U is empty. For large n, the set U is nonempty. The sandpile $n\delta_{\mathbf{0}}$ is invariant under permutations and sign changes of the coordinates, and for $e = \mathbf{e}_k$ with $1 \leq k \leq d$ and $e = \mathbf{e}_1 - \mathbf{e}_k$ with $2 \leq k \leq d$ it is decreasing in direction e on $\{x \in \mathbb{Z}^d : x \cdot e \geq 0\}$, because $x + e \neq \mathbf{0}$ there. Hence by Lemmas 9.33 and 9.34, u has the same invariance and

$$u(z) \geq u(z + \mathbf{e}_k) \quad \text{if } z_k \geq 0 \qquad \text{and} \qquad u(z) \geq u(z + \mathbf{e}_1 - \mathbf{e}_k) \quad \text{if } z_1 \geq z_k\,.$$

By symmetry there is $x \in U$ with $x_1 = K$ and $x_i \geq 0$ for every i. Lowering $x_2, \ldots, x_d$ to 0 one unit at a time and applying the first inequality at each step gives $u(K\mathbf{e}_1) \geq u(x) > 0$.

Now let $|y|_1 \leq K$. By symmetry we may assume $y_1 \geq y_2 \geq \cdots \geq y_d \geq 0$. Let $y' := y + (K - |y|_1)\mathbf{e}_1$, which still has nonincreasing coordinates and satisfies $|y'|_1 = K$. The first inequality gives $u(y) \geq u(y')$. We pass from $K\mathbf{e}_1$ to y' by moving one unit at a time from the first coordinate to one of the coordinates $2, \ldots, d$. Before a move $z \mapsto z - \mathbf{e}_1 + \mathbf{e}_k$, at least one unit remains to be moved and coordinate k is not yet full, so $z_1 \geq y'_1 + 1 \geq y'_k + 1 \geq z_k + 2$. The second inequality at the point $z - \mathbf{e}_1 + \mathbf{e}_k$ therefore gives $u(z - \mathbf{e}_1 + \mathbf{e}_k) \geq u(z)$. Hence $u(y) \geq u(y') \geq u(K\mathbf{e}_1) > 0$, and symmetry gives

$$\{y \in \mathbb{Z}^d : |y|_1 \leq K\} \subseteq U \subseteq \{y \in \mathbb{Z}^d : |y|_\infty \leq K\}\,. \tag{9.8}$$

Since a grain moves only when its current site topples, every grain ends at a site of U or at a neighbor of U, and every vertex can absorb at most $2d-1$ grains, so $n \le (2d-1)(2K+3)^d$. This gives $K \ge c(d)n^{1/d}$ for all sufficiently large n.

For the reverse inequality, note that U contains $\mathbf{0}$ and is connected, because a site other than the origin first topples only after receiving a grain from a neighbor that toppled earlier. At the end of the stabilization every site of U has toppled, so by Lemma 9.11 with $S = \emptyset$, at most n edges have an endpoint in U. Since U is connected, at least $|U|-1$ edges have both endpoints in U, and hence $|U|-1 \le n$. By the first inclusion in (9.8), the set U contains the cube $\{0,\dots,\lfloor K/d \rfloor\}^d$, so

$$(\lfloor K/d \rfloor + 1)^d \le n+1\,.$$

Thus $K \le C(d)n^{1/d}$.

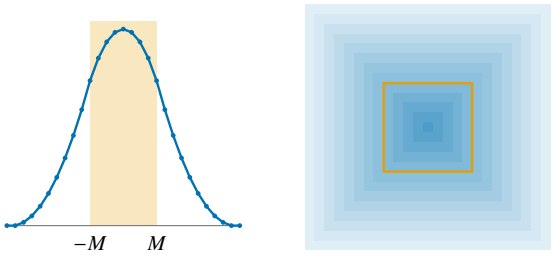
For the second step, fix $n \ge 0$ and a background $\eta \le b$. The idea is to "clear out" the background in a large cube by adding to u a function whose Laplacian is at most $-b$ there and at most 1 everywhere, which we obtain by stabilizing $b+1$ grains at each site of an interval of $\mathbb{Z}$ and taking the minimum over the coordinates. The final sandpile for $\eta \equiv 0$ is $(n\delta_{\mathbf{0}})^\circ = n\delta_{\mathbf{0}} + \Delta^{::}u$. Let $M := K+1$. Every grain ends at the origin or at a neighbor of a toppled site, so $(n\delta_{\mathbf{0}})^\circ$ vanishes outside the cube $\{x \in \mathbb{Z}^d : |x|_\infty \le M\}$.

On $\mathbb{Z}$, the sandpile $(b+1)\mathbf{1}_{[-M,M]}$ has $(b+1)(2M+1)$ grains. Let $R := M + (b+1)(2M+1)$. By Lemma 9.14, this sandpile stabilizes and no grain leaves the interval $[-R,R]$. Its odometer g is therefore a nonnegative integer-valued function that vanishes outside $[-R,R]$, and stability of the final sandpile on $\mathbb{Z}$ gives

$$g(j+1) + g(j-1) - 2g(j) \le 1 - (b+1)\mathbf{1}_{\{|j|\le M\}} \qquad \text{for } j \in \mathbb{Z}\,. \tag{9.9}$$

Let $w(x) := \min_{1\le i\le d} g(x_i)$ for $x \in \mathbb{Z}^d$, a nonnegative integer-valued function that vanishes outside the cube $\{|x|_\infty \le R\}$ (see Figure 9.16). If i is an index at which the minimum is attained, then $w(x) = g(x_i)$ and $w(y) \le g(y_i)$ for every neighbor y of x, so $\Delta^{::}w(x)$ is at most $g(x_i+1) + g(x_i-1) - 2g(x_i)$, as in (9.5). By (9.9), it follows that $\Delta^{::}w \le -b$ on the cube $\{|x|_\infty \le M\}$ and $\Delta^{::}w \le 1$ everywhere.

We check that $u+w$ stabilizes $\eta + n\delta_{\mathbf{0}}$. On the cube $\{|x|_\infty \le M\}$ we have $\eta + \Delta^{::}w \le 0$ and $(n\delta_{\mathbf{0}})^\circ \le 2d-1$. Outside this cube $(n\delta_{\mathbf{0}})^\circ$ vanishes, while $\eta \le b$ and $\Delta^{::}w \le 1$, so $\eta + \Delta^{::}w \le b+1 \le 2d-1$. Hence $\eta + n\delta_{\mathbf{0}} + \Delta^{::}(u+w) = \eta + (n\delta_{\mathbf{0}})^\circ + \Delta^{::}w \le 2d-1$ everywhere, and $u+w$ is a nonnegative, integer-valued, finitely supported stabilizing toppling function for $\eta + n\delta_{\mathbf{0}}$. By Corollary 9.30, $\eta + n\delta_{\mathbf{0}}$ stabilizes, which proves part (a), and its final odometer is at most $u+w$. Since U lies in the cube $\{|x|_\infty \le K\}$ and $K < R$, the final odometer vanishes outside the cube $\{|x|_\infty \le R\}$. Now



(a) (b)

Figure 9.16: The second step of the proof of Proposition 9.35 for $d = 2$, $b = 2$ and $M = 4$: (a) the odometer g of the sandpile $3\mathbf{1}_{[-M,M]}$ on $\mathbb{Z}$, and (b) the level sets of $w(x) = \min\{g(x_1), g(x_2)\}$, which are squares. The discrete Laplacian of w is at most $-b$ on the square $\{|x|_\infty \le M\}$ (orange) and at most 1 everywhere.

$R = (2b + 3)K + 3b + 4$ and $K \le C(d)n^{1/d}$ for all large n by the first step, so $R \le C(d, b)n^{1/d}$ for all large n, which proves part (b).

For the final step, suppose $\eta \ge 0$. Every legal toppling sequence for $n\delta_{\mathbf{0}}$ remains legal after adding η, and since $\eta + n\delta_{\mathbf{0}}$ stabilizes, by the least action principle (Lemma 9.2) its final odometer is at least u. The lower inclusion follows from (9.8). □

In Chapter 4 we obtained bounds by Euclidean balls for $\eta \equiv 0$ through the divisible sandpile, with explicit leading constants (Proposition 4.11), but only for $d \ge 2$. The proof above is combinatorial, and by (9.8), for $\eta \equiv 0$ the toppled set lies between a diamond and a cube of the same radius; parts (a) and (b) of Proposition 9.35 hold in every dimension and for every background at most $2d - 2$, and part (c) for every such background that is nonnegative.

In the special case when the background is the constant $2d - 2$ grains at every site, the support of the odometer can be identified as a cube (see Figure 9.17).

Proposition 9.36 *Let $d \ge 1$ and $n \ge 2$ be integers. The sandpile $(2d-2)+n\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$ stabilizes. If $u_\infty^{(n)}$ is its final odometer, then there exists a nonnegative integer K_n such that*

$$\{u_\infty^{(n)} > 0\} = \{x \in \mathbb{Z}^d : |x|_\infty \le K_n\}.$$

Proof. By part (a) of Proposition 9.35 with $b = 2d - 2$, the sandpile $(2d - 2) + n\delta_{\mathbf{0}}$ stabilizes, so by the least action principle every legal toppling sequence started from it is finite. The idea is to topple in an order in which the set of sites toppled so far is a box after each batch of topplings, so that the final toppled set is a box; by symmetry it is then a cube.

Topple the origin first, so that the toppled set is the box $\{\mathbf{0}\}$. Suppose that the set B of sites toppled so far is a box and that the current sandpile has

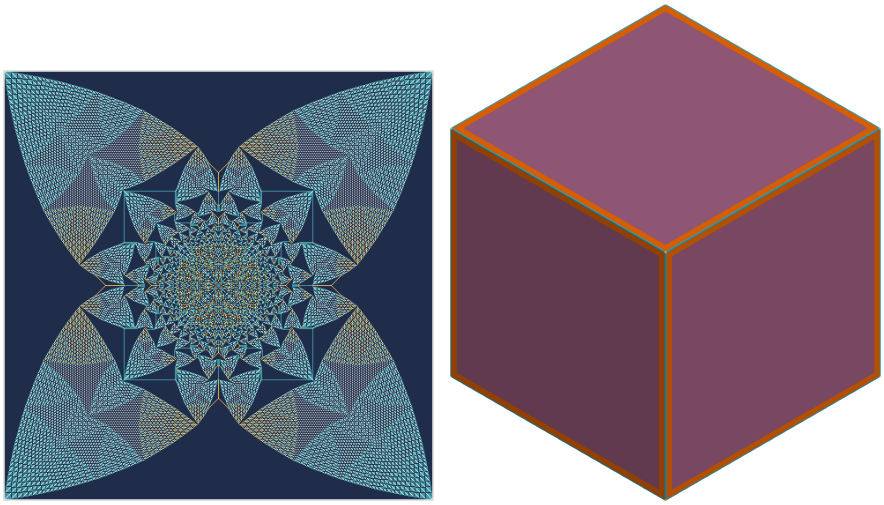


Figure 9.17: The stabilization of $(2d-2)+n\delta_{\mathbf{0}}$ for $d=2$ and $n=10^5$ (left) and for $d=3$ and $n=10^6$ (right), each toppled site colored by its final number of grains as in Figure 9.15. The sites that toppled form the square $\{|x|_\infty \le 242\}$ and the cube $\{|x|_\infty \le 98\}$.

an unstable site. If this site lies in B, then we topple it, and the toppled set does not change. Otherwise the unstable site v lies outside B. It was stable initially, so it has received a grain from a neighbor in B, and hence it lies in a box F obtained from B by fixing one coordinate one step beyond its interval in B and keeping the other coordinate intervals. Every site of F differs from the origin, has not toppled, and has received a grain from its neighbor in B, so it has at least $2d-1$ grains. Topple v, and then topple, one at a time, untoppled sites of F adjacent to a site of F already toppled; each has received the additional grain it needs. Since F is connected, all its sites topple, and the toppled set becomes the box $B \cup F$ (see Figure 9.18).

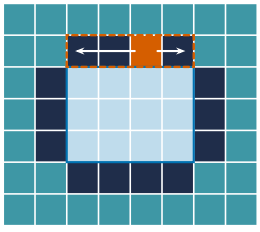


Figure 9.18: The proof of Proposition 9.36 in $\mathbb{Z}^2$: the toppled sites form a box (blue), and each site outside the box next to one of its sides holds three grains (navy). Once one of them is unstable (vermilion), the sites along that side (dashed) topple one after another (arrows), and the toppled set becomes a larger box.

This procedure consists of legal topplings, so it ends at a stable sandpile,

and the final toppled set $\{u_\infty^{(n)} > 0\}$ is a box. The sandpile $(2d-2) + n\delta_{\mathbf{0}}$ is invariant under reflections and permutations of the coordinates, and the final odometer does not depend on the toppling order, by Lemma 9.3(b), so this box is invariant under the same symmetries. It is therefore a cube $\{x \in \mathbb{Z}^d : |x|_\infty \le K_n\}$ with K_n a nonnegative integer. □

9.10 ROBUST AND EXPLOSIVE BACKGROUNDS

Let G be a locally finite digraph. Recall from Section 9.7 that a (possibly infinite) sandpile η on G is robust if $\eta+\xi$ stabilizes for every finitely supported sandpile $\xi \ge 0$. We say that η is **explosive** if there exists such a ξ for which parallel toppling of $\eta+\xi$ (in which every unstable vertex topples once in each round, as in Section 9.8) results in every vertex of G eventually toppling.

An infinite avalanche need not be an explosion. On $\mathbb{Z}^2$, let η equal 3 on the horizontal axis and 2 elsewhere. Then adding one grain at the origin makes each site of the horizontal axis topple once and no other site topple, so by the least action principle, $\eta + \delta_{\mathbf{0}}$ does not stabilize, although each site topples only finitely often. Hence η is not robust, but not every vertex topples.

The next lemma gives a simple test for robustness.

Lemma 9.37 *Let G be a strongly connected, locally finite digraph, and let v be a non-sink vertex of G.*

(a) Let η be a sandpile on G, and let $\xi \ge 0$ be finitely supported. There is an integer $n \ge 0$ and a finite legal toppling sequence from $\eta + n\delta_v$ whose final sandpile is at least $\eta + \xi$.

(b) A sandpile η on G is robust if and only if $\eta + n\delta_v$ stabilizes for every integer $n \ge 0$.

Proof. (a) Let Γ be the union of $\{v\}$ and one directed path from v to each vertex of $\mathrm{supp}(\xi)$. For each vertex of Γ other than v, keep an incoming edge on a shortest path from v in Γ. These edges form a finite directed tree rooted at v. Extend ξ by zero to the other vertices of the tree. Starting at the leaves and moving toward the root, choose nonnegative integers $q(x)$ so that, for every child y of x,

$$q(x) \ge \deg(y)\, q(y) + \max\{\xi(y), -\eta(y)\}\,.$$

For a leaf, take $q(x) = 0$. Finally, choose n so that

$$n \ge \deg(v)\, q(v) + \max\{\xi(v), -\eta(v)\}\,.$$

Starting at the root and moving outward, topple each vertex x exactly $q(x)$ times before proceeding to its children. A child y receives at least $q(x)$ grains

from its parent. Before its $(k+1)$st toppling, where $0 \le k < q(y)$, its height is at least

$$\eta(y) + q(x) - k \deg(y) \ge \deg(y)\,.$$

These topplings are therefore legal, and the final increase at y is at least $q(x) - \deg(y)\, q(y) \ge \xi(y)$. Similarly, if $0 \le k < q(v)$, then before the root's $(k+1)$st toppling its height is at least $\eta(v) + n - k \deg(v) \ge \deg(v)$, and the final increase at v is at least $n - \deg(v)\, q(v) \ge \xi(v)$. Grains received from other edges only increase the heights. This proves part (a).

(b) If η is robust, then $\eta + n\delta_v$ stabilizes, since $n\delta_v$ is finitely supported. Conversely, suppose $\eta + n\delta_v$ stabilizes for every $n \ge 0$, and let $\xi \ge 0$ be finitely supported. By part (a) there is a finite legal sequence from $\eta + n\delta_v$ to a sandpile β with $\beta \ge \eta + \xi$. Let r be the odometer of this sequence, and let u be the final odometer of $\eta + n\delta_v$. By the least action principle, $r \le u$. Moreover,

$$\beta - (u - r)\mathsf{L} = \eta + n\delta_v - u\mathsf{L}\,.$$

The sandpile $\eta + n\delta_v - u\mathsf{L}$ is stable, so $u - r$ is a stabilizing toppling function for β, and hence also for the smaller sandpile $\eta + \xi$. By Corollary 9.30, $\eta + \xi$ stabilizes, so η is robust. □

For G a strongly connected digraph, the simplest example of an explosive sandpile is one where every vertex is fully occupied, that is, has $\deg(v) - 1$ grains. In this case, adding a single grain causes every vertex to topple, since each out-neighbor of a toppled vertex becomes unstable and G is strongly connected. (Four grains at every site of a bounded domain, stabilized with and without a sink, are shown in Figure 9.19.)

We next observe that on $\mathbb{Z}^d$ a pointwise upper bound implies robustness.

Lemma 9.38 *Let $d \ge 1$. If a sandpile η on $\mathbb{Z}^d$ is bounded from above pointwise by $2d-2$, then η is robust.*

Proof. This is immediate from Lemma 9.37(b) and Proposition 9.35(a). □

See Figure 9.20 for the stabilization of a large pile over four constant backgrounds on $\mathbb{Z}^3$.

For intermediate background heights, both behaviors occur. We state the result for a random background and prove it in two dimensions (see Figure 9.21, and Figure 9.22 for stabilizations on robust random backgrounds). As we will see in Proposition 9.43, an analogue of the result holds in every dimension $d \ge 2$, for every i.i.d. background with values between d and $2d-1$ that equals $2d-1$ with positive probability, but the two-dimensional argument is shorter.

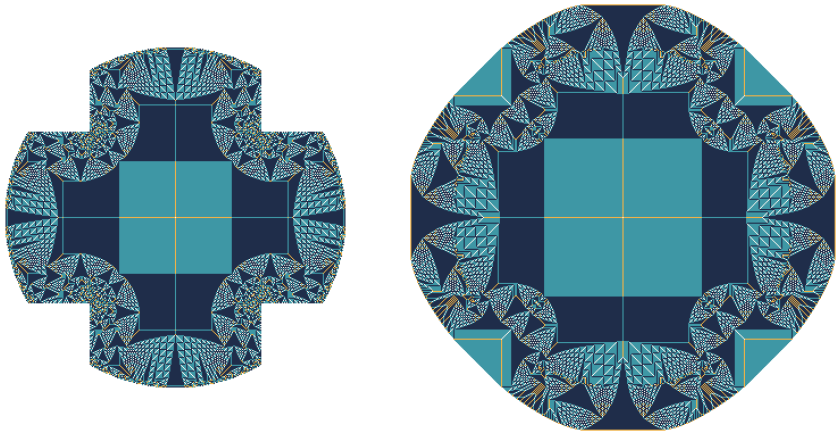


Figure 9.19: Sandpiles on $\mathbb{Z}^2$, drawn at the same scale. On the left, the initial condition is four grains at every site of the cross $\{x \in \mathbb{Z}^2 : |x| < 150 \text{ and } \min(|x_1|, |x_2|) \le 75\}$, with the sites outside the cross identified as the sink. On the right, the initial condition is the same, but the sink is removed and grains are free to spread. Sites with zero, one, two and three grains are off-white, orange, teal and navy.

Lemma 9.39 *Let $p \in (0, 1]$, and let η be the i.i.d. random background on $\mathbb{Z}^2$ given by*

$$\eta(x) := \begin{cases} 3, & \text{with probability } p, \\ 2, & \text{otherwise}. \end{cases}$$

Then η is explosive with probability one.

Proof. Let F_n be the event that every side of $[-n, n]^2$ contains a non-corner site x with $\eta(x) = 3$. We claim that for every $k \ge 1$ we have that

$$\bigcap_{n \ge k} F_n \subseteq \{\eta \text{ is explosive}\}.$$

To see this, consider the finitely supported sandpile $2\mathbf{1}_{[-k,k]^2}$. Every site of $[-k, k]^2$ then has at least four grains, and so it can be toppled. Once those sites have toppled, every non-corner site on the next square has at least three grains, and on F_{k+1} each side has a site with four grains. Starting from this site, legal topplings propagate in both directions along that side (see Figure 9.23), and each corner then receives one grain from each of its two side-neighbors and topples. Iterating this gives a sequence of legal topplings in which every site of $\mathbb{Z}^2$ topples, so by Lemma 9.32(a) every site eventually topples under parallel toppling, proving the claim.

We then observe, by the union bound, that for every $k \ge 1$

$$\mathbb{P}\Big\{\bigcap_{n \ge k} F_n\Big\} \ge 1 - \sum_{n \ge k} \mathbb{P}\{F_n^c\},$$

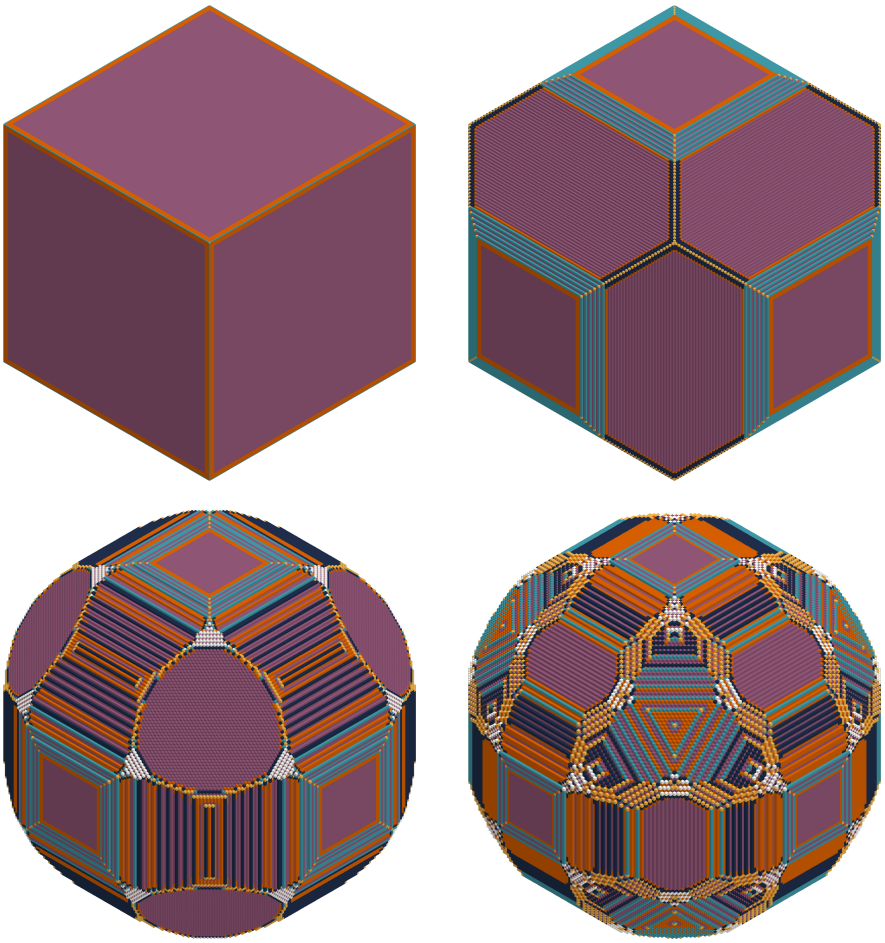


Figure 9.20: The stabilization of $4 \times 10^6 \delta_{\mathbf{0}} + (2d - k)$ on $\mathbb{Z}^3$ for $k = 2, 3, 4, 5$ (top left to bottom right), that is, on the constant backgrounds 4, 3, 2 and 1, drawn and colored as in Figure 9.15. Each solid is scaled to fill its panel: the cube for $k = 2$ has side 311, and the others are 201, 161 and 141 sites across.

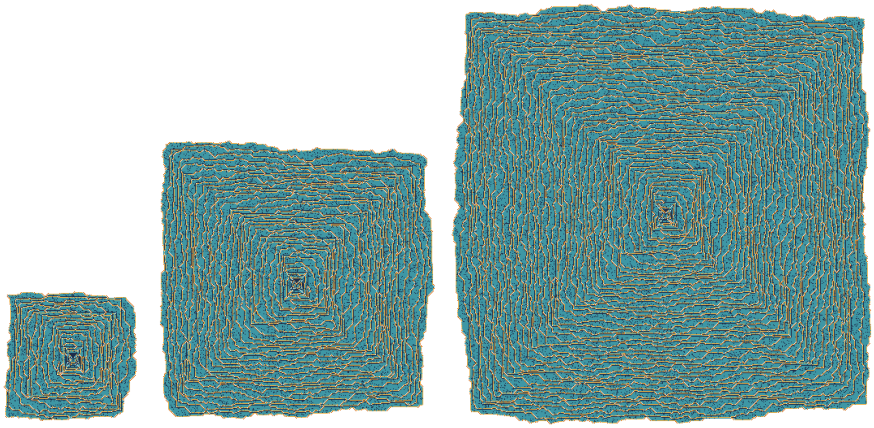


Figure 9.21: Snapshots of an exploding sandpile with background η under parallel toppling, where $\mathbb{P}\{\eta(v) = 3\} = 1/10$ and $\mathbb{P}\{\eta(v) = 2\} = 9/10$, after 400, 800 and 1200 rounds, drawn at the same scale, started from two grains at every site of $[-3, 3]^2$ added to η. Sites that have toppled are colored by their height, zero, one, two and three in off-white, orange, teal and navy, unstable sites are vermilion, and the other sites are white. By Lemma 9.39, this background is explosive with probability one.

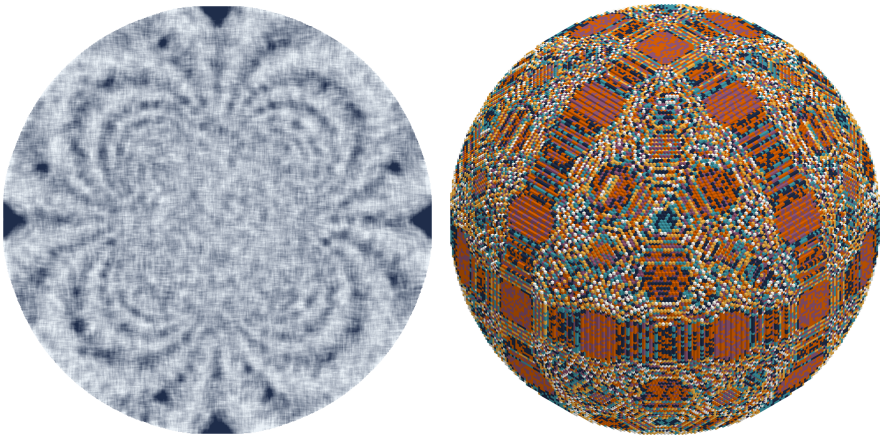


Figure 9.22: The stabilization of n grains at the origin on the i.i.d. background η with $\mathbb{P}\{\eta(v) = -1\} = \mathbb{P}\{\eta(v) = 0\} = 1/2$, for $d = 2$ and $n = 10^6$ (left) and for $d = 3$ and $n = 10^7$ (right). On the left each toppled site is shaded by the average number of grains in the 13×13 box around it, from light (2.0 or less) to navy (2.4 or more); on the right the sites are drawn and colored as in Figure 9.15. These backgrounds are robust by Lemma 9.38.

which completes the proof upon noting that $\mathbb{P}\{F_n^c\} \le 4(1-p)^{2n-1}$, so the right side tends to one as $k \to \infty$. □

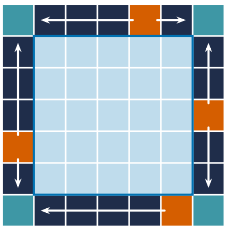


Figure 9.23: The proof of Lemma 9.39 on one square: once every site of $[-2,2]^2$ (blue) has toppled, each non-corner site of the boundary of $[-3,3]^2$ holds at least three grains (navy), and a site with background 3 on each side holds four (vermilion). Topplings spread from it along its side (arrows), and each corner (teal) topples once both of its neighbors on the square have toppled.

A stable background is robust if it is at most $2d-2$ on infinitely many of the sets $\{x \in \mathbb{Z}^d : |x|_\infty = n\}$.

Lemma 9.40 *Let $d \ge 1$, let η be a stable sandpile on $\mathbb{Z}^d$, and let $1 \le n_1 < n_2 < \cdots$ be integers. Suppose*

$$\eta(x) \le 2d-2 \qquad \text{when } |x|_\infty = n_i \text{ for some } i \ge 1\,.$$

Then η is robust.

Proof. By Lemma 9.37(b), it suffices to show that $\eta + k\delta_{\mathbf{0}}$ stabilizes for every $k \ge 1$. For an integer $N \ge 1$, let

$$g_N(x) := (N - |x|_\infty)^+\,.$$

We have $\Delta^{::} g_N(\mathbf{0}) = -2d$. If $0 < |x|_\infty < N$ and exactly one coordinate attains the maximum absolute value, then one inward and one outward contribution to $\Delta^{::} g_N(x)$ cancel. If several coordinates attain the maximum absolute value, then there is no inward contribution and there are several outward ones. Thus $\Delta^{::} g_N(x) \le 0$ in either case. When $|x|_\infty = N$, at most one neighbor has positive g_N, so $\Delta^{::} g_N(x) \le 1$. Finally, $\Delta^{::} g_N(x) = 0$ when $|x|_\infty > N$.

Let $g := g_{n_1} + \cdots + g_{n_k}$. Then $\Delta^{::} g(\mathbf{0}) = -2dk$. If $|x|_\infty = n_j$ for some $j \le k$, then only g_{n_j} can contribute positively, so $\Delta^{::} g(x) \le 1$, and elsewhere $\Delta^{::} g \le 0$. Since $\eta \le 2d-1$ everywhere and $\eta \le 2d-2$ where $|x|_\infty = n_i$, it follows that $\eta + k\delta_{\mathbf{0}} + \Delta^{::} g \le 2d-1$ everywhere; at the origin, $\eta(\mathbf{0}) + k - 2dk \le 2d-1$. By Corollary 9.30, $\eta + k\delta_{\mathbf{0}}$ stabilizes, with odometer at most g. □

Part (b) of the next lemma implies that if η is explosive on an Eulerian digraph, then adding a finite number of grains to it leads to every vertex toppling infinitely often. We proved this fact earlier in the finite undirected case (Lemma 9.10).

LEMMA 9.41 *Let σ be a sandpile on a locally finite Eulerian digraph G.*

(a) If σ stabilizes, then some vertex of G never topples.

(b) If, under the parallel toppling procedure, every vertex eventually topples, then all vertices topple infinitely often.

Proof. (a) Suppose for sake of contradiction that σ stabilizes and every vertex of G topples. Fix a stabilizing sequence; since it is finite, G is finite. Let w be the first vertex to stop toppling. During its last toppling, w receives one grain along each self-loop, and every other incoming edge comes from a vertex which topples after w finished toppling, so one grain arrives along each such edge afterward. Since $\text{indeg}(w) = \deg(w)$, the vertex w is unstable at the end, a contradiction.

(b) Suppose for sake of contradiction that every vertex eventually topples, but some vertex topples only finitely often. For each such vertex w, let t_w be the last time at which w topples, and let v be a vertex such that t_v is minimal. During its last toppling, v receives one grain along each self-loop. For every other incoming edge (w, v) into v, the vertex w topples at some time at least t_v (by minimality if w topples finitely often, and since w topples after time t_v otherwise). Since G is locally finite, all these grains have arrived by a finite time, and since G is Eulerian, their number is $\text{indeg}(v) = \deg(v)$, so v must topple again, a contradiction. □

With the methods of this chapter we can say little about the shapes in Figures 9.14 and 9.15. If grains are replaced by a continuous fluid, the fractal structure disappears, and in Chapter 4 we showed that the divisible sandpile started from mass n at the origin fills a Euclidean ball up to an error in the radius that does not grow with n.

9.11 BOOTSTRAP PERCOLATION*

Bootstrap percolation is a cellular automaton on $\mathbb{Z}^d$ with a random initial state and a deterministic update rule. Every site starts off as infected independently at random with probability p. Infected sites remain infected, and if an uninfected site has at least d neighbors which are infected, then it becomes infected. As we will see in the next section, there is a relationship between the sandpile and bootstrap percolation. A box in $\mathbb{Z}^d$ is a product of d finite intervals of integers, and its side lengths are the numbers of integers in these intervals. We say that a box is **internally spanned** if every site, under bootstrap percolation dynamics in the box, becomes infected. Enlarging a box can only add infected neighbors, so the dynamics on a box infect every site that the dynamics on a smaller box inside it infect.

The next lemma shows that a box whose side lengths are n or $n+1$ fails to be internally spanned with probability exponentially small in n. We allow both lengths because the layers added as a cube grows one coordinate at a time are boxes of this kind, one dimension lower.

Lemma 9.42 *For every integer $d \geq 1$ and every $p \in (0,1]$, there exists $c = c(d,p) > 0$ with the following property. Let $n \geq 1$ be an integer, and let $n_1, \ldots, n_d$ be integers in $\{n, n+1\}$. If the sites of the box $\{1,\ldots,n_1\} \times \cdots \times \{1,\ldots,n_d\}$ are initially infected independently with probability p, then the probability that this box is not internally spanned is at most* $\exp(-cn)$.

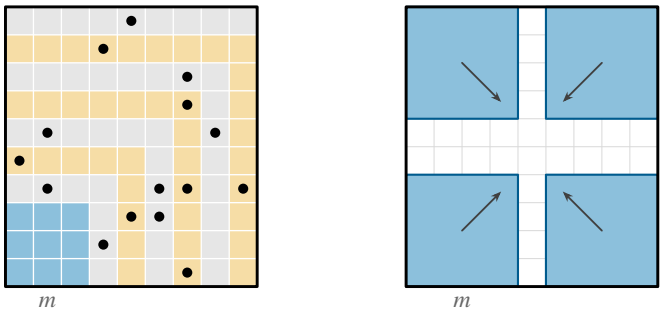
Proof. If $p = 1$, then every site is initially infected and the probability in the lemma is zero. Suppose $0 < p < 1$. For positive integers j and n, let $f_j(n)$ be the largest probability, over the boxes in $\mathbb{Z}^j$ with side lengths in $\{n, n+1\}$, that the box is not internally spanned under bootstrap percolation on $\mathbb{Z}^j$, whose threshold is j. We prove by induction on d that $f_d(n) \leq \exp(-cn)$ for all $n \geq 1$, where $c = c(d,p) > 0$. In dimension one, a box is an interval, and it is internally spanned if and only if it contains an initially infected site. Thus $f_1(n) = (1-p)^n$, which proves the base case with $c = \log(1/(1-p))$.

For the inductive step, suppose $d \geq 2$ and $f_{d-1}(k) \leq e^{-c_0 k}$ for every $k \geq 1$, where $c_0 > 0$. We grow an infected corner cube to a whole box one layer at a time, then bound $1 - f_d(n)$ away from zero, and finally improve this bound to exponential decay.

We first grow a corner cube (see Figure 9.24(a) for $d = 2$). Fix a box with side lengths $n_1, \ldots, n_d$ in $\{n, n+1\}$, let $1 \leq m \leq n$, and suppose every site of the cube of side m at one corner of the box is infected. For $k = m, m+1, \ldots, n-1$ in turn, we enlarge the current cube of side k to a cube of side $k+1$ by extending its first, second, …, dth coordinate interval by one integer toward the opposite corner of the box. Afterwards, we extend the ith coordinate interval once more for each i with $n_i = n+1$. Each extension adds a layer of sites with one fixed coordinate, and deleting that coordinate identifies the layer with a box in $\mathbb{Z}^{d-1}$ whose side lengths are k or $k+1$ (with $k = n$ for the last extensions). Every site of the layer has an infected neighbor in the box built so far, so a site with $d-1$ infected neighbors in the layer has d infected neighbors. Hence, if the layer is internally spanned under bootstrap percolation on $\mathbb{Z}^{d-1}$, then the dynamics on $\mathbb{Z}^d$ restricted to the box infect the whole layer. At most d layers are added for each $k \geq m$, so with $C := d/(1-e^{-c_0})$, by the union bound we have

$$\mathbb{P}\{\text{an added layer is not internally spanned}\} \leq d \sum_{k=m}^{\infty} e^{-c_0 k} = C e^{-c_0 m}. \tag{9.10}$$

The layers are disjoint from the corner cube, so if the corner cube and every added layer are internally spanned, then the box is internally spanned.



(a) (b)

Figure 9.24: (a) In $\mathbb{Z}^2$ an infected corner square of side $m = 3$ (blue) grows to a box with sides 9 and 10 by adding a column and then a row for each $k = 3, \dots, 8$ and a final row; each added layer is an interval, internally spanned when it contains an initially infected site (black). (b) The four corner squares of side $\lfloor 9/2 \rfloor = 4$ are disjoint, and each grows toward the opposite corner as in (a).

Next we bound $1 - f_d(n)$ away from zero. Choose an integer $K \geq 1$ with $Ce^{-c_0 K} \leq 1/2$, and let $\epsilon := p^{K^d}/2$. If $n \geq K$, then the corner cube of side K is entirely initially infected with probability p^{K^d}, independently of the added layers, which are all internally spanned with probability at least $1/2$ by (9.10). If $n < K$, then the box has at most K^d sites, so all of them are initially infected with probability at least p^{K^d}. In both cases the box is internally spanned with probability at least ϵ, so

$$f_d(n) \leq 1 - \epsilon \qquad \text{for every } n \geq 1\,. \tag{9.11}$$

It remains to obtain exponential decay. Let $n \geq 2$ and $m := \lfloor n/2 \rfloor$. Every side length of the box is at least $2m$, so the cubes of side m at distinct corners are disjoint, and since $d \geq 2$ there are at least four corners. Fix four corners, as in Figure 9.24(b). If the box is not internally spanned, then either none of the four corner cubes is internally spanned, or a layer in one of their four enlargements is not. By independence, (9.10) and the union bound give

$$f_d(n) \leq f_d(m)^4 + 4Ce^{-c_0 m}\,. \tag{9.12}$$

Choose an integer $M \geq 2$ with $4Ce^{-c_0 M/2} \leq \epsilon$, and let

$$c := \min\left\{\frac{c_0}{6}, \frac{\log(1/(1-\epsilon))}{2M}\right\}.$$

We prove by strong induction on n that $f_d(n) \leq e^{-cn}$. For $1 \leq n \leq 2M$, the bound (9.11) and the choice of c give $f_d(n) \leq 1 - \epsilon \leq e^{-2cM} \leq e^{-cn}$. For the

inductive step, let $n > 2M$ and assume that $f_d(k) \le e^{-ck}$ for $1 \le k < n$. Then $m \ge M$, and since $c_0 - 3c \ge c_0/2$, we have

$$4Ce^{-c_0 m} \le 4Ce^{-c_0 M/2} e^{-3cm} \le \epsilon e^{-3cm} .$$

We bound one factor of $f_d(m)^4$ by (9.11) and the other three by the inductive hypothesis, so by the recursion (9.12),

$$f_d(n) \le (1-\epsilon)e^{-3cm} + \epsilon e^{-3cm} = e^{-3cm} \le e^{-cn} ,$$

where the last inequality uses $3\lfloor n/2 \rfloor \ge n$ for $n \ge 3$. This completes the induction on n and hence on d. □

9.12 EXPLOSIVE BACKGROUNDS IN HIGHER DIMENSIONS*

Bootstrap percolation dynamics exactly match sandpile dynamics when the background takes values in $\{d, 2d\}$ and sites are constrained to topple at most once. Infected sites are those which have toppled, and sites with $2d$ grains start as infected. Indeed, a site beginning with d grains topples if and only if it has at least d neighbors which have toppled. Our proof of the next proposition uses this connection to bootstrap percolation, together with the large deviation bound of Lemma 9.42: starting from a finite cube whose sites have all toppled, we enlarge the cube one layer at a time, as in the proof of that lemma, and bootstrap percolation on each layer gives an order in which its sites can be toppled legally. (Recall from Section 9.10 that a background is explosive if there exists a nonnegative finitely supported sandpile whose addition makes every site eventually topple under parallel toppling.)

Proposition 9.43 *Let $d \ge 2$ be an integer, let $p \in (0, 1]$, and let η be an i.i.d. random background on $\mathbb{Z}^d$ such that, for every $x \in \mathbb{Z}^d$,*

$$d \le \eta(x) \le 2d - 1 \qquad \text{and} \qquad \mathbb{P}\{\eta(x) = 2d - 1\} = p .$$

Then η is explosive almost surely.

Proof. The proof has three steps. First, we fix a sequence of growing cubes and show how to topple the sites of one added layer. Next, we apply Lemma 9.42 in dimension $d - 1$ to all sufficiently large layers. Finally, we start the growth by adding grains to a finite cube.

Let $J_1 := \{0\}$. For $k \ge 1$, obtain the interval J_{k+1} from J_k by adding the integer just above J_k when k is odd, and the integer just below J_k when k is even. Then J_k has k elements, and the cubes J_k^d increase to $\mathbb{Z}^d$. We enlarge J_k^d to J_{k+1}^d by replacing, for $i = 1, \dots, d$ in turn, the ith factor J_k by J_{k+1}. The sites added at the ith replacement form a layer. Their ith coordinate is the

integer in $J_{k+1} \setminus J_k$, and deleting this coordinate identifies the layer with a box in $\mathbb{Z}^{d-1}$ whose side lengths are k or $k+1$. Every site of the layer has a neighbor in the box before the replacement.

We first consider one layer. Declare a site x initially infected when $\eta(x) = 2d-1$. Suppose every site of the box before the replacement has toppled once and no site of the new layer has toppled. Then every site of the layer still has its initial grains and has received at least one more. A site with $\eta(x) = 2d-1$ therefore has at least $2d$ grains and is unstable. Every other site of the layer has at least $d+1$ grains, so it becomes unstable after $d-1$ of its neighbors in the layer have toppled. Toppling the sites of the layer in an order in which bootstrap percolation on $\mathbb{Z}^{d-1}$ infects them is therefore legal. Thus, if $(d-1)$-dimensional bootstrap percolation dynamics internally span the layer, then every site of the box after the replacement has toppled once.

We next show that, almost surely, every layer added while J_k^d grows to J_{k+1}^d is internally spanned for all sufficiently large k. The events $\{\eta(x) = 2d-1\}$ are independent, and each has probability p. For each $k \geq 1$, the d layers added while J_k^d grows to J_{k+1}^d are boxes in $\mathbb{Z}^{d-1}$ with side lengths k or $k+1$. By Lemma 9.42 in dimension $d-1$, the probability that one of them is not internally spanned is at most $d\exp(-ck)$, where $c = c(d-1,p)$. These bounds are summable over k. By the Borel–Cantelli lemma, almost surely there is a positive integer K such that every layer added while J_k^d grows, for every $k \geq K$, is internally spanned.

Finally, fix a realization and such an integer K, and add d grains at each site of J_K^d. Each of these sites then has at least $2d$ grains, so toppling each of them once is legal. Toppling the d layers of each enlargement as above, for $k = K, K+1, \ldots$ in turn, gives for every $k \geq K$ a finite legal sequence that topples every site of J_k^d. Since the cubes J_k^d increase to $\mathbb{Z}^d$, by Lemma 9.32(a) parallel toppling of $\eta + d\mathbf{1}_{J_K^d}$ eventually topples every site. Hence η is explosive almost surely. □

EXERCISES

Exercise 9.1. The following reflection process is a real-valued analogue of the abelian property in Lemma 9.3. Assign to each vertex of a cycle with $N \geq 3$ vertices a real-valued label in such a way that the sum of all the labels is positive. A reflection replaces a triple of consecutive labels x, y, z where $y < 0$ by $x+y, -y, z+y$ respectively. Show that every sequence of reflections terminates with all labels nonnegative and that the order in which the reflections are applied does not affect the final labels. Hint: Consider the partial sums of labels along the cycle and observe that a reflection corresponds to sorting two adjacent partial sums.

Exercise 9.2. Let $G = (V, E)$ be a locally finite simple undirected graph. Let σ_0 be a sandpile on G, let σ_t be the sandpile after some topplings, and let u be the corresponding odometer, so $\sigma_t = \sigma_0 - u\mathsf{L}$. Show that for every finite set $U \subset V$,

$$\sum_{w \in U,\ v \in V \setminus U} (u(v) - u(w)) \mathbf{1}\{\{v, w\} \in E\} = \sum_{w \in U} (\sigma_t - \sigma_0)(w)\,.$$

This verifies (9.3) in the proof of Lemma 9.13.

Exercise 9.3. Let G be a finite digraph with a global sink. Show that the sandpile group of Section 9.4 is isomorphic to the additive group of harmonic functions modulo 1 on G that vanish at the sink. Hint: use Lemma 9.16.

Exercise 9.4. In the setting of Section 9.5, let $\boldsymbol{\delta}$ be the sandpile given by $\boldsymbol{\delta}(v) = \deg(v)$ at every non-sink vertex, and let $\alpha := \boldsymbol{\delta} - \boldsymbol{\delta}^\circ$. Show that every entry of α is at least 1 and that $(\sigma + \alpha)^\circ = \sigma$ for every recurrent sandpile σ on G.

Exercise 9.5. Let G be a finite digraph with a global sink. The **inverse** of a recurrent sandpile σ on G is the recurrent sandpile $\bar{\sigma}$ which satisfies $(\sigma + \bar{\sigma})^\circ = I$, where I is the identity element of the sandpile group. Show that the inverse can be computed via

$$\bar{\sigma} = (\xi - \xi^\circ - \sigma)^\circ\,,$$

where ξ is a sandpile satisfying $\xi(v) \geq 3\deg(v) - 3$ for all non-sink vertices v.

Exercise 9.6. Show that the condition of being strongly connected in the definition of a finite Eulerian digraph (Section 9.6) can be replaced by the condition that the underlying undirected graph is connected. That is, if G is a finite digraph with $\mathrm{indeg}(v) = \deg(v)$ at every vertex v, show that G is strongly connected if and only if its underlying undirected graph is connected. Give an infinite digraph for which the equivalence fails.

Exercise 9.7. Section 9.6 defines Eulerian tours and Eulerian digraphs. Let G be a finite digraph without isolated vertices. Show that G has an Eulerian tour if and only if G is strongly connected and $\mathrm{indeg}(v) = \deg(v)$ for every vertex v.

Exercise 9.8. Let G be the cube $\{0, \ldots, k\}^d$ in $\mathbb{Z}^d$, for integers $d \geq 1$ and $k \geq 2$, with the boundary (the vertices with a coordinate equal to zero or k) identified as the sink, keeping edge multiplicities and replacing each edge by two oppositely directed edges, as in Section 9.6. Show that the sandpile $\sigma \equiv d$ on the non-sink vertices is recurrent.

Exercise 9.9*. Let G be a finite connected plane graph, and let G^* be its plane dual. Replace every edge of G and G^* by two oppositely directed edges. The two digraphs are Eulerian in the sense of Section 9.6. Choose

a vertex of each digraph and delete its outgoing edges, so that it becomes a sink. Show that, whichever vertices are chosen, the sandpile groups of the two digraphs are isomorphic. Hint: use Exercise 9.3.

Exercise 9.10. In the setting of Section 9.7, let $p \le q$ be integers, and let $f : \mathbb{Z} \to \mathbb{Z}$ be supported on $[p, q]$ with

$$\sum_{y=p}^{q} f(y) = 0 \qquad \text{and} \qquad \sum_{y=p}^{q} y f(y) = 0\,.$$

(a) Show that there is an integer-valued function g such that $f = -g\mathsf{L}$ and $g(x) = 0$ unless $p < x < q$.

(b) Suppose that there are no integers $x_1 < x_2 < x_3$ such that $f(x_1) < 0$, $f(x_2) > 0$ and $f(x_3) < 0$. Show that $g \ge 0$.

Exercise 9.11. This exercise determines all robust zero-one backgrounds on $\mathbb{Z}$ (see Section 9.7). (a) Let $x \le y$ be integers, let $v \in [x, y] \cap \mathbb{Z}$, and let $\sigma = \mathbf{1}_{[x,y]}$ be a sandpile on $\mathbb{Z}$. What is the stabilization of $\sigma + \delta_v$?

(b) Let σ be a sandpile on $\mathbb{Z}$ taking only the values zero and one. Show that σ is robust if and only if $\sigma(p_i) = \sigma(m_i) = 0$ for two infinite sequences $p_i \to +\infty$ and $m_i \to -\infty$.

Exercise 9.12. Generalize Lemma 9.32(a) from $\mathbb{Z}^d$ to locally finite digraphs. Let σ be a sandpile on a locally finite digraph. Under the parallel toppling procedure, every unstable non-sink vertex topples once in each round. Show that every vertex toppled by a finite legal toppling sequence from σ is eventually toppled by the parallel toppling procedure.

Exercise 9.13. Extend Lemma 9.34 to the case where φ is reflection across the hyperplane $x \cdot \mathbf{e}_i = 1/2$ for some $1 \le i \le d$. That is, if a sandpile σ_0 on $\mathbb{Z}^d$ satisfies $\sigma_0 \circ \varphi = \sigma_0$ and is decreasing in direction $\mathbf{e}_i$ on the half-space $\{x \in \mathbb{Z}^d : x \cdot \mathbf{e}_i \ge 1/2\}$, show that the parallel toppling odometer u_t is also decreasing in direction $\mathbf{e}_i$ on that half-space for all $t \ge 0$.

Exercise 9.14. In the setting of Section 9.9, give a second proof of the case $d = 2$ of Proposition 9.36, one that does not choose a toppling order. Let $n \ge 2$ be an integer. Start an abelian sandpile on $\mathbb{Z}^2$ with $n + 2$ grains at the origin and two grains at every other site. Prove that there is a nonnegative integer k such that the set of sites that topple is

$$\{x \in \mathbb{Z}^2 : |x|_\infty \le k\}\,.$$

Exercise 9.15. Consider a background $\eta \le 3$ on $\mathbb{Z}^2$ for which there are horizontal lines $\mathbb{Z} \times \{y\}$, with y unbounded above and below, and vertical lines $\{x\} \times \mathbb{Z}$, with x unbounded above and below, on which $\eta \equiv 2$. Show that η is robust (see Lemma 9.37). Hint: We may assume that $\eta \equiv 3$ off these

lines. Apply Exercise 9.11 in the horizontal and vertical directions and take the minimum of the two stabilizing odometers.

Exercise 9.16. In the setting of Section 9.10, start a sandpile on $\mathbb{Z}^2$ with four grains at the origin and three grains at every other site, and apply parallel toppling, in which every unstable site topples once in each round. Prove that every site topples infinitely often. Use the least action principle, Lemma 9.2, to conclude that this sandpile does not stabilize.

Exercise 9.17. Let $d \geq 1$, and let $x_i = (x_{i1}, \ldots, x_{id})$ for $1 \leq i \leq d$ be linearly independent vectors in $\mathbb{Z}^d$ satisfying $\gcd(x_{1j}, \ldots, x_{dj}) = 1$ for all $1 \leq j \leq d$. Let $\Lambda := \mathbb{Z}x_1 + \cdots + \mathbb{Z}x_d$. Show that the background $(2d-2) + \mathbf{1}_\Lambda$ on $\mathbb{Z}^d$ is explosive (see Section 9.10).

NOTES

The abelian sandpile model was discovered several times by researchers in different communities operating independently. In the combinatorics literature, the name more commonly used is chip-firing. The model was proposed by Bak, Tang, and Wiesenfeld (1987, pp. 381–384) in two dimensions in 1987, as a prototype of self-organized criticality, and they predicted that its avalanche distributions follow power laws. Dhar (1990, pp. 1613–1614) extended the model to arbitrary finite systems with site-dependent toppling rules and proved many of its fundamental properties, including that the final stable sandpile does not depend on the toppling order and that the grain addition operators commute (Lemma 9.8). Diaconis and Fulton (1991, Proposition 4.1) proved order independence for a related game, following an argument of Eriksson, which shows that every legal sequence of moves is a permutation of a subsequence of each terminating legal sequence, as in Lemma 9.2. Their Lemma 2.2, on commuting operators, shows that the law of a related growth model does not depend on the order of the choices. Dhar and Majumdar (1990, Section 6) computed the avalanche mass and duration exponents on the Bethe lattice. Dhar and Ramaswamy (1989, pp. 1659–1661) solved the directed model and determined its critical exponents in every dimension.

Engel (1975) introduced an algorithm, the probabilistic abacus, to compute absorption probabilities and expected absorption times in finite Markov chains with a version of the abelian sandpile; Engel (1976, pp. 59–65) gave a self-contained description and proved its basic identities, apart from the recurrence of the critical load; a note added in proof (Engel, 1976, p. 69) announced a proof of the recurrence by L. Scheller and sketched its idea. A manuscript by Snell (2009, pp. 1 and 4–6), written circa 1979 and posted in 2009, completed the proof; its argument for the recurrence of the critical load (Snell, 2009, pp. 5–6) is due to Peter Doyle. Björner and Lovász (1992,

Section 6) identified the abacus as a chip-firing game and proved recurrence of the critical load. Gabrielov (1993, Sections 1–2 and 5–6) established the abelian property for continuous avalanche models and related avalanche distributions to Tutte polynomials.

Spencer (1986, Section 2) used a one-dimensional sandpile to balance vectors in the max norm, and this sandpile was later analyzed in depth in Sections 2–4 of Anderson, Lovász, Shor, Spencer, Tardos, and Winograd (1989). The sandpile group was also studied from the perspective of arithmetic geometry by Lorenzini (1989, pp. 481–484) and Lorenzini (1991, pp. 277–278), as a finite group attached to the graph Laplacian that was later identified with the sandpile group. Levine (2009, Theorems 3.3 and 1.2) gave exact sequences for wired trees and a full cyclic decomposition for wired regular trees.

Björner, Lovász, and Shor (1991) analyzed in detail the sandpile on finite graphs without sink; part (a) of Lemma 9.12 is their Theorem 3.3(c), and part (b) follows from their Theorem 3.3(b), or from their Theorem 3.3(a) when G is a tree. The polynomial bound for the number of topplings in Lemma 9.13 is due to Tardos (1988); our proof follows his Lemmas 4 and 5, and his Theorem 2 gives the resulting bound $2|V||E|\operatorname{diam}(G)$. Björner, Lovász, and Shor (1991, Theorem 4.2) gave a different bound on the number of topplings, namely $2|V|N/\lambda_1$, where N is the number of grains and λ_1 is the least positive eigenvalue of the graph Laplacian. For directed graphs, analogues of these termination and toppling bounds were proved by Björner and Lovász (1992, Sections 2 and 4).

The burning algorithm underlying Lemma 9.27 is due to Dhar (1990, p. 1615). Majumdar and Dhar (1992, pp. 131–133) matched the sandpiles that pass the test with the spanning trees of the graph, for symmetric toppling matrices. Speer (1993, Theorem 3 and Section 4) showed that the test can fail for asymmetric toppling matrices and introduced the script algorithm, which characterizes recurrence for the asymmetric toppling matrices considered there.

The representation of the sandpile group via recurrent sandpiles was developed by Dhar, Ruelle, Sen, and Verma (1995, Sections 2–3). Biggs (1999, Theorems 4.2 and 7.3), who called the group the critical group, and Biggs (1997, Sections 28–32) related it to the graph Laplacian and algebraic potential theory. Our exposition in Sections 9.4–9.6 relies heavily on Corollary 2.5 of Cori and Rossin (2000) and Sections 2 and 4 of the survey by Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008).

Fey, Levine, and Peres (2010, Lemma 2.3) proved the least action principle on $\mathbb{Z}^d$ for sandpiles that stabilize: the odometer is at most every $g : \mathbb{Z}^d \to \mathbb{Z}_{\geq 0}$ for which $\sigma - g\mathsf{L}$ is stable. Corollary 9.30 also gives stabilization, for finitely supported g, and Lemma 9.2 is a version for toppling sequences on general

digraphs. Part (b) of Lemma 9.41 extends to locally finite Eulerian digraphs their argument, in Section 1, that if every site of $\mathbb{Z}^d$ topples, then every site topples infinitely often. They defined robust and explosive backgrounds on $\mathbb{Z}^d$ in Section 1, through additions of grains at the origin, and Section 9.10 adapts their criteria for explosion and robustness, Theorems 4.1 and 4.2. The background with three grains on the horizontal axis is their example, in Section 1, of an infinite avalanche in which each site topples only finitely often.

Sections 9.8 and 9.9, including the growth bounds of Proposition 9.35, are adapted from Theorems 4 and 5 of Le Borgne and Rossin (2002), Lemma 5.3 of Babai and Gorodezky (2007), and Section 2.2 and Corollary 2.1 of Bou-Rabee (2022). The upper bound in Proposition 9.35 for backgrounds up to $2d-2$ adapts the background modification by which Fey, Levine, and Peres (2010) proved their Theorem 1.1 for constant backgrounds; Exercise 9.10 is their Lemma 3.3, and Exercise 9.17 is their Proposition 1.3. Proposition 9.36 is the part of Theorem 4.1 of Fey and Redig (2008) that identifies the toppled set as a cube. Theorem 1.1 of Pegden and Smart (2013) shows that the stabilizations of $n\delta_{\mathbf{0}}$ shown in Figures 9.14 and 9.15, rescaled by $n^{1/d}$, converge in the sense of local averages as $n \to \infty$. In $\mathbb{Z}^2$, Theorem 1.1 of Levine, Pegden, and Smart (2017) describes the constraint in the obstacle problem that determines this limit through an Apollonian circle packing, and Theorem 1.3 of Levine, Pegden, and Smart (2016b) uses it to construct piecewise quadratic solutions of the differential equation in this obstacle problem whose pieces form a fractal pattern.

Bootstrap percolation was introduced by Chalupa, Leath, and Reich (1979, p. L31). The proof of Lemma 9.42 adapts the induction on boxes for the modified basic model in Lemma 3.1 and Proposition 3.2 of Schonmann (1992); Theorem 3.1 there gives the corresponding exponential-relaxation result. In the modified basic model a site becomes infected when it has an infected neighbor in each coordinate direction, a more restrictive rule than the one here. Lemma 9.39 is the case $d = 2$ of Proposition 1.4 of Fey, Levine, and Peres (2010), which proves that the background $2d-2$ plus an i.i.d. Bernoulli field with positive parameter is explosive almost surely in every dimension $d \geq 2$, using the criterion of their Theorem 4.1; they remark on the similarity of their argument to Straley's argument for bootstrap percolation, which van Enter (1987) made rigorous: when $d = 2$ and each site is initially infected with probability $p > 0$, the bootstrap percolation of Section 9.11 almost surely infects every site of $\mathbb{Z}^2$. For stable backgrounds, Proposition 9.43 strengthens Theorem 1.4 of Bou-Rabee (2024). That paper calls a background explosive when it is not robust, and its theorem shows that every i.i.d. background that is at least d everywhere and equals $2d-1$ with positive probability is almost surely not robust; Proposition 9.43 shows

that adding finitely many grains can make every site topple. Proposition 9.43 includes Proposition 1.4 of Fey, Levine, and Peres (2010) for $d \geq 2$. Our proof adapts the argument in Section 7 of Bou-Rabee (2024). The proofs of Lemma 9.42 and Proposition 9.43 given here differ from the cited proofs.

The result of Exercise 9.3 appears in Section 2 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008), in the paragraph after their Lemma 2.17, and the result of Exercise 9.9 is Theorem 2 of Cori and Rossin (2000), but using Exercise 9.3 simplifies the proof.

A family of processes with the abelian property, the abelian networks, was introduced by Bond and Levine (2016, Section 2 and Theorem 4.6), who gave axioms under which the abelian distributed processors of Dhar (1999, Section 5) have the abelian property, and further analyzed by Chan and Levine (2022, Chapters 4–7), who developed their nonhalting dynamics. Reflection processes were studied earlier in Proposition 2.1 of Mozes (1990) and Propositions 1–3 of Alon, Krasikov, and Peres (1989), motivated by a 1986 IMO problem, and Bond and Levine (2016, Section 5.2) list them as a possible example for an extension of abelian networks with shared memory. Reflection processes appear in Exercise 9.1, and the hint is from a solution given on pp. 311–312 of Winkler (2024).

Chapter Ten

The Divisible Sandpile

In Chapter 4 we proved that the odometer and final configuration of the divisible sandpile do not depend on the order of sequential topplings (Section 4.1), and that a point source of mass fills a region close to a ball (Figures 4.3 and 4.4). In this chapter we consider the divisible sandpile model beyond $\mathbb{Z}^d$, and ask whether a random configuration on an infinite graph stabilizes at all. Throughout this chapter, graphs $G = (V, E)$ are undirected, locally finite, connected and simple, and have at least two vertices.

In Section 10.1 we characterize the final configuration by quadrature inequalities for superharmonic functions. In Section 10.2 we extend the least action principle to general toppling procedures, including parallel toppling, and in Section 10.3 we identify the odometer with the value of an optimal stopping problem for simple random walk.

In Section 10.4 we use this representation to prove a phase transition for bounded i.i.d. masses on an infinite connected graph of bounded degree: the sandpile stabilizes below mean 1 and explodes above it, and at mean 1 it explodes unless the masses are constant. (At mean 1 the proof combines a ball of small masses with a variance bound and a zero-one law.) On infinite vertex-transitive graphs, conservation of expected mass gives the conclusions away from mean 1, assuming only that the initial mass is integrable (Section 10.5).

10.1 DISCRETE QUADRATURE

The next theorem gives a system of inequalities that uniquely characterizes the final configuration σ_∞. It is in some sense dual to the least action principle, Theorem 4.3, which gives a system of inequalities that uniquely characterizes the odometer, and it extends the discrete inequality (1.1) and its continuum counterpart in Chapter 1.

For $\sigma, h : V \to \mathbb{R}$ with σ having finite support, we write

$$\langle \sigma, h \rangle := \sum_{v \in V} \sigma(v) h(v) \,.$$

For intuition, suppose that σ is a divisible sandpile and h is a discrete harmonic function. The inner product $\langle \sigma, h \rangle$ measures the total "weight"

of the sandpile, when we assign weight $h(v)$ to each unit of mass at v. Exercise 10.2 is a good warmup for the next theorem: it asks you to check that toppling at x leaves the total weight unchanged when h is harmonic at x, and does not increase it when h is superharmonic at x.

Theorem 10.1 (Quadrature Inequalities) *Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$ be a finitely supported configuration, and let σ_∞ be its final configuration. Then σ_∞ is the unique function $\eta : V \to (-\infty, 1]$ such that $\eta - \sigma$ has finite support and*

$$\langle \eta - \sigma, h \rangle \le 0 \qquad \text{for all } h : V \to \mathbb{R} \text{ such that } \Delta h \le 0 \text{ on } \{\eta = 1\}\,. \quad (10.1)$$

Proof. By Lemma 4.4(ii), the odometer u_∞ has finite support. From $\sigma_\infty - \sigma = \Delta u_\infty$ we obtain, by summation by parts (Exercise 10.1),

$$\langle \sigma_\infty - \sigma, h \rangle = \langle \Delta u_\infty, h \rangle = \langle u_\infty, \Delta h \rangle \le 0\,,$$

whenever $\Delta h \le 0$ on $\{\sigma_\infty = 1\}$, since $u_\infty \ge 0$ and $\{u_\infty > 0\} \subseteq \{\sigma_\infty = 1\}$ by Lemma 4.4(i). Thus $\eta = \sigma_\infty$ satisfies (10.1).

For uniqueness, suppose that distinct functions $\eta, \xi : V \to (-\infty, 1]$ satisfy (10.1) and that $\eta - \sigma$ and $\xi - \sigma$ have finite support. Interchanging η and ξ if necessary, we may assume that $\eta(z) > \xi(z)$ for some $z \in V$. Since σ has finite support, so does η, and $A := \{\eta = 1\} \cup \{z\}$ is a nonempty finite proper subset of V.

Let h solve the finite Dirichlet problem

$$\Delta h = -\delta_z \quad \text{on } A \qquad \text{and} \qquad h = 0 \quad \text{outside } A\,.$$

This square linear system has a unique solution, since by Lemma 3.1, applied to a solution and to its negative, the homogeneous system has only the zero solution, and the same lemma applied to $-h$ gives $h \ge 0$. The equation at z gives $\deg(z)h(z) = 1 + \sum_{y \sim z} h(y) \ge 1$, so $h(z) > 0$, and $\Delta h \ge 0$ outside A, because $h \ge 0$ and vanishes there.

Thus h is superharmonic on $\{\eta = 1\}$, and $-h$ is superharmonic on $\{\xi = 1\}$, since $\Delta h \ge 0$ off z and $\xi(z) < \eta(z) \le 1$. Applying (10.1) to these two functions, we obtain

$$\langle \eta - \sigma, h \rangle \le 0 \le \langle \xi - \sigma, h \rangle\,.$$

But h vanishes outside A, and $\eta = 1 \ge \xi$ on $A \setminus \{z\}$, so

$$\langle \eta - \xi, h \rangle \ge (\eta(z) - \xi(z))h(z) > 0\,.$$

This contradiction proves uniqueness. □

10.2 GENERAL TOPPLING PROCEDURES

We now generalize the sequential toppling of Section 4.1. The general formalism includes most of the natural ways to topple, including: parallel updates in discrete time, in which all unstable vertices topple simultaneously; toppling in nested volumes, in which we successively stabilize larger and larger finite regions of G; and toppling in continuous time, in which each vertex has a Poisson clock and topples whenever its clock rings. We will show that they all give the same odometer.

For a locally finite connected graph $G = (V, E)$ and a configuration $\sigma : V \to \mathbb{R}$, recall from Theorem 4.3 the family

$$\mathcal{F}_\sigma = \{f : V \to \mathbb{R} \mid f \geq 0 \text{ and } \sigma + \Delta f \leq 1\},$$

and write, with the usual convention $\inf \varnothing = \infty$,

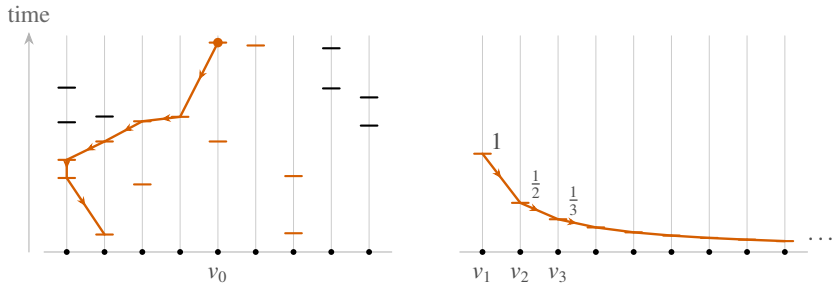
$$u_\infty^\sigma(v) := \inf\{f(v) \mid f \in \mathcal{F}_\sigma\} \qquad \text{for } v \in V. \tag{10.2}$$

(Members of $\mathcal{F}_\sigma$ above the odometer of a point source on $\mathbb{Z}$ are shown in Figure 4.1(b) in Chapter 4.) To motivate the general definition, here are two examples.

Example 10.2 (Toppling in Parallel) At each positive integer time n, all unstable vertices topple all of their excess mass simultaneously. This procedure is useful because it preserves symmetry: if the initial configuration is random and its law is invariant under a group of automorphisms of G (for example, translations in $\mathbb{Z}^d$), then the law of the configuration at time n is also invariant.

Example 10.3 (Poisson Clocks) Each vertex $v \in V$ has an independent rate-1 **Poisson clock**, which rings at times $t_1^v < t_2^v < \cdots$ such that, with $t_0^v = 0$, the increments $t_k^v - t_{k-1}^v$ are i.i.d. exponential random variables of mean 1. Whenever its clock rings, v topples all of its excess mass. In Example 10.13 we show that this procedure is well defined when G has bounded degree.

In a general toppling procedure, the mass that v has sent to each neighbor by time t may increase only by jumps (condition (ii) below). The mass that v emits at time t depends on the mass it has received before t from its neighbors, and we require that there be no "infinite backward chain of topplings" (condition (iii)): following the jumps of neighbors backward in time must end after finitely many steps. Jump times may accumulate from below, as in toppling in nested volumes (Example 10.12), where the times $n - 1/k$ for $k \geq 1$ increase to n, but a path $v_1, v_2, \ldots$ whose vertices jump at the times $1, \frac{1}{2}, \frac{1}{3}, \ldots$ is excluded (see Figure 10.1). By Exercise 10.9, mass leaves each vertex only in jumps. The technical details of the toppling



(a) (b)

Figure 10.1: (a) Rings of independent rate-1 Poisson clocks (ticks) at the vertices of a path, with time increasing upward. The mass emitted at the top ring at v_0 depends only on the initial masses and on the rings reached from it by steps to an earlier ring at the same or an adjacent vertex (vermilion); every such backward chain ends after finitely many steps. (b) Jumps at the times $1, \frac{1}{2}, \frac{1}{3}, \ldots$ at the adjacent vertices $v_1, v_2, v_3, \ldots$ form a backward chain that never ends; condition (iii) of Definition 10.4 excludes it.

procedures are tangential to Sections 10.3–10.5, which use only parallel toppling (Example 10.11 and Lemma 10.15), so the reader may wish to skim them.

Definition 10.4 Let $G = (V, E)$ be a locally finite connected graph. A **toppling procedure** is a Borel function

$$U : [0, \infty) \times V \times \mathbb{R}^V \longrightarrow [0, \infty) .$$

We write its value at (t, v, σ) as $U_t^\sigma(v)$. We require that $U_0^\sigma(v) = 0$ and that the following conditions hold for every $v \in V$ and $\sigma \in \mathbb{R}^V$.

(i) The function $t \mapsto U_t^\sigma(v)$ is right-continuous and nondecreasing.

(ii) If $0 \le s < r$ and $U_r^\sigma(v) > U_s^\sigma(v)$, then the function $t \mapsto U_t^\sigma(v)$ is discontinuous at a time in $(s, r]$.

(iii) There do not exist vertices $(v_i)_{i \ge 1}$ and strictly decreasing times $(t_i)_{i \ge 1}$ such that

- $t \mapsto U_t^\sigma(v_i)$ is discontinuous at t_i;
- $v_{i+1} = v_i$ or $v_{i+1} \sim v_i$.

The interpretation of a toppling procedure is that starting from an initial configuration σ, the total mass emitted by a vertex v to each of its neighbors during the time interval $[0, t]$ is $U_t^\sigma(v)$, so that the resulting configuration at time t is $\sigma_t^U = \sigma + \Delta U_t^\sigma$.

Definition 10.5 Let U be a toppling procedure and let $\sigma : V \to \mathbb{R}$. Its **odometer** is

$$U_\infty^\sigma(v) := \lim_{t\to\infty} U_t^\sigma(v) \qquad \text{for } v \in V\,,$$

where the limit exists in $[0,\infty]$ since $U_t^\sigma(v)$ is nondecreasing in t.

Note that if $U_\infty^\sigma < \infty$ on V, then by local finiteness the limit

$$\sigma_\infty^U := \lim_{t\to\infty} \sigma_t^U = \sigma + \Delta U_\infty^\sigma$$

exists.

Definition 10.6 A toppling procedure U is called **stabilizing** for $\sigma : V \to \mathbb{R}$ if U_∞^σ is finite and $\sigma_\infty^U \le 1$ pointwise.

A basic question arises: for which σ does there exist a stabilizing toppling procedure? For instance, one might expect (correctly) that there is no such procedure for $\sigma \equiv 2$ on $\mathbb{Z}^d$ (Exercise 10.6). We can rephrase this question in terms of the family $\mathcal{F}_\sigma$.

Proposition 10.7 *Let $G = (V, E)$ be a locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Then σ is stabilizable if and only if there exists a stabilizing toppling procedure for σ.*

Proof. By Theorem 4.3(iii), σ is stabilizable if and only if $\mathcal{F}_\sigma$ is nonempty, so it suffices to show that a stabilizing toppling procedure for σ exists if and only if $\mathcal{F}_\sigma \neq \varnothing$. If U is a stabilizing toppling procedure for σ, then $U_\infty^\sigma \in \mathcal{F}_\sigma$. Conversely, every $f \in \mathcal{F}_\sigma$ arises from a stabilizing toppling procedure for σ simply by setting

$$U_t^\eta(v) = f(v)\mathbf{1}_{[1,\infty)}(t) \qquad \text{for } \eta \in \mathbb{R}^V\,,$$

a single jump at time 1. □

Definition 10.8 Let U be a toppling procedure and let $\sigma : V \to \mathbb{R}$. We call U **legal** for σ if, for all $v \in V$ and all $t > 0$,

$$U_t^\sigma(v) - U_{t^-}^\sigma(v) \le \frac{(\sigma_{t^-}^U(v) - 1)_+}{\deg(v)}\,,$$

where $\sigma_t^U = \sigma + \Delta U_t^\sigma$ and $U_{t^-}^\sigma(v)$ denotes the left limit at t.

Thus, in a legal toppling procedure, a vertex with mass at most 1 cannot emit mass, while a vertex with mass greater than 1 must keep at least mass 1 for itself. The next proposition extends the least action principle from sequential toppling to all legal toppling procedures.

Proposition 10.9 (Least Action Principle and Abelian Property) *Let σ be a real-valued configuration on a locally finite connected graph $G = (V, E)$, and let L be a legal toppling procedure for σ. Then the following statements hold.*

(i) For all $f \in \mathcal{F}_\sigma$,

$$L^\sigma_\infty \le f\,.$$

(ii) If L is stabilizing for σ, then $L^\sigma_\infty = u^\sigma_\infty$.

Proof. (i) Let $f \in \mathcal{F}_\sigma$. For $v \in V$ let

$$t_v := \inf\{t \ge 0 : L^\sigma_t(v) > f(v)\}\,,$$

and suppose for a contradiction that $t_v < \infty$ for some $v \in V$. By conditions (ii) and (iii) of Definition 10.4, $L^\sigma(v)$ jumps across $f(v)$ at time t_v, since otherwise by condition (ii) there would be jump times decreasing strictly to t_v, contrary to condition (iii). Let $g := f - L^\sigma_{t_v^-}$. Now at time t_v^-, since L is legal for σ,

$$\sigma^L_{t_v^-}(v) \ge 1 + \deg(v)(L^\sigma_{t_v}(v) - L^\sigma_{t_v^-}(v)) > 1 + \deg(v)g(v)\,.$$

On the other hand,

$$\begin{aligned}\sigma^L_{t_v^-}(v) &= (\sigma + \Delta f)(v) - \Delta g(v)\\ &\le 1 + \deg(v)g(v) - \sum_{w\sim v} g(w)\,.\end{aligned}$$

It follows that $g(w) < 0$ for a neighbor w of v, so $t_w < t_v$, and by the same argument $L^\sigma(w)$ jumps at time t_w. Iterating gives adjacent vertices with strictly decreasing jump times, contrary to condition (iii) of Definition 10.4. Hence $L^\sigma_\infty \le f$.

(ii) This also follows from (i): if L is stabilizing for σ, then $L^\sigma_\infty \in \mathcal{F}_\sigma$, and by part (i) it attains the infimum in (10.2) simultaneously for all $v \in V$. □

In particular, if U is a stabilizing toppling procedure for σ, legal or not, then $U^\sigma_\infty \in \mathcal{F}_\sigma$, so $L^\sigma_\infty \le U^\sigma_\infty$ by part (i). By part (ii), the odometer u^σ_∞ and the final configuration $\sigma_\infty = \sigma + \Delta u^\sigma_\infty$ do not depend on the choice of legal stabilizing toppling procedure.

Increasing the initial mass can only increase the odometer.

Lemma 10.10 *Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma, \eta \in \mathbb{R}^V$ satisfy $\sigma \le \eta$. Then*

$$u^\sigma_\infty(v) \le u^\eta_\infty(v) \qquad \text{for every } v \in V\,.$$

Proof. Since $\sigma \le \eta$, every $f \in \mathcal{F}_\eta$ satisfies $\sigma + \Delta f \le \eta + \Delta f \le 1$, so $\mathcal{F}_\eta \subseteq \mathcal{F}_\sigma$, and taking the infimum in (10.2) over the smaller family gives $u^\sigma_\infty \le u^\eta_\infty$. □

By Theorem 4.3 and Lemma 4.1, u^σ_∞ is also the odometer of every thorough toppling sequence started from σ, so the lemma applies to these odometers as well.

Whenever we need to fix a particular toppling procedure, we will choose one of the following four.

Example 10.11 (Parallel Toppling as a Toppling Procedure) At each integer time $k \ge 1$, all unstable vertices topple all of their excess mass simultaneously, that is,

$$U^\sigma_k(v) - U^\sigma_{k-1}(v) = \frac{(\sigma^U_{k-1}(v) - 1)_+}{\deg(v)} \qquad \text{for } v \in V\,,$$

where $\sigma^U_k = \sigma + \Delta U^\sigma_k$, and $U^\sigma_t = U^\sigma_{\lfloor t \rfloor}$ between integer times. This is a legal toppling procedure. Recalling $\zeta(x) = (\sigma(x) - 1)/\deg(x)$ from Section 4.1, the update rule (4.2), applied simultaneously at all vertices, gives

$$U^\sigma_{k+1}(x) = \max\Big\{U^\sigma_k(x),\ \frac{1}{\deg(x)} \sum_{y \sim x} U^\sigma_k(y) + \zeta(x)\Big\} \qquad \text{for } x \in V\,. \tag{10.3}$$

Two further observations about parallel toppling will be useful later. First, if σ stabilizes, then $\mathcal{F}_\sigma \neq \varnothing$ by Theorem 4.3(iii), and $U^\sigma_\infty \le f$ for every $f \in \mathcal{F}_\sigma$ by Proposition 10.9(i), so U^σ_∞ is finite. Second, whenever U^σ_∞ is finite, U is stabilizing for σ: indeed, if

$$U^\sigma_\infty(v) = \frac{1}{\deg(v)} \sum_{k=0}^{\infty} (\sigma^U_k(v) - 1)_+ < \infty \qquad \text{for } v \in V\,,$$

then $(\sigma^U_k(v) - 1)_+ \to 0$ as $k \to \infty$, and hence $\sigma^U_\infty(v) \le 1$.

At each time step a vertex retains mass at most 1 and receives mass $(\sigma^U_k(y) - 1)_+/\deg(y)$ from each neighbor y, so

$$\sigma^U_{k+1}(v) = \min\{\sigma^U_k(v), 1\} + \sum_{y \sim v} \frac{(\sigma^U_k(y) - 1)_+}{\deg(y)} \ge \min\{\sigma^U_k(v), 1\}\,. \tag{10.4}$$

Hence once v has emitted positive mass, its mass is at least 1 at all later times, and induction gives $\sigma^U_k \ge \min\{\sigma, 1\}$ for every $k \ge 0$.

Example 10.12 (Toppling in Nested Volumes) Let $G = (V, E)$ be infinite, and let $\varnothing \neq V_1 \subset V_2 \subset \cdots$ be finite proper subsets of V with $\bigcup_{n\geq 1} V_n = V$. Between times $n-1$ and n we topple in parallel to stabilize all vertices in V_n: formally, we set

$$U^{\sigma}_{n-\frac{1}{k+1}}(v) - U^{\sigma}_{n-\frac{1}{k}}(v) := \frac{(\sigma^{U}_{n-\frac{1}{k}}(v) - 1)_+ \mathbf{1}_{V_n}(v)}{\deg(v)} \qquad \text{for } n, k \geq 1\,,$$

and $U^{\sigma}_n(v) := \lim_{k\to\infty} U^{\sigma}_{n-1/k}(v)$, and the odometer is constant between these times. This procedure is legal.

To see that V_n is stable at time n, consider toppling a configuration η in parallel inside a nonempty finite proper subset $A \subset V$. Since vertices outside A never topple, replacing η by $\eta\mathbf{1}_A$ leaves the odometers unchanged, and by Lemma 4.4(ii) and Theorem 4.3(ii), $\eta\mathbf{1}_A$ has a finite stabilizing odometer. Each finite sequence of rounds is legal for $\eta\mathbf{1}_A$, so by Proposition 10.9(i) its odometer is at most this stabilizing odometer. Thus the increasing odometer converges, so its increments, and hence the excesses on A, tend to zero, and by local finiteness the configurations converge to a limit that is stable on A. Now apply this with $A = V_n$ and $\eta = \sigma^{U}_{n-1}$.

Example 10.13 (Poisson Clocks as a Toppling Procedure) Suppose every vertex of G has degree at most an integer $\deg_{\max} \geq 1$, and use the independent Poisson clocks of Example 10.3, setting

$$U^{\sigma}_t(v) := \frac{1}{\deg(v)} \sum_{k\geq 1} (\sigma^{U}_{(t^v_k)-}(v) - 1)_+ \mathbf{1}\{t^v_k \leq t\} \qquad \text{for } v \in V \text{ and } t > 0\,.$$

For $v_0 \in V$, $t_0 > 0$ and $N \geq 1$, let $A_N(v_0, t_0)$ be the event that there are clock rings $t_0 > t_1 > \cdots > t_N > 0$ at vertices $v_1, \ldots, v_N$ with $v_j = v_{j-1}$ or $v_j \sim v_{j-1}$ for $1 \leq j \leq N$. A union bound over the possible vertex sequences gives

$$\mathbb{P}\{A_N(v_0, t_0)\} \leq \frac{(\deg_{\max} + 1)^N t_0^N}{N!} \longrightarrow 0\,,$$

and taking a countable union over $v_0 \in V$ and positive integer upper bounds for t_0 shows that condition (iii) in Definition 10.4 holds almost surely. Almost surely no two clocks ring simultaneously. To determine the mass at a fixed ring, we trace backward the rings it depends on; each ring has only finitely many earlier rings at its vertex and its neighbors, so if infinitely many rings were needed, then repeatedly choosing an earlier ring with infinitely many predecessors would give an infinite backward chain. Thus only finitely many rings are needed, and the masses are defined recursively from the

earliest ones. Each clock rings only finitely often in a bounded interval, so conditions (i) and (ii) of Definition 10.4 hold, and each value is obtained from finitely many coordinates of the initial configuration by finitely many sums and positive-part operations, so the resulting map is Borel. Thus the displayed formula defines a toppling procedure almost surely. Exercise 10.10 gives a sufficient condition for graphs of unbounded degree.

Example 10.14 (Toppling in Two Stages) In this procedure we are given a decomposition $\sigma = \eta + \xi$ of the initial configuration into two pieces, where η stabilizes and $\xi \geq 0$. In the first stage we ignore the extra mass ξ and stabilize η by toppling in parallel at times $1 - \frac{1}{k+1}$ for $k \geq 1$, obtaining $\eta + \Delta u^{\eta}_{\infty} + \xi$. The condition $\xi \geq 0$ ensures that all topplings that are legal for η are also legal for σ. Now we topple $\eta + \Delta u^{\eta}_{\infty} + \xi$ in parallel at times $2 - \frac{1}{k+1}$ for $k \geq 1$. If this second stage stabilizes, then by Proposition 10.9(ii) the combined odometer is u^{σ}_{∞}.

If σ stabilizes, then its odometer $u^{\sigma}_{\infty}(v)$ is the total amount of mass sent from v to each of its neighbors, in every legal stabilizing toppling procedure for σ. Next we observe that parallel toppling attains the infimum (10.2), so that the odometer can also be expressed as a pointwise supremum.

Lemma 10.15 *Let $G = (V, E)$ be a locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Then*

(i) either u^{σ}_{∞} is finite on V, or $u^{\sigma}_{\infty} \equiv \infty$;

(ii) parallel toppling has odometer u^{σ}_{∞}.

Proof. (i) If $\mathcal{F}_{\sigma} \neq \emptyset$, then every $f \in \mathcal{F}_{\sigma}$ is finite and bounds u^{σ}_{∞} by (10.2). Otherwise $u^{\sigma}_{\infty} \equiv \infty$, because $\inf \emptyset = \infty$.

(ii) Let P be parallel toppling. By Proposition 10.9(i), $P^{\sigma}_{\infty} \leq u^{\sigma}_{\infty}$. There are two cases. First, if P^{σ}_{∞} is finite on V, then by Example 10.11 it is stabilizing as well as legal, so $P^{\sigma}_{\infty} = u^{\sigma}_{\infty}$ by Proposition 10.9(ii). Second, if $P^{\sigma}_{\infty}(w) = \infty$ for a vertex w, then for every neighbor v of w the parallel update gives

$$P^{\sigma}_{k+1}(v) \geq \frac{\sigma(v) - 1 + \sum_{y \sim v} P^{\sigma}_{k}(y)}{\deg(v)} \geq \frac{P^{\sigma}_{k}(w) + \sigma(v) - 1}{\deg(v)},$$

and the right side tends to ∞ with k, so $P^{\sigma}_{\infty}(v) = \infty$. Since G is connected, it follows that $P^{\sigma}_{\infty} \equiv \infty$, and then $P^{\sigma}_{\infty} \leq u^{\sigma}_{\infty}$ gives $u^{\sigma}_{\infty} \equiv \infty = P^{\sigma}_{\infty}$. □

By Proposition 10.9(i), every procedure U that is legal for σ has $U^{\sigma}_{\infty} \leq u^{\sigma}_{\infty}$, and parallel toppling attains equality by Lemma 10.15(ii). Hence, for every $v \in V$,

$$u^{\sigma}_{\infty}(v) = \sup\{U^{\sigma}_{\infty}(v) : U \text{ is legal for } \sigma\}.$$

The next proposition relates the stabilizability of a particular configuration, one excess mass, to the transience of simple random walk on the underlying graph. Denote by $\mathbb{P}_w$ the law of the discrete time simple random walk $(X_n)_{n\ge 0}$ on G started at $X_0 = w$. Its Green function is

$$g^{::}(w,v) := \frac{1}{\deg(v)} \sum_{n\ge 0} \mathbb{P}_w\{X_n = v\} \qquad \text{for } w, v \in V\,, \tag{10.5}$$

where the sum may be infinite. Summing the first-step identity

$$\mathbb{P}_w\{X_{n+1} = v\} = \frac{1}{\deg(w)} \sum_{y\sim w} \mathbb{P}_y\{X_n = v\}$$

over $n \ge 0$ shows that, when $g^{::}$ is finite, $\Delta_w g^{::}(w,v) = -\delta_v(w)$ for $w, v \in V$. Reversing paths gives $\deg(w)\mathbb{P}_w\{X_n = v\} = \deg(v)\mathbb{P}_v\{X_n = w\}$, so $g^{::}$ is symmetric: $g^{::}(w,v) = g^{::}(v,w)$.

Proposition 10.16 *Let $G = (V,E)$ be a locally finite connected graph, let $o \in V$, let $\beta > 0$, and let $\sigma = 1 + \beta\delta_o$. Then the following statements hold.*

(a) If simple random walk on G is transient, then σ stabilizes and

$$u_\infty^\sigma(v) = \beta\, g^{::}(o,v) \qquad \text{for } v \in V\,.$$

(b) If simple random walk on G is recurrent, then σ does not stabilize.

Proof. We compute the odometer U_k^σ and the configuration σ_k^U at time k of parallel toppling. Setting $q_j(v) := \mathbb{P}_o\{X_j = v\}/\deg(v)$, observe that

$$\begin{aligned}\Delta q_j(v) &= \sum_{w\sim v} \left(\frac{\mathbb{P}_o\{X_j = w\}}{\deg(w)} - \frac{\mathbb{P}_o\{X_j = v\}}{\deg(v)} \right) \\ &= \deg(v)\big(q_{j+1}(v) - q_j(v)\big)\,. \end{aligned} \tag{10.6}$$

Let us show by induction on k that

$$U_k^\sigma(v) = \beta \sum_{j=0}^{k-1} q_j(v) \qquad \text{for } k \ge 0 \text{ and } v \in V\,. \tag{10.7}$$

The case $k = 0$ is the empty sum. If (10.7) holds at time k, then summing (10.6) we obtain

$$\sigma_k^U(v) = \sigma(v) + \Delta U_k^\sigma(v) = 1 + \beta \deg(v) q_k(v)\,,$$

whence $U_{k+1}^\sigma(v) - U_k^\sigma(v) = \beta q_k(v)$, so (10.7) holds at time $k+1$, completing the inductive step (see Figure 10.2). Taking $k \to \infty$ in (10.7) yields $U_\infty^\sigma(v) = \beta g^{::}(o,v)$.

(a) If the walk is transient, then this is finite, so Lemma 10.15(ii) gives $u^{\sigma}_{\infty} = U^{\sigma}_{\infty} = \beta g^{::}(o, \cdot)$. In particular $\mathcal{F}_{\sigma} \neq \varnothing$, and σ stabilizes by Theorem 4.3(iii).

(b) If the walk is recurrent, then $U^{\sigma}_{\infty} \equiv \infty$, so $u^{\sigma}_{\infty} \equiv \infty$ by Lemma 10.15(ii). Then $\mathcal{F}_{\sigma} = \varnothing$, and σ explodes by Theorem 4.3(iii). □

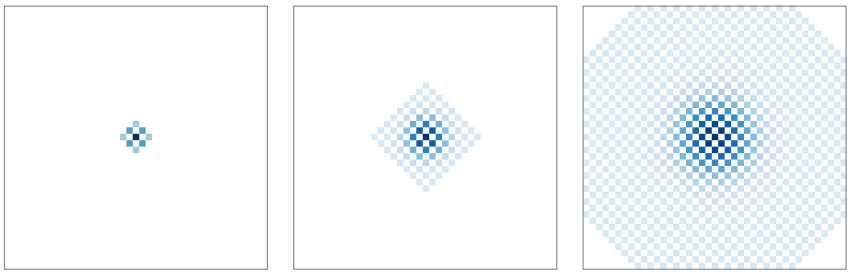


Figure 10.2: The configuration $1 + \beta\delta_o$ on $\mathbb{Z}^2$ after $k = 2$, 8 and 32 rounds of parallel toppling: the excess $\sigma^{U}_{k} - 1$ (shaded, darker for larger, each panel scaled to its maximum) is β times the distribution at time k of simple random walk from o, as in the proof of Proposition 10.16.

Exercise 10.14 asks you to prove, using killed Green functions, that stabilization of $1 + \beta\delta_o$ implies transience, and Exercise 10.12 extends the proposition to finitely many excess masses. In the next section we extend the computation in the proof to an arbitrary initial configuration.

10.3 RANDOM WALK REPRESENTATION

Throughout the section, $G = (V, E)$ is a locally finite connected graph, $\sigma : V \to \mathbb{R}$ is fixed, and $(X_k)_{k \geq 0}$ is simple random walk, with law $\mathbb{P}_x$ and expectation $\mathbb{E}_x$ when $X_0 = x$. All stopping times are for the natural filtration of the walk.

For one excess mass, we saw that the excess after k rounds of parallel toppling is β times the law of simple random walk at time k. For a general initial configuration, the odometer is the value of an optimal stopping problem for the walk, in which the walk gains $(\sigma(x) - 1)/\deg(x)$ at each visit to x. Recalling $\zeta(x) = (\sigma(x) - 1)/\deg(x)$ from Section 4.1, we denote the total gain before time n by

$$\mathsf{B}_n := \sum_{k=0}^{n-1} \zeta(X_k) \qquad \text{for } n \geq 1, \qquad \text{with } \mathsf{B}_0 := 0\,. \tag{10.8}$$

The next lemma shows that the parallel update (10.3) is the Bellman equation for this stopping problem.

Lemma 10.17 *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $(U_n^\sigma)_{n\geq 0}$ be the odometers produced by parallel toppling. Then, for every $n \geq 0$ and $x \in V$,*

$$U_{n+1}^\sigma(x) = \left(\frac{1}{\deg(x)} \sum_{y\sim x} U_n^\sigma(y) + \zeta(x) \right)_+ . \tag{10.9}$$

Proof. For $n \geq 0$ and $x \in V$, let

$$b_n(x) := \frac{1}{\deg(x)} \sum_{y\sim x} U_n^\sigma(y) + \zeta(x) ,$$

so that (10.3) reads $U_{n+1}^\sigma(x) = \max\{U_n^\sigma(x), b_n(x)\}$. Since the increments of parallel toppling are nonnegative, U_n^σ, and hence $b_n(x)$, is nondecreasing in n. We show by induction that $U_{n+1}^\sigma = \max\{0, b_n\}$ for every $n \geq 0$, which is (10.9). For $n = 0$ this holds since $U_0^\sigma = 0$, and if $U_n^\sigma = \max\{0, b_{n-1}\}$ for some $n \geq 1$, then

$$U_{n+1}^\sigma = \max\{0, b_{n-1}, b_n\} = \max\{0, b_n\} .$$

□

Solving this recursion, we obtain a formula for the odometer at time n.

Theorem 10.18 (Random Walk Representation) *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $(U_n^\sigma)_{n\geq 0}$ be the odometers produced by parallel toppling. Then, for every $x \in V$ and $n \geq 0$,*

$$U_n^\sigma(x) = \max_{T\leq n} \mathbb{E}_x[\mathsf{B}_T] ,$$

where the maximum is over stopping times bounded by n.

Proof. Let $w_n(x) := \max_{T\leq n} \mathbb{E}_x[\mathsf{B}_T]$. We prove by induction on n that $w_n = U_n^\sigma$ and that the maximum is attained. The case $n = 0$ is immediate, taking $T = 0$.

Suppose the claim holds at time n. A stopping time $T \leq n+1$ either equals 0 or is at least 1, and in the latter case, given X_1, the remaining stopping time is bounded by n, so by the Markov property

$$\mathbb{E}_x[\mathsf{B}_T] \leq \zeta(x) + \frac{1}{\deg(x)} \sum_{y\sim x} w_n(y) .$$

Allowing also $T = 0$, we obtain

$$w_{n+1}(x) \leq \left(\frac{1}{\deg(x)} \sum_{y\sim x} w_n(y) + \zeta(x) \right)_+ = U_{n+1}^\sigma(x) ,$$

by the inductive hypothesis and Lemma 10.17.

For the reverse inequality, if $U^{\sigma}_{n+1}(x) = 0$, then $T = 0$ attains equality. Otherwise, for each neighbor y of x choose an optimal stopping time $T_y \le n$ for the walk started at y, and consider the rule that takes one step and then, if $X_1 = y$, follows T_y. By the Markov property, its expected payoff is

$$\zeta(x) + \frac{1}{\deg(x)} \sum_{y \sim x} w_n(y) = U^{\sigma}_{n+1}(x)\,,$$

so this rule is optimal, which completes the induction. □

It is optimal to stop the walk as soon as no further gain is possible (Exercise 10.13). Letting $n \to \infty$, we obtain the following.

Corollary 10.19 *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let u^{σ}_{∞} be the divisible-sandpile odometer. Then, for every $x \in V$,*

$$u^{\sigma}_{\infty}(x) = \sup_{T} \mathbb{E}_x[\mathsf{B}_T]\,,$$

where the supremum is over bounded stopping times.

Proof. By Lemma 10.15(ii), $U^{\sigma}_n \uparrow u^{\sigma}_{\infty}$. Since every bounded stopping time is bounded by an integer n, the corollary follows from Theorem 10.18 by taking the supremum over n. □

When the stopping times are bounded by the exit time from a finite set, the supremum can be expressed in terms of killed Green functions.

10.3.1 Killed Green Functions

For a nonempty finite proper subset $A \subsetneq V$, the exit time from A is the hitting time $\tau_{A^c} = \inf\{k \ge 0 : X_k \notin A\}$ of its complement $A^c := V \setminus A$. Generalizing the killed Green function (3.18) on $\mathbb{Z}^d$, write

$$g^{\cdot\cdot}_A(y, v) := \frac{1}{\deg(v)} \sum_{k \ge 0} \mathbb{P}_y\{X_k = v \text{ and } k < \tau_{A^c}\} \qquad \text{for } y, v \in V\,, \tag{10.10}$$

that is, $1/\deg(v)$ times the expected number of visits to v before leaving A. Reversing paths gives

$$\deg(y)\mathbb{P}_y\{X_k = v \text{ and } k < \tau_{A^c}\} = \deg(v)\mathbb{P}_v\{X_k = y \text{ and } k < \tau_{A^c}\}\,,$$

so $g^{\cdot\cdot}_A(y, v) = g^{\cdot\cdot}_A(v, y)$. A first-step decomposition gives $\Delta_y g^{\cdot\cdot}_A(y, v) = -\delta_v(y)$ for $y \in A$ and $v \in V$, and by symmetry

$$\Delta_v g^{\cdot\cdot}_A(y, v) = -\delta_y(v) \qquad \text{for } y \in V \text{ and } v \in A\,, \tag{10.11}$$

where both sides vanish when $y \notin A$. Thus $g^{\cdots}_A(y,\cdot)$ vanishes outside A and is harmonic on $A \setminus \{y\}$.

If $o \in V$ and A is a finite proper subset containing o, then grouping the steps of the walk by the vertex visited gives

$$\mathbb{E}_o[\mathsf{B}_{\tau_{A^c}}] = \sum_{v \in A} g^{\cdots}_A(o,v)(\sigma(v) - 1)\,. \tag{10.12}$$

We first solve the stopping problem up to the exit time from a finite set.

Proposition 10.20 *Let $G = (V,E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $o \in V$, and let $K \subset V$ be a finite proper subset containing o. For $x \in V$, let*

$$w_K(x) := \sup_{T \le \tau_{K^c}} \mathbb{E}_x[\mathsf{B}_T]\,.$$

Then $w_K(x) < \infty$ for every $x \in V$, and

$$w_K(o) = \max\left\{0, \sup_{\substack{A \subseteq K \text{ with } o \in A \\ A \text{ connected}}} \sum_{v \in A} g^{\cdots}_A(o,v)(\sigma(v) - 1)\right\}. \tag{10.13}$$

Proof. Since K is finite and proper, $\mathbb{E}_x \tau_{K^c} < \infty$, and since the gain per step is bounded on K, we have $w_K < \infty$. Taking $T = 0$ gives $w_K \ge 0$, and $w_K = 0$ outside K. For every stopping time $T \le \tau_{K^c}$ and integer $n \ge 0$,

$$|\mathsf{B}_{T \wedge n}| \le \tau_{K^c} \max_{v \in K} \left|\frac{\sigma(v) - 1}{\deg(v)}\right|,$$

so by the dominated convergence theorem $\mathbb{E}_x[\mathsf{B}_{T\wedge n}] \to \mathbb{E}_x[\mathsf{B}_T]$ as $n \to \infty$, and hence

$$w_K(x) = \sup_{\substack{T \le \tau_{K^c} \\ T \text{ bounded}}} \mathbb{E}_x[\mathsf{B}_T] \qquad \text{for } x \in V\,. \tag{10.14}$$

Next, a stopping time $T \le \tau_{K^c}$ for the walk from $x \in K$ is 0 or at least 1, and in the latter case, given $X_1 = y$, the remaining stopping time is admissible from y, with expected payoff at most $w_K(y)$. Hence

$$w_K(x) \le \left(\frac{\sigma(x) - 1}{\deg(x)} + \frac{1}{\deg(x)} \sum_{y \sim x} w_K(y)\right)_+ \qquad \text{for } x \in K\,. \tag{10.15}$$

For every connected $A \subseteq K$ containing o, the stopping time τ_{A^c} is admissible, so by (10.12)

$$\sum_{v \in A} g^{\cdots}_A(o,v)(\sigma(v) - 1) \le w_K(o)\,.$$

Taking the supremum over A and then the maximum with 0, we see that the right side of (10.13) is at most $w_K(o)$, with equality when $w_K(o) = 0$.

For the reverse inequality, suppose $w_K(o) > 0$, and let D be the component of o in $\{w_K > 0\} \subseteq K$. On D, where $w_K > 0$, inequality (10.15) gives $\Delta w_K \geq 1 - \sigma$. Let $g := g^{\cdot\cdot}_D(o, \cdot)$, so that $\Delta g = -\delta_o$ on D by (10.11). Since g is nonnegative and vanishes outside D, while $w_K = 0$ at every vertex outside D adjacent to D, summation by parts gives

$$\begin{aligned}\sum_{v \in D} g(v)(\sigma(v) - 1) &\geq -\sum_{v \in V} g(v)\Delta w_K(v) \\ &= -\sum_{v \in V} w_K(v)\Delta g(v) = w_K(o)\,.\end{aligned}$$

(All sums are finite, and the middle equality follows by interchanging endpoints in the neighbor sum.) Since D is admissible in (10.13), this proves the reverse inequality. □

Letting K increase to V, we obtain the odometer as a supremum of the sums in (10.12).

Theorem 10.21 *Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $o \in V$, and let u^σ_∞ be the odometer of parallel toppling started from σ. Then*

$$u^\sigma_\infty(o) = \max\left\{0, \sup_{\substack{A \subset V \text{ finite with } o \in A \\ A \text{ connected}}} \sum_{v \in A} g^{\cdot\cdot}_A(o, v)(\sigma(v) - 1)\right\}.$$

Proof. For $m \geq 1$, let K_m be the set of vertices at graph distance less than m from o. Since $\tau_{K_m^c} \geq m$ from o, the union over m of the bounded stopping times $T \leq \tau_{K_m^c}$ is the family of all bounded stopping times. By (10.14) and Corollary 10.19, it follows that

$$u^\sigma_\infty(o) = \sup_{m \geq 1} w_{K_m}(o)\,.$$

Since every finite connected set containing o lies in some K_m, the theorem follows from Proposition 10.20. □

See Figure 10.3 for the set A that attains the supremum for a sample of i.i.d. masses on $\mathbb{Z}^2$, with a walk stopped on leaving it.

We now use the random-walk representation to study i.i.d. initial masses.

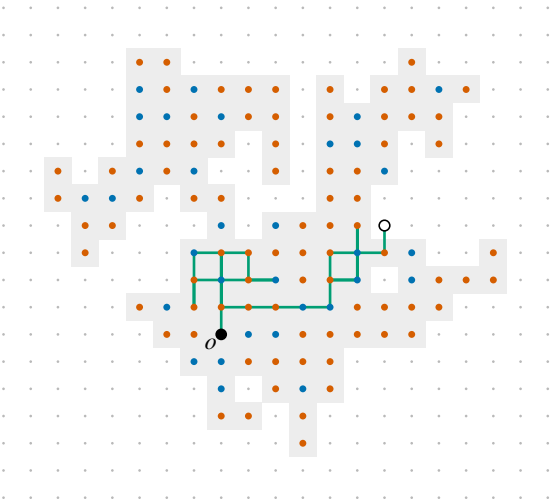


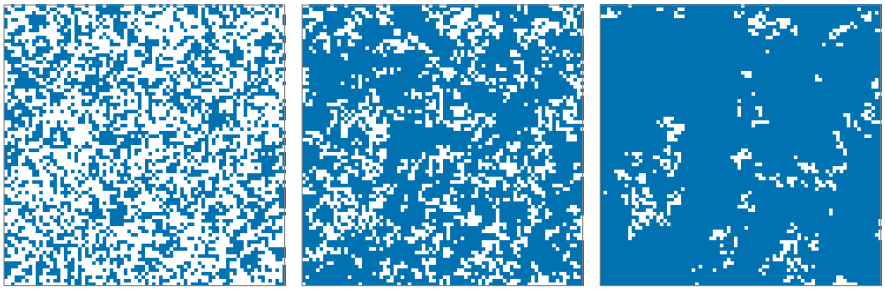
Figure 10.3: A component D (shaded) of the set where the odometer is positive, for i.i.d. masses uniform on $[0, 1.7]$ in $\mathbb{Z}^2$, and a simple random walk from o stopped at its first site outside D (hollow dot). The walk gains $\zeta(X_k)$ at each step, positive at the sites where $\sigma > 1$ (vermilion) and negative where $\sigma < 1$ (blue), and its expected gain is $u^{\sigma}_{\infty}(o)$, so $A = D$ attains the supremum in Theorem 10.21.

10.4 RANDOM SANDPILES ON BOUNDED-DEGREE GRAPHS

Throughout this section, $G = (V, E)$ is infinite and connected, with all degrees at most $\deg_{\max}$, and the initial masses $(\sigma(v))_{v \in V}$ are bounded i.i.d. real random variables with common mean μ. As in Section 10.3, $(X_k)_{k \geq 0}$ is simple random walk, now independent of the masses, with law $\mathbb{P}_x$ and expectation $\mathbb{E}_x$ when $X_0 = x$; recall that $\zeta(v) = (\sigma(v) - 1)/\deg(v)$, that $\mathsf{B}_n = \sum_{k<n} \zeta(X_k)$ is the payoff collected before time n, and that U^{σ}_n is the odometer after n rounds of parallel toppling (Example 10.11). We write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation over the masses. For $n \geq 1$ and $x, v \in V$, let

$$g^{\cdots}_{<n}(x, v) := \frac{1}{\deg(v)} \sum_{k=0}^{n-1} \mathbb{P}_x\{X_k = v\} \quad \text{and} \quad \|g^{\cdots}_{<n}(x, \cdot)\|_2^2 = \sum_{v \in V} g^{\cdots}_{<n}(x, v)^2 \,. \tag{10.16}$$

The random walk representation explains why, for bounded i.i.d. masses on a graph of bounded degree, the mean determines whether the sandpile stabilizes. If the walk stops at a fixed time, then its payoff is a weighted sum of the masses, whose mean has the sign of $\mu - 1$ and, when $\mu \neq 1$, grows linearly in time in absolute value. Above mean 1 this drift gives explosion, and below mean 1 it gives stabilization, because many visits to a single site are unlikely. At mean 1 there is no drift, and explosion comes from the fluctuations of the masses together with balls of small masses.



$\mu = 0.75$ $\mu = 0.90$ $\mu = 0.97$

Figure 10.4: The sites that topple (blue) when i.i.d. masses uniform on $[0, 2\mu]$ are stabilized on the 256×256 torus, shown in a window of 80×80 sites, for $\mu = 0.75$, 0.9 and 0.97. The three samples use the same uniform variables, so by Lemma 10.10 the sets increase with μ; they fill more of the torus as μ approaches the critical mean 1.

Theorem 10.22 (Phase Transition on Bounded-Degree Graphs) *Let $G = (V, E)$ be an infinite connected graph of bounded degree, and let $(\sigma(v))_{v \in V}$ be bounded i.i.d. real random variables with common mean μ. Then the following statements hold.*

(a) If $\mu > 1$, then the sandpile explodes almost surely.

(b) If $\mu = 1$ and $\sigma(v)$ is not almost surely constant, then the sandpile explodes almost surely.

(c) If $\mu < 1$, then the sandpile stabilizes almost surely.

In other words, bounded i.i.d. masses that are not almost surely constant stabilize if and only if their mean is less than 1. (Under the hypotheses of part (c), the proof also shows that $\sup_{x \in V} \mathbb{E}[u_\infty^\sigma(x)] < \infty$.) Boundedness is far from necessary: part (a) extends to every law of mean in $(1, \infty]$, and part (b) to every law of mean 1 with finite positive variance, and to every law of mean 1 that is symmetric about 1 and is not a point mass. For part (c) it is enough that the positive part of $\sigma(v)$ have a finite pth moment for some $p > 3$, but for each $p < 3$ there are graphs of bounded degree on which this is not enough (see the Notes). The sets of sites that topple for three means below 1 are shown in Figure 10.4.

Let

$$Z_n(x) := \mathbb{E}_x[\mathsf{B}_n] = \sum_{v \in V} g^{::}_{<n}(x, v)(\sigma(v) - 1)$$

be the expected payoff of the walk stopped at the fixed time n. Taking the

constant stopping times $T = n$ and $T = 0$ in Theorem 10.18, we obtain

$$U_n^{\sigma}(x) \geq \max\{0, Z_n(x)\}\,. \tag{10.17}$$

The mean of $Z_n(x)$ is $(\mu - 1)\sum_v g^{::}_{<n}(x,v)$. Since the sum lies between $n/\deg_{\max}$ and n, the absolute value of the mean is of order n when $\mu \neq 1$. Its variance is $\mathrm{Var}(\sigma(v))\,\|g^{::}_{<n}(x,\cdot)\|_2^2$, so we need an upper bound on $\|g^{::}_{<n}(x,\cdot)\|_2^2$. The next lemma shows that on every graph of bounded degree it is much smaller than n^2, because the largest value of $g^{::}_{<n}(x,\cdot)$ is at most $2\sqrt{n}$.

Lemma 10.23 *Let $G = (V,E)$ be an infinite connected graph with all degrees at most $\deg_{\max}$, let $x \in V$, let $n \geq 1$ be an integer, and let $M_n(x) := \max_{v\in V} g^{::}_{<n}(x,v)$. Then*

$$M_n(x)^3 \leq 4\|g^{::}_{<n}(x,\cdot)\|_2^2 \leq 4nM_n(x)\,.$$

In particular, $M_n(x) \leq 2\sqrt{n}$, and the expected number of visits to x before time n by the walk started at x is at most $2\deg_{\max}\sqrt{n}$.

Proof. The idea is to compare the maximum of $h := g^{::}_{<n}(x,\cdot)$ with its energy, along a path from a maximizer to a site where h vanishes. Since $\sum_v \mathbb{P}_x\{X_k = v\} = 1$ for each k, we have $\sum_v \deg(v)h(v) = n$. In particular h has finite support, $\sum_v h(v) \leq n$, and $\|h\|_2^2 \leq M_n(x)\sum_v h(v) \leq nM_n(x)$, which is the second inequality.

For the first inequality, we bound the energy of h. Decomposing the visits to v according to the previous step gives $\Delta h(v) = \mathbb{P}_x\{X_n = v\} - \delta_x(v)$ for every $v \in V$, so since h has finite support, summation by parts over the unoriented edges gives

$$\sum_{\{v,w\}\in E} (h(v) - h(w))^2 = -\sum_{v\in V} h(v)\Delta h(v) \leq h(x) \leq M_n(x)\,.$$

Now let $v_0, v_1, \ldots, v_L$ be a simple path from a maximizer v_0 of h to a vertex v_L outside the support of h; it exists because the support is finite and G is infinite and connected. Telescoping h^2 along the path and applying the Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} M_n(x)^2 &= \sum_{i<L} (h(v_i) - h(v_{i+1}))(h(v_i) + h(v_{i+1})) \\ &\leq \sqrt{M_n(x)}\Big(\sum_{i<L} (h(v_i) + h(v_{i+1}))^2\Big)^{1/2} \leq 2\sqrt{M_n(x)\,\|h\|_2^2}\,, \end{aligned}$$

where the last step uses $(\alpha+\beta)^2 \leq 2\alpha^2 + 2\beta^2$ and the fact that each vertex lies on at most two edges of the path. Squaring gives $M_n(x)^3 \leq 4\|h\|_2^2$. The two

inequalities together give $M_n(x)^2 \le 4n$, and the expected number of visits to x is $\deg(x)h(x) \le \deg_{\max} M_n(x) \le 2\deg_{\max}\sqrt{n}$. □

Proof of Theorem 10.22. Choose $C_0 \ge 1$ such that $|\sigma(v) - 1| \le C_0$ and $|\sigma(v) - \mu| \le C_0$ almost surely; then $|\zeta| \le C_0$. Throughout the proof, c and C denote positive constants that depend only on $\deg_{\max}$ and on the law of $\sigma(v)$.

(a) Suppose that $\mu > 1$. Then $Z_n(x)$ has mean of order n and fluctuations of order at most $n^{3/4}$. Indeed, since $\sum_v \deg(v) g^{::}_{<n}(x,v) = n$, independence of the masses and Lemma 10.23 give

$$\mathbb{E}Z_n(x) \ge \frac{\mu - 1}{\deg_{\max}} n \qquad \text{and} \qquad \mathrm{Var}Z_n(x) \le C_0^2 \|g^{::}_{<n}(x,\cdot)\|_2^2 \le 2C_0^2 n^{3/2}.$$

By Chebyshev's inequality, $\mathbb{P}\{Z_n(x) \le L\} \to 0$ as $n \to \infty$, for every L. Since $u^\sigma_\infty(x) \ge U^\sigma_n(x) \ge Z_n(x)$ for every n, by Lemma 10.15(ii) and (10.17), it follows that the odometer is infinite almost surely, so the sandpile explodes almost surely.

(b) Suppose that $\mu = 1$ and $s^2 := \mathrm{Var}(\sigma(v)) > 0$. Now $Z_n(x)$ has mean zero, so explosion must come from the fluctuations.

Step 1: We show that the expected odometers tend to infinity. Let $m_n(x) := \mathbb{E}U^\sigma_n(x)$, which is at most $C_0 n$, and let $a_n := \inf_{x\in V} m_n(x)$. Since the odometers U^σ_n increase in n, so does a_n. Suppose, for a contradiction, that a_n converges to a finite limit a. The idea is that if the masses in a ball around x are small, then the payoff of the first N steps is at most a fixed negative number, whereas the odometer is nonnegative, and this increases the mean odometer by a fixed amount. Since $\sigma(v)$ is not almost surely constant and has mean 1, there is $\epsilon > 0$ with $p_\epsilon := \mathbb{P}\{\sigma(v) \le 1 - \epsilon\} > 0$. Choose an integer N with $t := \epsilon N/\deg_{\max} > a$, and for $x \in V$ let A_x be the event that every mass at graph distance less than N from x is at most $1 - \epsilon$. This ball has at most $N\deg^N_{\max}$ vertices, so $\mathbb{P}(A_x) \ge q := p_\epsilon^{N\deg^N_{\max}} > 0$ for every x.

We use two facts about the event A_x. First, conditioning on A_x lowers the masses in the ball in the stochastic order and leaves the other masses unchanged. By induction on n in (10.9), U^σ_n is nondecreasing in the initial masses, so coupling the conditional and unconditional laws through their quantile functions gives $\mathbb{E}[U^\sigma_n(y) \mid A_x] \le m_n(y)$ for all y and n. Second, the walk started at x stays in the ball up to time $N - 1$ (see Figure 10.5), so on A_x we have $Z_N(x) \le -\epsilon \sum_v g^{::}_{<N}(x,v) \le -t$, and since $\mathbb{E}Z_N(x) = 0$, it follows that $\mathbb{E}[Z_N(x); A_x^c] = -\mathbb{E}[Z_N(x); A_x] \ge t\mathbb{P}(A_x)$.

Let P be the transition operator of simple random walk, given by $\mathsf{P}f(x) := \deg(x)^{-1}\sum_{y\sim x} f(y)$. Dropping the positive part in (10.9) gives $U^\sigma_{k+1} \ge \zeta + \mathsf{P}U^\sigma_k$, and iterating this N times gives $U^\sigma_{n+N} \ge Z_N + \mathsf{P}^N U^\sigma_n$.

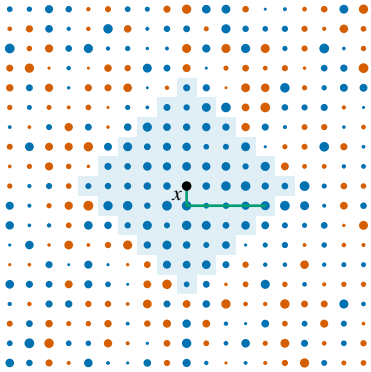


Figure 10.5: Step 1 of the proof of Theorem 10.22(b), with masses above 1 in vermilion and below 1 in blue, larger dots farther from 1: on the event A_x, every mass at graph distance less than N from x (shaded) is at most $1-\epsilon$, so the payoff B_N of the first N steps of the walk from x (green) is at most $-\epsilon N/\deg_{\max}$.

Using this inequality on A_x^c and $U_{n+N}^\sigma \ge 0$ on A_x, we obtain

$$\begin{aligned} m_{n+N}(x) &\ge \mathbb{E}[Z_N(x); A_x^c] + \mathbb{E}[\mathsf{P}^N U_n^\sigma(x); A_x^c] \\ &\ge t\mathbb{P}(A_x) + (1-\mathbb{P}(A_x))\mathsf{P}^N m_n(x) \ge a_n + q(t-a_n)\,, \end{aligned}$$

where the second inequality uses the two facts above, and the third uses $\mathsf{P}^N m_n \ge a_n$ and $t > a \ge a_n$. Taking the infimum over x and letting $n \to \infty$ gives $a \ge a + q(t-a) > a$, a contradiction. Hence $m_n(x) \to \infty$ for every x.

Step 2: We show that the odometer at a fixed vertex o is at least half its mean with probability bounded below. We first compare $m_n(o)$ with $\|g^{::}_{<n}(o,\cdot)\|_2$. The payoff $Z_n(o)$ is a weighted sum of the independent centered variables $\sigma(v)-1$, and when we expand its second and fourth powers, the terms in which an index occurs exactly once have mean zero, so

$$\mathbb{E}Z_n(o)^2 = s^2\|g^{::}_{<n}(o,\cdot)\|_2^2 \qquad \text{and} \qquad \mathbb{E}Z_n(o)^4 \le 3C_0^4\|g^{::}_{<n}(o,\cdot)\|_2^4\,.$$

Since $Z_n(o)$ is centered, (10.17) and the inequality $\mathbb{E}Z^2 \le (\mathbb{E}|Z|)^{2/3}(\mathbb{E}Z^4)^{1/3}$, a case of Hölder's inequality, give

$$m_n(o) \ge \mathbb{E}[Z_n(o)_+] = \frac{1}{2}\mathbb{E}|Z_n(o)| \ge \frac{(\mathbb{E}Z_n(o)^2)^{3/2}}{2(\mathbb{E}Z_n(o)^4)^{1/2}} \ge c\|g^{::}_{<n}(o,\cdot)\|_2\,.$$

Next, we bound the variance of $U_n^\sigma(o)$. Changing $\sigma(v)$ by δ changes $\mathbb{E}_o[\mathsf{B}_T]$ by at most $g^{::}_{<n}(o,v)|\delta|$ for every stopping time $T \le n$, and two possible values of a mass differ by at most $2C_0$, so changing one mass changes $U_n^\sigma(o)$ by at most $2C_0 g^{::}_{<n}(o,v)$. Since $U_n^\sigma(o)$ depends only on the finitely many masses within distance n of o, revealing them one at a time defines a martingale with increments bounded by $2C_0 g^{::}_{<n}(o,v)$, and orthogonality of its increments gives $\mathrm{Var}U_n^\sigma(o) \le 4C_0^2\|g^{::}_{<n}(o,\cdot)\|_2^2 \le Cm_n(o)^2$. Finally, since

$\mathbb{E}[U_n^\sigma(o); U_n^\sigma(o) \ge m_n(o)/2] \ge m_n(o)/2$, the Cauchy–Schwarz inequality gives

$$\mathbb{P}\{U_n^\sigma(o) \ge m_n(o)/2\} \ge \frac{m_n(o)^2}{4\mathbb{E}[U_n^\sigma(o)^2]} \ge c\,.$$

Step 3: We conclude, by the zero-one law, that the odometer is infinite almost surely. Let E be the event that $\limsup_n U_n^\sigma(o)/m_n(o) \ge 1/2$. By Step 2 and Fatou's lemma applied to the indicators of the complementary events, $\mathbb{P}(E) \ge c > 0$. Changing the masses at a finite set F of sites changes $U_n^\sigma(o)$ by at most $2C_0 \sum_{v\in F} g^{::}_{<n}(o,v) \le 2C_0|F|M_n(o)$, where $M_n(o) = \max_v g^{::}_{<n}(o,v)$ as in Lemma 10.23. Since $M_n(o)^3 \le 4\|g^{::}_{<n}(o,\cdot)\|_2^2 \le Cm_n(o)^2$ by that lemma and Step 2, and $m_n(o) \to \infty$ by Step 1, this change of $U_n^\sigma(o)/m_n(o)$ tends to zero, so it does not affect whether E occurs. Hence E is a tail event of the independent masses, and Kolmogorov's zero-one law gives $\mathbb{P}(E) = 1$. On E we have $U_n^\sigma(o) \ge m_n(o)/4$ for infinitely many n, and since $m_n(o) \to \infty$ and $u_\infty^\sigma(o) \ge U_n^\sigma(o)$, the odometer at o is infinite almost surely, so the sandpile explodes almost surely.

(c) Suppose that $\mu < 1$. Now the payoff has drift at most $-(1-\mu)/\deg_{\max}$ per step, and we show that it is positive at time n with probability at most of order $ne^{-cn^{1/4}}$, so the expected value of $\sup_n \mathsf{B}_n$ is finite. A few masses could outweigh the drift only if the walk spent a long time at a few sites, and by Lemma 10.23 this is unlikely.

Step 1: We show that the walk rarely spends more than $n^{3/4}$ steps at one site. Let $\ell_n(v) := \sum_{k<n} \mathbf{1}\{X_k = v\}$ be the number of visits to v before time n. By Lemma 10.23, $\mathbb{E}_v \ell_n(v) \le 2\deg_{\max}\sqrt{n}$, so with $r := \lceil 4\deg_{\max}\sqrt{n}\,\rceil$, Markov's inequality gives $\mathbb{P}_v\{\ell_n(v) \ge r\} \le 1/2$, and the strong Markov property at successive rth visits gives $\mathbb{P}_v\{\ell_n(v) \ge jr\} \le 2^{-j}$ for every integer $j \ge 1$. By the strong Markov property at the hitting time $\tau_v := \inf\{k \ge 0 : X_k = v\}$, it follows that

$$\mathbb{P}_x\{\ell_n(v) \ge jr\} \le \mathbb{P}_x\{\tau_v < n\}\, 2^{-j} \qquad \text{for every integer } j \ge 1\,.$$

Since the expected number of distinct sites visited before time n is at most n, we have

$$\sum_{v\in V} \mathbb{P}_x\{\tau_v < n\} = \mathbb{E}_x\big|\{X_0, \dots, X_{n-1}\}\big| \le n\,,$$

so by a union bound $\mathbb{P}_x\{\max_{v\in V} \ell_n(v) \ge jr\} \le n2^{-j}$ for $j \ge 1$. Taking $j = \lfloor n^{3/4}/r \rfloor$ and using $r \le (4\deg_{\max}+1)\sqrt{n}$, we obtain

$$\mathbb{P}_x\Big\{\max_{v\in V} \ell_n(v) > n^{3/4}\Big\} \le 2n \cdot 2^{-n^{1/4}/(4\deg_{\max}+1)}\,.$$

(If this choice gives $j = 0$, then the right side exceeds one, so the same bound is immediate.)

Step 2: We bound the probability that the payoff is positive at time n. If Y is centered with $|Y| \le C_0$ and λ is real, then convexity of $y \mapsto e^{\lambda y}$ on $[-C_0, C_0]$ gives $\mathbb{E}e^{\lambda Y} \le \cosh(\lambda C_0) \le e^{\lambda^2 C_0^2/2}$. Hence, for independent such variables Y_i, real weights a_i with $\sum_i a_i^2 > 0$, and $b > 0$, the Chernoff bound with $\lambda = b/(C_0^2 \sum_i a_i^2)$ gives

$$\mathbb{P}\Big\{\sum_i a_i Y_i > b\Big\} \le \exp\Big(-\frac{b^2}{2C_0^2 \sum_i a_i^2}\Big).$$

Now condition on the path of the walk, and let $a_v := \ell_n(v)/\deg(v)$. Since $\sum_v \ell_n(v) = n$, we have

$$\mathsf{B}_n = \sum_v a_v(\sigma(v) - 1) \le \sum_v a_v(\sigma(v) - \mu) - \frac{1-\mu}{\deg_{\max}} n\,.$$

If $\max_v \ell_n(v) \le n^{3/4}$, then $\sum_v a_v^2 \le n^{3/4} \sum_v \ell_n(v) = n^{7/4}$, so the inequality above with $Y_v := \sigma(v) - \mu$ shows that the conditional probability of $\{\mathsf{B}_n > 0\}$ given the path is at most $e^{-cn^{1/4}}$. Averaging over the walk and using Step 1, we obtain $\mathbb{E}\mathbb{P}_x\{\mathsf{B}_n > 0\} \le Cne^{-cn^{1/4}}$ for every $x \in V$ and $n \ge 1$.

Step 3: We bound the odometer. Each increment of the payoff is at most C_0 and $\mathsf{B}_0 = 0$, so $\max_{k \le n} \mathsf{B}_k$ increases in n only at times n with $\mathsf{B}_n > 0$, and then by at most C_0. Hence $\sup_n \mathsf{B}_n \le C_0 \sum_{n \ge 1} \mathbf{1}\{\mathsf{B}_n > 0\}$. By Corollary 10.19, $u_\infty^\sigma(x)$ is the supremum of $\mathbb{E}_x[\mathsf{B}_T]$ over bounded stopping times T, and each of these is at most $\mathbb{E}_x[\sup_n \mathsf{B}_n]$. Taking expectations over the masses and using Tonelli's theorem and Step 2, we obtain

$$\sup_{x \in V} \mathbb{E}u_\infty^\sigma(x) \le C_0 \sup_{x \in V} \sum_{n \ge 1} \mathbb{E}\mathbb{P}_x\{\mathsf{B}_n > 0\} < \infty\,.$$

In particular, the odometer is finite almost surely, and the sandpile stabilizes almost surely. □

On vertex-transitive graphs, we show in the next section that the conclusions of parts (a) and (c) hold assuming only that the initial mass is integrable.

10.5 RANDOM SANDPILES ON TRANSITIVE GRAPHS

A graph $G = (V, E)$ is **vertex-transitive** if it has a group Γ of automorphisms such that, for all vertices x and y, there is $\varphi \in \Gamma$ with $\varphi x = y$. We call the law

of σ **Γ-invariant** if

$$(\sigma(\varphi^{-1}v))_{v\in V} \overset{d}{=} (\sigma(v))_{v\in V} \qquad \text{for every } \varphi \in \Gamma\,,$$

and **Γ-ergodic** if every Γ-invariant event has probability 0 or 1. On vertex-transitive graphs, conservation of expected mass determines whether the sandpile stabilizes when the mean is not 1, for every integrable invariant ergodic law of the initial masses. (At mean 1 it gives no conclusion; Theorem 10.22(b) covers bounded i.i.d. masses, and the Notes describe what is known for other laws.)

For an invariant integrable law, parallel toppling preserves the expected mass.

Proposition 10.24 *Let $G = (V, E)$ be a locally finite connected graph, let Γ be a transitive group of automorphisms of G, and let $o \in V$. Let σ be an $\mathbb{R}^V$-valued random configuration whose law $\mathbb{P}$ is Γ-invariant and satisfies $\mathbb{E}|\sigma(o)| < \infty$. Let $(\sigma_k^U)_{k\ge0}$ and $(U_k^\sigma)_{k\ge0}$ be the configurations and odometers under parallel toppling. Then the following statements hold.*

(a) For every $k \ge 0$, the value $\mathbb{E}U_k^\sigma(v)$ does not depend on $v \in V$.

(b) For every $k \ge 0$, we have $\mathbb{E}\sigma_k^U(o) = \mathbb{E}\sigma(o)$.

(c) If $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, then $\mathbb{E}\sigma_\infty(o) = \mathbb{E}\sigma(o)$.

(d) If $\mathbb{E}\sigma(o) = 1$ and $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, then almost surely $\sigma_\infty(v) = 1$ for every $v \in V$ and $\Delta u_\infty^\sigma = 1 - \sigma$.

Proof. Since a transitive graph is regular, the parallel update (10.4) reads

$$\sigma_{k+1}^U(v) = \min\{\sigma_k^U(v), 1\} + \frac{1}{\deg(o)} \sum_{y\sim v} (\sigma_k^U(y) - 1)_+\,.$$

The right side is nondecreasing in each of the values $\sigma_k^U(y)$, so parallel toppling preserves the pointwise order of configurations, and it maps the constant configuration M to itself whenever $M \ge 1$. Parallel toppling also commutes with every graph automorphism, so for each k the law of σ_k^U is Γ-invariant, and by transitivity the law of $\sigma_k^U(v)$ does not depend on v. Since $|\min\{x, 1\}| \le |x| + 1$ and $(x-1)_+ \le |x| + 1$, induction on k shows that every $\sigma_k^U(v)$ is integrable.

(a) Since $U_k^\sigma(v) = \frac{1}{\deg(o)} \sum_{j=0}^{k-1} (\sigma_j^U(v) - 1)_+$ is integrable and its law does not depend on v, neither does its expectation.

(b) Each of the $\deg(o)$ neighbors y of o has $\mathbb{E}(\sigma_k^U(y)-1)_+ = \mathbb{E}(\sigma_k^U(o)-1)_+$, so taking expectations in the update at $v = o$ gives

$$\mathbb{E}\sigma_{k+1}^U(o) = \mathbb{E}\min\{\sigma_k^U(o), 1\} + \mathbb{E}(\sigma_k^U(o) - 1)_+ = \mathbb{E}\sigma_k^U(o)\,,$$

and induction on k proves $\mathbb{E}\sigma_k^U(o) = \mathbb{E}\sigma(o)$.

(c) Suppose $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, so that $\sigma_k^U(o) \to \sigma_\infty(o) \le 1$ almost surely. The idea is to compare σ with $\min\{\sigma, M\}$, whose configurations under parallel toppling stay below M. Let $M \ge 1$, and let $(\eta_k^U)_{k\ge 0}$ be the configurations under parallel toppling started from $\eta := \min\{\sigma, M\}$. Since $\eta \le \sigma$ and $\eta \le M$, preservation of order gives $\eta_k^U \le \sigma_k^U$ and $\eta_k^U \le M$, that is, $\eta_k^U \le \min\{\sigma_k^U, M\}$. The law of η is Γ-invariant and $\mathbb{E}|\eta(o)| < \infty$, so part (b), applied to η and to σ, gives

$$\mathbb{E}\min\{\sigma(o), M\} = \mathbb{E}\eta_k^U(o) \le \mathbb{E}\min\{\sigma_k^U(o), M\} \le \mathbb{E}\sigma_k^U(o) = \mathbb{E}\sigma(o)\,.$$

By (10.4) and Example 10.11, $\sigma_k^U \ge \min\{\sigma, 1\}$, so $\min\{\sigma_k^U(o), M\}$ lies between the integrable variable $\min\{\sigma(o), 1\}$ and M, and it tends to $\sigma_\infty(o)$ because $\sigma_\infty(o) \le 1 \le M$. By the dominated convergence theorem, letting $k \to \infty$ gives $\mathbb{E}\min\{\sigma(o), M\} \le \mathbb{E}\sigma_\infty(o) \le \mathbb{E}\sigma(o)$. Since $|\min\{\sigma(o), M\}| \le |\sigma(o)|$, the dominated convergence theorem again shows that the left side tends to $\mathbb{E}\sigma(o)$ as $M \to \infty$, so $\mathbb{E}\sigma_\infty(o) = \mathbb{E}\sigma(o)$.

(d) By part (c), $\mathbb{E}\sigma_\infty(o) = 1$. Since $\sigma_\infty(o) \le 1$, it follows that $\sigma_\infty(o) = 1$ almost surely. By invariance the same holds at every vertex, and since V is countable, $\sigma_\infty \equiv 1$ almost surely, and hence $\Delta u_\infty^\sigma = 1 - \sigma$. □

Conservation of expected mass and ergodicity now give the phase transition away from mean 1.

Proposition 10.25 *Let $G = (V, E)$ be a locally finite connected graph, let Γ be a transitive group of automorphisms of G, and let $o \in V$. Let σ be an $\mathbb{R}^V$-valued random configuration whose law $\mathbb{P}$ is Γ-invariant and Γ-ergodic and satisfies $\mathbb{E}|\sigma(o)| < \infty$. Then the following statements hold.*

(a) The probability that σ stabilizes is 0 or 1.

(b) If $\mathbb{E}\sigma(o) > 1$, then $\mathbb{P}\{\sigma \text{ stabilizes}\} = 0$.

(c) If $\mathbb{E}\sigma(o) < 1$, then $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$.

Proof. (a) The event that σ stabilizes is Γ-invariant, so it has probability 0 or 1 by ergodicity.

(b) If σ stabilized almost surely, then by Proposition 10.24(c) and $\sigma_\infty(o) \le 1$ we would have $\mathbb{E}\sigma(o) = \mathbb{E}\sigma_\infty(o) \le 1$. Hence if $\mathbb{E}\sigma(o) > 1$, then the probability of stabilization is not 1, so it is 0 by part (a).

(c) By part (a), it suffices to show that if $\mathbb{P}\{\sigma \text{ stabilizes}\} = 0$, then $\mathbb{E}\sigma(o) \ge 1$. In that case, by Theorem 4.3(iii) and Lemma 10.15, $U_n^\sigma(o) \uparrow \infty$ almost surely for parallel toppling, so $U_n^\sigma(o) > 0$ for all sufficiently large n. Since once o has emitted mass its mass remains at least 1, by (10.4), we have

$$\liminf_{n\to\infty} \sigma_n^U(o) \ge 1 \qquad \text{almost surely}\,.$$

Since $\sigma_n^U(o) \geq \sigma(o) \wedge 1$ by the same inequality, Fatou's lemma applied to $\sigma_n^U(o) - (\sigma(o) \wedge 1)$ and Proposition 10.24(b) give

$$1 \leq \mathbb{E}\Big[\liminf_{n\to\infty} \sigma_n^U(o)\Big] \leq \mathbb{E}\sigma(o)\,,$$

which proves part (c). □

Remark. Parts (b) and (c) extend by truncation to laws integrable on one side only. Suppose the law of σ is Γ-invariant and Γ-ergodic. If $\mathbb{E}[\sigma(o)_-] < \infty$ and $\mathbb{E}\sigma(o) \in (1, \infty]$, then choose M so large that $\mathbb{E}[\sigma(o) \wedge M] > 1$; by part (b), $\sigma \wedge M$ explodes almost surely, and since $\sigma \wedge M \leq \sigma$, so does σ, by Lemma 10.10. Likewise, if $\mathbb{E}[\sigma(o)_+] < \infty$ and $\mathbb{E}\sigma(o) \in [-\infty, 1)$, then choose M so large that $\mathbb{E}[\sigma(o) \vee (-M)] < 1$; by part (c), $\sigma \vee (-M)$ stabilizes almost surely, and since $\sigma \leq \sigma \vee (-M)$, so does σ. (The truncations are integrable, their means tend to $\mathbb{E}\sigma(o)$ by monotone convergence, and truncating coordinatewise preserves invariance and ergodicity.) Without transitivity, Proposition 10.24 can fail (Exercise 10.19).

EXERCISES

Exercise 10.1. Verify the summation identities used in the proof of Theorem 10.1. Let $G = (V, E)$ be a locally finite graph.

(a) Suppose $f : V \to \mathbb{R}$ has finite support. Show that

$$\sum_{v\in V} \Delta f(v) = 0\,.$$

(b) Let $f, h : V \to \mathbb{R}$, and suppose at least one of them has finite support. Show that both sums below converge absolutely and that

$$\sum_{v\in V} (\Delta f)(v)h(v) = \sum_{v\in V} f(v)(\Delta h)(v)\,.$$

Hint: Pair the two orientations of each edge.

Exercise 10.2. Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma : V \to \mathbb{R}$ be a divisible sandpile with finite support. Recall $\langle \sigma, h\rangle = \sum_{v\in V} \sigma(v)h(v)$ from Section 10.1.

(a) Show that if σ_1 is obtained from σ by performing a toppling at x, then

$$\langle \sigma_1, h\rangle \leq \langle \sigma, h\rangle$$

for all $h : V \to \mathbb{R}$ such that $\Delta h(x) \leq 0$.

(b) Deduce that if σ stabilizes to σ_∞ with odometer u_∞, then

$$\langle \sigma_\infty, h \rangle \le \langle \sigma, h \rangle$$

for all $h : V \to \mathbb{R}$ such that $\Delta h \le 0$ on $\{u_\infty > 0\}$.

(Compare Theorem 10.1. This extends Exercise 4.2 from $\mathbb{Z}^d$ to graphs.)

Exercise 10.3. Let $G = (V, E)$ be a finite connected graph with at least two vertices, and let $\sigma : V \to \mathbb{R}$. Use the least action principle, Theorem 4.3, to show that σ is stabilizable if and only if

$$\sum_{v \in V} \sigma(v) \le |V| \,.$$

Hint: For the converse, solve $\Delta f = 1 - \sigma - c$, where $c := 1 - |V|^{-1} \sum_v \sigma(v)$, and add a constant to make $f \ge 0$.

Exercise 10.4. Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma, \theta : V \to \mathbb{R}$. Generalize the divisible sandpile of Definition 4.2 so that each vertex v has **threshold** $\theta(v)$ and distributes all mass in excess of $\theta(v)$ equally among its neighbors. The usual divisible sandpile is the case $\theta \equiv 1$. Show that σ stabilizes for the generalized divisible sandpile if and only if $\sigma - \theta + 1$ stabilizes for the usual divisible sandpile.

Exercise 10.5. Consider the case $u_\infty \equiv \infty$ of Lemma 4.1. Let $G = (V, E)$ be a locally finite connected graph.

(a) Show that if $\sigma : V \to \mathbb{R}$ does not stabilize, but its configurations σ_k of (4.1) nevertheless converge pointwise along some thorough toppling sequence, then the limit is identically 1.

(b) Show that there is a toppling sequence as in part (a) when $G = \mathbb{Z}$ and

$$\sigma(v) = \begin{cases} 1, & v \ne \mathbf{0}, \\ 2, & v = \mathbf{0} \,. \end{cases}$$

Hint: Use the bipartition of $\mathbb{Z}$ to simulate parallel toppling by a thorough toppling sequence.

(c) Describe a thorough toppling sequence for the initial configuration σ in part (b) such that $\max_{v \in \mathbb{Z}}(\sigma_k(v) - 1) = O(k^{-\alpha})$ for a constant $\alpha > 0$.

(d) For the same initial configuration, find a thorough toppling sequence for which σ_k does not converge.

Exercise 10.6. Use the least action principle, Theorem 4.3, to study the constant configuration 2 on infinite graphs.

(a) Let G be an infinite connected graph of bounded degree. Show that the initial configuration $\sigma \equiv 2$ does not stabilize on G.

(b) Show that the bounded-degree hypothesis cannot be removed by exhibiting an infinite locally finite tree on which $\sigma \equiv 2$ stabilizes.

(c) Let G be an infinite locally finite connected graph. Show that $\sigma \equiv 2$ does not stabilize whenever, along every infinite path in G, the sum of $\deg(v)^{-1}$ diverges.

Exercise 10.7. Let $G = (V, E)$ be a locally finite connected graph. Use Theorem 4.3 to prove that if $\sigma : V \to \mathbb{R}$ stabilizes with odometer u_∞^σ, then $\inf_{x \in V} u_\infty^\sigma(x) = 0$. Is the infimum necessarily attained?

Exercise 10.8. Let $G = (V, E)$ be a locally finite connected graph, let $\sigma_0 : V \to \mathbb{R}$, and let $(v_i)_{i \geq 1}$ be a toppling sequence as in (4.1). Suppose the set V_* of vertices that occur infinitely often is finite and is not all of V.

(a) Show that the resulting odometer u_∞ is finite on V. Prove the same conclusion when each round topples a set of vertices in parallel and V_* is the set of vertices toppled in infinitely many rounds.

(b) Suppose only vertices in V_* are toppled. Show that u_∞ does not depend on their order and agrees with the odometer from parallel toppling on V_*.

Exercise 10.9. Let U be a toppling procedure on a locally finite connected graph $G = (V, E)$. Show that Definition 10.4 implies that $t \mapsto U_t^\sigma(v)$ is a pure-jump function for every $\sigma \in \mathbb{R}^V$ and $v \in V$. Thus there are a countable set $J \subset (0, \infty)$ of jump times and positive jump sizes $(a_s)_{s \in J}$ such that

$$U_t^\sigma(v) = \sum_{s \in J} a_s \mathbf{1}\{s \leq t\} \qquad \text{for } t > 0\,.$$

Exercise 10.10. We relax the bounded-degree assumption in Example 10.13. Let $G = (V, E)$ be a locally finite connected graph, and equip every vertex with an independent rate-1 Poisson clock. Suppose that there are a vertex $o \in V$ and a constant $C_o < \infty$ such that, for every $n \geq 1$, at most $C_o^n n!$ sequences $o = v_0, v_1, \ldots, v_n$ satisfy $v_j = v_{j-1}$ or $v_j \sim v_{j-1}$ for $1 \leq j \leq n$. Verify Definition 10.4(iii) for toppling with these clocks.

Hint: Prepend a fixed path from o to bound the number of such sequences from another vertex; then the argument of Example 10.13 works on every time interval of a fixed length less than $1/C_o$, and an infinite decreasing sequence of ring times has a limit, so one of its tails lies in such an interval.

Exercise 10.11. Let $G = (V, E)$ be a directed graph in which every vertex x has finite in-degree and finite positive out-degree $\deg(x)$. Extend the definition of the Laplacian to G, so that the divisible sandpile and the least action principle, Proposition 10.9, still hold.

Hint: Use

$$\Delta f(x) := \sum_{\substack{y \in V \\ (y,x) \in E}} f(y) - \deg(x)\, f(x)\,,$$

and let a toppling send equal mass along each outgoing edge. In Definition 10.4(iii), follow incoming edges backward.

Exercise 10.12. Extend Proposition 10.16 from one excess mass to finitely many excess masses. Let $G = (V, E)$ be an infinite locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Suppose $\sigma \geq 1$ and $\sigma - 1$ has finite nonempty support. Show that σ stabilizes if and only if simple random walk on G is transient.

Exercise 10.13. Identify an optimal stopping time in Theorem 10.18. Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $x \in V$, and let $n \geq 0$ be an integer. Let $(X_k)_{k \geq 0}$ be simple random walk started at x. Let U_n^σ be the odometer after n rounds of parallel toppling. For $0 \leq k \leq n$, let

$$\mathsf{B}_k := \sum_{j=0}^{k-1} \frac{\sigma(X_j) - 1}{\deg(X_j)}\,.$$

Show that $\max_{T \leq n} \mathbb{E}_x[\mathsf{B}_T]$ is attained by

$$T_n^* := \min\{0 \leq k \leq n : U_{n-k}^\sigma(X_k) = 0\}\,.$$

Here the maximum is over stopping times for the natural filtration of the walk that are bounded by n.

Exercise 10.14. Use killed Green functions to prove that stabilization implies transience in Proposition 10.16. Let $G = (V, E)$ be an infinite locally finite connected graph, let $o \in V$ and $\beta > 0$, and suppose that $\sigma = 1 + \beta\delta_o$ stabilizes. Prove that simple random walk on G is transient.

Hint: Compare $\beta^{-1} u_\infty^\sigma$ with the Green function killed on leaving a finite ball about o in the graph metric.

Exercise 10.15. This exercise extends the killed Green functions of Section 10.3 to a walk stopped when it hits a single vertex. Let $G = (V, E)$ be a locally finite connected recurrent graph, and let $y, z \in V$ be distinct. Let $(X_j)_{j \geq 0}$ be simple random walk, and write $\mathbb{P}_x$ for its law when $X_0 = x$. For $v \in \{y, z\}$, write τ_v for the hitting time of v and τ_v^+ for its first positive hitting

time. Let

$$f(x) := \frac{\mathbb{P}_x\{\tau_y < \tau_z\}}{\deg(y)\mathbb{P}_y\{\tau_z < \tau_y^+\}}\,.$$

(a) Show that

$$f(x) = \frac{1}{\deg(y)} \sum_{j=0}^{\infty} \mathbb{P}_x\{X_j = y \text{ and } \tau_z > j\}\,.$$

Thus $\deg(y)f(x)$ is the expected number of visits to y before hitting z.

(b) Show that $\Delta f = \delta_z - \delta_y$.

Hint for (b): Use a first-step decomposition and reverse paths between successive visits to $\{y, z\}$.

Exercise 10.16. Prove the abelian-sandpile analogue of Proposition 10.24(b). Let $d \geq 1$, and let $(\sigma_n)_{n\geq 0}$ be the configurations produced by parallel toppling (9.6) for the abelian sandpile on $\mathbb{Z}^d$. Suppose that the law of σ_0 is translation invariant and $\mathbb{E}|\sigma_0(\mathbf{0})| < \infty$. Show that

$$\mathbb{E}\sigma_n(\mathbf{0}) = \mathbb{E}\sigma_0(\mathbf{0}) \qquad \text{for } n \geq 0\,.$$

Exercise 10.17. The proof of Proposition 10.24(c) uses dominated convergence; this exercise concerns a related fact about uniform integrability. Let $(Z_n)_{n\geq 1}$ be identically distributed random variables with $\mathbb{E}|Z_1| < \infty$. Show that $(Z_n)_{n\geq 1}$ is uniformly integrable. Deduce that for every $\epsilon > 0$ there is $\delta > 0$ such that $\sup_n \mathbb{E}[|Z_n|\mathbf{1}_A] < \epsilon$ whenever $\mathbb{P}\{A\} < \delta$.

Exercise 10.18. Verify the form of Fatou's lemma used in the proof of Proposition 10.25(c). Let $(Z_n)_{n\geq 1}$ be random variables, and let Y be integrable. Show that if $Z_n \geq Y$ for every n, then

$$\mathbb{E}\left[\liminf_{n\to\infty} Z_n\right] \leq \liminf_{n\to\infty} \mathbb{E}Z_n\,.$$

Exercise 10.19. The **comb** C has vertex set $\mathbb{Z}^2$. For $x, y \in \mathbb{Z}$, its edges are

$$(x, 0) \sim (x \pm 1, 0) \qquad \text{and} \qquad (x, y) \sim (x, y \pm 1)\,.$$

The **spine** vertices $(x, 0)$ have degree 4, while the **tooth** vertices (x, y) with $y \neq 0$ have degree 2. This exercise shows that Proposition 10.24 can fail without transitivity.

Let $(\sigma(v))_{v\in\mathbb{Z}^2}$ be i.i.d. real-valued random variables with finite mean $\mathbb{E}\sigma(v) = \mu$ and

$$q := \mathbb{E}[(\sigma(v) - 1)_+] > 0\,.$$

Apply one round of parallel toppling to obtain $\sigma_1^U = \sigma + \Delta U_1^\sigma$, where $U_1^\sigma(v) = (\sigma(v) - 1)_+ / \deg(v)$.

(a) Show that, for every $x \in \mathbb{Z}$,

$$\mathbb{E}\sigma_1^U(x,0) = \mu + \frac{q}{2} \qquad \text{and} \qquad \mathbb{E}\sigma_1^U(x,\pm 1) = \mu - \frac{q}{4},$$

and $\mathbb{E}\sigma_1^U(x,y) = \mu$ for every $x \in \mathbb{Z}$ and $|y| \geq 2$.

(b) Deduce that the identity $\mathbb{E}\sigma_1^U(v) = \mathbb{E}\sigma(v)$ does not hold for every $v \in \mathbb{Z}^2$, and neither does $\mathbb{E}[\sigma_1^U(v)/\deg(v)] = \mathbb{E}[\sigma(v)/\deg(v)]$ for every $v \in \mathbb{Z}^2$.

NOTES

The divisible sandpile was introduced by Levine and Peres (2009), who proved that a point source fills a ball up to bounded error (Levine and Peres, 2009, Theorem 1.3); the Notes to Chapter 4 give the sources of the least action principle and of the description in Section 4.1 of the odometer as a least superharmonic majorant minus an obstacle. The multiple-source scaling limit is Theorem 1.4 of Levine and Peres (2010); its planar case with two equal sources is Theorem 7.1, and the limit is a quadrature domain (Levine and Peres, 2010, Proposition 6.6); Theorems 1.2–1.4 there also cover rotor-router aggregation and internal DLA, and Theorem 3.9 allows real-valued densities for the divisible sandpile. For an overview, see (Levine and Peres, 2017, Sections 2 and 5).

Proposition 10.9 extends Proposition 2.5 of Levine, Murugan, Peres, and Ugurcan (2016a), which assumes well-ordered toppling times. In the more general toppling procedures considered by Fey, Meester, and Redig (2009, Definition 2.1) for abelian sandpiles in infinite volume, from which Definition 10.4 is adapted, this assumption becomes a "no infinite backward chain" condition, and our proof of part (i) adapts their argument for the abelian property (Fey et al., 2009, Theorem 2.8). The supremum formula after Lemma 10.15 and Example 10.14 are Lemma 2.7 and Example 3 of Levine, Murugan, Peres, and Ugurcan (2016a). Fey, Meester, and Redig (2009, Lemma 2.10) proved conservation of density for translation-invariant abelian sandpiles on $\mathbb{Z}^d$ by an ergodic argument, and Levine, Murugan, Peres, and Ugurcan (2016a, Proposition 3.1 and Lemma 4.3) gave an elementary proof of Proposition 10.24. Lemma 10.17 is (Bou-Rabee et al., 2026, Lemma 3.1); in (Bou-Rabee et al., 2026, Section 1.2), parallel toppling is identified as a projected Jacobi iteration for the discrete obstacle problem.

The random-walk representation in Theorem 10.18, Proposition 10.20, the variational formula in Theorem 10.21 and Corollary 10.19 are Theorem 3.2,

Proposition 3.3, Theorem 3.4 and Corollary 3.5 of Bou-Rabee, Peres, and Sava-Huss (2026, pp. 11–12). Theorem 10.22 is the case of bounded masses of Theorems 1.2 and 1.3(i) there (Bou-Rabee et al., 2026, pp. 3–4). Part (a) is their Proposition 5.2, which needs only a mean in $(1, \infty]$. For part (b) they consider infinite and finite $\sum_v g^{::}(o, v)^2$ separately (Bou-Rabee et al., 2026, Propositions 6.4(b) and 6.6); using Lemma 10.23 and Kolmogorov's zero-one law, we avoid this split and their general zero-one law (Bou-Rabee et al., 2026, Proposition 4.1). They reduce the symmetric case to bounded laws by a convexity argument (Bou-Rabee et al., 2026, Proposition 6.7).

Boundedness of the masses is far from necessary in Theorem 10.22(c): Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 1.3(i), pp. 3–4) proved that if the mean lies in $[-\infty, 1)$ and the positive part of $\sigma(v)$ has a finite pth moment for some $p > 3$, then the qth moment of the odometer is bounded uniformly in the vertex for every $1 \le q < (p-1)/2$. Their proof (Bou-Rabee et al., 2026, Proposition 7.6, p. 24) uses the Fuk–Nagaev inequality (Bou-Rabee et al., 2026, Lemma 7.3, p. 23), of which inequality (1.7) of Rio (2017) gives a form attributed there to Fuk and to Nagaev, and moment bounds for local times (Bou-Rabee et al., 2026, Lemma 7.2, p. 22) from the on-diagonal bound $\sup_x \mathbb{P}_x\{X_n = x\} \le Cn^{-1/2}$ (Bou-Rabee et al., 2026, p. 4); summed over n, it gives the bound on visits in Lemma 10.23. The exponent 3 cannot be lowered: for every $p < 3$, Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 8.9, pp. 33–34) constructed a transient bounded-degree graph on which, for every $\mu < 1$, i.i.d. masses bounded below, with mean μ and a finite centered pth moment, explode almost surely, and Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 8.12, pp. 34–38) constructed, for every $d_f \in (\max\{p, 2\}, 3)$, a recurrent bounded-degree graph whose balls of radius r have at most a constant times r^{d_f} vertices, on which such masses explode, whereas on such graphs finite pth moments of the positive parts with $p > d_f$ suffice for stabilization below mean 1 (Bou-Rabee et al., 2026, Theorem 1.3(ii), p. 4). They left open whether i.i.d. masses of mean below 1 stabilize on every bounded-degree graph for $p = 3$ (Bou-Rabee et al., 2026, Remark 8.11, p. 34).

Proposition 10.16 appeared in (Levine et al., 2016a, Lemma 2.9). For integrable initial masses on an infinite vertex-transitive graph, with law invariant and ergodic under a transitive group of automorphisms, Levine, Murugan, Peres, and Ugurcan (2016a) proved explosion above mean 1 and stabilization below mean 1 (Levine et al., 2016a, Lemmas 4.1 and 4.2), parts (b) and (c) of Proposition 10.25. On such graphs, for i.i.d. masses of mean 1, they proved explosion when the variance is positive and finite (Levine et al., 2016a, Theorem 1.1) and, on recurrent graphs, for every law other than a point mass (Levine et al., 2016a, Lemma 4.4); after that lemma they extend this to invariant ergodic laws whose single-site conditional laws are not point

masses. Some such condition is needed even for bounded masses with a finite range of dependence: if $(\eta(x))_{x\in\mathbb{Z}^d}$ are i.i.d. Bernoulli$(1/2)$ variables, then $\sigma := 1 - \Delta(\eta/(2d))$ is such a stationary ergodic configuration, with values in $[0, 2]$, mean 1 and positive variance, yet σ stabilizes. This is a variant of (Bou-Rabee et al., 2026, Example 8.6, p. 30), where η is uniform on $[0, 1]$ and $\sigma = 1 - \Delta\eta$. On a finite connected graph, Levine, Murugan, Peres, and Ugurcan (2016a) computed the exact law of the odometer of the configuration

$$\sigma(v) = 1 + \xi(v) - \frac{1}{|V|}\sum_{w\in V}\xi(w)\,,$$

where the $\xi(v)$ are i.i.d. standard Gaussians. It stabilizes to the constant 1, and its odometer is a discrete bi-Laplacian Gaussian field shifted to have minimum 0 (Levine et al., 2016a, Proposition 1.3).

The preprint of Bou-Rabee and Panagiotis (2026) determined how fast the sandpile on $\mathbb{Z}^d$ explodes at mean 1. For i.i.d. masses of mean 1 with positive finite variance and an exponential moment, the expected mass sent from the origin to each neighbor during the first t rounds of parallel toppling is of order $t^{(4-d)/4}$ for $d \le 3$ and $\log t$ for $d = 4$, and at least a constant times $(\log t)^{2/d}$ for $d \ge 5$. For $d \ge 5$ the order depends on the lower tail of the masses: if $-\log \mathbb{P}\{\sigma(\mathbf{0}) \le 1 - s\}$ is of order s^{ν} as $s \to \infty$ for some $\nu \ge 1$ other than $d/2$, then the expected mass is of order $(\log t)^{1/\min\{\nu, d/2\}}$ (Bou-Rabee and Panagiotis, 2026, Theorem 1.3, pp. 5–6). The same paper showed that, for $d \ge 2$, the set of vertices that topple can contain an infinite connected component below mean 1: if the masses are i.i.d. with a law that depends on their mean and, uniformly over the means in an interval with right endpoint 1, the variance is at least a positive constant and a fixed exponential moment of the deviation from the mean stays bounded, then at every mean sufficiently close to 1 from below the set of vertices that topple contains an infinite connected component almost surely (Bou-Rabee and Panagiotis, 2026, Theorem 1.1, p. 3). This answers, for the divisible sandpile, a variant of a question that Fey, Meester, and Redig (2009, Section 5) raised for the abelian sandpile.

The variance bound in Step 2 of part (b) of the proof of Theorem 10.22 is a bounded-differences form of the Efron–Stein inequality. This is inequality (2.1), p. 754, of Steele (1986), for arbitrary functions of i.i.d. random variables; Efron and Stein (1981, inequality (1.6), p. 587) had proved a form of it for symmetric functions. Levin and Peres (2017, Exercise 13.12, p. 199) derive it for independent variables with values in a finite set from the spectral gap of the chain that resamples one coordinate at a time.

Aharonov and Shapiro (1976, Sections 1.1–1.2, pp. 39–40) studied do-

mains on which the integrals of analytic functions satisfy quadrature identities, and Sakai (1982, Chapters I–II) proved the existence and uniqueness of quadrature domains for subharmonic functions, for which the identity becomes an inequality as in Theorem 10.1; Gustafsson and Shapiro (2005, Sections 1, 2 and 5) surveyed both notions. Aleksanyan and Shahgholian (2019) studied a perturbed divisible sandpile started from mass n at the origin of $\mathbb{Z}^d$ with $d \geq 2$. A site with mass above a threshold m topples its excess above m, and a site with positive mass at most m topples all of its mass once its odometer exceeds $n^{2/d}/(2dm)$. (Their odometer is the total mass emitted, which is $2d$ times ours, so their threshold is $n^{2/d}/m$.) They proved that, for a suitable m growing slowly with n, the rescaled odometers converge to a spherically symmetric limit (Aleksanyan and Shahgholian, 2019, Theorem 4.3).

For the configuration displayed above with i.i.d. weights of mean 0 and variance 1 in place of Gaussians, Cipriani, Hazra, and Ruszel (2018b) proved on discrete tori that the rescaled odometer, viewed modulo additive constants, converges in law in a negative Sobolev space to the bi-Laplacian Gaussian field (Cipriani et al., 2018b, Theorems 1–2). Cipriani, Hazra, and Ruszel (2018a) proved that i.i.d. masses of mean 1 whose excesses are symmetric and in the domain of normal attraction of a stable law explode on infinite vertex-transitive graphs, under conditions on sums of powers of the Green function (Cipriani et al., 2018a, Theorems 2–3), and they found the scaling limit of the odometer for such weights in this configuration on discrete tori (Cipriani et al., 2018a, Theorem 5). For this configuration with i.i.d. weights of variance 1 on discrete tori, with mass moved by a long-range random walk, Chiarini, Jara, and Ruszel (2021) proved that the rescaled odometer converges in law to a fractional Gaussian field, with a logarithmic factor in the normalization when the jump law has tail exponent 2 (Chiarini et al., 2021, Theorem 3.4). In dimension two and for $1 < \alpha < 2$, Frómeta and Jara (2018) obtained the scaling limit of a divisible sandpile in which mass is redistributed by a truncated α-stable random walk with a nearest-neighbor correction, and identified the limiting odometer through an obstacle problem for a truncated fractional Laplacian (Frómeta and Jara, 2018, Theorems 1.2–1.3). Cipriani, de Graaff, and Ruszel (2020) replaced the i.i.d. weights by correlated centered Gaussian weights whose covariance on each torus is translation invariant, with positive Fourier transforms that are uniformly bounded and converge pointwise to a positive limit. They proved that the rescaled odometer on discrete tori, viewed modulo additive constants, converges in law to a Gaussian field determined by this limit (Cipriani et al., 2020, Theorem 1); suitable covariances give fractional Gaussian fields smoother than the bi-Laplacian field (Cipriani et al., 2020, Proposition 14).

For a uniformly bounded stationary ergodic initial configuration on lattice

approximations of a bounded Lipschitz domain, Bou-Rabee (2021) proved that, almost surely, the sup-norm distance between the rescaled odometer and that of the constant configuration with the same mean tends to 0 as the lattice spacing tends to 0 (Bou-Rabee, 2021, Theorem 8.2). Huss and Sava-Huss (2019) proved that on the doubly infinite Sierpinski gasket graph, for the divisible sandpile started from mass m at the origin, the set of vertices that topple lies between two consecutive graph-metric balls centered at the origin (Huss and Sava-Huss, 2019, Theorem 1.1), and computed the odometer exactly when $m = 3^{k+1}$ (Huss and Sava-Huss, 2019, Theorem 5.12). On the two-dimensional comb, Huss and Sava (2012) proved that the divisible sandpile started from mass n at the origin is close to an explicit anisotropic shape, and used it to obtain inner bounds for internal DLA and rotor-router aggregation (Huss and Sava, 2012, Theorems 3.5, 4.2 and 5.3).

For surveys, see Járai (2018, Sections 2–6) on abelian sandpiles and Ruszel (2020) on divisible-sandpile odometers, obstacle problems, and internal DLA (Ruszel, 2020, Sections 2.2, 4, and 6).

Chapter Eleven

The Rotor Model

To visualize a rotor walk, label each vertex of a graph by an arrow (a **rotor**) pointing toward one of its neighbors. At each time step, the walker first advances the rotor at its current location to point to the next neighbor in a prescribed periodic sequence, and then the walker moves to that neighbor.

Toppling is a way of routing grains through a directed graph in such a fashion that the grains emitted by each vertex travel in equal numbers along each of its outgoing edges. In order to ensure this equality, however, grains must wait at a vertex until sufficiently many additional grains have arrived to render it unstable. Rotor-routing is an alternative approach to distributing particles which dispenses with this waiting step. Since we cannot ensure exact equality without waiting, we settle for the condition that the particles emitted by each vertex travel in nearly equal numbers along each of its outgoing edges (at all times the numbers differ by at most one), and the rotors decide where each successive particle should be routed. Rotor walk was introduced under the name Eulerian walkers, in connection with self-organized criticality, and it was rediscovered several times: as a model of mobile agents exploring a territory, as an approach to load-balancing in multiprocessor systems, and as a way to derandomize random walk.

Like the sandpile, rotor walk has the abelian property: when finitely many particles are routed until they reach an absorbing set, neither the final rotors nor the number of departures from each vertex depends on the order in which the particles are moved. On a finite graph with a sink, this lets the sandpile group act on the rotor configurations that form spanning trees oriented toward the sink. The action is free and transitive, so there are exactly as many such trees as group elements, and this proves the matrix tree theorem (Section 11.2).

On an infinite graph, the first question is whether rotor walk keeps returning to its starting point. Unlike random walk, the answer depends on the initial rotors, and Section 11.3 shows how to choose them so that the walk is recurrent on $\mathbb{Z}^d$ in every dimension. When each vertex has as many incoming as outgoing edges, the walk splits into circuits (the stretches between successive full turns of the rotor at the starting vertex), and no circuit uses an edge twice. It follows that rotor walk on $\mathbb{Z}^d$ visits at least order $t^{d/(d+1)}$ sites in its first t steps, for all initial rotors (Section 11.5). With independent uniform

initial rotors, rotor walk is recurrent on the Manhattan and F-lattices, by a coupling with critical percolation (Section 11.6). On a transient graph, rotor walk escapes to infinity at most as often as random walk does (Section 11.7).

A path is live if, as the rotor at each internal vertex advances from its initial position, it points forward along the path before it points back. If no live path is infinite, then every circuit is complete and the walk is recurrent (Section 11.8). With independent uniform initial rotors, this criterion applies to every graph of maximum degree three and to $\mathbb{Z}^2$ with the clockwise mechanism (Section 11.9). On the comb in $\mathbb{Z}^d$ for $d \geq 2$, with independent uniform initial rotors and suitable cyclic orders, the sets of sites visited by the circuits are close to diamonds, and after t steps the walk has visited order $t^{d/(d+1)}$ sites, the smallest order that the bound of Section 11.5 allows (Section 11.10).

In Chapter 6 we used rotor walk on $\mathbb{Z}^d$ to grow a nearly spherical aggregate (Figure 6.1). In this chapter we study rotor walk itself, on general digraphs.

11.1 ROTOR WALK AND THE ABELIAN PROPERTY

In a rotor walk on a simple directed graph $G = (V, E)$, which is either finite or infinite, the exits from each vertex follow a prescribed periodic sequence. (A digraph is **simple** if it has no loops and at most one edge joins each ordered pair of vertices; all digraphs in this chapter are simple.) We assume that every vertex v has finite out-degree $\deg(v)$. As on $\mathbb{Z}^d$, a rotor configuration on G is a choice of an outgoing edge $\rho(v)$ for each vertex v of positive out-degree, and a rotor mechanism on G is a choice of a cyclic permutation π_v of the outgoing edges of each such vertex; vertices of out-degree zero have no rotors. For example, on $\mathbb{Z}^2$ one usually employs the clockwise cyclic ordering north, east, south, west.

A particle starts at a vertex X_0, with initial rotor configuration $\rho_0 = \rho$. In each step, if the particle is at v, the rotor at v is incremented according to π_v, and then the particle moves to where the rotor points. More formally, for all integer times $t \geq 0$,

$$\rho_{t+1}(v) = \begin{cases} \pi_v(\rho_t(v)), & v = X_t, \\ \rho_t(v), & v \neq X_t, \end{cases}$$

and the next particle location X_{t+1} is the head of the edge $\rho_{t+1}(X_t)$ (see Figure 11.1). The walk stops when it reaches a vertex of out-degree zero. The **path of rotors** from a vertex x follows the edges of a fixed configuration ρ without advancing them, and it stops at a vertex with no rotor.

Our first lemma says that on a finite digraph, rotor walk cannot avoid forever a set that every vertex can reach, so each routing in this section stops.

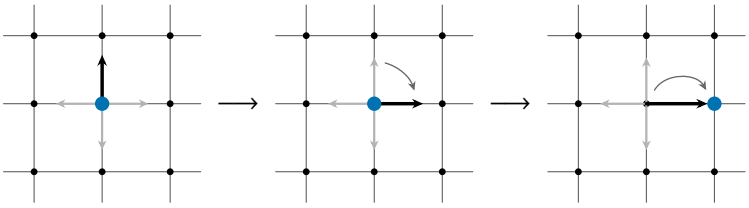


Figure 11.1: A step of rotor walk on $\mathbb{Z}^2$ with the clockwise mechanism. First the rotor moves from north to east, then the particle jumps.

Lemma 11.1 *Let $G = (V, E)$ be a finite digraph, and let $Z \subseteq V$ be nonempty. Suppose that every vertex outside Z has a directed path to Z. Then every rotor walk on G, stopped when it first enters Z, stops after finitely many steps.*

Proof. Suppose the particle never enters Z. Since G is finite, the particle visits a vertex v infinitely often. But if x is a vertex that is visited infinitely often and there is an edge from x to y, then y is also visited infinitely often, since each visit to x advances the rotor at x by one position. Inducting along a directed path from v to Z, we conclude that the particle eventually enters Z, a contradiction. □

Until the end of the proof of Proposition 11.3, fix a finite digraph $G = (V, E)$ and a nonempty set $Z \subseteq V$ such that every vertex outside Z has a directed path to Z. Vertices in Z are **absorbing**: they have no rotors, and they keep every particle that reaches them. A global sink, as in Chapter 9, is the case of a single absorbing vertex with no outgoing edges.

A **particle-and-rotor state** is a pair (σ, ρ) consisting of a particle configuration $\sigma : V \to \mathbb{Z}_{\geq 0}$ and a rotor configuration ρ on $V \setminus Z$. If $v \notin Z$ has at least one particle, then **actuating** v results in a new particle-and-rotor state, given by advancing the rotor at v and moving a single particle from v to the head of the new rotor. We say that this state is a **successor** of (σ, ρ). We say that (σ, ρ) is **stable** if no vertex can actuate, that is, all particles are in Z. If we route the particles one at a time, each until it enters Z, then by Lemma 11.1 we reach a stable state after finitely many actuations.

The rotor-router operation has the following abelian property, the rotor analogue of Lemma 9.3 for the sandpile; its proof follows that of the least action principle, Lemma 9.2. (The order of the actuations does not matter, because the sequence of edges along which a vertex sends particles is determined by its initial rotor and the mechanism.)

Lemma 11.2 *Let $G = (V, E)$ be a finite digraph, and let $Z \subseteq V$ be a nonempty set such that every vertex outside Z has a directed path to Z. Fix a rotor mechanism on $V \setminus Z$. Let $(\sigma_0, \rho_0), \ldots, (\sigma_n, \rho_n)$ and $(\widehat{\sigma}_0, \widehat{\rho}_0), \ldots, (\widehat{\sigma}_m, \widehat{\rho}_m)$ be sequences of particle-and-rotor states, each state a successor of the one*

before it, with $(\widehat{\sigma}_0, \widehat{\rho}_0) = (\sigma_0, \rho_0)$.

(a) If (σ_n, ρ_n) *is stable, then for each vertex, the number of times it actuates in the second sequence is at most the corresponding number in the first.*

(b) If (σ_n, ρ_n) *and* $(\widehat{\sigma}_m, \widehat{\rho}_m)$ *are both stable, then* $(\sigma_n, \rho_n) = (\widehat{\sigma}_m, \widehat{\rho}_m)$. *Moreover, each vertex actuates the same number of times in both sequences.*

Proof. (a) For a vertex $v \notin Z$, a vertex x and an integer $k \geq 0$, let $\nu_k(v, x)$ be the number of the first k departures from v that use the edge (v, x), with $\nu_k(v, x) = 0$ if there is no such edge. These departures are determined by the initial rotor at v and the mechanism, and $\nu_k(v, x)$ is nondecreasing in k. Let $N(v)$ be the number of actuations of v in the first sequence. Suppose, for a contradiction, that the second sequence actuates a vertex more often, and consider the first step at which it actuates a vertex x for the $(N(x) + 1)$st time. Just before this step, each vertex v has been actuated at most $N(v)$ times in the second sequence, and x has been actuated exactly $N(x)$ times. The number of particles at x is then at most

$$\sigma_0(x) + \sum_{v \notin Z} \nu_{N(v)}(v, x) - N(x) = \sigma_n(x) = 0\,.$$

The first equality is particle conservation at x in the first sequence, and $\sigma_n(x) = 0$ because (σ_n, ρ_n) is stable and $x \notin Z$. Thus x holds no particle and cannot be actuated, a contradiction.

(b) Applying (a) in both directions shows that each vertex is actuated equally often in the two sequences. The numbers of actuations determine how many particles cross each edge and where each rotor points at the end. Particle conservation then determines the final particle configuration, so $(\sigma_n, \rho_n) = (\widehat{\sigma}_m, \widehat{\rho}_m)$. □

Since each step of the two sequences actuates one vertex, it follows that $m \leq n$ in (a) and $m = n$ in (b).

We may also compare hitting probabilities of simple random walk and rotor walk. Recall from Chapter 9 that simple random walk on a digraph chooses an outgoing edge uniformly at each step.

Proposition 11.3 *Let* $G = (V, E)$ *be a finite digraph with a fixed rotor mechanism, and let* $Y \subseteq Z \subseteq V$ *be sets of vertices with* Z *nonempty. Assume that from each vertex outside* Z *there is a directed path to* Z. *Let* $\sigma : V \to \mathbb{Z}_{\geq 0}$ *be a particle configuration on* G. *If the particles perform independent simple random walks on* G *stopped on first hitting* Z, *let* $H(\sigma, Y)$ *be the expected number of particles that stop in* Y. *If the particles perform rotor walks*

starting at rotor configuration ρ on $V \setminus Z$ and stopped on first hitting Z, let $H_\rho(\sigma, Y)$ be the number of particles that stop in Y. For $v \in V$, let $h(v)$ be the probability that simple random walk started at v stops in Y. Then

$$\left|H_\rho(\sigma, Y) - H(\sigma, Y)\right| \le \frac{1}{2} \sum_{\substack{(w,v)\in E \\ w\notin Z}} |h(w) - h(v)| . \tag{11.1}$$

Note that the bound on the right side does not depend on ρ or σ.

Proof. We assign weight $h(v)$ to a particle located at v. Initially, the sum of all particle weights is $H(\sigma, Y)$. By Lemma 11.1, every particle reaches Z, and after all particles have stopped, the sum of the particle weights is $H_\rho(\sigma, Y)$, because h equals 1 on Y and 0 on $Z \setminus Y$. A move along an edge (w, v) changes the total weight by $h(v) - h(w)$, so $H_\rho(\sigma, Y) - H(\sigma, Y)$ is the sum of $h(v) - h(w)$ over all the moves.

Since h is harmonic on $V \setminus Z$ for simple random walk, for every $w \notin Z$

$$\sum_{v:(w,v)\in E} (h(w) - h(v)) = 0 . \tag{11.2}$$

The rotor at w sends the successive particles leaving w along its outgoing edges in cyclic order. Group these departures into consecutive blocks of $\deg(w)$. Each complete block uses every outgoing edge of w once, so it contributes zero by (11.2). The remaining departures use distinct outgoing edges, so they contribute the sum of a subset of the numbers $h(v) - h(w)$ with $(w, v) \in E$. These numbers sum to zero, so the sum of the positive ones and the absolute sum of the negative ones are both equal to half of $\sum_{v:(w,v)\in E} |h(w) - h(v)|$, and every subset sum is at most this half in absolute value. Summing over $w \notin Z$ proves (11.1). □

Recall from Chapter 9 that a digraph is Eulerian if it is strongly connected and each vertex has equal in-degree and out-degree. We will use the following lemma to study both recurrence and circuits.

Lemma 11.4 *Let G be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration on G. If the rotor walk started at a vertex x traverses a directed edge twice, then the first directed edge to be traversed twice is from x.*

Proof. When the first directed edge (v, w) is traversed twice, the vertex v has sent the particle along all its other outgoing edges exactly once, since its rotor turns through all $\deg(v)$ outgoing edges before it repeats one. Thus v has emitted the particle $\deg(v) + 1$ times. Since no other edge has been traversed twice, v has received the particle at most once along each incoming

edge, hence at most $\deg(v)$ times (its in-degree), and thus the particle started at v, that is, $v = x$. □

On a strongly connected digraph with finite out-degrees, a rotor walk is **recurrent** if it visits some (and hence every) vertex infinitely many times, and **transient** otherwise. Indeed, if the walk visits a vertex infinitely often, then, as in the proof of Lemma 11.1, it visits each out-neighbor of that vertex infinitely often, and, by strong connectivity, every vertex.

11.2 RELATING ROTOR WALK AND SANDPILES

Throughout this section, $G = (V, E)$ is a finite digraph with a global sink s, and we fix a rotor mechanism on $V \setminus \{s\}$. In this section we explore an important link between toppling and rotor-routing. A hint at this link comes from a straightforward count of configurations. Recall from Chapter 9 that a nonnegative stable sandpile assigns between 0 and $\deg(v) - 1$ grains to each vertex $v \neq s$. Thus the number of nonnegative stable sandpiles is exactly $\prod_{v \neq s} \deg(v)$, which is also the number of rotor configurations on G. Other connections become apparent when we let the sandpile group $\mathcal{S}(G)$ act on rotor configurations, which leads to a proof of the matrix tree theorem.

Given a vertex $v \neq s$, the **particle addition operator** $\mathcal{E}_v$ is the map on rotor configurations given by adding a particle at vertex v and routing it until it reaches the sink. Given a particle configuration σ on $V \setminus \{s\}$ and a rotor configuration ρ, write $\sigma(\rho)$ for the rotor configuration obtained by adding $\sigma(v)$ particles at each vertex v and routing them all to the sink. By Lemma 11.2(b) the order of these routings is immaterial. Thus the operators $\mathcal{E}_v$ commute (to see this, we can start with particles at v and w and then route them to the sink in either order), $\sigma(\rho)$ is obtained by applying each $\mathcal{E}_v$ exactly $\sigma(v)$ times in an arbitrary order, and clearly

$$\sigma_2(\sigma_1(\rho)) = (\sigma_1 + \sigma_2)(\rho)$$

for all particle configurations σ_1 and σ_2 and every rotor configuration ρ.

If, rather than running the particles until they reach the sink, each particle is run for a fixed number of steps, then the abelian property fails, as the example in Figure 11.2 illustrates. (The proof of Lemma 11.2 requires that particles be indistinguishable, and it is not possible to run each particle for a fixed number of steps without distinguishing between them.)

A rotor configuration is **acyclic** if the rotors do not form a directed cycle. Since G has a global sink, a rotor configuration is acyclic if and only if the rotors form an oriented spanning tree rooted at the sink. After a routing, the rotor at each vertex that sent a particle points along the edge by which a

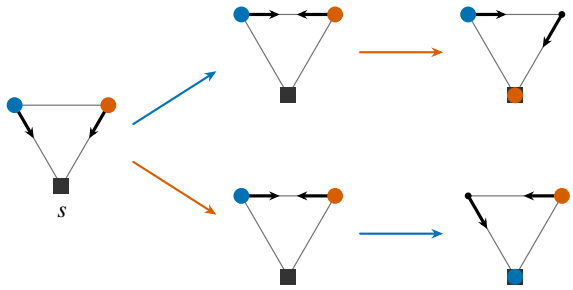


Figure 11.2: Failure of the abelian property for rotor walk stopped after two steps (or earlier at the sink s, a square) on the triangle. The bold arrows are the rotors, and each arrow between configurations is colored by the particle that moves, blue or vermilion. Moving blue first (upper row) or vermilion first (lower row) passes through the same configuration but ends in different ones.

particle last exited, and the next lemma uses this to locate the cycles of the final rotors.

Lemma 11.5 *Let $G = (V, E)$ be a finite digraph with a global sink s, and fix a rotor mechanism. Let ρ be a rotor configuration and σ a particle configuration on G. Then every directed cycle of the rotor configuration $\sigma(\rho)$ consists of vertices from which no particle departs while σ is routed.*

Proof. Suppose that a directed cycle of $\sigma(\rho)$ contains a vertex that sends a particle during the routing. Among such vertices of the cycle, choose the vertex x whose last departure is latest. The last departure from x uses the final rotor at x, so it sends a particle to the next vertex y on the cycle. At the end no particle remains at $y \neq s$, so y sends a particle after this time, and its last departure is later than that of x, contrary to the choice of x. □

A cycle of $\sigma(\rho)$ whose vertices never send a particle is also a cycle of ρ. Consequently, if ρ is acyclic, then so is $\sigma(\rho)$. Moreover, if every vertex other than s sends at least one particle while σ is routed, then $\sigma(\rho)$ has no cycle at all.

To show that each particle addition operator permutes the acyclic configurations, we temporarily give the sink an outgoing edge. The rotors then contain exactly one cycle, the walker lies on it, and the edge along which the walker arrived is determined, so that each step of rotor walk can be undone.

Lemma 11.6 *Let $\widehat{G}$ be a finite digraph in which every vertex has positive outdegree, and fix a rotor mechanism on $\widehat{G}$. Consider the pairs (ρ, x) in which ρ is a rotor configuration on $\widehat{G}$ whose edges contain exactly one directed cycle and x is a vertex of that cycle. Then one step of rotor walk from x permutes these pairs.*

Proof. We first show that the step maps a pair of this kind to a pair of this

kind. Let (ρ, x) be such a pair. Removing the rotor edge at x from ρ leaves an oriented spanning tree rooted at x, because x lies on the unique cycle. If the advanced rotor at x is the edge (x, y), then adding this edge to the tree creates exactly one directed cycle, and this cycle passes through y. Thus the step maps (ρ, x) to a pair of the same kind.

The set of pairs is finite, so it suffices to show that the step is injective. Suppose that it maps (ρ, x) to (ρ_1, y). The edge $\rho_1(x) = (x, y)$ lies on the cycle of ρ_1, so x is the vertex preceding y on that cycle. Moreover, ρ agrees with ρ_1 except at x, where $\rho(x)$ is the edge preceding $\rho_1(x)$ in the cyclic order at x. Thus (ρ, x) is determined by (ρ_1, y). □

For each $v \neq s$, once the edge (s, v) is added, particle addition at v is rotor walk from the sink, run until it returns to the sink (see Figure 11.3 for an example on the 3×3 grid).

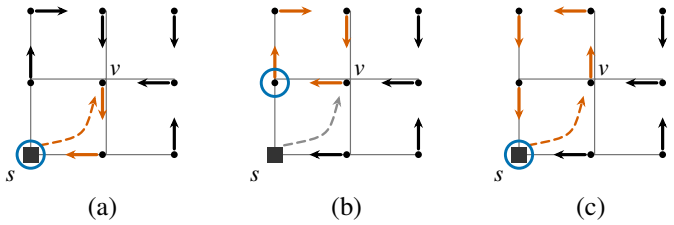


Figure 11.3: Particle addition at v on the 3×3 grid with sink s (square) and the clockwise mechanism, run as rotor walk with the added edge (s, v) (dashed): the rotors always contain exactly one cycle (vermilion), and the walker (circled) lies on it. The walk starts from an acyclic ρ with the walker at s (a), passes after two steps through a state whose cycle avoids s (b), and returns to s after seven steps with the acyclic rotors $\mathcal{E}_v(\rho)$ (c).

Lemma 11.7 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let $v \neq s$ be a vertex of G. The particle addition operator $\mathcal{E}_v$ is a permutation on the set of acyclic rotor configurations on G.*

Proof. Add the edge (s, v) to G, and let $\widehat{G}$ be the resulting digraph, in which every vertex has positive out-degree; the rotor at s is always this edge. A rotor configuration ρ on G, together with this rotor at s, contains exactly one cycle, passing through s, if and only if ρ is acyclic: the paths of rotors of an acyclic configuration end at s, and an additional cycle of ρ would avoid s. Thus the pairs of Lemma 11.6 on $\widehat{G}$ with the walker at s are exactly the acyclic configurations of G with the walker at s. By that lemma, rotor walk on $\widehat{G}$ permutes its pairs. Started from an acyclic ρ with the walker at s, the walk steps along (s, v) and then moves exactly as a particle routed from v in G, until it first returns to s with the rotors $\mathcal{E}_v(\rho)$. Hence $\mathcal{E}_v$ maps each pair with the walker at s to the next such pair on its cycle under this permutation. For a permutation of a finite set, the map taking each element of a subset to

the next element of the subset on its cycle is a permutation of the subset, so $\mathcal{E}_v$ permutes the acyclic configurations. □

In a full turn of the rotor at v, one particle is sent along each outgoing edge of v, just as a toppling of v sends one grain along each outgoing edge. Consequently, particle configurations that differ by topplings act in the same way on acyclic configurations. Recall from Chapter 9 that two configurations on $V \setminus \{s\}$ are equivalent modulo the reduced Laplacian L' if their difference is an integer combination of rows of L'.

Lemma 11.8 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let ρ be an acyclic rotor configuration on G. If particle configurations σ_1 and σ_2 on $V \setminus \{s\}$ are equivalent modulo the reduced Laplacian L', then $\sigma_1(\rho) = \sigma_2(\rho)$.*

Proof. By Lemma 11.7, the maps $\mathcal{E}_v$ are commuting permutations of the finite set of acyclic configurations. Their action therefore extends to integer-valued configurations, with negative values acting through inverse maps, and it suffices to show that each row of L' acts as the identity. Fix $v \neq s$, and add $\deg(v)$ particles at v. If we route (for one step) these $\deg(v)$ particles, then the rotor at v makes one full turn, back to its initial position, and one particle is sent along each outgoing edge from v; a particle sent to s stays there. Routing the remaining particles to s and applying Lemma 11.2(b) shows that $\deg(v)\mathbf{1}_v$ acts in the same way as the configuration with one particle at each out-neighbor of v other than s. The difference of these two configurations is the row of L' indexed by v, so this row acts as the identity. Integer combinations of rows therefore also act as the identity, which proves the lemma. □

It follows from Lemma 11.8 that the sandpile group of G acts on the set of oriented spanning trees of G rooted at the sink. Our next lemma shows that this action is transitive.

Lemma 11.9 *Let $G = (V, E)$ be a finite digraph with a global sink s, and fix a rotor mechanism. For every two acyclic rotor configurations ρ and ρ_1 on G, there exists a particle configuration σ on $V \setminus \{s\}$ such that $\sigma(\rho) = \rho_1$.*

Proof. For $v \neq s$, let $0 \leq \alpha(v) < \deg(v)$ denote the number of steps needed to turn $\rho(v)$ into $\rho_1(v)$, so that $\pi_v^{\alpha(v)}(\rho(v)) = \rho_1(v)$. Starting with rotor configuration ρ, and with $\alpha(v)$ particles at each vertex v, allow each particle to take just one step. The resulting rotor configuration is ρ_1; let β be the resulting particle configuration off the sink, so that $\alpha(\rho) = \beta(\rho_1)$ by Lemma 11.2(b).

By Lemma 9.18, choose a nonnegative stable sandpile σ equivalent to $\alpha - \beta$ modulo L'. Then $\sigma + \beta$ is equivalent to α, so Lemmas 11.2(b) and 11.8

give

$$\beta(\sigma(\rho)) = (\sigma + \beta)(\rho) = \alpha(\rho) = \beta(\rho_1)\,.$$

By Lemma 11.7, β acts as a permutation on the acyclic configurations. Cancelling it gives $\sigma(\rho) = \rho_1$. □

Next we define reachable rotor configurations on a digraph with a global sink, and show that the acyclic configurations are exactly the configurations reachable from every configuration.

Definition 11.10 Let G be a finite digraph with a global sink, and fix a rotor mechanism. Given rotor configurations ρ and ρ_0 on G, we say that ρ is **reachable** from ρ_0 if there is a particle configuration σ such that $\sigma(\rho_0) = \rho$.

By Lemma 11.9, every acyclic configuration is reachable from every acyclic configuration.

Lemma 11.11 *Let G be a finite digraph with a global sink, and fix a rotor mechanism. A rotor configuration on G is reachable from every rotor configuration on G if and only if it is acyclic.*

Proof. Suppose first that ρ is reachable from every rotor configuration. There is an acyclic configuration, because every vertex has a directed path to the sink, and by Lemma 11.5 every configuration reachable from an acyclic configuration must be acyclic, so ρ is acyclic.

Conversely, suppose that ρ is acyclic, and let ρ_0 be a rotor configuration. The configuration obtained by routing one particle from each vertex other than the sink, starting from ρ_0, is acyclic by Lemma 11.5, since the rotor at each vertex points along the edge by which a particle last exited. By Lemma 11.9, ρ is reachable from this acyclic configuration, and hence from ρ_0. □

Just as for the sandpile, there are several equivalent characterizations of acyclic rotor configurations.

Proposition 11.12 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let ρ be a rotor configuration on G. The following are equivalent.*

(1) The configuration ρ is acyclic.

(2) The configuration ρ is reachable from every rotor configuration.

(3) If a rotor configuration ρ_0 is reachable from ρ, then ρ is reachable from ρ_0.

(4) *The configuration ρ is reachable from every rotor configuration of the form $\mathcal{E}_v(\rho)$, where $v \neq s$.*

(5) *Each strongly connected component of the subgraph induced by $V \setminus \{s\}$ contains a vertex v such that ρ is reachable from $\mathcal{E}_v(\rho)$.*

Proof. By Lemma 11.11, (1) implies (2), and trivially (2) implies (3), which implies (4), which implies (5). It remains to show that (5) implies (1).

Assume (5), and suppose that ρ has a directed cycle. Its vertices are mutually reachable, so the cycle lies in one strongly connected component of the subgraph induced by $V \setminus \{s\}$. By (5) there are a vertex v of this component and a particle configuration σ with $\sigma(\mathcal{E}_v(\rho)) = \rho$, so $(\sigma + \mathbf{1}_v)(\rho) = \rho$ by Lemma 11.2(b). Route $\sigma + \mathbf{1}_v$ from ρ. The rotors end where they started, so each rotor makes a whole number of turns, and a vertex w that sends a particle sends at least $\deg(w)$ of them, one along each of its outgoing edges. The vertex v sends a particle, because the particle added at v must leave it, and every particle that reaches a vertex other than s leaves it again. Following a path in the component from v therefore shows that every vertex of the component sends a particle. By Lemma 11.5, every directed cycle of $(\sigma + \mathbf{1}_v)(\rho) = \rho$ avoids the vertices that send particles, so the cycle above is not a cycle of ρ, a contradiction. □

Next we show that the action of the sandpile group on the set of oriented spanning trees of G is free. (A group action is free if the only element of the group that has fixed points is the identity.)

Lemma 11.13 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let σ_1 and σ_2 be recurrent sandpiles on G. If there is an acyclic rotor configuration ρ on G such that $\sigma_1(\rho) = \sigma_2(\rho)$, then $\sigma_1 = \sigma_2$.*

Proof. For $i = 1, 2$, let $n_i(v)$ be the number of departures from v while σ_i is routed from ρ. Both routings start and end with the same rotors, so the rotor at v makes a whole number of extra turns in one routing compared with the other: $n_1(v) - n_2(v) = k_v \deg(v)$ for an integer k_v. Every full rotation results in $\deg(v)$ particles leaving v, one along each outgoing edge, so the numbers of particles sent along an edge (v, x) in the two routings differ by k_v. At each $x \neq s$, the particles present at the start plus those received equal those sent, because no particle remains at x. Subtracting these balance equations for the two routings gives

$$\sigma_1 - \sigma_2 = \sum_{v \neq s} k_v \mathsf{L}'_v \,,$$

where L'_v is the row of L' indexed by v. Thus σ_1 and σ_2 are equivalent modulo L'. By Lemma 9.23, an equivalence class contains at most one recurrent

sandpile, so $\sigma_1 = \sigma_2$. □

Because the action is free and transitive, the number of oriented spanning trees equals the order of the sandpile group, which is the determinant of the reduced Laplacian by Lemma 9.17. Recall from Section 9.1 that the Laplacian $\mathsf{L}(G)$ is the diagonal matrix of out-degrees minus the adjacency matrix.

Proposition 11.14 (Matrix Tree Theorem) *Let G be a finite digraph and v a vertex of G. The number of oriented spanning trees of G rooted at v is equal to the determinant of the matrix obtained by deleting from $\mathsf{L}(G)$ the row and column corresponding to v.*

Proof. If some vertex has no directed path to v, then there are no spanning trees rooted at v, and the set S of vertices from which v cannot be reached is nonempty. There are no edges from S to its complement, so listing the vertices of S first puts the reduced matrix in the block form

$$\begin{pmatrix} L_S & 0 \\ * & * \end{pmatrix},$$

where each row of the square block L_S indexed by S sums to zero, whence the reduced matrix has determinant zero.

If every vertex has a directed path to v, delete all outgoing edges from v to make v a global sink. Note that this changes neither the reduced matrix nor the set of oriented spanning trees rooted at v. Fix such a tree ρ, viewed as an acyclic rotor configuration. By Lemma 11.8, each equivalence class acts on ρ through its recurrent sandpile, which exists and is unique by Lemmas 9.22 and 9.23. The mapping $\sigma \mapsto \sigma(\rho)$ from the recurrent sandpiles to the oriented spanning trees rooted at v is onto by Lemma 11.9 and is one-to-one by Lemma 11.13. The recurrent sandpiles are in bijection with $\mathcal{S}(G)$, whose order is the determinant of the reduced Laplacian by Lemma 9.17. □

We can use rotor walk to define a family of bijections from recurrent sandpiles to oriented spanning trees. Indeed, the proof of Proposition 11.14 shows that for every oriented spanning tree ρ of G rooted at the sink, the map $\sigma \mapsto \sigma(\rho)$ is such a bijection (see Figure 11.4 for the cycle with five vertices, whose recurrent sandpiles are in Figure 9.8).

11.3 RECURRENCE OF ROTOR WALK

In this section, we consider a connected, locally finite undirected graph. We turn the graph into a directed one by giving each edge both orientations and denote the resulting digraph by $G = (V, E)$. This digraph is Eulerian, so

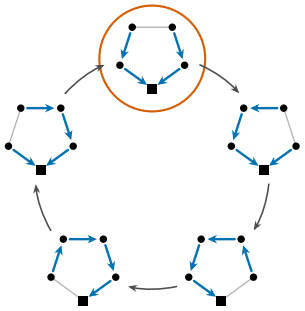


Figure 11.4: The five oriented spanning trees of the cycle with five vertices rooted at the sink (black square), in the positions of the recurrent sandpiles of Figure 9.8: the particle addition operator $\mathcal{E}_v$ at the vertex v just clockwise of the sink moves each tree to the next (gray arrows). Thus $\sigma \mapsto \sigma(\rho)$, with ρ the circled tree, matches each recurrent sandpile of Figure 9.8 with the tree in the same position, as in the proof of Proposition 11.14.

Lemma 11.4 applies to it. In contrast with random walk, whether rotor walk on G is recurrent depends on the initial rotors. The next lemma states that it does not depend on the choice of starting vertex.

Lemma 11.15 *Let G be a connected, locally finite undirected graph, with each edge replaced by its two orientations. Fix a rotor mechanism and an initial rotor configuration on G. The rotor walk is either recurrent for every starting vertex, or transient for every starting vertex.*

Proof. Since G is connected, it suffices to show that if the rotor walk started at a vertex x is recurrent, then so is the walk started at each neighbor y of x. We do so by showing that for every integer $m \geq 0$, the rotor walk started at y visits x at least m times.

Let S be the set of vertices visited by the rotor walk started at x until it has made $\deg(x) + m$ returns to x, which is finite since this walk is recurrent. Split x into two copies: a source, which has the rotor and the outgoing edges of x, and an absorbing copy, which has the incoming edges of x. Also make absorbing every vertex outside S adjacent to S. If we start $\deg(x) + m$ particles at the source and route them one at a time, then they reproduce the original walk up to its $(\deg(x) + m)$th return, so every particle is absorbed at the copy of x.

Now start with $\deg(x) + m$ particles at the source and move them in the following order. First, let $\deg(x)$ particles take one step each. This leaves the rotor at x unchanged, and one particle at each neighbor of x (including y). Let the particle at y walk until it is absorbed, then route the m particles still at the source one at a time, and finally route the remaining particles. By the abelian property, Lemma 11.2(b), every particle is again absorbed at the copy of x, regardless of the order in which they move. When the two copies

of x are identified, the route of the particle from y, followed by the routes of the m particles from the source, forms an initial segment of the rotor walk from y in the original configuration. This segment visits x at least m times, as claimed. □

Changing finitely many initial rotors does not affect recurrence either.

Lemma 11.16 *Let G be a connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism on G. If two rotor configurations on G differ at only finitely many vertices, then they are either both recurrent or both transient.*

Proof. Every finite change of the rotors, in either direction, is a finite sequence of single rotor increments, so it suffices to consider two rotor configurations ρ and $\widetilde{\rho}$ that differ at only a single vertex v, at which the rotor is incremented once in $\widetilde{\rho}$ compared with ρ. Suppose that ρ is recurrent, and start a particle at v. After one step, the configuration becomes $\widetilde{\rho}$ and the particle moves to a neighbor w of v. This implies that the rotor walk started at w is recurrent for $\widetilde{\rho}$, and hence, by Lemma 11.15, for every starting vertex. Conversely, ρ is obtained from $\widetilde{\rho}$ by $\deg(v) - 1$ further increments at v, so the same argument shows that if $\widetilde{\rho}$ is recurrent, then so is ρ. □

A similar result holds if two rotor mechanisms differ at only finitely many vertices (Exercise 11.2).

For a set of vertices $S \subseteq V$, the **outer boundary** of S is

$$\partial S := \{y \in V \setminus S : (x, y) \in E \text{ for an } x \in S\}. \tag{11.3}$$

We say that S has a **reflecting boundary** for a given rotor mechanism and initial rotor configuration if, for every vertex $y \in \partial S$, the rotor at y will send the particle to each of the neighbors of y in S before sending it to a neighbor of y in $V \setminus S$ (see Figure 11.5 for an example in two dimensions).

A reflecting boundary that the walk has not yet reached forces one more return to the start, as the next lemma shows.

Lemma 11.17 *Let $G = (V, E)$ be a connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism and an initial rotor configuration on G. Let $S \subset V$ be a finite set with reflecting boundary, and let $x \in S$ satisfy $\deg(x) \geq 1$. Then the rotor walk started at x will return to x before making a step from ∂S to $V \setminus S$.*

Proof. Suppose the first such step, from a vertex $y \in \partial S$ to a vertex $z \notin S$, occurs at time t. At this time, by the definition of reflecting boundary, y has previously sent the particle to each of its neighbors in S. However, by the minimality of t, the vertex y has received the particle only from its neighbors

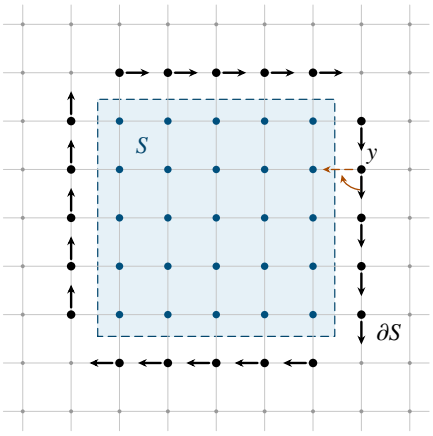


Figure 11.5: A square S (shaded) with a reflecting boundary for the clockwise mechanism: each vertex of ∂S (black) has one neighbor in S, and the next advance of its rotor points to that neighbor, as shown at y. The four vertices diagonal to the corners of S have no neighbor in S, so they are not in ∂S; their rotors, like all rotors off ∂S, are unconstrained and not drawn.

in S, and the walk did not start at $y \notin S$; therefore it must have received it twice from one such neighbor. By Lemma 11.4, the particle must have already returned to x. If the walk never takes such a step, then it stays in the finite set $S \cup \partial S$, so it eventually traverses an edge twice, and Lemma 11.4 again gives a return to x. □

Proposition 11.18 *Let $G = (V, E)$ be an infinite, connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism and an initial rotor configuration on G. If every finite set of vertices is a subset of a finite set with reflecting boundary, then the rotor walk starting from every vertex is recurrent.*

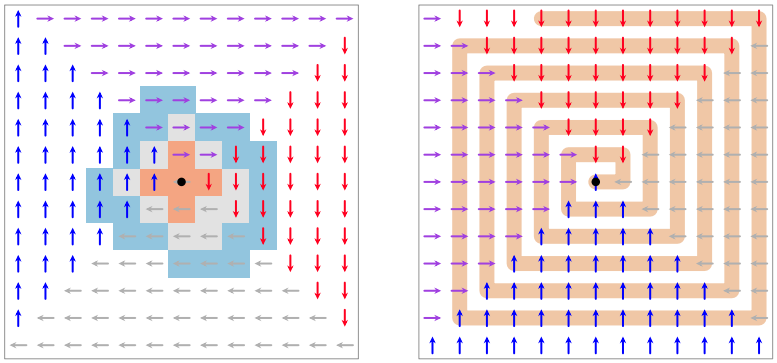
Proof. Fix $x \in V$, and suppose for a contradiction that the walk from x visits x for the last time at time t. Choose a finite set S with reflecting boundary that contains every vertex visited by time t. The walk has not visited ∂S by time t, so the rotors on ∂S are still in their initial positions, and S also has a reflecting boundary for the rotor configuration at time t. Since G is infinite and connected, $\deg(x) \geq 1$. At time t the walk is at x, so Lemma 11.17, applied to the walk from time t on, gives a later return to x, a contradiction. Hence the walk visits x infinitely often, so it is recurrent. □

On $\mathbb{Z}^d$, the cubes centered at the origin provide such sets for a suitable choice of initial rotors, for every mechanism.

Lemma 11.19 *For every integer $d \geq 1$ and every rotor mechanism on $\mathbb{Z}^d$, there is a rotor configuration for which the rotor walk is recurrent.*

Proof. By Proposition 11.18, it suffices to show that there is an initial rotor configuration which makes all the cubes $S_k := \{x \in \mathbb{Z}^d : |x|_\infty \le k\}$ for $k \ge 0$ have reflecting boundary. At each vertex x, choose the initial rotor so that its first advance points to a neighbor of x with the smallest ℓ^∞ norm. Each vertex of ∂S_k has exactly one coordinate equal to $\pm(k+1)$, so it has a unique neighbor in S_k, and this neighbor is its unique neighbor of smallest ℓ^∞ norm. Therefore S_k has a reflecting boundary. □

With the clockwise mechanism on $\mathbb{Z}^2$, these rotors are shown in Figure 11.6(a). Advancing each rotor once makes every path of rotors lead to the origin, and the walk then never returns (Figure 11.6(b)); by Theorem 11.30(i) below, such a walk is always transient.



(a) (b)

Figure 11.6: (a) The initial rotors of Lemma 11.19 near the origin **0** (dot), colored by direction (North blue, East purple, South red, West gray), and the sites visited by the first three circuits, shaded by the circuit of first visit; these circuits end after 8, 44 and 140 steps. (b) The same rotors, each advanced once, so that all paths of rotors lead to **0**: the first 140 steps of the walk (shaded path) spiral outward and visit a new site at every step, and the walk is transient by Theorem 11.30(i).

11.4 CIRCUITS IN EULERIAN DIGRAPHS

Throughout this section, $G = (V, E)$ is a locally finite Eulerian digraph, and we fix a rotor mechanism, an initial rotor configuration ρ, and a vertex o with $\deg(o) \ge 1$. Let $(X_t)_{t\ge 0}$ be the rotor walk started at o. We cut the walk at the times when the rotor at o completes a full turn, and each resulting segment, called a circuit, uses each directed edge at most once. The **odometer** at time $t \ge 0$ is the function

$$u_t(x) := \#\{0 \le r < t : X_r = x\}, \tag{11.4}$$

the number of departures from x before time t. For $n \geq 0$ let

$$T(n) := \inf\{t \geq 0 : X_t = o \text{ and } u_t(o) \geq n \deg(o)\},$$

where $\inf \varnothing = \infty$, be the time it takes the rotor walk to complete n circuits from o: at time $T(n)$ the rotor at o has completed its nth full turn and the walk is back at o. If $T(n-1) < \infty$, then the **nth circuit** is the segment of the walk from time $T(n-1)$ to time $T(n)$. The circuit is **complete** if $T(n) < \infty$ and incomplete otherwise. Its departures are the steps at the times r with $T(n-1) \leq r < T(n)$; when $T(n) = \infty$, they form the entire remaining walk (see Figure 11.7 for a first circuit).

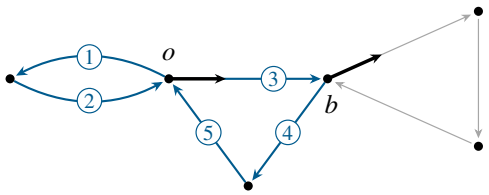


Figure 11.7: The first circuit of the rotor walk from o on an Eulerian digraph, from the initial rotors at o and b (black); the edges are numbered in the order of traversal. The walk returns to o at steps 2 and 5, so the circuit ends at $T(1) = 5$ because $\deg(o) = 2$, and it uses no directed edge twice and none of the gray edges.

Our first lemma says that each $x \in V$ is visited at most $\deg(x)$ times per circuit. The assumption that G is Eulerian is crucial here.

Lemma 11.20 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$. Then every circuit of the rotor walk from o, complete or incomplete, has at most $\deg(x)$ departures from each vertex x.*

Proof. It suffices to show that no directed edge is traversed twice within a circuit, because a circuit with more than $\deg(x)$ departures from x uses an outgoing edge of x twice. Consider the walk from the start of a circuit, which is again a rotor walk from o. By Lemma 11.4, the first edge that it traverses twice has tail o. Such a traversal is at least the $(\deg(o)+1)$st departure from o after the start of the circuit, so it happens after the walk has returned to o following its $\deg(o)$th departure from o, that is, after the circuit has ended. □

For certain initial rotor configurations, the bound in Lemma 11.20 is tight.

Lemma 11.21 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Let o be a vertex with $\deg(o) \geq 1$, and suppose that the first circuit of the rotor walk from o is*

complete. If the path of rotors of ρ from a vertex x reaches o, then the first circuit has exactly $\deg(x)$ *departures from x.*

Proof. We argue by induction on the length of the path of rotors from x to o (here and below, we only consider the first circuit from o). If this length is zero, then $x = o$, and o is visited exactly $\deg(o)$ times by definition.

Otherwise, let y be the first vertex on the path of initial rotors from x to o. By the inductive hypothesis, y is visited exactly $\deg(y)$ times, and since the circuit starts and ends at o, it also enters y exactly $\deg(y)$ times. Each incoming edge to y is traversed at most once by the proof of Lemma 11.20, so in fact each incoming edge to y is traversed exactly once. In particular, the edge (x, y) is traversed. Since $\rho(x) = (x, y)$ and the rotor at x advances before each departure, the edge (x, y) is first traversed at the $\deg(x)$th departure from x, so x is visited at least $\deg(x)$ times, and Lemma 11.20 gives equality. □

If G is finite, then every circuit is complete by Lemma 11.20. If G is infinite, then depending on the rotor mechanism and the initial rotor configuration ρ, rotor walk may or may not complete a circuit from o. In particular, Lemma 11.21 implies the following.

Lemma 11.22 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Suppose that there are distinct vertices $x_0, x_1, x_2, \ldots$ with $\deg(x_0) \geq 1$ and $\rho(x_k) = (x_k, x_{k-1})$ for every $k \geq 1$. Then the first circuit of the rotor walk from x_0 is incomplete.*

Proof. For every $k \geq 1$, the path of rotors from x_k reaches x_0. If the first circuit were complete, then by Lemma 11.21 it would have departures from each of the infinitely many vertices x_k, which is impossible for a finite segment of walk. □

Whenever $T(n-1) < \infty$, let A_n, the nth **circuit range**, be the set of sites visited during the nth circuit (including both of its endpoints if it is complete and its entire infinite tail otherwise), and define $\mathsf{A}_0 := \{o\}$. The next lemma shows that during each complete circuit, the rotor makes a full turn at every vertex visited by the previous circuit.

Lemma 11.23 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, and let $n \geq 0$ be an integer such that the $(n+1)$st circuit of the rotor walk from o is complete. Then:*

(i) the $(n+1)$st circuit has at most $\deg(x)$ departures from every $x \in V$;

(ii) the $(n+1)$st circuit has exactly $\deg(x)$ departures from every $x \in \mathsf{A}_n$.

Proof. (i) This is immediate from Lemma 11.20.

(ii) For $n = 0$, we have $\mathsf{A}_0 = \{o\}$, and the circuit has exactly $\deg(o)$ departures from o by definition. Let $n \geq 1$. In the rotor configuration at time $T(n)$, the rotor at each $x \in \mathsf{A}_n \setminus \{o\}$ points along the edge traversed most recently from x. Following these rotors from x leads to $X_{T(n)} = o$, since the times of the last departures from the successive vertices increase strictly until the path reaches o. Part (ii) now follows from Lemma 11.21, applied to the walk restarted at time $T(n)$. □

By part (ii), the $(n+1)$st circuit traverses each outgoing edge from every $x \in \mathsf{A}_n$, so in particular it visits all vertices in A_n and in the outer boundary (11.3) of A_n:

$$\mathsf{A}_{n+1} \supseteq \mathsf{A}_n \cup \partial\mathsf{A}_n \qquad \text{whenever the } (n+1)\text{st circuit is complete}. \tag{11.5}$$

See Figure 11.8 for the first three circuit ranges of a rotor walk on $\mathbb{Z}^2$ and the rotors used in the proof of Lemma 11.23(ii).

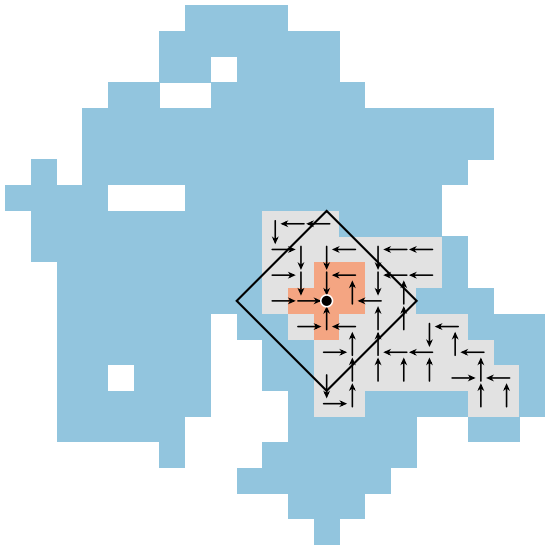


Figure 11.8: The circuit ranges $\mathsf{A}_1 \subseteq \mathsf{A}_2 \subseteq \mathsf{A}_3$ of rotor walk on $\mathbb{Z}^2$ from $\mathbf{0}$ (dot) with the clockwise mechanism and independent uniform initial rotors, each site colored light red, gray or blue by the first of them that contains it; the black outline is $B(\mathbf{0}, 3)$, which lies in A_3 by Lemma 11.25. At time $T(2)$ the rotors on A_2 (arrows) lead to $\mathbf{0}$, and the third circuit visits A_2 and its outer boundary, as (11.5) asserts.

Summing the departures in Lemma 11.23 gives the length of each circuit in terms of the degrees of the vertices it visits.

Proposition 11.24 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex*

with $\deg(o) \geq 1$*, and suppose that every circuit of the rotor walk from* o *is complete. Then, for every integer* $n \geq 0$*,*

$$\sum_{x \in \mathsf{A}_n} \deg(x) \leq T(n+1) - T(n) \leq \sum_{x \in \mathsf{A}_{n+1}} \deg(x)\,.$$

Proof. The length $T(n+1) - T(n)$ of the $(n+1)$st circuit is its number of departures. By Lemma 11.23(ii), it has exactly $\deg(x)$ departures from each $x \in \mathsf{A}_n$, and by Lemma 11.23(i), it has at most $\deg(x)$ departures from each $x \in \mathsf{A}_{n+1}$ and none from other vertices. □

Summing the bounds of Proposition 11.24 over the first n circuits gives

$$\sum_{j=0}^{n-1} \sum_{x \in \mathsf{A}_j} \deg(x) \leq T(n) \leq \sum_{j=1}^{n} \sum_{x \in \mathsf{A}_j} \deg(x)\,.$$

Suppose now that there are real numbers $\beta > 0$ and $q \geq 0$ such that

$$\sum_{x \in \mathsf{A}_n} \deg(x) = \beta n^q + o(n^q) \qquad \text{as } n \to \infty\,.$$

Since $\sum_{j=1}^{n} j^q = n^{q+1}/(q+1) + o(n^{q+1})$, both sides of the previous bound on $T(n)$ then equal $\beta n^{q+1}/(q+1) + o(n^{q+1})$, and hence

$$T(n) = \frac{\beta}{q+1} n^{q+1} + o(n^{q+1}) \qquad \text{as } n \to \infty\,.$$

For $x \in V$ and an integer $r \geq 0$, denote by $B(x, r)$ the set of vertices reachable from x by a directed path of length at most r. Inducting on n using (11.5), we obtain the following.

Lemma 11.25 *Let* $G = (V, E)$ *be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let* o *be a vertex with* $\deg(o) \geq 1$*, and let* $n \geq 0$ *be an integer with* $T(n) < \infty$*. Then* $B(o, n) \subseteq \mathsf{A}_n$*.*

Proof. We argue by induction on n. For $n = 0$, both sets equal $\{o\}$. For the inductive step, let $n \geq 1$ and suppose that $B(o, n-1) \subseteq \mathsf{A}_{n-1}$. The nth circuit is complete, so (11.5) gives $\mathsf{A}_n \supseteq \mathsf{A}_{n-1} \cup \partial\mathsf{A}_{n-1}$. Since $B(o, n-1) \subseteq \mathsf{A}_{n-1}$, the head of every edge leaving $B(o, n-1)$ lies in A_{n-1} or in $\partial\mathsf{A}_{n-1}$, so $\mathsf{A}_{n-1} \cup \partial\mathsf{A}_{n-1}$ contains $B(o, n-1) \cup \partial B(o, n-1)$, which equals $B(o, n)$. □

Observe that rotor walk is recurrent if every circuit is complete: by Lemma 11.25, every vertex lies in A_n for all large n, and by Lemma 11.23(ii), each later circuit visits it. Moreover, each vertex $x \in \mathsf{A}_n$ is visited exactly $\deg(x)$ times during the Nth circuit for each $N \geq n+1$, so we obtain the following.

Proposition 11.26 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, suppose that the rotor walk from o completes every circuit, and write ρ_t for its rotor configuration at time t. Then, for all integers $0 \leq n \leq N$,*

$$\rho_{T(N)}(x) = \rho_{T(n)}(x) \qquad \text{for every } x \in \mathsf{A}_n\,.$$

Proof. By Lemma 11.23(ii) and the inclusions $\mathsf{A}_n \subseteq \mathsf{A}_{n+1} \subseteq \cdots$, the rotor at each $x \in \mathsf{A}_n$ makes one full turn in each of the circuits $n+1, \ldots, N$, so it returns to its position at time $T(n)$. □

The following proposition is a kind of converse to Lemma 11.23 in the case of undirected graphs. Recall the outer boundary ∂S and reflecting boundaries from Section 11.3.

Proposition 11.27 *Let $G = (V, E)$ be a connected, locally finite undirected graph, with each edge replaced by its two orientations. Let $S_1, S_2, \ldots \subset V$ be finite connected sets with $S_{n+1} \supseteq S_n \cup \partial S_n$ for every $n \geq 1$, and let $o \in S_1$ satisfy $\deg(o) \geq 1$. Then there exist a rotor mechanism and initial rotors such that the nth circuit for rotor walk started at o traverses each edge incident to S_n exactly once in each direction, and no other edges.*

Proof. **Step 1:** We first choose a rotor mechanism and initial rotors so that S_n has a reflecting boundary for all $n \geq 1$. This imposes a constraint on the rotor mechanism and configuration only on the boundaries ∂S_n, which are pairwise disjoint, because $\partial S_n \subseteq S_{n+1}$ while ∂S_m is disjoint from $S_m \supseteq S_{n+1}$ for $m > n$. At each $y \in \partial S_n$, choose the cyclic order so that the edges from y into S_n are consecutive, and let the initial rotor at y be the edge just before them. We then define the initial rotors so that the vertices in $S_1 \setminus \{o\}$ form a directed forest pointing towards o and for all $n \geq 1$, the vertices in $S_{n+1} \setminus (S_n \cup \partial S_n)$ form a directed forest pointing towards $S_n \cup \partial S_n$. This is possible because S_{n+1} is connected. Choose all remaining rotors and cyclic orders arbitrarily.

Step 2: We claim that at time $T(n-1)$, the rotors on $S_n \setminus \{o\}$ form paths leading to o and the rotors on ∂S_n are in their initial positions, and that the nth circuit then uses exactly the asserted edges. This follows by induction on $n \geq 1$. For $n = 1$, the first part follows from Step 1, since $T(0) = 0$. Assume the first part at time $T(n-1)$. As in the proof of Lemma 11.17, a step from ∂S_n to $V \setminus S_n$ requires an edge to be traversed twice, so by Lemma 11.4 the walk returns to o for the $\deg(o)$th time, completing the nth circuit, before such a step. The circuit therefore stays in the finite set $S_n \cup \partial S_n$, and by Lemma 11.20 it is complete. By Lemma 11.21, it has exactly $\deg(x)$ departures from every vertex $x \in S_n$, so it uses every edge

with tail in S_n once. Each vertex of ∂S_n is entered once from each of its neighbors in S_n and, since S_n has a reflecting boundary, the walk leaves it once toward each. Hence the circuit uses exactly the edges incident to S_n, once in each direction.

At time $T(n)$, the rotors on S_n are back in their positions at time $T(n-1)$, and the rotors on ∂S_n point into S_n. Together with the unvisited forest on $S_{n+1} \setminus (S_n \cup \partial S_n)$, they form paths from $S_{n+1} \setminus \{o\}$ leading to o. The circuit stayed in $S_n \cup \partial S_n \subseteq S_{n+1}$, so the rotors on ∂S_{n+1} are still in their initial positions. This completes the induction. □

11.5 RANGE OF ROTOR WALK

Throughout this section, $G = (V, E)$ is an infinite, locally finite Eulerian digraph, and we fix a rotor mechanism, an initial rotor configuration ρ, and a starting vertex o with $\deg(o) \geq 1$. The **range** of rotor walk at time t is the set $\mathsf{R}_t := \{X_0, \dots, X_t\}$ of distinct sites visited by the walk. Recall from Section 11.4 the odometer u_t of (11.4), the completion time $T(n)$ of the nth circuit, the circuit ranges A_n, and the set $B(x, r)$ of vertices reachable from x by a directed path of length at most r. In this section we investigate the growth rate of $\#\mathsf{R}_t$ and prove the following theorem.

Theorem 11.28 *Let $c, d > 0$ and $M \geq 1$ be real numbers. Let $G = (V, E)$ be an infinite Eulerian digraph in which every vertex has degree at most M, and suppose that*

$$\#B(x, r) \geq c r^d \qquad \text{for every } x \in V \text{ and every integer } r \geq 1$$

(for example, $\mathbb{Z}^d$). Then the number of distinct sites visited by a rotor walk in t steps satisfies

$$\#\mathsf{R}_t \geq c_1 t^{d/(d+1)} \tag{11.6}$$

for every integer $t \geq 1$, where $c_1 = c_1(c, d, M) > 0$ is a constant (not depending on the starting vertex, the rotor configuration or the mechanism).

Exercise 11.6 asks you to construct initial rotors on $\mathbb{Z}^2$ for which the range grows like $t^{2/3}$, so the exponent cannot be increased. Proposition 11.27, applied on $\mathbb{Z}^d$ with $S_n = [-n, n]^d \cap \mathbb{Z}^d$, shows the same in every dimension: the nth circuit then traverses each of the $2d(n+1)(2n+1)^{d-1}$ edges incident to S_n once in each direction, so the range after t steps has order $t^{d/(d+1)}$. We deduce Theorem 11.28 as a consequence of a more general result.

Theorem 11.29 *Let $G = (V, E)$ be an infinite, locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a*

vertex with $\deg(o) \geq 1$*, and consider the rotor walk from* o*. For integers* $r \geq 0$*, let*

$$w(r) := \sum_{x \in B(o,r)} \deg(x) \qquad \text{and} \qquad W(r) := \sum_{n=0}^{r-1} w(n)\,.$$

For integers $t \geq 1$*, let* $W^{-1}(t)$ *be the smallest integer* $r \geq 0$ *with* $W(r) > t$*, and let* $\deg_{\max}(t)$ *be the largest degree of a vertex of* $B(o,t)$*. Then, for every integer* $t \geq 1$*, the following bounds hold.*

(i) $u_t(o) \leq \deg(o) W^{-1}(t)$*.*

(ii) $u_t(x) \leq \deg(x) W^{-1}(t)$ *for every* $x \in V$*.*

(iii) The range satisfies

$$\#\mathsf{R}_t \geq \frac{t}{W^{-1}(t)\ \deg_{\max}(t)}\,. \tag{11.7}$$

Proof. If the walk completes r circuits, then by Lemmas 11.25 and 11.23(ii), for each $0 \leq n < r$ the $(n+1)$st circuit from o has $\deg(x)$ departures from every vertex x of $B(o,n) \subseteq \mathsf{A}_n$, so the total length of the first r circuits is at least $W(r)$. Therefore, if $t < W(r)$, then the rotor walk has not yet completed its rth circuit at time t. Taking $r = W^{-1}(t)$, we see that the walk has completed fewer than $W^{-1}(t)$ circuits by time t.

(i) and (ii) By Lemma 11.20, each circuit, complete or not, has at most $\deg(x)$ departures from a vertex x. Before time t, the walk has started at most $W^{-1}(t)$ circuits, so $u_t(x) \leq \deg(x) W^{-1}(t)$. Taking $x = o$ gives (i).

(iii) By time t, the walk has visited only vertices of $\mathsf{R}_t \subseteq B(o,t)$. We use part (ii) and write

$$t = \sum_{x \in \mathsf{R}_t} u_t(x) \leq \#\mathsf{R}_t\ W^{-1}(t)\ \deg_{\max}(t)\,,$$

which proves part (iii). □

Proof of Theorem 11.28. Fix a starting vertex o. The volume hypothesis gives $w(n) \geq \#B(o,n) \geq cn^d$ for $n \geq 1$, and $w(0) \geq 1$. Comparing the sum with an integral gives a constant $c_2 = c_2(c,d) > 0$ with $W(r) \geq c_2 r^{d+1}$ for every integer $r \geq 1$, and hence a constant $C = C(c,d)$ with $W^{-1}(t) \leq C t^{1/(d+1)}$ for every integer $t \geq 1$. Substituting this bound and $\deg_{\max}(t) \leq M$ into (11.7) gives

$$\#\mathsf{R}_t \geq \frac{t^{d/(d+1)}}{CM}\,,$$

which is (11.6) with $c_1 = 1/(CM)$. □

On the other hand, if G has exponential volume growth, then the range is at least of order $t/\log t$. Indeed, if there are constants $c_0 > 0$ and $a > 1$ with $\#B(o,r) \geq c_0 a^r$ for every $r \geq 1$, then there is a constant C, depending only on c_0 and a, with $W^{-1}(t) \leq C \log t$ for $t \geq 2$. If moreover every vertex has degree at most M, then this gives $\#\mathsf{R}_t \geq t/(CM \log t)$.

Part (i) of the next theorem gives a sufficient condition for rotor walk to be transient. Part (iii) shows that on a graph of bounded degree, the number of visited sites $\#\mathsf{R}_t$ of a transient rotor walk grows linearly in t.

Theorem 11.30 *Let $G = (V, E)$ be an infinite, locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Let o be a vertex with $\deg(o) \geq 1$, consider the rotor walk from o, and let $\deg_{\max}(t)$ be the largest degree of a vertex of $B(o,t)$.*

(i) If there are distinct vertices $x_0 = o, x_1, x_2, \ldots$ with $\rho(x_k) = (x_k, x_{k-1})$ for every $k \geq 1$, then the walk is transient.

(ii) If the first circuit is not complete by an integer time $t \geq 1$, then $\#\mathsf{R}_t \geq t/\deg_{\max}(t)$.

(iii) If the walk is transient, then there is an integer $t_0 \geq 0$ such that

$$\#\mathsf{R}_t \geq \frac{t - t_0}{\deg_{\max}(t)} \qquad \text{for every integer } t \geq 1\,.$$

Proof. (i) By Lemma 11.22, rotor walk never completes its first circuit from o (see Figure 11.6(b) for an example). By Lemma 11.20, it then visits each vertex x at most $\deg(x)$ times in total, so it is transient.

(ii) If rotor walk has not completed its first circuit by time t, then it visits each vertex x at most $\deg(x) \leq \deg_{\max}(t)$ times before time t by Lemma 11.20. Summing over $x \in \mathsf{R}_t$ gives $t \leq \#\mathsf{R}_t \deg_{\max}(t)$.

(iii) If rotor walk is transient, then for some n it does not complete its nth circuit, since otherwise it would be recurrent, as noted after Lemma 11.25. Let t_0 be the total length of the first $n-1$ circuits, the time at which the nth circuit starts at o. Apply (ii) to the walk from time t_0 on, which is a rotor walk from o whose first circuit never completes, and use $B(o, t-t_0) \subseteq B(o,t)$. For $t \leq t_0$, the asserted lower bound is at most zero. □

11.6 ROTOR WALK ON DIRECTED LATTICES

In this section we consider two different orientations of the square grid $\mathbb{Z}^2$, displayed in Figure 11.9. The ***F*-lattice** has outgoing vertical arrows (north and south) at even sites, those (x, y) with $x + y$ even, and outgoing horizontal

arrows (east and west) at odd sites. The **Manhattan lattice** has every even row pointing east, every odd row pointing west, every even column pointing south and every odd column pointing north. In these two lattices every vertex has two incoming and two outgoing edges, so they are Eulerian, and there is a unique rotor mechanism on each lattice (namely, exits from a given vertex alternate between the two outgoing edges). In **uniform rotor walk**, the initial rotors are independent, and each is uniform on the outgoing edges of its vertex. We show (quantitative) recurrence of uniform rotor walk on both the Manhattan and the F-lattice, by relating it to critical percolation and the Lorentz mirror model.

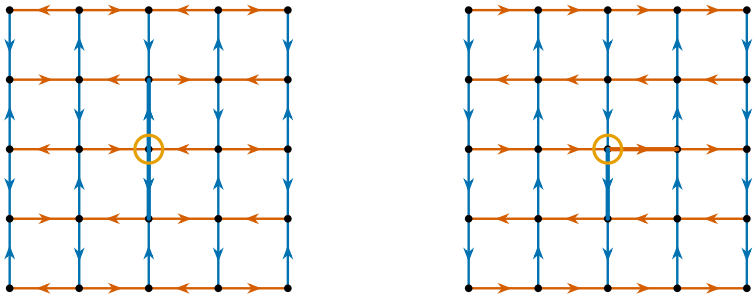


Figure 11.9: The F-lattice (left) and the Manhattan lattice (right), with vertical edges blue and horizontal edges vermilion. Each vertex has two incoming and two outgoing edges. The outgoing edges, thickened at one vertex of each lattice, are parallel in the F-lattice and perpendicular in the Manhattan lattice.

Consider the **half-dual lattice** $\mathbb{L}$, a square grid whose vertices are the points $(x+\frac{1}{2}, y+\frac{1}{2})$ for $x, y \in \mathbb{Z}$ with $x+y$ odd, and whose edges join vertices at Euclidean distance $\sqrt{2}$. We study critical bond percolation on $\mathbb{L}$: each edge of $\mathbb{L}$ is either open or closed, independently with probability $\frac{1}{2}$. Note that each vertex v of $\mathbb{Z}^2$ is the midpoint of a unique edge e_v of $\mathbb{L}$. We consider two different rules for placing two-sided mirrors at the vertices of $\mathbb{Z}^2$.

- Manhattan lattice: If e_v is closed, then v has a mirror oriented parallel to e_v; otherwise v has no mirror.

- F-lattice: Each vertex v has a mirror, which is oriented parallel to e_v if e_v is closed, and perpendicular to e_v if e_v is open.

Consider now the **first-glance mirror walk**: starting at the origin $\mathbf{0}$, it travels along an outgoing edge chosen uniformly at random, independently of the mirrors. On its first visit to each vertex $v \neq \mathbf{0}$, the walker behaves like a light ray: if there is a mirror at v, then the walker reflects by a right angle, and if there is no mirror, then the walker continues straight. At this point the rotor at v is set to the edge used. On all subsequent visits to v, the walker follows

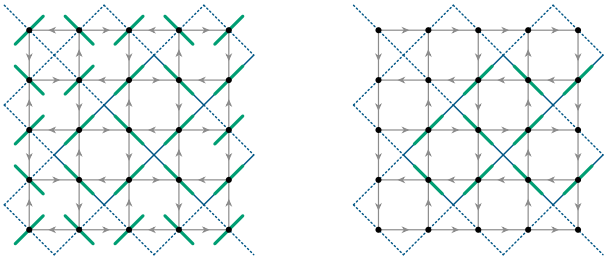


Figure 11.10: A sample of the percolation on the half-dual lattice $\mathbb{L}$ (closed edges solid, open edges dotted) and the resulting mirrors (green) on the F-lattice (left) and on the Manhattan lattice (right). Each vertex v is the midpoint of the edge e_v; on the F-lattice its mirror is parallel to e_v if e_v is closed and perpendicular if it is open, and on the Manhattan lattice only closed edges give mirrors, parallel to them.

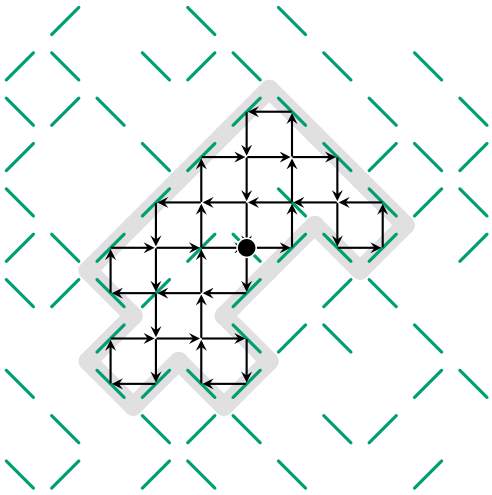


Figure 11.11: Mirror walk on the Manhattan lattice: the first circuit of a first-glance mirror walk, which ends at the origin (large dot) after $T(1) = 44$ steps: first visits follow the two-sided mirrors (green), and later visits follow the rotor mechanism. The mirrors along a cycle of closed edges of $\mathbb{L}$ around the origin (shaded) reflect the walk inward at first visits, so the circuit ends before the walk crosses the cycle, as Lemma 11.32 asserts.

the usual rules of rotor walk (see Figure 11.10 for the mirror assignments and Figure 11.11 for a sample path).

The mirrors are placed so that at a first visit the walk is equally likely to leave along either outgoing edge, and this makes the first-glance mirror walk a uniform rotor walk.

Lemma 11.31 *With the mirror assignments described above, uniform rotor walk started at* $\mathbf{0}$ *on the Manhattan lattice or the F-lattice has the same law as the first-glance mirror walk.*

Proof. The mirror placements are such that the first-glance mirror walk must

follow a directed edge of the corresponding lattice. When the walk first visits a vertex $v \neq \mathbf{0}$, the edge by which it leaves is uniform on the outgoing edges from v; this remains true even if we condition on the past, because the edge e_v has not been inspected before, and changing the status of e_v (open or closed) changes the edge by which the walk leaves. Applying the inverse of the rotor mechanism at v to this edge gives the initial rotor at v, which is therefore uniform and independent of the past. The first departure from $\mathbf{0}$ has the same property, and later departures follow the rotor mechanism, as in uniform rotor walk. □

Given the states of the edges of $\mathbb{L}$, we have described how to set up mirrors and run a rotor walk, using the mirrors to reveal the initial rotors as needed. The next lemma holds for every choice of these states. A cycle of closed edges around the origin acts like a reflecting boundary: the walk cannot cross it before completing a circuit (see Figure 11.11).

Lemma 11.32 *Let $\Lambda_1, \dots, \Lambda_m$ be disjoint cycles of closed edges in $\mathbb{L}$ that surround $\mathbf{0}$, with Λ_j enclosed by Λ_{j+1} for $1 \le j < m$. If the first-glance mirror walk visits a vertex of $\mathbb{Z}^2$ in the unbounded component of $\mathbb{R}^2 \setminus \Lambda_m$, then it completes at least m circuits before that visit.*

Proof. We first show that the walk completes a circuit before it crosses a single closed cycle that it has not yet touched, and then we induct on the number of nested cycles.

Step 1: Let Λ be a cycle of closed edges surrounding $\mathbf{0}$, and suppose that the walk has not yet visited a vertex v with $e_v \in \Lambda$. To cross Λ outward, the walker must leave a vertex v with $e_v \in \Lambda$. Since e_v is closed, the walker reflects off the mirror at v back inward the first time it visits v, so only on a later visit to v does it leave outward. Moreover, the two incoming edges to v are on opposite sides of the mirror, and before the first outward crossing only the incoming edge on the inner side can have been used. Therefore the walker must use this incoming edge twice before crossing. The first edge to be used twice is from the origin by Lemma 11.4, so the walk has completed a circuit before the crossing.

Step 2: For $1 \le j \le m$, let t_j be the time of the first visit to the unbounded component of $\mathbb{R}^2 \setminus \Lambda_j$. We prove by induction on j that $T(j) < t_j$, where, as in Section 11.4, $T(j)$ is the completion time of the jth circuit. For $j = 1$, this is Step 1. For the inductive step, suppose that $T(j-1) < t_{j-1}$. At time $T(j-1)$, the walk is at $\mathbf{0}$ and has not yet visited a vertex v with $e_v \in \Lambda_j$, since Λ_j lies outside Λ_{j-1}. Applying Step 1 to the walk restarted at time $T(j-1)$ gives $T(j) < t_j$. Taking $j = m$ proves the lemma. □

To make a quantitative statement, consider the probability of finding a closed cycle within a given annulus. For $r > 0$, write $Q(\mathbf{0}, r) := [-r, r]^2$.

Theorem 11.33 *There is a constant $\delta > 0$ such that, for every integer $\ell \geq 1$, critical bond percolation on $\mathbb{L}$ contains a cycle of closed edges surrounding the origin in $Q(\mathbf{0}, 3\ell) \setminus Q(\mathbf{0}, \ell)$ with probability at least δ.*

This is a consequence of the Russo–Seymour–Welsh estimate and the Harris inequality, for annuli that are rotated by 45° relative to the edges of $\mathbb{L}$. Exercise 11.7 asks you to prove the theorem from the Harris inequality, planar duality and a lower bound, uniform in the size, on the probability that a rectangle of bounded aspect ratio has a path of open edges joining its two shorter sides; its solution gives the complete argument. Together with Lemma 11.32, the theorem gives the recurrence of uniform rotor walk on both lattices.

Theorem 11.34 *Uniform rotor walk on the Manhattan lattice and on the F-lattice is recurrent almost surely.*

Theorem 11.34 follows from a quantitative bound on returns. Recall from (11.4) that $u_t(\mathbf{0})$ is the number of visits to $\mathbf{0}$ by the first t steps of uniform rotor walk.

Theorem 11.35 *There are constants $a, c > 0$ and $t_0 \geq 1$ with the following property. For uniform rotor walk started at $\mathbf{0}$ on the Manhattan lattice or on the F-lattice, and for every integer $t \geq t_0$,*

$$\mathbb{P}\{u_t(\mathbf{0}) < c \log t\} \leq t^{-a} .$$

Proof. By Lemma 11.31, we may work with the first-glance mirror walk. Both lattices are infinite Eulerian digraphs of degree 2. Let $\delta > 0$ be the constant of Theorem 11.33, fix an integer $L \geq 1$ with $(1 - \delta)^L \leq 1/16$, and for an integer $t \geq 2$ let $J := \lfloor (\log t)/(8L \log 9) \rfloor$. We group the annuli $Q(\mathbf{0}, 3 \cdot 9^j) \setminus Q(\mathbf{0}, 9^j)$ for $1 \leq j \leq LJ$ into J groups of L. The annuli are disjoint, so the events that the groups contain no closed cycle surrounding $\mathbf{0}$ are independent, and by Theorem 11.33 each has probability at most $(1 - \delta)^L \leq 1/16$.

Suppose that at least $J/2$ groups contain such a cycle. One cycle in each of $\lceil J/2 \rceil$ of them gives disjoint nested cycles in $Q(\mathbf{0}, 3 \cdot 9^{LJ})$, so by Lemma 11.32 the walk stays in this square until it completes $\lceil J/2 \rceil$ circuits. By Lemma 11.20, during these circuits it departs at most twice per circuit from each of the $(6 \cdot 9^{LJ} + 1)^2 \leq 49 t^{1/4}$ lattice points of the square, so the circuits finish within $98 J t^{1/4} \leq 98 t^{1/4} \log t < t$ steps for large t. Each completed circuit has two departures from $\mathbf{0}$, so $u_t(\mathbf{0}) \geq J$.

Hence if $u_t(\mathbf{0}) < J$, then there is a set of more than $J/2$ groups that all fail. Each such set fails with probability at most $16^{-J/2}$, and there are at most 2^J

sets, so

$$\mathbb{P}\{u_t(\mathbf{0}) < J\} \le 2^J 16^{-J/2} = 2^{-J}\,.$$

Choose t_0 so that $98t^{1/4}\log t < t$ and $J \ge (\log t)/(16L\log 9)$ for $t \ge t_0$. Take $c := 1/(16L\log 9)$ and $a := c\log 2$; since $J \ge c\log t$, the last bound proves the theorem. □

Proof of Theorem 11.34. For integers $m \ge 0$ and $t \ge t_0$ with $c\log t > m$, Theorem 11.35 gives

$$\mathbb{P}\Big\{\sup_{r\ge 0} u_r(\mathbf{0}) \le m\Big\} \le t^{-a}\,.$$

Taking $t \to \infty$, we see that the left side, which does not depend on t, is zero. By countable subadditivity over m, the walk departs from $\mathbf{0}$ infinitely often almost surely. A vertex visited infinitely often is left along each of its outgoing edges infinitely often, so by strong connectivity the walk is recurrent. □

Although we used the same technique to show that uniform rotor walk on these two lattices is recurrent, experiments suggest that the behavior of the two walks is rather different: the number of distinct sites visited in t steps appears to be of order $t^{2/3}$ on the Manhattan lattice but of order t on the F-lattice. This difference is clearly visible in Figure 11.12 (on the Manhattan lattice the bands of equal increments of $t^{1/3}$ have nearly equal widths, as they would if the radius of the range grew like $t^{1/3}$), but it remains unproved.

11.7 ESCAPE RATE OF ROTOR WALK

Throughout this section, $G = (V, E)$ is a strongly connected digraph with at least two vertices and finite in-degree and out-degree at every vertex, and we fix a vertex o, a rotor mechanism and an initial rotor configuration ρ.

To quantify the degree of transience of ρ, consider the following experiment, which we call the **infinite experiment**: let each of n particles in turn perform rotor walk starting from o until either returning to o or escaping to infinity. (Importantly, we do not reset the rotors in between trials.) If a particle visited a vertex infinitely often, then it would leave along every outgoing edge of that vertex infinitely often, and by strong connectivity it would return to o. Thus a particle that never returns visits each vertex finitely often, so its rotors converge pointwise, and the next particle starts from these limiting rotors. Denote by $I(\rho, n)$ the number of these particles that escape to infinity, that is, never return, and let $N_n(v)$ count all departures from $v \in V$ by the n particles.

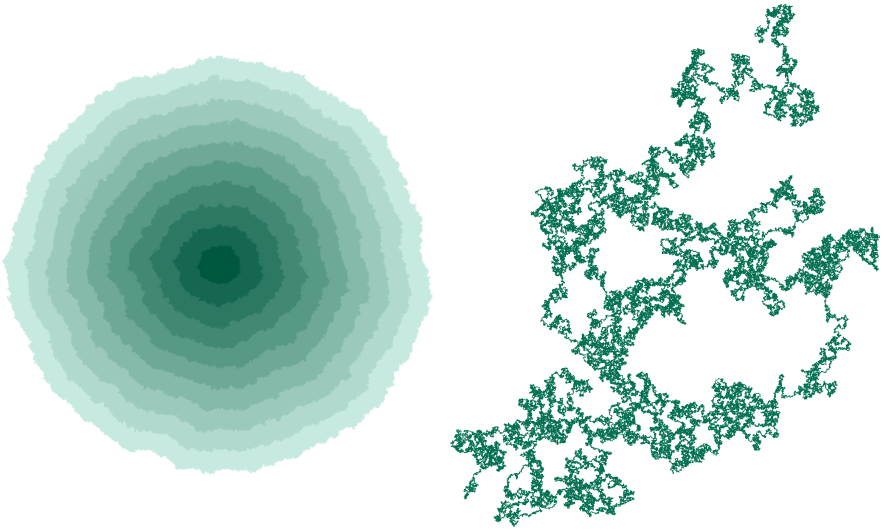


Figure 11.12: Set of sites visited by uniform rotor walk run for 5×10^{10} steps on the Manhattan lattice (left) and 10^7 steps on the F-lattice (right), in windows of about $18{,}300 \times 17{,}800$ and $7{,}800 \times 9{,}500$ sites. On the left each visited site is shaded from dark to light by the cube root of its first visit time, in ten bands of equal width; on the right every visited site is dark green.

Theorem 11.36 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. Let $(X_k)_{k\geq 0}$ be simple random walk on G started at o. Then*

$$\limsup_{n\to\infty} \frac{I(\rho, n)}{n} \leq \mathbb{P}_o\{X_k \neq o \text{ for every } k \geq 1\}\,. \tag{11.8}$$

We bound the numbers of arrivals using the rotor mechanism and compare them with expected visits of random walk stopped at its first return.

Proof of Theorem 11.36. For $v \in V$, let $Y_n(v)$ count arrivals at v by the first n particles, excluding their initial placements. These counts are finite by the construction above. Each particle leaves o once and leaves every other vertex whenever it arrives, so $N_n(o) = n$ and $N_n(v) = Y_n(v)$ for $v \neq o$. The arrivals at o are the returns, so $Y_n(o) = n - I(\rho, n)$.

Let $p(w, v)$ be the random-walk transition probability from w to v; sums over w below include only in-neighbors of v. Since rotors are never reset, each outgoing edge of w is used $N_n(w)/\deg(w)$ times up to an error less than one. Summing over incoming edges at v gives

$$\left| Y_n(v) - n\,p(o, v) - \sum_{w\neq o} Y_n(w)\,p(w, v) \right| \leq \operatorname{indeg}(v)\,,$$

where $\mathrm{indeg}(v)$ is the in-degree of v. Let $\alpha(v) := \liminf_{n\to\infty} Y_n(v)/n \in [0,\infty]$. Since the lower limit of a finite sum is at least the sum of its lower limits, dividing by n gives

$$\alpha(v) \geq p(o,v) + \sum_{w\neq o} \alpha(w)\, p(w,v) \qquad \text{for every } v \in V\,. \tag{11.9}$$

Next we compare α with expected visits up to the first return. Let $\tau_o^+ := \inf\{k \geq 1 : X_k = o\}$, and for $m \geq 0$ let

$$g_m(v) := \sum_{k=1}^{m} \mathbb{P}_o\{X_k = v,\ k \leq \tau_o^+\}\,.$$

For $k \geq 2$, an arrival at time $k \leq \tau_o^+$ comes from $w \neq o$. The Markov property therefore gives

$$g_{m+1}(v) = p(o,v) + \sum_{w\neq o} g_m(w)\, p(w,v)\,.$$

We prove by induction on m that $\alpha \geq g_m$ for every $m \geq 0$. The base case is $\alpha \geq 0 = g_0$. For the inductive step, if $\alpha \geq g_m$, then the nonnegativity of $p(w,v)$, the recurrence for g_{m+1}, and (11.9) give $\alpha \geq g_{m+1}$. At $v = o$ this reads

$$1 - \limsup_{n\to\infty} \frac{I(\rho,n)}{n} = \alpha(o) \geq g_m(o) = \mathbb{P}_o\{\tau_o^+ \leq m\}\,.$$

By continuity of probability along the increasing events $\{\tau_o^+ \leq m\}$, letting $m \to \infty$ proves (11.8). □

11.7.1 Comparison of the Finite and Infinite Experiments*

One way to estimate the number of escapes to infinity is to look at how many particles exit a large ball before returning to o. The proof of Theorem 11.36 did not need this, and the rest of the chapter does not use this subsection, except for Exercise 11.9.

Recall from Section 11.4 that $B(o,r)$ is the set of vertices reachable from o by a directed path of length at most r, and from (11.3) that $\partial B(o,r)$ is the set of heads of the edges that leave $B(o,r)$. For an integer $r \geq 1$, starting from the initial rotor configuration ρ, let each of n particles in turn perform rotor walk starting from o until either returning to o or first visiting $\partial B(o,r)$; we call this the **finite experiment** (see Figure 11.13 for its final rotors on $\mathbb{Z}^2$). Denote by $I_r(\rho,n)$ the number of particles that visit $\partial B(o,r)$, and for $x \in V$ let $N_{n,r}(x)$ be the total number of departures from x made by the n particles.

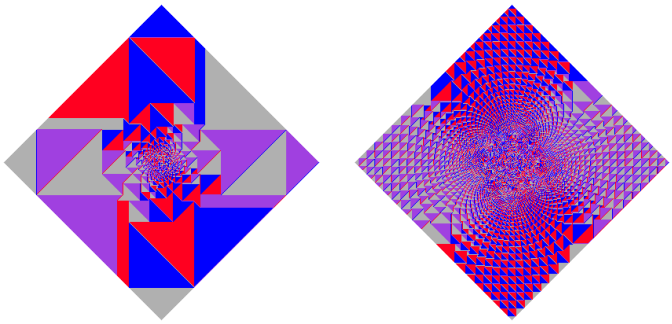


Figure 11.13: Final rotors in $B(\mathbf{0}, 200) \subset \mathbb{Z}^2$ after n particles started from the origin performed rotor walk with the clockwise mechanism until returning to the origin or exiting $B(\mathbf{0}, 200)$. Initial rotors pointed west. Rotors are colored by direction (North blue, East purple, South red, West gray). Left: $n = 3000$, of which 694 exit. Right: $n = 30000$, of which 7090 exit.

To compare the two experiments, we count departures. A particle that never returns to o visits each vertex finitely often, so $N_n(x)$ is finite. Since $B(o, r)$ is finite (out-degrees are finite), a particle of the finite experiment that never stopped would visit some vertex of $B(o, r)$ infinitely often and hence return to o. So every particle of the finite experiment stops in finitely many steps, and $N_{n,r}(x)$ is finite.

Lemma 11.37 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. For all integers $n, r \geq 1$, the numbers of departures in the finite and the infinite experiment satisfy $N_{n,r} \leq N_n$ at every vertex.*

Proof. Suppose, for a contradiction, that $N_{n,r}(x) > N_n(x)$ for a vertex x, and consider the first departure of the finite experiment after which its count at the tail x of the departure exceeds $N_n(x)$. In both experiments each particle departs from o exactly once, so $x \neq o$. Before this departure, each vertex has been left at most as often as in the infinite experiment. Each vertex sends particles along the same sequence of edges in both experiments, so x has received at most as many particles as it receives in the infinite experiment, namely $N_n(x)$, because every particle that enters $x \neq o$ in the infinite experiment also leaves it. But $N_n(x)$ particles have already left x, so no particle is present at x, a contradiction. □

The number of particles that return to o is the number of particles sent along the edges into o. Each vertex sends particles along the same sequence of edges in both experiments, so Lemma 11.37, applied at the finitely many in-neighbors of o, shows that at most as many particles return in the finite experiment as in the infinite one. In the finite experiment each particle either

returns or is counted by $I_r(\rho, n)$, and in the infinite experiment each either returns or is counted by $I(\rho, n)$. Hence

$$I(\rho, n) \le I_r(\rho, n) \qquad \text{for all integers } n, r \ge 1 .$$

Next we show that, for fixed n, the finite experiment agrees with the infinite one near o once the ball is large.

Lemma 11.38 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. For every integer $n \ge 0$ and every finite set $K \subset V$, the numbers of departures in the finite and the infinite experiment satisfy $N_{n,r} = N_n$ on K for all sufficiently large r.*

Proof. We argue by induction on n. For $n = 0$, both counts vanish. For the inductive step, let $n \ge 1$ and assume the claim for $n - 1$ and every finite set. In the infinite experiment, the nth particle departs from K only finitely often. Choose a finite initial segment of its route that contains all these departures, and let L be the union of K and the vertices of this segment. By the inductive hypothesis, applied to L, the two experiments have made the same numbers of departures from every vertex of L after $n - 1$ particles, for all large r. Their rotors on L therefore agree at that time, and if also $L \subseteq B(o, r)$, then the nth particle of the finite experiment follows the chosen segment. Hence $N_{n,r} \ge N_n$ on K for all large r, and Lemma 11.37 gives equality. □

The next lemma gives the limit of $I_r(\rho, n)$ as $r \to \infty$.

Lemma 11.39 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. For every integer $n \ge 1$, we have $I_r(\rho, n) = I(\rho, n)$ for all sufficiently large r.*

Proof. The number of particles that return to o is determined by the numbers of departures from the in-neighbors of o, because each vertex sends particles along a fixed sequence of edges. Lemma 11.38, applied to the finite set of in-neighbors of o, shows that these numbers agree in the two experiments for all large r. Hence the same number of particles returns, and $I_r(\rho, n) = I(\rho, n)$. □

11.8 LIVE PATHS AND RECURRENCE

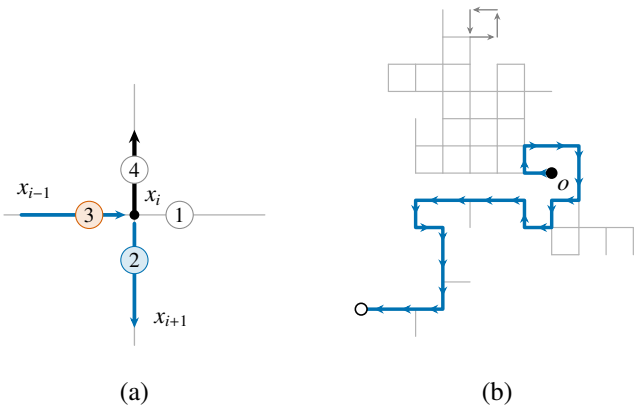
Throughout this section, G is a connected, locally finite, simple undirected graph with at least two vertices, with each edge replaced by its two orientations, and we fix a rotor mechanism and initial rotors. For neighbors v and w,

we write $r_v(w)$ for the position of the edge (v, w) in the cyclic order of the outgoing edges of v that starts immediately after the initial rotor. The walk leaves v along these edges in increasing order of r_v, one edge per visit.

A finite or infinite path $x_0, x_1, \ldots$ of distinct vertices is **live** if

$$r_{x_i}(x_{i+1}) < r_{x_i}(x_{i-1})$$

at every internal vertex x_i. In words, the rotor at each internal vertex points forward along the path before it points back. The absence of infinite live paths gives a criterion for recurrence: if the walk never returns to its start, then it crosses each edge at most once in each direction, and what remains after cancelling the opposite crossings contains an infinite live path (see Figure 11.14).



(a) (b)

Figure 11.14: (a) With the clockwise mechanism and the initial rotor at x_i pointing North (bold), the positions r_{x_i} (numbers) of East, South, West and North are $1, 2, 3, 4$, and the path is live at x_i because $r_{x_i}(x_{i+1}) = 2 < 3 = r_{x_i}(x_{i-1})$. (b) A rotor walk on $\mathbb{Z}^2$ from o with the clockwise mechanism and independent uniform initial rotors, stopped after 138 steps without returning to o: cancelling the edges crossed in both directions (gray) leaves crossings (arrows) that contain a live path (blue) from o to the walker (circle).

Proposition 11.40 *Let G be a connected, locally finite, simple undirected graph with at least two vertices. Fix a rotor mechanism and initial rotors such that there is no infinite live path. Then rotor walk from every vertex is recurrent and completes every circuit.*

Proof. We first show that the walk returns to its starting vertex o. Suppose that it never returns. By Lemma 11.4, the first directed edge that the walk traverses twice leaves o. Since the walk never returns to o, no directed edge is traversed twice, and every vertex is visited finitely often.

Cancel the opposite traversals of each edge. In the directed graph of the remaining traversals, the number of outgoing edges minus the number of

incoming edges is 1 at o and 0 at every other vertex. Infinitely many vertices are reachable from o in this graph. Otherwise, summing these differences over the finite set of reachable vertices gives 1, whereas no edge of the graph leaves that set, so the sum is at most 0.

Consequently this graph contains an infinite simple directed path from o. Indeed, the finite simple directed paths from o have unbounded lengths and finitely many extensions at each step, so we can repeatedly choose an extension that has arbitrarily long continuations. At each internal vertex of the resulting path, the walk traversed the outgoing edge of the path but not the reverse of its incoming edge. The walk leaves a vertex v in increasing order of r_v, so the outgoing edge precedes the reverse of the incoming edge, and the path is live. This contradicts the hypothesis, so the walk returns to o.

Changing finitely many rotors does not create an infinite live path, because an infinite simple path has a tail that avoids the changed vertices. After each return to o, only finitely many rotors have changed, so the same argument shows that the walk returns again. Hence the walk returns to o infinitely often and completes every circuit. A vertex visited infinitely often is left toward each of its neighbors infinitely often, so by connectedness every vertex is visited infinitely often, and the walk is recurrent. □

Next we rule out infinite live paths for random initial rotors.

11.9 RECURRENCE OF UNIFORM ROTOR WALK

Recall from Section 11.6 that in uniform rotor walk the initial rotors are independent, and each is uniform among the outgoing edges of its vertex. We show that uniform rotor walk has almost surely no infinite live path, and is therefore recurrent, on every graph of maximum degree at most three and on $\mathbb{Z}^2$ with the clockwise mechanism. At a vertex v of degree at most three and for a fixed neighbor u, the position $r_v(u)$ of Section 11.8 is uniform, so a live path arriving from u has on average at most one way to continue. We explore these continuations with a queue, and we show that the exploration ends by comparing the length of the queue with a random walk on $\mathbb{Z}$ (see Figure 11.15 for uniform rotor walk on the square lattice and on two lattices of degree three).

Corollary 11.41 *Let G be a connected, simple undirected graph with at least two vertices and maximum degree at most three. Fix a rotor mechanism and take independent uniform initial rotors. Then almost surely there is no infinite live path, and rotor walk from every vertex is recurrent.*

Proof. Explore a fixed edge and queue its two endpoints. To process the first vertex v in the queue, remove all its entries, examine the rotor at v if it has not

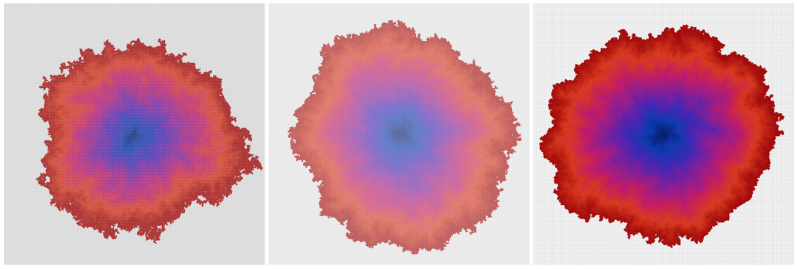


Figure 11.15: The sites visited by uniform rotor walk with the clockwise mechanism on the square, honeycomb and square–octagon lattices after 4×10^6 steps, each colored by the circuit in which it was first visited, from blue for the early circuits to red for the late ones. The last two lattices have degree three, so these walks are recurrent by Corollary 11.41, and the first is recurrent by Theorem 11.42.

been examined, and explore every unexplored edge $\{v, w\}$ with $r_v(w) < r_v(u)$ for an explored edge $\{v, u\}$. Then queue the other endpoints of the newly explored edges. Thus one entry is queued for each endpoint of each explored edge, except the vertex processed when the edge was explored, so v receives at most $\deg(v)$ entries and is processed at most $\deg(v)$ times.

Step 1: We show that the queue becomes empty almost surely. Let $U_1, U_2, \ldots$ be independent uniform random variables on $[0, 1)$, and let $Z_i := \lfloor 3U_i \rfloor - 1$, which is uniform on $\{-1, 0, 1\}$. Let v be the ith vertex processed for the first time, and use U_i to sample the rotor at v. At this processing, the k removed entries correspond to the k explored edges at v. If $k = 1$, the position $r_v(u)$ of the explored edge $\{v, u\}$ is uniform, so we may sample it as $\lfloor \deg(v)U_i \rfloor + 1$, and the queue length changes by $\lfloor \deg(v)U_i \rfloor - 1 \le Z_i$. If $k \ge 2$, at most $\deg(v) - k$ edges are newly explored, so it changes by at most $\deg(v) - 2k \le -1 \le Z_i$. At a later processing of v, there are an old explored edge at v and a newly explored one, so at most one entry is added, at least one is removed, and the queue length does not increase. Thus after m first processings the queue length is at most $2 + Z_1 + \cdots + Z_m$, a random walk on $\mathbb{Z}$ started at 2.

For $j \ge 0$ let h_j be the probability that this walk started at j reaches 0. Then $h_0 = 1$, and conditioning on the first step gives $2h_j = h_{j-1} + h_{j+1}$ for $j \ge 1$. So h_j is affine in j, and being bounded, it is constant, equal to $h_0 = 1$. If the queue never became empty, there would be infinitely many first processings, because each vertex is processed finitely often, and the walk would stay positive forever, which has probability 0. Hence the queue becomes empty almost surely, after finitely many edges have been explored.

Step 2: We show that live paths from the fixed edge use only explored edges. After v is processed, the explored edges at v form an initial segment in the order of r_v, and v is queued whenever an edge at v is explored while a neighbor is processed. So when the queue is empty, the explored edges at

every vertex v form an initial segment in the order of r_v. Let $x_0, x_1, \ldots$ be a live path whose first edge is the fixed edge, which is explored. If $\{x_{i-1}, x_i\}$ is explored for $i \geq 1$, then so is $\{x_i, x_{i+1}\}$, because $r_{x_i}(x_{i+1}) < r_{x_i}(x_{i-1})$. By induction on i, the path uses only explored edges, so it is finite (Figure 11.16). The graph has countably many edges, since it is connected and locally finite. Hence almost surely there is no infinite live path, and Proposition 11.40 completes the proof. □

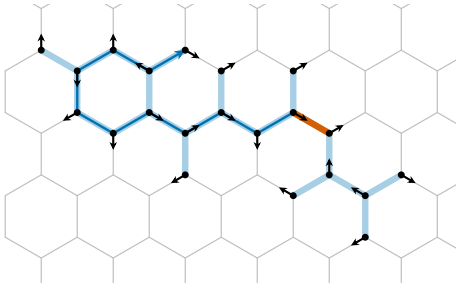


Figure 11.16: The exploration in the proof of Corollary 11.41 on the honeycomb lattice with the clockwise mechanism and independent uniform initial rotors (black arrows, at the processed vertices): from the fixed edge (vermilion), the queue becomes empty after 20 further edges are explored (light blue). Every live path whose first edge is the fixed edge uses only explored edges, and the blue arrows show a longest one.

This applies to every graph of maximum degree at most three, periodic or not. On the square lattice, where vertices have degree four, a live path has on average $3/2$ ways to continue, and we use cycles that surround the origin instead.

11.9.1 The Square Lattice*

On $\mathbb{Z}^2$ with the clockwise mechanism, a live path cannot cross a suitable clockwise cycle from inside to outside. We find such cycles around every finite set by transferring the closed cycles of Theorem 11.33 to the rotors.

Theorem 11.42 *Equip $\mathbb{Z}^2$ with the clockwise rotor mechanism and independent uniform initial rotors. Then almost surely there is no infinite live path, and rotor walk from every vertex is recurrent.*

Proof. We use only the outgoing edges in positions 1 and 4 at each vertex. For a uniform initial rotor these are one horizontal edge and one vertical edge, each direction chosen by a fair coin, the two coins independent, and the choices at different vertices independent. We show that, almost surely, infinitely many clockwise cycles of such edges surround the origin, and that a live path cannot leave the region bounded by such a cycle.

Step 1: We find surrounding clockwise cycles of these edges. The affine map

$$(s,t) \longmapsto \Big(\frac{s-t+1}{2}, \frac{s+t}{2}\Big)$$

identifies the half-dual lattice $\mathbb{L}$ of Section 11.6 with $\mathbb{Z}^2$. Recall that $Q(\mathbf{0},\ell) = [-\ell,\ell]^2$; the map takes this square to the diamond $\{(x,y) : |x-\frac{1}{2}| + |y| \le \ell\}$. For integers $j \ge 1$, consider the image of the annulus

$$Q(\mathbf{0}, 3\cdot 9^j) \setminus Q(\mathbf{0}, 9^j)\,.$$

These annuli are disjoint, and their inner diamonds exhaust the plane. Theorem 11.33 gives a constant $\delta > 0$, independent of j. Critical bond percolation is unchanged when open and closed bonds are exchanged, so each of these annuli contains, with probability at least δ, a simple open bond cycle that surrounds its inner diamond.

Fix one annulus. We pass from bond percolation to rotor edges in two replacements. Start from critical bond percolation on the edges incident to the annulus, and orient each open bond in both directions. First, replace the two identical bits of each bond, one for each orientation, by two independent fair bits. Then, at each vertex of the annulus, replace its two opposite horizontal bits by two complementary fair bits, and likewise its two vertical bits. After both replacements, the outgoing edges chosen at each vertex have the law of its edges in positions 1 and 4.

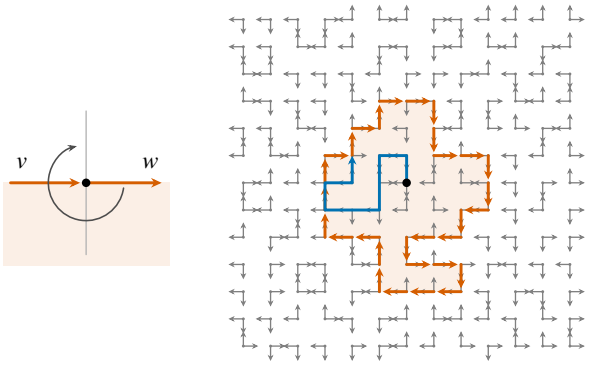
Neither replacement decreases the probability that the annulus contains a simple clockwise cycle that surrounds its inner diamond. To see this, condition on all bits outside one pair being replaced. A simple cycle uses at most one bit of the pair, so the conditional event either is constant, or requires one particular bit, or requires at least one of the two bits. A particular bit has probability $1/2$ before and after the replacement. The probability that at least one of the two bits is set increases from $1/2$ to $3/4$ in the first replacement and from $3/4$ to 1 in the second. Hence, with probability at least δ, the annulus contains such a cycle of edges in positions 1 and 4. These events depend on the rotors in disjoint sets of vertices, so they are independent, and the Borel–Cantelli lemma gives infinitely many such cycles almost surely.

Step 2: We show that a live path cannot leave the region bounded by such a cycle. At a vertex of such a cycle, let v be its predecessor and w its successor on the cycle. The clockwise order of the directions at the vertex is

$$w, \quad \text{the directions inside the cycle}, \quad v, \quad \text{the directions outside the cycle}\,.$$

Since w is in position 1 or 4, every inside direction precedes v, and v precedes every outside direction. A live path that arrives at the vertex from inside, or

from v, must therefore leave toward the inside or toward w. So a live path that starts strictly inside the cycle stays in the closed region it bounds, which contains only finitely many vertices (Figure 11.17). The cycles surround every finite set, so an infinite live path would start inside one of them, a contradiction. Hence there is almost surely no infinite live path, and Proposition 11.40 gives recurrence. □



(a) (b)

Figure 11.17: (a) In Step 2 of the proof of Theorem 11.42, at a vertex of a clockwise cycle (vermilion) with predecessor v and successor w, the directions in clockwise order (gray arrow) are w, the inside direction (shaded), v and the outside direction. (b) The edges in positions 1 and 4 (gray) of independent uniform initial rotors near the origin (dot), a clockwise cycle of 30 of them around it, and a longest live path from the origin (blue); every live path from the origin stays in the shaded region.

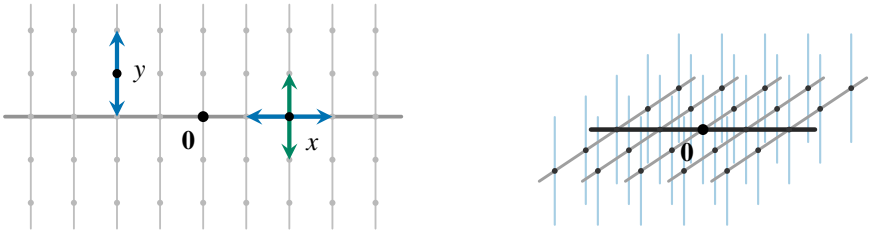
11.10 UNIFORM ROTOR WALK ON THE COMB

Throughout this section, $d \geq 2$ is an integer. On the comb in $\mathbb{Z}^d$ defined below, the ball of radius r about the root contains of order r^d vertices, so by Theorem 11.29, rotor walk from the root visits at least of order $t^{d/(d+1)}$ sites in t steps. We show that, with suitable cyclic orders, uniform rotor walk attains this order and its range is close to a diamond; Exercise 11.11 concerns other cyclic orders. The comb is a tree, and on a tree the walk is easy to follow. The rotor at a newly visited vertex sends the walker to the children that precede the parent, in the cyclic order after the initial rotor, and then back to the parent; on each later visit from the parent, it sends the walker to all the children. We use this to label each vertex by the circuit in which it is first visited. Under the assumption on cyclic orders below, each nonroot label is one plus a sum of independent indicators, all but at most $d - 1$ of which are fair coins, and by concentration of these sums, every circuit is complete and the range is close to a diamond.

For $x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$, let $j(x)$ be the largest index with $x_{j(x)} \neq 0$, and let

$$\mathrm{par}(x) := x - \mathrm{sgn}(x_{j(x)})\, \mathbf{e}_{j(x)} .$$

The **comb** is the graph on $\mathbb{Z}^d$ that joins each $x \neq \mathbf{0}$ to $\mathrm{par}(x)$. Each step from x to $\mathrm{par}(x)$ decreases $\|x\|_1$ by one, so following these steps from x leads to $\mathbf{0}$, and a cycle would contain a vertex of largest norm with two parents. Thus the comb is a tree with root $\mathbf{0}$ and parent map par, and its graph distance from $\mathbf{0}$ to x is $\|x\|_1$. In the plane it is the x-axis with a vertical tooth through each of its lattice points. The children of $x \neq \mathbf{0}$ are $2x - \mathrm{par}(x)$ and $x \pm \mathbf{e}_k$ for $j(x) < k \leq d$, and the children of the root are its $2d$ lattice neighbors. Along a path away from the root, the coordinate direction changes at most $d - 1$ times (see Figure 11.18).



(a) (b)

Figure 11.18: (a) The comb in $\mathbb{Z}^2$. At y and at x, blue arrows point toward the parent and toward the child that continues the same line; green arrows begin the tooth through x. (b) The comb in $\mathbb{Z}^3$, with a tooth in direction $\mathbf{e}_3$ attached to each vertex of the comb in $\mathbb{Z}^2$.

Since the comb distance from $\mathbf{0}$ to x is $\|x\|_1$, the ball $B(\mathbf{0}, R)$ of Section 11.4 is the set $\{x \in \mathbb{Z}^d : \|x\|_1 \leq R\}$, and we use this description for every real R, so $B(\mathbf{0}, R) = \varnothing$ for $R < 0$. Comparing unit cubes centered at its points with real diamonds whose radii differ by $d/2$ gives

$$\#B(\mathbf{0}, R) = \frac{2^d}{d!} R^d + O((R+1)^{d-1}) \qquad \text{for } R \geq 0 ,$$

since the real diamond of radius R has volume $2^d R^d / d!$. Recall from Section 11.4 that a circuit ends when the rotor at the root has made a full turn and the walker has returned to the root, that $T(n)$ is the completion time of the nth circuit, and that A_n is the set of vertices it visits, with $T(0) = 0$ and $\mathsf{A}_0 = \{\mathbf{0}\}$. An outgoing edge of a vertex precedes another if it comes earlier in the cyclic order that starts just after the initial rotor and ends with it.

We now define the labels. Let $\mathsf{h}(\mathbf{0}) := 0$, and for $x \neq \mathbf{0}$ with path

$\mathbf{0} = v_0, v_1, \dots, v_r = x$ from the root, let

$$\mathsf{h}(x) := 1 + \sum_{i=1}^{r-1} \mathbf{1}\{(v_i, v_{i-1}) \text{ precedes } (v_i, v_{i+1})\}\,. \tag{11.10}$$

These numbers are read off the initial rotors, whether or not the walk ever returns. The neighbors of the root have label 1, and every other child has the label of its parent or that label plus one, so the labels are nondecreasing along paths away from the root. The next lemma shows that, as long as the sets $\{\mathsf{h} \le n\}$ are finite, $\mathsf{h}(x)$ is the index of the circuit in which x is first visited.

Lemma 11.43 *Let $d \ge 2$ be an integer. Fix a rotor mechanism and initial rotors on the comb in $\mathbb{Z}^d$, and let $n \ge 1$ be an integer such that the set $\{x : \mathsf{h}(x) \le n\}$ is finite. Then the first n circuits are complete,*

$$\mathsf{A}_k = \{x : \mathsf{h}(x) \le k\} \qquad \textit{for } 0 \le k \le n\,, \tag{11.11}$$

at the completion of circuit n every visited vertex other than the root has its rotor pointing to its parent, and

$$T(n) = 2 \sum_{k=1}^{n} (\#\mathsf{A}_k - 1)\,. \tag{11.12}$$

Proof. We prove by induction on k, for $0 \le k \le n$, that circuit k is complete, that it traverses each edge of the finite tree $\{\mathsf{h} \le k\}$ once in each direction and no other edge, and that at its completion every visited vertex other than the root has its rotor pointing to its parent. The case $k = 0$ is immediate, since then only the root has been visited.

For the inductive step, let $1 \le k \le n$ and assume the claim for $k - 1$. To complete circuit k, the walk must leave the root once toward each of its children and return after each departure. At a vertex $v \ne \mathbf{0}$ reached from its parent during this circuit, there are two cases. If $\mathsf{h}(v) < k$, then by the inductive hypothesis its rotor points to its parent, so the departures from v before the next one to its parent go to each child once, and each child has label at most $\mathsf{h}(v) + 1 \le k$. If $\mathsf{h}(v) = k$, then v has not been visited before, its rotor is untouched, and its departures before the first one to its parent go exactly to the children with label k; its other children have label $k + 1$. In both cases the departures from v before the next one to its parent go exactly to its children in $\{\mathsf{h} \le k\}$, once each. We now induct on the height of the finite subtree of $\{\mathsf{h} \le k\}$ rooted at v. If v is a leaf of this subtree, then its next departure is to its parent. For the inductive step, each excursion from v into a child crosses the edge to that child and, by the inductive hypothesis on the height, traverses the child's subtree once in each direction before returning to

v. After these excursions, the walk moves from v to its parent, and the rotor at v points there. Consequently the rotor at the root completes its turn, circuit k is complete, and its range is $\{\mathsf{h} \le k\}$, which proves the claim for k.

Finally, the finite tree A_k has $\#\mathsf{A}_k - 1$ edges, so circuit k has length $2(\#\mathsf{A}_k - 1)$, and summing over k gives (11.12). □

See Figure 11.19 for the labels of a sample of initial rotors.

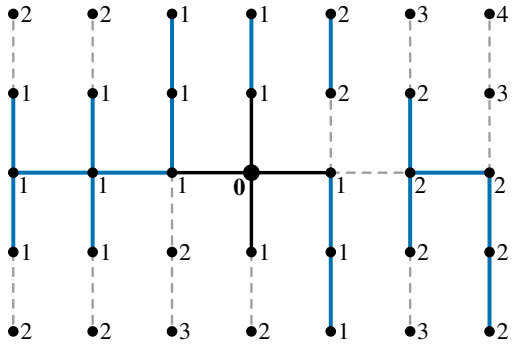


Figure 11.19: The labels $\mathsf{h}(x)$ for a sample of uniform initial rotors on the comb in $\mathbb{Z}^2$ with the clockwise mechanism. Away from the root, a blue edge from a vertex to a child precedes the edge from that vertex to its parent, so its endpoints have equal labels; a dashed edge increases the label by one. The root has label 0 and its four neighbors have label 1.

To show that the sets $\{\mathsf{h} \le n\}$ are finite and close to diamonds, we use the following concentration estimate for sums of bounded independent variables.

Lemma 11.44 *Let $N \ge 1$ be an integer, and let Y be a sum of N independent random variables with values in $[0, 1]$. Then, for every real $u \ge 0$,*

$$\mathbb{P}\{|Y - \mathbb{E}Y| > u\} \le 2e^{-u^2/(4N)}.$$

Proof. We bound exponential moments and apply Markov's inequality. Let $Z_1, \dots, Z_N$ be the independent centered summands of Y. Then $Y - \mathbb{E}Y = \sum_{i=1}^{N} Z_i$, and each Z_i has mean zero and absolute value at most one. If $u \ge N$, then the left side vanishes, because $|Y - \mathbb{E}Y| \le N$, so suppose $u < N$. For $|z| \le 1$ and $0 \le \lambda \le 1$, the exponential series gives $e^{\lambda z} \le 1 + \lambda z + \lambda^2$, because the terms of degree at least two have total absolute value at most $\lambda^2 \sum_{k \ge 2} 1/k! < \lambda^2$. Hence $\mathbb{E}e^{\lambda Z_i} \le 1 + \lambda^2 \le e^{\lambda^2}$. Take $\lambda := u/(2N) < 1/2$. By independence and Markov's inequality,

$$\mathbb{P}\{Y - \mathbb{E}Y > u\} \le e^{-\lambda u} \prod_{i=1}^{N} \mathbb{E}e^{\lambda Z_i} \le e^{N\lambda^2 - \lambda u} = e^{-u^2/(4N)}.$$

The same argument applied to $-Z_i$ bounds $\mathbb{P}\{Y - \mathbb{E}Y < -u\}$, and adding the two bounds proves the lemma. □

Unless we say otherwise, the initial rotors are independent and uniform, and at each vertex $x \neq \mathbf{0}$ the edges $(x, \mathrm{par}(x))$ and $(x, 2x - \mathrm{par}(x))$ are half a turn apart in the cyclic order. The clockwise mechanism on the comb in $\mathbb{Z}^2$ has this property. The edge $(x, \mathrm{par}(x))$ then has a uniform position k in the cyclic order after the initial rotor, with $1 \le k \le \deg(x)$, and the edge $(x, 2x - \mathrm{par}(x))$ precedes it if and only if $k > \deg(x)/2$. Hence

$$\mathbb{P}\{(x, 2x - \mathrm{par}(x)) \text{ precedes } (x, \mathrm{par}(x))\} = \frac{1}{2}\,, \tag{11.13}$$

and these events are independent at distinct vertices.

Theorem 11.45 (Shape Theorem on the Comb) *Let $d \ge 2$ be an integer and $a > 0$ real. Run rotor walk on the comb in $\mathbb{Z}^d$ from the root, with independent uniform initial rotors, and suppose that at each vertex x other than the root the edges $(x, \mathrm{par}(x))$ and $(x, 2x - \mathrm{par}(x))$ are half a turn apart in the cyclic order. Then:*

(a) Almost surely every circuit is complete.

(b) There are constants $c, C > 0$, depending only on d and a, such that for every integer $n \ge 2$, with probability at least $1 - Cn^{-a}$,

$$B(\mathbf{0}, 2n - c\sqrt{n \log n}) \subseteq \mathsf{A}_n \subseteq B(\mathbf{0}, 2n + c\sqrt{n \log n})$$

and

$$\left| T(n) - \frac{2^{2d+1}}{(d+1)!} n^{d+1} \right| \le c n^{d+1/2} \sqrt{\log n}\,.$$

In other words, after n circuits the walk has visited, up to an error of order $\sqrt{n \log n}$, exactly the vertices within graph distance $2n$ of the root. This is twice the radius that Lemma 11.25 guarantees for arbitrary initial rotors. We prove both statements through the labels. By concentration, $\{\mathsf{h} \le n\}$ is finite and close to $B(\mathbf{0}, 2n)$, and by Lemma 11.43 it equals A_n.

Proof. (a) **Step 1:** We estimate the labels. Fix an integer $n \ge 2$. For $x \neq \mathbf{0}$, write $r := \|x\|_1$. By (11.10), $\mathsf{h}(x)$ is a sum of r independent $[0,1]$-valued variables, counting the constant 1. Apart from this constant and at most $d - 1$ changes of coordinate direction, these variables are fair coins by (11.13), so $|\mathbb{E}\mathsf{h}(x) - r/2| \le d$. Choose $C_1 > 0$ with $C_1^2 \ge 24(a + d)$, and let $b := C_1\sqrt{n \log n} + d$. For $1 \le r \le 6n$, Lemma 11.44 gives

$$\mathbb{P}\{|\mathsf{h}(x) - r/2| > b\} \le 2\exp\Big(-\frac{C_1^2 n \log n}{4r}\Big) \le 2n^{-C_1^2/24}\,.$$

The ball $B(\mathbf{0}, 6n)$ has $O(n^d)$ vertices, and the root satisfies the bound trivially, so a union bound gives

$$\mathbb{P}\Big\{ \max_{x \in B(\mathbf{0},6n)} \Big| \mathsf{h}(x) - \frac{\|x\|_1}{2} \Big| \le b \Big\} \ge 1 - Cn^{-a} . \tag{11.14}$$

Step 2: We show that on this event the first n circuits are complete. For large n we have $1 \le b \le n$. On the event in (11.14), every vertex of norm $6n$ has $\mathsf{h}(x) \ge 3n - b \ge 2n > n$, and since the labels are nondecreasing along paths from the root, $\{\mathsf{h} \le n\} \subseteq B(\mathbf{0}, 6n)$. By Lemma 11.43, the first n circuits are complete. Hence, for each fixed $m \ge 1$, the probability that $\{\mathsf{h} \le m\}$ is infinite is at most Cn^{-a} for every large $n \ge m$, so it is zero. A countable union over m shows that almost surely every set $\{\mathsf{h} \le m\}$ is finite, and Lemma 11.43 then shows that every circuit is complete. This proves (a).

(b) **Step 1:** We obtain the ranges of the first n circuits. On the event in (11.14), the identity (11.11) and the bound on h give, for $0 \le k \le n$,

$$B(\mathbf{0}, \max\{2k - 2b, 0\}) \subseteq \mathsf{A}_k \subseteq B(\mathbf{0}, 2k + 2b) .$$

For $k < b$ the inner inclusion only says that the root is visited; otherwise both inclusions follow by comparing $\mathsf{h}(x)$ with k. The case $k = n$ is the asserted shape estimate.

Step 2: We sum the lengths of the circuits. All the radii above are at most $4n$, so the count of $\#B(\mathbf{0}, R)$ and the mean value theorem for R^d give

$$\#\mathsf{A}_k = \frac{2^{2d}}{d!} k^d + O(bn^{d-1}) \qquad \text{for } 1 \le k \le n .$$

By (11.12) and $\sum_{k=1}^{n} k^d = n^{d+1}/(d+1) + O(n^d)$,

$$T(n) = \frac{2^{2d+1}}{(d+1)!} n^{d+1} + O(bn^d) .$$

Since $b = O(\sqrt{n \log n})$, this proves both bounds for all large n, and increasing C makes the asserted probability bound trivial for the finitely many remaining $n \ge 2$. □

The theorem also determines the size of the range at arbitrary times. Recall from Section 11.5 that $\mathsf{R}_t = \{X_0, \dots, X_t\}$ is the set of vertices visited by time t.

Corollary 11.46 *Let $d \ge 2$ be an integer. Run rotor walk on the comb in $\mathbb{Z}^d$ from the root, with independent uniform initial rotors, and suppose that at each vertex x other than the root the edges $(x, \mathrm{par}(x))$ and $(x, 2x - \mathrm{par}(x))$ are half a turn apart in the cyclic order. Then, almost surely,*

$$\frac{\#\mathsf{R}_t}{t^{d/(d+1)}} \longrightarrow \frac{(2(d+1)!)^{d/(d+1)}}{d!} \qquad \text{as } t \to \infty .$$

Proof. Apply Theorem 11.45 with $a = 2$. The probabilities Cn^{-2} are summable, so by the Borel–Cantelli lemma, almost surely the bounds in part (b) of the theorem apply for all large n, and the count of $\#B(\mathbf{0}, R)$ gives

$$\#\mathsf{A}_n = \frac{2^{2d}}{d!} n^d (1 + o(1)) \qquad \text{and} \qquad T(n) = \frac{2^{2d+1}}{(d+1)!} n^{d+1} (1 + o(1)) .$$

If $T(n) \le t < T(n+1)$, then $\mathsf{A}_n \subseteq \mathsf{R}_t \subseteq \mathsf{A}_{n+1}$ by (11.11). Consecutive range sizes and consecutive completion times have the same asymptotics, so squeezing $\#\mathsf{R}_t$ between $\#\mathsf{A}_n$ and $\#\mathsf{A}_{n+1}$ and dividing by $t^{d/(d+1)}$ gives the asserted limit. □

Thus on the comb the lower bound of order $t^{d/(d+1)}$ that Theorem 11.29 gives at the root cannot be improved (see Figure 11.20 for successive circuit ranges); for $d = 2$, the limiting constant in Corollary 11.46 is $12^{2/3}/2 = 18^{1/3} \approx 2.62$.

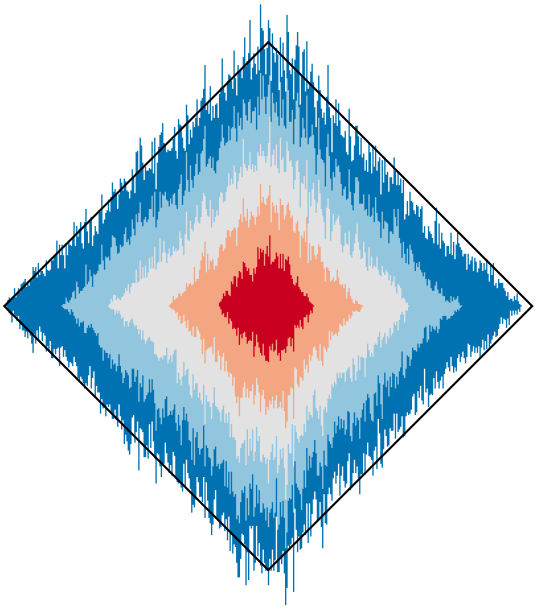


Figure 11.20: The set of sites visited by a rotor walk with uniform initial rotors on the comb in $\mathbb{Z}^2$ after 100 circuits. The vertices first visited in successive blocks of twenty circuits are colored from red (first block) to blue (last block), and the black outline is $B(\mathbf{0}, 200)$.

On a periodic graph such as $\mathbb{Z}^2$, a vertex can be reached along many paths, and the circuit in which it is first visited is no longer read off a single path. Still, the range of uniform rotor walk on $\mathbb{Z}^2$ with the clockwise mechanism has a deterministic limit shape, proved by a subadditivity argument; see the Notes.

EXERCISES

Exercise 11.1. Proposition 11.3 concerns the numbers of particles that reach a set; this exercise concerns numbers of visits. Let G be a finite connected undirected graph, regarded as a digraph by replacing each edge with its two orientations, and designate a vertex s as the sink. Fix a rotor mechanism, an initial rotor configuration ρ, and a non-sink vertex $o \in V(G)$. For every integer $n \geq 1$ and every non-sink vertex x, let $\ell_n(\rho, x)$ be the total number of visits to x made before absorption by n rotor walks starting at o, where the rotors are not reset between runs. Let $(X_k)_{k\geq 0}$ be simple random walk on G, and let τ_s be its hitting time of s. Show that

$$\left| \ell_n(\rho, x) - n\, \mathbb{E}_o \sum_{k=0}^{\tau_s - 1} \mathbf{1}\{X_k = x\} \right| \leq \deg(x)\, \mathbb{E}_x \tau_s \,,$$

for every $n \geq 1$ and every $x \neq s$, and deduce that $\ell_n(\rho, x)/n$ converges to $\mathbb{E}_o \sum_{k=0}^{\tau_s - 1} \mathbf{1}\{X_k = x\}$.

Exercise 11.2. Extend Lemma 11.16 from rotor configurations to rotor mechanisms. Let G be an infinite, connected, locally finite undirected graph, regarded as a digraph by replacing each edge with its two orientations. Show that if two rotor mechanisms π and $\widetilde{\pi}$ differ at only finitely many vertices, then for every rotor configuration ρ such that (π, ρ) is recurrent, $(\widetilde{\pi}, \rho)$ is also recurrent. Hint: If the two mechanisms differ only at a single vertex v, consider a multiple of $\deg(v)$ returns to v.

Exercise 11.3. Construct the boundaries used in Proposition 11.18. Let G be a locally finite undirected graph, regarded as a digraph by replacing every edge with its two orientations. Show that for every finite set of vertices $S \subset V(G)$, there exist a rotor configuration and mechanism so that S has a reflecting boundary.

Exercise 11.4. Show that for every infinite, connected, locally finite undirected graph G, regarded as a digraph by replacing every edge with its two orientations, there is a rotor mechanism and configuration for which the rotor walk is recurrent. (This is a weaker form of Lemma 11.19, in which the mechanism is arbitrary; whether every rotor mechanism admits such a configuration is an open question, see the Notes.)

Exercise 11.5. (Eulerian Walker) Let G be a finite connected undirected graph with at least one edge, viewed as an Eulerian digraph by replacing each edge with its two orientations. Fix a rotor mechanism and an initial rotor at every vertex, and start the walk at a vertex o. Show that the rotor walk is eventually periodic and that its eventual period traverses every directed edge exactly once.

Hint: Apply Lemma 11.4 to a longest block of consecutive distinct edges and to its shifts.

Exercise 11.6. Consider the rotor walk on $\mathbb{Z}^2$ started at the origin, with the clockwise rotor mechanism. Construct an initial rotor configuration for which, for every integer $n \geq 1$, the nth circuit visits exactly the diamond $B(\mathbf{0}, n) = \{x \in \mathbb{Z}^2 : \|x\|_1 \leq n\}$. Let $T(n)$ be the completion time of the nth circuit, and let R_t be the range through time t. Show that

$$T(n) = \frac{8}{3}n^3 + O(n^2) \qquad \text{and} \qquad \#\mathsf{R}_{T(n)} = 2n^2 + O(n)\,,$$

and deduce that

$$\frac{\#\mathsf{R}_{T(n)}}{T(n)^{2/3}} \longrightarrow \left(\frac{9}{8}\right)^{1/3}.$$

In particular, the exponent $2/3$ in Theorem 11.28 cannot be increased. Hint: Use the proof of Proposition 11.27 with $S_n = B(\mathbf{0}, n-1)$; at a vertex $y \neq \mathbf{0}$, the edges into $B(\mathbf{0}, \|y\|_1 - 1)$ are consecutive in the clockwise order.

Exercise 11.7*. This exercise proves Theorem 11.33. Consider independent bond percolation on $\mathbb{Z}^2$ in which each edge is open or closed with probability $\frac{1}{2}$ each. A **rectangle** is the subgraph of $\mathbb{Z}^2$ consisting of all vertices and edges, boundary edges included, in a product of two closed intervals with integer endpoints. A **crossing** of a rectangle is a path of open edges of the rectangle joining two opposite sides. For integers $m, n \geq 1$, let $\Pi(m, n)$ be the probability that $[0, m] \times [0, n]$ has a crossing joining its left and right sides.

(a) Let $\beta_1, \ldots, \beta_N$ be independent Bernoulli variables, and call an event determined by them **increasing** if changing a variable from 0 to 1 cannot destroy it. Prove the Harris inequality $\mathbb{P}(A \cap B) \geq \mathbb{P}(A)\mathbb{P}(B)$ for increasing events A and B. Deduce that $\mathbb{P}(A_1 \cap \cdots \cap A_k) \geq \mathbb{P}(A_1) \cdots \mathbb{P}(A_k)$ for every integer $k \geq 2$ and all increasing events $A_1, \ldots, A_k$.

(b) Call an edge of the dual lattice $\mathbb{Z}^2 + (\frac{1}{2}, \frac{1}{2})$ open if the edge of $\mathbb{Z}^2$ that it crosses is closed, and define crossings in the dual lattice in the same way. For $R := [0, m] \times [0, n]$ with integers $m, n \geq 1$, let R^* be the dual rectangle $[\frac{1}{2}, m - \frac{1}{2}] \times [-\frac{1}{2}, n + \frac{1}{2}]$, keeping only the duals of edges of R. Prove that either R has a crossing joining its left and right sides or R^* has a crossing joining its bottom and top, but not both. Deduce that $\Pi(n, n) \geq \frac{1}{2}$ for every integer $n \geq 1$.

(c) Prove that $\Pi(2m - n, 2n) \geq \Pi(m, 2n)^2/32$ for all integers $n \geq 1$ and $m \geq 2n$. Deduce that there is $c > 0$ such that every rectangle whose shorter side has length at least 2 and whose longer side is at most ten

times its shorter side has a crossing joining its two shorter sides with probability at least c. Hint for the inequality: Reflect in the line $y = n$ a vertical crossing of $[0, n]^2$ whose choice depends only on the edges on it and to its left.

(d) Let $\Theta(x, y) := (\frac{x-y}{2}, \frac{x+y}{2})$, which maps the half-dual lattice $\mathbb{L}$ of Section 11.6 onto a translate of $\mathbb{Z}^2$. Show that crossings of four rectangles of this translate that form a square annulus around the origin contain a simple cycle surrounding the origin. Use this and part (c) to prove Theorem 11.33, noting that percolation on $\mathbb{L}$ is unchanged when open and closed edges are exchanged.

Exercise 11.8*. Consider uniform rotor walk started at $\mathbf{0}$ on the F-lattice, with the odometer u_t of (11.4). Show that there is a constant C such that

$$\mathbb{P}\{u_t(\mathbf{0}) = 1\} \geq t^{-C} \qquad \text{for every integer } t \geq 2\,.$$

Deduce that if $a > C$ and $c > 0$, then the inequality of Theorem 11.35 fails for all large t. Hint: Put independent mirrors on a cylinder with $2t - 1$ columns and $t - 1$ rows, and note that an odd number of bottom ends cannot all be joined in pairs.

Exercise 11.9. In the setting of Section 11.7, $I(\rho, n)$ is the number of particles among the first n that never return to o, and $I_r(\rho, n)$ is the number that visit $\partial B(o, r)$ when the particles are stopped there. Consider the following incorrect "proof" of Lemma 11.39:

"Let r_1 be such that all particles which return to o do not leave $B(o, r_1)$, and let r_2 be such that all particles which escape to infinity never return to $B(o, r_1)$ after they leave $B(o, r_2)$. If $r \geq r_2$, then stopping particles when they hit $\partial B(o, r)$ has no effect on the rotors inside $B(o, r_1)$, and therefore has no effect on whether future particles return to o. Hence $I_r(\rho, n) = I(\rho, n)$ for all $r \geq r_2$." Find the mistake in the above "proof".

Exercise 11.10. Let G be an infinite, connected, locally finite undirected graph, regarded as a digraph by replacing each edge with its two orientations. Fix a rotor mechanism, a vertex o, and an initial rotor configuration ρ. Run rotor walks successively from o. After a transient walk escapes, start the next walk from the pointwise limiting rotor configuration left by the transient walk. If a walk is recurrent, then stop the experiment. For every integer $n \geq 1$, let $\ell_n(\rho, o)$ be the total number of visits to o made by the first n walks when all are transient, and let $\ell_n(\rho, o) = \infty$ otherwise. Show that

$$\liminf_{n\to\infty} \frac{\ell_n(\rho, o)}{n} \geq \mathbb{E}_o \sum_{k=0}^{\infty} \mathbf{1}\{X_k = o\}\,.$$

Here $(X_k)_{k\ge 0}$ is simple random walk on G started at o. Hint: This is a reformulation of Theorem 11.36.

Exercise 11.11. Extend Theorem 11.45 and Corollary 11.46 to cyclic orders in which the edges $(x, \mathrm{par}(x))$ and $(x, 2x - \mathrm{par}(x))$ are not half a turn apart. For both parts of the exercise, let $T(n)$ be the completion time of the nth circuit, A_n the nth circuit range, and R_t the range through time t.

(a) Run rotor walk on the comb in $\mathbb{Z}^2$ from the root, with independent uniform initial rotors and cyclic order north, east, west, south at every vertex of the x-axis. For every $0 < \epsilon < 1$, show that almost surely, for all sufficiently large n, the range A_n contains every lattice point of $(1-\epsilon)P_n$ and is contained in $(1+\epsilon)P_n$, where P_n is the quadrilateral with horizontal vertices $(4n, 0)$ and $(-\frac{4n}{3}, 0)$ and vertical vertices $(0, \pm 2n)$.

(b) Let $d \ge 2$ be an integer. Choose a cyclic order at the root, and suppose that the cyclic order at each $x \ne \mathbf{0}$ depends only on $j(x)$, the largest index with $x_{j(x)} \ne 0$, as a cyclic order of the directions $\pm\mathbf{e}_k$ with $k \ge j(x)$. Run rotor walk from the root with independent uniform initial rotors. For $1 \le j \le d$ and $s \in \{+, -\}$, let $m_{j,s}$ be the unique integer satisfying $1 \le m_{j,s} < 2(d - j + 1)$ such that $m_{j,s}$ advances of the rotor take $(x, \mathrm{par}(x))$ to $(x, 2x - \mathrm{par}(x))$ for every vertex x with $j(x) = j$ and x_j of sign s. Let

$$q_{j,s} := 1 - \frac{m_{j,s}}{2(d - j + 1)}\,.$$

For $x \in \mathbb{R}^d$ and $1 \le j \le d$, let s_j be the sign of x_j. Either sign may be used when $x_j = 0$ because $q_{j,s_j}|x_j| = 0$. Let

$$K := \left\{ x \in \mathbb{R}^d : \sum_{j=1}^{d} q_{j,s_j}|x_j| \le 1 \right\}.$$

Show that for every $0 < \epsilon < 1$, almost surely for all sufficiently large n,

$$(1 - \epsilon)nK \cap \mathbb{Z}^d \subseteq \mathsf{A}_n \subseteq (1 + \epsilon)nK \cap \mathbb{Z}^d\,.$$

Show also that

$$\mathcal{L}(K) = \frac{1}{d!} \prod_{j=1}^{d} \left(q_{j,+}^{-1} + q_{j,-}^{-1} \right)$$

and that, almost surely,

$$T(n) = \frac{2}{d+1}\mathcal{L}(K)n^{d+1}(1 + o(1)) \qquad \text{as } n \to \infty\,.$$

Deduce that, almost surely,

$$\frac{\#\mathsf{R}_t}{t^{d/(d+1)}} \longrightarrow \left(\frac{d+1}{2}\right)^{d/(d+1)} \mathcal{L}(K)^{1/(d+1)}.$$

Hint: In part (a), the east edge precedes the west edge with probability 3/4 at a vertex of the x-axis whose parent lies to the west.

NOTES

The rotor-router model was first introduced by Priezzhev, Dhar, Dhar, and Krishnamurthy (1996) in 1996 under the name Eulerian walkers, in connection with self-organized criticality. In the same year, Wagner, Lindenbaum, and Bruckstein (1996, Section 2.1) introduced ANT-WALK-1, which is the same walk when there is one walker and ties are broken by a fixed rule. The relation of the model to sandpiles was developed further in Section II of the later paper Povolotsky, Priezzhev, and Shcherbakov (1998). The model was rediscovered several times: by Rabani, Sinclair, and Wanka (1998) as an approach to load-balancing in multiprocessor systems, by Propp as a way to derandomize internal DLA, and by Dumitriu, Tetali, and Winkler (2003) (the whirling tours of their Section 9.2) as part of their analysis of a graph-based game; see the Introduction of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008). The Introduction of Florescu, Ganguly, Levine, and Peres (2014) states that Propp proposed rotor walk as a deterministic analogue of random walk in a lecture in 2003.

Much of Sections 11.1 and 11.2 is based on Sections 3–4 of the survey Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008): Lemma 11.2 is a form of their Lemma 3.9, and the proof of the matrix tree theorem (Proposition 11.14) through the action of the sandpile group on rotor configurations is their Corollary 3.18. Proposition 11.3 is due to Holroyd and Propp (2010): it is the finite-set, many-particle form of their Theorem 1; Proposition 3.21 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) proves the weaker bound with the sum over all edges and without the factor $\frac{1}{2}$, and Proposition 5.7 of Levine and Peres (2009), which they credit to Holroyd and Propp, gives the undirected case with the constant 1.

Section 11.3 is based on Angel and Holroyd (2012). Lemma 11.15 and Lemma 11.19 correspond to Theorems 1 and 2 of Angel and Holroyd (2012), and Exercise 11.4 is their Theorem 4. For a locally finite tree, Theorem 3 of Angel and Holroyd (2011) proves that the limiting proportion of escapes equals the random-walk escape probability when only finitely many rotors are positioned so that their next advance points toward the root. In

Open Question (i) of Angel and Holroyd (2012) they also raise the question whether there exist a graph and rotor mechanism such that for every initial rotor configuration the rotor walk is transient; equivalently, whether every rotor mechanism admits an initial rotor configuration for which the walk is recurrent, the question mentioned in Exercise 11.4.

Section 2 of Angel and Holroyd (2011) introduces live paths, and their Proposition 8 identifies the total number of escapes of successive particles started at the root of a tree with the number of live ends, that is, infinite paths from the root whose tails beyond some vertex are live paths. In Section 11.8 we use the same condition, together with the cancellation of opposite traversals, to prove recurrence.

Sections 11.4–11.6 are based on Florescu, Levine, and Peres (2016); our theorem and lemma numbers for that paper refer to arXiv:1408.5533. Proposition 11.27 is their Proposition 2.7, which they credit to Proposition 11 of Angel and Holroyd (2012). Theorem 11.29 is a corrected form of their Theorem 3.1, whose part (ii), $u_t(x) \le u_t(o) + \deg(x)$, fails at vertices of degree larger than $\deg(o)$. Theorem 11.28 is a polynomial-growth version of their Theorem 1.1, Theorem 11.30 is their Theorem 3.3, and Theorem 11.34 is their Theorem 1.5. Section 5 of Florescu, Levine, and Peres (2016) takes the vertices of $\mathbb{L}$ with $x + y$ even; with that choice a mirror parallel to a closed edge e_v would reflect the walk on the Manhattan lattice onto an incoming edge of v, so we take $x + y$ odd. The growth rates of the range on the two directed lattices suggested by experiment, at the end of Section 11.6, are reported at the end of their Section 5. Theorem 11.35 gives the bound for one small exponent a, which suffices for recurrence. It corresponds to their Theorem 5.4, which asserts the bound for every $a > 0$ on both lattices; Exercise 11.8 shows that on the F-lattice the bound fails for large a. The precursor Florescu, Ganguly, Levine, and Peres (2014) proves that on $\mathbb{Z}^2$ at most $(\pi/2 + o(1))\, n/\log n$ of the first n particles escape, for every initial rotor configuration (their Theorem 2), and that when every vertex sends its first particle in the direction $\mathbf{e}_d$, the number of escapes is at least of order $n/\log n$ in $d = 2$ and of order n in $d \ge 3$ (their Theorems 2 and 1). Theorem 1 of Cooper and Spencer (2006) shows that for Propp's P-machine, a rotor walk on $\mathbb{Z}^d$ in which all chips move at once, the number of chips at each site after n steps differs from its expected value for independent random walks by at most a constant depending only on d, for every initial configuration of chips on the even sites and every setting of the rotors. Early results on rotor-router aggregation in one dimension appear in Sections 2–3 of Levine (2004). Theorem 11.33 adapts the open-circuit bound (9.72) on p. 220 of Grimmett (1989), which he deduces from the Russo–Seymour–Welsh (RSW) theorem (9.70)–(9.71) on p. 219, together with its restatement (9.100) with an unspecified constant and its dual closed-circuit form (9.101) on p. 230, to

the annuli $Q(\mathbf{0}, 3\ell) \setminus Q(\mathbf{0}, \ell)$, which are rotated by 45° relative to the edges of $\mathbb{L}$; the adaptation is Exercise 11.7. Its part (b) corresponds to Lemma 3 and Corollary 5 of Bollobás and Riordan (2006). Its proof of the RSW estimate for rectangles follows their reflection argument in Lemma 6 and Corollary 7, which concern $2\rho n$ by $2n$ rectangles for integers $\rho > 1$; the exercise extends it to every rectangle of shorter side at least 2 and longer side at most ten times the shorter. The four rectangles of part (d) are those in their proof of Theorem 8 (numbers as in arXiv:math/0410359). The fact (P) used in the solution is the one that Bollobás and Riordan (2006) use in the proofs of Lemma 3 and Corollary 7, remarking that only the easy part of the Jordan curve theorem is needed.

Oded Schramm proved Theorem 11.36 for regular graphs while attending a talk in 2003 (unpublished). See Theorem 10 of Holroyd and Propp (2010) for an extension to irreducible Markov chains with rational transition probabilities; Theorem 11.36 is its specialization to simple random walk on a strongly connected, locally finite digraph. Theorem 1 of Angel and Holroyd (2011) also states the bound, for every graph with finite degrees. An alternative proof and generalization can be found in Lemma 5.1 of Chan (2019), which on undirected graphs bounds below the number of visits of successive rotor walks to each vertex; a special case of this is Exercise 11.10. The convergence conclusion of Exercise 11.1 is Lemma 3.5 of Chan (2019).

He (2014) gave examples of rotor configurations on $\mathbb{Z}^d$ for which the bound in Theorem 11.36 is sharp when $d \geq 3$: for a rotor mechanism with the same cyclic order at every vertex, in which, for some $i < d$, the directions $\pm\mathbf{e}_i$ separate $\mathbf{e}_d$ from $-\mathbf{e}_d$, he constructs an initial rotor configuration for which the proportion of escapes converges to the escape probability of random walk when $d \geq 3$ (his Theorem 1.3), and the number of escapes is $(\pi/2 + o(1))\, n/\log n$, the upper bound of Florescu, Ganguly, Levine, and Peres (2014), when $d = 2$ (his Theorem 1.2). Theorem 1 of Chan (2020) extends the existence of such a configuration to every connected, simple, locally finite transient graph. Theorem 1.4 of Chan (2021) proves that a rotor configuration sampled from the wired spanning forest has maximal escape rate when every tree of the wired spanning forest is almost surely one-ended.

In Florescu, Levine, and Peres (2016), the rotor walk on the comb graph was also investigated: the two-dimensional case of Theorem 11.45 is, up to the corrections below, their Theorem 4.1, proved by viewing the walk along the x-axis and each tooth as rotor walks on $\mathbb{Z}$. Our proof identifies the circuit ranges with the finite trees $\{\mathsf{h} \leq n\}$, then uses concentration and counts their edges; it applies in every dimension. Their Theorem 4.1 states diamonds of radius $n \pm c\sqrt{n \log n}$ at time $\frac{16}{3}n^3$, and states the limit $(3/2)^{2/3}$; the calculation in their Lemma 4.3 and in the proof of their Lemma 4.4 gives radius $2n \pm c\sqrt{n \log n}$ after n circuits, as in Theorem 11.45, and the limit

$18^{1/3}$ of Corollary 11.46.

Theorem 4.1 of Florescu, Levine, and Peres (2016) is stated for every rotor mechanism, but its proof uses the probability $1/2$ of (11.13) at every vertex of the x-axis, which holds only when at each such vertex x the edges $(x, \mathrm{par}(x))$ and $(x, 2x - \mathrm{par}(x))$ are half a turn apart in the cyclic order, as Theorem 11.45 assumes; Exercise 11.11 concerns other cyclic orders.

Sections 11.8 and 11.9 follow Sections 2, 4 and 5 of Bou-Rabee and Peres (2026). Proposition 11.40 is the recurrence part of their Proposition 2.8. Theorem 11.42 is the square-lattice case of the recurrence conclusion (i) of their Theorem 1.1, which also covers doubly periodic graphs in $\mathbb{R}^2$ of maximum degree three with a doubly periodic rotor mechanism. The infinite-graph case of Corollary 11.41 follows from their Proposition 4.1, which proves recurrence without periodicity by dominating the live paths that start with a given edge by a critical Galton–Watson tree; we instead compare the queue length of our exploration with a random walk on $\mathbb{Z}$. Figure 11.15 is reproduced from their paper.

On $\mathbb{Z}^2$ they show that there are constants $c, C > 0$ such that, for each directed edge and every integer $R \geq 1$, the probability that a live path starting with that edge reaches graph distance R from its first vertex is at most Ce^{-cR}, by comparing a dual configuration with subcritical percolation (their Proposition 5.1); for recurrence it suffices that no live path is infinite, and we compare the closed cycles of Theorem 11.33 with clockwise rotor cycles by replacing pairs of bits. By their Theorem 1.1, the range has a deterministic limit shape: there are a compact convex set $\mathfrak{B} \subset \mathbb{R}^2$, with $\mathbf{0}$ in its interior, and a constant $\lambda > 0$ such that, almost surely, $t^{-1/3}\mathsf{R}_t$ converges to $\lambda\mathfrak{B}$ in Hausdorff distance, and $\#\mathsf{R}_t/t^{2/3}$ converges to a deterministic positive constant, so the order $t^{2/3}$ of Theorem 11.28 is attained. Their proof bounds from below the index of the circuit in which each vertex is first visited by counting the vertices of a path at which the live condition fails, applies the subadditive ergodic theorem to obtain the limit shape of the circuit ranges, and sums the circuit lengths to pass from circuit number to elapsed time (their Section 3); we do not prove these results here. Earlier, Monte Carlo simulations of Kapri and Dhar (2009) had indicated that the set of sites visited by this walk on $\mathbb{Z}^2$ is asymptotically a disk of radius of order $t^{1/3}$; whether $\mathfrak{B}$ is a disk is Problem 3 of Bou-Rabee and Peres (2026).

A Preview of Volume II

Every shape theorem in this volume starts from a point source of mass, particles or grains at the origin of $\mathbb{Z}^d$ (or at two sites of $\mathbb{Z}^2$, in Chapter 7), and says how closely the region they fill approaches a ball (or Neumann's oval). In Volume II we go further in three directions. We spread the initial mass over many sites, so that the limit shape is found by solving a free boundary problem; we find the scaling limit of the abelian sandpile and prove that in the plane its limit shape is not a disk; and we show that the clusters of internal DLA and rotor-router aggregation deviate from a ball by at most a power of the logarithm of its radius.

In the first part we study the Dirichlet and obstacle problems in $\mathbb{R}^d$. Recall from Chapter 2 that the Dirichlet problem can fail to have a solution (on a punctured disk, no harmonic function takes the value 0 on the circle and 1 at the center). On every bounded domain the infimum of all superharmonic functions lying above the boundary data is a candidate solution (Perron's method). We prove that it is harmonic and that a boundary point is regular (this function attains the boundary value there for all continuous data) if and only if it has a barrier, a superharmonic function that tends to zero at the point and stays positive near the rest of the boundary. By the Beurling projection theorem of Chapter 2, every boundary point of a bounded simply connected planar domain has one. We also extend the convergence of discrete harmonic functions proved in Chapter 3, for limits with uniformly continuous second derivatives, to continuous boundary values on domains from which a random walk started near the boundary is unlikely to go far before leaving.

The obstacle problem comes next. In Chapter 4 the odometer of the divisible sandpile from a point source was the gap between an explicit obstacle and its least superharmonic majorant. For an initial density spread over $\mathbb{R}^d$ the same recipe defines an odometer, but the set where it is positive is no longer explicit. We prove that this set is bounded, and that if the density is continuous almost everywhere and takes no values strictly between a number $\lambda < 1$ and 1, then the boundary of this set has Lebesgue measure zero and mass is conserved, that is, the volume of this set equals the initial mass inside it.

In the second part we start the three models of Chapters 4–6 from a density spread over the lattice $\delta\mathbb{Z}^d$, whose spacing δ tends to zero. Their

odometers satisfy the same discrete Poisson equation up to an error (zero for the divisible sandpile, the divergence of a bounded field on the edges for rotor-router aggregation, and an error controlled by martingales for internal DLA) that is small in a negative Sobolev norm, so all three converge to the solution of the same obstacle problem; when the density is either zero or at least one at each point, the occupied sets converge as well (for internal DLA, in probability). For example, if one particle starts at every site of a set A and one at every site of a set B, then the limit is the smash sum $A \oplus B$, whose volume is the sum of the volumes of A and B. We also return to point sources in the plane. If k distinct points have positive masses, and a bounded open set has the property that defines the limit shape of Chapter 7 (the logarithmic potential of the sources minus that of the uniform measure on the set is nonnegative, and positive precisely on the set), then its boundary lies on an algebraic curve of degree $2k$ and on none of lower degree; for two sources it is the quartic of Neumann's oval or the pair of circles bounding two disjoint disks.

In Chapter 9 we developed the combinatorics of the abelian sandpile, and in Section 4.3 we showed that the sites toppled by n grains at the origin of $\mathbb{Z}^d$ lie between two Euclidean balls, but neither explains the patterns of the frontispiece. We prove the theorem of Pegden and Smart (2013) that the odometer, scaled by $n^{1/d}$ in space and divided by $n^{2/d}$, converges, and so do the final sandpiles in the sense of local averages. The limit obeys a constraint on its curvature. Wherever the limiting odometer is positive and twice differentiable, its Hessian lies on the boundary of the closed set of symmetric matrices whose quadratic forms, up to an affine function, lie below an integer-valued function on $\mathbb{Z}^d$ with Laplacian at most $2d - 1$ (the most grains a stable site can hold). Each periodic pattern of grains corresponds to a matrix on this boundary, and in the plane the boundary is described by an Apollonian circle packing, which explains the fractal patterns of the frontispiece. The same method gives the scaling limit of the identity element of the sandpile group of a large box.

Is the limit shape of the planar sandpile a disk? In the frontispiece, at the top, at the bottom and on both sides, the boundary runs straight for a stretch, and the triangle just inside each of these segments holds three grains at every site. We prove that both features persist in the limit. For n grains at the origin of $\mathbb{Z}^2$, with no other sand, the set where the limiting odometer is positive is convex. Near each of the four points where its boundary crosses a coordinate axis, the boundary is a segment of the line through that point perpendicular to the axis, and inside the set the limiting odometer equals three halves of the square of the distance to that line, so that the limiting density of grains there is 3. A disk has no straight pieces of boundary, so the limit shape is neither a disk nor an ellipse.

A second theorem concerns the sets of sites that topple when n grains are added at the origin of $\mathbb{Z}^d$ to a stable background of h grains at every site, where $h \le 2d-2$ is an integer (a negative h is a deficit of $|h|$ grains at every site). These sets, scaled by $n^{1/d}$, converge in Hausdorff distance to the set of points whose projection onto each coordinate subspace of dimension $\min(2d-1-h, d)$ lies in the projection of the support of the limiting odometer. For $h \le d-1$ this set is the support itself. For $d \le h \le 2d-2$ the boundary of the limit has flat pieces of dimension $h-d+1$, so the limit is again neither a ball nor an ellipsoid, and for $h = 2d-2$ it is a cube, as the toppled set is for every $n \ge 2$ by Proposition 9.36.

Following the topplings in time leads to a scaling limit of another kind. The parallel toppling procedure of Section 9.8 topples every unstable site once in each round. Start it from four grains at every site of the square $\{1, \dots, N\}^2$, and let grains that leave the square disappear. A site topples at most once per round, which suggests scaling time by N^2 and space by N, as for the heat equation. We prove that, after this rescaling, the number of times each site has toppled, divided by N^2, converges to a unique continuous function of time and position in the unit square. Which partial differential equation the limit solves is open.

In the third part we sharpen the shape theorems of Chapters 5 and 6. Let r be the radius of the ball of volume n. In Chapter 5 we showed that the internal DLA cluster of n particles deviates from this ball by at most a power of r smaller than one. We prove that, almost surely for all large n, the cluster lies between the balls of radii $r \pm C\sqrt{\log r}$ when $d \ge 3$, and between the balls of radii $r \pm C \log r \, (\log\log r)^{3/2}$ when $d = 2$, where C depends only on d. In the proof, a suitable function of the position of each walk is a martingale, and we control the deviation of the cluster from the ball by a large deviation estimate for the sum of these martingales. For rotor-router aggregation in $\mathbb{Z}^d$ with $d \ge 2$, we prove that, for all initial rotors and rotor mechanisms and all large n, the aggregate of n particles lies between the balls of radii $r \pm C \log r$, which removes a factor $\log r$ from the outer bound of Chapter 6.

In Chapter 11 we showed that rotor walk on $\mathbb{Z}^d$ visits at least of order $t^{d/(d+1)}$ sites in its first t steps, for all initial rotors, and that rotor walk with random initial rotors is almost surely recurrent when long live paths (at whose internal vertices the rotor points forward before back) are exponentially unlikely, as they are for independent uniform initial rotors on $\mathbb{Z}^2$ turning clockwise and on periodic graphs of maximum degree three. We prove that, on a periodic graph with independent initial rotors whose laws are periodic, the same condition implies that, almost surely, the set of sites visited by time t, scaled by $t^{1/(d+1)}$, converges to a deterministic convex body, so that the lower bound is attained. (The proof follows first-passage percolation, with the passage time between two sites replaced by the number of circuits that

the walk needs to get from one to the other.) For uniform initial rotors on $\mathbb{Z}^2$ this confirms the radius of order $t^{1/3}$ seen in the simulations of Kapri and Dhar (2009), which suggested that the visited set is close to a disk; whether the limit shape is a disk is open.

In a shorter chapter we survey, without full proofs, what changes when the lattice or the rule changes. The balls of this volume come from the symmetry of the walk rather than from the rule of aggregation. On a variant of $\mathbb{Z}^2$ called the layered square lattice, internal DLA and rotor-router aggregation both grow diamonds, and with suitably chosen rotors the rotor-router aggregate is exactly a diamond infinitely often. Internal DLA driven by a walk with a constant drift, rescaled differently along the drift, fills a heat ball, the region on which solutions of the heat equation obey a mean value property. Dissipation slows the growth of the planar sandpile from radius of order $\sqrt{n}$ to radius of order $\log n$, and the Manna model, in which each toppling sends grains to random neighbors, keeps its abelian property once the random choices are fixed in advance. The chapter ends with abelian networks, in which the abelian properties of Chapters 5, 9 and 11 become instances of one local-to-global principle.

Volume II ends with open problems. One of them concerns the two sandpiles of Figure 9.17, on the background of two grains in $\mathbb{Z}^2$ and of four grains in $\mathbb{Z}^3$. Except in a region near the origin and at a few sites, a slice through the origin of the sandpile formed by n grains on the second background appears to agree with a sandpile of suitably many grains on the first, each site of the slice holding two grains more; this is the dimensional reduction conjecture.

Solutions to Exercises

This chapter gives the solutions of the exercises that the text uses, and of a few others close to it, grouped by chapter. A separate document gives the solutions of all the exercises.

CHAPTER 1. MODELS OF LAPLACIAN GROWTH

1.3. (a) Given A_n, the next particle performs simple random walk from $\mathbf{0}$ until it first leaves A_n, that is, until it first visits $-a_n - 1$ or $b_n + 1$. The probability of visiting $-a_n - 1$ first is a discrete harmonic function of the starting point on A_n, equal to 1 at $-a_n - 1$ and to 0 at $b_n + 1$. On an interval of $\mathbb{Z}$ such a function is linear, so from $\mathbf{0}$ this probability is

$$\frac{b_n + 1}{a_n + b_n + 2} = \frac{b_n + 1}{n + 1} .$$

(b) Let $Y_n = a_n - b_n$. Since $a_n + b_n = n - 1$, part (a) gives

$$\mathbb{P}\{Y_{n+1} = Y_n + 1 \mid A_n\} = \frac{b_n + 1}{n + 1} = \frac{1}{2} - \frac{Y_n}{2(n + 1)} ,$$

and otherwise $Y_{n+1} = Y_n - 1$. The conditional mean of $Y_{n+1} - Y_n$ is therefore $-Y_n/(n + 1)$, and

$$\mathbb{E}[Y_{n+1}^2 \,\big|\, A_n] = Y_n^2 - \frac{2Y_n^2}{n + 1} + 1 = \frac{n - 1}{n + 1} Y_n^2 + 1 .$$

Thus $V_n = \mathbb{E}[Y_n^2]$ satisfies $V_{n+1} = \frac{n-1}{n+1} V_n + 1$. Since $A_1 = \{\mathbf{0}\}$ we have $V_1 = 0$ and hence $V_2 = 1 = (2 + 1)/3$, and if $V_n = (n + 1)/3$ for some $n \geq 2$, then $V_{n+1} = (n - 1)/3 + 1 = (n + 2)/3$. By induction, $V_n = (n + 1)/3$ for all $n \geq 2$.

CHAPTER 2. POTENTIAL THEORY IN $\mathbb{R}^d$

2.3. Let $m = \inf_K u \in [-\infty, \infty]$, and choose $x_n \in K$ with $u(x_n) \to m$. By compactness a subsequence converges to a point $x \in K$, and by lower

semicontinuity

$$u(x) \le \liminf_{n\to\infty} u(x_n) = m\,.$$

Since $m \le u(x)$ by definition, equality holds, and in particular $m > -\infty$.

2.4. Let $x_n \to x_0$. For each $f \in \mathcal{F}$, lower semicontinuity of f at x_0 and $f \le s$ give

$$f(x_0) \le \liminf f(x_n) \le \liminf s(x_n)\,.$$

Taking the supremum over $f \in \mathcal{F}$ gives $s(x_0) \le \liminf s(x_n)$.

2.5. The following argument proves both parts and does not use compactness. Let $\psi = \arctan \circ u$, with $\arctan(+\infty) = \pi/2$. Since arctan is continuous and increasing on $(-\infty, +\infty]$, the function $\psi : K \to (-\pi/2, \pi/2]$ is lower semicontinuous in the relative topology. For $n \ge 1$, let

$$f_n(x) := \inf_{y\in K}\{\psi(y) + n|x - y|\}\,.$$

Each f_n is n-Lipschitz, and $-\pi/2 \le f_n \le \psi$. Given $x \in K$ and $\epsilon > 0$, choose $\delta > 0$ with $\psi \ge \psi(x) - \epsilon$ on $K \cap \mathrm{B}(x, \delta)$. Since the terms with $|y - x| \ge \delta$ are at least $-\pi/2 + n\delta$, we have $f_n(x) \ge \psi(x) - \epsilon$ once $n\delta \ge \pi$, hence $f_n(x) \uparrow \psi(x)$. Moreover $f_n(x) > -\pi/2$, since otherwise there are points $y_k \in K$ with $\psi(y_k) + n|x - y_k| \to -\pi/2$, which forces $y_k \to x$ and $\psi(y_k) \to -\pi/2 < \psi(x)$, contradicting lower semicontinuity at x. Therefore

$$g_n := \tan(\min\{f_n, \pi/2 - 1/n\})$$

is continuous and real-valued on K, satisfies $g_n \le u$, and $g_n(x) \to u(x)$ for every $x \in K$, with $\tan(\pi/2) := +\infty$. The family $\{g_n : n \ge 1\}$ has the required property in both parts.

2.6. The function $f \circ h$ is continuous and hence locally integrable. For every ball $\mathrm{B}(y, r)$ in the neighborhood of Definition 2.2, the mean value property and Jensen's inequality give

$$(f \circ h)(y) = f(\mathbf{A}_r h(y)) \le \mathbf{A}_r(f \circ h)(y)\,,$$

so $f \circ h$ is subharmonic. The choices $f(t) = |t|$ and $f(t) = t^2$ give the two stated cases.

2.36. Let $x_0 \in \Omega_1$. For each $x \in \overline{\Omega}_1$, consider a path in Ω_2 from x_0 to x, and choose $\rho > 0$ so that the closed 4ρ-neighborhood of the path lies in Ω_2. We cover the path by a chain of balls. By uniform continuity there are finitely many successive points along the path, consecutive ones at distance less than $\rho/2$, so the ball of radius 2ρ about each point lies in Ω_2 and the next point

lies in its concentric ball of radius $\rho/2$. Applying Theorem 2.30 (scaled, with ratio $1/4$) to each ball along the path, the values of h at successive points are comparable. One more application in the ball of radius 2ρ about x gives a neighborhood U_x of x and a constant $C_x < \infty$ such that

$$C_x^{-1} h(x_0) \le h(y) \le C_x h(x_0) \qquad \text{for } y \in U_x \,.$$

Finitely many of the neighborhoods U_x cover $\overline{\Omega}_1$, and taking the largest corresponding constant gives $C_0 < \infty$, depending only on the two domains, such that $C_0^{-1} h(x_0) \le h \le C_0 h(x_0)$ on Ω_1. Therefore $\sup_{\Omega_1} h \le C_0^2 \inf_{\Omega_1} h$.

2.39. False: let

$$g(x) := d\omega_d (\mathcal{G}(x + \mathbf{e}_1) - \mathcal{G}(\mathbf{e}_1)) + x_1 \,.$$

By Corollary 2.34, g is superharmonic, since the added affine function is harmonic. Also $g(\mathbf{0}) = 0$, and $g(x) \to +\infty$ as $x \to -\mathbf{e}_1$. To bound g below, note that by (2.14) the function $t \mapsto d\omega_d \mathcal{G}(\sqrt{t}\mathbf{e}_1)$ has derivative $-\frac{1}{2} t^{-d/2}$ for $t > 0$, so it is convex. Its tangent line at $t = 1$, evaluated at $t = |x + \mathbf{e}_1|^2$, gives

$$g(x) \ge -\frac{1}{2}(|x + \mathbf{e}_1|^2 - 1) + x_1 = -\frac{1}{2}|x|^2$$

(also at the pole, where $g = +\infty$).

Now let $f(x) = |x|^2 + \min\{101, g(x)\}$. The truncation at 101 makes f finite and continuous, also at $-\mathbf{e}_1$, and by the lower bound $f(x) \ge |x|^2/2$ and $f(\mathbf{0}) = 0$. Since the minimum of two superharmonic functions is superharmonic, $f(x) - |x|^2$ is superharmonic. For sufficiently small $0 < s < 1$, the point $x = -(1 - s)\mathbf{e}_1$ lies in the unit ball and satisfies $g(x) \ge 101$, hence $f(x) = |x|^2 + 101 > 100|x|^2$.

2.40. True. Since f has a minimum at the origin, $\nabla f(\mathbf{0}) = \mathbf{0}$, so by Taylor's formula along the segment from $\mathbf{0}$ to x

$$f(x) = \int_0^1 (1 - t) x^T (D^2 f)(tx) x \, \mathrm{d}t \le |x|^2 \int_0^1 (1 - t) \, \mathrm{d}t = \frac{1}{2}|x|^2 \,.$$

2.42. Let $1 \le i \le n$ and consider the function

$$W(x) := q_i \sum_{j \ne i} q_j \mathcal{G}(x - y_j) \,,$$

which is harmonic in a neighborhood of y_i. If it had a local minimum at y_i, then by the strong minimum principle (Lemma 2.8) it would be constant in a ball about y_i. Since the kernels in (2.14) are real analytic away from their poles, W is real analytic on the connected open set $\mathbb{R}^d \setminus \{y_j : j \ne i\}$, so by

the identity theorem W would be constant on this set. But if $q_k \neq 0$ for some $k \neq i$, then

$$W(x) = q_i q_k \mathcal{G}(x - y_k) + O(1) \qquad \text{as } x \to y_k \,,$$

which is unbounded since $q_i q_k \neq 0$.

2.44. Let $B = \mathrm{B}(\mathbf{0}, r)$. If $|x| \geq r$, then by Lemma 2.33, integrating over B in spherical shells, we obtain

$$\mathbf{G1}_B(x) = \int_0^r d\omega_d s^{d-1} \mathcal{G}(x)\, \mathrm{d}s = \frac{r^2}{d(d-2)} \left(\frac{|x|}{r}\right)^{2-d} .$$

When $0 < |x| < r$, we have by Lemma 2.33

$$\begin{aligned} \mathbf{G1}_B(x) &= \int_0^{|x|} d\omega_d s^{d-1} \mathcal{G}(x)\, \mathrm{d}s + \int_{|x|}^r d\omega_d s^{d-1} \mathcal{G}(s\mathbf{e}_1)\, \mathrm{d}s \\ &= \frac{|x|^2}{d(d-2)} + \frac{r^2 - |x|^2}{2(d-2)} \\ &= \frac{r^2}{2(d-2)} - \frac{|x|^2}{2d} \,. \end{aligned}$$

At $x = \mathbf{0}$, direct integration gives $r^2/(2(d-2))$.

2.45. Let $B = \mathrm{B}(\mathbf{0}, r)$. If $|x| \geq r$, then by Lemma 2.33

$$\mathbf{G1}_B(x) = \pi r^2 \mathcal{G}(x) = -\frac{r^2}{2} \log |x| \,.$$

When $0 < |x| < r$, we have

$$\begin{aligned} \mathbf{G1}_B(x) &= \pi |x|^2 \mathcal{G}(x) + \int_{|x|}^r 2\pi s \mathcal{G}(s\mathbf{e}_1)\, \mathrm{d}s \\ &= -\frac{|x|^2}{2} \log |x| - \int_{|x|}^r s \log s \, \mathrm{d}s \\ &= \frac{r^2}{4}(1 - 2\log r) - \frac{|x|^2}{4} \,. \end{aligned}$$

Direct integration gives the same formula at $x = \mathbf{0}$.

2.46. With y_n and m_n as in Example 2.41, let

$$v(x) := \sum_{n=1}^{\infty} m_n \mathcal{G}(x - y_n) \,.$$

The assumption there implies $\sum_n m_n < \infty$. Given a compact ball D, since (y_n) is bounded there is a constant $C_D < \infty$ such that $\mathcal{G}(x - y_n) + C_D \geq 0$ for $x \in D$ and every n. By local integrability of $\mathcal{G}$,

$$\sup_n \int_D (\mathcal{G}(x - y_n) + C_D)\, \mathrm{d}x < \infty\,.$$

By Tonelli's theorem it follows that v is locally integrable, and by monotone convergence, applied to these nonnegative functions, the mean value inequality for v follows from the one for each term.

Since $\min(1, v)$ is at most 1 and at most v,

$$\mathbf{A}_r \min(1, v)(x) \leq \min(1, \mathbf{A}_r v(x)) \leq \min(1, v(x))\,,$$

so $u = \min(1, v)$ is superharmonic. Moreover v is lower semicontinuous, as used in Exercise 2.33, since the increasing functions

$$\sum_{n=1}^{N} m_n(\mathcal{G}(\cdot - y_n) + C_D)$$

are lower semicontinuous on D, and their limit differs from v by the finite constant $C_D \sum_n m_n$. Finally, $u(y_n) = 1$ for every n whereas $u(\mathbf{0}) < 1$, so u is discontinuous at the origin.

2.47. Let u be a lower semicontinuous superharmonic function on $\mathbb{R}$, let $a < b$, and let ℓ be affine with $\ell(a) \leq u(a)$ and $\ell(b) \leq u(b)$. The restriction of u to $[a, b]$ is lower semicontinuous on $[a, b]$ and superharmonic in (a, b), so by Lemma 2.12(b) we have $u \geq \ell$ on $[a, b]$. Thus u lies above its chords, so u is concave as an $\mathbb{R} \cup \{+\infty\}$-valued function.

By local integrability, u is finite almost everywhere. So if $u(x_0) = +\infty$ at some point x_0, we can choose $x_0 < z < y$ with u finite at z and y, and concavity applied to z (a strict convex combination of x_0 and y) would give $u(z) = +\infty$. Thus u is finite everywhere, hence continuous, since every finite concave function on $\mathbb{R}$ is continuous.

The one-sided derivatives of a concave function exist and are nonincreasing, and at each point of nondifferentiability the open interval between the right and left derivatives is nonempty. These intervals are pairwise disjoint, so choosing a rational number from each interval shows that there are at most countably many such points. Finally, $u(x) = -|x|$ is superharmonic but not differentiable at the origin.

2.48. Let $y \in \partial B$, and for $0 < r \leq 1$ let

$$h_r(x) := \mathcal{G}(x - ry) - \mathcal{G}(rx - y)\,.$$

The first term is harmonic in x away from ry, and for $r < 1$ the second is harmonic in x on a neighborhood of $\overline{B}$, since its only pole y/r lies outside $\overline{B}$. Moreover $h_1 = 0$, and since

$$\nabla \mathcal{G}(w) = -\frac{w}{d\omega_d |w|^d} ,$$

differentiating at $r = 1$ gives, for $x \in B$,

$$\left.\frac{\partial h_r(x)}{\partial r}\right|_{r=1} = -\nabla \mathcal{G}(x-y)\cdot(x+y) = \frac{|x|^2-1}{d\omega_d |x-y|^d} = -\frac{p(x)}{d\omega_d} .$$

For $r < 1$ sufficiently close to 1, the difference quotient $h_r/(r-1)$ is harmonic on each fixed compact subset of B, and by the formula for h_r these quotients converge locally uniformly to $-p/(d\omega_d)$ as $r \uparrow 1$. Applying Lemma 2.24 along a sequence $r_n \uparrow 1$ to these quotients on $\mathrm{B}(\mathbf{0}, r_n)$ and to their negatives, it follows that p is harmonic in B, so $\Delta p = 0$ in B.

CHAPTER 3. POTENTIAL THEORY IN $\mathbb{Z}^d$

3.5. The cross term vanishes, since $v = 0$ off A, so summation by parts over the edges meeting A gives

$$\sum_{\{x,y\}} (h(x)-h(y))(v(x)-v(y)) = -\sum_{x\in A} v(x)\Delta^{\ddot{}} h(x) = 0 .$$

If v has energy zero, then it is constant along every edge meeting A, so a path from a site of A to ∂A shows that $v = 0$.

3.8. In part (a) let $p = 1/2$, and in both parts let $q = \mathbb{P}_0\{X_\tau = r\}$. In both parts, $\ell + r$ consecutive positive increments force the walk to hit r, so by the Markov property

$$\mathbb{P}_0\{\tau > k(\ell+r)\} \le \left(1 - p^{\ell+r}\right)^k \longrightarrow 0 ,$$

so $\tau < \infty$ almost surely.

(a) For simple random walk, the stopped process $X_{t\wedge\tau}$ is a bounded martingale, so by optional stopping

$$0 = rq - \ell(1-q) ,$$

hence $q = \ell/(\ell+r)$.

(b) For the biased walk, let $\rho = (1-p)/p$. Then $\rho^{X_{t\wedge\tau}}$ is a bounded martingale, hence

$$1 = q\rho^r + (1-q)\rho^{-\ell} ,$$

which gives

$$q = \frac{1-\rho^{\ell}}{1-\rho^{\ell+r}}\,.$$

3.16. Let $\mathcal{S}_n = \sum_{j=1}^{n} 1/j$, so that

$$\sum_{j=1}^{n} \frac{1}{2j-1} = \mathcal{S}_{2n} - \frac{1}{2}\mathcal{S}_n\,,$$

and let

$$r_m := \mathcal{S}_m - \log m - \frac{1}{2m}\,.$$

By Taylor expansion $r_{m+1} - r_m = O(m^{-3})$, and since $r_m \to \gamma_E$, summing the increments from m to infinity gives

$$\mathcal{S}_m = \log m + \gamma_E + \frac{1}{2m} + O(m^{-2})\,.$$

Substituting, the terms of order n^{-1} cancel, leaving

$$\mathcal{S}_{2n} - \frac{1}{2}\mathcal{S}_n = \frac{1}{2}\log n + \log 2 + \frac{\gamma_E}{2} + O(n^{-2})\,.$$

CHAPTER 4. DIVISIBLE SANDPILE IN $\mathbb{Z}^d$: SINGLE SOURCE

4.2. If a toppling at x moves mass t to each neighbor, it changes the configuration μ to $\mu + t\Delta^{::}\delta_x$, and summing the finitely many nonzero terms gives

$$\sum_{y\in\mathbb{Z}^d} h(y)((\mu + t\Delta^{::}\delta_x)(y) - \mu(y)) = t\Delta^{::}h(x) \le 0\,,$$

so a toppling cannot increase the total weight $\sum_y h(y)\mu(y)$.

Let u be the odometer, which has finite support by Lemma 4.4(ii) (a divisible sandpile with finite initial support on an infinite graph stabilizes, with an odometer of finite support). Since $\sigma_\infty = \sigma + \Delta^{::}u$, by summation by parts

$$\sum_{x\in\mathbb{Z}^d} h(x)\sigma_\infty(x) - \sum_{x\in\mathbb{Z}^d} h(x)\sigma(x) = \sum_{x\in\mathbb{Z}^d} u(x)\Delta^{::}h(x) \le 0\,,$$

since $u \ge 0$ and $\Delta^{::}h \le 0$, which gives (1.1). If h is discrete harmonic, then applying this inequality to both h and $-h$ gives equality.

CHAPTER 7. INTERNAL AGGREGATION IN $\mathbb{Z}^2$: TWO SOURCES

7.2. The function h is smooth by Theorem 2.3. Choose a smooth function χ supported in the neighborhood where h is harmonic, with $\chi = 1$ near $\overline{\Omega(q)}$, and let $\psi = \chi h$, extended by zero outside that neighborhood. By Corollaries 2.39 and 2.40,

$$\int U_q(z)\Delta\psi(z)\,\mathrm{d}A(z) = \int_{\Omega(q)} h(z)\,\mathrm{d}A(z) - h(\mathbf{0}) - h(q)\,,$$

where all integrals exist since logarithmic singularities are locally integrable. The left side is zero, since $\Delta\psi = \Delta h = 0$ near $\overline{\Omega(q)}$ and $U_q = 0$ outside $\Omega(q)$ by Theorem 7.2, which proves the identity, including when $q = \mathbf{0}$.

CHAPTER 10. THE DIVISIBLE SANDPILE

10.9. Let $F(t) = U_t^{\sigma}(v)$. Since F is monotone, the set J of its discontinuities is countable. Let $a_s = F(s) - F(s^-) > 0$ for $s \in J$, and let

$$H(t) \coloneqq \sum_{\substack{s\in J\\ s\le t}} a_s\,.$$

For $a < b$, the increase $F(b) - F(a)$ is at least the sum of the jumps in $(a, b]$, so $F - H$ is nondecreasing, and since its jump at each point is zero, $F - H$ is continuous.

Suppose that $F - H$ is not constant, and choose $a < b$ such that $(F - H)(a) < (F - H)(b)$. Let

$$t \coloneqq \sup\{s \in [a, b] : (F - H)(s) = (F - H)(a)\}\,.$$

By continuity and monotonicity, $(F - H)(t + \epsilon) > (F - H)(t)$ for every $\epsilon > 0$, so by condition (ii) of Definition 10.4 there is a jump time $s_1 > t$, and given s_i, applying the same condition between t and $(t + s_i)/2$ gives a jump time s_{i+1} satisfying

$$s_i > s_{i+1} > t\,.$$

Taking $v_i = v$ contradicts condition (iii). Thus $F - H$ is constant, and since $F(0) = H(0) = 0$, we have $F = H$.

10.10. Let o and C_o be as in the exercise. To bound the number of such sequences from a vertex w, let $k = \mathrm{dist}(o, w)$ and fix a path from o to w of length k. Prepending this path maps distinct sequences $w = v_0, \ldots, v_N$ with

$v_j = v_{j-1}$ or $v_j \sim v_{j-1}$ to distinct such sequences of length $N+k$ from o, so there are at most $C_o^{N+k}(N+k)!$ of them.

Let I be an interval of length $a < 1/C_o$. For a fixed such sequence $w = v_0, \dots, v_N$, the expected number of times $t_1 > \cdots > t_N$ in I with t_j a ring of the clock at v_{j-1} is $a^N/N!$, by the factorial moment formula for Poisson processes. Summing over these sequences,

$$\begin{aligned}\mathbb{P}\{\text{a decreasing chain of length } N \text{ from } w \text{ in } I\} &\le C_o^{N+k}(N+k)!\frac{a^N}{N!} \\ &= C_o^k\frac{(N+k)!}{N!}(C_o a)^N .\end{aligned}$$

For fixed k the ratio $(N+k)!/N!$ grows polynomially in N while $(C_o a)^N$ decays geometrically, so the bound tends to zero, and almost surely there is no infinite decreasing chain from w inside I.

Taking the union over the countably many vertices w and the countably many intervals with rational endpoints of length less than $1/C_o$ proves condition (iii) of Definition 10.4 almost surely, since an infinite strictly decreasing sequence of positive times converges, so one of its tails lies in such an interval, and that tail has its first ring at a vertex w.

10.12. Suppose first that σ stabilizes, and choose w with $\beta = \sigma(w) - 1 > 0$. Since $1 + \beta\delta_w \le \sigma$, by Lemma 10.10 $u_\infty^{1+\beta\delta_w} \le u_\infty^\sigma < \infty$, so $1 + \beta\delta_w$ stabilizes by Theorem 4.3(iii), and hence by Proposition 10.16(b), simple random walk is transient.

Conversely, if the walk is transient, then the Green function is finite, so

$$f(x) := \sum_{w \in \operatorname{supp}(\sigma - 1)} (\sigma(w) - 1) g^{\cdot\cdot}(x, w)$$

is a finite nonnegative function, and since $\Delta_x g^{\cdot\cdot}(x, w) = -\delta_w(x)$ for every $w \in V$,

$$\sigma + \Delta f = 1 ,$$

so σ stabilizes by the least action principle.

10.19. (a) We have $\mathbb{E}U_1^\sigma(v) = q/\deg(v)$. At a spine vertex, the two spine neighbors contribute $q/4$ each and the two tooth neighbors contribute $q/2$ each, while the vertex emits $4(q/4)$, so its expected mass increases by $q/2$. At $(x, \pm 1)$, the expected mass received is $q/4 + q/2$, while the expected mass emitted is $2(q/2)$, giving a change of $-q/4$. Farther along a tooth, the expected masses received and emitted are both q.

(b) By part (a), $\mathbb{E}\sigma_1^U(v)$ is not identically $\mathbb{E}\sigma(v) = \mu$. Moreover, $\mathbb{E}\sigma_1^U(x, 0)/4 = \mu/4 + q/8$ and $\mathbb{E}\sigma_1^U(x, \pm 1)/2 = \mu/2 - q/8$, and since $q > 0$ these differ from their time-zero values $\mu/4$ and $\mu/2$.

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